PHRASEBOOK

The Language of Maths

for people who were never taught the words

$2 + 3 = 5$ ⟷ *two add three is five*

Before any maths: a promise

If maths has always looked like a wall of strange marks, that is **not** because you are bad at it. Maths is written in a language, a very squashed-up, shorthand language, and almost every book expects you to speak it already. Nobody ever sat you down and taught you the words.

This part does exactly that. Think of it as a **phrasebook** for a country you are about to visit. You do not need to be fluent. You only need to be able to read the signs, and to know what to say when someone points at one.

There are no sums to do in this part. There is only reading.

How the rest of this book works

- Each section starts with a boxed RULE (or a numbered PROCEDURE) that says exactly what to do, then worked examples, then a picture only if it helps. Every chapter ends with a rules page listing all its rules, and the Formula Pages at the back collect them all.
- In worked examples, each line changes only one thing. The part that changed is marked, and a handwritten note says why you are allowed to do it.
- Before you are asked to trust a rule, it is tested with real numbers so you can watch it work.
- This is a Xournal++ file, so write on it. Circle what you don't get. Highlight what you do.
- Take it slowly. Ten minutes on one line is normal. Mathematicians do it too.

Lesson 1 · Maths is shorthand

Here is a sentence in ordinary English. Read it slowly. Then follow the arrows: each coloured chunk gets turned into maths shorthand.

I think of a number, → x (x is a letter standing for the mystery number)

double it, → $2x$ (2x means 2 times x)

add three, → $+3$ (add 3)

and I get eleven. → $= 11$ (is the same value as 11)

Put the pieces together and the whole sentence becomes:

$$2x + 3 = 11$$

That is all an equation is: a sentence, written in shorthand. The maths version is shorter, but it says **exactly the same thing** as the English. Every time you see maths, there is an English sentence hiding inside it. Your job is only to find that sentence.

It works the other way too. Here is a piece of maths you might see in a shop:

$$C = p \times n$$

C — the total cost

p — the price of one

n — how many you buy

So $C = p \times n$ is read out as: "the total cost is the price of one thing times the number of things you buy". The letters are just the first letters of the words.

Why bother with shorthand? Because the long English sentences get very tiring very quickly, and because the short version is easier to move around. Later in this book you will learn to shuffle shorthand sentences around to find missing numbers. You cannot do that easily with paragraphs of English.

Lesson 2 · Doing things to numbers

Every line below is one phrase. The English is on the left, the **maths** is on the right, and underneath is how people actually *say it out loud*.

Put 3 and 4 together. $3 + 4$

say: three plus four

Start with 7 and take 2 away. $7 - 2$

say: seven minus two

Three lots of four (4 + 4 + 4). 3×4

say: three times four

The times sign has disguises. All of these mean 3 times 4: 3×4, $3 \cdot 4$, $3 * 4$ (on computers), $3(4)$ and $(3)(4)$.

Share 12 equally into 3 groups. $12 \div 3$

say: twelve divided by three

Division also has disguises: $12 \div 3$, $12/3$ and $\frac{12}{3}$. The line in a fraction **is** a division sign.

---

5 multiplied by itself (5 times 5). 5^2

say: five squared

The small raised 2 means "multiply two copies together": $5^2 = 5 \times 5 = 25$. It is **not** 5 times 2. (All of Chapter 9 is about these.)

Which number, times itself, makes 9? $\sqrt{9}$

say: the square root of nine

The answer is 3, because $3 \times 3 = 9$. So $\sqrt{9} = 3$. A square root **undoes** squaring.

Four below zero (like a temperature of minus 4 degrees). -4

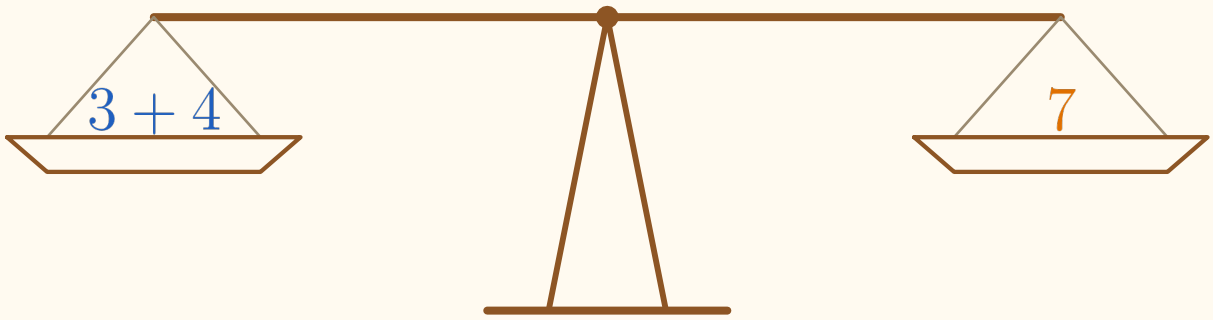
say: minus four / negative four

Watch out: the letter x and the times sign $\times$ look almost the same. That is why, once letters appear, people usually leave the times sign out. You will see this on the next pages.

Lesson 3 · The = sign is a balance

Most people learn at school that "=" means "and the answer is...". That idea causes a huge amount of confusion later on. Here is what it really means:

= means **"the thing on the left has exactly the same value as the thing on the right"**. It is a **balanced see-saw**.



same value on both sides, so it balances

Because it is a balance, it does not matter which side is which:

$3 + 4 = 7$ ✓ $\quad 7 = 3 + 4$ ✓ $\quad 3 + 4 = 2 + 5$ ✓

All three are true. The last one surprises people: there is no "answer" on the right at all, just two different ways of making 7. They balance, so the = sign is correct.

Why this matters later: when you rearrange formulas (Chapter 14) and solve equations (Chapter 15), the whole trick is to keep the see-saw balanced. Whatever you do to one side, you must do to the other side too.

The rest of the balance family

not the same value ..... $3 \neq 5$

say: three is not equal to five

smaller than ..... $3 < 8$

say: three is less than eight

bigger than ..... $8 > 3$

say: eight is greater than three

smaller than, or maybe equal to ..... $x \leq 10$

say: x is less than or equal to ten

bigger than, or maybe equal to ..... $x \geq 0$

say: x is greater than or equal to zero

roughly the same ..... $\pi \approx 3.14$

say: pi is approximately three point one four

A trick for < and >

The sign is a mouth. It always opens wide towards the bigger number and points its narrow tip at the smaller one. (It is "greedy": it wants to eat the bigger number.)

3 < 8

wide end at the 8

8 > 3

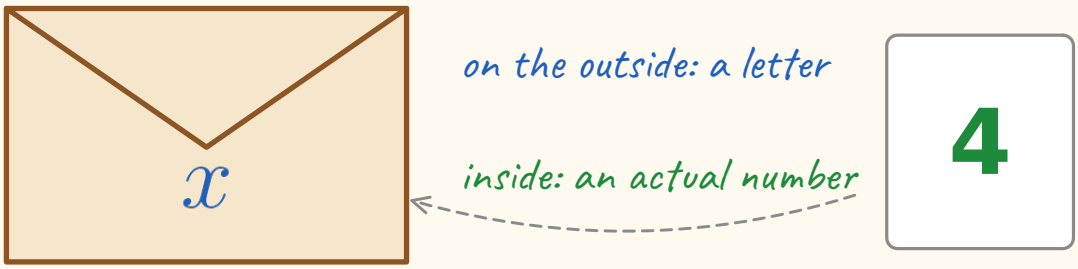
wide end at the 8

Your Lecture 1 slides also use three rarer members of the family:

- $1000000 \gg 2$ *say: is much greater than.* Far, far bigger.
- $0.001 \ll 5$ *say: is much less than.* Far, far smaller.
- $a + b \equiv b + a$ *say: is identically equal to.* Three lines instead of two means "this is not just true for one number, it is **always** true, whatever numbers you use".

This is the step that loses most people, so let's go very slowly.

A **letter** in maths is **a number wearing a disguise**. Picture a sealed envelope with a number written on a card inside. We cannot see the number, so we write a letter on the outside of the envelope to talk about it.



Letters do one of two jobs:

1. A mystery number. In $2x + 3 = 11$ there is exactly one number hiding in the envelope. (It's 4: double 4 is 8, add 3 is 11.) Finding it is called solving.

2. A number that can change. In $C = p \times n$ the letters change from shop to shop and day to day. The shorthand is a recipe that works whatever the numbers are. A letter used like this is called a variable (because it can vary).

How letters are written (the rules nobody tells you)

3 times x ---------- $3x$

say: three x

The times sign is left out. The number is written **first**: $3x$, never $x3$.

---

x times y ---------- xy

say: x y

Two letters side by side are multiplied.

---

x times x ---------- x^2

say: x squared

Not $2x$. Check with a number: if $x = 5$, then $x^2 = 5 \times 5 = 25$ but $2x = 2 \times 5 = 10$. Different!

Opening the envelope

If someone tells you what number is inside, you can replace the letter with that number and work out the value. This is called substituting. Let's say $x = 4$.

$3x \longrightarrow 3 \times 4 \longrightarrow 12$ *3x means 3 times x, and x is 4*

$x^2 \longrightarrow 4 \times 4 \longrightarrow 16$ *x squared means x times x*

$2x + 3 \longrightarrow 2 \times 4 + 3 \longrightarrow 11$ *double 4 is 8, then add 3*

$\frac{x}{2} \longrightarrow \frac{4}{2} \longrightarrow 2$ *4 shared into 2 is 2*

The orange 4s show where the envelope was opened. Nothing else changed. That will be true all through this book: orange marks the part of a line that just changed.

Why use letters at all?

Because a letter lets you say something is true **for every number at once**. For example, try adding two numbers in either order:

$$2 + 5 = 5 + 2 \qquad 10 + 1 = 1 + 10 \qquad 7 + 30 = 30 + 7$$

You could write examples forever. Or you can write it once, with letters:

$$a + b = b + a$$

and that one short line says: "whatever two numbers you pick, adding them in either order gives the same answer". The letters are envelopes that can hold **any** number.

Round brackets $( )$ are a fence around part of a calculation. Whatever is inside the fence gets worked out **first**, and then treated as a single number.

Watch the same three numbers give different answers:

$$(2+3)\times 4$$

fence first: 2+3 is 5

$$\downarrow$$

$$5\times 4$$

$$\downarrow$$

$$20$$

$$2+3\times 4$$

$$\downarrow$$

no fence: times goes before plus

$$2+12$$

$$\downarrow$$

$$14$$

Without a fence there is an agreed order ("times and divide before plus and minus"). The full set of rules, and the famous internet argument about $8 \div 2(2+2)$, are in Chapter 1.

A number touching a bracket means multiply

4 times the result of (2 + 3) ..... $4(2+3)$

say: four, bracket, two plus three

There is an invisible times sign between the 4 and the bracket: $4(2+3) = 4 \times (2+3) = 4 \times 5 = 20$.

3 times the result of (x + 1) ..... $3(x+1)$

say: three, bracket, x plus one

Same idea with a letter. (How to "open" a bracket like this is in Chapter 11.)

A **formula** is a shorthand recipe that connects some quantities with letters. You will meet lots of them. Here is how to read them. Say each one out loud as you read.

$$y = mx + c$$

y — m times x — plus c

say: *"y equals m x plus c" (m and x are touching, so they are multiplied)*

$$v = u + at$$

v — u — plus a times t

say: *"v equals u plus a t" (this one is from physics: speed after some time)*

A Greek letter or two will turn up. The first is π, said "pie". It is not a variable: it is one fixed number, about 3.14159, that has to do with circles (Chapter 5 explains where it comes from).

$$A = \pi r^2$$

A — pi times — r squared

say: *"A equals pie r squared" (the area of a circle with radius r)*

Two more reading tips

Small letters written low, like x_1 and x_2, are just name tags. x_1 is said "x one" and means "the first x". It is **not** x times 1.

A letter with a little mark, like a' ("a dash" or "a prime") or $\bar{x}$ ("x bar"), is a different quantity related to a or x: $\bar{x}$ is the mean of the x values (Chapter 24), and $f'(x)$ is the gradient function of f (Chapter 34). Each mark is explained when it first appears.

Formulas with layers: read from the inside out

Some formulas are built in layers, like an onion. Your Lecture 2 slides use this one, for how long a pendulum (a swinging weight on a string) takes to swing:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Don't read it left to right. Find the innermost piece and work outwards:

layer 1 (the middle): $\frac{l}{g}$ *"l divided by g"*

layer 2: $\sqrt{\frac{l}{g}}$ *"the square root of that"*

layer 3: $2\pi\sqrt{\frac{l}{g}}$ *"two pi times that"*

outside: $T = 2\pi\sqrt{\frac{l}{g}}$ *"T is equal to all of that"*

So the whole thing is read: *"T equals two pi times the square root of l over g"*.

You now know how to read it. How to rearrange it, for example to find the length of string needed for a given swing time, is exactly what Chapter 14 teaches, one layer at a time.

Lesson 7 · How to read a page of maths

Maths is not read like a novel. A single line can take five minutes, and that is normal: people who are good at maths read it slowly too. Here is a route to follow whenever a line looks scary. Walk it one stepping stone at a time.

1

Say it out loud in English.

Use the phrases in this part. If you can say it, you can start to understand it.

2

Find the words you don't know.

The Word Finder at the back of the book lists every maths word with the page where it is properly explained.

3

Find the rule first.

Each section opens with a boxed RULE or PROCEDURE; read it before the example. Every chapter's rules are listed together on its rules page, and the Formula Pages at the back gather them all, with tables of standard derivatives and integrals.

4

Put real numbers in.

Replace each letter with a small, easy number such as 2 or 3 and work it out. Watching what happens teaches you more than staring.

5

Compare each line with the one above.

Ask: what ONE thing changed? Why was that allowed? In this book the change is marked and the reason is written beside it.

6

Cover the answer and try.

Hide the rest of a worked example with a sheet of paper and uncover one line at a time. Guess the next line before you look.

7

Stuck? Circle it and move on.

Mark the line in red on this page. Understanding often arrives on the second or third visit, after your brain has slept on it.

Your map of the book

The book is a railway network. Each coloured line is a Part; each stop is a chapter. Chapters only use ideas from stops earlier on the network, so if you get lost you can always travel back. Part 4 only needs Parts 1 and 2, so you can jump to it early.

Part 0 (this part) is the ticket office. Everything starts here.

Got a lecture coming up?

Each of your NTU maths lectures, and the chapters of this book that prepare you for it. The numbers in circles are chapter numbers, listed in the order to read them. The colour tells you which Part of the book (and which line on the railway map) they are in.

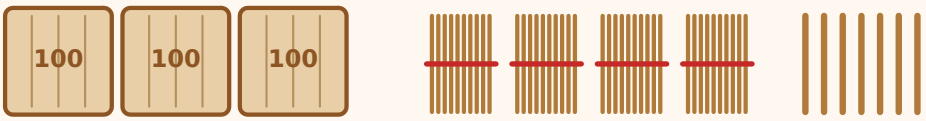
L01	Combining fractions and mathematical writing	read: 1 2 3
L02	Algebraic manipulation	read: 8 11 12 14
L03	Ratios and proportions	read: 4 15
L04	Data and descriptive statistics	read: 24
L05	Graphical analysis	read: 23
L06	Trigonometry	read: 27 28 29 30 31
L07	Powers, logs and graphing	read: 9 19 20 21 22
L08	Probability and Bayes	read: 25
L09	The Normal distribution	read: 26
L10	Vectors and scalars	read: 32
L11	Polar coordinates	read: 30 33
L12	Matrices and systems of equations	read: 15 16
L13	Functions and series	read: 6 17 18
L14	Differentiation and curves	read: 15 20 34
L15	Further differentiation	read: 35
L16	Introduction to integration	read: 36
L17	Further integration	read: 37

A note about your slides. A few slides contain small mistakes (typos in formulas). When a chapter meets one, it says so and shows the correct version. They are all collected in the Slide Errata at the back.

Chapter 1

Arithmetic of whole numbers

the four things you can do to numbers, done with sticks, lifts and dots



3 hundreds 4 tens 7 ones

347

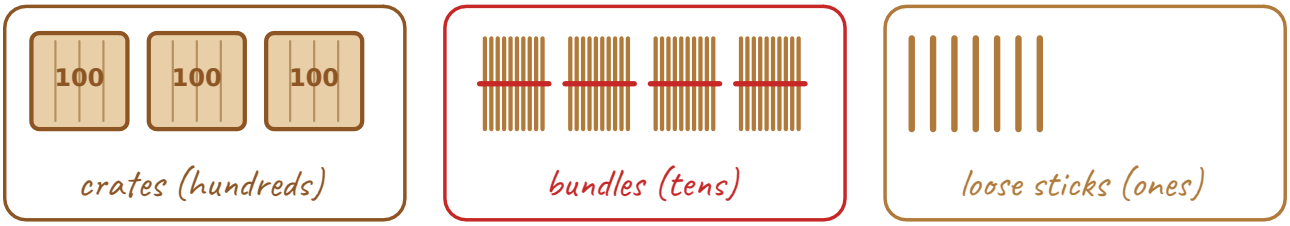
Everything in maths is built on four actions: adding, taking away, multiplying and dividing. You already do these every day with money. This chapter slows them right down so you can **see** what is happening inside them, because every later chapter, from fractions to calculus, is these four actions in disguise.

It also covers negative numbers, the order you do things in (the famous BODMAS), and the factor and multiple ideas that fractions (Chapter 2) depend on.

Goes with **Lecture 1** (mathematical writing and BODMAS).

Numbers are bundles of ten

Why does 347 mean three hundred and forty-seven? Imagine counting sticks. Every time you reach **ten sticks**, you tie them into a **bundle**. Every time you have **ten bundles**, you pack them in a **crate** (a crate holds 100 sticks).



A written number is just a report of what is in each box, in order: 3 crates, 4 bundles, 7 loose sticks is written 347. The **position** of a digit tells you its size. That is called **place value**.

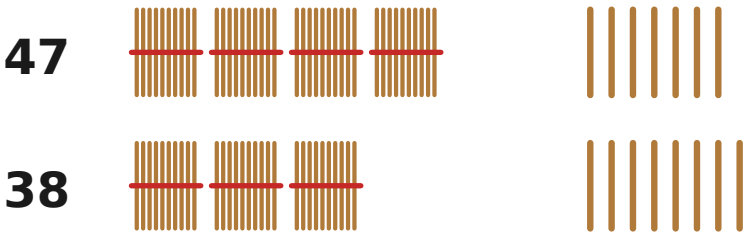
Zero is a place-holder. 307 means 3 crates, **no** bundles, 7 sticks. Without the 0 you'd write 37, which is a completely different number. Zero says "this box is empty".

A car mileage counter works the same way. When the ones wheel goes past 9, it rolls back to 0 and pushes the tens wheel on by one:

19 plus 1: the ones wheel rolls from 9 to 0 and the tens wheel clicks from 1 to 2. That "roll over and push the next one" is exactly what **carrying** means in adding.

Adding: put the piles together, re-bundle

Work out $47 + 38$. Put 47 sticks and 38 sticks side by side, each in its columns:



ones: 7 + 8 = 15 loose sticks. That's more than ten, so tie ten of them into a new bundle: 15 = 1 bundle and 5 sticks

Step by step, column by column, starting with the ones:

$$\begin{array}{r} {\scriptstyle 1} \\ 47 \\ +\,38 \\ \hline 85 \end{array}$$

1. ones column: 7 + 8 = 15. Write the 5, and carry the new bundle (1 ten) over to the tens column.

2. tens column: 4 + 3 = 7 bundles, plus the 1 carried bundle = 8 bundles. Write 8.

3. So 47 + 38 = 85 (8 bundles and 5 sticks).

Carrying is just re-bundling: ten ones become one ten, ten tens become one hundred. Adding also doesn't care about order: $47 + 38 = 38 + 47$.

Taking away: sometimes you must open a bundle

Work out $52 - 17$. Start with 52: five bundles and 2 loose sticks. You need to take away 7 loose sticks... but you only have 2!

52 = 5 bundles, 2 sticks

open one bundle: now 4 bundles and 2 + 10 = 12 loose sticks

$$\begin{array}{r} {\scriptstyle 4\ \ 12} \\ 5\!\!\not{2} \\ -\ 17 \\ \hline 35 \end{array}$$

1. ones: 2 - 7? Not enough. Open a bundle: the 5 tens become 4, the 2 ones become 12.

2. ones: 12 - 7 = 5. tens: 4 - 1 = 3.

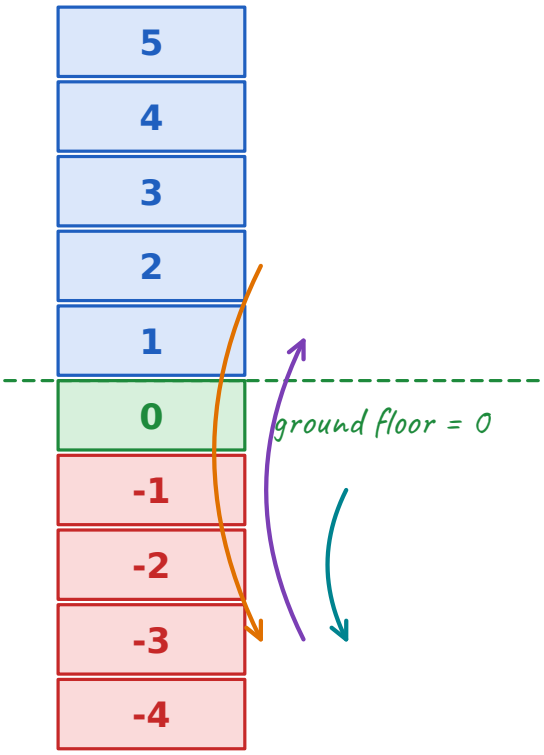
3. So 52 - 17 = 35.

Check by adding back: $35 + 17 = 52$. ✓ Taking away is the undo button for adding.

Order matters here! $52 - 17$ is not the same as $17 - 52$. (The second one goes below zero, which is the next idea.)

Below zero: the lift in a building

A lift has floors above the ground (1, 2, 3...) and basement floors below it. Call the ground floor 0 and the basement floors **negative**: -1, -2, -3 (said "minus one" or "negative one").



$$2 - 5 = -3$$

start at 2, go DOWN 5 floors

$$-3 + 4 = 1$$

start at -3, go UP 4 floors

$$-1 - 2 = -3$$

start at -1, go DOWN 2 floors

Adding a positive number moves you **up**; **taking away** a positive number moves you **down**. The rules below cover every other case. The **size** of a number is how far it is from 0, ignoring the sign: the size of -7 is 7.

RULE: adding and subtracting with negative numbers

Adding a negative is the same as subtracting: $a + (-b) = a - b$.
E.g. $7 + (-3) = 7 - 3 = 4$.

Subtracting a negative is the same as adding: $a - (-b) = a + b$.
E.g. $5 - (-2) = 5 + 2 = 7$.

Two signs, different: subtract the smaller size from the bigger size, and give the answer the **sign of the number with the bigger size**. E.g. $-9 + 4 = -5$ (9 is bigger, it was negative); $-3 + 8 = 5$ (8 is bigger, it was positive).

Both negative: add the sizes, the answer is negative. E.g. $-4 - 6 = -10$, $-4 + (-6) = -10$.

$7 + (-3) = 7 - 3 = 4$ ← + (−) becomes −

$-9 + 4 = -5$ ← sizes 9 and 4: 9 − 4 = 5; the 9 was negative

$-2 - (-5) = -2 + 5 = 3$ ← − (−) becomes +, then sizes 5 − 2 = 3, sign of the 5

$-4 + (-6) = -4 - 6 = -10$ ← both negative: add sizes, keep −

Why subtracting a negative adds. Think of money: -2 is a debt of £2. If someone **takes away** a debt you owe, you are **better off** by £2, so $5 - (-2) = 7$. Two minus signs **next to each other** make a plus.

Multiplying and dividing negatives

RULE: signs when multiplying and dividing

Work out the sizes as usual, then fix the sign.

Same signs give a positive answer: $(+) \times (+) = +$ and $(-) \times (-) = +$.

Different signs give a negative answer: $(+) \times (-) = -$ and $(-) \times (+) = -$.

Division follows exactly the same sign rules as multiplication.

Several numbers: count the minus signs. **Even** count gives $+$, **odd** count gives $-$. E.g. $(-2) \times (-3) \times (-1) = -6$ (three minus signs).

Signs	×	÷	Answer
$+$ and $+$	$3 \times 4 = 12$	$12 \div 3 = 4$	**positive**
$+$ and $-$	$3 \times (-4) = -12$	$12 \div (-3) = -4$	**negative**
$-$ and $+$	$(-3) \times 4 = -12$	$-12 \div 3 = -4$	**negative**
$-$ and $-$	$(-3) \times (-4) = 12$	$-12 \div (-3) = 4$	**positive**

Why is negative times negative positive? Watch a pattern. Each line below multiplies by one less than the line above, so the answer changes by 3 each time:

$$3 \times 2 = 6$$
$$3 \times 1 = 3$$
$$3 \times 0 = 0$$
$$3 \times (-1) = -3$$
$$3 \times (-2) = -6$$

down 3 each time ↓

$$-3 \times 2 = -6$$
$$-3 \times 1 = -3$$
$$-3 \times 0 = 0$$
$$-3 \times (-1) = 3$$
$$-3 \times (-2) = 6$$

UP 3 each time ↓

On the left the answers go 6, 3, 0, then keep going down: -3, -6. On the right they go -6, -3, 0, then keep going **up**: 3, 6. The pattern forces it: a negative times a negative is **positive**.

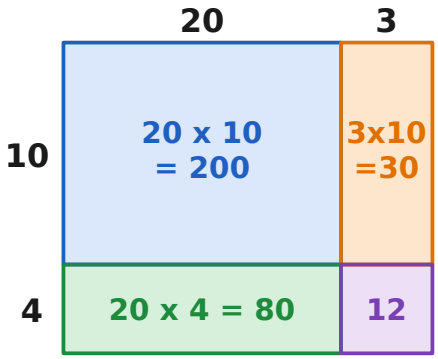
Careful: the "two minuses make a plus" rule for × and ÷ is about **two numbers**. For adding it is different: $-3 + (-4) = -7$, not $+7$.

Multiplying: rows of dots

4×6 means "4 lots of 6". Draw it as 4 rows of 6 dots: a rectangle.

4 rows of 6 = 24 dots.
Turn the page sideways and it's
6 rows of 4: still 24. Order doesn't matter.

Bigger numbers: split the rectangle. 23×14 is a rectangle 23 wide and 14 tall. Split 23 into 20 + 3 and 14 into 10 + 4. Now it's four easy rectangles (the **grid method**):



Add the four pieces:
200 + 30 + 80 + 12 = 322

$$23 \times 14 = 322$$

RULE: laws of multiplication (and which operations can be swapped)

Commutative: order does not matter for $\times$ and $+$: $a \times b = b \times a$, $a + b = b + a$. **It does matter for $-$ and $\div$:** $7 - 3 \neq 3 - 7$, $20 \div 4 \neq 4 \div 20$.

Distributive law: $a(b + c) = ab + ac$. Multiplying a sum means multiplying **each part** and adding. The grid method above is this law: $23 \times 14 = 23 \times 10 + 23 \times 4 = 230 + 92 = 322$.

Also works with minus: $a(b - c) = ab - ac$. E.g. $6 \times 19 = 6 \times 20 - 6 \times 1 = 114$.

This "split into easy rectangles" picture comes back in Chapter 11 , where it's used to multiply out brackets like $(x + 3)(x + 2)$.

Times tables. Quick recall of the times tables up to 10 x 10 makes everything later much less tiring. If they are shaky, the grid on the next page is worth pinning up.

The times-table square

Find a row and a column; where they meet is the answer. The diagonal holds the **square numbers** (1×1, 2×2, ...), which come up constantly later.

x	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

square numbers on the diagonal

Dividing: sharing out, and what's left over

$20 \div 4$ asks: **share 20 sweets between 4 people. How many each?** Deal them out one at a time, like cards:

Division is the undo button for multiplying: $4 \times 5 = 20$, so $20 \div 4 = 5$. The same sum can be asked the other way round: "how many groups of 4 can I make from 20?" (also 5).

RULE: division

Division undoes multiplication: if $a \times b = c$ then $c \div b = a$ and $c \div a = b$. Use this to check every division.

Order matters: $20 \div 4 = 5$ but $4 \div 20 = \frac{4}{20} = \frac{1}{5}$. Division is not commutative.

With a remainder: number = quotient × divisor + remainder, and the remainder is always smaller than the divisor. E.g. $23 = 5 \times 4 + 3$.

You can never divide by 0.

Remainders. Share 23 sweets between 4: each gets 5 (that uses 20), and **3 are left over**. We write $23 \div 4 = 5$ remainder 3, or "5 r 3".

Bigger division, sharing crate by crate ("short division"): $156 \div 6$.

$$\begin{array}{r} 26 \\ 6\overline{)\,15^{3}6} \end{array}$$

1. 6 into 1 (hundred)? Can't. Look at 15 (tens) instead.
2. 6 into 15 = 2, remainder 3. Write 2 on top.
   The 3 tens left over become 30 ones: write a small 3.
3. 6 into 36 = 6. Write 6. So 156 ÷ 6 = 26.

Check by multiplying back: $26 \times 6 = 120 + 36 = 156$. ✓

Dividing by zero is not allowed. "Share 5 sweets between 0 people" has no sensible answer. Calculators give an error, and so does maths.

The order of operations (BODMAS)

PROCEDURE: order of operations

1. **Brackets** first. With brackets inside brackets, work from the **innermost** outwards.
2. **Orders**: powers and roots (5^2, $\sqrt{9}$).
3. **Division and Multiplication**: **equal priority**. Work **left to right**, doing whichever comes first.
4. **Addition and Subtraction**: **equal priority**. Work **left to right**.

RULE: hidden brackets

A **fraction bar** acts as brackets round the whole top and the whole bottom: $\frac{4+8}{5-2} = (4+8) \div (5-2)$.

A power applies only to what it touches: $-3^2 = -(3^2) = -9$, but $(-3)^2 = (-3) \times (-3) = 9$.

Like getting dressed (socks before shoes), doing the steps in a different order gives a different result. The order is often remembered as **BODMAS**:

B **Brackets**
( ) first, always

O **Orders**
powers and roots, like 5^2 or $\sqrt{9}$

DM **Division and Multiplication**
equal rank: work left to right

AS **Addition and Subtraction**
equal rank: work left to right

(Some people learned **BIDMAS** or **PEMDAS**: same rules, different words.) Note that D and M are **equal**: do whichever comes first reading left to right. Same for A and S.

Example, one step per line: $3 + 4 \times 2^2 - (6 - 1)$

$$3 + 4 \times 2^2 - (6 - 1) \quad \longleftarrow \text{B: brackets first: } 6 - 1 = 5$$

$$3 + 4 \times 2^2 - 5$$ ← O: powers: 2 squared = 4

$$3 + 4 \times 4 - 5$$ ← M: multiply: 4 x 4 = 16

$$3 + 16 - 5$$ ← A/S left to right: 3 + 16 = 19

$$19 - 5 = 14$$ ← then 19 - 5

Equal priority means left to right:

$$12 \div 3 \times 2 = 12 \div 3 \times 2 = 4 \times 2 = 8$$

÷ comes first reading left to right

$$10 - 4 + 3 = 10 - 4 + 3 = 6 + 3 = 9$$

– comes first reading left to right

Nested brackets, innermost first: $2 \times [3 + (8 - 2)^2]$

$$2 \times [3 + (8 - 2)^2] = 2 \times [3 + 6^2]$$ ← inner bracket: 8 – 2 = 6

$$= 2 \times [3 + 36] = 2 \times 39 = 78$$ ← power, then the outer bracket

Fraction bar and powers of negatives:

$$\frac{4 + 8}{5 - 2} = \frac{12}{3} = 4$$ ← top and bottom first, then divide

$$(-3)^2 = 9, \qquad -3^2 = -9$$ ← the square touches –3 in the first, only 3 in the second

✗ Common mistakes

✗ $12 \div 3 \times 2 = 12 \div 6 = 2$ ✓ $12 \div 3 \times 2 = 4 \times 2 = 8$

× is not done before ÷: they are equal, so go left to right

✗ $10 - 4 + 3 = 10 - 7 = 3$ ✓ $10 - 4 + 3 = 6 + 3 = 9$

same for + and –: left to right

✗ $-3^2 = 9$ ✓ $-3^2 = -(3^2) = -9$

without brackets the power does not include the minus sign

The internet argument: 8 ÷ 2(2+2)

Your Lecture 1 slides show this famous puzzle. People get **16** or **1**, and argue for days. Here are the two readings side by side:

Reading A: strict left to right

$$8 \div 2(2+2)$$

$$8 \div 2 \times 4$$

$$4 \times 4$$

$$16$$

brackets first, then ÷ and x left to right

Reading B: "2(2+2)" is one lump

$$8 \div 2(2+2)$$

$$8 \div (2 \times 4)$$

$$8 \div 8$$

$$1$$

treat 2(4) as a single number, because it's glued

The honest answer: the expression is **badly written**. The rules above give 16, but plenty of calculators and scientists treat "$2(2+2)$" as glued together and give 1. The lesson from your lecture is: **when you write maths, remove the doubt.** Use extra brackets, and use a fraction line instead of ÷:

$$\frac{8}{2}(2+2) = 16 \qquad \frac{8}{2(2+2)} = 1$$

Nobody can argue with either of those. That's why, from now on, this book (and your lecturer) almost never uses the ÷ sign once letters appear.

Factors: every way to make a rectangle

A **factor** of 12 is a number that divides into 12 exactly, with no remainder. Here's a picture of it: take 12 dots and find **every rectangle** you can make with them.

1 x 12

2 x 6

3 x 4

(4 x 3, 6 x 2 and 12 x 1 are the same rectangles turned sideways)

The side lengths are the factors: **1, 2, 3, 4, 6, 12**. Factors come in pairs that multiply to 12: 1 & 12, 2 & 6, 3 & 4.

Primes are the numbers bigger than 1 that **only** make a single line of dots:

7 dots: only 1 x 7 works. No other rectangle. 7 is prime.

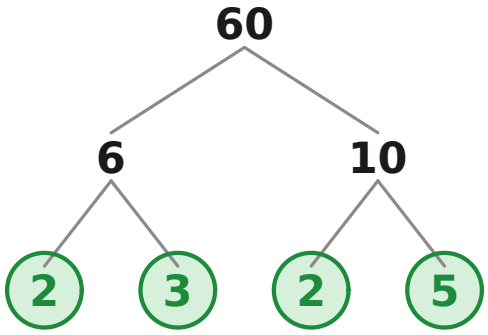
8 dots makes 2 x 4: 8 is NOT prime

The first primes are **2, 3, 5, 7, 11, 13, 17, 19, 23, 29**. (1 is not counted as prime.) A whole number bigger than 1 that isn't prime is called **composite** (it has a factor other than 1 and itself). 0 and 1 are neither prime nor composite.

Multiples are the times table of a number: the multiples of 4 are 4, 8, 12, 16, 20... "12 is a multiple of 4" and "4 is a factor of 12" say the same thing from opposite ends.

Breaking a number into primes: a factor tree

Every whole number bigger than 1 can be built by multiplying primes together, and in only one way. To find the primes, keep splitting into any factor pair until only primes are left. Take 60:



60 = 6 x 10

6 = 2 x 3, 10 = 2 x 5

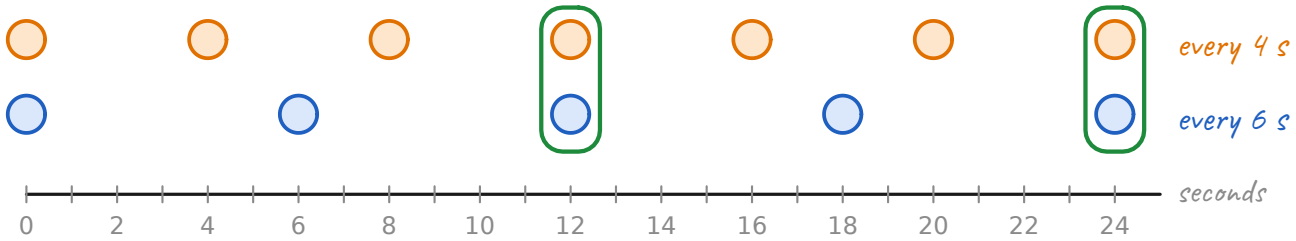
circled numbers are prime: stop

$$60 = 2 \times 2 \times 3 \times 5$$
$$= 2^2 \times 3 \times 5$$

You could start with $60 = 4 \times 15$ instead; you'd still end with the same primes. This list of primes is the number's **prime factorisation**.

LCM: when do two blinking lights blink together?

One traffic light blinks every 4 seconds; another blinks every 6 seconds. They both blink at the start. When do they next blink **at the same time**?



Together at **12 seconds** (and again at 24, 36...). 12 is the smallest number in **both** times tables: the **Lowest Common Multiple**, or **LCM**, of 4 and 6.

Chapter 2 uses the LCM to find the smallest piece-size two fractions can be cut into.

HCF: the biggest number that fits into both

The **Highest Common Factor** (HCF) of 12 and 18 is the **biggest** number that divides into both. List the factors and find the biggest one they share:

Factors of 12: 1, 2, 3, 4, **6**, 12

Factors of 18: 1, 2, 3, **6**, 9, 18

HCF = 6

Chapter 2 uses the HCF to simplify fractions in one step: $\frac{12}{18}$ becomes $\frac{2}{3}$ by dividing top and bottom by 6.

Don't mix them up: **M**ultiples get bigger (LC**M** is at least as big as both numbers). **F**actors get smaller (HC**F** is no bigger than either number).

HCF and LCM from prime factorisations

Listing factors or multiples is slow for big numbers. Prime factorisations give both in one go.

PROCEDURE: HCF and LCM using primes

1. Write each number as a product of primes, using powers: $24 = 2^3 \times 3$, $36 = 2^2 \times 3^2$.
2. **HCF**: take only the primes that appear in **every** number, each with its **lowest** power.
3. **LCM**: take **every** prime that appears in any of the numbers, each with its **highest** power.
4. Multiply out. Check: for two numbers, $\text{HCF} \times \text{LCM} =$ the two numbers multiplied together.

Example: HCF and LCM of 24 and 36.

$$24 = 2^3 \times 3, \qquad 36 = 2^2 \times 3^2$$ ← *factor trees*

$$\text{HCF} = 2^2 \times 3^1 = 12$$ ← *shared primes 2 and 3, lowest powers*

$$\text{LCM} = 2^3 \times 3^2 = 8 \times 9 = 72$$ ← *all primes, highest powers*

$$12 \times 72 = 864 = 24 \times 36 \checkmark$$ ← *the check*

Word problems: which one do I need? "Biggest thing that fits into both" (cutting, sharing into equal groups) is the **HCF**. "When do they next happen together" (timetables, repeating patterns) is the **LCM**.

Example. A floor is 84 cm by 60 cm. It is covered exactly with identical square tiles, as large as possible. What size are the tiles?

$$84 = 2^2 \times 3 \times 7, \qquad 60 = 2^2 \times 3 \times 5$$

tile side must divide both: HCF

$$\text{HCF} = 2^2 \times 3 = 12 \text{ cm}$$ ← *84 ÷ 12 = 7 tiles by 60 ÷ 12 = 5 tiles*

Example. Buses leave every 12 minutes and trams every 18 minutes. Both leave at 9:00. When do they next leave together?

$$12 = 2^2 \times 3, \quad 18 = 2 \times 3^2 \;\Rightarrow\; \text{LCM} = 2^2 \times 3^2 = 36$$

"together again": LCM

$$9{:}00 + 36 \text{ min} = 9{:}36$$

Find the missing number

PROCEDURE: finding an unknown

1. Write the unknown as $?$ (later in the book: a letter such as x).
2. **Undo** what was done to it, using the **inverse** operation: $+$ is undone by $-$, $-$ by $+$, $\times$ by $\div$, $\div$ by $\times$.
3. Apply the sign rules for negatives as you go.
4. **Check** by putting your answer back into the original line.

Example. $? - 7 = -3$

$$? = -3 + 7 = 4$$ ← undo − 7 by adding 7

$$4 - 7 = -3 \checkmark$$ ← check

Example. $-4 \times ? = 28$

$$? = 28 \div (-4) = -7$$ ← undo × (−4) by dividing; different signs: negative

$$-4 \times (-7) = 28 \checkmark$$ ← same signs: positive

Example. $? \div 3 = -6$

$$? = -6 \times 3 = -18$$ ← undo ÷ 3 by multiplying by 3

Example. The HCF of 12 and n is 4, and n is from 20 to 30. What could n be?

$$4 \mid n: \quad n \in \{20, 24, 28\}$$ ← 4 must divide n: multiples of 4 in range

$$\text{HCF}(12, 24) = 12 \neq 4$$ ← 24 shares the factor 3 with 12 as well: rule it out

$$\text{HCF}(12, 20) = 4, \quad \text{HCF}(12, 28) = 4 \Rightarrow n = 20 \text{ or } 28$$

(The "|" sign means "divides exactly": $4 \mid 20$ reads "4 divides 20".)

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Place value	each column is worth 10 times the column to its right; 0 holds an empty place
Adding a negative	$a + (-b) = a - b$ e.g. $7 + (-3) = 4$
Subtracting a negative	$a - (-b) = a + b$ e.g. $5 - (-2) = 7$
Different signs (+, −)	subtract the sizes; the answer takes the sign of the number with the bigger size: $-9 + 4 = -5$
× and ÷ signs	$(-a) \times (-b) = ab, \quad (-a) \times b = -ab$ same signs give $+$, different signs give $-$; the same for $\div$
Order of operations	brackets (innermost first), powers, × and ÷ left to right, + and − left to right
Fraction bar	$\dfrac{a+b}{c+d} = (a+b) \div (c+d)$ the bar acts as brackets
Powers of negatives	$(-3)^2 = 9, \qquad -3^2 = -9$
Commutative	$a + b = b + a, \quad a \times b = b \times a$ **not** for $-$ or $\div$
Distributive law	$a(b + c) = ab + ac$
Division	$a \times b = c \iff c \div b = a$ never divide by 0
Remainder	number = quotient × divisor + remainder (remainder < divisor)
Prime factorisation	every whole number above 1 is one product of primes, e.g. $60 = 2^2 \times 3 \times 5$
HCF	primes common to all the numbers, lowest powers
LCM	every prime that appears, highest powers
HCF and LCM	$\mathrm{HCF}(a, b) \times \mathrm{LCM}(a, b) = a \times b$
Find an unknown	undo the operation with its inverse, then check

Number detective: your turn

Try each one on paper or straight onto this page. Answers are underneath each group, written small and upside-down... well, small and grey. Cover them with your hand!

Bundles

1. What does the 6 in 3602 stand for?
2. 68 + 57
3. 81 - 36

1. 6 hundreds (600) 2. 125 (ones: 8+7 = 15, carry 1) 3. 45 (open a bundle: 11 - 6, 7 - 3)

The lift

4. -4 + 9
5. 3 - 8
6. -2 - (-5)
7. (-4) x (-5)

4. 5 5. -5 6. 3 (taking away a debt) 7. 20 (same signs: positive)

Getting dressed

8. 10 - 2 x 3
9. (10 - 2) x 3
10. $2 + 3^2$

8. 4 (multiply first) 9. 24 (brackets first) 10. 11 (power first: 9, then + 2)

Dots and lights

11. All factors of 18
12. Is 21 prime?
13. LCM of 6 and 8
14. HCF of 24 and 36

11. 1, 2, 3, 6, 9, 18 12. No: 21 = 3 x 7 13. 24 14. 12

Signs, order and primes

15. $-6 + (-9)$
16. $-12 \div 3 \times 2$
17. $(-5)^2$ and -5^2
18. $2 \times [10 - (1 + 2)^2]$
19. $(18 - 3) / (2 + 3)$
20. HCF and LCM of 18 and 24, using prime factors

15. -15 (both negative: add sizes) 16. -8 (left to right: -4×2) 17. 25 and -25 18. 2 (inner bracket 3, then $10 - 9 = 1$) 19. 3 ($15 \div 5$) 20. $18 = 2 \times 3^2$, $24 = 2^3 \times 3$: HCF $= 2 \times 3 = 6$, LCM $= 2^3 \times 3^2 = 72$

Find the unknown

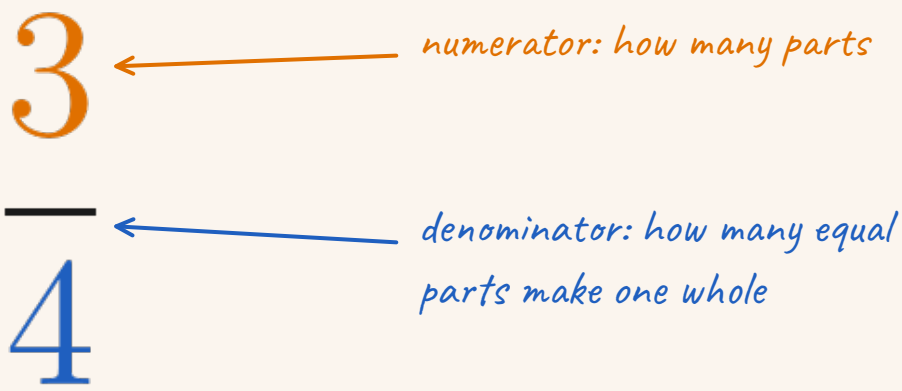
21. $? - 7 = -3$
22. $-4 \times ? = 28$
23. $? \div (-6) = 5$
24. $15 + ? = 9$
25. The HCF of 12 and n is 4, and n is from 20 to 30. Find n.
26. The LCM of 6 and n is 30, and n is less than 10. Find n.

21. 4 ($? = -3 + 7$) 22. -7 ($? = 28 \div (-4)$) 23. -30 ($? = 5 \times (-6)$) 24. -6 ($? = 9 - 15$) 25. 20 or 28 (multiples of 4; 24 is out because HCF(12, 24) = 12) 26. 5 ($30 = 2 \times 3 \times 5$ and $6 = 2 \times 3$, so n must supply the 5; 10 is too big)

Chapter 2

Fractions

the rules for each operation, and why they work



Everything you do with two fractions comes from **three formulas**: one for adding and subtracting, one for multiplying, one for dividing. They are on the next pages. Put your numbers in for a, b, c, d and work it out; the rest of the chapter is examples.

Goes with **Lecture 1** (Combining fractions). Uses factors, HCF and LCM from Chapter 1.

What a fraction means

DEFINITION: reading a fraction

$\frac{a}{b}$ means: split one whole into **b equal parts** and take **a** of them.

a is the **numerator** (top). **b** is the **denominator** (bottom). b can never be 0.

The fraction line **is** a division: $\frac{a}{b} = a \div b$. So $\frac{12}{3} = 4$ and $\frac{3}{4} = 0.75$.

Any whole number can be written over 1: $5 = \frac{5}{1}$.

$$\frac{3}{4}$$

3 of 4 equal parts

The parts must be **equal**. The denominator gives the parts their name: halves, thirds, quarters, fifths... The numerator counts how many of those parts you have.

- **Proper fraction**: top smaller than bottom, so less than 1 (e.g. $\frac{3}{4}$).
- **Improper fraction**: top bigger than bottom, so more than 1 (e.g. $\frac{7}{4}$).
- **Mixed number**: a whole number plus a proper fraction (e.g. $1\frac{3}{4}$, "one and three quarters").

The three formulas

These are the whole lesson. Put your numbers in for a, b, c, d and work it out.

THE RULES

Add / subtract

same for + and −

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

Multiply

multiply across

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Divide

flip the 2nd, then multiply across

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

With numbers ($a = 2$, $b = 3$, $c = 4$, $d = 5$: Lecture 1's examples):

$$\frac{2}{3} + \frac{4}{5} = \frac{2 \times 5 + 3 \times 4}{3 \times 5} = \frac{22}{15}$$ ← *ad + bc over bd*

$$\frac{2}{3} - \frac{4}{5} = \frac{2 \times 5 - 3 \times 4}{3 \times 5} = -\frac{2}{15}$$ ← *ad − bc over bd*

$$\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$$ ← *ac over bd*

$$\frac{2}{3} \div \frac{4}{5} = \frac{2 \times 5}{3 \times 4} = \frac{10}{12}$$ ← *ad over bc*

INPUTS: getting the numbers into a/b form

A whole number goes over 1: $4 = \frac{4}{1}$.

A mixed number becomes an improper fraction first: $2\frac{1}{3} = \frac{2\times3+1}{3} = \frac{7}{3}$.

Simplifying the answer is an optional tidy-up: $\frac{10}{12}$ and $\frac{5}{6}$ are the same number. Do it when a question asks for the simplest form.

Where the add/subtract formula comes from

Multiplying a fraction's top and bottom by the same number doesn't change its value. So multiply the first fraction by $\frac{d}{d}$ and the second by $\frac{b}{b}$:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad}{bd} + \frac{bc}{bd} = \frac{ad + bc}{bd}$$

Both fractions are now over bd, so their tops add. That's the whole derivation. When a textbook says "find a common denominator", it means this step; the formula has already done it for you with bd.

Multiplying needs no such step: $\frac{a}{b} \times \frac{c}{d}$ is just $a \times c \div (b \times d)$. Dividing by $\frac{c}{d}$ is multiplying by $\frac{d}{c}$, because $\frac{c}{d} \times \frac{d}{c} = 1$.

Adding and subtracting

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

Example: $\frac{3}{4} + \frac{1}{6}$.

$$= \frac{3 \times 6 + 4 \times 1}{4 \times 6} = \frac{18 + 4}{24} = \frac{22}{24}$$

a = 3, b = 4, c = 1, d = 6

$$= \frac{11}{12}$$ ← *optional: simplify, ÷ 2*

Example: $\frac{5}{6} - \frac{1}{4}$.

$$= \frac{5 \times 4 - 6 \times 1}{6 \times 4} = \frac{20 - 6}{24} = \frac{14}{24} = \frac{7}{12}$$

ad – bc over bd

Example: the bottoms are already the same, $\frac{1}{5} + \frac{2}{5}$.

$$= \frac{1 \times 5 + 5 \times 2}{5 \times 5} = \frac{15}{25} = \frac{3}{5}$$ ← *the formula still works*

Shortcut when $b = d$: the formula's answer always simplifies to "add the tops, keep the bottom", $\frac{1+2}{5} = \frac{3}{5}$. Use it or ignore it; the answer is the same.

Optional, for smaller numbers: instead of bd you may use any number both bottoms divide into, such as the lowest common multiple. For $\frac{3}{4} + \frac{1}{6}$ that is 12: $\frac{9}{12} + \frac{2}{12} = \frac{11}{12}$, the same answer as above without the simplifying at the end.

Slide error alert. On the Lecture 1 slide for **subtraction**, the left-hand side is printed with a **+** sign. The correct formula has a minus on both sides.

Sense check (your lecturer's habit): $\frac{2}{3}$ and $\frac{4}{5}$ are each a bit under 1, so $\frac{2}{3}+\frac{4}{5}=\frac{22}{15}=1\frac{7}{15}$, a bit under 2, is sensible. ✓

Negative fractions

The lecture's subtraction example is $\frac{2}{3}-\frac{4}{5}$, which comes out **negative**: you take away more than you started with.

RULE: negative fractions

The minus sign can sit in front, on top, or on the bottom: $-\frac{a}{b}=\frac{-a}{b}=\frac{a}{-b}$. Put it on **top** while you work.

The three formulas work unchanged; the tops follow the rules for negative numbers (Chapter 1).

Multiplying and dividing: same sign gives **+**, different signs give **−**.

Two minus signs cancel: $\frac{-a}{-b}=\frac{a}{b}$.

$$\frac{2}{3}-\frac{4}{5}=\frac{2\times5-3\times4}{3\times5}$$ ← ad − bc over bd

$$=\frac{10-12}{15}=-\frac{2}{15}$$ ← 10 − 12 = −2

Sense check: you took away $\frac{4}{5}=0.8$ from $\frac{2}{3}\approx0.67$, so the answer must be a little below zero. $-\frac{2}{15}\approx-0.13$ ✓

$$\frac{-1}{4}+\frac{2}{3}=\frac{-1\times3+4\times2}{4\times3}=\frac{5}{12}$$

−3 + 8 = 5

$$\left(-\frac{2}{3}\right)\times\frac{9}{4}=-\frac{18}{12}=-\frac{3}{2}$$ ← different signs: negative

$$\left(-\frac{1}{2}\right) \div \left(-\frac{1}{4}\right) = \frac{1}{2} \times \frac{4}{1} = 2$$ ← same signs: positive

Multiplying

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Multiply across. Whole numbers go over 1; mixed numbers become improper fractions first. "$\times$" with fractions means "of": $\frac{2}{3} \times \frac{4}{5}$ is two thirds of four fifths.

Example: Lecture 1, $\frac{2}{3} \times \frac{4}{5}$.

$$\frac{2}{3} \times \frac{4}{5} = \frac{2 \times 4}{3 \times 5} = \frac{8}{15}$$ ← ac over bd

Optional shortcut (cross-cancelling): before multiplying you may divide any top and any bottom by a common factor, to keep the numbers small. Only across $\times$, never $+$ or $-$.

Example: cross-cancelling, $\frac{3}{4} \times \frac{2}{9}$.

$$\frac{\cancel{3}^{1}}{\cancel{4}_{2}} \times \frac{\cancel{2}^{1}}{\cancel{9}_{3}} = \frac{1 \times 1}{2 \times 3} = \frac{1}{6}$$ ← 3 with 9 (÷3); 2 with 4 (÷2)

Example: a whole number, $3 \times \frac{2}{5}$.

$$\frac{3}{1} \times \frac{2}{5} = \frac{6}{5} = 1\frac{1}{5}$$ ← 3 = 3/1

Example: mixed numbers, $1\frac{1}{2} \times 2\frac{1}{2}$.

$$= \frac{3}{2} \times \frac{5}{2} = \frac{15}{4} = 3\frac{3}{4}$$ ← change to improper first

Not $1 \times 2 + \frac{1}{2} \times \frac{1}{2} = 2\frac{1}{4}$. Multiplying the wholes and the parts separately misses the cross terms (like expanding brackets, Chapter 11).

RULE: a fraction of an amount

"$\frac{a}{b}$ of N" means $\frac{a}{b} \times N$: divide N by the bottom, then multiply by the top.

$$\frac{3}{5} \text{ of } 40 = \frac{40}{5} \times 3 = 8 \times 3 = 24$$

÷ bottom, × top

Dividing

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

Change ÷ to ×, flip the **second** fraction only, then multiply across. Whole numbers go over 1 ($4 = \frac{4}{1}$, flipped $\frac{1}{4}$); mixed numbers become improper fractions first.

The **reciprocal** of a fraction is the fraction flipped upside down: $\frac{2}{3} \to \frac{3}{2}$, $4 \to \frac{1}{4}$. A number times its reciprocal is always 1.

Example: Lecture 1, $\frac{2}{3} \div \frac{4}{5}$.

$$\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4}$$ ← keep, change, flip

$$= \frac{10}{12} = \frac{5}{6}$$ ← multiply, then simplify ÷ 2

Example: dividing by a whole number, $\frac{2}{3} \div 4$.

$$\frac{2}{3} \div \frac{4}{1} = \frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$$ ← 4 = 4/1, flipped 1/4

Example: mixed numbers, $2\frac{1}{2} \div 1\frac{1}{4}$.

$$= \frac{5}{2} \div \frac{5}{4} = \frac{5}{2} \times \frac{4}{5} = \frac{20}{10} = 2$$ ← improper first

Why flipping works: $\frac{c}{d} \times \frac{d}{c} = 1$, so multiplying by $\frac{d}{c}$ undoes multiplying by $\frac{c}{d}$, which is exactly what dividing by $\frac{c}{d}$ does. E.g. $3 \div \frac{1}{4} = 3 \times 4 = 12$: twelve quarters in 3.

Sense check: dividing by a number less than 1 makes the answer bigger; dividing by a number more than 1 makes it smaller.

Simplifying an answer (equivalent fractions)

The formulas often give answers like $\frac{10}{12}$. That is already correct; simplifying just writes the same number with smaller numbers.

RULE: equivalent fractions

Multiplying or dividing the top **and** the bottom by the **same** non-zero number gives the same value: $\frac{a}{b} = \frac{a \times k}{b \times k}$.

Adding or subtracting the same number to top and bottom does NOT keep the value.

$$\frac{3}{4} = \frac{3 \times 5}{4 \times 5} = \frac{15}{20}$$ ← top AND bottom × 5

$$\frac{1}{2} \neq \frac{1+1}{2+1} = \frac{2}{3}$$ ← adding 1 to both changes the value

PROCEDURE: simplifying (cancelling)

1. Find the **highest common factor** (HCF) of the top and bottom (Chapter 1).
2. Divide the top and the bottom by it.
3. Check: if no number except 1 divides both, the fraction is in **simplest form**. (Smaller steps are fine: divide by any common factor, repeat until none is left.)

Example: simplify $\frac{12}{18}$.

$$\mathrm{HCF}(12, 18) = 6$$ ← factors of 12: 1,2,3,4,6,12; of 18: 1,2,3,6,9,18

$$\frac{12 \div 6}{18 \div 6} = \frac{2}{3}$$ ← divide top and bottom by 6

Cancel factors, never terms. In $\frac{6}{8}$ you divide the **whole** top and the **whole** bottom. In $\frac{2+4}{2+6}$ you may **not** cross out the 2s: work out the top and bottom first ($\frac{6}{8}$), then cancel.

Mixed numbers and improper fractions

RULE: converting

Mixed to improper: whole × bottom, add the top, keep the bottom: $p\frac{a}{b} = \frac{p \times b + a}{b}$.

Improper to mixed: divide top by bottom. The quotient is the whole number, the remainder is the new top, the bottom stays the same.

$$2\frac{1}{3} = \frac{2 \times 3 + 1}{3} = \frac{7}{3}$$ ← 2 wholes = 6 thirds, plus 1 third

$$\frac{11}{4} : \; 11 \div 4 = 2 \text{ r } 3$$ ← 4 goes into 11 twice, 3 left over

$$\frac{11}{4} = 2\frac{3}{4}$$ ← 2 wholes and 3 quarters

Negative mixed numbers: $-1\frac{3}{4}$ means $-(1 + \frac{3}{4}) = -\frac{7}{4}$. The minus sign applies to the whole thing, **not** just the 1.

Mixed numbers

Change each mixed number to an improper fraction, then use the formula as normal.

Example: $1\frac{1}{2} + 2\frac{3}{4}$.

$$= \frac{3}{2} + \frac{11}{4} = \frac{3 \times 4 + 2 \times 11}{2 \times 4} = \frac{34}{8}$$

improper, then the formula

$$= \frac{17}{4} = 4\frac{1}{4}$$ ← *simplify: 17 ÷ 4 = 4 r 1*

Example: $3\frac{1}{4} - 1\frac{2}{3}$.

$$= \frac{13}{4} - \frac{5}{3} = \frac{13 \times 3 - 4 \times 5}{4 \times 3} = \frac{19}{12} = 1\frac{7}{12}$$

improper, then the formula

Comparing fractions

$$\frac{a}{b} > \frac{c}{d} \iff ad > bc \qquad (\text{positive } b,\ d)$$

Cross-multiply and compare ad with bc. It's the top of the subtraction formula: $\frac{a}{b} - \frac{c}{d} = \frac{ad-bc}{bd}$ is positive exactly when $ad > bc$. Or turn both into decimals (top ÷ bottom).

Example: Lecture 1, $\frac{9}{17}$ or $\frac{49}{97}$?

$$9 \times 97 = 873 \qquad 49 \times 17 = 833$$

each top × the OTHER bottom

$$873 > 833 \ \Rightarrow\ \frac{9}{17} > \frac{49}{97}$$

The lecture does it with decimals: $0.529 > 0.505$.

Order of operations and whole numbers

RULE: order of operations with fractions

The usual order applies (Chapter 1): **brackets, powers, × and ÷ (left to right), + and − (left to right)**.

A fraction bar groups like brackets: in $\frac{1+2}{3+4}$, work out the top and the bottom first.

"Of" means ×.

$$\frac{1}{2}+\frac{1}{3}\times\frac{3}{4}=\frac{1}{2}+\frac{1}{4}=\frac{3}{4}$$ ← × first, then +

$$\left(\frac{1}{2}+\frac{1}{3}\right)\times\frac{3}{4}=\frac{5}{6}\times\frac{3}{4}=\frac{15}{24}=\frac{5}{8}$$

brackets first

Whole numbers with fractions. Write the whole number over 1, then use the formula:

$$3+\frac{2}{5}=\frac{3}{1}+\frac{2}{5}=\frac{3\times5+1\times2}{1\times5}=\frac{17}{5}$$

3 = 3/1

$$2-\frac{3}{4}=\frac{2}{1}-\frac{3}{4}=\frac{2\times4-1\times3}{1\times4}=\frac{5}{4}$$

2 = 2/1

Fractions or decimals? $\frac{1}{3}=0.333...$ never ends, so any decimal you write is slightly wrong. For exact work (algebra), **keep fractions**. For measured data (statistics), decimals are fine (Lecture 1).

Find the missing number

PROCEDURE: finding an unknown fraction

1. Write the unknown as $?$ (or x).
2. **Undo** the operation that was done to it, using the opposite operation on the known values: + is undone by −, × is undone by ÷ (and the other way round).
3. Then use the three formulas to work out the answer.
4. Check by putting your answer back into the original question.

Example: $\frac{3}{4} = \frac{?}{20}$.

$$4 \times 5 = 20 \Rightarrow ? = 3 \times 5 = 15$$

← the bottom was × 5, so the top is × 5

Example: $\frac{2}{5} + ? = \frac{9}{10}$.

$$? = \frac{9}{10} - \frac{2}{5} = \frac{9 \times 5 - 10 \times 2}{10 \times 5} = \frac{25}{50} = \frac{1}{2}$$

undo + with −

Example: $? \times \frac{3}{4} = \frac{1}{2}$.

$$? = \frac{1}{2} \div \frac{3}{4} = \frac{1}{2} \times \frac{4}{3} = \frac{4}{6} = \frac{2}{3}$$

← undo × with ÷

Example: $\frac{5}{6} \div ? = \frac{5}{3}$.

$$? = \frac{5}{6} \div \frac{5}{3} = \frac{5}{6} \times \frac{3}{5} = \frac{15}{30} = \frac{1}{2}$$

← the unknown is the divisor: ? = 5/6 ÷ 5/3

Example: $\frac{3}{8}$ of a number is 15. What is the number?

$$\frac{3}{8} \times ? = 15 \Rightarrow ? = 15 \div \frac{3}{8} = 15 \times \frac{8}{3} = 40$$

undo × with ÷

The mistakes to avoid

✗ Common mistakes

✗ $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$

✓ $\frac{1}{2} + \frac{1}{3} = \frac{1 \times 3 + 2 \times 1}{2 \times 3} = \frac{5}{6}$

adding tops and adding bottoms isn't the formula: use (ad + bc)/bd

✗ $\frac{\cancel{3}}{4} + \frac{1}{\cancel{3}}$

✓ $\frac{3}{4} + \frac{1}{3} = \frac{9+4}{12} = \frac{13}{12}$

cross-cancelling only works across ×

✗ $\frac{2}{3} \div \frac{4}{5} = \frac{3}{2} \times \frac{4}{5}$

✓ $\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4}$

flip the SECOND fraction only

✗ $1\frac{1}{2} \times 2\frac{1}{2} = 2\frac{1}{4}$

✓ $\frac{3}{2} \times \frac{5}{2} = \frac{15}{4} = 3\frac{3}{4}$

change mixed numbers to improper fractions before × or ÷

✗ $\frac{2+4}{2+6} = \frac{4}{6}$

✓ $\frac{2+4}{2+6} = \frac{6}{8} = \frac{3}{4}$

cancel factors of the whole top and bottom, not single terms

Fraction rules

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Fraction = division	$\frac{a}{b} = a \div b, \quad n = \frac{n}{1}$
Equivalent fractions	$\frac{a}{b} = \frac{a \times k}{b \times k} = \frac{a \div k}{b \div k}$ same non-zero k on top and bottom; never + or −
Simplify	divide top and bottom by their HCF
Mixed to improper	$p\frac{a}{b} = \frac{pb + a}{b}$
Add / subtract	$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$ plug in; simplify at the end if asked
Multiply	$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$ multiply across
Divide	$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$ flip the second fraction only, then multiply across
Reciprocal	$\frac{a}{b} \to \frac{b}{a}, \qquad \frac{a}{b} \times \frac{b}{a} = 1$
Compare	$\frac{a}{b} > \frac{c}{d} \iff ad > bc$ for positive bottoms
Negative fractions	$-\frac{a}{b} = \frac{-a}{b} = \frac{a}{-b}$
Fraction of an amount	$\frac{a}{b} \text{ of } N = \frac{N}{b} \times a$
Whole and mixed numbers	whole number over 1; mixed number to improper; then the formula

Your turn

Answers are in grey under each group; full working is on the next page.

Simplify and convert

1. Simplify 18/24
2. Write 23/5 as a mixed number
3. Write 3 2/7 as an improper fraction
4. Which is bigger, 5/8 or 7/11?

1. 3/4 2. 4 3/5 3. 23/7 4. 7/11 (5×11 = 55 < 56 = 7×8)

Add and subtract

5. 2/9 + 5/9
6. 3/4 − 2/5
7. 5/6 + 3/8
8. 1/3 − 3/4
9. 4 1/5 − 2 2/3

5. 7/9 6. 7/20 7. 29/24 = 1 5/24 8. −5/12 9. 23/15 = 1 8/15

Multiply and divide

10. 4/9 × 3/8
11. 2 1/3 × 1 1/2
12. 5/6 ÷ 10/9
13. 3/4 ÷ 6
14. 1 1/4 ÷ 2 1/2

10. 1/6 11. 7/2 = 3 1/2 12. 3/4 13. 1/8 14. 1/2

Mixed operations

15. 1/2 + 2/3 × 3/4
16. (1/2 + 2/3) × 3/4
17. 3/5 of 45
18. −2/3 × 3/8

15. 1 16. 7/8 17. 27 18. −1/4

Find the unknown

19. 5/6 = ?/30
20. ? + 1/4 = 7/8
21. ? × 2/3 = 4/9
22. 3/4 of a number is 18. Find it.
23. 2/5 ÷ ? = 4/5

19. 25 20. 5/8 21. 2/3 22. 24 23. 1/2

Answers, worked

1. $\dfrac{18 \div 6}{24 \div 6} = \dfrac{3}{4}$

2. $23 \div 5 = 4 \text{ r } 3 \Rightarrow 4\frac{3}{5}$

3. $\dfrac{3 \times 7 + 2}{7} = \dfrac{23}{7}$

4. $5 \times 11 = 55 < 56 = 7 \times 8 \Rightarrow \dfrac{7}{11}$ is bigger

5. $\dfrac{2 + 5}{9} = \dfrac{7}{9}$

6. $\dfrac{3 \times 5 - 4 \times 2}{4 \times 5} = \dfrac{7}{20}$

7. $\dfrac{5 \times 8 + 6 \times 3}{6 \times 8} = \dfrac{58}{48} = \dfrac{29}{24} = 1\frac{5}{24}$

8. $\dfrac{1 \times 4 - 3 \times 3}{3 \times 4} = -\dfrac{5}{12}$

9. $\dfrac{21}{5} - \dfrac{8}{3} = \dfrac{63 - 40}{15} = \dfrac{23}{15} = 1\frac{8}{15}$

10. $\dfrac{4 \times 3}{9 \times 8} = \dfrac{12}{72} = \dfrac{1}{6}$

11. $\dfrac{7}{3} \times \dfrac{3}{2} = \dfrac{21}{6} = \dfrac{7}{2} = 3\frac{1}{2}$

12. $\dfrac{5}{6} \times \dfrac{9}{10} = \dfrac{45}{60} = \dfrac{3}{4}$

13. $\dfrac{3}{4} \times \dfrac{1}{6} = \dfrac{3}{24} = \dfrac{1}{8}$

14. $\dfrac{5}{4} \div \dfrac{5}{2} = \dfrac{5}{4} \times \dfrac{2}{5} = \dfrac{10}{20} = \dfrac{1}{2}$

15. $\frac{1}{2} + \frac{6}{12} = \frac{1}{2} + \frac{1}{2} = 1$

16. $\frac{7}{6} \times \frac{3}{4} = \frac{21}{24} = \frac{7}{8}$

17. $\frac{45}{5} \times 3 = 9 \times 3 = 27$

18. $-\frac{6}{24} = -\frac{1}{4}$

19. $6 \times 5 = 30 \Rightarrow ? = 5 \times 5 = 25$

20. $? = \frac{7}{8} - \frac{1}{4} = \frac{28-8}{32} = \frac{20}{32} = \frac{5}{8}$

21. $? = \frac{4}{9} \div \frac{2}{3} = \frac{4}{9} \times \frac{3}{2} = \frac{12}{18} = \frac{2}{3}$

22. $? = 18 \div \frac{3}{4} = 18 \times \frac{4}{3} = 24$

23. $? = \frac{2}{5} \div \frac{4}{5} = \frac{2}{5} \times \frac{5}{4} = \frac{10}{20} = \frac{1}{2}$

FRACTION RULES CARD

cut it out, keep it next to you

ADD / SUBTRACT

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

$$\frac{1}{2} + \frac{1}{3} = \frac{1 \times 3 + 2 \times 1}{2 \times 3} = \frac{5}{6}$$

MULTIPLY

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

$$\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$$

DIVIDE

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

$$\frac{2}{3} \div \frac{4}{5} = \frac{2}{3} \times \frac{5}{4} = \frac{10}{12}$$

SIMPLIFY (OPTIONAL)

$$\frac{a \times k}{b \times k} = \frac{a}{b}$$

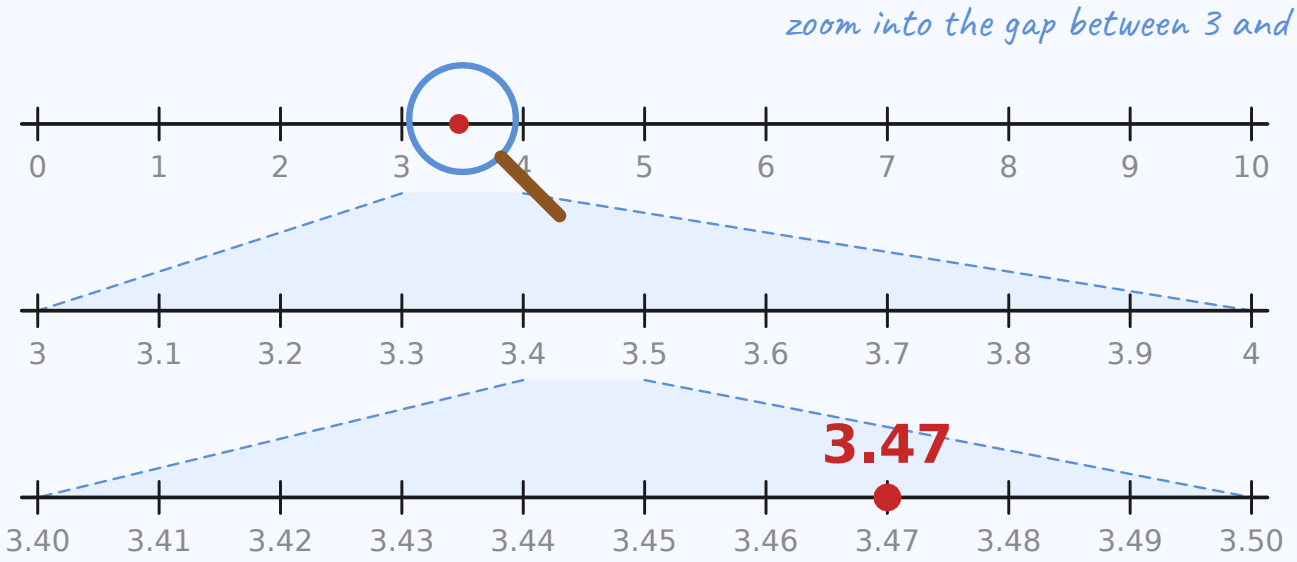
$$\frac{10}{12} = \frac{10 \div 2}{12 \div 2} = \frac{5}{6}$$

whole numbers over 1, mixed numbers improper, then plug in

Chapter 3

Decimal fractions

numbers with a dot in them, seen through a magnifying glass



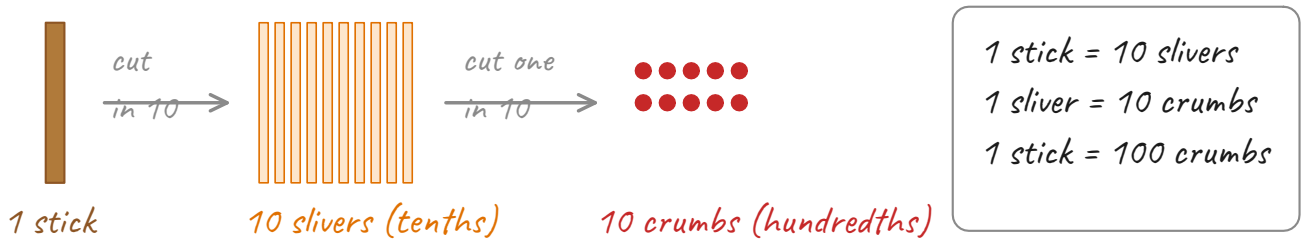
A whole number counts whole things. But life is full of bits: £3.47, 1.8 metres, 37.5 °C. The **dot** in these numbers lets you say "and a bit" precisely. This chapter shows that a decimal is nothing new: it's the same place-value idea from Chapter 1, carried on **to the right**, and it's another way to write a fraction (Chapter 2).

You'll learn to read, compare, add, multiply and divide decimals, turn fractions into decimals with a comic-strip division, and round numbers sensibly (decimal places and significant figures, both on your Lecture 1 slides).

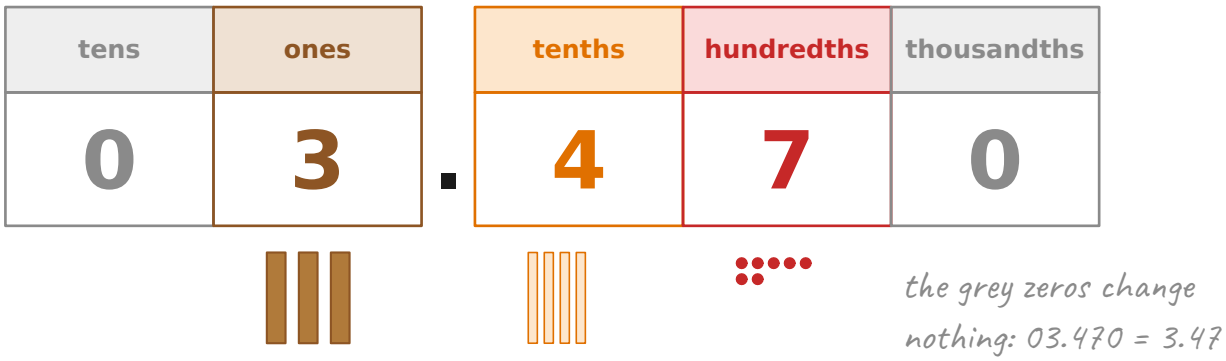
Goes with **Lecture 1** (significant figures, decimal places, recurring decimals).

🔍 Cutting a stick into ten

In Chapter 1, ten sticks made a bundle. Now go the other way. Take **one stick** and cut it into **ten equal slivers**. Each sliver is a tenth of a stick. Cut one sliver into ten crumbs: each crumb is a hundredth of a stick.



Now **3 sticks, 4 slivers and 7 crumbs** is written like this. The **dot** (called the **decimal point**) is a fence: whole sticks on the left, pieces on the right.

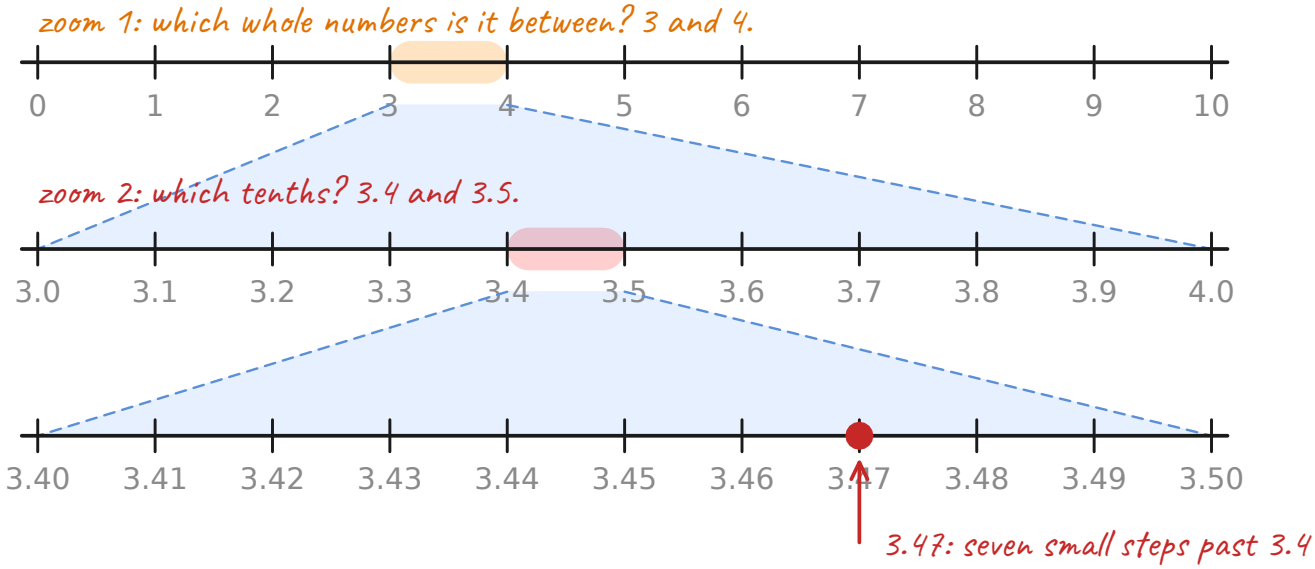


Read it as **"three point four seven"**. (Not "three point forty-seven": after the dot you read the digits one at a time.)

Each column is **ten times smaller** than the column to its left, on **both** sides of the dot. That's the only rule. Hundreds, tens, ones, tenths, hundredths, thousandths... one long ladder, and the dot just marks where the whole ones stop.

The magnifying glass: where does 3.47 live?

A number line is a ruler that goes on forever. Decimals live **in the gaps** between whole numbers. To find 3.47, zoom in twice:



Every zoom shows one more digit. The first digit after the dot says which tenth, the next says which hundredth, and so on. You could keep zooming forever, which is why some numbers need endless digits (you'll meet them later in this chapter).

Which is bigger: 0.5 or 0.45?

Lots of people say 0.45, because 45 is bigger than 5. **That's the trap.** Compare column by column, from the left, like a race where the first difference wins:

With slivers you can see it: half a stick (5 slivers) is more than 4 slivers and a few crumbs. **Line the dots up, pad with zeros, compare from the left.**

A decimal is a fraction wearing a disguise

PROCEDURE: decimal to fraction

1. Count the digits after the point: 1 digit means tenths, 2 hundredths, 3 thousandths.
2. Write the digits after the point over 10, 100 or 1000 (one 0 per digit), then simplify.
3. A whole-number part stays in front (mixed number): $2.6 = 2\frac{6}{10} = 2\frac{3}{5}$.

The last column you use tells you the bottom number. Read the digits after the dot as a whole number, and put it over 10, 100 or 1000:

$$0.7 = \frac{7}{10}$$ *one digit after the dot: tenths*

$$0.25 = \frac{25}{100} = \frac{1}{4}$$ *two digits: hundredths, then simplify*

$$0.125 = \frac{125}{1000} = \frac{1}{8}$$ *three digits: thousandths*

$$2.6 = 2\frac{6}{10} = 2\frac{3}{5}$$ *the whole 2 just rides along*

Worth knowing by heart (they come up constantly):

$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{8}$	$\frac{1}{3}$
0.5	0.25	0.75	0.2	0.1	0.125	$0.\dot{3}$

(The dot over the 3 in the last one is explained later in this chapter.)

🔍 Adding and taking away: line up the dots

PROCEDURE: adding and subtracting decimals

1. Write the numbers in columns with the **decimal points in a vertical line**.
2. Fill empty places with zeros so every number has the same number of decimal places ($5 = 5.00$).
3. Add or subtract column by column from the right, exactly as for whole numbers.
4. The point in the answer goes directly under the other points.

A shopping receipt already does this. Pence line up under pence, pounds under pounds, so the dots form a straight vertical line:

CORNER SHOP

milk	1.35
bread	0.99
cheese	12.50
TOTAL	14.84

the dots sit in one straight column.

Then add exactly like Chapter 1:

hundredths: 5 + 9 + 0 = 14, write 4 carry 1

tenths: 3 + 9 + 5 + 1 = 18, write 8 carry 1

ones: 1 + 0 + 2 + 1 = 4

tens: 1

the dot drops straight down into the answer

Taking away: fill the empty spaces with zeros first. Work out $5 - 1.37$:

$$\begin{array}{r} 5.00 \\ -\ 1.37 \\ \hline 3.63 \end{array}$$

5 is the same as 5.00 (5 pounds = 5 pounds 0 pence).

Write the zeros in so every column has a digit.

then take away exactly as in Chapter 1, opening bundles when a column is too small.

check: 3.63 + 1.37 = 5.00 ✓

Times 10: the dot stays still, the digits slide

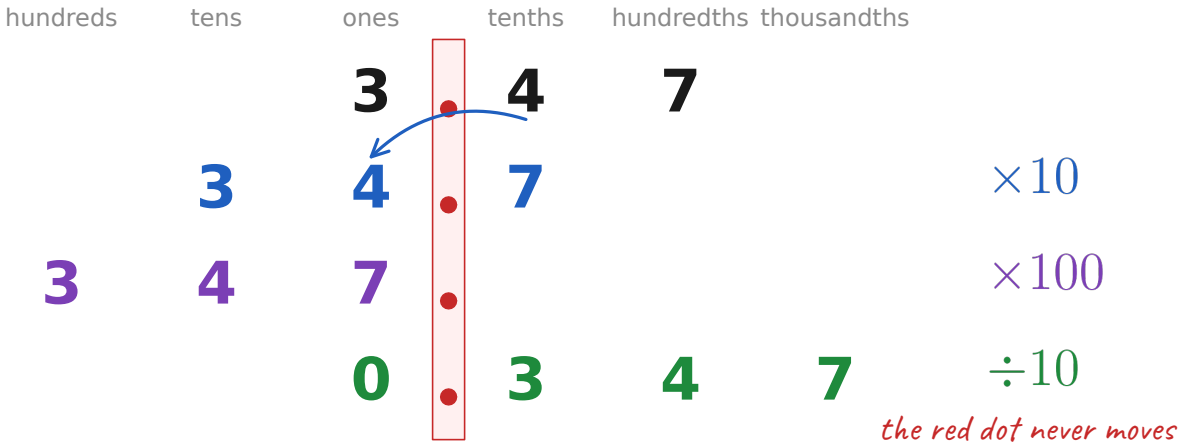
RULE: multiplying and dividing by 10, 100, 1000

$\times 10$, $\times 100$, $\times 1000$: every digit moves **1, 2 or 3 places left** (the number gets bigger). $3.47 \times 100 = 347$.

$\div 10$, $\div 100$, $\div 1000$: every digit moves **1, 2 or 3 places right** (the number gets smaller). $3.47 \div 10 = 0.347$.

Fill any gap next to the point with 0: $0.6 \div 100 = 0.006$.

Multiplying by 10 makes every stick into a bundle, every sliver into a stick, every crumb into a sliver. Each digit moves **one column to the left**. Picture the dot nailed to the page and the digits sliding past it:

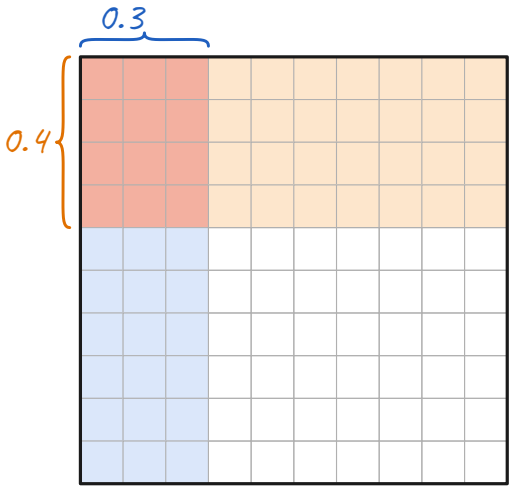


× 10: digits slide 1 place left. **× 100**: 2 places. **× 1000**: 3 places. **÷ 10**: 1 place right, and so on. Gaps next to the dot get filled with zero (0.347 needs a 0 in the ones column so the dot isn't left floating).

Shortcut people say: "move the decimal point". It gives the same answer, but it's really the digits moving; the columns stay where they are.

🔍 Multiplying decimals: a square cut into 100

What is 0.3×0.4? Think of "of": 0.3 **of** 0.4. Take a square that is 1 by 1, cut into a 10 × 10 grid. Shade 3 columns (0.3 of the width) and 4 rows (0.4 of the height):



The overlap (red) is 3 × 4 = 12 little squares.

Each little square is 1/100 of the whole.

$$0.3 \times 0.4 = \frac{12}{100} = 0.12$$

the answer is SMALLER than both. Multiplying by something less than 1 shrinks, like taking a part of a part.

PROCEDURE: multiplying decimals

1. Ignore the decimal points and multiply the whole numbers.
2. Count the digits after the point in **both** numbers together.
3. Give the answer that many digits after the point (count from the right; add 0s at the front if needed: $0.03 \times 0.2 = 0.006$).
4. **Then** you may drop zeros at the end: 0.5×0.4: $5 \times 4 = 20$, 2 places, $0.20 = 0.2$.

2.5×1.3 ← *1. ignore the dots for a moment*

$25 \times 13 = 325$ ← *2. multiply whole numbers (Chapter 1 grid method)*

$2.5 \rightarrow 1$ digit, $1.3 \rightarrow 1$ digit ← *3. count digits after the dots: 1 + 1 = 2*

3.25 ← *4. give the answer 2 digits after the dot*

$0.5 \times 0.4 = 0.20 = 0.2$ ← *5 × 4 = 20; 2 places gives 0.20; the end 0 can go*

Place the point before you drop the zero. Write 20, count 2 places from the right: 0.20. Dropping the 0 first and then counting 2 places gives the wrong 0.02.

Sanity check: 2.5 × 1.3 is roughly 2.5 × 1 = 2.5, a bit more. 3.25 is sensible; 32.5 or 0.325 would be silly. Always ask "is the answer the right size?"

Dividing by a decimal: make it a whole number first

How many 15p coins fit into £4.50? That's $4.5 \div 0.15$. Dividing by 0.15 feels impossible, but in **pence** the question is easy: how many 15s in 450?

... 30 coins make 450p

$$\frac{4.5}{0.15} = \frac{4.5 \times 100}{0.15 \times 100} = \frac{450}{15} = 30$$

multiply top and bottom by the same number: that's an equivalent fraction (Chapter 2), so the answer can't change

PROCEDURE: dividing by a decimal

1. Multiply **both** numbers by the same power of 10 (10, 100, ...) so that the number you divide **by** becomes a whole number.
2. Divide normally (short division). The answer is unchanged, because $\frac{a}{b} = \frac{10a}{10b}$.
3. Check by multiplying back: answer × divisor = the number you started with.

$$0.36 \div 0.4 = \frac{3.6}{4} = 0.9$$ ← *× 10 top and bottom; check 0.9 × 0.4 = 0.36*

Dividing by a number **less than 1** makes the answer **bigger** (lots of small coins fit). That's the mirror of multiplying on the previous page.

Fraction to decimal: division as a comic strip

PROCEDURE: fraction to decimal

1. $\frac{a}{b} = a \div b$: divide the top by the bottom.
2. Write the top as $a.000\ldots$ and keep dividing; each remainder carries to the next column.
3. Stop when the remainder is 0 (**terminating** decimal) or a remainder repeats (**recurring** decimal: the digits repeat from there on).

A fraction is a division (Chapter 2): $\frac{3}{8}$ means $3 \div 8$. Eight people share 3 pizzas. Follow the strip one panel at a time:

1

$3 \div 8$

8 into 3? Too small. No whole pizzas each.

0.

→

2

$30 \div 8 = 3 \text{ r } 6$

cut the 3 pizzas into tenths: 30 slices. 3 each, 6 left.

0.3

→

3

$60 \div 8 = 7 \text{ r } 4$

cut the 6 leftover tenths into 60 hundredths. 7 each, 4 left.

0.37

→

4

$40 \div 8 = 5 \text{ r } 0$

40 thousandths. 5 each, nothing left. STOP.

0.375

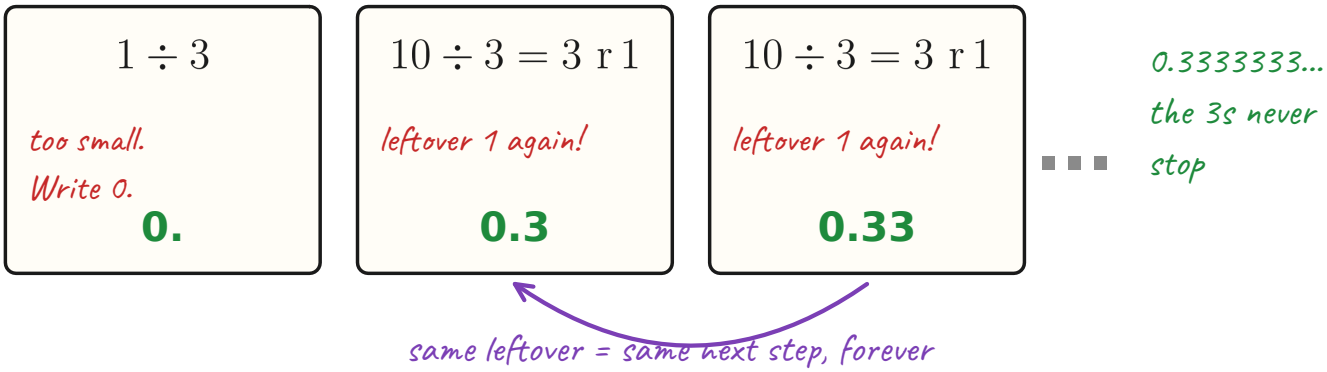
Written the usual compact way ("bus stop" division), the same four panels look like this. The small red numbers are the leftovers carried to the next column:

```
      0.3 7 5
   ┌──────────
8  │ 3.0 0 0
     3 6 4
```

the carried 3, 6 and 4 are the leftovers from panels 1, 2 and 3. Add as many 0s after the dot as you need: 3 = 3.000

When the strip never ends: recurring decimals

Try $\frac{1}{3} = 1 \div 3$. Panel after panel, the same leftover comes back:



A decimal that repeats forever is called **recurring**. We can't write infinitely many 3s, so we put a **dot over the repeating digit**:

$\frac{1}{3} = 0.\dot{3}$ *0.3333...*

$\frac{1}{6} = 0.1\dot{6}$ *only the 6 repeats: 0.16666...*

$\frac{2}{11} = 0.\dot{1}\dot{8}$ *dots on first and last of the repeating block: 0.181818...*

$\frac{1}{7} = 0.\dot{1}42857\dot{}$ *block 142857 repeats*

Your Lecture 1 slide writes $\frac{1}{3} = 0.3\dot{3} \approx 0.33$. The wavy equals sign $\approx$ means **"is roughly"**: 0.33 is close to a third, but not exactly it.

Why the lecture says keep fractions when you want exact answers: $\frac{1}{3} \times 3 = 1$ exactly, but $0.33 \times 3 = 0.99$. The rounding error sneaks in. Use fractions for algebra, decimals when an approximate answer is fine (statistics, measurements).

Which fractions stop, and recurring back to fractions

RULE: terminating or recurring?

Simplify the fraction first.

If the denominator's prime factors are **only 2s and 5s**, the decimal **terminates** (stops). Reason: it can be scaled to a denominator of 10, 100, 1000, ... (and $10 = 2 \times 5$).

Any other prime factor (3, 7, 11, ...) in the denominator gives a **recurring** decimal.

Fraction	Denominator	Primes	Decimal
$\frac{3}{8}$	8	2^3	**terminates**: 0.375
$\frac{7}{20}$	20	$2^2 \times 5$	**terminates**: 0.35
$\frac{5}{12}$	12	$2^2 \times 3$	**recurs**: $0.41\dot{6}$
$\frac{9}{15}$	$15 \to 5$ after simplifying to $\frac{3}{5}$	5	**terminates**: 0.6

PROCEDURE: recurring decimal to fraction

1. Call the decimal x. Write out several repeats: $x = 0.272727\ldots$
2. Multiply by 10 if one digit repeats, 100 if two digits repeat, 1000 if three, ... so the repeating part lines up: $100x = 27.2727\ldots$
3. Subtract: the endless tails cancel. $100x - x = 27$, so $99x = 27$.
4. Divide and simplify: $x = \frac{27}{99} = \frac{3}{11}$.

Example. Write $0.\dot{3}$ as a fraction.

$$x = 0.3333\ldots$$ ← one digit repeats: use 10x

$$10x = 3.3333\ldots$$ ← × 10

$$10x - x = 3 \;\Rightarrow\; 9x = 3$$ ← the tails .3333... cancel

$$x = \frac{3}{9} = \frac{1}{3}$$ ← ÷ 9, simplify

Example. $0.1\dot{6}$ (only the 6 repeats): use $10x$ and $100x$, which both end in $.666\ldots$

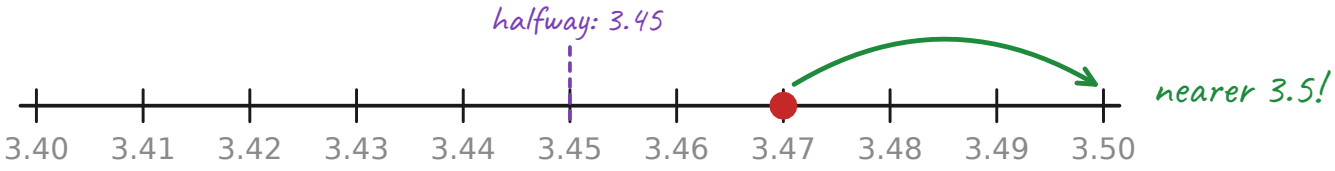
$$100x = 16.666\ldots, \quad 10x = 1.666\ldots$$

both tails are .666...

$$90x = 15 \;\Rightarrow\; x = \frac{15}{90} = \frac{1}{6}$$ ← *subtract, then ÷ 90*

Rounding: which mark on the ruler is nearer?

A price of £3.47 on a sign that only shows 10p steps: is it nearer £3.40 or £3.50? Zoom in with the magnifying glass:



So 3.47 **rounded to 1 decimal place** is 3.5. "Decimal places" (d.p.) just means digits after the dot. You don't have to draw a ruler every time. The rule is:

PROCEDURE: rounding to a number of decimal places

1. Find the last digit you are keeping (the **cut**).
2. Look at the **next** digit only (the **decider**).
3. Decider **5 or more**: increase the kept digit by 1. Decider **4 or less**: leave it.
4. Remove every digit after the cut. Keep end zeros that are inside the required places (3.96 to 1 d.p. is 4.0, not 4).

Round 2.71828 to 2 decimal places:

2. 71|**8**28 → **2.72**

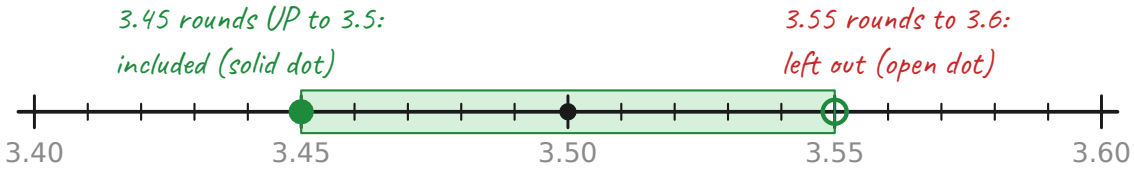
cut *decider 8*

8 is 5 or more: the 1 bumps up to 2

Bumping can ripple: 3.96 to 1 d.p. The decider 6 bumps the 9 up, 9 + 1 = 10, so it carries: 3.96 becomes **4.0**. Keep the 0: it shows you rounded to 1 d.p.

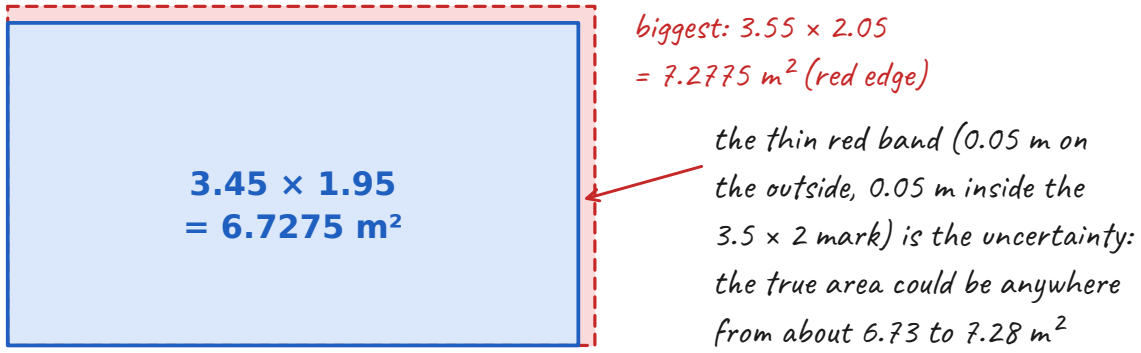
Going backwards: what could it have been? A tape measure says a rug is **3.5 m** long, rounded to 1 d.p. The true length isn't exactly 3.5.

> ✎ **YOUR GO (pencil first)**
>
> *On a ruler from 3.4 to 3.6, shade every true length that would round to 3.5. Where does the shading start, and where does it stop?*



So "3.5 to 1 d.p." really means **anywhere from 3.45 up to (but not including) 3.55**: half a step either side of the mark. Those two ends are the **lower and upper bounds**.

Why it matters: the rug's area. It's 3.5 m by 2.0 m, both to 1 d.p., so "7 m²". But the true rug is somewhere between the smallest and the biggest possible rectangles:



A thin half-step on each side becomes a whole band of area: about 0.28 m² either way. Multiplying rounded numbers multiplies their uncertainty too, so don't quote "7.000 m²" from measurements that were only good to 1 d.p.

Upper and lower bounds: the rules

PROCEDURE: bounds of a rounded number

1. Find the step size of the rounding: 1 d.p. means steps of 0.1, 2 d.p. steps of 0.01, nearest 10 means steps of 10.
2. **Half a step** either side: lower bound $=$ value $-$ $\frac{1}{2}$ step, upper bound $=$ value $+$ $\frac{1}{2}$ step.
3. Write $\text{lower} \le x < \text{upper}$: the lower bound is included, the upper bound is not (it would round up).

$$7.8 \text{ (1 d.p.)}: \quad 7.75 \le x < 7.85$$ ← step 0.1, half step 0.05

$$70 \text{ (nearest 10)}: \quad 65 \le x < 75$$ ← step 10, half step 5

RULE: bounds of a calculation

Add: biggest $=$ upper $+$ upper; smallest $=$ lower $+$ lower. Same idea for $\times$.

Subtract: biggest $=$ upper $-$ **lower**; smallest $=$ lower $-$ **upper**.

Divide: biggest $=$ upper $\div$ **lower**; smallest $=$ lower $\div$ **upper**. (A big top over a small bottom gives the biggest answer.)

Example (division). A plank is 2.4 m long and is cut into pieces 0.3 m long, both measured to 1 d.p. Find the bounds for the number of pieces, $2.4 \div 0.3$.

$$2.35 \le \text{plank} < 2.45, \qquad 0.25 \le \text{piece} < 0.35$$

half a step (0.05) each side

$$\text{max} = \frac{2.45}{0.25} = 9.8$$ ← upper ÷ lower

$$\text{min} = \frac{2.35}{0.35} \approx 6.71$$ ← lower ÷ upper

So "8 pieces" ($2.4 \div 0.3$) could really be anything from about 6.7 to 9.8: rounding in a division can matter a lot.

🔍 Rounding versus chopping (truncating)

Your Lecture 1 slide says 0.123456789 "truncated" to 2 d.p. is 0.12. To **truncate** means to just **chop off** the extra digits without looking at the decider:

0.678 *round* ⟶ **0.68** *decider 8 → bump the 7 up*

0.678 *truncate* ⟶ **0.67** *just chop after 2 digits*

For 0.123456789 both give 0.12 (the decider is 3, so nothing bumps). They only differ when the decider is 5 or more. Unless told to truncate, **round**.

Significant figures: a rough ruler and a fine ruler

RULE: which digits are significant?

Non-zero digits always count. 4.72 has 3 s.f.

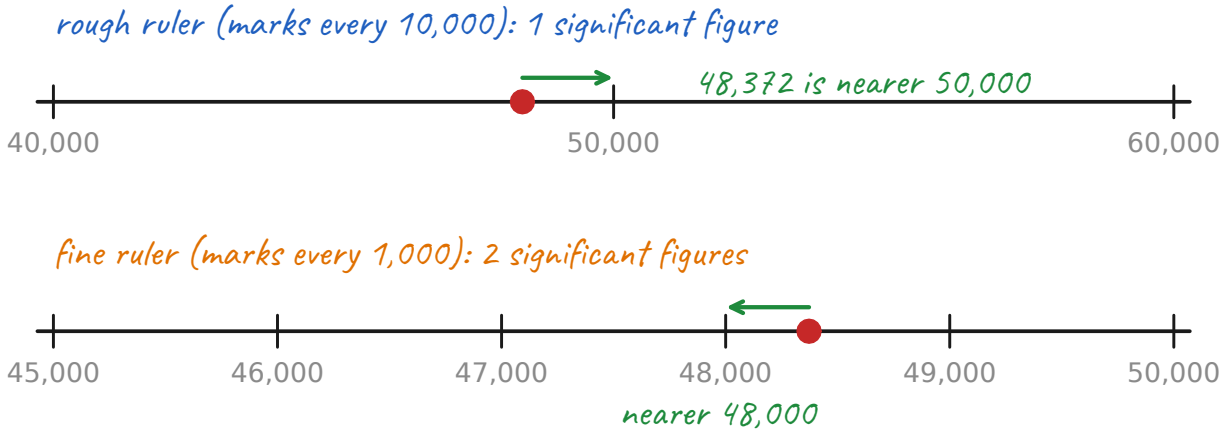
Leading zeros never count: zeros before the first non-zero digit only hold place. 0.0047 has 2 s.f. (the 4 and the 7).

Zeros between non-zero digits always count: 3.07 has 3 s.f.; 2005 has 4 s.f.

Trailing zeros after the decimal point count: they were written on purpose to show precision. 0.030 has 2 s.f. (the 3 and the final 0); 4.0 has 2 s.f.

Trailing zeros in a whole number are ambiguous: $48\,000$ could be 2, 3, 4 or 5 s.f. Say which ("48 000 to 2 s.f.") when it matters.

A crowd of **48,372** people. A newspaper says "about 50,000"; a report says "48,000". Both are rounding, just with rulers of different fineness:



Significant figures (s.f.) count digits **from the first non-zero digit**. They say how fine your ruler is, whatever the size of the number. Zeros at the end of a big number stay, because they hold the size (48,000 must not shrink to 48).

Small numbers: the zeros at the front are only place-holders, so they don't count. Your Lecture 1 table, with the counting shown:

0. 123456789
1 2 3 4 5 6 7 8 9
leading 0s: not counted

1 s.f.	**0.1**
2 s.f.	**0.12**
3 s.f.	**0.123**
4 s.f.	**0.1235**

4 s.f.: decider 5, so 4 bumps to 5!

Heads-up: the slide table shows 0.1234 for 4 s.f.; that is the **truncated** value. Rounded properly it's 0.1235, because the next digit is 5.

One more: 0.004 72 to 2 s.f. Skip the zeros, the first significant digit is 4. Keep 4 and 7, decider 2: **0.0047**.

PROCEDURE: rounding to n significant figures

1. Start counting at the **first non-zero digit**. Count n digits (zeros in between count). That is the cut.
2. Round at the cut using the decider, as for decimal places.
3. **Whole-number part:** replace the dropped digits with zeros to keep the size ($48\,372 \to 48\,000$). **After the point:** just drop them.

Number	s.f.	Why
3.07	3	the 0 is between 3 and 7: counts
0.030	2	leading zeros don't count; the final 0 after the point does
0.0405	3	4, 0, 5 (the zero between counts)
12.50	4	trailing zero after the point counts
5200	2, 3 or 4	trailing zeros in a whole number: ambiguous

$$3.0496 \to 3.05 \ (3 \text{ s.f.}) \longleftarrow \textit{cut after 3, 0, 4; decider 9: 4 → 5}$$

$$2045 \to 2000 \ (2 \text{ s.f.}) \longleftarrow \textit{keep 2, 0; decider 4; refill with zeros}$$

$$0.06996 \to 0.0700 \ (3 \text{ s.f.}) \longleftarrow \textit{6, 9, 9; decider 6 ripples: keep both zeros}$$

Estimating: round everything to 1 s.f. to check

Before trusting a calculator, round each number to 1 significant figure and do the easy sum in your head. If the real answer is nowhere near, you pressed a wrong button.

$$4.87 \times 21.3$$ ← the real question

$$\approx 5 \times 20$$ ← 1 s.f. each: 4.87 → 5, 21.3 → 20

$$= 100$$ ← easy!

The calculator says 103.731. Close to 100 ✓. If it said 1037.31 you'd know a dot went astray.

Find the unknown

PROCEDURE: finding an unknown decimal

1. Undo the operation with its inverse ($\times$ with $\div$, $+$ with $-$), then use the decimal rules above.
2. For "it was rounded to ..." questions, the unknown is a **range**: give the lower and upper bounds.
3. Check by putting the answer back.

Example. $? \times 0.4 = 0.36$

$$? = 0.36 \div 0.4 = 3.6 \div 4 = 0.9$$ ← undo × 0.4; × 10 top and bottom

$$0.9 \times 0.4 = 0.36 \checkmark$$ ← 9 × 4 = 36, 2 places

Example. $? + 2.75 = 5.1$

$$? = 5.10 - 2.75 = 2.35$$ ← undo + with −; pad 5.1 to 5.10

Example. A length rounded to 1 d.p. is 2.7 cm. What could it really be?

$$2.7 - 0.05 = 2.65, \quad 2.7 + 0.05 = 2.75$$

half of the 0.1 step each side

$$2.65 \le x < 2.75$$ ← 2.75 itself would round to 2.8

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Place value	each column is $\frac{1}{10}$ of the one to its left: tenths, hundredths, thousandths
Compare	line up the points, pad with zeros, compare from the left
Decimal to fraction	$0.abc = \dfrac{abc}{1000}$ one 0 per decimal place, then simplify
Fraction to decimal	$\dfrac{a}{b} = a \div b$
+ and −	line up the points; fill gaps with 0
× and ÷ by 10, 100, 1000	digits move 1, 2, 3 places (left for ×, right for ÷)
Multiply	multiply without points; answer has as many decimal places as both numbers together
Divide	$\dfrac{a}{b} = \dfrac{10a}{10b}$ scale both until the divisor is whole
Terminating?	simplified denominator has only the primes 2 and 5
Recurring to fraction	$x = 0.\dot{2}\dot{7} :\ \ 100x - x = 27 \Rightarrow x = \dfrac{27}{99}$ multiply by 10, 100, ... (one 0 per repeating digit), subtract
Rounding	decider 5 or more: round up; 4 or less: leave
Significant figures	count from the first non-zero digit; zeros between count; trailing zeros after the point count; leading zeros never
Bounds	$\text{value} - \frac{1}{2}\text{step} \le x < \text{value} + \frac{1}{2}\text{step}$
Bounds of a ÷	$\max \dfrac{a}{b} = \dfrac{a_{\max}}{b_{\min}}, \quad \min \dfrac{a}{b} = \dfrac{a_{\min}}{b_{\max}}$
Estimate	round every number to 1 s.f. first

🔍 Magnifying glass: your turn

Answers are at the bottom of each group, small and grey. Cover them first!

Reading and comparing

1. What is the 6 worth in 2.068?
2. Which is bigger: 0.7 or 0.69?
3. Put in order: 0.3, 0.33, 0.303
4. Write 0.35 as a fraction (simplest)

1. 6 hundredths 2. 0.7 3. 0.3, 0.303, 0.33 4. 7/20

Working with them

5. 2.4 + 13.75
6. 10 - 3.62
7. 0.6 × 0.7
8. 1.2 × 3.5
9. 7.2 ÷ 0.08

5. 16.15 6. 6.38 7. 0.42 8. 4.2 9. 90

Fraction to decimal

10. 5/8
11. 2/3
12. 7/20

10. 0.625 11. 0.6 recurring 12. 0.35

Rounding

13. 6.748 to 1 d.p.
14. 6.748 to 2 d.p.
15. 0.03049 to 2 s.f.
16. 72,650 to 2 s.f.
17. Estimate 9.8 × 51

13. 6.7 14. 6.75 15. 0.030 16. 73,000 17. 10 × 50 = 500

Significant figures, recurring, bounds

18. How many s.f. in 3.07? In 0.030? In 0.0405?
19. Round 0.04096 to 3 s.f.
20. Write 0.4 recurring (0.444...) as a fraction
21. Does 7/40 terminate? Does 5/6?
22. Bounds of 12.6 (rounded to 1 d.p.)
23. a = 8 and b = 2, both to the nearest whole number. Largest a ÷ b?

18. 3, 2, 3 19. 0.0410 (4, 0, 9; decider 6 bumps the 9) 20. 4/9 ($10x - x = 4$) 21. yes ($40 = 2^3 \times 5$); no (6 has a 3) 22. $12.55 \le x < 12.65$ 23. 8.5 ÷ 1.5 = 5.67 (upper ÷ lower)

Find the unknown

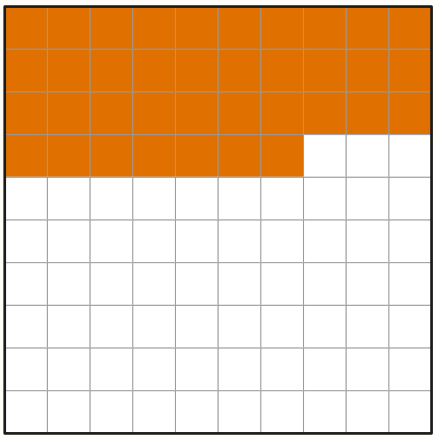
24. ? × 0.4 = 0.36
25. ? − 1.8 = 2.65
26. ? ÷ 0.2 = 15
27. A number rounded to 1 d.p. is 4.3. What are its bounds?
28. A number rounded to 2 s.f. is 0.56. What are its bounds?

24. 0.9 (0.36 ÷ 0.4 = 3.6 ÷ 4) 25. 4.45 (2.65 + 1.8) 26. 3 (15 × 0.2) 27. $4.25 \le x < 4.35$ (± 0.05) 28. $0.555 \le x < 0.565$ (± 0.005)

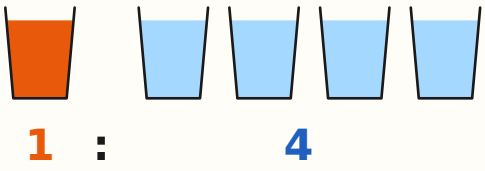
Chapter 4

Percentage and ratio

comparing amounts: out of a hundred, and parts of a recipe



37 out of 100 = 37%



squash to water 1 : 4

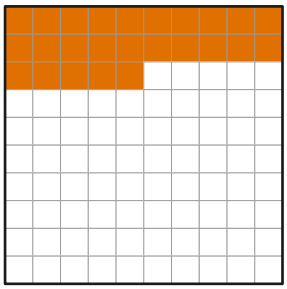
"20% off". "Mix 1 part squash with 4 parts water". "The model is 1 : 72 scale". Both **percentages** and **ratios** answer the same kind of question: how big is one amount compared to another? They're fractions (Chapter 2) dressed for everyday life.

This chapter goes through all of Lecture 3: percentages of amounts, going up and down by a percentage, working backwards, ratios, sharing in a ratio, scaling with the "one of them" table, and mixing solutions.

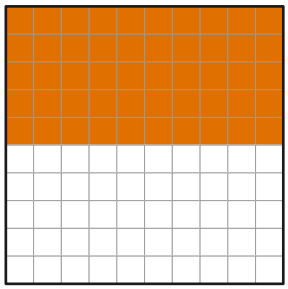
Goes with **Lecture 3** (ratios and proportions).

Per cent means "out of a hundred"

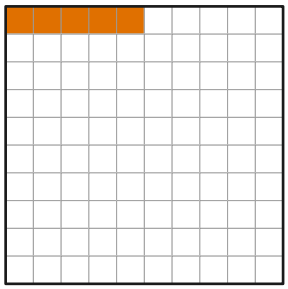
"Cent" means hundred (a century is 100 years). So **per cent** means **per hundred**: chop the whole thing, whatever it is, into 100 equal pieces, and count pieces. The symbol **%** is a tiny drawing of "/100" (a slash between two zeros).



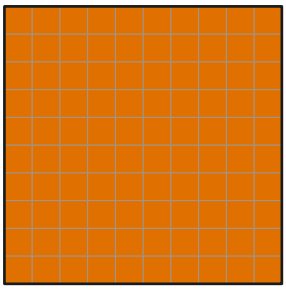
25% = 1/4



50% = 1/2

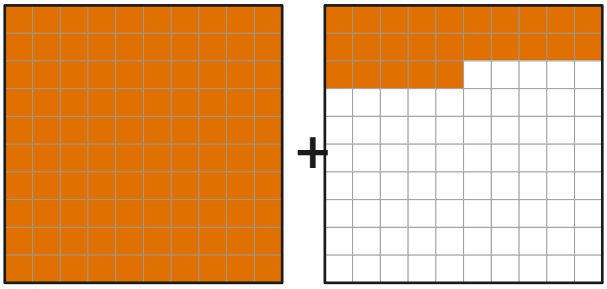


5% = 1/20



100% = all of it

More than 100% just means more than one whole. **125%** is one full square and another quarter of a square:



125% = 125/100 = 1.25

(one and a quarter wholes)

Three outfits for the same number

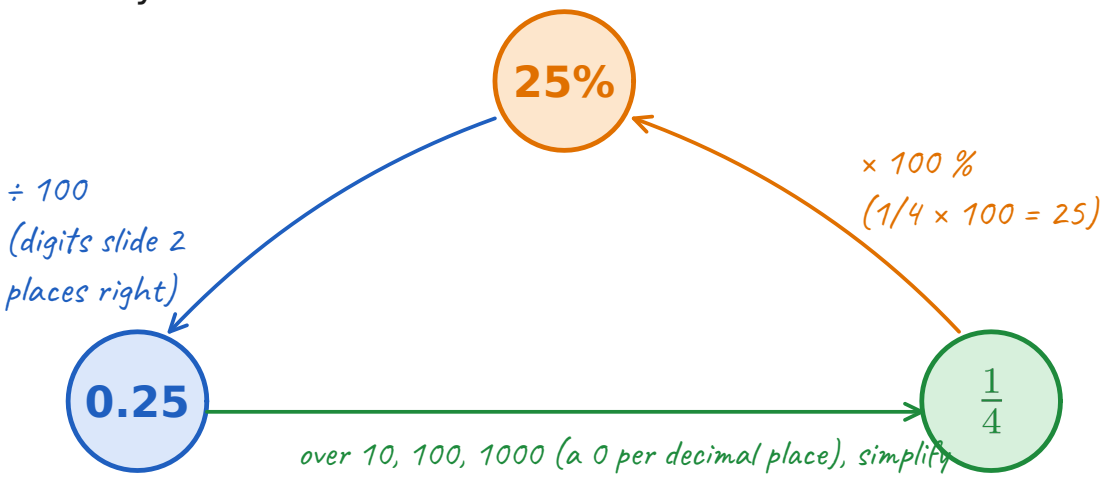
RULE: converting percentages, decimals and fractions

Percentage to decimal: divide by 100. $35\% = 0.35$.

Percentage to fraction: write over 100 and simplify. $35\% = \frac{35}{100} = \frac{7}{20}$.

Decimal or fraction to percentage: multiply by 100. $0.07 = 7\%$; $\frac{3}{8} \times 100 = 37.5\%$.

A quarter can wear three outfits: $\frac{1}{4}$, 0.25 and 25%. Moving between them is always one of these arrows:



Your Lecture 3 examples, one arrow each:

$$5\% = \frac{5}{100} = 0.05$$ ← percent → divide by 100

$$125\% = \frac{125}{100} = 1.25$$ ← same arrow, bigger number

$$0.75 = \frac{75}{100} = 75\%$$ ← decimal → multiply by 100

$$1.583 = 158.3\%$$ ← decimal → multiply by 100

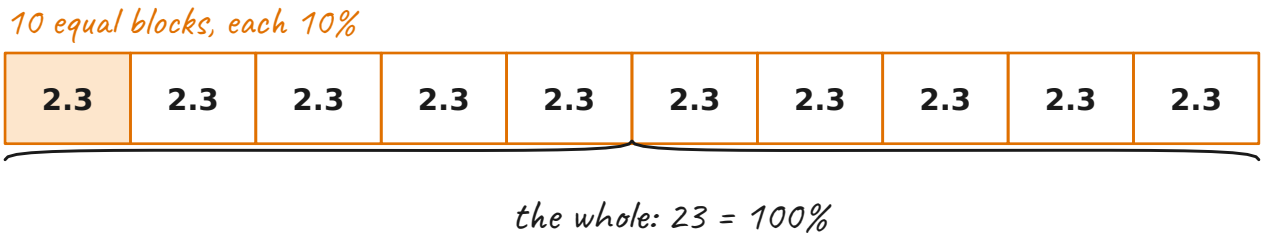
Finding 15% of 23: the 10% ladder

RULE: percentage of an amount

$x\%$ of $y = \frac{x}{100} \times y$ ("of" means $\times$).

Without a calculator: build it from 10% ($\div 10$), 5% (half of 10%) and 1% ($\div 100$).

You don't need a formula to begin with. **10%** is easy: it's a tenth, so divide by 10 (digits slide one place, Chapter 3). Build everything else from 10% and 1%:



$$10\% \text{ of } 23 = 2.3 \quad \leftarrow \text{divide by 10}$$

$$5\% \text{ of } 23 = 1.15 \quad \leftarrow \text{half of 10\%}$$

$$15\% \text{ of } 23 = 2.3 + 1.15 = 3.45 \quad \leftarrow \text{10\% + 5\%}$$

The one-line version from the lecture: "of" means multiply, and $x\%$ means $\frac{x}{100}$:

$$x\% \text{ of } y = \frac{x}{100} \times y$$

x is the percentage,
y is the whole amount

$$15\% \text{ of } 23 = 0.15 \times 23 = 3.45$$

And above 100%: 125% of 20 is $1.25 \times 20 = 25$. Bigger than 20, as it should be.

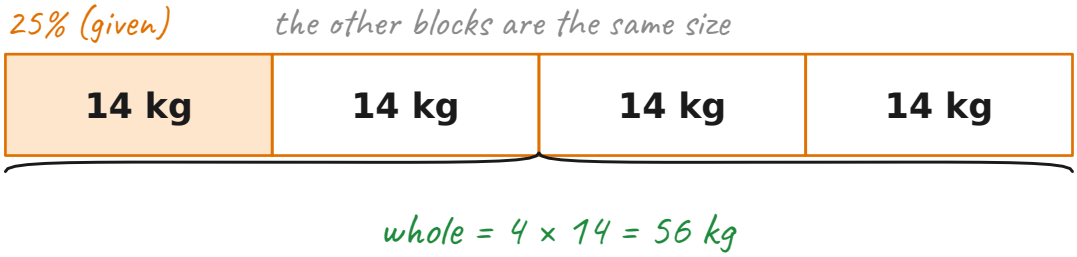
Working backwards: 25% of A is 14 kg

RULE: finding the whole from a percentage of it

If $p\%$ of the whole is N, then 1% is $N \div p$ and the whole (100%) is $\frac{N}{p} \times 100$.

Equivalently: whole $= N \div \frac{p}{100}$. E.g. 20% of a number is 9: $9 \div 0.2 = 45$.

You know a **part** and want the **whole**. Draw the whole as a bar. 25% is a quarter, so cut it into 4 blocks. One block is 14 kg:



The lecture's way (go through 1%, it always works, even for ugly percentages):

$$25\% \rightarrow 14 \quad \leftarrow \text{what we know}$$

$$1\% \rightarrow \frac{14}{25} \quad \leftarrow \div 25 \text{ on both sides}$$

$$100\% \rightarrow 100 \times \frac{14}{25} = 56 \quad \leftarrow \times 100 \text{ on both sides}$$

Common slip: 25% of A is 14, so A is 14 + 25% of 14? No! The 25% is of the **unknown whole**, not of 14. Always ask "percent of **what**?"

Going up and down: the multiplier

RULE: percentage increase and decrease (multipliers)

Increase by r%: new $=$ old $\times \left(1 + \frac{r}{100}\right)$. Up 12% means $\times 1.12$.

Decrease by r%: new $=$ old $\times \left(1 - \frac{r}{100}\right)$. Down 15% means $\times 0.85$.

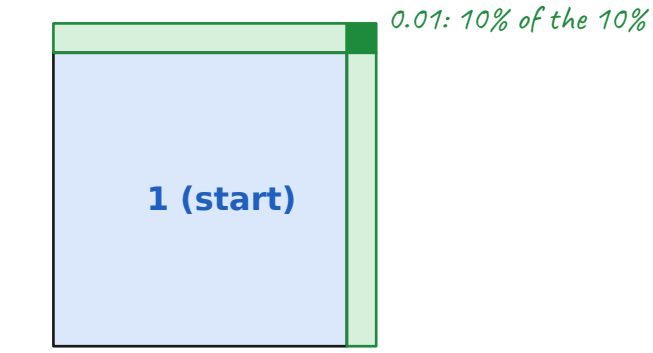
Several changes in a row: multiply the multipliers. Up 10% then down 10% is $\times 1.1 \times 0.9 = \times 0.99$, a 1% fall overall (not 0%).

Increase 56.0 ml by 10%. The old amount is 100%. Adding 10% makes it **110%** of the old amount, so multiply by $\frac{110}{100} = 1.10$. That number is the **multiplier**.

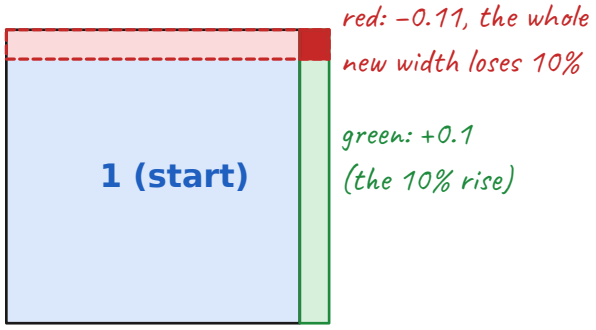
Up 10% then down 10% does NOT get you back to 56! The second 10% is 10% of a bigger number (61.6), so it takes away more (6.16) than was added (5.6).

(The lecture rounds 55.44 to 55.4.) Decreasing by 10% leaves **90%**, so the multiplier is 0.90, exactly as the rule box says: $1 - \frac{10}{100} = 0.9$.

Where did the missing 1% go? Draw it. Two multipliers in a row multiply, and multiplying is the area of a rectangle (Chapter 1). Let the starting amount be a 1 by 1 square:



up 10%, then up 10% again:
1 + 0.1 + 0.1 + 0.01 = 1.21



up 10%, then down 10%:
1 + 0.1 − 0.1 − 0.01 = 0.99

On the right, the 10% cut is taken off the **new, wider** amount, so it also slices off the little dark corner: 10% of the 10% that was added. That corner, 0.01, is the lost 1%. $56 \times 0.99 = 55.44$ ✓. On the left the same corner is a bonus: that's **interest on interest**, which Chapter 21 turns into compound growth.

A shop receipt: a £40 jacket in a 30% sale. Multiplier $\frac{70}{100} = 0.7$, so you pay $40 \times 0.7 =$ **£28**.

Percentage change: always compare to where you started

RULE: percentage change

Percentage change $= \dfrac{\text{new} - \text{original}}{\text{original}} \times 100.$

Divide by the **original** (the value before the change), never by the new value.

A positive answer is an increase; a negative answer is a decrease.

A price goes from £40 to £50. By what percentage did it rise? Take the **change** and compare it with the **original**:

$$\frac{50 - 40}{40} \times 100 = \frac{10}{40} \times 100 = 25\% \quad \leftarrow \text{a rise of 25\%}$$

$$\frac{40 - 50}{50} \times 100 = \frac{-10}{50} \times 100 = -20\% \quad \leftarrow \text{back down: only 20\% (and a minus sign means a fall)}$$

Same £10 gap, different percentages, because the "original" is different each time.

Reverse percentages: finding the original

A price **after** a percentage change is known, and you want the price **before** it.

PROCEDURE: reverse percentages

1. Write the multiplier for the change: $1 + \frac{r}{100}$ for an increase, $1 - \frac{r}{100}$ for a decrease.
2. new = original × multiplier, so **original = new ÷ multiplier**.
3. Check by applying the percentage change to your answer.

Example. After a 20% rise a jacket costs £60. What did it cost before?

$$\text{multiplier} = 1 + \frac{20}{100} = 1.2$$ ← a 20% increase

$$\text{original} = 60 \div 1.2 = £50$$ ← undo × 1.2 by ÷ 1.2

$$50 \times 1.2 = 60 \checkmark$$ ← check

Example. In a 15% sale a lamp costs £102. What was the full price?

$$102 \div 0.85 = £120$$ ← decrease of 15%: multiplier 0.85

$$120 \times 0.85 = 102 \checkmark$$ ← check

Example. A salary of £27 600 includes a 15% rise. What was it before?

$$27\,600 \div 1.15 = £24\,000$$

✗ Common mistakes

✗ $60 - 20\%$ of $60 = 48$ ✓ $60 \div 1.2 = 50$

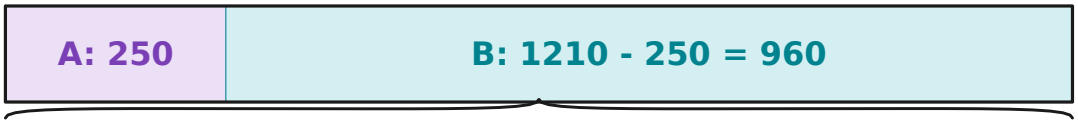
the 20% was of the ORIGINAL price, not of £60. Check: 48 + 20% of 48 = 57.60, not 60

✗ $102 + 15\%$ of $102 = 117.30$ ✓ $102 \div 0.85 = 120$

same mistake the other way: the 15% was taken off the unknown full price, not off £102

What percentage of the mixture is B?

250 mg of A is mixed with some B to make 1.21 g of mixture. What percentage is B? **First, same units**: 1.21 g is 1210 mg (1 g = 1000 mg).



whole mixture 1210 mg

$$\frac{960}{1210} \times 100 \approx 79.3\%$$

Part over whole, times 100 is the percentage every time.

Ratio: a recipe

RULE: what a ratio tells you

$a : b$ means: for every a parts of the first thing there are b parts of the second. **Order matters:** $1 : 4$ is not $4 : 1$.

There are $a + b$ parts altogether, so the first thing is $\frac{a}{a+b}$ of the total and the second is $\frac{b}{a+b}$. In squash $1 : 4$, squash is $\frac{1}{5}$ of the drink.

Simplify by dividing every part by the same number (ideally the HCF): $12 : 18 = 2 : 3$.

Same units first, then simplify: $50\ \text{cm} : 2\ \text{m} = 50 : 200 = 1 : 4$.

Squash bottles say **"1 part squash to 4 parts water"**. A "part" can be any size of cup: an egg-cup or a bucket. The ratio only tells you the **shape of the recipe**, not the amount. We write it **1 : 4**, read "one **to** four".

egg-cups: 1 + 4

mugs: 1 + 4

jugs: 1 + 4

Three very different amounts, all the same taste: every one is 1 : 4.

Double the parts, same ratio: 2 : 8 tastes identical to 1 : 4. So ratios **simplify** exactly like fractions (Chapter 2): divide every number by the same thing.

$$2 : 8 \;=\; 1 : 4 \qquad 260 : 100 : 20 \;=\; 13 : 5 : 1$$

÷ 2 ÷ 20 (the HCF): a ratio can have 3 parts!

Order matters: squash : water 1 : 4 is not water : squash 1 : 4. That second one would be very strong squash!

Mixed units. Simplify $1.2\ \text{kg} : 300\ \text{g}$.

$$1.2\ \text{kg} = 1200\ \text{g}$$ ← 1. same units (grams)

$$1200 : 300 = 4 : 1$$ ← 2. ÷ 300 (the HCF)

Reading the lecture's notation: a : b :: 1 : 72

A model aeroplane: 1 cm on the model stands for 72 cm on the real plane. The scale is **1 : 72**. If the model is a cm long and the real plane is b cm long:

$$a : b \;::\; 1 : 72$$

say: "a is to b as 1 is to 72"
the :: means "is in the same ratio as"

Every ratio statement can be rewritten as a fraction statement. These all say the same thing (the lecture lists them):

$$\frac{a}{b} = \frac{1}{72} \qquad b = 72a \qquad a = \frac{b}{72}$$

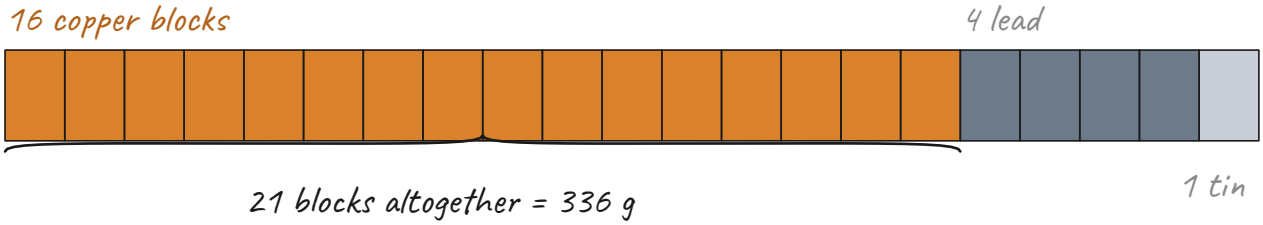
Say it in words: the real thing is 72 times longer than the model; the model is one 72nd of the real thing. A 30 cm model means a $30 \times 72 = 2160$ cm (21.6 m) plane.

Sharing in a ratio: count the parts

PROCEDURE: sharing an amount in a ratio

1. **Add the parts** to get the total number of parts: for $a : b : c$ that is $a + b + c$.
2. **Divide** the total amount by the number of parts: this is the size of **one part**.
3. **Multiply** one part by each number in the ratio.
4. Check that the shares add back up to the total.

Lecture 3, Example 3. 336 g of alloy is copper, lead and tin in the ratio 16 : 4 : 1. How much of each? The lecture's advice: **count the parts**.



$$16 + 4 + 1 = 21 \text{ parts}$$ ← 1. count the parts

$$336 \div 21 = 16 \text{ g per part}$$ ← 2. how heavy is one part?

$$16 \times 16 = 256, \quad 4 \times 16 = 64, \quad 1 \times 16 = 16$$

3. each metal

$$256 + 64 + 16 = 336 \checkmark$$ ← 4. check they add back up

Then 4 g copper, 36 g lead and 4 g tin are added: 260 g, 100 g and 20 g (380 g in all). The new ratio is 260 : 100 : 20, which simplifies (÷ 20) to **13 : 5 : 1**.

PROCEDURE: when you are given one share or the difference, not the total

1. Whatever you are told, first turn it into **"so many parts = this amount"**.
2. Given one share: that share is its own number of parts. Given a difference: subtract the parts.
3. Divide to find one part, then multiply up to whatever is asked for.

Example (one share given). Money is shared A : B = 3 : 5. B gets £40. Find A's share and the total.

$$5 \text{ parts} = £40 \;\Rightarrow\; 1 \text{ part} = £8$$ ← *B is 5 parts*

$$A = 3 \times 8 = £24, \quad \text{total} = 8 \times 8 = £64$$

3 + 5 = 8 parts in all

Example (difference given). A : B = 5 : 3, and A gets £20 more than B. How much does each get?

$$5 - 3 = 2 \text{ parts} = £20 \;\Rightarrow\; 1 \text{ part} = £10$$

the difference is 2 parts

$$A = 5 \times 10 = £50, \quad B = 3 \times 10 = £30$$

check: 50 − 30 = 20

Ratios as percentages (and back)

RULE: converting between ratios and percentages

Ratio to percentage: each part's percentage $= \dfrac{\text{its number}}{\text{sum of all the numbers}} \times 100.$

Percentages to ratio: write the percentages as a ratio and simplify: $30\% : 70\% = 30 : 70 = 3 : 7.$

The percentages of all the parts add up to 100% (check, allowing for rounding).

Lecture 3. The alloy is copper : lead : tin $= 16 : 4 : 1$. What percentage of it is each metal?

$$16 + 4 + 1 = 21 \text{ parts}$$ ← *the whole is 21 parts*

$$\text{copper} = \frac{16}{21} \times 100 \approx 76.2\%$$

$$\text{lead} = \frac{4}{21} \times 100 \approx 19.0\%$$

$$\text{tin} = \frac{1}{21} \times 100 \approx 4.8\%$$ ← *76.2 + 19.0 + 4.8 = 100 ✓*

Back again. A solution is 35% acid and the rest water. Acid : water $= 35 : 65 = 7 : 13$ (÷ 5).

The "one of them" table (unitary method)

Lecture 3, Example 2. 2.504 g of copper oxide contains 2 g of copper. How much copper is in 0.939 g of copper oxide? Make a two-column table and always pass through **one**:

copper oxide	copper
2.504 g	2 g
1 g	$\dfrac{2}{2.504}$ g
0.939 g	$0.939 \times \dfrac{2}{2.504} = 0.75$ g

÷ 2.504 on both sides

× 0.939 on both sides

Whatever you do to one column, do to the other. That keeps the ratio the same, which is what **proportion** means: the two amounts grow and shrink together.

Example 1 with the same table: speeds U and V are in the ratio 8 : 3, and U = 5 m/s. Find V.

$$8 \text{ parts} \to 5$$ ← U is 8 parts

$$1 \text{ part} \to \frac{5}{8}$$ ← ÷ 8

$$3 \text{ parts} \to 3 \times \frac{5}{8} = \frac{15}{8} = 1.875$$ ← × 3: V = 1.875 m/s

The slide does the same examples by **cross-multiplying**; that rule is stated in the next section.

Proportions: cross-multiplying, and inverse proportion

RULE: cross-multiplication

$a : b = c : d$ (also written $a : b :: c : d$) means $\dfrac{a}{b} = \dfrac{c}{d}$.

If $\dfrac{a}{b} = \dfrac{c}{d}$ **then** $a \times d = b \times c$: multiply each top by the **other** bottom.

Why: multiply both sides by b and by d; the bottoms cancel. (Chapter 15 uses this for equations.)

Then divide to get the unknown on its own.

Lecture 3, Example 1. $U : V = 8 : 3$ and $U = 5$ m/s. Find V.

$$\frac{5}{V} = \frac{8}{3}$$ ← U/V = 8/3 with U = 5

$$5 \times 3 = 8 \times V \;\Rightarrow\; 15 = 8V$$ ← cross-multiply

$$V = \frac{15}{8} = 1.875 \text{ m/s}$$ ← ÷ 8

Lecture 3, Example 2a. 2.504 g of copper oxide contains 2 g of copper. How much copper is in 0.939 g? Call it x g.

$$\frac{2}{x} = \frac{2.504}{0.939}$$ ← copper 2 : x matches oxide 2.504 : 0.939

$$2 \times 0.939 = 2.504x \;\Rightarrow\; x = \frac{1.878}{2.504} = 0.75 \text{ g}$$

cross-multiply, ÷ 2.504

Missing term in a proportion. $3 : 8 = x : 20$

$$\frac{3}{8} = \frac{x}{20} \Rightarrow 8x = 60 \Rightarrow x = 7.5$$ ← 3 × 20 = 8 × x

RULE: direct and inverse proportion

Direct proportion: both amounts go up together; double one and the other doubles. The **ratio** $\frac{y}{x}$ stays the same (the unitary method and cross-multiplying work).

Inverse proportion: one goes up as the other goes down; double one and the other **halves**. The **product** $x \times y$ stays the same.

Typical inverse case: more workers, fewer days; faster speed, less time.

Example (inverse). 4 people paint a fence in 6 hours. How long would 3 people take (same speed each)?

$$4 \times 6 = 24 \text{ person-hours}$$ ← *the product stays fixed*

$$24 \div 3 = 8 \text{ hours}$$ ← *fewer people, more time*

Not $6 \times \frac{3}{4} = 4.5$ hours: that would be treating it as direct proportion. Ask "does more of one mean more or less of the other?" first.

Ugly ratios: the lecture's tidy-up trick

Measured ratios are rarely neat, e.g. 262 : 96.7 : 18.1. To make one easy to read:

$$262 : 96.7 : 18.1$$ ← *the raw ratio*

$$2620 : 967 : 181$$ ← *1. × 10 to remove the decimals*

$$14.48 : 5.34 : 1$$ ← *2. ÷ the smallest (181): now it's "per 1 of the last"*

$$1448 : 534 : 100$$ ← *3. × 100 (optional, whole numbers)*

Step 2 is the useful one: "14.48 of the first for every 1 of the last".

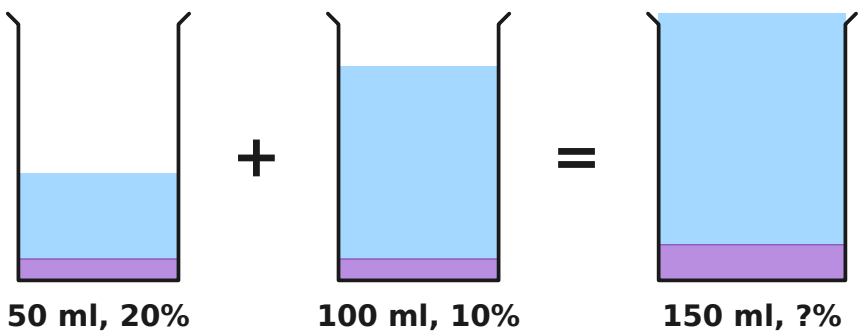
Mixing solutions: beakers

A "20% solution of A" means 20% of the liquid is chemical A and the rest is water. This percentage is called the **concentration** or **strength**.

PROCEDURE: mixing solutions

1. Amount of A in a solution = concentration × volume (e.g. 20% of 50 ml = 10 ml).
2. Add the amounts of A, and add the volumes.
3. New concentration $= \dfrac{\text{total A}}{\text{total volume}} \times 100$. **Never average the percentages.**

Lecture 3, Mixing 1: 50 ml of 20% solution is added to 100 ml of 10% solution. How strong is the result? **Track the A, not the percentages**:



A in first: 20% of 50 = 10 ml
A in second: 10% of 100 = 10 ml
A in total: 20 ml

purple = chemical A, blue = water. (Real chemistry check from the slides: 100 ml alcohol + 100 ml water gives only about 190 ml!)

$$\frac{20}{150} \times 100 = 13.\dot{3}\% \quad \longleftarrow \textit{A over whole, × 100}$$

The tempting wrong answer is to average: (20 + 10) ÷ 2 = 15%. That ignores that there is twice as much of the weaker one, so the answer is pulled towards 10%.

Find the unknown

RULE: which rule do I need?

Value **after** a % change known, original wanted: **divide** by the multiplier.

One share (or the difference between shares) known, total wanted: turn it into "so many parts $=$ amount", find one part, multiply up.

Three of the four numbers in $a : b = c : d$ known: cross-multiply, then divide.

Part and percentage known, whole wanted: whole $=$ part $\div \frac{p}{100}$.

Example. A car loses 30% of its value and is now worth £8 400. What was it worth?

$$8400 \div 0.7 = £12\,000$$ ← decrease 30%: multiplier 0.7

Example. Sand and cement are mixed 5 : 2. 14 kg of cement is used. How much mortar is made?

$$2 \text{ parts} = 14 \text{ kg} \Rightarrow 1 \text{ part} = 7 \text{ kg}$$ ← cement is 2 parts

$$7 \times 7 = 49 \text{ kg}$$ ← 5 + 2 = 7 parts in total

Example. $\frac{x}{12} = \frac{5}{8}$

$$8x = 60 \Rightarrow x = 7.5$$ ← cross-multiply: x × 8 = 12 × 5

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Per cent	$x\% = \dfrac{x}{100}$ % → decimal: ÷ 100; decimal or fraction → %: × 100
Percentage of	$x\%$ of $y = \dfrac{x}{100} \times y$
Whole from a part	$\text{whole} = \text{part} \div \dfrac{p}{100}$
Increase / decrease by r%	$\text{new} = \text{old} \times \left(1 \pm \dfrac{r}{100}\right)$ + for an increase, − for a decrease; repeated changes: multiply the multipliers
Percentage change	$\dfrac{\text{new} - \text{old}}{\text{old}} \times 100$ divide by the original (old) value
Reverse percentage	$\text{original} = \dfrac{\text{new}}{1 \pm \frac{r}{100}}$ divide by the multiplier; **not** new − r% of new
Ratio as fractions	$a : b \Rightarrow \dfrac{a}{a+b}$ and $\dfrac{b}{a+b}$ of the total
Simplify a ratio	same units first, then divide every part by the HCF
Share in a ratio	add the parts, divide the total, multiply each part
One share or difference	"so many parts = amount", then one part, then multiply
Ratio to percentage	$\dfrac{\text{part}}{\text{sum of parts}} \times 100$
Cross-multiplication	$\dfrac{a}{b} = \dfrac{c}{d} \Rightarrow ad = bc$ direct proportion: the ratio stays the same
Inverse proportion	$x \times y = \text{constant}$
Mixing	$\dfrac{\text{total A}}{\text{total volume}} \times 100$ never average the percentages

Recipe cards: your turn

Answers under each group, small and grey. Draw a bar or a table for every one!

Harder mixtures ("how much 15% solution must be added to...?" and the two-solution "compounds" problems) need an equation with a letter in it. They're solved step by step in Chapter 15.

Percentages

1. Write 35% as a decimal and a fraction
2. Write 0.07 as a percentage
3. Find 20% of 65
4. Find 17.5% of 80
5. 30% of a number is 12. Find the number

1. 0.35 = 7/20 2. 7% 3. 13 4. 14 5. 40

Up, down and change

6. Increase 250 by 12%
7. Decrease 80 by 15%
8. A price falls from £60 to £45. Percentage change?
9. What single multiplier is +20% then -20%?

6. 280 7. 68 8. -25% 9. 1.2 × 0.8 = 0.96 (a 4% fall)

Ratio

10. Simplify 45 : 60
11. Share £84 in the ratio 3 : 4
12. Squash 2 : 7. 450 ml of drink: how much squash?
13. Map scale 1 : 25,000. 4 cm on the map is how many km?

10. 3 : 4 11. £36 and £48 12. 100 ml 13. 1 km

Mixtures

14. 200 ml of 30% solution + 300 ml of 5% solution. Strength?

14. (60 + 15)/500 = 15%

Ratio rules

15. Simplify 50 cm : 2 m
16. Write the ratio 9 : 3 : 3 as percentages
17. A solution is 40% acid, the rest water. Write acid : water
18. 6 workers take 10 days. How long would 4 workers take?
19. A : B = 7 : 2 and A gets £35 more than B. Find each share

15. 1 : 4 (50 : 200) 16. 60%, 20%, 20% (9/15, 3/15, 3/15) 17. 2 : 3 (40 : 60) 18. 15 days (inverse: 6 × 10 = 60, ÷ 4) 19. A £49, B £14 (5 parts = £35, 1 part = £7)

Find the unknown

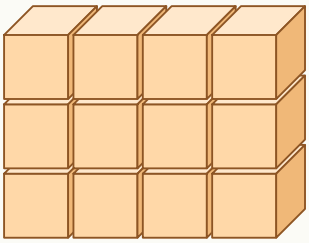
20. After a 10% rise a price is £55. What was it before?
21. After a 25% discount a coat costs £45. What was the full price?
22. Boys : girls = 4 : 5 and there are 20 girls. How many pupils altogether?
23. x : 15 = 4 : 6. Find x
24. 2.5 / x = 5 / 8. Find x
25. 12% of a number is 18. Find the number

20. £50 (55 ÷ 1.1) 21. £60 (45 ÷ 0.75) 22. 36 (5 parts = 20, 1 part = 4, 9 parts) 23. 10 (6x = 15 × 4 = 60) 24. 4 (5x = 2.5 × 8 = 20) 25. 150 (18 ÷ 0.12)

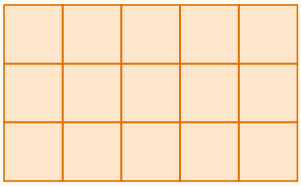
Chapter 5

Measurement

how long, how much space, how much stuff, how fast



12 little cubes: a volume



15 squares: an area

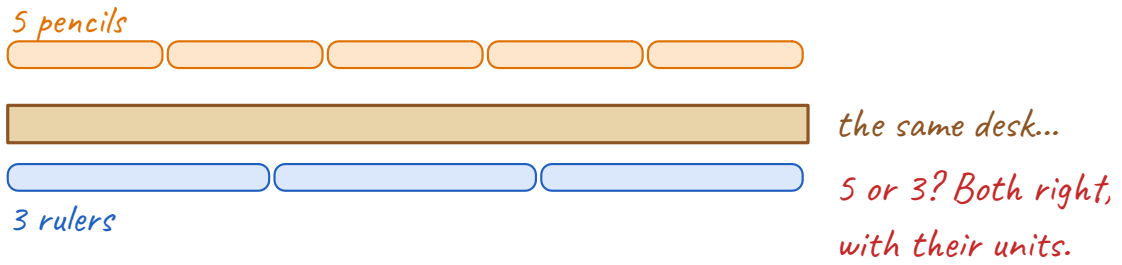
Measuring is counting how many **units** fit. A length counts little rulers, an area counts little squares, a volume counts little cubes. Almost every mistake in this topic comes from mixing up units, so this chapter slows right down on **changing units**.

It also covers perimeter, area and volume of the everyday shapes, time (which is sneakily not decimal), and "per" units like metres per second, which appear all over your lectures (the slides write speeds as m s^{-1}).

Used throughout the lectures (units in L03, L09, L10, L14-L16).

A number without a unit means nothing

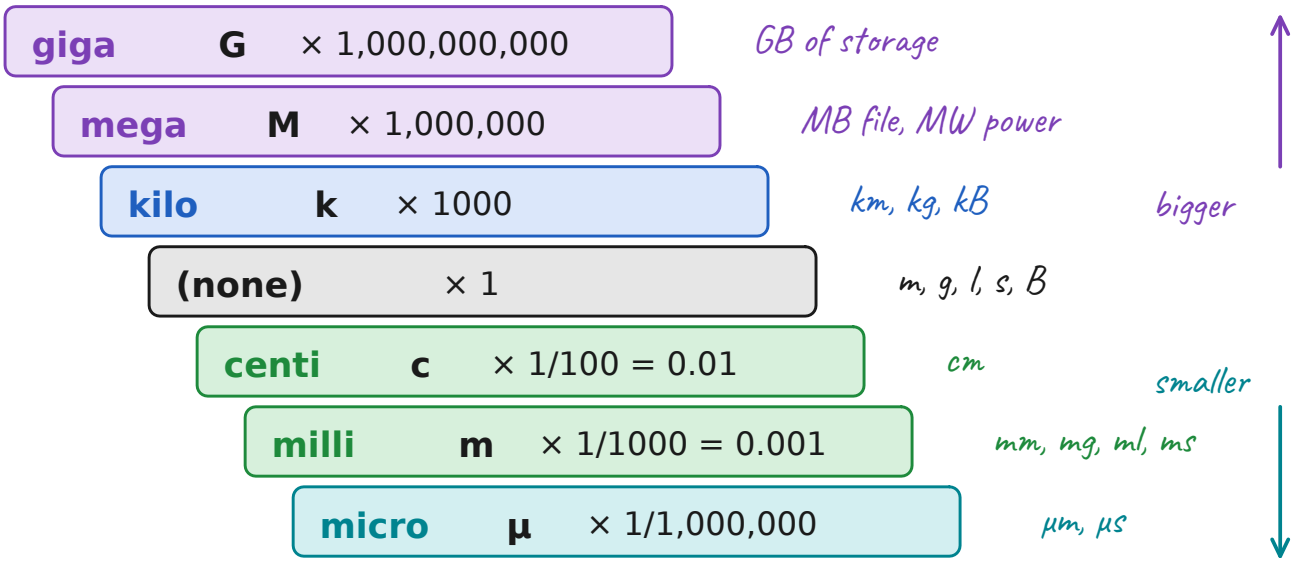
"The desk is 3 long." 3 what? Pencils? Hands? Metres? The **unit** is the thing you counted. **The number and the unit always travel together.**



Scientists agreed on standard units so everyone gets the same number: the **metre** (m) for length, the **kilogram** (kg) for mass, the **second** (s) for time, the **litre** (l) for liquid volume.

The prefix ladder: kilo, centi, milli...

Instead of inventing new names, the metric system sticks a **prefix** (a word-start) on the front. The prefix is a multiplier (Chapter 4). Each one means the same thing on **every** unit:



So a **kilo**gram is 1000 grams, a **milli**litre is a thousandth of a litre, a **centi**metre is a hundredth of a metre. **Careful:** capital M (mega) and small m (milli) differ by a factor of a thousand million!

Computer aside: in computing, a kilobyte is sometimes 1024 bytes instead of 1000, for reasons you'll see in Chapter 10 (number bases).

Changing units is like swapping coins

RULE: changing length, mass and capacity units

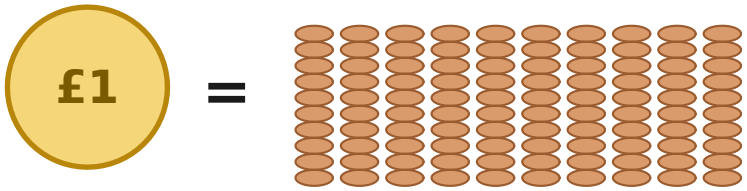
Know the swap rate: $1\ \text{km} = 1000\ \text{m}$, $1\ \text{m} = 100\ \text{cm}$, $1\ \text{cm} = 10\ \text{mm}$, $1\ \text{kg} = 1000\ \text{g}$, $1\ \text{l} = 1000\ \text{ml}$.

Big unit to small unit: multiply by the rate (the number gets bigger).

Small unit to big unit: divide by the rate (the number gets smaller).

Several steps: do one swap at a time along the chain.

A £1 coin is always worth exactly 100 pennies. Not roughly, not today's rate: **always**, by definition. Swap one pound for pennies and the **number gets bigger** (1 becomes 100), but you have exactly the same money, just counted in smaller pieces. Units work the same way: **1 m is always exactly 100 cm**.

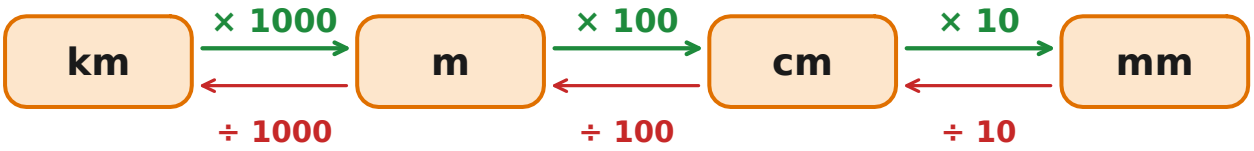


100 pennies (10 stacks of 10)

same amount of money, bigger number, because each piece is smaller. 1 m → 100 cm works identically.

(Currency exchange between countries is a bad model for this, because those rates change every day. Unit swaps never change: 1 km is 1000 m today, tomorrow and forever.)

going to smaller units: multiply (you need more of them)



going to bigger units: divide (you need fewer of them)

Example: 2.5 km in centimetres. Walk along the chain, one swap at a time:

$$2.5\ \text{km} \quad \longleftarrow \text{start}$$

$$2.5 \times 1000 = 2500\ \text{m} \quad \longleftarrow \text{swap km → m: × 1000}$$

$$2500 \times 100 = 250{,}000\ \text{cm} \quad \longleftarrow \text{swap m → cm: × 100}$$

Backwards: 350 g in kg. Grams are small, kilograms big, so divide: $350 \div 1000 = 0.35$ kg.

Sense check every time: is the new number bigger or smaller? Metres to centimetres must get bigger (a small unit needs lots). If it shrank, you went the wrong way.

Perimeter: walk round the fence

RULE: perimeter

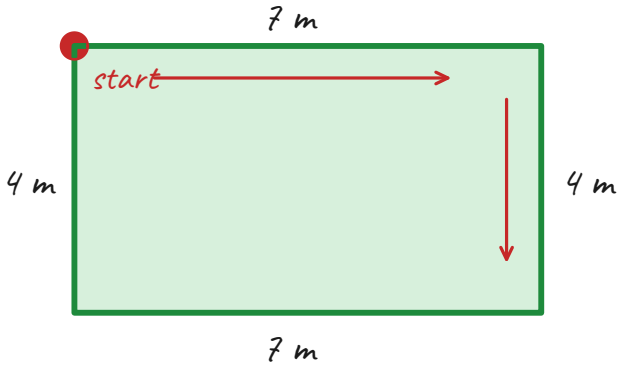
Perimeter = the sum of the lengths of **all** the sides. Its unit is a length (m, cm).

Rectangle, length l and width w: $P = 2l + 2w = 2(l + w)$. Square, side s: $P = 4s$.

Circle (circumference): $C = \pi d = 2\pi r$.

Shapes with missing sides: work the missing lengths out from the given ones first.

The **perimeter** is the distance all the way round the edge. Imagine walking along a garden fence and adding up every stretch until you're back at the start:

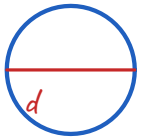


$$7 + 4 + 7 + 4 = 22 \text{ m}$$

every side counted once;
the answer is a length (m)

With the formula: $P = 2(l + w) = 2(7 + 4) = 2 \times 11 = 22$ m. Same answer, fewer additions.

Circles: the perimeter of a circle is called the **circumference**. Wrap a string round any circle and it's always about 3.14 diameters long; that number is π (Part 0):



$$C = \pi d = 2\pi r$$

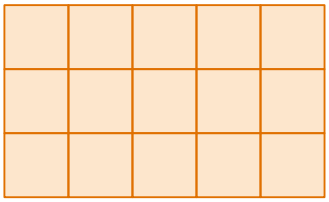
d = diameter (across), r = radius (centre to edge, half of d)

Area: count the squares

Area is how much flat surface a shape covers. It is measured in **square units** (cm², m²).

Shape	Area	Notes
Rectangle	$A = l \times w$	length × width
Square	$A = s^2$	side × side
Triangle	$A = \frac{1}{2}bh$	h is the **perpendicular** height, not the slant side
Parallelogram	$A = bh$	perpendicular height again
Trapezium	$A = \frac{1}{2}(a+b)h$	a, b = the two parallel sides
Circle	$A = \pi r^2$	r = radius (half the diameter)

A rectangle 5 cm by 3 cm holds 3 rows of 5 squares:

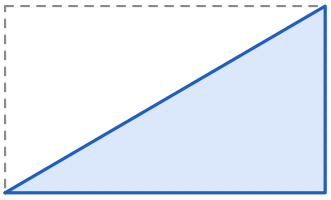


$$5 \times 3 = 15 \text{ cm}^2$$

cm² ("square centimetres"): each little square is 1 cm by 1 cm

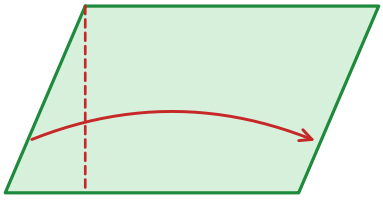
Other shapes are rectangles in disguise. **Cut and move pieces**; the area can't change:

triangle = half a rectangle



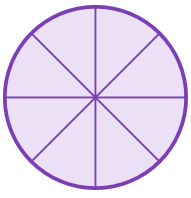
$$A = \frac{1}{2} \times b \times h$$

slide the corner across



$$A = b \times h$$

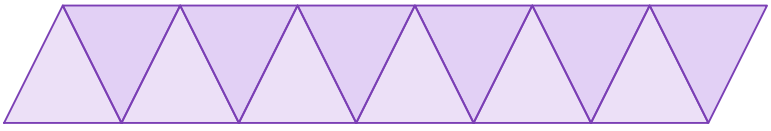
circle



$$A = \pi r^2$$

b = base, h = height measured straight up (at right angles to the base), not along the slope

Why πr² for a circle? Cut it into many thin pizza slices and lay them alternately point-up, point-down. They form an almost-rectangle: its height is the radius r and its length is half the circumference, πr. Area ≈ πr × r = πr².



height r, length πr

Trapezium and compound shapes

RULE: area of a trapezium

A trapezium has one pair of parallel sides, lengths a and b, a distance h apart.

$A = \frac{1}{2}(a+b)h$: average the parallel sides, multiply by the height.

Why: two copies, one turned upside down, make a parallelogram with base $a+b$ and height h.

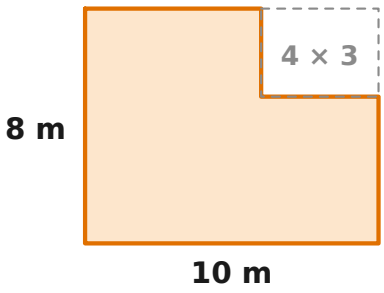
$$A = \tfrac{1}{2}(6+10) \times 4 = \tfrac{1}{2} \times 16 \times 4 = 32 \text{ cm}^2$$

parallel sides 6 cm and 10 cm, height 4 cm

PROCEDURE: compound shapes

1. Split the shape into rectangles, triangles, … (draw the cut lines).
2. Work out any missing side lengths from the ones you are given.
3. **Add** the areas of the pieces. For a hole or a corner cut out, **subtract** its area from the whole.
4. Perimeter: add only the **outside** edges (the cut lines are not part of it).

Example. The L-shape below: a 10 m by 8 m rectangle with a 4 m by 3 m corner removed.



whole rectangle $10 \times 8 = 80$ m^2

corner cut out $4 \times 3 = 12$ m^2

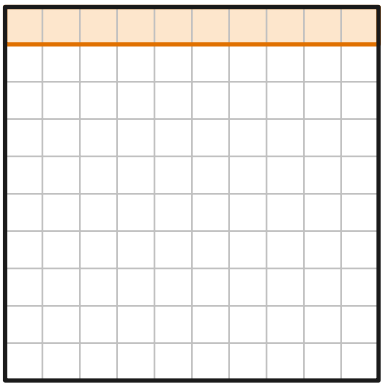
$$A = 80 - 12 = 68 \text{ m}^2$$

perimeter: still $2(10+8) = 36$ m (the step just moves two edges inward)

The big trap: 1 m² is NOT 100 cm²

It feels as if it should be. 1 m = 100 cm, so surely 1 m² = 100 cm²? Test it on a square small enough to see every piece: **one square centimetre**, blown up, with millimetres marked. 1 cm = 10 mm.

1 cm = 10 mm



1 cm
= 10 mm

orange: ONE row of 10 little squares. That's only 10 mm².

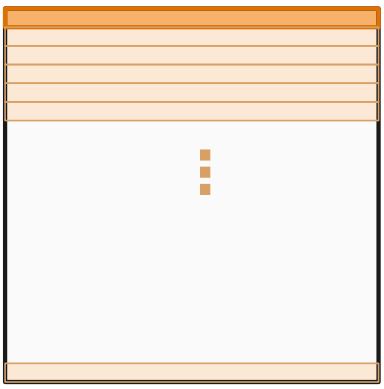
the whole square needs 10 rows of 10 = 100 little squares.

$$1 \text{ cm}^2 = 100 \text{ mm}^2$$

not 10 mm²: the 10 happens twice, once across and once down

Now the square metre. Same idea, bigger numbers. 100 cm² is **one ribbon** 100 cm long and 1 cm tall, running along the top edge. The square metre is 100 cm tall, so it needs **100 of those ribbons** stacked up:

1 m = 100 cm



1 m
= 100 cm
(100 ribbons)

top ribbon = 100 cm²
(100 little 1 cm squares in a row)

100 ribbons × 100 cm² each:

$$1 \text{ m}^2 = 10{,}000 \text{ cm}^2$$

ribbons drawn thicker than real, so you can see them

Why the little ² catches everyone. "m²" is short for m × m. So swap **each** m for 100 cm:

$$1 \text{ m}^2 = 1 \text{ m} \times 1 \text{ m} = 100 \text{ cm} \times 100 \text{ cm} = 10{,}000 \text{ cm}^2$$

The trap is mixing up two different things that look alike:

100 cm^2 ← *a hundred little 1 cm squares (one ribbon)*

$(100 \text{ cm})^2$ ← *a square whose SIDE is 100 cm: 10,000 little squares*

1 m² is the second one. The ² squares everything inside the bracket: the 100 **and** the cm.

RULE: converting area and volume units

If $1\ (\text{big unit}) = k\ (\text{small units})$ for length, then for **area** the rate is k^2 and for **volume** it is k^3.

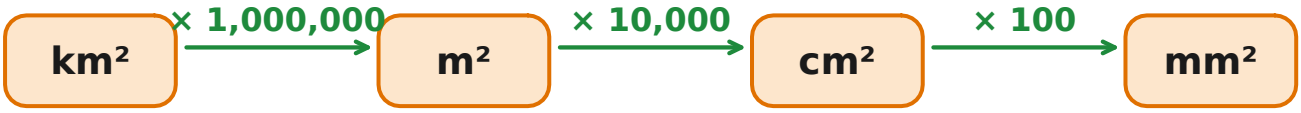
$1\ \text{m} = 100\ \text{cm}$, so $1\ \text{m}^2 = 100^2 = 10\,000\ \text{cm}^2$ and $1\ \text{m}^3 = 100^3 = 1\,000\,000\ \text{cm}^3$.

$1\ \text{cm} = 10\ \text{mm}$, so $1\ \text{cm}^2 = 100\ \text{mm}^2$ and $1\ \text{cm}^3 = 1000\ \text{mm}^3$ (so mm³ → cm³ is $\div 1000$).

Liquids: $1\ \text{l} = 1000\ \text{cm}^3$, $1\ \text{ml} = 1\ \text{cm}^3$, $1\ \text{m}^3 = 1000\ \text{l}$.

As before: big to small, multiply; small to big, divide.

The chain for areas:



Example: a room of 12 m² in cm²: $12 \times 10\,000 = 120\,000$ cm².

$$4500\ \text{mm}^3 = 4500 \div 1000 = 4.5\ \text{cm}^3$$ ← small to big: ÷ 10^3

$$0.25\ \text{m}^3 = 0.25 \times 1000 = 250\ \text{l}$$ ← $1\ \text{m}^3 = 1000$ litres

Volume: stack the cubes

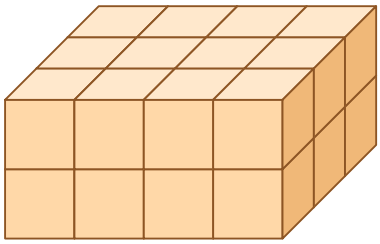
RULE: volume

Volume is measured in **cubic units** (cm³, m³).

Cuboid: $V = l \times w \times h$.

Any prism (same cross-section all the way through, e.g. a cylinder): $V = \text{area of cross-section} \times \text{length}$. Cylinder: $V = \pi r^2 h$.

Volume is how much 3-D space something fills, counted in unit cubes. A box 4 cm long, 3 cm wide and 2 cm tall: one layer on the floor holds 4 × 3 = 12 cubes, and there are 2 layers.



$$4 \times 3 \times 2 = 24 \text{ cm}^3$$

cm³ ("cubic centimetres"): count of 1 cm cubes

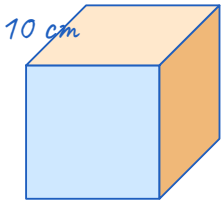
The layer idea works for any "straight-up" shape: volume = area of the floor × height. A cylinder (a tin can) is a stack of circles, like a pile of coins:

$$V = \pi r^2 \times h$$

floor area (a circle) × height.

r = 3 cm, h = 10 cm: V = π × 9 × 10 ≈ 282.7 cm³

Volume units, and litres. A 1 m cube holds 100 × 100 × 100 = 1,000,000 cm³ (the swap rate is cubed). Liquids use litres:



a 10 cm × 10 cm × 10 cm cube = 1000 cm³ = 1 litre

so 1 cm³ = 1 ml (a millilitre, a thousandth of a litre)

RULE: surface area of a cuboid

Surface area is the total area of all the faces (cm², m²): the cardboard needed to make the box.

A cuboid has 6 rectangular faces in 3 matching pairs:
$S = 2(lw + lh + wh)$.

A cube of side s: $S = 6s^2$ (and $V = s^3$).

Example. A box 8 cm by 5 cm by 2 cm.

$$V = 8 \times 5 \times 2 = 80 \text{ cm}^3$$

$$S = 2(8 \times 5 + 8 \times 2 + 5 \times 2) = 2(40 + 16 + 10) = 132 \text{ cm}^2$$

top/bottom, front/back, sides

Time is not decimal!

RULE: time

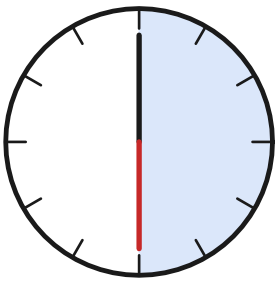
$1 \text{ h} = 60 \text{ min}$, $1 \text{ min} = 60 \text{ s}$, so $1 \text{ h} = 3600 \text{ s}$.

Decimal hours to h and min: keep the whole hours; minutes = decimal part $\times 60$. $2.75 \text{ h} = 2 \text{ h } 45 \text{ min}$.

h and min to decimal hours: hours + minutes $\div 60$.
$2 \text{ h } 15 \text{ min} = 2 + \frac{15}{60} = 2.25 \text{ h}$.

Adding times / durations: add minutes and hours separately; every 60 minutes carries 1 hour. Never use a calculator's decimal result as "hours.minutes".

An hour has **60** minutes, not 100. So 1.5 hours is **not** 1 hour 50 minutes. The ".5" means half an hour, which is 30 minutes. Look at the clock:



0.5 hour = half the clock = 30 min

0.5 h	0.5 × 60 = 30 min
0.25 h	0.25 × 60 = 15 min
0.1 h	0.1 × 60 = 6 min
2.75 h	2 h and 0.75 × 60 = 45 min

Hours → minutes: multiply the decimal part by 60. Minutes → hours: divide by 60, so 40 minutes is $\frac{40}{60} = \frac{2}{3} \approx 0.67$ hours.

Time arithmetic. A train leaves at 09:45 and the journey takes 2 h 35 min. When does it arrive?

$$9\text{ h }45\text{ min} + 2\text{ h }35\text{ min} = 11\text{ h }80\text{ min}$$ ← add hours, add minutes

$$80\text{ min} = 1\text{ h }20\text{ min} \Rightarrow 12{:}20$$ ← carry 60 min as 1 hour

Duration. From 10:50 to 13:25: count up to the next hour, then on.

$$10{:}50 \to 11{:}00 \to 13{:}00 \to 13{:}25: \quad 10 + 120 + 25 = 155\text{ min}$$

$$155\text{ min} = 2\text{ h }35\text{ min} = 2\tfrac{35}{60} \approx 2.58\text{ h}$$ ← 155 ÷ 60 = 2 r 35

✗ Common mistakes

✗ $2\text{ h }15\text{ min} = 2.15\text{ h}$ ✓ $2\text{ h }15\text{ min} = 2.25\text{ h}$

15 minutes is 15/60 = 0.25 of an hour, not 0.15

✗ $1.5\text{ h} = 1\text{ h }50\text{ min}$ ✓ $1.5\text{ h} = 1\text{ h }30\text{ min}$

0.5 h × 60 = 30 min

"Per" means divide: speed, density and friends

RULE: speed, distance, time (and density)

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \text{ so } \text{distance} = \text{speed} \times \text{time} \text{ and } \text{time} = \frac{\text{distance}}{\text{speed}}.$$

$$\text{density} = \frac{\text{mass}}{\text{volume}}, \text{ so } \text{mass} = \text{density} \times \text{volume} \text{ and } \text{volume} = \frac{\text{mass}}{\text{density}}.$$

Units must match: km and hours give km/h; m and s give m/s. Convert minutes to hours (÷ 60) before using km/h.

A sprinter runs 100 m in 12.5 s. **Speed** is how many metres she covers **per** second: share the 100 m out equally among the 12.5 seconds.

$$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{100 \text{ m}}{12.5 \text{ s}} = 8 \text{ m/s}$$

read m/s as "metres per second": the units divide too!

Three ways to write the same unit. Your lecture slides write metres per second as "ms^{-1}" (Lecture 3 even writes "5ms-1"). It means the same as m/s, **but strictly it needs a space**: "m s^{-1}". Without the space, "ms" is the symbol for **milliseconds**, so "ms^{-1}" would mean "per millisecond". Write it with a space (or as m/s) in your own work:

$$\text{m/s} \qquad \frac{\text{m}}{\text{s}} \qquad \text{m s}^{-1}$$

the little $^{-1}$ means "per"; Chapter 9 explains why

Changing "per" units: one swap at a time. 8 m/s in km/h (kilometres per hour):

$$8 \text{ m per second}$$ ← *start*

$$8 \times 3600 = 28{,}800 \text{ m per hour}$$ ← *1 hour = 3600 seconds: 3600 times as far*

$$28{,}800 \div 1000 = 28.8 \text{ km per hour}$$ ← *metres → km: ÷ 1000*

Density is mass per volume. A 50 cm³ block weighing 135 g has density $\frac{135}{50} = 2.7$ g/cm³ (that's aluminium). The other two versions of each formula are in the rule box above; Chapter 14 shows how to rearrange any formula.

Working backwards: find the missing measurement

PROCEDURE: finding an unknown length, radius or time

1. Write the formula that links what you know to what you want.
2. Put in the known numbers.
3. **Undo** the operations around the unknown: $\times$ by $\div$, squaring by square-rooting, π by dividing by π.
4. Check the units, and that the size is sensible.

Know	Want	Use
area A of a square	side	$s = \sqrt{A}$
area A and width w of a rectangle	length	$l = A \div w$
area A, base b of a triangle	height	$h = 2A \div b$
circumference C	radius	$r = \dfrac{C}{2\pi}$
area A of a circle	radius	$r = \sqrt{\dfrac{A}{\pi}}$
volume V, length l, width w	height	$h = V \div (l \times w)$
distance d, speed s	time	$t = d \div s$

Examples.

$s^2 = 49 \text{ cm}^2 \Rightarrow s = \sqrt{49} = 7 \text{ cm}$ ← square: undo squaring

$l \times 4 = 36 \Rightarrow l = 36 \div 4 = 9 \text{ m}$ ← rectangle 36 m², width 4 m

$2\pi r = 31.4 \Rightarrow r = \dfrac{31.4}{2\pi} \approx 5.0 \text{ cm}$ ← circumference 31.4 cm

$\pi r^2 = 50 \Rightarrow r = \sqrt{\dfrac{50}{\pi}} \approx 3.99 \text{ cm}$ ← circle area 50 cm²

$$t = \frac{150\ \text{km}}{60\ \text{km/h}} = 2.5\ \text{h} = 2\ \text{h}\ 30\ \text{min}$$ ← *150 km at 60 km/h*

$$d = 8\ \text{m/s} \times 45\ \text{s} = 360\ \text{m}$$ ← *distance = speed × time*

✗ Common mistakes

✗ $\pi r^2 = 50 \Rightarrow r = \frac{50}{\pi}$

✓ $r = \sqrt{\frac{50}{\pi}}$

dividing by π gives r^2, not r: take the square root last

✗ $\text{time} = \text{speed} \div \text{distance}$

✓ $\text{time} = \text{distance} \div \text{speed}$

check the units: km ÷ (km/h) = h

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Changing units	big unit to small: multiply; small to big: divide. 1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm, 1 kg = 1000 g, 1 l = 1000 ml
Perimeter	$P = 2(l + w)$ sum of all the outside edges; circle $C = \pi d = 2\pi r$
Rectangle, triangle	$A = lw, \qquad A = \frac{1}{2}bh$ h perpendicular to b
Parallelogram, trapezium	$A = bh, \qquad A = \frac{1}{2}(a + b)h$
Circle	$A = \pi r^2$
Compound shapes	split up and add; subtract holes and cut-out corners
Volume	$V = lwh, \qquad V = \pi r^2 h$ prism: cross-section area × length
Surface area (cuboid)	$S = 2(lw + lh + wh)$
Area and volume units	square the length rate for area, cube it for volume: $1\ \text{m}^2 = 10^4\ \text{cm}^2$, $1\ \text{m}^3 = 10^6\ \text{cm}^3$
Litres	$1\ \text{l} = 1000\ \text{cm}^3$, $1\ \text{ml} = 1\ \text{cm}^3$, $1\ \text{m}^3 = 1000\ \text{l}$
Time	decimal part × 60 = minutes; minutes ÷ 60 = decimal hours; 1 h = 3600 s
Speed	$s = \frac{d}{t}, \quad d = st, \quad t = \frac{d}{s}$
Density	$\rho = \frac{m}{V}, \quad m = \rho V, \quad V = \frac{m}{\rho}$ ρ (rho) = density
Working backwards	write the formula, substitute, undo the operations around the unknown

📏 Tape measure: your turn

Answers under each group in grey. Draw the chain or the picture first!

Changing units

1. 3.2 km in m
2. 450 mm in m
3. 2.5 kg in g
4. 750 ml in litres
5. 0.004 s in ms

1. 3200 m 2. 0.45 m 3. 2500 g 4. 0.75 l 5. 4 ms

Shapes

6. Perimeter and area of a 9 cm by 6 cm rectangle
7. Area of a triangle, base 10 cm, height 7 cm
8. Circumference and area of a circle, radius 5 cm (1 d.p.)

6. 30 cm, 54 cm² 7. 35 cm² 8. 31.4 cm, 78.5 cm²

Area and volume units

9. 3 m² in cm²
10. Volume of a 5 × 4 × 3 cm box
11. 2 litres in cm³
12. 0.5 m³ in cm³

9. 30,000 cm² 10. 60 cm³ 11. 2000 cm³ 12. 500,000 cm³

Time and per

13. 2.4 hours in hours and minutes
14. 150 km in 2.5 h: speed?
15. 20 m/s in km/h

13. 2 h 24 min 14. 60 km/h 15. 72 km/h

More shapes, units and time

16. Area of a trapezium: parallel sides 5 cm and 9 cm, height 4 cm
17. A 12 m by 9 m rectangle with a 5 m by 4 m corner cut out: area?
18. Surface area of a 6 cm × 3 cm × 2 cm box
19. 6500 mm^3 in cm^3
20. 3 h 40 min in decimal hours (2 d.p.)
21. A bus leaves at 08:35 and takes 1 h 50 min. Arrival time?

16. 28 cm^2 (½ × 14 × 4) 17. 88 m^2 (108 − 20) 18. 72 cm^2 (2(18 + 12 + 6)) 19. 6.5 cm^3 (÷ 1000) 20. 3.67 h (3 + 40/60) 21. 10:25 (9 h 85 min = 10 h 25 min)

Find the unknown

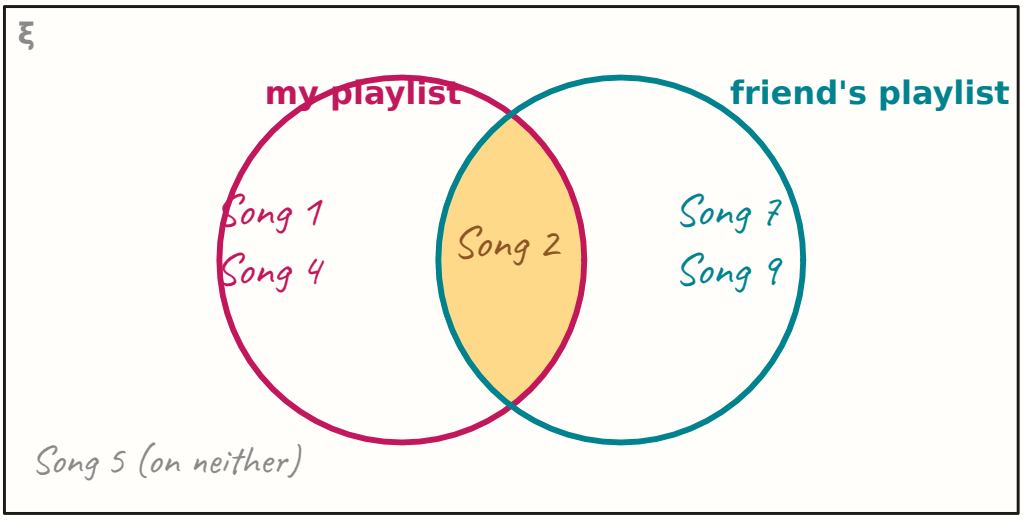
22. A square has area 64 cm^2. How long is a side?
23. A rectangle has area 54 m^2 and width 6 m. How long is it?
24. A circle has circumference 62.8 cm. Find the radius (1 d.p.)
25. A circle has area 154 cm^2. Find the radius (1 d.p.)
26. How long does 210 km take at 84 km/h?
27. A box has volume 120 cm^3, length 6 cm and width 4 cm. Find its height

22. 8 cm (√64) 23. 9 m (54 ÷ 6) 24. 10.0 cm (62.8 ÷ 2π) 25. 7.0 cm (√(154 ÷ π)) 26. 2.5 h = 2 h 30 min (210 ÷ 84) 27. 5 cm (120 ÷ (6 × 4))

Chapter 6

Sets

collections of things, drawn as hula hoops on the floor



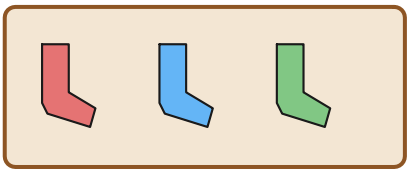
A **set** is just a collection of things. That's all. Your sock drawer is a set of socks; a playlist is a set of songs. Sets give us a tidy way to talk about groups, what's in both groups, what's in either and what's left out.

Your Lecture 13 uses sets to describe the kinds of numbers (natural, integer, rational, real) and the inputs a function accepts (Chapter 18). This chapter decodes every symbol, including the scary-looking $\{x \mid 1 \leq x \leq 4\}$.

Goes with **Lecture 13** (types of numbers, sets, domains).

A set is a sock drawer

Open the drawer: a red sock, a blue sock and a green sock. Write the set with **curly brackets** (called **braces**) and commas:



$$S = \{\text{red},\ \text{blue},\ \text{green}\}$$

S is the set's name; each sock is an element

Two rules make a set different from a list:

$$\{\text{red}, \text{blue}, \text{green}\} = \{\text{green}, \text{red}, \text{blue}\}$$

1. order doesn't matter (the drawer isn't sorted)

$$\{\text{red}, \text{red}, \text{blue}\} = \{\text{red}, \text{blue}\}$$

2. repeats don't count ("is red in there?" yes)

Is it in the drawer? The symbol $\in$ (a curly E, for "element") means "is in"; with a slash, $\notin$ means "is not in":

$\text{blue} \in S$ *"blue is in S" (true)* $\text{pink} \notin S$ *"pink is not in S"*

Sets can hold numbers too: $A = \{2, 4, 6, 8\}$ is the set of even numbers from 2 to 8, and $3 \notin A$. The number of elements is written $n(A)$; here $n(A) = 4$.

Two playlists: in both, in either, only in one

You and a friend each have a playlist. Songs are numbered 1 to 9 in the whole music library. The whole library is called the **universal set** ξ ("ksee", a Greek letter) or U: everything we're allowed to talk about.

$A = \{1, 2, 3, 4\}$ $B = \{2, 3, 7, 9\}$ $\xi = \{1, 2, \ldots, 9\}$

my playlist *friend's playlist*

$A \cap B = \{2, 3\}$

intersection, "A and B"

songs on BOTH playlists. ∩ looks like a bridge (or an upside-down cup) sitting on both.

$A \cup B = \{1, 2, 3, 4, 7, 9\}$

union, "A or B"

merge the playlists, no duplicates. ∪ looks like a cup that holds everything.

$A - B = \{1, 4\}$

difference, "A minus B"

songs only on mine: take away anything your friend also has.

$A' = \{5, 6, 7, 8, 9\}$

complement, "not A"

everything in the library that's NOT on my playlist. (Some books write A with a bar over it.)

Memory hook: ∪ is a **U** for **U**nion; ∩ is the other one. "Or" in maths always includes "both" (song 2 is in A or B, it's in both, so it counts).

RULE: the four set operations

$A \cap B$: elements in A **and** in B. $A \cup B$: elements in A **or** B (or both), each listed once.

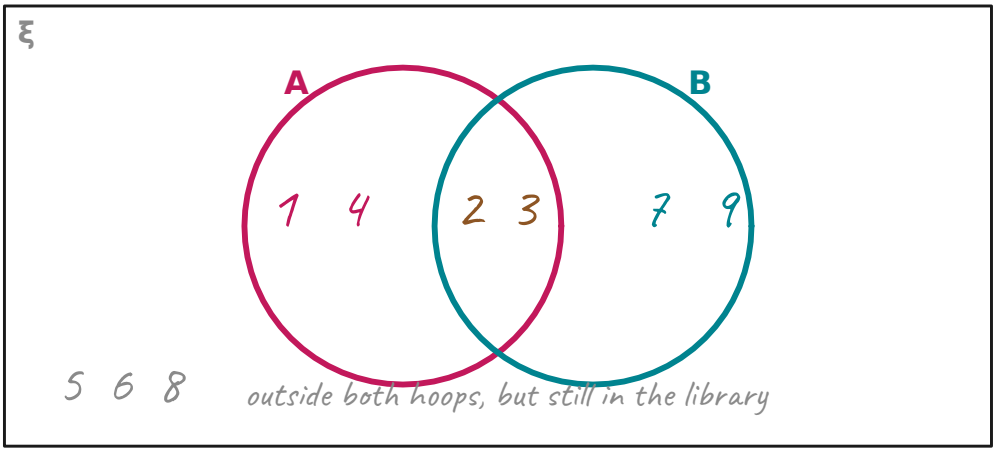
A': elements of ξ that are **not** in A. Counting: $n(A') = n(\xi) - n(A)$.

$A - B$: elements of A that are not in B. Same thing as $A \cap B'$ ("in A and not in B").

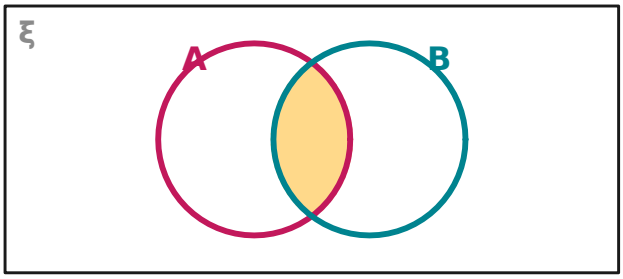
$$n(A') = n(\xi) - n(A) = 9 - 4 = 5$$ ← *matches A' = {5, 6, 7, 8, 9}*

Hula hoops on the floor: Venn diagrams

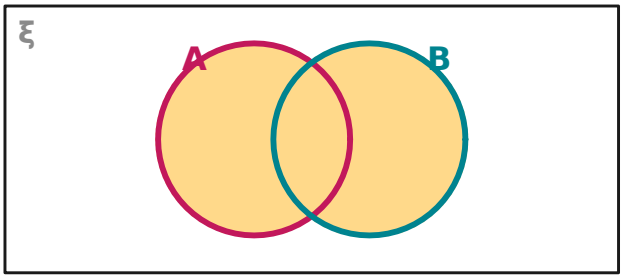
Lay two hula hoops on the floor so they overlap, inside a rectangle (the whole library). Put each song where it belongs. Songs on both playlists stand in the overlap. This picture is a **Venn diagram**:



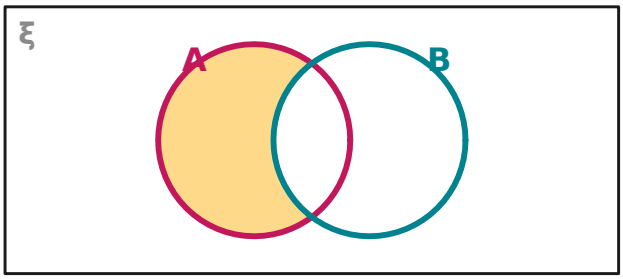
Each operation from the last page is a **region** of the floor. Shade it and read it off:



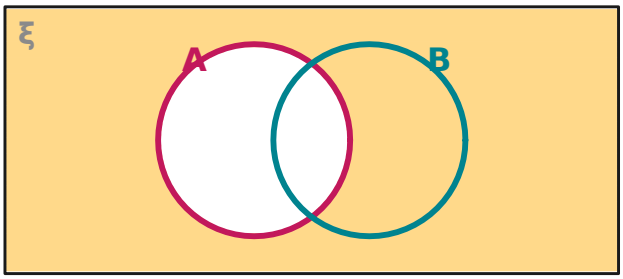
$A \cap B$ the overlap only



$A \cup B$ anything inside a hoop



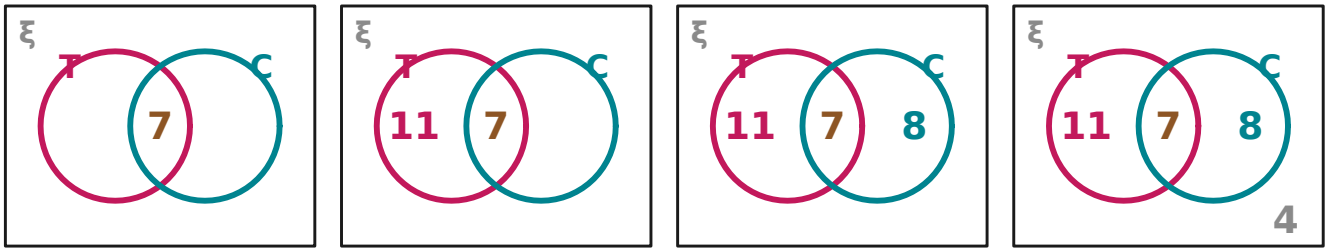
$A - B$ A's hoop, minus the overlap



A' everything outside A

Counting with a Venn diagram: fill it from the middle

A survey of **30** students: **18** drink tea, **15** drink coffee, **7** drink both. How many drink neither? The trick: **always start with the overlap**, then work outwards.



1. both: 7 goes in the middle

2. tea only: 18 - 7 = 11

3. coffee only: 15 - 7 = 8

4. neither: 30 - 11 - 7 - 8 = 4

Check: 11 + 7 + 8 + 4 = 30 students ✓ *(T = tea, C = coffee)*

The classic mistake: 18 + 15 = 33 is more than 30 students! That's because the 7 who drink both got counted twice, once with tea and once with coffee. The fix, as a rule:

RULE: counting a union (inclusion-exclusion)

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$: add the two sets, then take off the overlap you counted twice.

Everyone is somewhere: $n(\xi) = n(A \cup B) + n\big((A \cup B)'\big)$, i.e. total = at least one + neither.

Any three of $n(A)$, $n(B)$, $n(A \cap B)$, $n(A \cup B)$ give you the fourth.

$$n(T \cup C) = 18 + 15 - 7 = 26$$ ← *drink at least one; 30 - 26 = 4 drink neither*

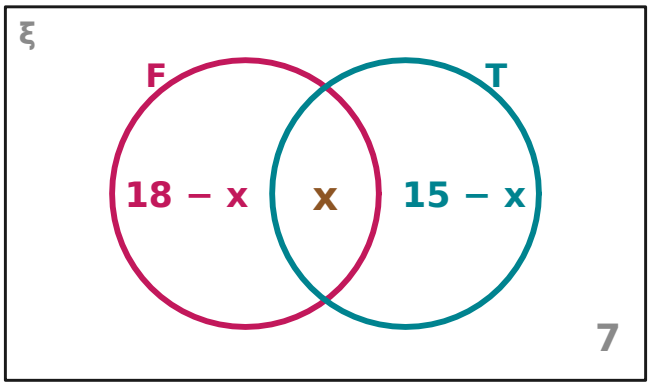
When the overlap is the unknown: call it x

Often you're told the totals and asked for the overlap. You can't start in the middle, so put an **x** there and let the algebra do it.

PROCEDURE: Venn diagram with an unknown

1. Let x be the number in the region you don't know (usually the overlap).
2. Fill every other region in terms of x: A only $= n(A) - x$, B only $= n(B) - x$.
3. Put the "neither" number outside both circles.
4. All regions add up to the total: write that equation and solve for x.
5. Check: put x back in. No region may come out negative.

Example. 30 students: 18 play football, 15 play tennis, 7 play neither. How many play both?



$$(18 - x) + x + (15 - x) + 7 = 30$$

$$40 - x = 30$$

$$x = 10$$

check: 8 + 10 + 5 + 7 = 30 ✓

Same answer from the rule. At least one sport: $30 - 7 = 23$. Then

$$23 = 18 + 15 - n(F \cap T) \;\Rightarrow\; n(F \cap T) = 33 - 23 = 10$$

Finding a set's size instead. $n(A \cup B) = 50$, $n(A) = 32$, $n(A \cap B) = 12$. Find $n(B)$:

$$50 = 32 + n(B) - 12 \;\Rightarrow\; n(B) = 50 - 20 = 30$$

Three sets: same idea, fill from the middle out

With three circles there are **8 regions**: the middle (all three), three "exactly two" regions, three "only" regions, and outside. The method is the same as with two: start in the middle.

PROCEDURE: filling a three-set Venn diagram

1. Middle: write $n(A \cap B \cap C)$.
2. Each pair: $n(A \cap B)$ includes the middle, so "exactly A and B" $= n(A \cap B) -$ middle. Do all three pairs.
3. Each "only": $n(A)$ minus the three numbers already inside circle A.
4. Outside: total minus everything inside the circles.
5. Check: all 8 regions add to the total; none is negative.

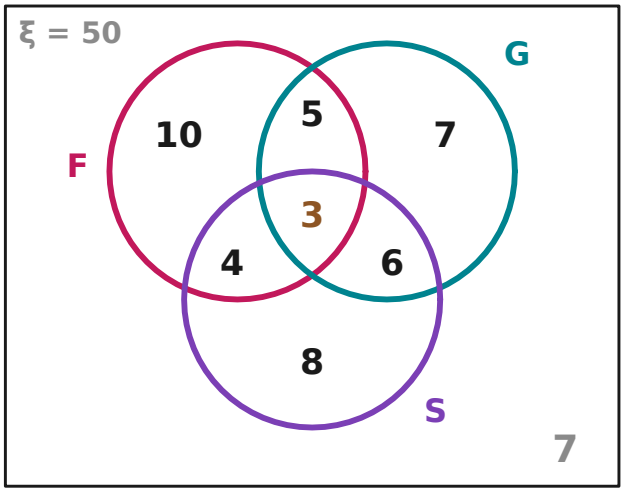
RULE: counting a union of three sets

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Adding the three sets counts the pairs twice, so take them off; that removes the middle three times, so add it back once.

Read carefully: "8 study French and German" normally includes anyone who also studies Spanish. "Only French and German" does not.

Example. 50 students: 22 study French, 21 German, 21 Spanish; 8 study French and German, 7 French and Spanish, 9 German and Spanish; 3 study all three. How many study none? How many study French only?



$$\text{all three} = 3$$
$$F \cap G \text{ only} = 8 - 3 = 5$$
$$F \cap S \text{ only} = 7 - 3 = 4$$
$$G \cap S \text{ only} = 9 - 3 = 6$$
$$F \text{ only} = 22 - 5 - 4 - 3 = 10$$
$$G \text{ only} = 21 - 5 - 6 - 3 = 7$$
$$S \text{ only} = 21 - 4 - 6 - 3 = 8$$
$$\text{none} = 50 - 43 = 7$$

43 = 3 + 5 + 4 + 6 + 10 + 7 + 8, everyone inside

Same answer from the rule:

$$n(F \cup G \cup S) = 22 + 21 + 21 - 8 - 7 - 9 + 3 = 43$$

so 50 - 43 = 7 study none

Complements of unions and intersections (De Morgan)

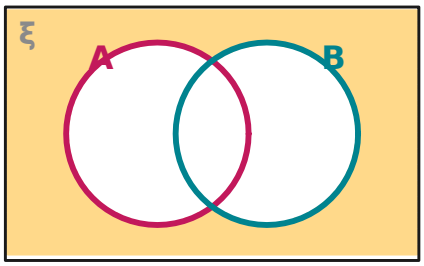
"Not in A or B" is easy to misread. Shade it and you'll see it's the region outside both hoops, which is the same as "not in A **and** not in B".

RULE: De Morgan's laws for sets

$(A \cup B)' = A' \cap B'$: outside the union = outside A and outside B.

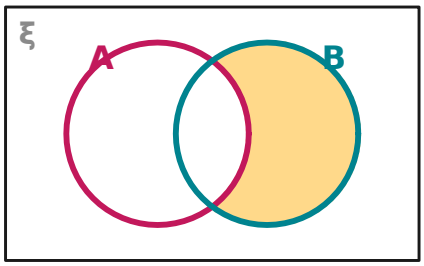
$(A \cap B)' = A' \cup B'$: outside the overlap = outside A or outside B.

Taking the complement swaps $\cup$ and $\cap$ and puts a dash on each set.



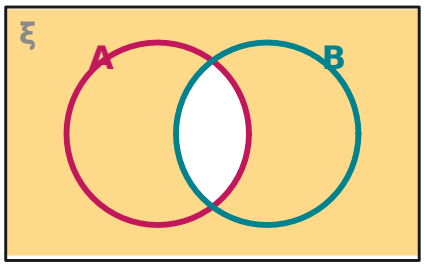
$(A \cup B)'$

outside both



$A' \cap B$

B only



$(A \cap B)'$

all but the overlap

Check on the playlists (ξ = {1, ..., 9}, A = {1, 2, 3, 4}, B = {2, 3, 7, 9}):

$$(A \cup B)' = \xi - \{1,2,3,4,7,9\} = \{5,6,8\}$$

$$A' \cap B' = \{5,6,7,8,9\} \cap \{1,4,5,6,8\} = \{5,6,8\}$$

the same set

$$A' \cap B = \{7,9\}$$ ← *in B, not in A: same as B - A*

✗ Common mistakes

✗ $(A \cup B)' = A' \cup B'$ ✓ $(A \cup B)' = A' \cap B'$

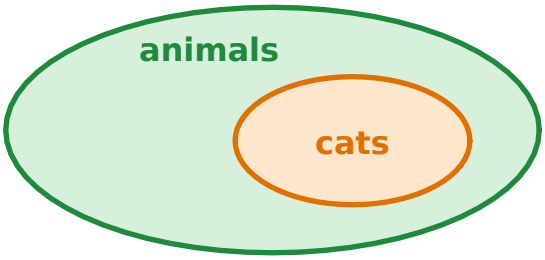
the symbol flips when the dash goes inside

✗ $A' \cap B = (A \cap B)'$ ✓ $A' \cap B = B - A$

only B's own part, not everything outside the overlap

Subsets and the empty set

Every cat is an animal, so the set of cats sits **entirely inside** the set of animals. We say cats are a **subset** of animals, and draw one hoop inside another:



$$\text{Cats} \subset \text{Animals}$$

$\subset$ reads "is a (proper) subset of". It looks like a C for "contained in".

The empty set has nothing in it at all, written $\{\}$ or $\varnothing$. Your lecture's example: the set of fish that drive BMWs. It counts as a subset of **every** set, because it has no element that could fail to be inside.

RULE: subsets

$A \subseteq B$: every element of A is also in B (A may equal B). Every set is $\subseteq$ itself.

$A \subset B$ (**proper** subset): $A \subseteq B$ and $A \neq B$, so B has at least one extra element.

$\varnothing$ is a subset of every set.

A set with n elements has 2^n subsets (each element is either in or out), and $2^n - 1$ proper subsets.

The set of **all** subsets of S is called the **power set** $P(S)$ or $\mathcal{P}(S)$. So $|P(S)| = 2^{|S|}$. For $S = \{a, b, c\}$, $P(S)$ has 8 members and always includes $\varnothing$ and S itself.

Example. The subsets of $\{a, b, c\}$:

$$\varnothing,\ \{a\}, \{b\}, \{c\},\ \{a, b\}, \{a, c\}, \{b, c\},\ \{a, b, c\}$$

$2^3 = 8$ subsets, $\quad 8 - 1 = 7$ proper ← *the last one is the set itself*

Decoding set-builder notation

Listing works for small sets. For big ones, maths describes the **rule for getting in**, like a club's entry rules. This is from your Lecture 13 slides:

$$\{\ x \mid 1 \leq x \leq 4\ ,\ x \in \mathbb{Z}\ \}$$

- x — numbers x
- $1 \leq x \leq 4$ — x is between 1 and 4, 1 and 4 allowed
- $x \in \mathbb{Z}$ — and x is a whole number
- $\{$ … $\mid$ … $\}$ — the set of all … such that …

$= \{1, 2, 3, 4\}$

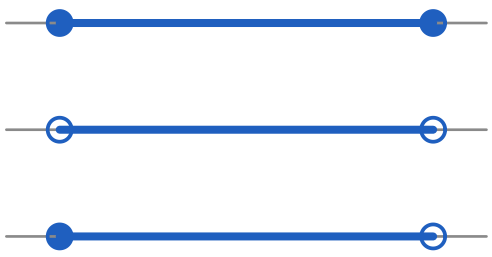
Why the "$x \in \mathbb{Z}$" part matters (it's missing on the slide): without it, the rule $1 \leq x \leq 4$ also lets in 2.5, 3.14 and every other number in between, which is an endless set, not {1, 2, 3, 4}.

Intervals. When you do want every number between two ends, there's a shorter way. The square bracket means the end is **included**; the round bracket means it's **not**:

$$[-1, 1] = \{x \mid -1 \leq x \leq 1\}$$

$$(0, 3) = \{x \mid 0 < x < 3\}$$

$$[0, 3)$$



filled dot = end included, hollow dot = end left out

RULE: interval notation

$[a, b] = \{x \mid a \leq x \leq b\}$ (both ends in); $(a, b) = \{x \mid a < x < b\}$ (both ends out); $[a, b)$ and $(a, b]$ mix the two.

No upper end: $[a, \infty) = \{x \mid x \geq a\}$, $(a, \infty) = \{x \mid x > a\}$. No lower end: $(-\infty, b] = \{x \mid x \leq b\}$.

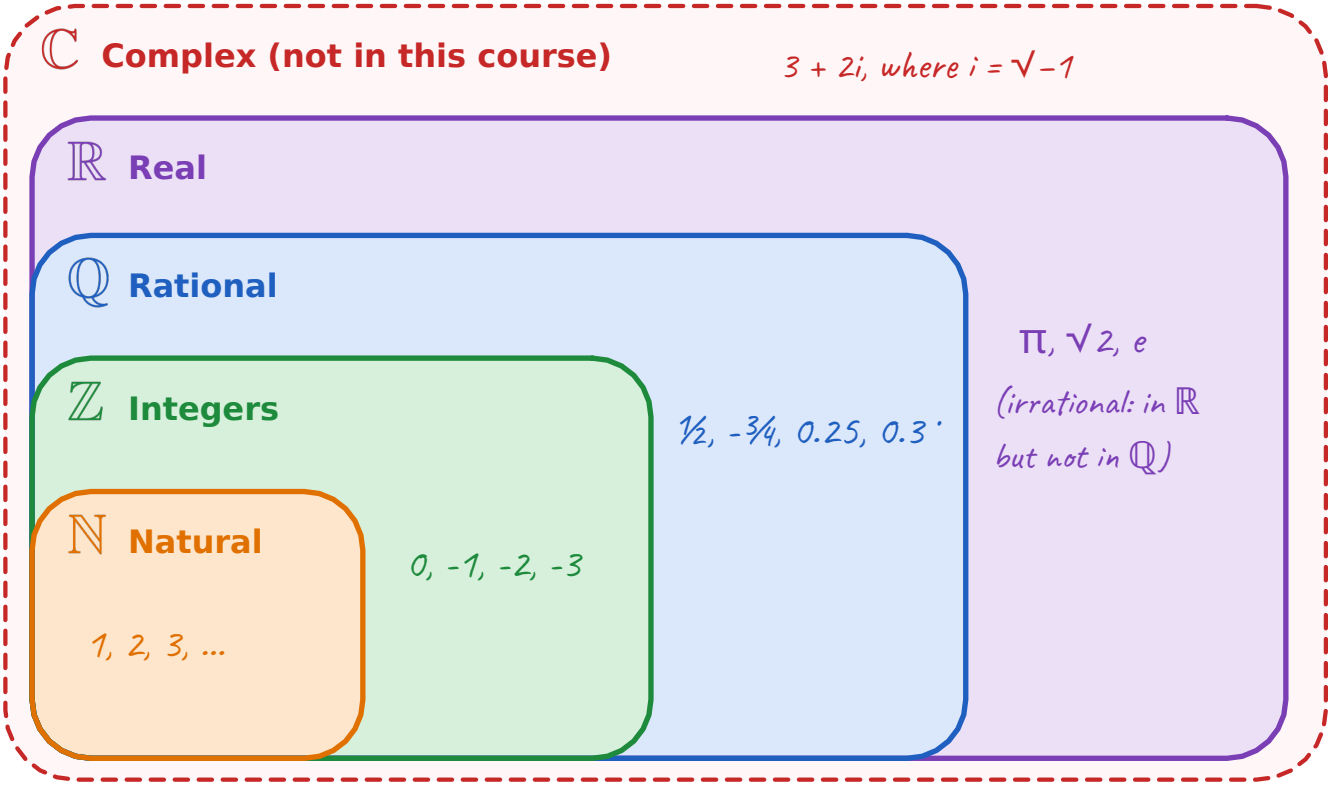
∞ is not a number you can reach, so it always takes a **round** bracket.

Intervals are sets of real numbers: $[0, 3)$ contains 2.9 and 0, but not 3.

$$[0, \infty) = \{x \mid x \geq 0\}$$ ← zero and every positive number

Nesting dolls: the families of numbers

Numbers come in families, and each family fits inside a bigger one, like Russian nesting dolls. Each has a fancy double-struck letter (from your Lecture 13):



ℕ natural: the counting numbers 1, 2, 3, ... (some books include 0).

ℤ integers: whole numbers, negatives and zero too: ..., -2, -1, 0, 1, 2, ... (Z is from the German "Zahlen", numbers.)

ℚ rational: any fraction of two integers ("ratio-nal"; Q for quotient): ½, 6/1, 3/7, -0.25. Their decimals either stop or repeat (Chapter 3).

ℝ real: every point on the number line, including numbers whose decimals never stop and never repeat: π = 3.14159..., √2 = 1.41421...; these are called irrational.

ℂ complex: the dashed outer doll. No real number squares to give a negative (2^2 and $(-2)^2$ are both +4), so mathematicians invented a new number **i** with $i^2 = -1$. Numbers like 3 + 2i are not on the number line at all, which is why your Lecture 13 defines the reals as "any number that could be used as a distance on a line (so excludes complex numbers)". You won't need them in this course; they turn up later in electronics and waves.

So $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$: each doll fits in the next. The slides write the irrationals as $\mathbb{R} - \mathbb{Q}$ (real minus rational), which is set difference from earlier in this chapter.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Membership	$x \in A$, $x \notin A$; order and repeats don't matter; $n(A)$ = number of elements
Intersection, union	$A \cap B$: in both; $A \cup B$: in either (or both)
Complement	$n(A') = n(\xi) - n(A)$ A' = everything in ξ not in A
Difference	$A - B = A \cap B'$: in A and not in B
Counting a union	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$ don't count the overlap twice
Venn with an unknown	put x in the unknown region, fill the rest in terms of x, add all regions to the total, solve
Three sets	$n(A \cup B \cup C) = \text{singles} - \text{pairs} + n(A \cap B \cap C)$ singles $= n(A) + n(B) + n(C)$; pairs $= n(A \cap B) + n(A \cap C) + n(B \cap C)$. Venn: fill from the middle out
De Morgan	$(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$
Subsets	$A \subseteq B$: may be equal; $A \subset B$: proper (B has more); $\varnothing$ is a subset of every set
Number of subsets / power set	$\lvert P(S)\rvert = 2^{\lvert S\rvert}$ $P(S)$ is the set of all subsets of S (including $\varnothing$ and S); $2^n - 1$ proper
Set-builder	$\{x \mid \text{rule}\}$: the set of all x such that the rule holds
Intervals	$[a, b]$, (a, b), $[a, \infty)$: square = end included, round = end left out; ∞ always round
Number families	$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$; irrationals $= \mathbb{R} - \mathbb{Q}$

Hula hoops: your turn

ξ = {1, 2, …, 10}, A = {1, 2, 3, 4, 5}, B = {2, 4, 6, 8, 10}. Answers in grey.

Reading sets

1. Is 6 ∈ A?
2. List A ∩ B
3. List A ∪ B
4. List A - B
5. List B'
6. Find n(A ∪ B)

1. no 2. {2, 4} 3. {1, 2, 3, 4, 5, 6, 8, 10} 4. {1, 3, 5} 5. {1, 3, 5, 7, 9} 6. 8

Venn counting

7. 40 people: 25 have a cat, 20 a dog, 12 both. How many have neither?
8. How many have a cat but no dog?

7. 40 - (13 + 12 + 8) = 7 8. 13

Decoding

9. Write {x | x ∈ ℤ, -2 ≤ x < 3} as a list
10. Which family is the smallest that contains -5?
11. Is 0.75 rational? Is √2?
12. Draw [2, 5) on a number line

9. {-2, -1, 0, 1, 2} 10. ℤ 11. yes (3/4); no 12. filled dot at 2, hollow at 5

Set rules

13. List (A ∪ B)′
14. List A′ ∩ B
15. List (A ∩ B)′ and A′ ∪ B′. Are they equal?
16. How many subsets does {a, b, c, d} have?
17. Is A ⊆ A? Is A ⊂ A?
18. Write {x | x ≥ 4} in interval notation

13. {7, 9} 14. {6, 8, 10} 15. both {1, 3, 5, 6, 7, 8, 9, 10}; yes (De Morgan) 16. $2^4 = 16$ 17. yes; no (a proper subset must be smaller) 18. [4, ∞)

Find the unknown

19. 50 students: 28 play football, 22 play tennis, 9 play neither. How many play both?

20. $n(A \cup B) = 40$, $n(A) = 25$, $n(B) = 21$. Find $n(A \cap B)$

21. $n(A \cup B) = 35$, $n(A) = 20$, $n(A \cap B) = 8$. Find $n(B)$

22. $n(\xi) = 60$ and $n(A') = 42$. Find $n(A)$

23. 32 students: 20 study French, 15 Spanish, 4 neither. How many study both?

24. A set has 64 subsets. How many elements does it have?

19. $(28 - x) + x + (22 - x) + 9 = 50$, so $59 - x = 50$, $x = 9$ 20. $25 + 21 - 40 = 6$ 21. $35 = 20 + n(B) - 8$, $n(B) = 23$ 22. $60 - 42 = 18$ 23. $39 - x = 32$, $x = 7$ 24. $2^n = 64$, $n = 6$

Three sets

25. 100 people: 45 like A, 40 like B, 35 like C; 15 like A and B, 12 A and C, 10 B and C; 5 like all three. How many like none?

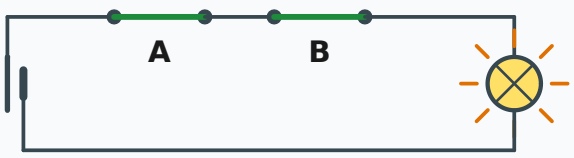
26. In the same survey, how many like A only?

25. $45 + 40 + 35 - 15 - 12 - 10 + 5 = 88$ like at least one, so $100 - 88 = 12$ 26. $45 - (15 - 5) - (12 - 5) - 5 = 23$

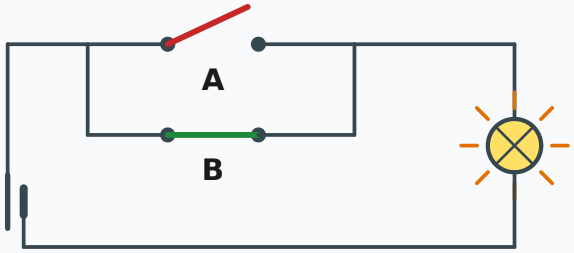
Chapter 7

Elementary logic

true or false, wired up like light switches



both switches on: lamp on



either switch on: lamp on

Logic is the maths of **true** and **false**. It's how computers make decisions (every if in a program), how search engines combine words, and how you check whether an argument actually holds up.

There are only a handful of ideas: **not**, **and**, **or**, and **if... then**. We'll wire each one as a light switch circuit, then write it as a **truth table** (a list of every possible case), and finally as Python code.

Not on a lecture slide, but underneath Chapter 6 (sets) and every program you'll write.

Statements: sentences that are true or false

A **statement** is a sentence that is definitely **true** or definitely **false**, even if you don't know which yet. Which of these are statements?

"Paris is in France."	✓	*statement: true*
"7 is an even number."	✓	*statement: false (still a statement!)*
"Shut the door."	✗	*an order: can't be true or false*
"Is it raining?"	✗	*a question: not a statement*
"x > 3"	?	*depends on x: a statement once x has a value*

Being true or false is called the statement's **truth value**, written **T** or **F**. Computers write 1 and 0, or True and False. We'll use letters like P and Q as names for statements, just as x names a number.

NOT: flip it

RULE: NOT

$\neg P$ is true when P is false, and false when P is true.

Two NOTs cancel: $\neg\neg P$ has the same value as P.

NOT turns true into false and false into true, like flicking a switch. If P is "it is raining", then $\neg P$ ("not P") is "it is not raining".

P	$\neg P$
T	F
F	T

Only 2 rows: P can only be T or F.

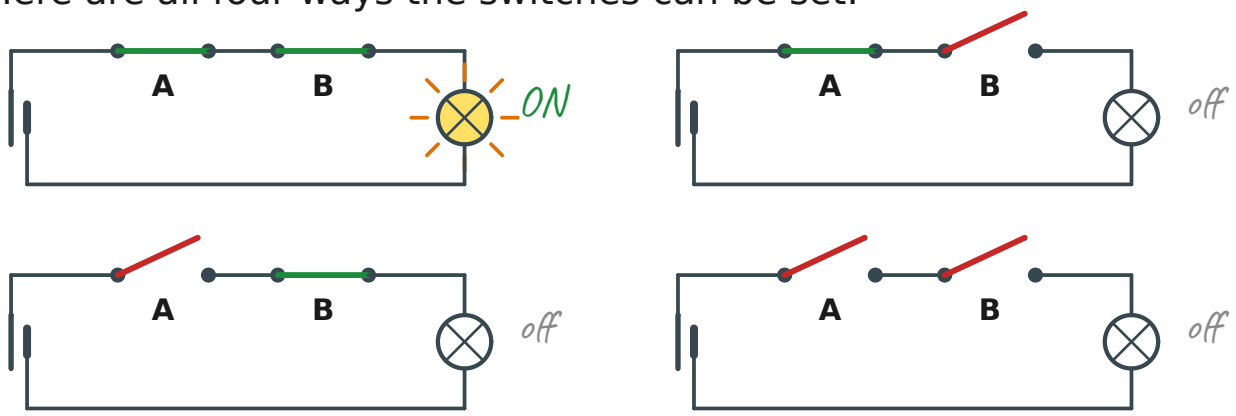
NOT just swaps them.

Set version (Chapter 6): the complement A'

Other ways books write it: ~P, P̄, !P (in C and Java), not P (Python).

AND: two switches in a row

Wire two switches one after the other ("in series"). Electricity must get through **both**, so the lamp lights only when A **and** B are both on. Here are all four ways the switches can be set:



Replace "on" by T and "off" by F, and you have the **truth table** for AND. We write "A and B" as $A \wedge B$:

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

just one T in the answer column: the row where both are T

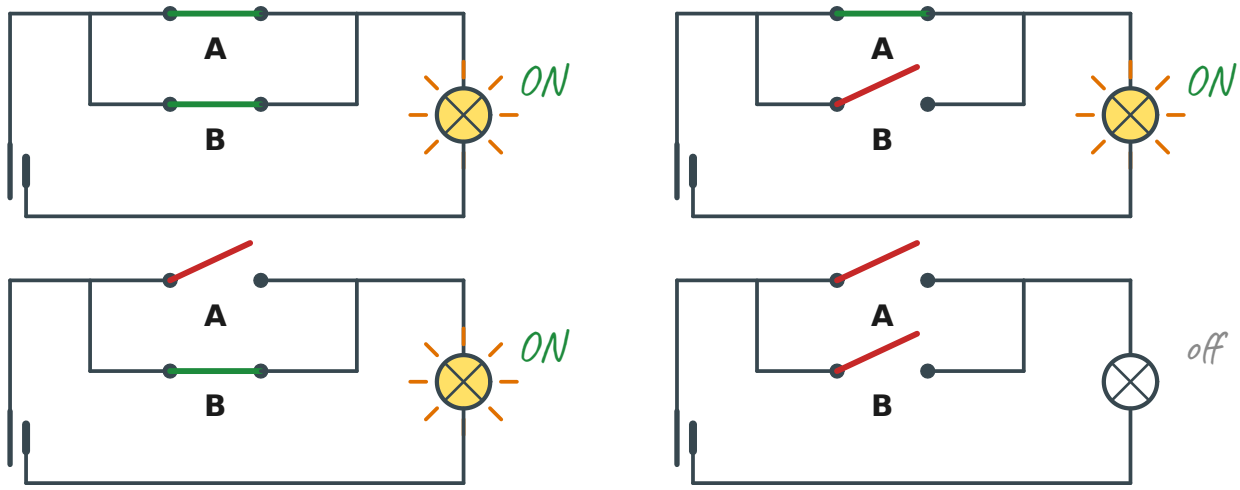
$\wedge$ looks like the $\cap$ of sets: $A \cap B$ = in A AND in B

RULE: AND

$A \wedge B$ is true **only** when A and B are both true. One F anywhere makes it F.

OR: two switches side by side

Now wire the switches side by side ("in parallel"). Electricity can take **either** path, so the lamp lights if A is on, or B is on, **or both**.



A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

just one F: the row where both are off

∨ is like the ∪ of sets: A ∪ B = in A OR in B (or both)

RULE: OR

$A \vee B$ is false **only** when A and B are both false. One T anywhere makes it T.

Maths "or" includes both. In everyday speech "or" often means one but not the other. Maths has a separate name for that, on the next page.

Exclusive or: the landing light

A menu says "tea **or** coffee included". You can't have both! That's **exclusive or** (XOR, written $\oplus$): true when exactly one is true.

A	B	$A \oplus B$
T	T	F
T	F	T
F	T	T
F	F	F

The light on a staircase has a switch at the bottom and one at the top. Flip either one and the light changes. Light on = the two switches are in different positions: that's XOR.

XOR is everywhere in computing: it's how a computer adds two binary digits (Chapter 10) without the carry, and a basic tool in encryption.

How many rows, and in what order

With one letter there are 2 cases, with two letters 4. Every new letter doubles the count, because each old row splits into a T version and an F version.

RULE: rows of a truth table

n different letters give 2^n rows: 1 letter 2 rows, 2 letters 4, 3 letters 8, 4 letters 16.

Standard order: the **first** column is the top half T, the bottom half F (TTTTFFFF for 3 letters).

Each next column halves the blocks: TTFFTTFF, then TFTFTFTF. The **last** letter column always alternates T, F.

This order lists every case exactly once, and makes your table match everyone else's.

RULE: order of operations (precedence)

$\neg$ first, then $\wedge$, then $\vee$, then $\Rightarrow$, then $\Leftrightarrow$. Brackets override all of these.

Like BODMAS: $\neg$ behaves like a minus sign, $\wedge$ like $\times$, $\vee$ like $+$.

$\neg P \vee Q \wedge R$ means $(\neg P) \vee (Q \wedge R)$; $P \wedge Q \Rightarrow R$ means $(P \wedge Q) \Rightarrow R$.

PROCEDURE: building a truth table

1. Count the letters n; draw 2^n rows.
2. Fill the letter columns in the standard order above.
3. Add one column for each piece, innermost brackets first, following the precedence.
4. The last column is the answer for the whole statement.

Example. Build the table for $(P \wedge Q) \vee \neg R$. Three letters, so $2^3 = 8$ rows:

P	Q	R	$P \wedge Q$	$\neg R$	answer
T	T	T	T	F	T
T	T	F	T	T	T
T	F	T	F	F	F
T	F	F	F	T	T
F	T	T	F	F	F
F	T	F	F	T	T
F	F	T	F	F	F
F	F	F	F	T	T

P: TTTTFFFF
Q: TTFFTTFF
R: TFTFTFTF

answer = (P∧Q) ∨ ¬R
F only when R is T
and P∧Q is F

Building bigger tables, column by column

Is "not (A and B)" the same as "(not A) or (not B)"? Don't guess: build a table. Add one column for each small piece, then combine:

A	B	$A \wedge B$	$\neg(A \wedge B)$	$\neg A$	$\neg B$	$\neg A \vee \neg B$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

The two outlined columns match in every row, so the statements always agree. This is **De Morgan's law**. In words: "it's not true that both happened" means "at least one of them didn't happen".

$$\neg(A \wedge B) = \neg A \vee \neg B \qquad \neg(A \vee B) = \neg A \wedge \neg B$$

NOT flips AND into OR... *...and OR into AND*

Programming use: "not (logged_in and paid)" is the same test as "not logged_in or not paid". Handy when tidying up conditions.

DEFINITION: equivalence, tautology, contingency, contradiction

Two statements are **logically equivalent** ($\equiv$) when their final columns match in every row.

A **tautology** is true in every row, whatever the letters are. Example: $A \vee \neg A$.

A **contradiction** is false in every row. Example: $A \wedge \neg A$.

A **contingency** is neither: the final column has at least one T and at least one F. Most everyday statements are contingencies ("it rains today").

De Morgan: $\neg(A \wedge B) \equiv \neg A \vee \neg B$ and $\neg(A \vee B) \equiv \neg A \wedge \neg B$. NOT goes inside and swaps $\wedge$ with $\vee$.

Example. One letter, so 2 rows:

A	$\neg A$	$A \vee \neg A$	$A \wedge \neg A$	A
T	F	T	F	T
F	T	T	F	F

all T: tautology

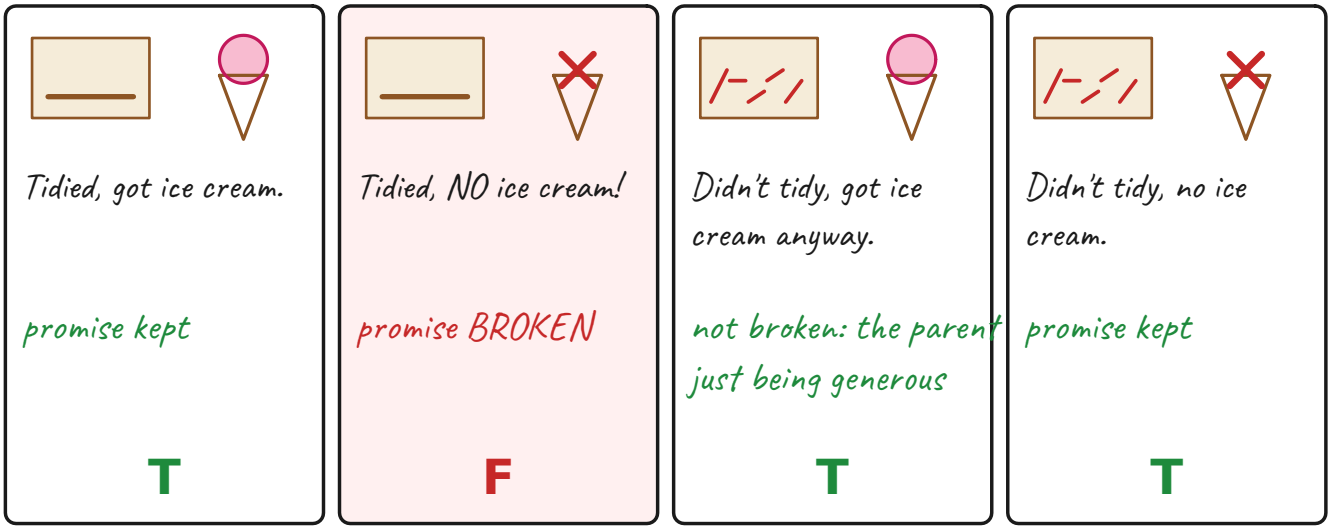
all F: contradiction

mix of T/F: contingency

Exam habit: build the truth table, then classify the final column: all T / all F / mixed.

IF... THEN: a parent's promise

A parent says: **"If you tidy your room, then you get ice cream."** When has the parent **broken** the promise? Look at the four things that can happen:



Only panel 2 breaks the promise. So "if P then Q", written $P \Rightarrow Q$, is **false only when P is true and Q is false**. Everything else counts as true:

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

P = "you tidy", Q = "you get ice cream"

rows 3 and 4 feel odd, but the promise said nothing about what happens if you DON'T tidy, so it can't be broken there

RULE: implication

$P \Rightarrow Q$ is false **only** when P is T and Q is F.

$P \Rightarrow Q \equiv \neg P \vee Q$: "either P doesn't happen, or Q does". Same table, row for row.

Negation: $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$. To break a promise, P must happen **and** Q must not.

Example. Negate "if you tidy, you get ice cream":

$$\neg(T \Rightarrow I) \equiv T \wedge \neg I \longleftarrow \text{"you tidied and you did not get ice cream"}$$

✗ Common mistakes

✗ $\neg(P \Rightarrow Q) \equiv \neg P \Rightarrow \neg Q$ ✓ $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$

the negation of an "if" is not another "if"

Turning an "if" around

Start with a true statement: **"If it's a cat, then it's an animal."** Now shuffle it in three ways. Which versions are still true?

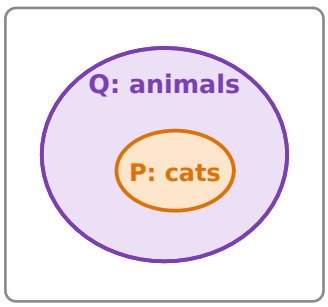
original	If it's a cat, then it's an animal. *true*
converse (swap)	If it's an animal, then it's a cat. *FALSE: a dog is an animal*
contrapositive	If it's not an animal, then it's not a cat. *true: swap AND put NOT on both; always matches the original*
inverse	If it's not a cat, then it's not an animal. *FALSE: a dog again (NOT on both, no swap)*

The picture that settles it. Draw everything that makes P true as a region, and the same for Q. "If P then Q" says **the P region sits entirely inside the Q region**.

✎ YOUR GO (pencil first)

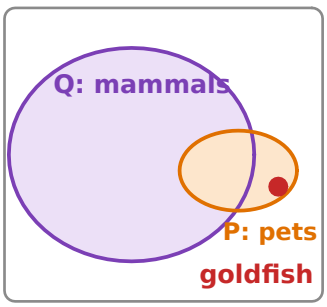
Picture a small blob "cats" inside a big blob "animals". Where would a dot have to be to break "if it's a cat, then it's an animal"? Is there any room for it?

if cat, then animal



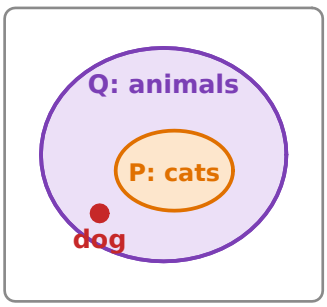
true: every cat is an animal

if pet, then mammal



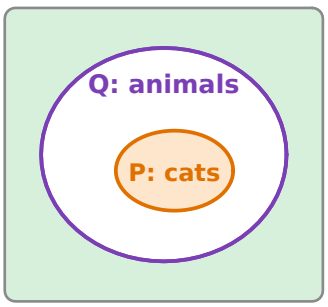
FALSE: a goldfish is a pet, not a mammal

converse: if animal, then cat



FALSE: a dog is an animal, not a cat

contrapositive: if not animal, then not cat



true: nothing outside Q is inside P

A **counterexample** to "if P then Q" is a dot **inside P but outside Q**. When P sits wholly inside Q there's nowhere to put one, so the statement is true. The **converse** asks the opposite (is Q inside P?), which is a different question. The **contrapositive** looks at the outside: everything outside Q is automatically outside P, the same picture read from the other side.

When a statement and its converse are **both** true, we say "P **if and only if** Q", written $P \Leftrightarrow Q$. Example: "a number is even if and only if it divides by 2". The two sides always stand or fall together.

A very common reasoning error is assuming the converse: "all cats are animals, Tom is an animal, so Tom is a cat". The last step uses the converse, which isn't true.

If and only if, necessary, sufficient

RULE: if and only if

$P \Leftrightarrow Q$ is true when P and Q have the **same** value (both T or both F).

$P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P)$: the statement and its converse both hold.

It is the opposite of XOR: $P \Leftrightarrow Q \equiv \neg(P \oplus Q)$.

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

T when the two match,
F when they differ

RULE: the same "if" in different words

All of these mean $P \Rightarrow Q$: "if P then Q", "Q if P", "P **only if** Q", "P is **sufficient** for Q", "Q is **necessary** for P".

Only if points forward: "you can vote only if you are 18+" is vote $\Rightarrow$ 18+. Being 18 doesn't force you to vote.

Sufficient: P is enough to guarantee Q. **Necessary**: without Q, P can't happen (that is the contrapositive $\neg Q \Rightarrow \neg P$).

"Necessary **and** sufficient" means $P \Leftrightarrow Q$.

$\text{rain} \Rightarrow \text{clouds}$ ← *rain is sufficient for clouds; clouds are necessary for rain*

The same logic in Python

Programs are full of truth tables in disguise. Python spells the symbols out as words:

```
age = 20
has_ticket = False

if age >= 18 and has_ticket:
    print('Come in')
elif age >= 18 or has_ticket:
    print('Nearly!')
if not has_ticket:
    print('Buy a ticket')
```

age >= 18 is True (20 ≥ 18)
has_ticket is False

True and False → False: skip 'Come in'

True or False → True: prints 'Nearly!'

not False → True: prints 'Buy a ticket'

maths	$\neg$	$\wedge$	$\vee$	$\oplus$
Python	`not`	`and`	`or`	`^` `or` `!=`

Python bonus: "and" stops early if the left side is False (the answer must be False anyway). That's why "if x != 0 and 10 / x > 2:" never divides by zero.

"For all" and "there exists"

Two more symbols appear in proofs and in later maths courses:

$\forall$ *"for all" (an upside-down A, for All)*

$\exists$ *"there exists" (a backwards E, for Exists)*

$\forall x \in \mathbb{R},\ x^2 \geq 0$ reads "for every real number x, x squared is at least 0" (true). $\exists x \in \mathbb{Z},\ x + 5 = 2$ reads "there is an integer x with x + 5 = 2" (true: x = -3).

To prove a "for all" statement false, one counterexample is enough. "Every prime is odd"? 2 is prime and even. Done: false.

RULE: negating a quantifier

$\neg\,\forall x\, P(x) \equiv \exists x\, \neg P(x)$: "not all are" = "at least one isn't".

$\neg\,\exists x\, P(x) \equiv \forall x\, \neg P(x)$: "none is" = "every one isn't".

NOT swaps $\forall$ and $\exists$ and moves inside, like De Morgan swaps $\wedge$ and $\vee$.

$$\neg\,(\forall p \text{ prime},\ p \text{ odd}) \equiv \exists p \text{ prime},\ p \text{ even}$$ ← *witness: p = 2*

✗ Common mistakes

✗ $\neg\,(\text{all pass}) = \text{all fail}$

one failure is enough to make "all pass" false

✓ $\neg\,(\text{all pass}) = \text{some fail}$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

NOT, AND, OR	$\neg P$ flips; $A \wedge B$ T only if both T; $A \vee B$ F only if both F; $A \oplus B$ T if exactly one T
Rows	$\text{rows} = 2^n$ first letter TTTT FFFF, next TTFF TTFF, last TFTF TFTF
Precedence	$\neg$, then $\wedge$, then $\vee$, then $\Rightarrow$, then $\Leftrightarrow$; brackets first
Equivalent	same final column in every row ($\equiv$)
Tautology / contingency / contradiction	final column all T / mixed T and F / all F. Exams ask you to classify after the table.
De Morgan	$\neg(A \wedge B) = \neg A \vee \neg B$ and $\neg(A \vee B) = \neg A \wedge \neg B$
Implication	$P \Rightarrow Q \equiv \neg P \vee Q$ F only when P is T and Q is F
Negating an if	$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$
Converse, contrapositive	converse $Q \Rightarrow P$ (may be false); contrapositive $\neg Q \Rightarrow \neg P$ (always matches $P \Rightarrow Q$)
If and only if	$P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P)$ T when P and Q have the same value
Same "if", other words	P only if Q; P is sufficient for Q; Q is necessary for P: all mean $P \Rightarrow Q$
Quantifiers	$\neg\, \forall x\, P(x) \equiv \exists x\, \neg P(x)$ and $\neg\, \exists x\, P(x) \equiv \forall x\, \neg P(x)$; one counterexample disproves "for all"

Switchboard: your turn

Answers in grey. Draw the circuit or build the table for each one.

Statements

1. Is "x + 1 = 4" a statement? When?
2. Give the truth value of "3 > 5 or 2 < 4"
3. Give the truth value of "3 > 5 and 2 < 4"

1. only once x has a value (true if x = 3) 2. T 3. F

Tables

4. Build the truth table for ¬A ∧ B
5. Build the table for A ∨ ¬A. What do you notice?
6. Use a table to check ¬(A ∨ B) = ¬A ∧ ¬B

4. T only in the row A = F, B = T 5. always T (called a tautology) 6. the columns match

If... then

7. "If it rains, the match is off." It didn't rain and the match was off. Promise broken?
8. Write the converse and contrapositive of "if n is divisible by 4, n is even". Which are true?

7. no (row 4) 8. converse: if even then divisible by 4 (false: 6); contrapositive: if odd then not divisible by 4 (true)

Python

9. x = 5. Is (x > 3 and x < 10) True?
10. x = 5. Is not (x == 5 or x == 7) True?

9. True 10. False

Rules and bigger tables

11. How many rows does a table for P, Q, R and S need?
12. Write the R column of a P, Q, R table in the standard order
13. Put brackets into ¬P ∨ Q ∧ R to show the order
14. Rewrite P ⇒ Q using only ¬ and ∨
15. Negate "every bus is late"
16. "You can drive only if you have a licence." Write it as P ⇒ Q

11. $2^4 = 16$ 12. T F T F T F T F 13. (¬P) ∨ (Q ∧ R) 14. ¬P ∨ Q 15. some bus is not late 16. drive ⇒ licence (a licence is necessary, not sufficient)

Find the unknown

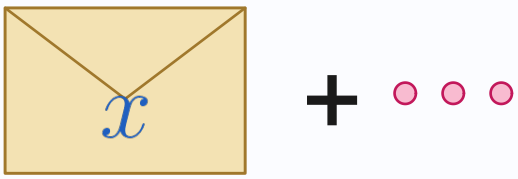
17. $P \Rightarrow Q$ is false. What are P and Q?
18. $P \vee Q$ is false. What are P and Q?
19. $P \Rightarrow Q$ is true and Q is false. What is P?
20. $P \Leftrightarrow Q$ is true and P is false. What is Q?
21. $P \wedge Q$ is true. What is $P \Rightarrow \neg Q$?
22. A truth table has 32 rows. How many letters are in it?

17. P = T, Q = F (the only F row) 18. both F (the only F row of OR) 19. F (P = T would make T ⇒ F, which is F) 20. F (⇔ needs them to match) 21. F (P = T, Q = T, so T ⇒ F) 22. $2^n = 32$, n = 5

Chapter 8

Algebra: letters for numbers

a letter is a sealed envelope with a number inside



$x + 3$

a sealed envelope of sweets, and 3 loose ones

Algebra scares people because suddenly there are letters in the sums. But a letter is just a **number you can't see yet**, like sweets in a sealed envelope. Everything you know about numbers still works; you just can't do the final bit of arithmetic until someone opens the envelope.

This chapter teaches the **grammar** of algebra: how things are written, what each part of an expression is called, how to put numbers in (substitution), and how to turn sentences into algebra. Chapters 11 to 15 then teach you to move algebra around.

Goes with **Lecture 2** (algebra, letters as numbers, Greek letters).

A letter is a sealed envelope

I hand you an envelope with some sweets in it, and 3 loose sweets. How many do you have? You can't say a number, but you **can** describe it exactly: "whatever is in the envelope, plus 3". Name the envelope x, and that's $x + 3$.

$x + 3$ — *an envelope and 3 more sweets*

$2x$ — *TWO envelopes with the same letter. Same letter = same number inside, always.*

$x + y$ — *different letters: might be different numbers (might be the same, by luck)*

$3x - 1$ — *three envelopes, then eat one sweet*

Opening the envelopes. If it turns out $x = 5$, then $x + 3 = 8$, $2x = 10$ and $3x - 1 = 14$. Until then, the algebra is a **recipe** waiting for its ingredient.

The grammar: what's hidden, what's shortened

RULE: algebraic notation

A number next to a letter, or two letters side by side, are **multiplied**: $3x = 3 \times x$, $xy = x \times y$. A letter on its own has coefficient 1: $x = 1x$.

Division is written as a fraction: $\frac{x}{2} = x \div 2$.

A power applies only to the thing it touches: $x^2 = x \times x$ (not $2x$), $3x^2 = 3 \times x \times x$, but $(3x)^2 = 3x \times 3x = 9x^2$.

Write the number first, then the letters in alphabetical order: $3ab$, not $a3b$.

Algebra is written in a shorthand. Each rule below is tested with $x = 5$, so you can **see** it's true rather than take it on trust:

$3x$	means	$3 \times x$	$3 \times 5 = 15$	*the × between a number and a letter is invisible*
xy	means	$x \times y$		*letters side by side are multiplied too*
x	means	$1x$	$1 \times 5 = 5$	*a lonely x means one x; we never write the 1*
$\frac{x}{2}$	means	$x \div 2$	$5 \div 2 = 2.5$	*division is written as a fraction*
x^2	means	$x \times x$	$5 \times 5 = 25$	*x squared: NOT 2 × x (that's 10)*
$3x^2$	means	$3 \times x \times x$	$3 \times 25 = 75$	*the power sits on x only, not on the 3*
$(3x)^2$	means	$3x \times 3x$	$15 \times 15 = 225$	*brackets put the power on everything inside*

Order: write the number first, then letters (usually alphabetical): $3ab$, not $a3b$ or $ba3$. They mean the same (multiplying in any order gives the same answer) but $3ab$ is the tidy way.

An x-ray of $3x^2 + 5x - 7$

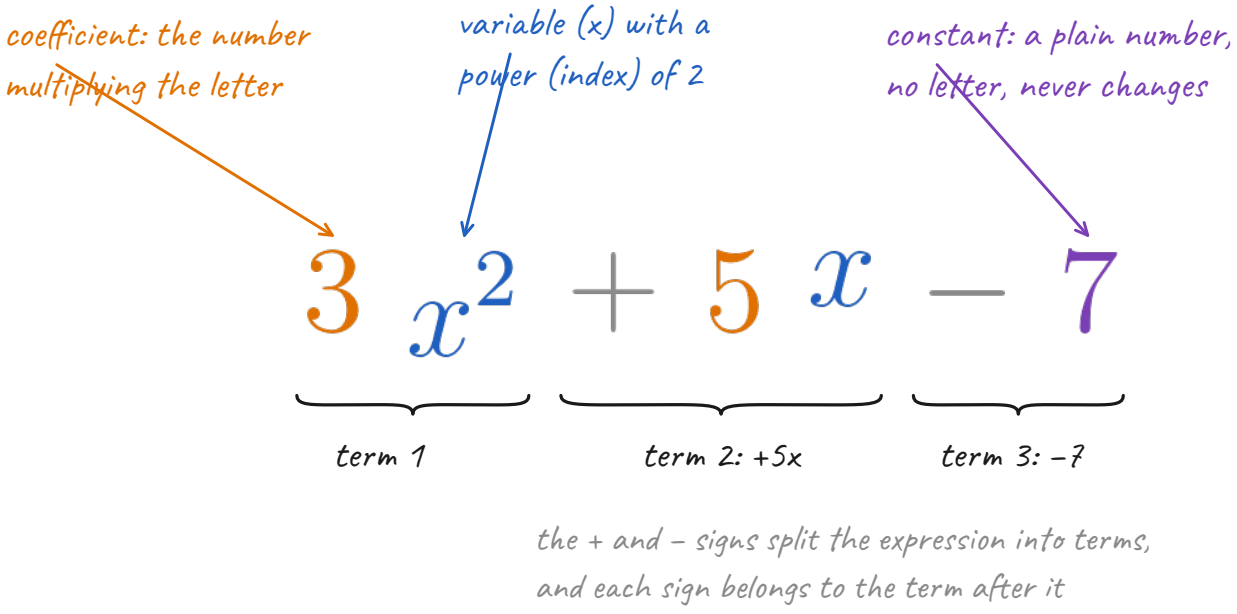
RULE: the parts of an expression

Terms are the pieces separated by $+$ and $-$ signs. Each sign belongs to the term after it.

The **coefficient** of a term is the number multiplying its letters, **sign included**: in $x^2 - x$ the coefficients are 1 and -1.

A **constant** is a term with no letter.

Your lectures will use words like **term**, **coefficient** and **constant** without stopping. Here's every part labelled on one example:



Lonely letters have invisible coefficients. In $x^2 - x$, the coefficient of x^2 is 1 and the coefficient of x is **−1** (the minus sign is part of it).

The whole thing $3x^2 + 5x - 7$ is called an **expression**: a mathematical phrase. It has no equals sign, so it doesn't claim anything; it's just a quantity.

Expression, equation, formula, identity

RULE: four kinds of algebraic statement

Expression: no equals sign. It can be simplified or evaluated, but not solved.

Equation: two expressions joined by $=$, true only for particular values. Solving it means finding those values.

Formula: an equation linking real quantities; substitute the known ones to find the other.

Identity: true for **every** value of the letters, written with $\equiv$.

These four words sound alike but do different jobs. Think of them as types of sentence:

expression $2x + 3$

a phrase: no equals sign, no claim. Like "two coffees and a cake".

equation $2x + 3 = 11$

a claim that's true only for some values. Here only x = 4. Solving means finding them (Chapter 15).

formula $A = lw$

a rule linking real quantities (area = length × width). Plug numbers in to get the answer.

identity $2(x + 3) \equiv 2x + 6$

true for EVERY value of x. The ≡ (three lines) says "identical": two ways of writing the same thing.

Substitution: open the envelopes

PROCEDURE: substitution

1. Replace every letter by its value, written **in brackets**: $x = -3$ turns $2x^2$ into $2(-3)^2$.
2. Work out the result in the order of operations (BODMAS, Chapter 1): **B**rackets, then **O**rders (powers and roots), then **D**ivision and **M**ultiplication left to right, then **A**ddition and **S**ubtraction left to right.
3. A fraction line groups like brackets: work out the **whole top** and the **whole bottom** first, then divide.

RULE: signs when multiplying and powering

$(+) \times (+) = +$, $(-) \times (-) = +$, $(+) \times (-) = -$, $(-) \times (+) = -$.

So a negative number to an **even** power is positive: $(-3)^2 = 9$. To an **odd** power it stays negative: $(-2)^3 = -8$.

-3^2 means $-(3^2) = -9$: without brackets the power is done first and the minus after. That is why you always bracket a substituted number.

Substitution means replacing each letter with its number. Find $3x^2 + 5x - 7$ when $x = 2$:

$$3x^2 + 5x - 7$$ ← the expression

$$3(2)^2 + 5(2) - 7$$ ← rub out each x, write (2)

$$3 \times 4 + 10 - 7$$ ← powers first: $(2)^2 = 4$; and $5 \times 2 = 10$

$$12 + 10 - 7 = 15$$ ← multiply, then add and take away left to right

Now the one everyone gets wrong: $x = -1$. The brackets save you:

$$3(-1)^2 + 5(-1) - 7$$ ← brackets around the −1

$$3 \times 1 - 5 - 7$$ ← $(-1)^2 = (-1) \times (-1) = +1$ (negative × negative, Ch. 1)

$$3 - 5 - 7 = -9$$

Without brackets you'd type -1^2 into a calculator and get -1, because the calculator squares the 1 first and then applies the minus. With brackets, $(-1)^2 = 1$. Always bracket negative numbers when substituting.

Example: $2x^2$ when $x = -3$.

$$2(-3)^2 = 2 \times 9 = 18$$ ← *power first: $(-3)^2 = +9$, then × 2*

$$\text{not } 2 \times (-9) = -18$$ ← *that squares 3 and forgets the minus*

Example: $\dfrac{x+y}{x-y}$ when $x = 5$ and $y = -3$.

$$\frac{(5) + (-3)}{(5) - (-3)}$$ ← *brackets round both values*

$$= \frac{5-3}{5+3} = \frac{2}{8}$$ ← *top: + (−3) = − 3; bottom: − (−3) = + 3*

$$= \frac{1}{4}$$ ← *simplify the fraction (Chapter 2)*

Not $\frac{5+3}{5-3}$: the minus sign of y must come in with it. And never cancel the 5s: the top and bottom are sums, so each must be finished before dividing.

Formulas work the same way. Physics: $v = u + at$ with $u = 3$, $a = 2$, $t = 5$ (the letters are starting speed, acceleration and time):

$$v = (3) + (2)(5) = 3 + 10 = 13$$ ← *at means a × t*

Turning words into algebra

PROCEDURE: forming an expression from words

1. Choose a letter for each unknown quantity and say what it stands for ("let n be the number").
2. Translate one phrase at a time, in the order the sentence gives them.
3. If an operation applies to **everything so far**, put everything so far in brackets: "add 3, then multiply by 5" is $5(n+3)$, not $5n+3$.
4. Test with a number: pick $n=2$, follow the words, and check the expression gives the same.

Read the sentence slowly and translate one phrase at a time, like a phrasebook:

"think of a number"	$\longrightarrow\ n$	*choose a letter for the unknown*
"double it"	$\longrightarrow\ 2n$	*× 2*
"add 3"	$\longrightarrow\ 2n+3$	*+ 3 to everything so far*
"then multiply by 5"	$\longrightarrow\ 5(2n+3)$	*brackets: the WHOLE thing gets × 5*

More translations, with the tricky words highlighted:

2 coffees at c pounds and 3 cakes at k pounds	$2c+3k$
a rectangle l long and w wide: perimeter	$2l+2w$
5 **less than** a number	$n-5 \quad (\text{not } 5-n)$
a number **shared equally** between 4	$\frac{n}{4}$
the **square** of a number, plus the number	n^2+n

"less than" reverses the order: "5 less than 12" is $12-5=7$. Test with a number whenever you're unsure.

From words to an equation, and solving it

When the words also tell you the **result**, the expression becomes an **equation**, and the unknown can be found. Chapter 15 covers solving in full; simple ones need only one idea:

RULE: keeping an equation balanced

An equation stays true if you do **the same thing to both sides**: add, subtract, multiply or divide (but not divide by 0).

To get the unknown alone, undo what was done to it in **reverse order**: the last operation applied is the first one you undo. $+$ is undone by $-$, $\times$ by $\div$.

PROCEDURE: word problem to answer

1. Let a letter stand for the unknown.
2. Write an expression for the quantity the words describe.
3. Set it equal to the value given: that is the equation.
4. Solve by undoing operations in reverse order, the same on both sides.
5. Check the answer in the **words** of the question, not just the equation.

Example: I think of a number, double it and add 3. The answer is 17. What was the number?

$2n + 3 = 17$ ← let n be the number; double, then + 3

$2n = 14$ ← undo the last step first: − 3 from both sides

$n = 7$ ← undo × 2: ÷ 2 both sides

Check in words: double 7 is 14, add 3 is 17 ✓.

Example: 4 concert tickets plus a £3 booking fee cost £35. How much is one ticket?

$4t + 3 = 35$ ← let t be the price of one ticket in £

$4t = 32$ ← − 3 from both sides

$t = 8$ ← ÷ 4 both sides: £8 a ticket

Example: think of a number, add 5, then multiply by 3. The answer is 27.

$3(n+5) = 27$ ← brackets: the whole $n + 5$ is tripled

$n + 5 = 9$ ← the last step was × 3, so ÷ 3 first

$n = 4$ ← then − 5

Check: 4 + 5 = 9, and 9 × 3 = 27 ✓.

Find the unknown: formulas with one letter missing

A very common question type gives you **every value but one** and asks for the missing one. It is substitution followed by a small equation:

PROCEDURE: finding a missing value

1. Write down the formula or equation.
2. Substitute every value you know, in brackets.
3. Tidy each side (do the arithmetic you can).
4. Undo the operations round the unknown, in reverse order, doing the same to both sides.
5. Check by substituting **all** the values back into the original.

Example: $v = u + at$. Given $v = 20$, $u = 5$ and $t = 3$, find a.

$20 = (5) + a(3)$ ← substitute what you know

$20 = 5 + 3a$ ← tidy: $a \times 3$ is written $3a$

$15 = 3a$ ← − 5 from both sides

$a = 5$ ← ÷ 3 both sides

Check: $5 + 5 \times 3 = 20$ ✓.

Example: $3p - 4 = 11$. Find p.

$3p = 15$ ← + 4 both sides (undo the − 4)

$p = 5$ ← ÷ 3 both sides

Example: the area of a triangle is $A = \frac{1}{2}bh$. $A = 24$ and $b = 8$: find h.

$$24 = \tfrac{1}{2}(8)h = 4h$$ ← half of 8 is 4

$$h = 6$$ ← ÷ 4 both sides

Example: when $x = 2$, the expression $3x + k$ equals 10. Find k.

$$3(2) + k = 10 \;\Rightarrow\; 6 + k = 10$$ ← substitute x = 2

$$k = 4$$ ← − 6 both sides

The unknown can be any letter, even one that looks like a "given" in the formula. Whichever letter has no value is the one you solve for.

Mistakes to avoid

✗ Common mistakes

✗ $x = -3 :\; 2x^2 = -18$ ✓ $2(-3)^2 = 18$

bracket the substituted value; the square makes it positive

✗ $-3^2 = 9$ ✓ $-3^2 = -9,\; (-3)^2 = 9$

without brackets only the 3 is squared

✗ $x = 2 :\; 3x^2 = 36$ ✓ $3(2)^2 = 12$

the power is on x only; $(3x)^2$ would be 36

✗ $x^2 = 2x$ ✓ $x^2 = x \times x$

squared means times itself, not times 2

✗ $5 \text{ less than } n = 5 - n$ ✓ $n - 5$

"less than" reverses the order

✗ $x = -3 :\; \dfrac{x+5}{x} = \dfrac{5}{1}$ ✓ $\dfrac{(-3)+5}{(-3)} = -\dfrac{2}{3}$

finish the top and the bottom before dividing; never cancel part of a sum

The Greek-letter phrasebook

When mathematicians run out of English letters (or want a letter with a traditional job), they use Greek ones. Your Lecture 2 lists the common ones. Here they are with how to say them and where you'll meet them:

letter	name	say it	you'll meet it in...
α	**alpha**	*AL-fa*	angles; radiation (α particles)
β	**beta**	*BEE-ta*	angles; radiation
γ	**gamma**	*GAM-a*	angles; gamma rays
δ, Δ	**delta**	*DEL-ta*	a small change (Δx: change in x) (Ch. 20, 34)
ε	**epsilon**	*EP-si-lon*	a tiny amount; errors
θ	**theta**	*THEE-ta*	the most common angle letter (Ch. 27-31)
ϕ	**phi**	*FYE (or fee)*	another angle
λ	**lambda**	*LAM-da*	wavelength; eigenvalues
μ	**mu**	*MEW*	the mean in statistics (Ch. 24, 26)
σ, Σ	**sigma**	*SIG-ma*	σ: spread (Ch. 24); Σ: add up (Ch. 17)
π	**pi**	*PIE*	3.14159..., the circle number
ω	**omega**	*oh-MEE-ga*	angular speed; spinning
χ	**chi**	*KYE*	the chi-squared test in statistics

Don't panic when you see one: a Greek letter is just another envelope with a number in it. θ = 30° is no different from x = 30.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Hidden ×	$3x = 3 \times x, \quad xy = x \times y$ a letter alone has coefficient 1
Division	$\frac{x}{2} = x \div 2$
Powers	$3x^2 = 3 \times x \times x, \quad (3x)^2 = 9x^2$ a power applies only to what it touches
Terms	pieces separated by $+$ and $-$; each sign belongs to the term after it
Coefficient	the number multiplying the letters, sign included (-1 in $-x$)
Constant	a term with no letter
Expression / equation	an expression has no $=$; an equation is true only for some values
Formula / identity	$2(x+3) \equiv 2x+6$ a formula links quantities; an identity ($\equiv$) is true for all values
Substitution	replace each letter by its value **in brackets**, then BODMAS
Sign rules	$(-) \times (-) = +, \quad (+) \times (-) = -$
Powers of negative numbers	$(-3)^2 = 9, \quad -3^2 = -9, \quad (-2)^3 = -8$ even power: positive; odd power: negative
Fractions	finish the whole top and the whole bottom, then divide
Forming	letter for the unknown; translate phrase by phrase; bracket "everything so far"
Balancing	do the same to both sides; undo operations in reverse order
Find the unknown	substitute the known values, tidy, undo, check

Envelopes: your turn

Answers in grey. For every one, test your answer by putting a number in!

Reading algebra

1. What does 4ab mean in full?
2. Is x^2 the same as 2x? Test with x = 3
3. In $7y^3 - y + 2$, name the coefficient of y^3, the coefficient of y, and the constant

1. $4 \times a \times b$ 2. no: 9 vs 6 3. 7, −1, 2

Substitution (x = 3, y = −2)

4. 2x + 5
5. $x^2 - y$
6. $3y^2$
7. $xy - 4$
8. $(x + y)^2$
9. y^3
10. $(x + y)/(x - y)$
11. $(x - y)/(2y)$

4. 11 5. 11 6. 12 7. −10 8. 1 9. −8 10. 1/5 11. 5/(−4) = −5/4

Words to algebra

12. A number, times 4, minus 7
13. £p shared equally between 3 people
14. The cost of 5 pens at q pence each and a pad at 90p
15. 7 more than twice a number

12. 4n − 7 13. p/3 14. 5q + 90 15. 2n + 7

Types

16. Expression, equation, formula or identity? (a) 3x − 1 (b) 3x − 1 = 8 (c) $E = mc^2$ (d) $x + x \equiv 2x$

16. (a) expression (b) equation (c) formula (d) identity

Find the unknown

17. v = u + at with v = 20, u = 5, t = 3. Find a.

18. 3p − 4 = 11. Find p.

19. P = 2l + 2w with P = 30, l = 9. Find w.

20. A = ½bh with A = 24, h = 6. Find b.

21. I think of a number, multiply it by 4 and subtract 7. The answer is 21. Find the number.

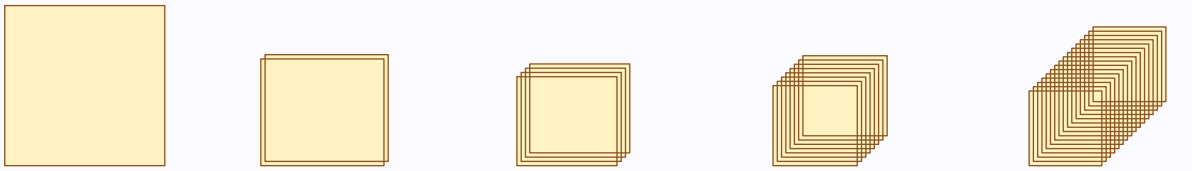
22. When x = 3, 2x + k = 15. Find k.

17. a = 5 (20 = 5 + 3a) 18. p = 5 (3p = 15) 19. w = 6 (30 = 18 + 2w) 20. b = 8 (24 = 3b) 21. 7 (4n − 7 = 21, 4n = 28) 22. k = 9 (6 + k = 15)

Chapter 9

Indices (powers and roots)

multiplying a number by itself, again and again



$2^0 = 1$ — 0 folds

$2^1 = 2$ — 1 fold

$2^2 = 4$ — 2 folds

$2^3 = 8$ — 3 folds

$2^4 = 16$ — 4 folds

Fold a sheet of paper in half: 2 layers. Again: 4. Again: 8. Each fold **doubles** the layers. After 5 folds you have $2 \times 2 \times 2 \times 2 \times 2 = 32$ layers. Writing all those 2s is tiring, so maths has a shorthand: 2^5. That little raised number is a **power** (also called an **index** or **exponent**).

Powers are everywhere: areas (cm²), volumes (cm³), computer memory (2^{10} bytes), tiny and huge numbers (standard form), and later exponential growth and logarithms (Chapters 21-22). This chapter builds every rule by **counting**, so you never have to memorise blindly.

Goes with **Lecture 7** (power terminology, combining powers, standard form).

A power is repeated multiplication

RULE: what a power means

$a^n = a \times a \times \cdots \times a$ with n copies of a. a is the **base**; n is the **power** (also called the **index** or **exponent**).

$a^1 = a$, a^2 is "a squared", a^3 is "a cubed". a^n is **not** $a \times n$.

Your lecture puts it neatly. **Repeated adding** has a shortcut called multiplying; **repeated multiplying** has a shortcut called a power:

$$3 + 3 + 3 + 3 = 4 \times 3 = 12$$ *four 3s added*

$$3 \times 3 \times 3 \times 3 = 3^4 = 81$$ *four 3s multiplied*

3^4 ← *the power (index, exponent): how many copies of the base*

say: "three to the power four" or "three to the four"

the base: the number being multiplied

Special names: power 2 is "squared" ($5^2 = 25$), power 3 is "cubed" ($2^3 = 8$). Power 1 is just the number itself: $7^1 = 7$ (one copy).

The trap: 3^4 is **not** 3×4. $3^4 = 81$ but $3 \times 4 = 12$. The power counts copies; it doesn't multiply by itself.

Typing powers: calculators have an x^y or ^ key; computers write 3^4, and Python writes 3**4. **Your slides mention "x^2" for when you can't type a raised number.**

$$10^1 = 10 \qquad 10^2 = 100 \qquad 10^3 = 1000 \qquad 10^6 = 1{,}000{,}000$$

Powers of 10 are easy: the power counts the zeros.

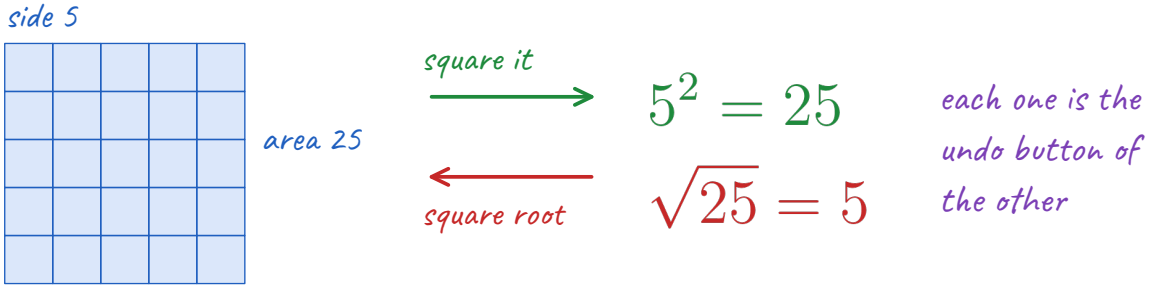
Roots: running a power backwards

RULE: roots

$\sqrt[n]{a}$ (the n-th root of a) is the number whose n-th power is a: $\sqrt[3]{8} = 2$ because $2^3 = 8$. $\sqrt{a}$ means $\sqrt[2]{a}$.

The $\sqrt{\ }$ sign means the **positive** root: $\sqrt{25} = 5$. But $x^2 = 25$ has two solutions, $x = \pm 5$.

A square with side 5 has area $5^2 = 25$. Going backwards, "which side gives area 25?", is the **square root**: $\sqrt{25} = 5$. The √ sign started life as a letter r, for radix (root).



Cube roots undo cubing: a cube of volume 8 has side 2, so $\sqrt[3]{8} = 2$. In general $\sqrt[n]{x}$ (the **n-th root**) undoes the power n: $\sqrt[4]{81} = 3$ because $3^4 = 81$.

Two answers hiding: $(-5)^2$ is also 25 (negative × negative is positive). The √ sign always means the **positive** one, but when solving $x^2 = 25$ (Chapter 15) remember $x = 5$ or $x = -5$, written $x = \pm 5$.

Negative bases, and roots of negative numbers

RULE: powers of a negative number

A negative base to an **even** power is **positive**: $(-2)^4 = 16$. Every pair of minus signs multiplies to $+$.

A negative base to an **odd** power is **negative**: $(-2)^3 = -8$. One minus sign is left over.

Brackets matter: $(-3)^2 = (-3) \times (-3) = 9$, but $-3^2 = -(3 \times 3) = -9$. Without brackets the power is done first and only touches the 3.

RULE: roots of a negative number

Even roots ($\sqrt{}$, $\sqrt[4]{}$, ...) of a negative number **do not exist** as real numbers: no real number squared gives -4, so $\sqrt{-4}$ is undefined (your calculator says Error).

Odd roots of a negative number are negative: $\sqrt[3]{-8} = -2$ because $(-2)^3 = -8$.

$(-2)^4 = (-2)(-2)(-2)(-2) = 16$ ← *two pairs of minuses: positive*

$(-2)^5 = 16 \times (-2) = -32$ ← *one more minus: negative*

$-2^4 = -(2^4) = -16$ ← *no brackets: the minus waits until the end*

$\sqrt[3]{-27} = -3$ ← *check: $(-3)^3 = -27$*

The first law: multiplying powers

RULE: multiplying powers of the same base

$a^m \times a^n = a^{m+n}$: keep the base, **add** the powers.

Only for the **same base**: $x^2 \times y^3$ and $2^3 \times 3^2$ cannot be combined this way.

Only for **multiplying**: $x^2 + x^3$ is not x^5; it can't be simplified.

Numbers in front (coefficients) are multiplied as ordinary numbers: $3x^2 \times 2x^3 = 6x^5$.

Why? What is $x^2 \times x^3$? **Write out every x and count them**:

$$x^2 \times x^3 = \underbrace{x\,x}_{\text{2 x's}} \times \underbrace{x\,x\,x}_{\text{3 x's}} = \underbrace{x^5}_{\text{5 x's altogether}}$$

The powers just **add**. Test with numbers: $2^2 \times 2^3 = 4 \times 8 = 32$ and $2^5 = 32$ ✓.

Example: $3x^2 \times 2x^3$.

$$(3 \times 2)(x^2 \times x^3) = 6x^5 \quad \longleftarrow \text{numbers × numbers; add the powers}$$

Only for the same base. $x^2 \times y^3$ can't be squashed (different envelopes), and $x^2 + x^3$ is **not** x^5 (that's adding, not multiplying: test $x = 2$: $4 + 8 = 12$, but $2^5 = 32$).

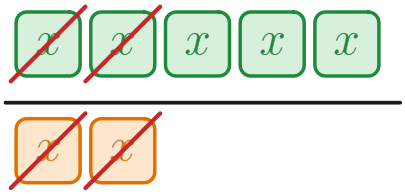
Dividing: cancel in pairs

RULE: dividing powers of the same base

$a^m \div a^n = \dfrac{a^m}{a^n} = a^{m-n}$: keep the base, **subtract** the powers (top minus bottom).

Coefficients are divided as ordinary numbers: $6x^5 \div 2x^2 = 3x^3$.

Why? $\frac{x^5}{x^2}$: write the x's out on top and bottom. Each x on the bottom cancels one on top (like simplifying a fraction, Chapter 2):



$= x^3$

two pairs cancel;
5 − 2 = 3 x's are left

Test: $\frac{2^5}{2^2} = \frac{32}{4} = 8 = 2^3$ ✓.

Example: $6x^5 \div 2x^2$.

$$\frac{6x^5}{2x^2} = \frac{6}{2} \times \frac{x^5}{x^2} = 3x^3$$ ← *divide the numbers; subtract the powers*

Example: $\dfrac{10a^4b^3}{5ab}$.

$$\frac{10}{5}\, a^{4-1}\, b^{3-1} = 2a^3b^2$$ ← *each letter separately; a = a¹*

A power of a power: groups of groups

RULE: powers of powers and of brackets

$(a^m)^n = a^{mn}$: **multiply** the powers.

A power outside a bracket applies to **every factor inside**: $(ab)^n = a^n b^n$ and $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$.

That includes the number in front: $(2x^3)^2 = 2^2 x^6 = 4x^6$, **not** $2x^6$.

Why? $(x^2)^3$ means three copies of x^2. Each copy is a pair of x's, so count the pairs:

$$(x^2)^3 = \boxed{x\,x}\ \boxed{x\,x}\ \boxed{x\,x} = x^6$$

3 groups of 2 = 6 x's: the powers multiply

Brackets with several things inside (the last two laws on your slides). The power applies to **everything** in the bracket:

$$(xy)^3 = xy \times xy \times xy = x^3y^3$$ ← *rearrange: all x's together, all y's together*

$$\left(\frac{x}{y}\right)^3 = \frac{x}{y} \times \frac{x}{y} \times \frac{x}{y} = \frac{x^3}{y^3}$$ ← *top × top, bottom × bottom*

$(2x)^3 = 8x^3$ ← *the 2 gets cubed too! ($2^3 = 8$)*

$(2x^3)^2 = 2^2 \times x^{3\times 2} = 4x^6$ ← *square the 2; multiply the powers*

$(-3a^2b)^3 = (-3)^3 a^6 b^3 = -27a^6b^3$ ← *odd power of −3 stays negative*

The staircase: zero and negative powers

RULE: zero and negative powers

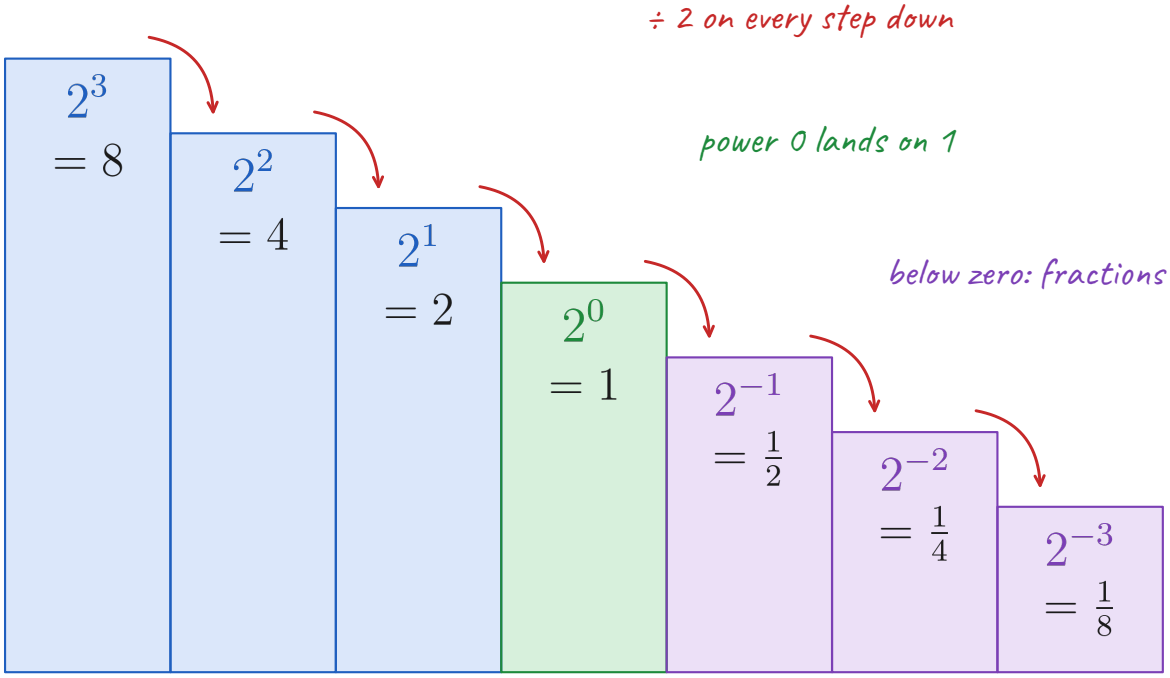
$a^0 = 1$ for any $a \neq 0$.

$a^{-n} = \dfrac{1}{a^n}$: a negative power means **one over** (the reciprocal). It does **not** make the answer negative: $2^{-3} = \frac{1}{8}$.

Turned the other way: $\dfrac{1}{a^{-n}} = a^n$, so $\dfrac{1}{x^{-2}} = x^2$. A power can move between top and bottom of a fraction by changing its sign.

For a fraction, flip it and make the power positive: $\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^{n}$.

Why? What could 2^0 or 2^{-1} possibly mean? "Zero copies of 2 multiplied"? Walk down a staircase instead. Every step down **divides by 2** (the opposite of one more fold):



So **anything to the power 0 is 1**, and a **negative power means "one over"** the positive power. "One over" is the **reciprocal**: flip the number upside down. A whole number is secretly over 1, so $8 = \frac{8}{1}$ flips to $\frac{1}{8}$ (Chapter 2, "Dividing"). Read the staircase: 2^{-3} is $2^3 = 8$ flipped, which is $\frac{1}{8}$.

$$\frac{1}{x^{-2}} = \frac{1}{\frac{1}{x^2}} = x^2$$ ← one over a reciprocal flips it back

$$\left(\frac{2}{3}\right)^{-2} = \left(\frac{3}{2}\right)^{2} = \frac{9}{4}$$ ← flip, then square

$$5x^{-3} = \frac{5}{x^3}$$ ← only the x has the power; the 5 stays on top

The dividing law agrees, which is how we know the staircase isn't a trick: $\frac{x^3}{x^3} = x^{3-3} = x^0$, but anything divided by itself is 1. And $\frac{x^2}{x^3} = x^{-1}$, but cancelling leaves $\frac{1}{x}$.

A negative power does NOT make the number negative: $2^{-3} = \frac{1}{8}$, a small positive number.

Fraction powers are roots

RULE: fraction powers

$a^{\frac{1}{n}} = \sqrt[n]{a}$: the **bottom** of the fraction is the **root**.

$a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}$: the **top** is an ordinary **power**.

PROCEDURE: evaluating a (negative) fraction power

1. If the power is negative, write one over the same thing with a positive power.
2. Take the root given by the bottom number n (smaller numbers this way round).
3. Raise to the power given by the top number m.

Why? What would $9^{\frac{1}{2}}$ mean? Use the power-of-a-power law as a detective. Square it:

$$\left(9^{\frac{1}{2}}\right)^2 = 9^{\frac{1}{2}\times 2} = 9^1 = 9$$ ← *powers multiply: ½ × 2 = 1*

so $9^{\frac{1}{2}}$ is the number whose square is 9 ← *that's the definition of √*

$$9^{\frac{1}{2}} = \sqrt{9} = 3$$

That is the rule in the box: the bottom of the fraction is the root, the top an ordinary power.

$$8^{\frac{2}{3}} = \left(\sqrt[3]{8}\right)^2 = 2^2 = 4$$ ← *root first (cube root of 8 is 2), then square*

$$16^{-\frac{1}{2}} = \frac{1}{\sqrt{16}} = \frac{1}{4}$$ ← *minus: one over. Half: square root.*

$$27^{-\frac{2}{3}} = \frac{1}{27^{\frac{2}{3}}} = \frac{1}{\left(\sqrt[3]{27}\right)^2} = \frac{1}{3^2} = \frac{1}{9}$$ ← *the procedure: flip, root, power*

Why roots first? Smaller numbers: $(\sqrt[3]{8})^2 = 2^2$ is easier than $\sqrt[3]{64}$ (same answer, 4).

All the index laws in one table

Every law in this chapter, with one example each. Learn the table; the rest of the chapter explains where each line comes from.

Law	Rule	Example
Multiplying	$a^m \times a^n = a^{m+n}$	$x^2 \times x^3 = x^5$
Dividing	$a^m \div a^n = a^{m-n}$	$x^7 \div x^3 = x^4$
Power of a power	$(a^m)^n = a^{mn}$	$(x^2)^3 = x^6$
Zero power	$a^0 = 1 \quad (a \neq 0)$	$5^0 = 1$
Negative power	$a^{-n} = \dfrac{1}{a^n}$	$2^{-3} = \dfrac{1}{8}$
One over a negative power	$\dfrac{1}{a^{-n}} = a^n$	$\dfrac{1}{x^{-2}} = x^2$
Root	$a^{\frac{1}{n}} = \sqrt[n]{a}$	$8^{\frac{1}{3}} = \sqrt[3]{8} = 2$
Fraction power	$a^{\frac{m}{n}} = (\sqrt[n]{a})^m$	$8^{\frac{2}{3}} = 2^2 = 4$
Power of a product	$(ab)^n = a^n b^n$	$(2x^3)^2 = 4x^6$
Power of a fraction	$\left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$	$\left(\dfrac{x}{3}\right)^2 = \dfrac{x^2}{9}$
Negative power of a fraction	$\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^n$	$\left(\dfrac{2}{3}\right)^{-2} = \dfrac{9}{4}$

RULE: when the laws apply

The first three laws need the **same base**. $2^3 \times 2^4 = 2^7$, but $2^3 \times 3^4$ can only be worked out as numbers (8×81). Never multiply the bases: $2^3 \times 2^4$ is not 4^7.

There are **no laws for adding**: $x^2 + x^3$ and $2^3 + 2^4$ stay as they are (or are worked out as numbers).

Coefficients are ordinary numbers: multiply or divide them separately, but a bracket's power applies to them too.

Find the unknown: equations with a power

Exam questions often give you the answer and hide the power (or the base). Two rules handle almost all of them:

PROCEDURE: unknown power: make the bases match

1. Write both sides as powers of the **same base**: $\frac{1}{9} = 3^{-2}$, $32 = 2^5$, $4 = 2^2$, $8 = 2^3$.
2. If $a^p = a^q$ (same base, not 0, 1 or -1), then the powers are equal: $p = q$.
3. Solve that ordinary equation, then check in the original.

RULE: unknown base: undo the power

If $x^{\frac{m}{n}} = c$, raise both sides to the **flipped** power $\frac{n}{m}$: $x = c^{\frac{n}{m}}$, because $\left(x^{\frac{m}{n}}\right)^{\frac{n}{m}} = x^1$.

Example: solve $3^n = \frac{1}{9}$.

$$\frac{1}{9} = \frac{1}{3^2} = 3^{-2}$$ ← write 1/9 as a power of 3

$$3^n = 3^{-2} \Rightarrow n = -2$$ ← same base: powers equal

Example: solve $2^x = 32$.

$$2^x = 2^5 \Rightarrow x = 5$$ ← 2 × 2 × 2 × 2 × 2 = 32

Example: solve $4^x = 8$. (4 and 8 are both powers of 2.)

$$(2^2)^x = 2^3$$ ← $4 = 2^2$, $8 = 2^3$

$$2^{2x} = 2^3$$ ← power of a power: multiply

$$2x = 3 \Rightarrow x = \frac{3}{2}$$ ← powers equal

Check: $4^{\frac{3}{2}} = (\sqrt{4})^3 = 2^3 = 8$ ✓.

Example: solve $x^{\frac{3}{2}} = 8$.

$$x = 8^{\frac{2}{3}}$$ ← raise both sides to the flipped power 2/3

$$x = \left(\sqrt[3]{8}\right)^2 = 2^2 = 4$$ ← root first, then square

Example: find k if $x^k \times x^3 = x^{10}$.

$$x^{k+3} = x^{10} \Rightarrow k + 3 = 10 \Rightarrow k = 7$$ ← first law, then match powers

Example: solve $5^{x+1} = 125$.

$$5^{x+1} = 5^3 \Rightarrow x + 1 = 3 \Rightarrow x = 2$$ ← $125 = 5^3$

Test before you trust: four famous mistakes

The square-root one is on Lecture 2's "common mistakes" slide; the other three are just as common. Each one is caught instantly by trying small numbers:

$(a+b)^2 \neq a^2+b^2$ — a = 3, b = 4: $(7)^2 = 49$, but $9 + 16 = 25$

the square is on the whole bracket; Chapter 11 shows $(a+b)^2 = a^2 + 2ab + b^2$

$\sqrt{a^2+b^2} \neq a+b$ — a = 3, b = 4: $\sqrt{25} = 5$, but $3 + 4 = 7$

roots don't split over +

$2x^3 \neq (2x)^3$ — x = 1: $2 \times 1 = 2$, but $2^3 = 8$

the power only sits on what it touches

$x^2+x^3 \neq x^5$ — x = 2: $4 + 8 = 12$, but 32

laws are for ×, not +

✗ Common mistakes

✗ $2^3 \times 2^4 = 4^7$ ✓ $2^3 \times 2^4 = 2^7$

same base: keep the base, add the powers

✗ $(2x^3)^2 = 2x^6$ ✓ $(2x^3)^2 = 4x^6$

the bracket's power applies to the 2 as well

✗ $3^{-2} = -9$ ✓ $3^{-2} = \frac{1}{9}$

a negative power means one over, not a negative answer

✗ $-3^2 = 9$ ✓ $-3^2 = -9, \quad (-3)^2 = 9$

without brackets the power touches only the 3

✗ $x^6 \div x^2 = x^3$ ✓ $x^6 \div x^2 = x^4$

dividing subtracts the powers; it doesn't divide them

✗ $x^0 = 0$

anything (except 0) to the power 0 is 1

✓ $x^0 = 1$

Standard form: KB, MB, GB and tiny things

RULE: standard form

A number in standard form is $a \times 10^n$ with $1 \leq a < 10$ and n a whole number (positive, negative or zero).

Numbers ≥ 10 have a positive power; numbers between 0 and 1 have a negative power; numbers from 1 to 9.99... have power 0.

A 1 GB file holds about 1,000,000,000 bytes. A bacterium is 0.000002 m long. Counting zeros is error-prone, so we write a **small number × a power of 10**:

1 kB	1000 bytes	1×10^3
1 MB	1,000,000 bytes	1×10^6
1 GB	1,000,000,000 bytes	1×10^9
57		5.7×10^1
0.0024		2.4×10^{-3}
0.000001		1×10^{-6}

big numbers: positive power

small numbers: negative power

The recipe. Write the digits with the dot after the first non-zero digit (that's a, between 1 and 10). Then count how many places the digits had to slide (Chapter 3): that's the power. **Digits slid left** (the number was small, less than 1) → **negative** power. Digits slid right (the number was big) → positive power.

$0.0024 \rightarrow 2.4$ ← *digits slide 3 places left, so the dot seems to move right*

$0.0024 = 2.4 \times 10^{-3}$ ← *× 10^{-3} (÷ 1000) puts it back*

Slide note: Lecture 7 writes $1 \leq a \leq 10$; it should be $1 \leq a < 10$. The number 10 itself is written 1×10^1, never 10×10^0.

Multiplying in standard form uses the first law on the powers of 10: $(3 \times 10^4) \times (2 \times 10^5) = 6 \times 10^9$. Calculators show 2.4×10^{-3} as 2.4E-3 (E for exponent).

Calculating in standard form

PROCEDURE: multiplying and dividing

1. Multiply (or divide) the front numbers.
2. Add the powers of 10 when multiplying; subtract them when dividing.
3. **Re-normalise**: if the front number is 10 or more, divide it by 10 and add 1 to the power ($20 \times 10^5 = 2 \times 10^6$); if it is less than 1, multiply it by 10 and take 1 off the power ($0.5 \times 10^3 = 5 \times 10^2$).

PROCEDURE: adding and subtracting

1. Rewrite the numbers with the **same power of 10** first (or write them out in full).
2. Add or subtract the front numbers; keep the power.
3. Re-normalise if needed.

Example: $(6 \times 10^5) \times (5 \times 10^3)$.

$$(6 \times 5) \times 10^{5+3} = 30 \times 10^8$$ ← multiply fronts, add powers

$$= 3 \times 10^9$$ ← 30 is too big: 30 = 3 × 10^1, so add 1 to the power

Example: $(8 \times 10^7) \div (2 \times 10^3)$.

$$\frac{8}{2} \times 10^{7-3} = 4 \times 10^4$$ ← divide fronts, subtract powers

Example: $(3 \times 10^4) \div (6 \times 10^{-2})$.

$$\frac{3}{6} \times 10^{4-(-2)} = 0.5 \times 10^6$$ ← 4 − (−2) = 6

$$= 5 \times 10^5$$ ← 0.5 is less than 1: × 10 and take 1 off the power

Example: $(3 \times 10^4) + (5 \times 10^3)$.

$$3 \times 10^4 + 0.5 \times 10^4$$ ← $5 \times 10^3 = 0.5 \times 10^4$: same power now

$$= 3.5 \times 10^4$$ ← add the fronts, keep the power

Check by writing them out: $30\,000 + 5000 = 35\,000$ ✓. **Never** add the powers when adding: $3 \times 10^4 + 5 \times 10^3$ is not 8×10^7.

Example: $(7.2 \times 10^{-3}) - (4 \times 10^{-4})$.

$$7.2 \times 10^{-3} - 0.4 \times 10^{-3} = 6.8 \times 10^{-3}$$ ← $4 \times 10^{-4} = 0.4 \times 10^{-3}$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Power	$a^n = a \times a \times \cdots \times a$ n copies of a; $a^1 = a$
Multiplying	$a^m \times a^n = a^{m+n}$ same base only; coefficients multiply
Dividing	$\dfrac{a^m}{a^n} = a^{m-n}$ same base only; coefficients divide
Power of a power	$(a^m)^n = a^{mn}$
Brackets	$(ab)^n = a^n b^n, \quad \left(\dfrac{a}{b}\right)^n = \dfrac{a^n}{b^n}$ the number in front is raised too: $(2x^3)^2 = 4x^6$
Zero power	$a^0 = 1 \quad (a \neq 0)$
Negative power	$a^{-n} = \dfrac{1}{a^n}, \quad \dfrac{1}{a^{-n}} = a^n$ one over; never a negative answer
Negative power of a fraction	$\left(\dfrac{a}{b}\right)^{-n} = \left(\dfrac{b}{a}\right)^n$
Fraction power	$a^{\frac{1}{n}} = \sqrt[n]{a}, \quad a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$ bottom = root, top = power
Negative base	$(-2)^4 = 16, \ (-2)^3 = -8, \ -3^2 = -9$ even power positive, odd power negative
Roots of negatives	$\sqrt[3]{-8} = -2$ odd roots fine; even roots don't exist (real numbers)
No adding law	$x^2 + x^3$ and $(a+b)^2$ don't follow the index laws
Unknown power	$a^p = a^q \Rightarrow p = q$ write both sides as powers of the same base
Unknown base	$x^{\frac{m}{n}} = c \Rightarrow x = c^{\frac{n}{m}}$
Standard form	$a \times 10^n, \quad 1 \leq a < 10$ × / ÷: fronts, then powers, then re-normalise
Adding in standard form	make the powers of 10 equal first, then add the fronts

Folding paper: your turn

Answers in grey. Write out the letters longhand if you're not sure!

Values

1. 2^6
2. 5^3
3. 10^0
4. 4^{-1}
5. 3^{-2}
6. 25^(1/2)
7. 27^(2/3)
8. √(49)

1. 64 2. 125 3. 1 4. 1/4 5. 1/9 6. 5 7. 9 8. 7

Simplify

9. $x^4 \times x^5$
10. $y^9 \div y^3$
11. $(a^3)^4$
12. $(3x)^2$
13. $x^{-2} \times x^5$
14. $(x^2y^3)^2$

9. x^9 10. y^6 11. a^{12} 12. $9x^2$ 13. x^3 14. x^4y^6

Standard form

15. Write 45,000 in standard form
16. Write 0.00061 in standard form
17. $(4 \times 10^3) \times (2 \times 10^{-5})$

15. 4.5×10^4 16. 6.1×10^{-4} 17. 8×10^{-2}

Spot the error

18. Tom says $3^{-2} = -9$. Why is he wrong?

18. negative power means one over: 1/9

Coefficients and negatives

19. $6x^5 \div 2x^2$
20. $(2x^3)^2$
21. $12a^6b^2 \div 4a^2b$
22. $(-2)^5$
23. -2^4
24. $\sqrt[3]{(-27)}$
25. $1/x^{-3}$
26. $(3/4)^{-2}$

19. $3x^3$ 20. $4x^6$ 21. $3a^4b$ 22. -32 23. -16 24. -3 25. x^3 26. $16/9$

Calculating in standard form

27. $(6 \times 10^5) \times (5 \times 10^3)$
28. $(9 \times 10^8) \div (3 \times 10^{-2})$
29. $(4 \times 10^3) + (2 \times 10^2)$
30. Write 25×10^4 in standard form

27. 3×10^9 28. 3×10^{10} 29. 4.2×10^3 30. 2.5×10^5

Find the unknown

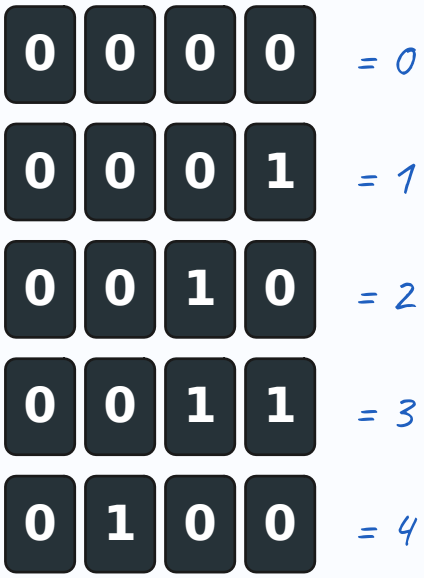
31. $3^n = 1/9$
32. $2^x = 32$
33. $4^x = 8$
34. x^(3/2) = 8
35. $x^k \times x^3 = x^{10}$
36. 5^(x+1) = 125

31. $n = -2$ ($1/9 = 3^{-2}$) 32. $x = 5$ ($32 = 2^5$) 33. $x = 3/2$ ($2^{2x} = 2^3$) 34. x = 4 (x = 8^(2/3))
35. $k = 7$ ($k + 3 = 10$) 36. $x = 2$ ($125 = 5^3$, $x + 1 = 3$)

Chapter 10

Number bases

counting with fewer (or more) than ten digits



a mileage counter whose wheels only have 0 and 1 on them: this is binary, the way every computer counts

We count in tens because we have ten fingers. Nothing else about ten is special. A computer's switches have only two states (off/on), so computers count in **twos**: binary. Programmers also use **sixteens** (hexadecimal) because it's a compact way to write binary.

The good news: every base works exactly like the place value you already know from Chapter 1. Only the size of the bundles changes.

Not a lecture topic, but essential for Computer Science modules.

101 Our usual numbers, taken apart column by column

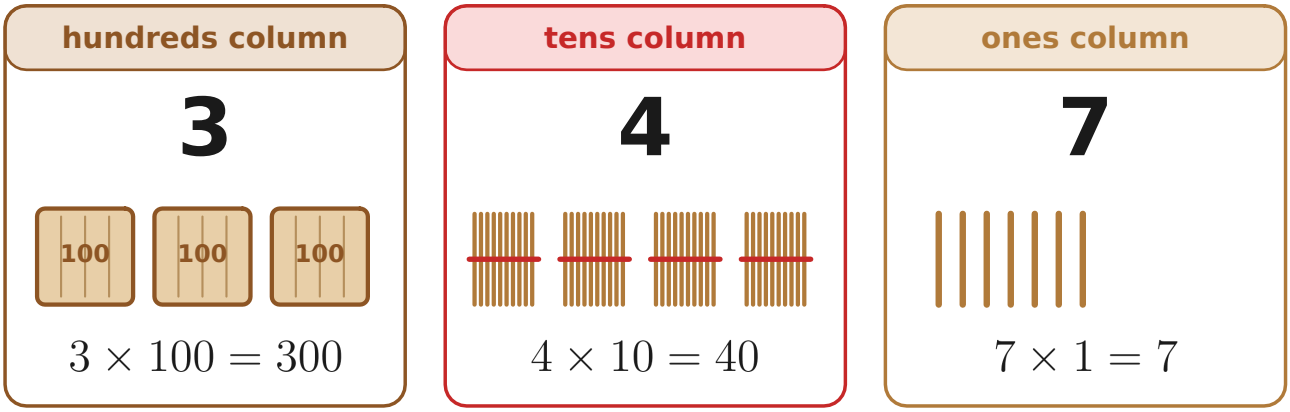
RULE: place value in any base

In base b there are b digits: $0, 1, \ldots, b-1$.

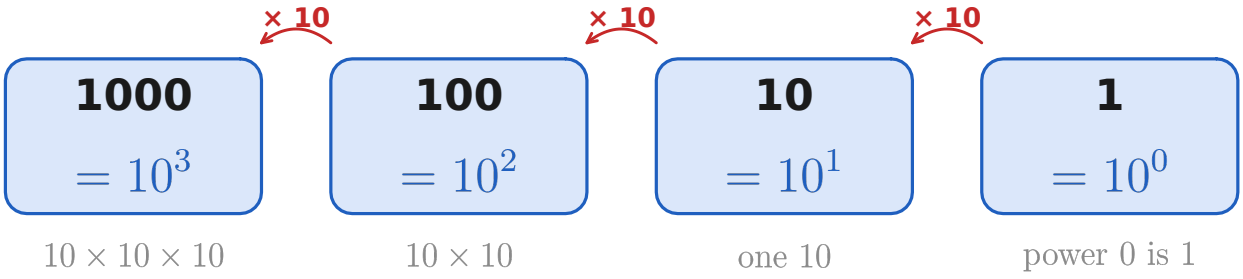
Columns, from the right, are worth $b^0 = 1,\ b^1,\ b^2,\ b^3, \ldots$: each column is worth b times the one to its right.

The value of a number is the sum of (digit × column value). A subscript gives the base: 347_{10}, 10110_2, $2F_{16}$.

Take the number 347. It has three digits, and each digit sits in its own slot, called a **column** (like the columns of a table). Reading from the right: the **ones** column, the **tens** column, the **hundreds** column. The 3, 4 and 7 are not worked out from anything: they are simply the digits of 347, read left to right. What the **column** changes is how much each digit is worth (Chapter 1's crates, bundles and sticks):



The **column value** is what **one** of something is worth in that column: 1 in the ones column, 10 in the tens column, 100 in the hundreds column. Each step to the left is **ten times** bigger, because ten sticks make a bundle and ten bundles make a crate. Now write those column values with powers (Chapter 9). 10^2 just means two tens multiplied:



Put it together. Write the column values as powers over the digits, then multiply each digit by its column value and add:

$10^2 = 100$	$10^1 = 10$	$10^0 = 1$
3	**4**	**7**

$$3 \times 100 + 4 \times 10 + 7 \times 1 = 347$$

The number 10 is called the **base**. A base-10 system needs exactly ten digits (0 to 9) because when a column reaches ten it **rolls over** to 0 and pushes a 1 into the next column, like the mileage counter.

Change the base, change the bundles. In base 2, bundles are made of 2; in base 16, of 16. Everything else stays the same.

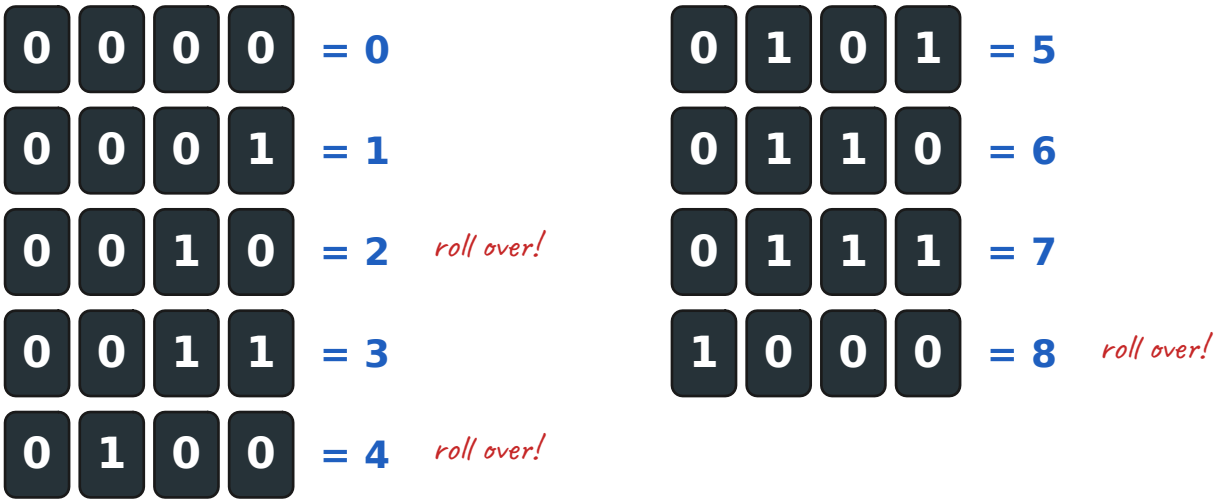
101 Counting in binary: a counter with only 0 and 1

RULE: binary (base 2)

Two digits only: 0 and 1 (called **bits**).

Column values from the right: $1, 2, 4, 8, 16, 32, 64, 128, \ldots$ (the powers of 2).

Each wheel has just two digits. Add 1 and the right wheel goes 0 → 1. Add 1 again: it **rolls over** to 0 and pushes the next wheel on. Watch:



Binary has two digits, 0 and 1, called **bits** (binary digits). The columns are worth 1, 2, 4, 8, 16, ...: each is **double** the one to its right, because two of one column make one of the next.

101 Binary → ordinary: add up the column values

PROCEDURE: base b → decimal

1. Write the column values $1, b, b^2, \ldots$ over the digits, starting from the **right**.
2. Multiply each digit by its column value (in binary: keep the columns with a 1).
3. Add the results.

To read 10110_2 (the little 2 means "in base 2"), write the column values over the digits and add up the ones that have a 1 under them:

16	8	4	2	1
1	**0**	**1**	**1**	**0**

$$16 + 4 + 2 = 22$$

So $10110_2 = 22_{10}$. Read aloud as "one zero one one zero", never "ten thousand one hundred and ten" (that's a base-10 number).

101 Ordinary → binary: making change with coins

RULE: decimal → binary (two methods)

Biggest power first: list the powers of 2 up to the number. Working from the biggest, write 1 and subtract if it fits, 0 if it doesn't, until nothing is left.

Repeated halving: divide by 2, write the remainder (0 or 1), and repeat on the quotient until it is 0. Read the remainders from the **bottom up**.

Imagine a country with coins worth **16, 8, 4, 2 and 1**, and you may use **at most one** of each. Pay 22 using the biggest coins first:

16	8	4	2	1
1	**0**	**1**	**1**	**0**
22: use the 16 → 6 left	*6: 8 is too big → skip*	*6: use the 4 → 2 left*	*2: use the 2 → 0 left*	*0: nothing left → skip*

Used = 1, skipped = 0. Reading the row: **10110**. That's 22 in binary.

Method 2: keep halving (handy for big numbers). Divide by 2 and write down each remainder (Chapter 1); then read the remainders **from the bottom up**:

22 ÷ 2 = 11	**remainder 0**
11 ÷ 2 = 5	**remainder 1**
5 ÷ 2 = 2	**remainder 1**
2 ÷ 2 = 1	**remainder 0**
1 ÷ 2 = 0	**remainder 1**

read upwards: 1 0 1 1 0

101 Base 16 (hexadecimal): running out of digits

RULE: hexadecimal (base 16) and octal (base 8)

Hex digits: 0 to 9, then **A = 10, B = 11, C = 12, D = 13, E = 14, F = 15**. Columns: 1, 16, 256, 4096 (powers of 16).

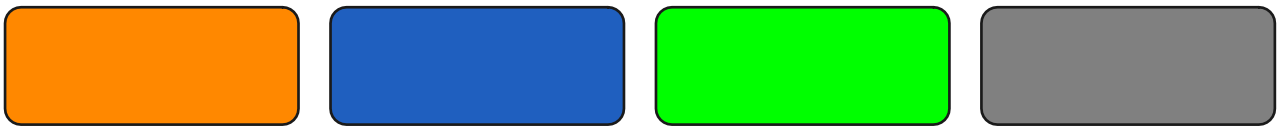
Octal digits: 0 to 7. Columns: 1, 8, 64, 512 (powers of 8).

In base 16 a column can hold up to fifteen before rolling over. But we only have ten digit symbols! So we borrow letters: **A = 10, B = 11, C = 12, D = 13, E = 14, F = 15**.

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Columns are worth 1, 16, 256, ... (powers of 16). So $2F_{16} = 2 \times 16 + 15 = 47$.

Where you've seen hex: colour codes. A web colour like #FF8800 is three hex pairs: how much **red**, **green** and **blue** light, each from 00 (none) to FF (full):



#FF8800	#1F5FBF	#00FF00	#808080
R255 G136 B0	R31 G95 B191	R0 G255 B0	R128 G128 B128

FF = 15 × 16 + 15 = 255, the biggest two hex digits can hold. #FF8800 is full red, a bit over half green, no blue: orange.

Why programmers love hex: each hex digit is exactly 4 bits, so binary converts in chunks with no arithmetic:

1111	1000	1000	0000
↓	↓	↓	↓
F	**8**	**8**	**0**

11111000 10000000
= F8 80

Octal (base 8, digits 0-7, columns 1, 8, 64) works the same way with 3-bit chunks; you'll see it in Linux file permissions like chmod 755.

101 Decimal → any base: divide and read upwards

PROCEDURE: decimal → base b

1. Divide the number by b. Write down the whole-number quotient and the **remainder** (a remainder is always between 0 and $b-1$).
2. Divide the quotient by b again, and keep going until the quotient is **0**.
3. Write each remainder as a single base-b digit (in hex, 10 to 15 become A to F).
4. Read the remainders from the **bottom up**: the last remainder is the leftmost digit.
5. Check by converting back: digit × column value, added up.

This is the halving method from binary with 2 replaced by b. It works because each division peels off the rightmost digit: the remainder is what doesn't fill a whole bundle of b.

Example: write 200 in hexadecimal.

200 ÷ 16 = 12 **remainder 8**
12 ÷ 16 = 0 **remainder 12 (= C)**

read upwards: C 8

$200 = C8_{16}$ ← check: 12 × 16 + 8 = 192 + 8 = 200 ✓

Example: write 100 in octal (base 8).

100 ÷ 8 = 12 **remainder 4**
12 ÷ 8 = 1 **remainder 4**
1 ÷ 8 = 0 **remainder 1**

read upwards: 1 4 4

$100 = 144_8$ ← check: 1 × 64 + 4 × 8 + 4 = 100 ✓

Example: write 45 in base 3.

45 ÷ 3 = 15 **remainder 0**
15 ÷ 3 = 5 **remainder 0**
5 ÷ 3 = 1 **remainder 2**
1 ÷ 3 = 0 **remainder 1**

read upwards: 1 2 0 0

$45 = 1200_3$ ← check: 1 × 27 + 2 × 9 = 45 ✓

RULE: binary ↔ hex and binary ↔ octal

One hex digit is exactly **4 bits** (because $16 = 2^4$); one octal digit is exactly **3 bits** ($8 = 2^3$).

Binary → hex: split the bits into groups of 4 **from the right**, pad the left group with zeros, and convert each group (0000 = 0 up to 1111 = F). For octal use groups of 3.

Hex → binary: replace each hex digit by its 4 bits (keep leading zeros inside the number).

Example: write 1101011_2 in hex and in octal.

$$\underbrace{0110}_{6}\ \underbrace{1011}_{B} \Rightarrow 6B_{16}$$

← groups of 4 from the right; pad with a 0

$$\underbrace{001}_{1}\ \underbrace{101}_{5}\ \underbrace{011}_{3} \Rightarrow 153_8$$

← groups of 3 from the right

Example: write $2F_{16}$ in binary.

$$2 \to 0010, \quad F \to 1111 \Rightarrow 101111_2$$

← drop the leading zeros at the very front

Check: $101111_2 = 32 + 8 + 4 + 2 + 1 = 47$, and $2F_{16} = 47$ ✓.

101 Adding in binary: 1 + 1 = 10

RULE: binary addition

Line the numbers up on the right and add column by column, **right to left**, as in base 10.

$0+0=0$, $0+1=1$, $1+1=10_2$ (write 0, carry 1), $1+1+1=11_2$ (write 1, carry 1).

Check by converting everything to decimal.

There are only four single-digit sums. The last one rolls over, just as 9 + 1 does in base 10:

0 + 0 = 0 **0 + 1 = 1** **1 + 0 = 1** **1 + 1 = 10**

write 0, carry 1

Work out $1011_2 + 110_2$, column by column from the right, exactly like Chapter 1:

```
  1  1  1
     1  0  1  1
+       1  1  0
-----------------
  1  0  0  0  1
```

1s column: 1 + 0 = 1
2s column: 1 + 1 = 10 → write 0, carry 1
4s column: 0 + 1 + 1 (carry) = 10 → write 0, carry 1
8s column: 1 + 1 (carry) = 10 → 0, carry 1

Check in base 10: 11 + 6 = 17, and $10001_2 = 16+1 = 17$ ✓.

Link to Chapter 7, carefully. A column with **two** bits and no carry coming in is simple: digit = A XOR B, carry = A AND B (a **half adder**). But look at the 4s column above: it adds **three** bits, A, B and the carry that came in. Every column after the first can have three inputs, so a chip uses a **full adder**: the digit is 1 when an **odd** number of the three are 1, and the carry out is 1 when **two or more** are 1. Here's every column of the sum:

column	A	B	carry in	digit (odd?)	carry out (2+?)
1s	1	0	0	**1**	**0**
2s	1	1	0	**0**	**1**
4s	0	1	1	**0**	**1**
8s	1	0	1	**0**	**1**
16s	0	0	1	**1**	**0**

Read the green digits from the bottom up: 10001 ✓. Only the 1s column can never receive a carry, so it can be a half adder. Every other column **might** get one (here the 2s column happened not to), so each needs a full adder. In Chapter 7's words, with C the carry in:

digit = A XOR B XOR C **carry out** = (A AND B) OR (C AND (A XOR B))

101 Subtracting in binary: borrowing a 2

RULE: binary subtraction

Work column by column, right to left: $0-0=0$, $1-0=1$, $1-1=0$.

$0-1$ needs a **borrow**: take 1 from the next column to the left. It arrives as **2** (one of the next column is worth two of this one), so the column becomes $10_2-1=1$.

If the next column is also 0, borrow from further left; every 0 passed over becomes 1.

Example: 1101_2-110_2 (that is $13-6$).

	0	**2**	**2**	
	1	1	0	1
−		1	1	0
	0	1	1	1

1s: 1 − 0 = 1
2s: 0 − 1, borrow from the 4s (its 1 becomes 0):
2 − 1 = 1
4s: now 0 − 1, borrow from the 8s (its 1 becomes 0):
2 − 1 = 1
8s: 0 − 0 = 0

So $1101_2-110_2=111_2$. Check in base 10: $13-6=7=4+2+1$ ✓.

Computers don't actually borrow: they add a negative number stored in "two's complement" form. You'll meet that in Computer Science; on paper, borrowing is fine.

101 Bits, bytes and why a kilobyte might be 1024

A group of **8 bits** is a **byte**. With 8 wheels of 0/1 you can make $2^8 = 256$ different patterns (00000000 to 11111111, i.e. 0 to 255). That's why colour values stop at 255.

Computers like powers of 2, and $2^{10} = 1024$ is close to 1000. So memory sizes were traditionally counted in 1024s: 1 KB = 1024 bytes. Newer notation calls that a **kibibyte** (KiB) and keeps kB = 1000 bytes (the metric prefix from Chapter 5).

101 Any base at all

In base b the digits go from 0 to b − 1, and the columns are worth $b^0 = 1$, b^1, b^2, … . For example, base 3 has digits 0, 1, 2 and columns 1, 3, 9, 27:

$$2101_3 = 2 \times 27 + 1 \times 9 + 0 \times 3 + 1 \times 1$$

← column values are powers of 3

$$= 54 + 9 + 0 + 1 = 64$$

Watch out: a digit can never equal the base. "2" can't appear in binary, and "8" can't appear in octal, just as there's no single digit for ten in base 10.

101 Find the unknown: missing digits and missing bases

Some questions give the value and hide a digit, or hide the base itself. Both are solved by writing the place-value sum with a letter in it:

PROCEDURE: missing digit or base

1. Write the number as (digit × column value) added up, using a letter for the unknown digit, or b for an unknown base (columns $1, b, b^2, \ldots$).
2. Set that sum equal to the value you are given, and solve.
3. Check the answer makes sense: a digit must be a whole number from 0 to $b-1$; a base must be a whole number **bigger than every digit** in the number.

Example: $101x_2 = 11_{10}$. Find the digit x.

$$1 \times 8 + 0 \times 4 + 1 \times 2 + x \times 1 = 11$$ ← columns 8, 4, 2, 1

$$10 + x = 11 \;\Rightarrow\; x = 1$$ ← a valid binary digit ✓

Example: in which base is $21 = 7_{10}$?

$$2 \times b + 1 \times 1 = 7$$ ← columns b and 1

$$2b = 6 \;\Rightarrow\; b = 3$$ ← digits 2 and 1 are both less than 3 ✓

Example: in which base is $100 = 49_{10}$?

$$1 \times b^2 = 49 \;\Rightarrow\; b = 7$$ ← a base is positive, so not −7

Example: $3x_{16} = 59_{10}$. Find the hex digit x.

$$3 \times 16 + x = 59 \;\Rightarrow\; x = 11 = B$$ ← 11 is written B in hex

Example: $25_b = 17_{10}$. Find b.

$$2b + 5 = 17 \;\Rightarrow\; b = 6$$ ← 6 is bigger than the digits 2 and 5 ✓

101 Mistakes to avoid

✗ Common mistakes

✗ $200 \rightarrow 8C_{16}$ ✓ $200 = C8_{16}$

read the remainders from the bottom up

✗ $12_{10} = 12_{16}$ ✓ $12_{10} = C_{16}$

10 to 15 are single hex digits A to F

✗ $1_2 + 1_2 = 2_2$ ✓ $1_2 + 1_2 = 10_2$

there is no digit 2 in binary: write 0, carry 1

✗ $1101011_2 \rightarrow 1101\,011$ ✓ $0110\,1011 = 6B_{16}$

group the bits from the right, not the left

✗ $101_2 = 101$ ✓ $101_2 = 4 + 1 = 5$

the columns are worth 4, 2, 1, not 100, 10, 1; say "one zero one"

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Place value in base b	$(d_2\, d_1\, d_0)_b = d_2 b^2 + d_1 b + d_0$ digits 0 to $b-1$; columns $1, b, b^2, \ldots$ from the right
Binary	digits 0, 1; columns 1, 2, 4, 8, 16, 32, ...
Hexadecimal	digits 0-9, A = 10 ... F = 15; columns 1, 16, 256, ...
Octal	digits 0-7; columns 1, 8, 64, ...
Base b → decimal	multiply each digit by its column value and add
Decimal → base b	divide by b repeatedly until the quotient is 0; read the remainders bottom-up
Decimal → binary	or: take the biggest power of 2 that fits, write 1, subtract, repeat
Binary ↔ hex	groups of 4 bits from the right ($16 = 2^4$)
Binary ↔ octal	groups of 3 bits from the right ($8 = 2^3$)
Binary addition	$1 + 1 = 10_2, \quad 1 + 1 + 1 = 11_2$ right to left, carrying
Binary subtraction	$10_2 - 1 = 1$ 0 − 1: borrow from the left; it arrives as 2
Bits and bytes	n bits $\rightarrow 2^n$ values 8 bits = 1 byte = 256 values (0-255)
Missing digit or base	write the place-value sum with a letter; solve; check digits < base

🔢 Counters and coins: your turn

Answers in grey.

Binary to ordinary

1. 1101_2
2. 100000_2
3. 11111111_2

1. 13 2. 32 3. 255

Ordinary to binary

4. 9
5. 25
6. 100

4. 1001 5. 11001 6. 1100100

Hexadecimal

7. $1A_{16}$ in base 10
8. 200 in hex
9. What colour is #0000FF?
10. Write 10110111_2 in hex

7. 26 8. C8 9. pure blue 10. B7

Arithmetic

11. $101_2 + 11_2$
12. $1111_2 + 1_2$
13. How many values can 4 bits store?

11. 1000_2 (5 + 3 = 8) 12. 10000_2 13. $2^4 = 16$

Other bases

14. 122_3 in base 10
15. Why is 128_8 not a valid octal number?

14. 17 15. octal has no digit 8

More conversions

16. 100 in octal
17. $1110_2 - 101_2$
18. $3C_{16}$ in binary
19. 1101011_2 in octal

16. 144_8 (100 ÷ 8 = 12 r 4, 12 ÷ 8 = 1 r 4, 1 ÷ 8 = 0 r 1) 17. 1001_2 (14 − 5 = 9) 18. 111100_2 (0011 1100) 19. 153_8 (001 101 011)

Find the unknown

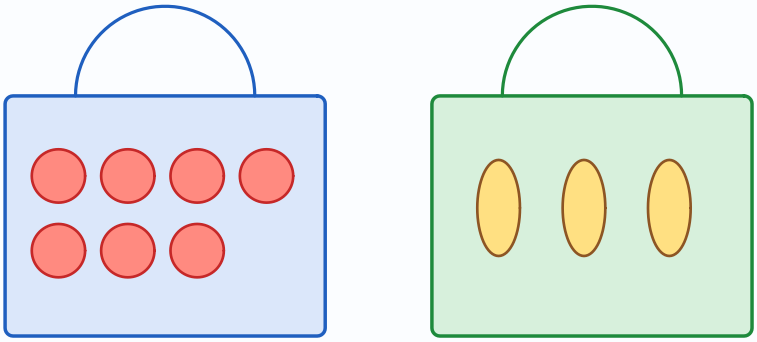
20. $101x_2 = 11$. Find the digit x.
21. In which base is 21 = 7?
22. In which base is 100 = 36?
23. $3x_{16} = 59$. Find the hex digit x.
24. 25 in base b equals 17. Find b.
25. $1x1_2 = 7$. Find x.

20. x = 1 (8 + 2 + x = 11) 21. base 3 (2b + 1 = 7) 22. base 6 ($b^2 = 36$) 23. x = B (48 + x = 59, x = 11) 24. b = 6 (2b + 5 = 17) 25. x = 1 (4 + 2x + 1 = 7)

Chapter 11

Simplifying algebraic expressions

sorting the shopping, and multiplying out brackets with a grid



$7a$ $3b$

apples in one bag, bananas in the other. Never "10ab"!

Simplifying means rewriting an expression so it's **shorter but equal**: the same value for every number you could put in. Two skills do almost all of the work: **collecting like terms** (sorting the shopping into bags) and **expanding brackets** (multiplying everything inside by what's outside).

Every rule is tested with numbers, so you'll never have to wonder "is this allowed?". That same habit catches two mistakes on the Lecture 2 slides, near the end of the chapter.

Goes with **Lecture 2** (expanding brackets, common mistakes).

Like terms: same fruit, same bag

RULE: collecting like terms

Like terms have exactly the **same letters** raised to the **same powers**. Only the number in front (the coefficient) may differ: $3x^2$ and $-5x^2$ are like; x and x^2 are not.

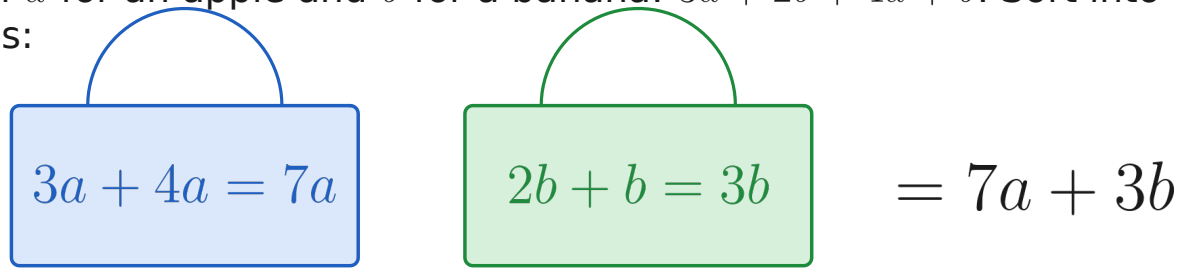
To collect like terms, add or subtract their **coefficients**; the letters and powers stay exactly as they are: $3x^2 - 5x^2 = -2x^2$.

Unlike terms cannot be combined by adding: $7a + 3b$ and $x + x^2$ are already as simple as they get.

The order of letters inside a term doesn't matter: xy and yx are like terms.

Terms	Like?	Why
$3x$ and $5x$	**yes**	same letter, same power: $3x + 5x = 8x$
x and x^2	**no**	same letter, different powers
$2xy$ and $5yx$	**yes**	yx is xy: $2xy + 5yx = 7xy$
$4a^2b$ and $3ab^2$	**no**	the powers sit on different letters
7 and -2	**yes**	both constants: $7 - 2 = 5$

Your shopping: **3 apples, 2 bananas, 4 apples, 1 banana**. Written with a for an apple and b for a banana: $3a + 2b + 4a + b$. Sort into bags:



Terms with **exactly the same letters and powers** are **like terms**: they can be added by adding their coefficients. Different letters stay separate: $7a + 3b$ can't be squashed into $10ab$ (10 apple-bananas?).

Powers count as a different fruit. x and x^2 are **not** like terms. Test with $x = 3$: $x + x^2 = 3 + 9 = 12$, but "$2x^2$" would be 18 and "$2x$" would be 6. So $x + x^2$ stays as it is.

A bigger example, colour-coded by kind:

$$5x^2 + 3x - 2x^2 + 7 - x + 1$$

purple = x^2 bag, blue = x bag, orange = plain numbers

$$= (5-2)x^2 + (3-1)x + (7+1) = 3x^2 + 2x + 8$$

Test with $x = 1$: the original gives $5 + 3 - 2 + 7 - 1 + 1 = 13$; the answer gives $3 + 2 + 8 = 13$ ✓.

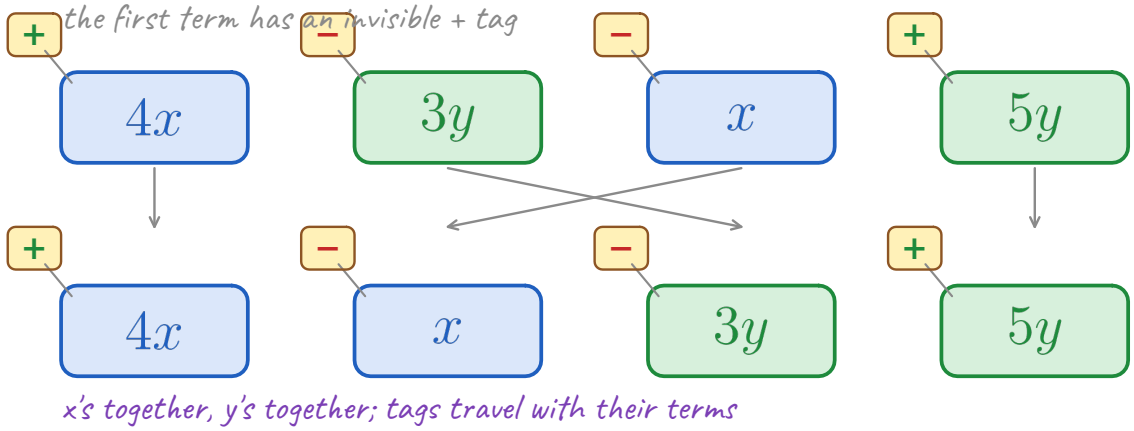
The minus sign is a luggage tag

RULE: signs belong to terms

Each term owns the sign **directly in front of it**. A first term with no sign is $+$.

You may reorder terms, but each sign moves **with** its term: $4x - 3y - x = 4x - x - 3y$.

When you sort terms, each one carries the sign **in front of it**, like a luggage tag tied to a suitcase. Move the suitcase and the tag comes too. Simplify $4x - 3y - x + 5y$:



$$4x - x - 3y + 5y \quad \longleftarrow \text{ rearranged, tags attached}$$

$$= 3x + 2y \quad \longleftarrow \text{ 4 − 1 = 3 x's; −3 + 5 = +2 y's}$$

Classic slip: reading $-3y + 5y$ as $-8y$ by "adding the 3 and 5". Use the lift from Chapter 1: start 3 floors down, go up 5 → floor 2.

Multiplying terms is different from adding them

PROCEDURE: multiplying terms

1. Multiply the **coefficients** (using the sign rules: same signs give $+$, different signs give $-$).
2. Multiply the **letters**: the same letter's powers **add** ($x \times x^2 = x^3$); different letters sit side by side ($a \times b = ab$).
3. Any terms can be multiplied; they don't need to be like terms.

To **multiply** terms you don't need like terms at all. Multiply the numbers, then the letters (using the power law $x \times x = x^2$ from Chapter 9):

$$3x \times 2x = (3 \times 2)(x \times x) = 6x^2$$ ← *numbers × numbers, letters × letters*

$$2a \times 5b = 10ab$$ ← *different letters just sit side by side*

$$-4x \times 3x^2 = -12x^3$$ ← *negative × positive = negative; $x^1 \times x^2 = x^3$*

Your Lecture 2 "common mistakes" slide , each caught by testing $x = 3$, $y = 2$:

$2x + x \neq 2x^2$	6 + 3 = 9, but 2 × 9 = 18	$2x + x = 3x$
$2x \times x \neq 3x$	6 × 3 = 18, but 3 × 3 = 9	$2x \times x = 2x^2$
$2x + 2y \neq 2xy$	6 + 4 = 10, but 2 × 6 = 12	stays $2x + 2y$
$2x + 2x^2 \neq 4x^2$	6 + 18 = 24, but 36	stays $2x + 2x^2$

The pattern: adding needs like terms and keeps the power; multiplying works on anything and adds the powers.

Expanding one bracket: everything inside gets multiplied

RULE: expanding a single bracket

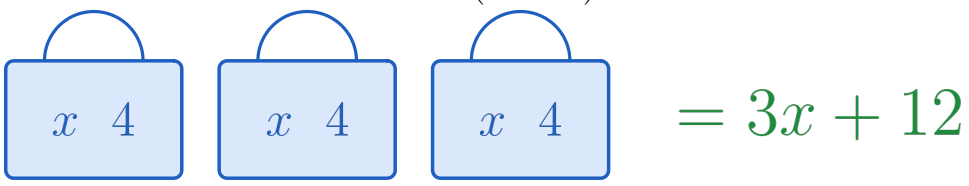
Multiply **every** term inside the bracket by the term outside, **sign included**: $a(b+c) = ab + ac$.

A minus sign in front of a bracket is -1 outside it: it **changes the sign of every term** inside. $-(x-3) = -x+3$.

Then collect like terms.

Outside	Inside	Product	Example
$+$	$+$	+	$3 \times 4 = 12$
$+$	$-$	−	$3 \times (-4) = -12$
$-$	$+$	−	$-2 \times x = -2x$
$-$	$-$	+	$-2 \times (-5) = +10$

$3(x+4)$ means three bags, each holding x and 4. Tip them all out: three x's and three 4s. So $3(x+4) = 3x + 12$.



The **grid method** does the same thing and works for everything that follows. The outside goes down the side, the inside across the top; fill each box by multiplying:

×	x	$+4$
3	$3x$	$+12$

3(x + 4) = 3x + 12

×	x	-5
-2	$-2x$	$+10$

−2 × −5 = +10 (tags multiply!)

Letters outside work the same: $x(x+3) = x^2 + 3x$.

Example: $-(x-3)$. The minus in front means $-1\times$ the bracket.

$$-1 \times x + (-1) \times (-3)$$ ← every term inside gets × (−1)

$$= -x + 3$$ ← (−1)(x) = −x; (−1)(−3) = +3

Example: $-x(2x-5)$.

$$(-x)(2x) + (-x)(-5)$$ ← the −x multiplies both terms

$$= -2x^2 + 5x$$ ← −x × 2x = −2x²; −x × −5 = +5x

Test with $x = 2$: $-2(4-5) = 2$ and $-8+10 = 2$ ✓.

Expand then collect. $3(x+2) - 2(x-1)$. The -2 (with its minus tag) multiplies **both** things in its bracket:

$$3(x+2) - 2(x-1)$$

$$= 3x + 6 - 2x + 2$$ ← −2 × x = −2x; −2 × −1 = +2

$$= x + 8$$ ← collect: 3x − 2x = x; 6 + 2 = 8

Two brackets: the grid is a rectangle's area

PROCEDURE: expanding two brackets

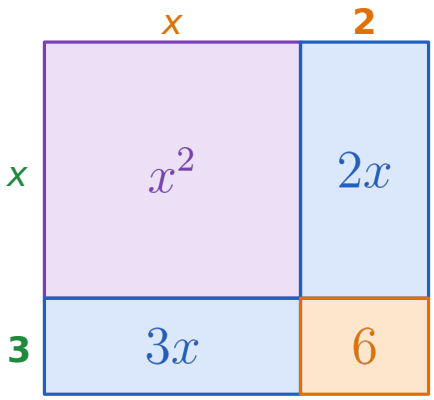
1. Multiply **each** term in the first bracket by **each** term in the second (2 × 2 = 4 products; a grid keeps track).
2. Each product takes the sign rules: watch the minus signs.
3. Collect like terms (usually the two middle x-terms).
4. Test with a number, e.g. $x = 2$, in the original and in your answer.

Your slides start with numbers: $(2+3)(4+5) = 5 \times 9 = 45$. The long way, multiplying **everything in the first bracket by everything in the second**:

$$2 \times 4 + 2 \times 5 + 3 \times 4 + 3 \times 5 = 8 + 10 + 12 + 15 = 45$$

same answer ✓

Why does that work? Draw a rectangle $2+3$ tall and $4+5$ wide and cut it into four pieces. Now do the same with letters: $(x+3)(x+2)$:



×	x	$+2$
x	x^2	$+2x$
$+3$	$+3x$	$+6$

the grid is the rectangle, tidied

$$(x+3)(x+2) = x^2 + 2x + 3x + 6 = x^2 + 5x + 6$$

collect the two blue x-terms

Test $x = 1$: $(4)(3) = 12$ and $1 + 5 + 6 = 12$ ✓. With a minus: $(x+4)(x-3)$: the grid gives $x^2 - 3x + 4x - 12 = x^2 + x - 12$ (the -3 tag makes its row of boxes negative).

Perfect squares: the two missing rectangles

RULE: three special products

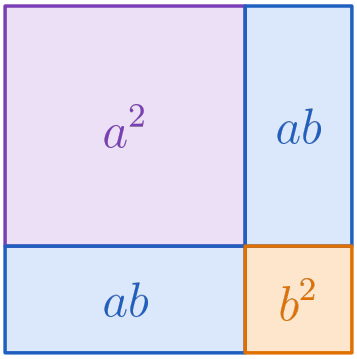
$(a+b)^2 = a^2 + 2ab + b^2$

$(a-b)^2 = a^2 - 2ab + b^2$

$(a+b)(a-b) = a^2 - b^2$ (the **difference of two squares**: the middle terms cancel)

Squaring a bracket is **not** squaring each term: $(a+b)^2 \neq a^2 + b^2$.

A very common mistake is $(a+b)^2 = a^2 + b^2$. The picture shows why it's wrong. A square of side $a+b$ has **four** pieces, not two:



$a^2 + b^2$ only counts the purple and orange squares. The two blue rectangles are forgotten, and together they're $2ab$.

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

Using it (Lecture 2 example): expand $(2x-3y)^2$. Replace a by $2x$ and b by $3y$, in brackets, exactly like substitution:

$$(2x)^2 - 2(2x)(3y) + (3y)^2$$ ← *a → 2x, b → 3y*

$$= 4x^2 - 12xy + 9y^2$$ ← *$(2x)^2 = 4x^2$, $2 \times 2 \times 3 = 12$, $(3y)^2 = 9y^2$*

Difference of two squares. $(a-b)(a+b)$: the grid gives $a^2 + ab - ab - b^2$. The middle terms **cancel each other**, leaving $a^2 - b^2$. Chapter 12 runs this backwards.

Three brackets, and catching a slide mistake

PROCEDURE: three or more brackets

1. Expand **two** of the brackets and collect like terms.
2. Multiply that result by the next bracket: every term by every term (a bigger grid).
3. Collect like terms again, and test with $x = 1$ **and** $x = 2$.

$(x+1)^3 = (x+1)(x+1)(x+1)$. Do it in two stages: first two brackets, then multiply the result by the last one with a bigger grid:

$$(x+1)(x+1) = x^2 + 2x + 1$$ ← *stage 1: a perfect square*

$\times$	x^2	$+2x$	$+1$
x	x^3	$+2x^2$	$+x$
$+1$	$+x^2$	$+2x$	$+1$

stage 2: $(x+1) \times (x^2 + 2x + 1)$
same colours = like terms

$$(x+1)^3 = x^3 + 3x^2 + 3x + 1$$ ← *collect: $2x^2 + x^2 = 3x^2$, $x + 2x = 3x$*

Slide check. Lecture 2 writes $(x+1)^3 = x^3 + 4x^2 + 3x + 1$. Test with $x = 1$: the left side is $2^3 = 8$. The slide's version gives $1 + 4 + 3 + 1 = 9$ ✗, ours gives $1 + 3 + 3 + 1 = 8$ ✓. (The slide's own working has $2x^2 + x^2 + x^2$, one x^2 too many.)

The "MASSIVE example" $(x+4)(x+3)(x+2)(x+1)$: the slide ends with $44x^2$. With $x = 1$ the left side is $5 \times 4 \times 3 \times 2 = 120$, but $1 + 10 + 44 + 50 + 24 = 129$ ✗. The correct answer is $x^4 + 10x^3 + 35x^2 + 50x + 24$ (it sums to 120 ✓). On the slide, one "$12x^2$" was copied as "$21x^2$".

The lesson: putting a number into both sides takes ten seconds and catches most expanding mistakes, including your lecturer's. $x = 1$ is quick, but it can't tell x^2 from x^3 (both are 1), so use $x = 2$ as well when powers are involved.

Find the unknown: matching coefficients

A common question gives an expansion with a letter missing, like "$(x+a)(x+3) = x^2 + 7x + 12$: find a". The equals sign here means **identical** (true for every x, Chapter 8's $\equiv$), so each kind of term must match:

PROCEDURE: matching coefficients

1. Expand and simplify the side containing the unknown.
2. Collect it into x^2 terms, x terms and constants.
3. The two sides are identical, so the **coefficients must match**: x^2 with x^2, x with x, constant with constant. Each match is a small equation.
4. Solve one of them for the unknown, then use the others as a **check**.

Example: $(x+a)(x+3) = x^2 + 7x + 12$. Find a.

$$x^2 + 3x + ax + 3a$$ ← expand (grid: x^2, $3x$, ax, $3a$)

$$= x^2 + (a+3)x + 3a$$ ← collect the x-terms

$$x: \ a + 3 = 7 \ \Rightarrow \ a = 4$$ ← match the x coefficients

$$\text{constant: } 3a = 3(4) = 12 \ \checkmark$$ ← the constants agree too

Example: $2(x+k) - 3 = 2x + 5$. Find k.

$$2x + 2k - 3 = 2x + 5$$ ← expand the bracket

$$2k - 3 = 5 \ \Rightarrow \ k = 4$$ ← x-terms already match; match the constants

Example: $(x+p)^2 = x^2 + 10x + 25$. Find p.

$$x^2 + 2px + p^2$$ ← perfect square

$$2p = 10 \ \Rightarrow \ p = 5; \quad p^2 = 25 \ \checkmark$$ ← match x, then check the constant

Example: $3(x+2)-a(x-1)=x+8$. Find a.

$$3x+6-ax+a=(3-a)x+(6+a)$$ ← −a × −1 = +a

$$3-a=1 \;\Rightarrow\; a=2; \quad 6+a=8 \;\checkmark$$

If the check fails, either you expanded wrongly or no value works. Go back to the expansion first.

Mistakes to avoid

✗ Common mistakes

✗ $-(x-3)=-x-3$

✓ $-(x-3)=-x+3$

the minus changes the sign of EVERY term inside

✗ $3(x+4)=3x+4$

✓ $3(x+4)=3x+12$

the 3 multiplies every term, not just the first

✗ $-x(2x-5)=-2x^2-5x$

✓ $-2x^2+5x$

−x × −5 = +5x

✗ $(x+3)^2=x^2+9$

✓ $(x+3)^2=x^2+6x+9$

two brackets: don't forget the 2 × 3x

✗ $\dfrac{2x+3}{x}=2+3$

✓ $\dfrac{2x+3}{x}=2+\dfrac{3}{x}$

dividing by x divides every term; 3 has no x to cancel

✗ $2(1+3)=\dfrac{16}{x}$

✓ $2x=13, \quad x=6.5$

Lecture 2's "dividing by x" mistake, solving 2x + 3 = 16: x was taken out of 2x only. Undo the + 3 first, then ÷ 2

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Like terms	same letters AND same powers; only the coefficient may differ
Collecting	$3x^2 - 5x^2 = -2x^2$ add the coefficients; letters and powers unchanged
Unlike terms	$x + x^2, \quad 7a + 3b$ cannot be combined by adding
Signs	each term owns the sign in front of it; the sign moves with the term
Multiplying terms	$-4x \times 3x^2 = -12x^3$ coefficients × coefficients; add powers of the same letter
Sign rules	$(+)(+) = +, \ (-)(-) = +, \ (+)(-) = -$
One bracket	$a(b + c) = ab + ac$ every term inside, sign included
Minus outside	$-(x - 3) = -x + 3$ changes the sign of every term inside
Two brackets	$(a + b)(c + d) = ac + ad + bc + bd$ each term by each term, then collect
Perfect squares	$(a \pm b)^2 = a^2 \pm 2ab + b^2$ never just $a^2 + b^2$
Difference of two squares	$(a + b)(a - b) = a^2 - b^2$
Three brackets	two at a time, collecting in between
Dividing by x	$\frac{2x + 3}{x} = 2 + \frac{3}{x}$ divide every term
Matching coefficients	in an identity, match x^2, x and constant terms to find a missing letter
Check	test with $x = 1$ and $x = 2$

Shopping bags: your turn

Answers in grey. Test each answer with x = 1 or x = 2!

Collect like terms

1. $5a + 3b - 2a + b$
2. $4x^2 - x + 3x^2 + 5x - 2$
3. $7p - 3q - 9p + q$

1. $3a + 4b$ 2. $7x^2 + 4x - 2$ 3. $-2p - 2q$

Multiply

4. $4x \times 3y$
5. $-2a \times 5a$
6. $3x^2 \times 2x^3$

4. $12xy$ 5. $-10a^2$ 6. $6x^5$

One bracket

7. $5(2x - 3)$
8. $-3(x - 4)$
9. $x(2x + 7)$
10. $2(x + 5) - 3(x - 1)$

7. $10x - 15$ 8. $-3x + 12$ 9. $2x^2 + 7x$ 10. $-x + 13$

Two brackets

11. $(x + 5)(x + 2)$
12. $(x - 3)(x + 7)$
13. $(2x + 1)(x - 4)$
14. $(x - 6)^2$
15. $(3x + 2)(3x - 2)$

11. $x^2 + 7x + 10$ 12. $x^2 + 4x - 21$ 13. $2x^2 - 7x - 4$ 14. $x^2 - 12x + 36$ 15. $9x^2 - 4$

Minus signs outside

16. $-(2x - 7)$
17. $-x(3x - 4)$
18. $4 - 2(x - 5)$

16. $-2x + 7$ 17. $-3x^2 + 4x$ 18. $-2x + 14$

Find the unknown

19. $(x + a)(x + 3) = x^2 + 7x + 12$. Find a.

20. $2(x + k) - 3 = 2x + 5$. Find k.

21. $(x + p)^2 = x^2 + 10x + 25$. Find p.

22. $a(x - 2) + 3x = 7x - 8$. Find a.

23. $(2x + 1)(x + b) = 2x^2 + 7x + 3$. Find b.

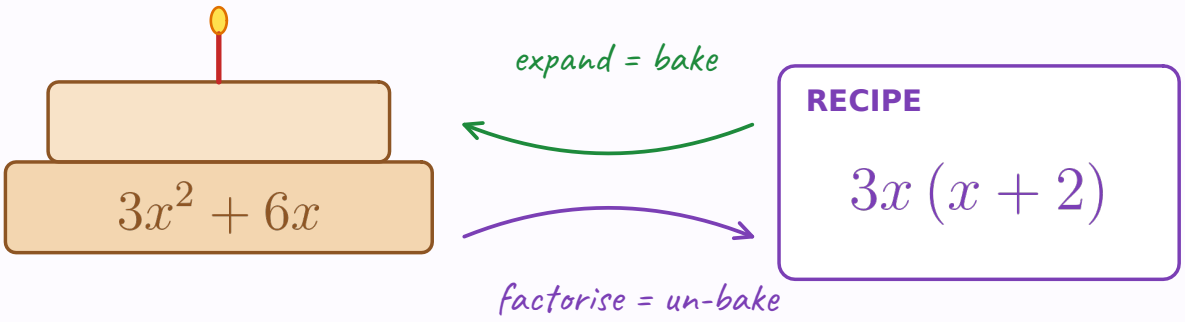
24. $5x - 3(x - c) = 2x + 12$. Find c.

19. a = 4 ($a + 3 = 7$; check $3a = 12$) 20. k = 4 ($2k - 3 = 5$) 21. p = 5 ($2p = 10$; $p^2 = 25$) 22. a = 4 ($a + 3 = 7$; $-2a = -8$) 23. b = 3 (constants: $b = 3$; check $2b + 1 = 7$) 24. c = 4 ($3c = 12$)

Chapter 12

Factorisation

un-baking: reading the recipe back off the cake



In Chapter 11 you **expanded**: brackets in, long expression out. **Factorising** is the same trip backwards. You're handed the long expression (the cake) and asked for the brackets that made it (the recipe).

It's harder in this direction, the way un-baking is harder than baking, so every shape gets a **rule stated in a box first**: take out common factors first; for three terms, find two numbers that **add** to one coefficient and **multiply** to another; for two terms, look for squares. A formula (the grid formula) is kept as a back-up that always finds the numbers, a map says which rule to use, and a summary page lists them all. Every answer can be checked by baking it again: expand it and see if you get back where you started.

Goes with **Lecture 2** (factorising, difference of two squares, perfect squares, cubes).

What "factor" means (numbers first)

A **factor** is something you multiply by. $12 = 3 \times 4$, so 3 and 4 are factors of 12 (Chapter 1). **Factorising** means rewriting something **as a multiplication**.

Try it with a sum of numbers: $6 + 9$. Both numbers are in the 3 times table:

$$6 + 9 = 3 \times 2 + 3 \times 3$$ ← *split each into 3 × something*

$$= 3 \times (2 + 3)$$ ← *the 3 is shared, so pull it out front*

$$= 3 \times 5 = 15$$ ← *same as 6 + 9 ✓*

That's all factorising is: spot what every term **shares**, write it once outside a bracket, and put what's **left over** inside. Your slides do the same with fractions:

$$\frac{2}{7} + \frac{2}{9} = 2\left(\frac{1}{7} + \frac{1}{9}\right)$$ ← *each term is 2 × something*

Common factors: break every term into atoms

PROCEDURE: taking out a common factor (always the first step)

1. Find the **highest common factor** of every term: the biggest number that divides all the numbers, times each letter to the **lowest** power it appears in every term.
2. Write it once outside a bracket. Inside, write what is left of each term (each term divided by it).
3. Check: the terms inside the bracket share nothing more, and expanding gives the question back.

Lecture 2 example: $3x^2 + 6x$. Break each term into the smallest things multiplied together (its "atoms"). Numbers split into primes, and x^2 splits into $x \times x$:

$$3x^2 = 3 \times x \times x$$

$$6x = 2 \times 3 \times x$$

highlighted = in BOTH rows (3 and x)

shared atoms go outside: 3 × x = 3x. left over: x (top row) and 2 (bottom row)

$$3x^2 + 6x = 3x\,(x + 2)$$ ← *outside: shared; inside: leftovers*

Bake it again to check: $3x \times x = 3x^2$ and $3x \times 2 = 6x$ ✓. And with a number, $x = 2$: $12 + 12 = 24$ and $6 \times 4 = 24$ ✓.

Take out the biggest shared thing. $3x^2 + 6x = 3(x^2 + 2x)$ is true, but the bracket still has a shared x inside, so you're not finished. Keep going until the terms in the bracket share nothing.

Three terms, same idea (your slides): $3x + 4qx + 12wx$. Every term has an x atom; the numbers 3, 4, 12 share nothing (4 isn't a multiple of 3). So only x comes out:

$$3x + 4qx + 12wx = x\,(3 + 4q + 12w)$$ ← *x × 3, x × 4q, x × 12w*

Why bother? Lecture 2 asks you to make x the subject of $3x + xy = z$. The x is stuck in two places; taking it out glues the two copies into one, and then one division finishes the job:

$$3x + xy = z \;\Rightarrow\; x(3+y) = z \;\Rightarrow\; x = \frac{z}{3+y}$$

one x, then divide

More common factors: chunks and pairs

Lecture 2: $(x+y)^3 + 7(x+y)^2$. It looks frightening, but $(x+y)$ appears whole every time. Cover it with a box and call it $\square$:

$$\square^3 + 7\,\square^2$$ ← *□ is just a name for (x + y)*

$$= \square^2(\square + 7)$$ ← *atoms: □ 3 = □ × □ × □; shared: □ 2*

$$= (x+y)^2(x+y+7)$$ ← *take the box off again*

Four terms: group them in pairs. The slide's example should read $x^2 + 2y + 2x^3 + 4xy$. Look at the last two terms: $2x^3 + 4xy$ share $2x$.

$$(x^2 + 2y) + (2x^3 + 4xy)$$ ← *two pairs*

$$= (x^2 + 2y) + 2x\,(x^2 + 2y)$$ ← *the second pair: 2x × (x² + 2y)*

$$= (x^2 + 2y)(1 + 2x)$$ ← *(x² + 2y) is shared; the first pair is 1 × it*

Slide check. The slide prints $2y^2$ instead of $2y$ and finishes with $(x^2+2y)(x^2+2y+2x)$. Test with $x = y = 1$: the slide's start is $1+2+2+4 = 9$, but its answer is $3 \times 5 = 15$ ✗. The version above gives $3 \times 3 = 9$ ✓.

Three terms: find two numbers that add and multiply

RULE: factorising x² + bx + c (the number in front of x² is 1)

Find two numbers that **ADD to b** and **MULTIPLY to c** . Call them p and q.

Then $x^2 + bx + c = (x + p)(x + q)$.

Signs: if c is **positive**, both numbers have the same sign as b. If c is **negative**, one is positive and one is negative, and the bigger one (ignoring signs) takes the sign of b.

Search the **factor pairs of c** (the multiply condition is easier to list), and test each pair's sum.

Why it works: expand $(x + p)(x + q) = x^2 + px + qx + pq = x^2 + (p + q)x + pq$. The middle number is the **sum** and the last number is the **product**. Lecture 2 says it in words: "the 1 and 3 are each multiplied by x, giving $x + 3x = 4x$; the 1 and 3 are multiplied together to give 3."

Example 1: $x^2 + 7x + 12$. Need: add to 7, multiply to 12.

$$12 = 1 \times 12, \quad 2 \times 6, \quad 3 \times 4 \leftarrow \text{factor pairs of 12}$$

$$1 + 12 = 13, \quad 2 + 6 = 8, \quad 3 + 4 = 7 \checkmark \leftarrow \text{test each sum: 3 and 4 work}$$

$$x^2 + 7x + 12 = (x + 3)(x + 4)$$

Example 2: $x^2 - x - 6$. Here $b = -1$ and $c = -6$: the numbers have opposite signs (c is negative) and the bigger one is negative (b is negative).

$$-6 = 1 \times (-6), \quad 2 \times (-3), \quad (-1) \times 6, \quad (-2) \times 3$$

pairs that multiply to −6

$$1 - 6 = -5, \quad 2 - 3 = -1 \checkmark \leftarrow \text{2 and −3 add to −1}$$

$$x^2 - x - 6 = (x + 2)(x - 3)$$

Bake it to check: $(x + 2)(x - 3) = x^2 - 3x + 2x - 6 = x^2 - x - 6$ ✓.

PROCEDURE: factorising $ax^2 + bx + c$ when a is not 1: split the middle

1. Take out any common factor first. Then work out $a \times c$.
2. Find two numbers that **MULTIPLY to ac** and **ADD to b**.
3. **Split the middle term** bx into two x terms using those numbers.
4. **Factorise in pairs**: a common factor from the first two terms, and from the last two.
5. The two brackets now match: take that bracket out as a common factor.

Example 3: $6x^2 + 7x - 3$. So $a = 6$, $b = 7$, $c = -3$.

$$ac = 6 \times (-3) = -18$$ ← 1. a × c

$$9 \times (-2) = -18, \quad 9 + (-2) = 7$$ ← 2. multiply to −18, add to 7

$$6x^2 + 9x - 2x - 3$$ ← 3. split 7x into 9x − 2x

$$= 3x(2x + 3) - 1(2x + 3)$$ ← 4. factorise each pair

$$= (2x + 3)(3x - 1)$$ ← 5. (2x + 3) is shared

In step 4, take out −1 (not +1) from $-2x - 3$, so that the second bracket matches the first: $-2x - 3 = -1(2x + 3)$. If the brackets don't match, check the signs or swap the order of the two middle terms.

✗ Common mistakes

✗ $x^2 + 5x + 6 = (x + 6)(x + 1)$ ✓ $x^2 + 5x + 6 = (x + 2)(x + 3)$

6 and 1 multiply to 6 but add to 7: the numbers must do BOTH

✗ $x^2 - 2x - 8 = (x + 2)(x + 4)$ ✓ $x^2 - 2x - 8 = (x + 2)(x - 4)$

c is negative, so one number is negative; the bigger one takes the sign of b

✗ $2x^2 + 8x + 6 = (x + 1)(x + 3)$ ✓ $2x^2 + 8x + 6 = 2(x + 1)(x + 3)$

don't lose the common factor you took out

When the search is slow (big numbers, or no whole numbers work), the next section gives a formula that **computes** the two numbers directly instead of searching for them.

Another way for three terms: the grid formula

This is an **alternative** to the search. It finds the very same pair of numbers (they add to b and multiply to ac), but by formula, so it also works when the numbers aren't whole. Chapter 11 filled a grid from its edges. Factorising $ax^2 + bx + c$ runs it **backwards**: the boxes are the question, the edges are the answer, and the edges have a formula:

THE GRID FORMULA for $ax^2 + bx + c$

$\times$	ax	$+m$
x	ax^2	$m\,x$
$+\dfrac{n}{a}$	$n\,x$	c

box = side edge × top edge

1. $\Delta = b^2 - 4ac$
2. $m,\ n = \dfrac{b \pm \sqrt{\Delta}}{2}$
3. $\left(x + \dfrac{n}{a}\right)(ax + m)$

Δ is the Greek D, "delta". The two numbers m and n are **the pair**. (Take out any common factor first.)

Worked example: $2x^2 + 7x + 3$. Read off the letters: $a = 2$, $b = 7$, $c = 3$.

$$\Delta = 7^2 - 4 \times 2 \times 3 = 49 - 24 = 25 \quad \leftarrow \text{1. } 25 = 5^2$$

$$m = \frac{7+5}{2} = 6, \qquad n = \frac{7-5}{2} = 1 \quad \leftarrow \text{2. the pair}$$

3. Put the numbers into the grid, in the places the letters were:

$\times$	$2x$	$+6$
x	$2x^2$	$+6x$
$+\dfrac{1}{2}$	$+x$	$+3$

Top edges: $ax = 2x$ and $m = 6$.

Side edges: x and $\frac{n}{a} = \frac{1}{2}$.

Boxes: $2x \times x = 2x^2$, then $6x + x = 7x$, then $\frac{1}{2} \times 6 = 3$. That's the question ✓

$$2x^2 + 7x + 3 = \left(x + \tfrac{1}{2}\right)(2x + 6) \quad \leftarrow \text{read the edges}$$

$$= (2x + 1)(x + 3) \quad \leftarrow \text{clear the fraction: } \times 2 \text{ and } \div 2$$

Doubling one bracket and halving the other changes nothing, like $3 \times 4 = 6 \times 2$. **Bake it to check:** $(2x+1)(x+3) = 2x^2+6x+x+3 = 2x^2+7x+3$ ✓.

When $a = 1$ there's no fraction: the answer is just $(x+n)(x+m)$. For $x^2+7x+12$, $\Delta = 49-48 = 1$ and the pair is $\frac{7\pm1}{2} = 4, 3$, so it's $(x+3)(x+4)$.

> ✎ **YOUR GO (pencil first)**
>
> *$3x^2 + 10x + 8$. Work out Δ, then the pair, then put them into the grid formula and clear the fraction.*

Answer: $\Delta = 100-96 = 4$, pair $\frac{10\pm2}{2} = 6, 4$. Top edges $3x$ and 6, side edges x and $\frac{4}{3}$. So $\left(x+\frac{4}{3}\right)(3x+6)$; × 3 and ÷ 3 gives $(3x+4)(x+2)$.

Why the grid formula always works

Multiply the card's edges back out, box by box. Every box comes out as the question, whatever a, b and c are:

$$x \times ax = ax^2 \longleftarrow \text{top-left ✓}$$

$$mx + nx = (m+n)\,x = bx \longleftarrow \text{middle ✓ : the } \sqrt{\Delta} \text{ parts cancel}$$

$$\frac{n}{a} \times m = \frac{mn}{a} = \frac{ac}{a} = c \longleftarrow \text{bottom-right ✓ (needs } mn = ac\text{)}$$

And $mn = ac$ is where Δ comes in. Multiply the pair; the top is a difference of two squares (Chapter 11):

$$mn = \frac{(b+\sqrt{\Delta})(b-\sqrt{\Delta})}{4} = \frac{b^2-\Delta}{4} = \frac{4ac}{4} = ac$$

because $\Delta = b^2 - 4ac$

That's the whole reason Δ is b^2-4ac: it's the number that makes the corner come out as c. Chapter 15 sets each bracket to zero, $x = -\frac{n}{a}$ or $x = -\frac{m}{a}$, and that turns into the quadratic formula.

What Δ tells you before you start

Step 1 sorts every quadratic before you touch the grid:

quadratic	Δ = b² − 4ac	√Δ	pair	
$2x^2 + 7x + 3$	$49 - 24 = 25$	5	$6,\ 1$	*a square: whole brackets*
$x^2 + 4x + 4$	$16 - 16 = 0$	0	$2,\ 2$	*zero: (x + 2) twice*
$x^2 + 4x + 2$	$16 - 8 = 8$	$\sqrt{8}$	$2 \pm \sqrt{2}$	*not a square: square roots*
$6x^2 + 4x + 3$	$16 - 72 = -56$	–	none	*negative: no brackets*

Not a square: the grid formula still works; the edges just contain square roots. There are no whole-number brackets, so we say it **doesn't factorise with whole numbers** (the add-and-multiply search finds nothing). With square roots allowed, it does:

$$x^2 + 4x + 2 = (x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$$

pair 2 ± √2, a = 1

Negative: no number squared is negative, so $\sqrt{\Delta}$ doesn't exist: no pair, **no brackets at all**. You haven't made a mistake; there's nothing to find. Exam questions are picked so that Δ is a perfect square. Chapter 15 shows the three cases as graphs that cross the x-axis twice, touch it once, or miss it.

Minus signs: keep each sign with its number

$6x^2 - x - 2$: $a = 6$, $b = -1$, $c = -2$. (Split the middle would search for two numbers that multiply to $ac = -12$ and add to -1: that's 3 and -4.) The grid formula's steps:

$$\Delta = (-1)^2 - 4 \times 6 \times (-2) = 1 + 48 = 49$$ ← 1. 49 = 7²

$$m = \frac{-1+7}{2} = 3, \qquad n = \frac{-1-7}{2} = -4$$ ← 2. the pair

$\times$	$6x$	$+3$
x	$6x^2$	$+3x$
$-\frac{2}{3}$	$-4x$	-2

3. **Top:** $6x$ and $+3$. **Side:** x and $\frac{n}{a} = \frac{-4}{6} = -\frac{2}{3}$.

Boxes: $3x - 4x = -x$ and $-\frac{2}{3} \times 3 = -2$ ✓. You never chose a minus sign: the formula handed it to you.

$$6x^2 - x - 2 = \left(x - \tfrac{2}{3}\right)(6x + 3)$$ ← read the edges

$$= (3x - 2)(2x + 1)$$ ← clear the fraction: × 3 and ÷ 3

Same answer, different look. $(3x - 2)(2x + 1)$ and $(2x + 1)(3x - 2)$ are the same, like $3 \times 4 = 4 \times 3$. Lecture 2 writes $2x^2 + 5x + 3 = 2(x + 1.5)(x + 1)$ and says it "isn't the only solution": moving the 2 into the first bracket gives $(2x + 3)(x + 1)$, the whole-number version an examiner expects.

The grid's two rows, written in a line, are exactly **split the middle** from the start of this part (the pair 6, 1 multiplies to $ac = 6$ and adds to $b = 7$):

$$2x^2 + 7x + 3 = (2x^2 + 6x) + (x + 3)$$ ← the grid's two rows

$$= 2x(x + 3) + 1(x + 3) = (2x + 1)(x + 3)$$ ← each row's side edge, then the shared bracket

Difference of two squares: cut and slide

RULE: difference of two squares

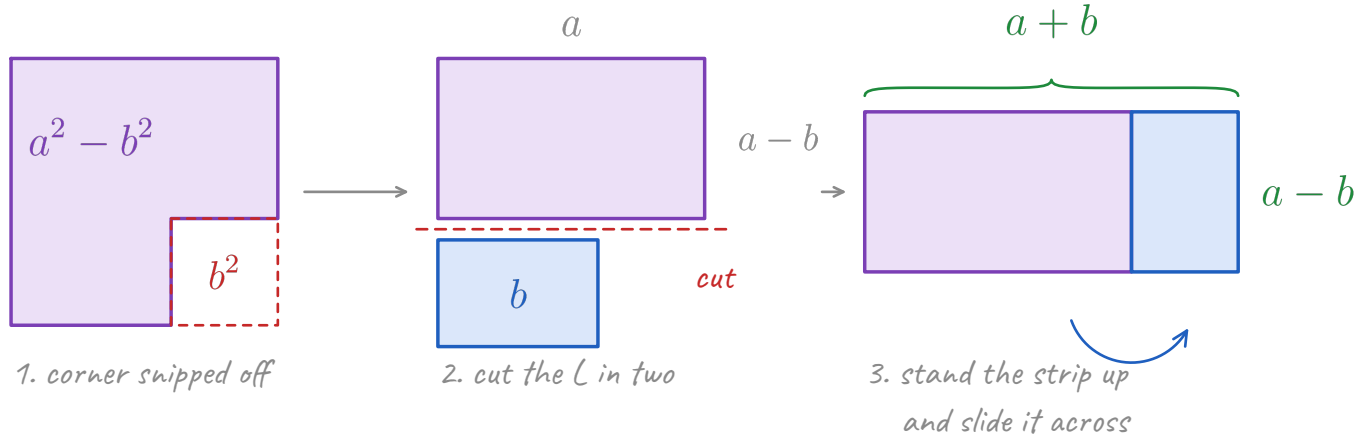
$a^2 - b^2 = (a - b)(a + b)$.

Use it when there are exactly **two** terms, a **minus** between them, and **each term is a square** (of a number, a letter, or both). Ask "a square of what?" to find a and b.

Take out a common factor first if there is one: $2x^2 - 18 = 2(x^2 - 9)$.

A sum of two squares, $a^2 + b^2$, does not factorise.

Why it's true: take a square of paper, a by a, and snip a small b by b square off one corner. What's left has area $a^2 - b^2$. Now watch:



Same paper, now a rectangle $(a + b)$ wide and $(a - b)$ tall:

$$a^2 - b^2 = (a - b)(a + b)$$

How to spot it: exactly **two** terms, a **minus** between them, and each term is a square. Then ask "a square of what?"

$$x^2 - 9 = x^2 - 3^2 = (x - 3)(x + 3)$$ ← 9 is 3 squared

$$x^2 - 4y^2 = x^2 - (2y)^2 = (x - 2y)(x + 2y)$$

$4y^2$ is $(2y)$ squared

It's also just the method with $b = 0$: for $x^2 - 9$, $\Delta = 0 + 36 = 36$ and the pair is ± 3. Spotting the two squares is simply faster. (The slide says "Expand: $x^2 - 4y^2$" but then factorises it; "factorise" is the right word.)

Party trick: $51 \times 49 = (50+1)(50-1) = 50^2 - 1^2 = 2499$, no calculator needed. **Warning:** $a^2 + b^2$ (a **plus**) does not factorise this way.

Perfect squares: a bracket times itself

RULE: perfect squares

$a^2 + 2ab + b^2 = (a+b)^2$ and $a^2 - 2ab + b^2 = (a-b)^2$.

How to spot it: the **first and last terms are squares** (a^2 and b^2, last term positive) and the **middle term is 2 × a × b**. The sign of the middle term goes in the bracket.

If the middle isn't $2ab$, it is not a perfect square: use the add-and-multiply rule instead.

It's the add-and-multiply rule with the two numbers equal: for $x^2 + 6x + 9$ the numbers are 3 and 3 (add to 6, multiply to 9), so both brackets are the same.

$$x^2 + 6x + 9 = x^2 + 2 \cdot x \cdot 3 + 3^2 = (x+3)^2$$

a = x, b = 3; middle 2·x·3 = 6x ✓

$$4x^2 - 20x + 25 = (2x)^2 - 2 \cdot 2x \cdot 5 + 5^2 = (2x-5)^2$$

a = 2x, b = 5; middle 20x ✓

Find the missing number. "What must k be for $x^2 + 10x + k$ to be a perfect square?" The bracket must be $(x+5)^2$, because $2 \times x \times 5 = 10x$. So:

$$k = 5^2 = 25$$

half of 10, squared: $(x+5)^2 = x^2 + 10x + 25$

✗ Common mistakes

✗ $x^2 - 9 = (x-3)^2$ ✓ $x^2 - 9 = (x-3)(x+3)$

$(x-3)^2 = x^2 - 6x + 9$: two terms with a minus is a difference of squares

✗ $x^2 + 9 = (x+3)(x-3)$ ✓ $x^2 + 9$ doesn't factorise

$(x+3)(x-3) = x^2 - 9$; a sum of squares has no brackets

✗ $x^2 + 4x + 16 = (x+4)^2$

the middle must be 2 × 4 × x = 8x

✓ $(x+4)^2 = x^2 + 8x + 16$

Cubes: a bigger grid

Lecture 2 lists two cube patterns. The same grid proves them, with three columns instead of two:

$(a-b)(a^2+ab+b^2)$

×	a^2	$+ab$	$+b^2$
a	a^3	$+a^2b$	$+ab^2$
$-b$	$-a^2b$	$-ab^2$	$-b^3$

Same colour, opposite sign: the **blue** boxes cancel and the **red** boxes cancel. Only the two **green** corners are left: $a^3 - b^3$.

$$a^3 - b^3 = (a-b)(a^2+ab+b^2)$$ ← a = 2, b = 1: 8 − 1 = 1 × 7

Swap b for $-b$ everywhere and the same grid gives the plus version:

$$a^3 + b^3 = (a+b)(a^2-ab+b^2)$$ ← a = 2, b = 1: 8 + 1 = 3 × 3

Remembering the signs, "SOAP": the first bracket has the **S**ame sign as the question, the middle of the second bracket is **O**pposite, and the last term is **A**lways **P**ositive.

Using it: spot two cubes, find a and b, and fill in the pattern.

$$8x^3 - 27 = (2x)^3 - 3^3$$ ← $8x^3$ is (2x) cubed, 27 is 3 cubed

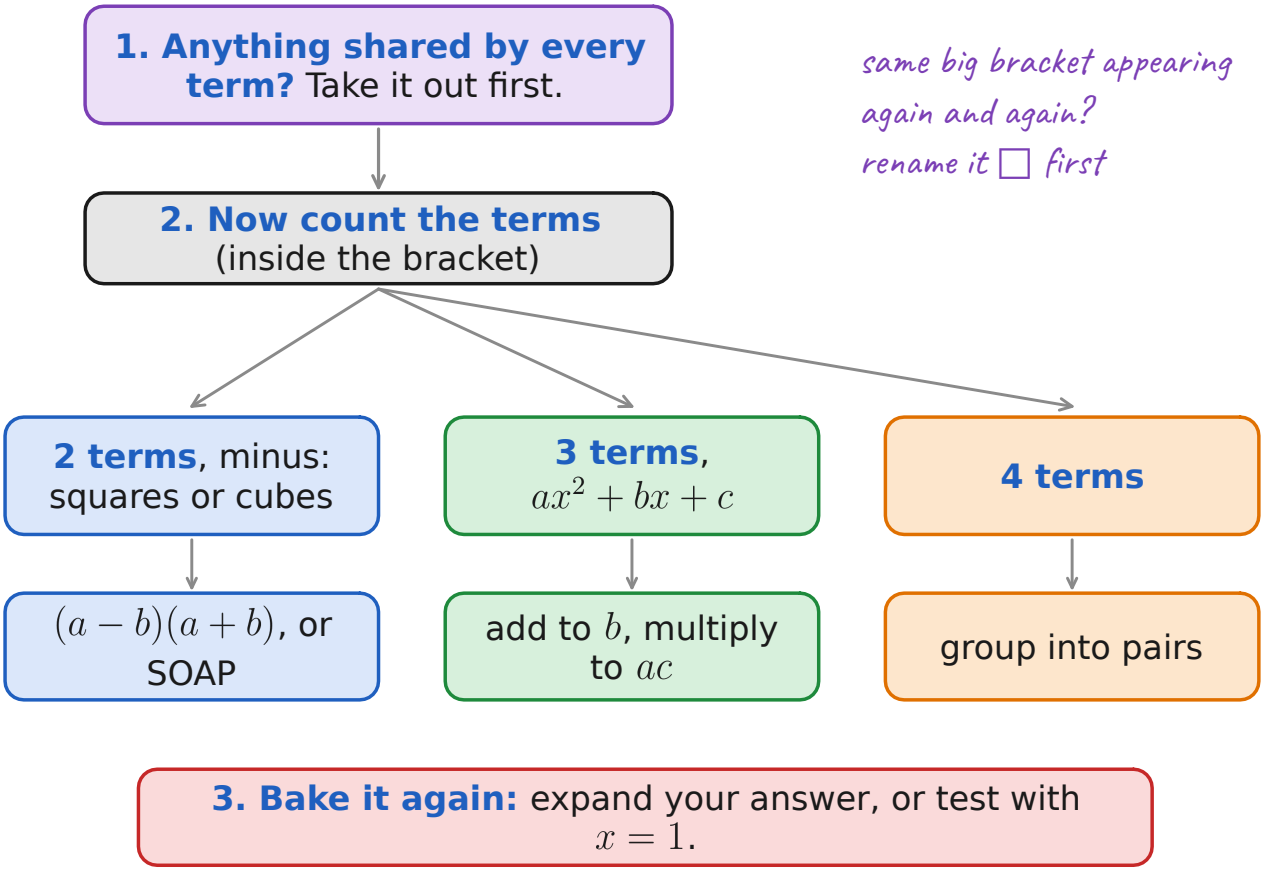
$$= (2x-3)(4x^2+6x+9)$$ ← a = 2x, b = 3 in the minus pattern

The map: which tool do I use?

PROCEDURE: factorising anything

1. **Always take out common factors first.** Look for a number, letter or bracket in every term.
2. Count the terms left in the bracket.
3. **Two terms:** two squares with a minus give $(a-b)(a+b)$; two cubes, plus **or** minus, use SOAP. A sum of two squares doesn't factorise.
4. **Three terms:** a perfect square $(a \pm b)^2$? If not: x^2+bx+c, find two numbers adding to b and multiplying to c; ax^2+bx+c, split the middle (multiply to ac, add to b). Stuck? Use the grid formula, or Δ to see if whole-number brackets exist at all.
5. **Four terms:** group into pairs and take out the shared bracket.
6. Check each bracket can't be factorised further, then expand to check the whole answer.

The same procedure as a map. Most questions need two stops: a common factor first, then one of the others.



Factorising rules

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Common factor first	take out the HCF of every term before anything else
x² + bx + c	$(x+p)(x+q): \quad p+q=b, \quad pq=c$ two numbers that **add** to b and **multiply** to c
ax² + bx + c (split the middle)	$pq=ac, \quad p+q=b: \quad ax^2+px+qx+c$ then factorise in pairs and take out the shared bracket
Grid formula (alternative)	$m,\, n=\dfrac{b\pm\sqrt{b^2-4ac}}{2}, \quad \left(x+\dfrac{n}{a}\right)(ax+m)$ computes the same pair; clear any fraction at the end
Does it factorise?	$\Delta=b^2-4ac$ perfect square: whole brackets; not a square: not with whole numbers; negative: not at all
Difference of two squares	$a^2-b^2=(a-b)(a+b)$ a^2+b^2 doesn't factorise
Perfect squares	$a^2\pm 2ab+b^2=(a\pm b)^2$ first and last square, middle $2ab$
Grouping (4 terms)	$ax+ay+bx+by=(a+b)(x+y)$ pairs, then the shared bracket
Difference of cubes	$a^3-b^3=(a-b)(a^2+ab+b^2)$
Sum of cubes	$a^3+b^3=(a+b)(a^2-ab+b^2)$ SOAP: Same, Opposite, Always Positive
Check	expand your answer: you must get the question back

Un-baking: your turn

Answers in grey. Bake each answer again (expand it) to check yourself.

Common factor

1. $4x + 12$ 2. $6x^2 - 9x$ 3. $5ab + 10a$ 4. $2x^3 + 4x^2 + 8x$

1. $4(x + 3)$ 2. $3x(2x - 3)$ 3. $5a(b + 2)$ 4. $2x(x^2 + 2x + 4)$

Three terms: the pair adds to b and multiplies to ac (Δ shows whether it exists)

5. $x^2 + 7x + 12$ 6. $x^2 - x - 20$ 7. $3x^2 + 6x - 24$
8. $2x^2 + 11x + 5$ 9. $6x^2 + 7x + 2$ 10. $4x^2 - 4x - 3$
11. $5x^2 + 13x - 6$

5. $\Delta = 1$, pair 4, 3: $(x + 3)(x + 4)$ 6. $\Delta = 81$, pair 4, −5: $(x - 5)(x + 4)$
7. take out 3 first; $\Delta = 36$, pair 4, −2: $3(x + 4)(x - 2)$ 8. $\Delta = 81$, pair 10, 1: $(2x + 1)(x + 5)$
9. $\Delta = 1$, pair 4, 3: $(2x + 1)(3x + 2)$ 10. $\Delta = 64$, pair 2, −6: $(2x - 3)(2x + 1)$
11. $\Delta = 289$, pair 15, −2: $(5x - 2)(x + 3)$

Whole, square roots, or nothing? (Δ first)

12. $x^2 + 6x + 8$ 13. $x^2 + 6x + 7$ 14. $x^2 + 2x + 3$

12. $\Delta = 4 = 2^2$: pair 4, 2, $(x + 2)(x + 4)$ 13. $\Delta = 8$: pair $3 \pm \sqrt{2}$, $(x + 3 - \sqrt{2})(x + 3 + \sqrt{2})$
14. $\Delta = -8$: no brackets

Difference of two squares

15. $x^2 - 25$ 16. $4x^2 - 49$ 17. $2x^2 - 18$
18. 97×103 (use the trick)

15. $(x - 5)(x + 5)$ 16. $(2x - 7)(2x + 7)$ 17. $2(x - 3)(x + 3)$ 18. $100^2 - 3^2 = 9991$

Grouping and chunks

19. $ax + ay + bx + by$
20. $(x - 1)^2 + 3(x - 1)$
21. Make x the subject: $ax - b = cx$

19. $(a + b)(x + y)$ 20. $(x - 1)(x + 2)$ 21. $x = b/(a - c)$

Cubes

22. $x^3 - 8$ 23. $27x^3 + 1$

22. $(x - 2)(x^2 + 2x + 4)$ 23. $(3x + 1)(9x^2 - 3x + 1)$

Find the unknown

24. $x^2 + kx + 12 = (x + 3)(x + p)$. Find p and k.
25. Find k so that $x^2 + 6x + k$ is a perfect square.
26. $x^2 - 5x + c = (x - 2)(x - q)$. Find q and c.
27. $(x + 4)$ is a factor of $x^2 + x + k$. Find k and the other factor.
28. Find the positive k so that $4x^2 + kx + 9$ is a perfect square.
29. $2x^2 + bx - 3 = (2x - 1)(x + 3)$. Find b.

24. $p = 4$, $k = 7$ ($3p = 12$, $k = 3 + p$) 25. $k = 9$ ($(6 \div 2)^2 = 9$)
26. $q = 3$, $c = 6$ ($2 + q = 5$, $c = 2q$) 27. $k = -12$, $(x - 3)$ ($4 + p = 1$, so $p = -3$; $k = 4p$)
28. $k = 12$ ($(2x + 3)^2$, middle $2 \times 2x \times 3$) 29. $b = 5$ (expand: $6x - x = 5x$)

Chapter 13

Algebraic fractions

nothing new: the same fraction moves, done twice side by side

numbers

$$\frac{2}{3}+\frac{4}{5}=\frac{2\cdot 5+3\cdot 4}{3\cdot 5}$$

letters

$$\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}$$

identical moves, the letters just sit where the numbers were

An **algebraic fraction** is a fraction with letters in it, like $\frac{3}{x}$ or $\frac{x+1}{x-2}$.

That's the only difference from Chapter 2, so the **rules are Chapter 2's rules**: the same three formulas for add/subtract, multiply and divide, and each section states its rule in a box before any example. Many moves are shown in **two columns**: numbers you already trust on the left, the same move with letters on the right.

One new idea does most of the work here: which parts of a fraction are **glued** together (multiplied) and which are only **loose** (added). You may cancel glued parts, and must never cancel loose ones.

Goes with **Lecture 1** (combining fractions) and **Lecture 2** (common mistakes, the 1/x = 2/y − 3/z example).

÷ Letters in a fraction, and the one number it hates

$\frac{x+1}{x-2}$ means **(the whole top) ÷ (the whole bottom)**. The fraction bar acts like a pair of brackets around the top and another around the bottom:

$$\frac{x+1}{x-2} = (x+1) \div (x-2)$$ ← *the bar groups each part*

To get a value, substitute (Chapter 8). With $x = 4$: $\frac{4+1}{4-2} = \frac{5}{2} = 2.5$.

The one number it hates. With $x = 2$ the bottom becomes $2 - 2 = 0$, and dividing by zero means nothing (Chapter 1: share 3 sweets among 0 people?). So this fraction only makes sense when $x \neq 2$. Whenever a letter sits on the bottom, ask: **which value makes the bottom zero?** That value is banned.

$$\frac{3}{x}$$
$x \neq 0$

$$\frac{5}{x+4}$$
$x \neq -4$

$$\frac{x}{(x-1)(x+3)}$$
$x \neq 1,\ x \neq -3$

÷ The three formulas, with letters

Exactly Chapter 2's formulas. Now a, b, c, d can be expressions like $x+1$; put each one in **brackets** when you plug it in.

Operation	Formula	Example with letters
Add / subtract	$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$	$\frac{3}{x} - \frac{2}{y} = \frac{3y-2x}{xy}$
Multiply	$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$	$\frac{x+1}{x} \times \frac{x^2}{x+1} = \frac{(x+1)x^2}{x(x+1)} = x$
Divide	$\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$	$\frac{x}{4} \div \frac{x^2}{6} = \frac{6x}{4x^2} = \frac{3}{2x}$

RULE: what's new with letters

Brackets: in $ad - bc$, b and c may be whole expressions; the minus applies to all of bc.

Banned values: any value that makes a bottom zero is not allowed.

Tidying the answer: factorise top and bottom, then cancel common **factors** (never terms): $\frac{x^2+3x}{x+3} = \frac{x(x+3)}{x+3} = x$.

÷ Cancelling: only glued parts come off

RULE: simplifying (cancelling)

Factorise the top and the bottom completely first.

Cancel a factor only if it **multiplies the whole top and the whole bottom**. Cancel factors, **never terms**: in $\frac{x+3}{3}$ the 3s do not cancel.

A banned value of the original fraction stays banned after cancelling.

Simplifying a fraction means dividing top and bottom by the **same thing**. Break both into atoms (as in Chapter 12) and cross out matching pairs:

with numbers

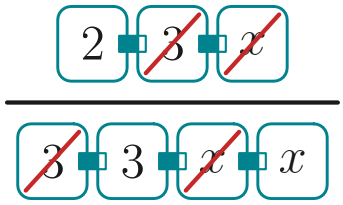
$$\frac{12}{18} = \frac{2 \cdot \cancel{6}}{3 \cdot \cancel{6}} = \frac{2}{3}$$

with letters

$$\frac{6x}{9x^2} = \frac{2 \cdot \cancel{3}\cancel{x}}{3x \cdot \cancel{3}\cancel{x}} = \frac{2}{3x}$$

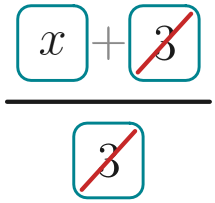
the same 6 (or 3x) on top and bottom

Here's the **glue** picture. The teal bars mean "multiplied", so those atoms are stuck to the whole top or the whole bottom. Dividing top and bottom by one of them is fine:



every atom is glued (x)
so a matching pair can be
lifted off: left with 2/(3x) ✓

Now a **loose** part. Can the 3s in $\frac{x+3}{3}$ cancel?



NO. The top 3 is only added on (loose).
It isn't multiplying the whole top.
Test x = 6: (6 + 3)/3 = 3, but "x" = 6 ✗

Fix: factorise first, to turn loose parts into glued ones.

with numbers

with letters

$$\frac{6+9}{3} = \frac{3(2+3)}{3} = 5$$

$$\frac{15}{20} = \frac{3 \cdot \cancel{5}}{4 \cdot \cancel{5}}$$

$$= \frac{3}{4}$$

$$\frac{x^2+3x}{x+3} = \frac{x\cancel{(x+3)}}{\cancel{(x+3)}} = x$$

factorise the top: now (x + 3) is glued on

$$\frac{x^2-9}{x^2+x-6} = \frac{(x-3)\cancel{(x+3)}}{(x-2)\cancel{(x+3)}}$$

top: squares; bottom: Δ = 25, pair 3, −2

$$= \frac{x-3}{x-2}$$

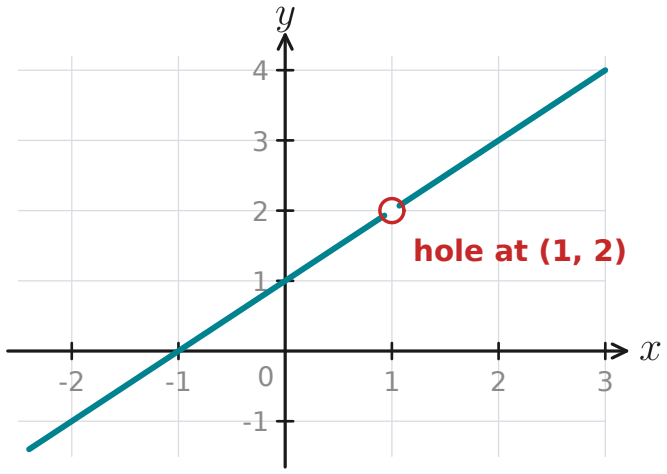
Rule of thumb: if there's a + or − showing on the top or bottom, factorise before you cancel anything.

Cancelling doesn't un-ban a number. Take $\frac{x^2-1}{x-1}$. The top is a difference of squares, so it cancels:

$$\frac{x^2-1}{x-1} = \frac{(x+1)\cancel{(x-1)}}{\cancel{(x-1)}} = x+1$$

✎ YOUR GO (pencil first)

The original fraction still has x − 1 on the bottom. Which x is banned? On the line y = x + 1, mark the one point the fraction can't give.



x	(x² − 1) ÷ (x − 1)
0.9	1.9
0.99	1.99
1	**0 ÷ 0: no value**
1.01	2.01
1.1	2.1

creeps up to 2 from both sides, but never actually lands on it

So the honest answer is $\frac{x^2-1}{x-1} = x+1$ **for** $x \neq 1$. The graph is the whole straight line with **one point punched out**, drawn as an open circle. Cancelling hides the $x-1$, but the ban stays with the original fraction.

÷ Back-to-front brackets: a − b = −(b − a)

RULE: reversing a subtraction

$a - b = -(b - a)$. Swapping the order of a subtraction only changes the sign: $5 - 3 = 2$ and $3 - 5 = -2$.

So $\frac{a-b}{b-a} = -1$ (as long as $a \neq b$).

When a top and bottom bracket are the same apart from order, take out -1 from one of them, then cancel.

with numbers

$$\frac{7-3}{3-7} = \frac{4}{-4} = -1$$

with letters

$$\frac{x-3}{3-x} = \frac{-(3-x)}{3-x} = -1$$

the top is −1 × the bottom

Example: $\frac{2-x}{x^2-4}$. The bottom factorises to $(x-2)(x+2)$, and $2 - x = -(x - 2)$:

$$\frac{2-x}{x^2-4} = \frac{-(x-2)}{(x-2)(x+2)}$$ ← *take out −1 on top; factorise the bottom*

$$= \frac{-1}{x+2} = -\frac{1}{x+2}$$ ← *cancel (x − 2); x ≠ 2, x ≠ −2*

Test $x = 0$: $\frac{2}{-4} = -0.5$ and $-\frac{1}{2} = -0.5$ ✓.

✗ Common mistakes

✗ $\frac{x+3}{3} = x$

3 is not a factor of the whole top: no cancelling

✓ $\frac{x+3}{3} = \frac{x}{3} + 1$

✗ $\frac{x^2+x}{x} = x^2 + 1$

factorise first: x is a factor of the WHOLE top

✓ $\frac{x(x+1)}{x} = x + 1$

✗ $\dfrac{x-3}{3-x}=1$

reversed brackets differ by a factor of −1

✓ $\dfrac{x-3}{3-x}=-1$

÷ Multiplying and dividing: the easy pair

RULE: multiplying and dividing algebraic fractions

Multiply: factorise every top and bottom, cancel any factor that appears on a top and a bottom (even in different fractions), then multiply tops together and bottoms together.

Divide: keep the first fraction, flip the second (its reciprocal), then multiply as above. **Flip before you cancel.**

Multiply: tops times tops, bottoms times bottoms. Divide: flip the second fraction (its reciprocal) and multiply. Exactly Chapter 2:

with numbers

$$\frac{2}{3}\times\frac{4}{5}=\frac{8}{15}$$

$$\frac{2}{3}\div\frac{4}{5}=\frac{2}{3}\times\frac{5}{4}$$

with letters

$$\frac{a}{b}\times\frac{c}{d}=\frac{ac}{bd}$$

tops × tops, bottoms × bottoms

$$\frac{a}{b}\div\frac{c}{d}=\frac{a}{b}\times\frac{d}{c}$$

flip the second, then multiply

A full example in tiny steps: $\frac{2x}{3}\times\frac{9}{4x^2}$

$$\frac{2x\cdot 9}{3\cdot 4x^2}$$ ← tops together, bottoms together

$$=\frac{18x}{12x^2}$$ ← 2 × 9 = 18; 3 × 4x² = 12x²

$$=\frac{3\cdot \cancel{6x}}{2x\cdot \cancel{6x}}=\frac{3}{2x}$$ ← cancel the glued 6x

Dividing: $\frac{x}{4}\div\frac{x^2}{6}$

$$= \frac{x}{4} \times \frac{6}{x^2} = \frac{6x}{4x^2}$$ ← flip the second fraction and multiply

$$= \frac{3}{2x}$$ ← cancel 2x from top and bottom

Test with $x = 3$: $\frac{3}{4} \div \frac{9}{6} = 0.75 \div 1.5 = 0.5$, and $\frac{3}{6} = 0.5$ ✓.

Dividing when you must factorise first: $\frac{x^2-9}{x+2} \div \frac{x-3}{x^2+2x}$

$$= \frac{x^2 - 9}{x + 2} \times \frac{x^2 + 2x}{x - 3}$$ ← flip the second, then ×

$$= \frac{(x-3)(x+3)}{x+2} \times \frac{x(x+2)}{x-3}$$ ← factorise every top and bottom

$$= \frac{\cancel{(x-3)}(x+3)}{\cancel{x+2}} \times \frac{\cancel{x(x+2)}}{\cancel{x-3}} = x(x+3)$$

cancel across the ×

Test $x = 1$: $\frac{-8}{3} \div \frac{-2}{3} = 4$ and $1 \times 4 = 4$ ✓.

÷ Adding and subtracting

RULE: adding and subtracting

$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$, with each top and bottom in brackets. This always works.

If the bottoms share a factor (like x and $2x$), the answer will contain that factor to cancel at the end. The optional procedure below avoids that by using a smaller bottom.

PROCEDURE: optional: the lowest common denominator (LCD)

1. **Factorise every bottom.**
2. The **LCD** is made of every different factor, each taken the **most times it appears in any one bottom**. (If the bottoms share no factor, the LCD is just the bottoms multiplied.)
3. Multiply the top and bottom of each fraction by whatever its bottom is **missing** from the LCD.
4. Add or subtract the tops (second top in brackets when subtracting). Keep the LCD.
5. Factorise the new top and cancel if you can.

with numbers

$$\frac{1}{3} + \frac{1}{4} = \frac{1 \cdot 4 + 3 \cdot 1}{3 \cdot 4}$$

$$= \frac{7}{12}$$

with letters

$$\frac{1}{x} + \frac{1}{y} = \frac{1 \cdot y + x \cdot 1}{xy}$$

ad + bc over bd

$$= \frac{y + x}{xy}$$

That's Lecture 1's formula. (Its subtraction slide prints a + where the − should be; the appendix lists it.) When the bottoms **share a factor** it gives a bigger bottom than needed and you cancel at the end; the LCD is the optional way to skip that.

Shared factor: $\frac{1}{x} + \frac{1}{2x}$. The bottoms are x and $2 \times x$. The factors are 2 (at most once) and x (at most once), so the LCD is $2x$, not $2x^2$:

$$\frac{1}{x}+\frac{1}{2x}=\frac{1\times 2}{x\times 2}+\frac{1}{2x}$$ ← x is missing a 2

$$=\frac{2+1}{2x}=\frac{3}{2x}$$ ← same bottoms: add the tops

(Multiplying the bottoms gives $\frac{2x+x}{2x^2}=\frac{3x}{2x^2}$, which still has to be cancelled down to $\frac{3}{2x}$.)

Factorise first: $\frac{1}{x+1}+\frac{2}{x^2-1}$. The second bottom is a difference of two squares, $(x-1)(x+1)$, and it already contains $x+1$:

$$\text{LCD}=(x-1)(x+1)$$ ← (x + 1) at most once, (x − 1) at most once

$$\frac{1(x-1)}{(x+1)(x-1)}+\frac{2}{(x-1)(x+1)}$$ ← the first is missing (x − 1)

$$=\frac{x-1+2}{(x-1)(x+1)}=\frac{x+1}{(x-1)(x+1)}$$ ← add the tops

$$=\frac{1}{x-1}$$ ← cancel (x + 1); x ≠ ±1

Test $x=2$: $\frac{1}{3}+\frac{2}{3}=1$ and $\frac{1}{1}=1$ ✓.

Bottoms with brackets: $\frac{2}{x+1}+\frac{3}{x-2}$. Treat each bottom as one glued lump.

$$\frac{2(x-2)}{(x+1)(x-2)}+\frac{3(x+1)}{(x+1)(x-2)}$$ ← each × the other bottom

$$=\frac{2x-4+3x+3}{(x+1)(x-2)}$$ ← add the tops; expand

$$=\frac{5x-1}{(x+1)(x-2)}$$ ← collect like terms

Test $x=0$: $\frac{2}{1}+\frac{3}{-2}=0.5$ and $\frac{-1}{(1)(-2)}=0.5$ ✓.

÷ The minus-sign trap

RULE: subtracting a fraction

The minus applies to the **whole top** of the fraction after it. Put that top in **brackets**, then multiply out: $-(x-1) = -x+1$.

When you subtract a fraction, the minus applies to its **whole top**. Keep the top in a bracket until the minus has been multiplied in:

$\frac{x+3}{2} - \frac{x-1}{3}$

$$= \frac{3(x+3) - 2(x-1)}{6}$$ ← common bottom 6; brackets kept!

$$= \frac{3x + 9 - 2x + 2}{6}$$ ← −2 × −1 = +2 (the trap is writing −2)

$$= \frac{x + 11}{6}$$

Test $x = 1$: $\frac{4}{2} - \frac{0}{3} = 2$ and $\frac{12}{6} = 2$ ✓. Writing $-2x - 2$ would give $\frac{x+7}{6}$, which is $\frac{8}{6}$ at $x = 1$ ✗.

÷ Whole numbers, and fractions inside fractions

RULE: a whole number (or expression) with a fraction

Write the whole number over the other fraction's bottom: $1 = \frac{x+1}{x+1}$, $2 = \frac{2x}{x}$. Then add or subtract as usual.

with numbers

$$1 - \frac{3}{4} = \frac{4}{4} - \frac{3}{4} = \frac{1}{4}$$

with letters

$$1 - \frac{3}{x+1} = \frac{x+1}{x+1} - \frac{3}{x+1}$$

1 written with the same bottom

$$= \frac{x+1-3}{x+1} = \frac{x-2}{x+1}$$

PROCEDURE: a fraction inside a fraction (a complex fraction)

1. Find the LCD of **all** the small fractions inside.
2. Multiply the **top and the bottom of the big fraction** by it. Every small fraction clears.
3. Simplify: factorise and cancel.

Example: $\frac{1+\frac{1}{x}}{1-\frac{1}{x}}$. The only small bottom is x, so multiply top and bottom by x:

$$\frac{\left(1+\frac{1}{x}\right)\times x}{\left(1-\frac{1}{x}\right)\times x} = \frac{x+1}{x-1}$$ ← *x × 1 = x and x × 1/x = 1*

Example: $\frac{\frac{1}{x}+\frac{1}{y}}{\frac{1}{xy}}$. The LCD of x, y and xy is xy:

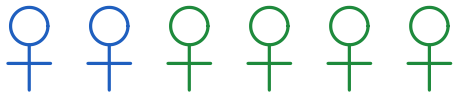
$$\frac{\left(\frac{1}{x}+\frac{1}{y}\right)\times xy}{\frac{1}{xy}\times xy} = \frac{y+x}{1} = x+y$$ ← *each small bottom cancels*

Test $x = 2$, $y = 3$: $\frac{1/2+1/3}{1/6} = \frac{5/6}{1/6} = 5$ and $2+3=5$ ✓.

÷ Splitting a fraction: the top may split, the bottom may not

Lecture 2's "common mistakes (2)" slide warns: $\frac{a}{b+c} \neq \frac{a}{b} + \frac{a}{c}$. Think of sweets:

12 sweets



2 friends arrive, then 4 more

$$\frac{12}{2+4} = \frac{12}{6} = 2 \text{ each} \qquad \frac{12}{2} + \frac{12}{4} = 6 + 3 = 9 \quad \times$$

Nine sweets each would need 54 sweets! Sharing among a **sum** of people is not the same as sharing among each group separately. But splitting the **top** is fine: 12 red + 6 blue sweets among 3 people is 4 red + 2 blue each.

allowed: split the top

$$\frac{a+b}{c} = \frac{a}{c} + \frac{b}{c}$$

not allowed: split the bottom

$$\frac{a}{b+c} \neq \frac{a}{b} + \frac{a}{c}$$

The same slide's other warning is the root version: $\sqrt{a^2+b^2} \neq a+b$. Test $a=3$, $b=4$: $\sqrt{25}=5$, not 7.

÷ The lecture's letters-only example

Lecture 2 asks you to solve $\frac{1}{x} = \frac{2}{y} - \frac{3}{z}$ for z. The heart of it is one subtraction of fractions, which is this chapter's skill. On the way, you meet $\frac{2}{y} - \frac{1}{x}$:

with numbers

$$\frac{2}{5} - \frac{1}{3} = \frac{2 \cdot 3 - 5}{15}$$

with letters

$$\frac{2}{y} - \frac{1}{x} = \frac{2x - y}{xy}$$

a/b − c/d = (ad − bc)/bd

Chapter 14 finishes the job (flip both sides, multiply by 3) and gets $z = \frac{3xy}{2x-y}$. With $x=1$, $y=1$: $z=3$, and indeed $\frac{1}{1} = \frac{2}{1} - \frac{3}{3}$ ✓.

÷ Equations with algebraic fractions

An **equation** has an = sign, so you may multiply **both sides** by the same thing (Part 0's balance: do the same to both sides). That clears every fraction in one go. Chapters 14 and 15 do rearranging and solving in full; this is the fraction part.

PROCEDURE: solving an equation with fractions

1. Write down the **banned values** (anything that makes a bottom zero).
2. Multiply **every term on both sides** by the LCD of all the bottoms.
3. Solve what's left (no fractions now).
4. **Reject any answer that is a banned value**, then check the rest in the original equation.

Example: $\frac{3}{x} + \frac{1}{2} = 2$. Banned: $x = 0$. LCD: $2x$.

$$2x \cdot \frac{3}{x} + 2x \cdot \frac{1}{2} = 2x \cdot 2 \quad \leftarrow \times 2x, \text{ every term}$$

$$6 + x = 4x \;\Rightarrow\; 6 = 3x \;\Rightarrow\; x = 2 \quad \leftarrow \text{solve}$$

Check: $\frac{3}{2} + \frac{1}{2} = 2$ ✓, and 2 is not banned.

Example with a rejected answer: $\frac{x}{x-2} = \frac{2}{x-2} + 3$. Banned: $x = 2$.

$$x = 2 + 3(x - 2) \quad \leftarrow \times (x - 2), \text{ every term}$$

$$x = 3x - 4 \;\Rightarrow\; x = 2 \quad \leftarrow \text{solve}$$

$$x = 2 \text{ is banned: no solution} \quad \leftarrow \text{it makes the bottom 0}$$

Multiplying by $x - 2$ is multiplying by 0 when $x = 2$, which can create a fake answer. That's why step 4 is not optional.

✗ Common mistakes

✗ $\frac{3}{x} + \frac{1}{2} = 2 \;\rightarrow\; 6 + x = 2$ ✓ $6 + x = 4x$

multiply EVERY term by the LCD, including the 2 on the right

Algebraic fraction rules

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Banned values	a fraction is undefined where its bottom is 0; cancelling doesn't remove the ban
Simplify	$\dfrac{x^2+3x}{x+3} = \dfrac{x\cancel{(x+3)}}{\cancel{(x+3)}} = x$ factorise top and bottom first; cancel **factors**, never terms
Back-to-front brackets	$a-b=-(b-a), \qquad \dfrac{a-b}{b-a}=-1$
Add / subtract	$\dfrac{a}{b}\pm\dfrac{c}{d}=\dfrac{ad\pm bc}{bd}$ always works; optional LCD (factorise the bottoms, each factor the most times it appears) gives a smaller bottom
Subtracting	$\dfrac{x+3}{2}-\dfrac{x-1}{3}=\dfrac{3(x+3)-2(x-1)}{6}$ second top in brackets
Multiply	$\dfrac{a}{b}\times\dfrac{c}{d}=\dfrac{ac}{bd}$ multiply across; factorise and cancel first
Divide	$\dfrac{a}{b}\div\dfrac{c}{d}=\dfrac{a}{b}\times\dfrac{d}{c}=\dfrac{ad}{bc}$ flip the second only
Splitting	$\dfrac{a+b}{c}=\dfrac{a}{c}+\dfrac{b}{c}, \qquad \dfrac{a}{b+c}\neq\dfrac{a}{b}+\dfrac{a}{c}$ the top may split, the bottom may not
Whole number with a fraction	$1-\dfrac{3}{x+1}=\dfrac{x+1-3}{x+1}$
Fraction inside a fraction	multiply the big top and big bottom by the LCD of the small fractions
Equations with fractions	× every term by the LCD, solve, **reject banned values**, check

÷ Twin columns: your turn

Answers in grey. Check each by substituting a number such as x = 3.

When is it undefined?

1. $7/(x - 5)$
2. $x/(x^2 - 4)$

1. x = 5 2. x = 2 or x = −2

Cancel (factorise first!)

3. $10x^2/(15x)$
4. $(4x + 8)/4$
5. $(x^2 - 1)/(x + 1)$
6. $(x^2 + 5x + 6)/(x + 2)$
7. $(x + 5)/5$

3. 2x/3 4. x + 2 5. x − 1 6. x + 3 7. can't cancel: it's x/5 + 1

Multiply and divide

8. $(3x/4) \times (8/x^2)$
9. $(x/2) \div (x/10)$
10. $((x + 1)/x) \times (x^2/(x + 1))$

8. 6/x 9. 5 10. x

Add and subtract

11. $1/x + 2/x$
12. $1/x + 1/(2x)$
13. $3/x - 2/y$
14. $1/(x + 1) + 1/(x - 1)$
15. $(x + 1)/2 - (x - 3)/4$

11. 3/x 12. 3/(2x) (LCD 2x) 13. (3y − 2x)/(xy) 14. 2x/((x + 1)(x − 1)) 15. (x + 5)/4

Signs, nested fractions, equations

16. $(x - 5)/(5 - x)$
17. $1 - 2/(x + 3)$
18. $(1 + 2/x)/(1 - 4/x^2)$
19. $((x^2 - 4)/(x + 1)) \div ((x - 2)/(x^2 + x))$
20. Solve $4/x - 1/3 = 1$

16. −1 17. (x + 1)/(x + 3) 18. x/(x − 2) (× x², then cancel x + 2) 19. x(x + 2) 20. x = 3 (× 3x: 12 − x = 3x)

Find the unknown

21. $\frac{3}{(x-1)} + \frac{A}{(x+1)} = \frac{(5x+1)}{(x^2-1)}$. Find A.

22. $\frac{(x^2+kx+6)}{(x+2)}$ simplifies to $x + 3$. Find k.

23. $\frac{1}{x} + \frac{1}{(kx)} = \frac{5}{(4x)}$. Find k.

24. $\frac{(x+2)}{(x-3)}$: for which x is it undefined, and for which is it 0?

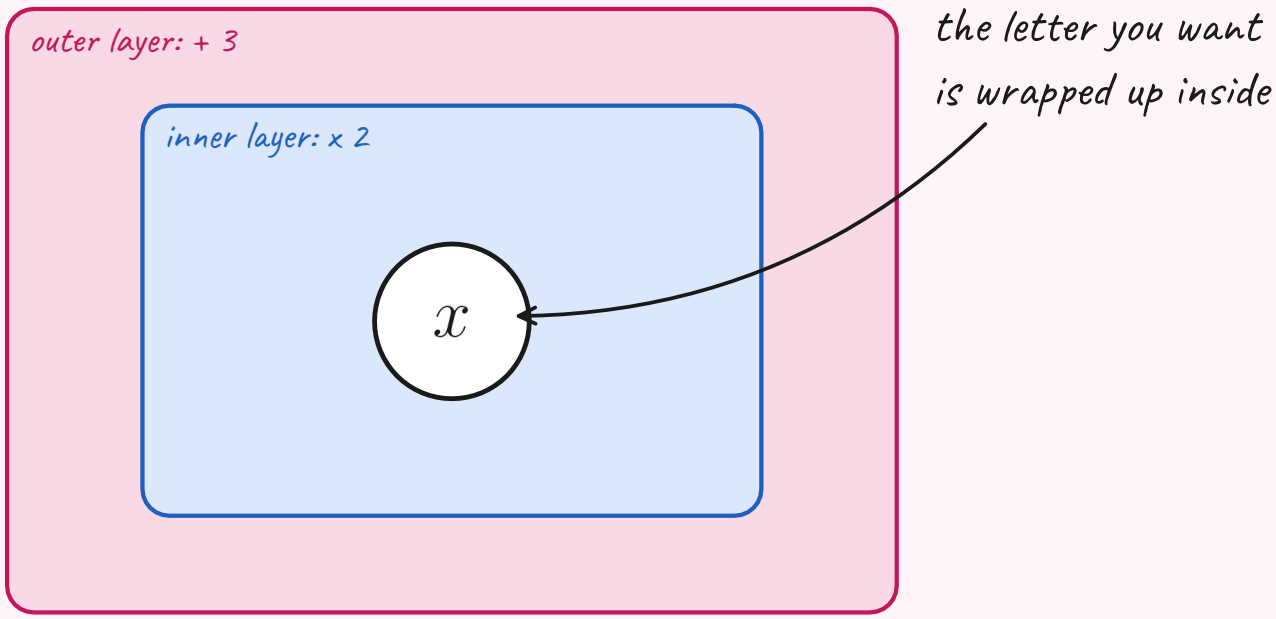
25. $\frac{a}{(x-2)} - \frac{1}{x} = \frac{(x+2)}{(x(x-2))}$. Find a.

21. A = 2 (3(x + 1) + A(x − 1) = 5x + 1) 22. k = 5 ((x + 2)(x + 3) = x² + 5x + 6) 23. k = 4 (× x: 1 + 1/k = 5/4) 24. undefined at x = 3 (bottom 0), zero at x = −2 (top 0) 25. a = 2 (ax − (x − 2) = x + 2)

Chapter 14

Transposing formulae

or: unwrapping a present to get at the letter inside



A formula is usually written to work out **one** particular thing. $F = \frac{9}{5}C + 32$ turns a temperature in Celsius into Fahrenheit. But what if you know the Fahrenheit and want the Celsius? You need to **rearrange** the formula so that C is on its own.

Rearranging a formula is called **transposing** it. The trick in this chapter is to see every formula as a present: the letter you want is in the middle, and the formula has wrapped it in layers ("times 2", "add 3", "square root"...). To get the letter out, you take the layers off, **outside layer first**.

Goes with **Lecture 2** (Algebraic manipulation) and parts of Lectures 7 and 12. Uses: letters and substituting (Chapter 8), squares and roots (Chapter 9), brackets (Chapter 11), factorising (Chapter 12) and algebraic fractions (Chapter 13).

What "make x the subject" means

The **subject** of a formula is the letter that is **on its own** on one side of the = sign. In

$$y = 2x + 3$$

the subject is y. The formula is a machine for working out y: give it an x and it tells you y. **"Make x the subject"** means: rearrange it so it looks like

$$x = (\text{something with } y \text{ in it})$$

so that it becomes a machine for working out x instead. The words **transpose**, **rearrange**, **change the subject**, **make x the subject** and **solve for x** all mean the same thing.

The undo buttons

RULE: inverse operations (undo buttons)

$+$ is undone by $-$, and $-$ by $+$.

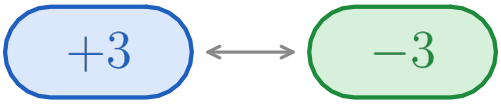
$\times$ is undone by $\div$, and $\div$ by $\times$ (**never divide by something that could be 0**).

Squaring is undone by square-rooting (giving $\pm$), and $\sqrt{}$ by squaring.

Cubing is undone by the cube root $\sqrt[3]{}$ (one answer, no $\pm$), and $\sqrt[3]{}$ by cubing.

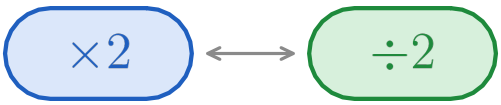
Flipping a fraction (taking the reciprocal) is undone by flipping it back.

Every operation has an **undo button**: another operation that exactly cancels it. The proper name is an **inverse operation**. Test each pair with a number: do it, then undo it, and you are back where you started.



$$7 \xrightarrow{+3} 10 \xrightarrow{-3} 7$$

plus is undone by minus (and minus by plus)



$$7 \xrightarrow{\times 2} 14 \xrightarrow{\div 2} 7$$

times is undone by divide (and divide by times)

square ⟷ $\sqrt{\ }$

$7 \xrightarrow{\text{sq}} 49 \xrightarrow{\sqrt{\ }} 7$

squaring is undone by square-rooting

flip ⟷ flip

$\frac{2}{7} \xrightarrow{\text{flip}} \frac{7}{2} \xrightarrow{\text{flip}} \frac{2}{7}$

flipping a fraction is undone by flipping it back

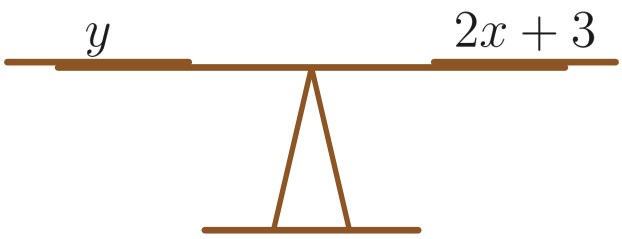
The only rule: keep the see-saw balanced

Part 0 said = is a balanced see-saw. If you take 3 off the left pan, it tips... unless you take 3 off the right pan as well. So the one and only rule of rearranging is:

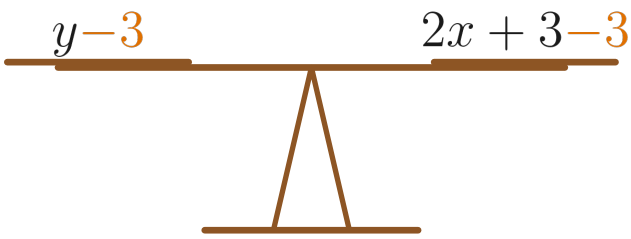
RULE: the balance rule

Whatever you do to one side, do exactly the same to the whole of the other side.

"The whole side": dividing $2x+3$ by 2 gives $\frac{2x+3}{2}$, which is $x+\frac{3}{2}$, not $x+3$.



balanced



take 3 from BOTH pans: still balanced

"The whole of the other side" matters. If the other side is $2x+3$ and you divide by 2, you must divide **all** of $2x+3$, not just the $2x$. More on this mistake later.

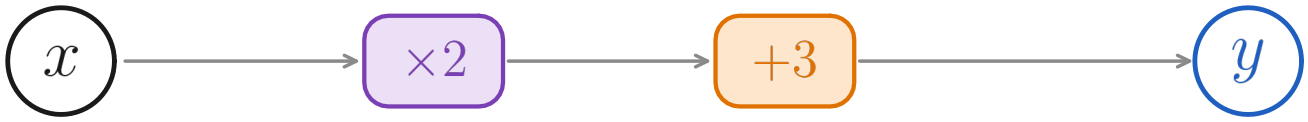
Your first unwrapping: y = 2x + 3

PROCEDURE: making a letter the subject

1. Write down what happens to the letter, **in order**, to build the formula (its layers).
2. Undo the layers in **reverse order** (last on, first off), using the undo buttons.
3. Do every move to the **whole** of both sides, one move per line.
4. Swap sides so the letter is on the left. Test with numbers.

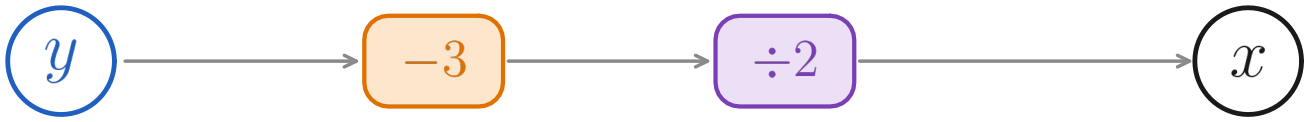
Step 1: work out how the present was wrapped. Start with x and ask: what happens to it, in order, to make y? x is first multiplied by 2 (that makes $2x$), and then 3 is added (that makes $2x + 3$).

wrapping (building the formula):

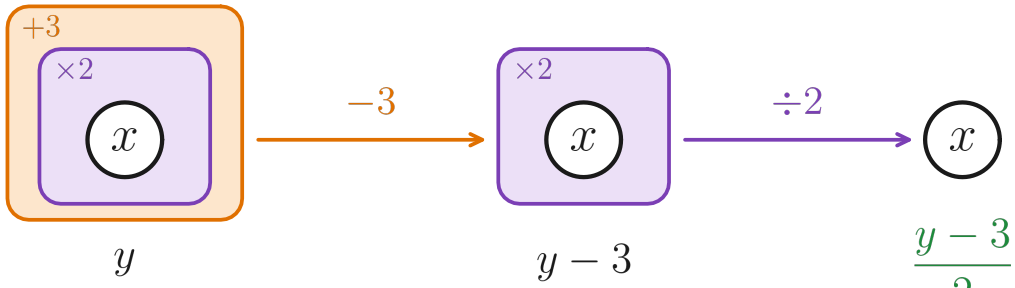


Step 2: unwrap in the reverse order, using the undo buttons. The last layer on was $+3$, so it comes off first, using -3. Then $\times 2$ comes off, using $\div 2$.

unwrapping (reverse order, undo buttons):



Picture it: peel the present one layer per step. Whatever each peel does to the x side, it does to the y side too:



underneath: the other side of the =

Step 3: do it on the formula, one layer per line, to both sides.

$$y = 2x + 3 \quad \longleftarrow \text{start}$$

$$y-3 = 2x + 3-3$$ ← take the outer layer off: subtract 3 from BOTH sides

$$y - 3 = 2x$$ ← +3 and -3 cancel out on the right

$$\frac{y-3}{2} = \frac{2x}{2}$$ ← next layer: divide BOTH sides by 2

$$\frac{y-3}{2} = x$$ ← 2x divided by 2 is just x

$$x = \frac{y-3}{2}$$ ← swap sides so x is on the left (allowed: = is a balance)

Test it with a number. Put $x = 4$ into the original: $y = 2 \times 4 + 3 = 11$. Now put $y = 11$ into the new formula: $x = \frac{11-3}{2} = \frac{8}{2} = 4$. We got the 4 back. The unwrapping worked. ✓

Last on, first off. Think of socks and shoes. You put socks on first and shoes second. To undress, the shoes come off first. Formulas are the same.

Why the order matters

What if you took the layers off in the wrong order, dividing by 2 first? Try it with numbers, where we know the answer should be $x = 4$ when $y = 11$:

Right order

$$11 \xrightarrow{-3} 8 \xrightarrow{\div 2} 4$$ ✓

Wrong order

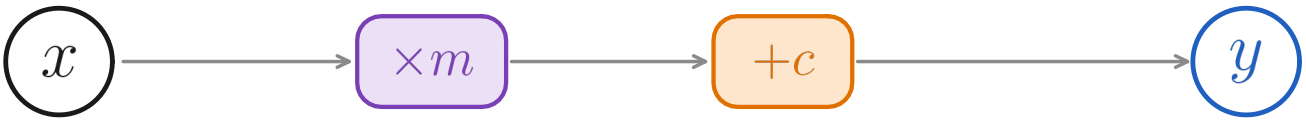
$$11 \xrightarrow{\div 2} 5.5 \xrightarrow{-3} 2.5$$ ✗

The wrong order gives 2.5, which is wrong. Undoing has to happen in the **reverse** order of doing. That is why Step 1 (working out how the formula was built) matters so much.

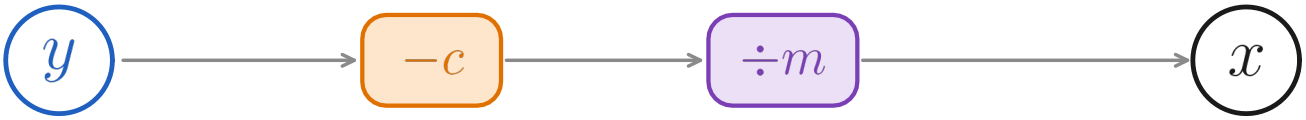
The lecture example: y = mx + c

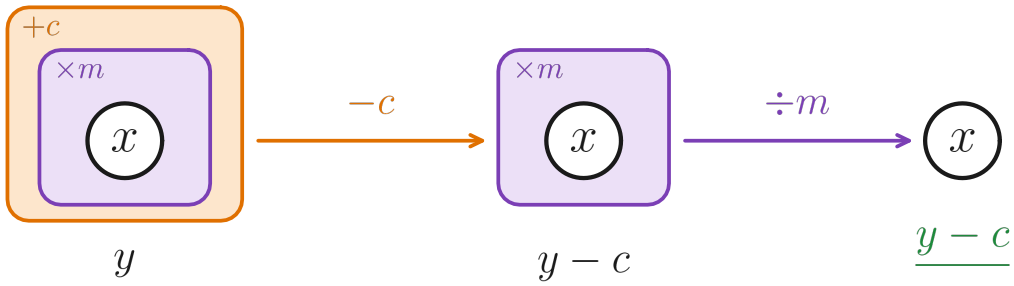
Lecture 2 rearranges the straight-line formula (you will meet it properly in Chapter 20). It is exactly the same present, with letters where the numbers were: $\times m$ instead of $\times 2$, and $+c$ instead of $+3$.

wrapping:



unwrapping:





underneath: the other side of the =

$$y - c = mx$$ ← *subtract c from both sides*

$$\frac{y - c}{m} = x$$ ← *divide both sides by m*

$$x = \frac{y - c}{m}$$ ← *swap sides*

When the letter has a minus in front

RULE: subject with a minus sign

If the letter's term is being **subtracted**, **add it to both sides** first. That moves it to the other side as a positive term.

Then unwrap as usual. (Or multiply both sides by -1, which flips every sign.)

Example: make x the subject of $y = 5 - 2x$.

$$y + 2x = 5$$ ← add 2x to both sides: now 2x is positive

$$2x = 5 - y$$ ← subtract y from both sides

$$x = \frac{5 - y}{2}$$ ← divide both sides by 2

Test: $x = 1$ gives $y = 5 - 2 = 3$. New formula: $x = \frac{5-3}{2} = 1$ ✓.

When the letter is on the bottom

RULE: subject in the denominator

Multiply both sides by the denominator first, so the letter comes up onto a top line.

Then unwrap as usual: the letter is now in an ordinary layer.

(If each side is a single fraction you may instead flip both sides; see the lecture example later.)

Example: make R the subject of $P = \frac{V}{R}$.

$$PR = V$$ ← multiply both sides by R

$$R = \frac{V}{P}$$ ← divide both sides by P

Test: $V = 12$, $R = 4$ gives $P = 3$, and $\frac{12}{3} = 4$ ✓.

Example: make v the subject of $t = \frac{d}{v+2}$. The bottom is a whole bracket:

$$t(v+2) = d$$ ← multiply both sides by (v + 2)

$$v + 2 = \frac{d}{t}$$ ← divide both sides by t

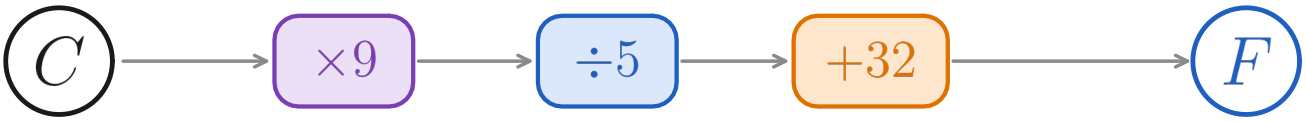
$$v = \frac{d}{t} - 2$$ ← subtract 2 from both sides

Test: $v = 3$, $d = 10$ gives $t = 2$; then $\frac{10}{2} - 2 = 3$ ✓.

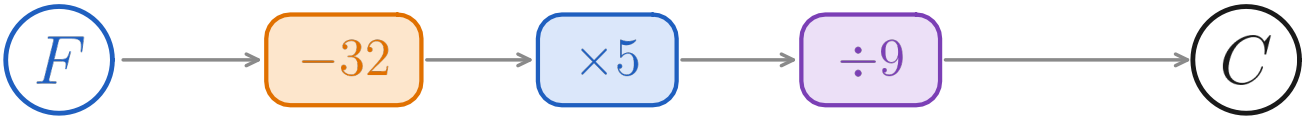
Three layers: Fahrenheit to Celsius

$F = \frac{9}{5}C + 32$ is the same as $F = \frac{9C}{5} + 32$. Make C the subject.

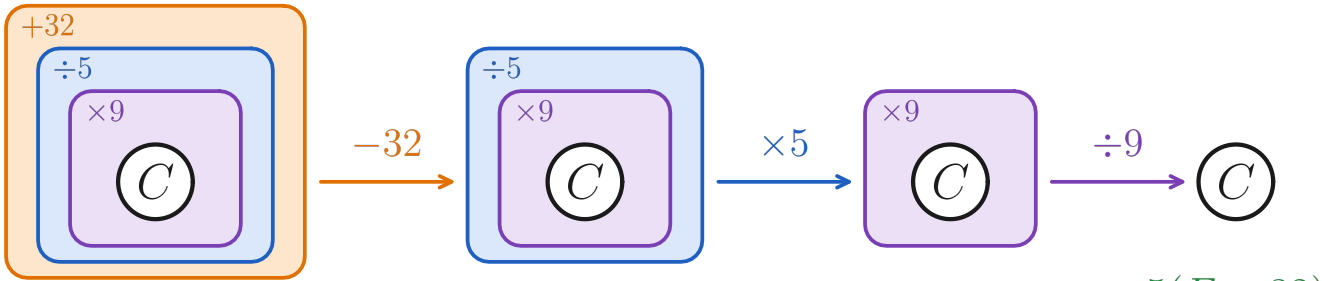
How it was wrapped: C is multiplied by 9, then divided by 5, then 32 is added.



Unwrap, last on first off: subtract 32, then **multiply** by 5 (undoes $\div 5$), then divide by 9.



Three layers, three peels:



F $\quad F - 32$ $\quad 5(F-32)$ $\quad \frac{5(F-32)}{9}$

underneath: the other side of the =

$$F = \frac{9C}{5} + 32$$ ← start

$$F - 32 = \frac{9C}{5}$$ ← outer layer: subtract 32 from both sides

$$5(F - 32) = 9C$$ ← next layer: multiply both sides by 5. Brackets: ALL of F - 32 gets x5

$$\frac{5(F - 32)}{9} = C$$ ← last layer: divide both sides by 9

$$C = \frac{5(F - 32)}{9}$$ ← swap sides

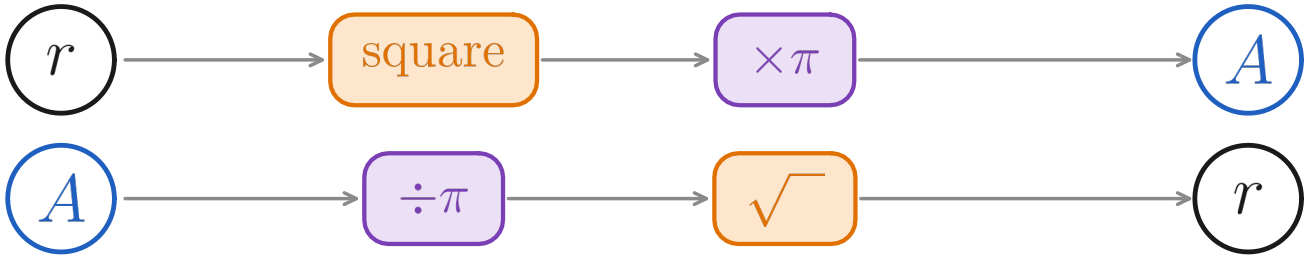
Test: water boils at 212 °F. $C = \frac{5(212-32)}{9} = \frac{5\times180}{9} = \frac{900}{9} = 100$ °C. ✓ Correct!

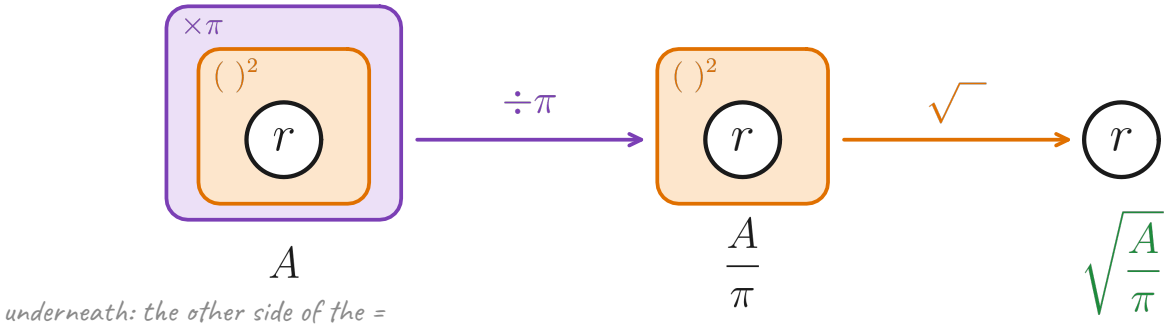
Why the bracket? Without it, $5F - 32$ would mean "multiply only F by 5, then take 32". But the whole left side, $F - 32$, has to be multiplied by 5. The bracket keeps it together.

Layers with squares and roots

The area of a circle is $A = \pi r^2$. Given the area, find the radius r.

How it was wrapped: r is squared first (that's r^2), then multiplied by π. (Squaring happens before multiplying: powers come before times in the order of operations.)





underneath: the other side of the =

$$A = \pi r^2 \quad \leftarrow \text{start}$$

$$\frac{A}{\pi} = r^2 \quad \leftarrow \text{divide both sides by pi}$$

$$\sqrt{\frac{A}{\pi}} = r \quad \leftarrow \text{square root both sides (undoes the squaring)}$$

$$r = \sqrt{\frac{A}{\pi}} \quad \leftarrow \text{swap sides}$$

Test: a circle of radius 3 has $A = \pi \times 9 \approx 28.27$. Then $\sqrt{28.27/\pi} = \sqrt{9} = 3$. ✓

A small print note: $(-3)^2$ is also 9, so strictly r could be $+3$ or -3 (written ± 3). A radius can't be negative, so here we keep the positive one. When a question isn't about a length, remember the $\pm$.

RULE: cubes and cube roots

A cube is undone by a **cube root**: if $r^3 = k$ then $r = \sqrt[3]{k}$.

There is only **one** cube root, and negatives are fine: $\sqrt[3]{-8} = -2$ because $(-2)^3 = -8$. So no $\pm$ here.

Example: the volume of a sphere is $V = \frac{4}{3}\pi r^3$. Make r the subject.

Wrapping: cube, then $\times\frac{4}{3}\pi$ (that is $\times 4$, $\div 3$, $\times\pi$). Unwrap in reverse:

$$3V = 4\pi r^3$$ ← multiply both sides by 3

$$\frac{3V}{4\pi} = r^3$$ ← divide both sides by 4 pi

$$r = \sqrt[3]{\frac{3V}{4\pi}}$$ ← cube root both sides, swap

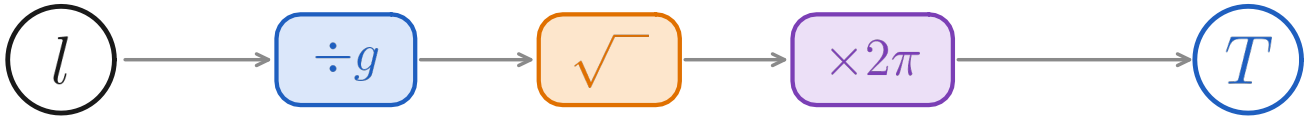
Test: $r = 2$ gives $V = \frac{4}{3}\pi \times 8 \approx 33.51$; then $\sqrt[3]{3 \times 33.51/(4\pi)} = \sqrt[3]{8} = 2$ ✓.

The pendulum, one layer at a time

This is the big example from Lecture 2 (and the one from Part 0, where you learned to read it). Make l (the string length) the subject:

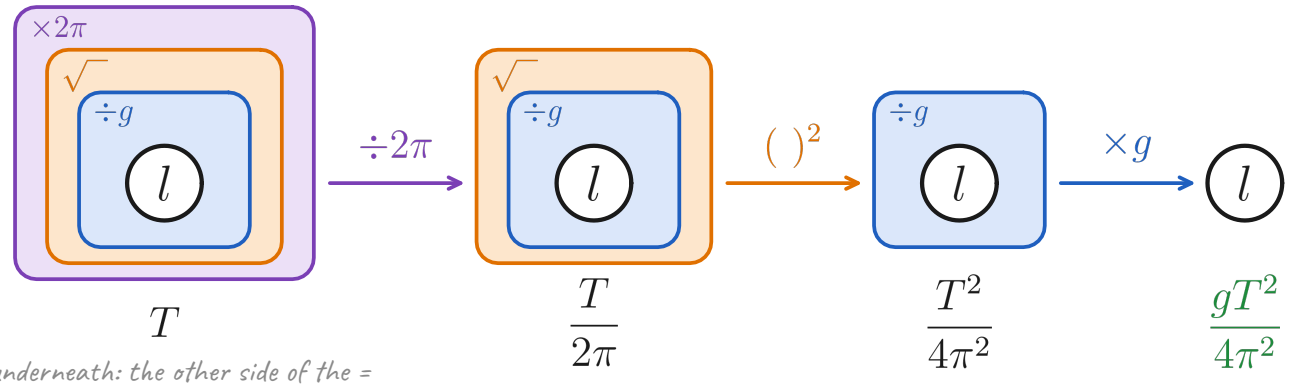
$$T = 2\pi\sqrt{\frac{l}{g}}$$

How it was wrapped, reading from the inside out: l is divided by g, then square-rooted, then multiplied by 2π.



Unwrapping (reverse order, undo buttons): divide by 2π, then square, then multiply by g.

The present, peeled from the outside in:



underneath: the other side of the =

$$T = 2\pi\sqrt{\frac{l}{g}} \quad \longleftarrow \text{start}$$

$$\frac{T}{2\pi} = \sqrt{\frac{l}{g}} \quad \longleftarrow \text{outer layer: divide both sides by 2 pi}$$

$$\left(\frac{T}{2\pi}\right)^2 = \frac{l}{g}$$

middle layer: square both sides (squaring undoes the root)

$$\frac{T^2}{4\pi^2} = \frac{l}{g}$$

tidy up: square the top (T x T) and the bottom (2 pi x 2 pi = 4 pi squared)

$$g \times \frac{T^2}{4\pi^2} = l$$

inner layer: multiply both sides by g

$$l = \frac{gT^2}{4\pi^2}$$

swap sides and tidy up. This matches your lecture slide!

Test: with $g = 9.81$ and a 1 metre string, $T = 2\pi\sqrt{1/9.81} \approx 2.006$ seconds. Put $T = 2.006$ into the new formula: $l = 9.81 \times 2.006^2/(4\pi^2) \approx 1.00$ metre. ✓

When the letter is on the bottom of a fraction

Lecture 2's hardest example: make z the subject of

$$\frac{1}{x} = \frac{2}{y} - \frac{3}{z}$$

RULE: flipping both sides

If **each side is one single fraction**, you may flip both sides: $\frac{a}{b} = \frac{c}{d}$ gives $\frac{b}{a} = \frac{d}{c}$.

Never flip term by term. Combine each side into one fraction first (Chapter 13).

This isn't a simple chain of layers, because there are several fractions. Plan: (1) get the z fraction alone on one side, (2) combine the other side into one fraction, (3) flip both sides.

$$\frac{1}{x} + \frac{3}{z} = \frac{2}{y}$$

← add 3/z to both sides (it was being taken away)

$$\frac{3}{z} = \frac{2}{y} - \frac{1}{x}$$

← subtract 1/x from both sides: now the z fraction is alone

$$\frac{3}{z} = \frac{2x}{yx} - \frac{1y}{xy}$$

← re-cut to a common bottom, xy (Chapter 13)

$$\frac{3}{z} = \frac{2x - y}{xy}$$

← same bottoms: combine the tops

$$\frac{z}{3} = \frac{xy}{2x - y}$$

← flip BOTH sides upside down. Allowed: if two things are equal, their reciprocals are equal too

$$z = \frac{3xy}{2x - y}$$ ← multiply both sides by 3

Test: pick $x = 1$, $y = 1$. The new formula gives $z = \frac{3}{2-1} = 3$. Check in the original: $\frac{1}{1} = \frac{2}{1} - \frac{3}{3}$ means $1 = 2 - 1$. ✓

Only flip when each side is ONE single fraction. Flipping $\frac{2}{y} - \frac{1}{x}$ term by term (into $\frac{y}{2} - \frac{x}{1}$) is wrong. Test with $x = 1, y = 1$: $\frac{2}{1} - \frac{1}{1} = 1$, flipped is 1; but $\frac{1}{2} - 1 = -\frac{1}{2}$. Not the same!

When the letter appears twice

PROCEDURE: subject appearing more than once

1. Clear fractions (multiply both sides by the bottom).
2. Expand brackets, then collect every term containing the letter on **one** side, everything else on the other.
3. **Factorise**: take the letter out as a common factor, so it appears once.
4. Divide both sides by the bracket (**that bracket must not be 0**).

Sometimes the letter you want is in two places, so it isn't wrapped in one neat present. Make x the subject of

$$y = \frac{x + 1}{x - 1}$$

Plan: (1) get rid of the fraction, (2) collect every x on one side, (3) take x out as a common factor, (4) divide.

$y(x - 1) = x + 1$ ← multiply both sides by (x - 1): the fraction disappears

$yx - y = x + 1$ ← open the bracket (Chapter 11)

$yx - x = 1 + y$ ← move the x terms left, the rest right: subtract x, add y, on both sides

$$x(y-1) = 1+y$$ ← take x out as a common factor (Chapter 12): now x appears ONCE

$$x = \frac{1+y}{y-1}$$ ← divide both sides by (y - 1)

Test: $x = 3$ gives $y = \frac{4}{2} = 2$. New formula with $y = 2$: $x = \frac{3}{1} = 3$. ✓

The factorising step is the key idea. Once all the x's are on one side, factorising turns "x in two places" into "x in one place", and then it's an ordinary one-layer unwrap.

Two safety rules

RULE: don't divide by something that could be zero

Dividing both sides by an expression is only allowed when that expression is **not 0**. In the last example, dividing by $(y-1)$ needs $y \neq 1$; write that condition down.

If you divide by a letter in an **equation**, you can lose an answer. Move everything to one side and factorise instead.

Example: $x^2 = 3x$. Dividing by x gives $x = 3$ and **loses** $x = 0$. Safe way:

$$x^2 - 3x = 0 \;\Rightarrow\; x(x-3) = 0$$ ← everything to one side, factorise

$$x = 0 \quad \text{or} \quad x = 3$$ ← either factor can be 0

RULE: squaring both sides can add false answers

Squaring both sides is allowed, but it can create answers that **don't fit the original**, because 3^2 and $(-3)^2$ are both 9.

So after squaring, **check every answer in the original** and throw out any that fail.

Example: $\sqrt{x} = -3$. Squaring gives $x = 9$, but $\sqrt{9} = 3$, not -3. So there is **no solution**: a square root (the $\sqrt{}$ sign) is never negative.

Example: $\sqrt{x+3} = x - 3$. Square: $x + 3 = x^2 - 6x + 9$, so $x^2 - 7x + 6 = 0$, $(x-1)(x-6) = 0$. Check $x = 6$: $\sqrt{9} = 3 = 6 - 3$ ✓. Check $x = 1$: $\sqrt{4} = 2$ but $1 - 3 = -2$ ✗. Only $x = 6$ is an answer.

Using a rearranged formula: find the missing value

Exams rarely stop at the rearranged formula. Usually you're given **some** of the values and asked for the one that's missing:

PROCEDURE: finding an unknown from a formula

1. Circle the letter you need to find. That's the subject you want.
2. **Rearrange** the formula to make it the subject (letters only, no numbers yet).
3. **Substitute** the values you know. Put negative numbers in brackets.
4. Calculate, with units. **Check** by putting every value back into the original formula.

(Substituting first and then solving also works. Rearranging first gives a formula you can reuse, and the steps are easier to check.)

Example 1 (motion): $v^2 = u^2 + 2as$. Find s when $v = 10$, $u = 4$, $a = 3$.

$$v^2 - u^2 = 2as$$ ← rearrange: subtract u^2

$$s = \frac{v^2 - u^2}{2a}$$ ← divide by 2a, swap

$$s = \frac{10^2 - 4^2}{2 \times 3} = \frac{100 - 16}{6} = \frac{84}{6}$$ ← substitute

$$s = 14$$ ← calculate

Check: $u^2 + 2as = 16 + 2 \times 3 \times 14 = 16 + 84 = 100 = 10^2$ ✓.

Example 2 (pendulum): $T = 2\pi\sqrt{\frac{l}{g}}$. How long must the string be for $T = 2$ seconds, with $g = 9.81$?

$$l = \frac{gT^2}{4\pi^2}$$ ← rearranged earlier in this chapter

$$l = \frac{9.81 \times 2^2}{4\pi^2} = \frac{39.24}{39.48}$$ ← substitute

$$l \approx 0.994 \text{ m}$$ ← about 99 cm

Check: $2\pi\sqrt{0.994/9.81} = 2\pi \times 0.3183 \approx 2.00$ s ✓.

Example 3 (energy): $E = \frac{1}{2}mv^2$. Find v when $E = 100$ J and $m = 8$ kg.

$$v = \sqrt{\frac{2E}{m}} = \sqrt{\frac{200}{8}} = \sqrt{25}$$ ← rearrange (×2, ÷m, √), substitute

$$v = 5 \text{ m/s}$$ ← a speed: keep the + root

Check: $\frac{1}{2} \times 8 \times 5^2 = 4 \times 25 = 100$ ✓.

Three classic slip-ups (tested with numbers)

Each one starts from the same kind of formula, shows the slip, then the correct move.

Dividing only part of a side.

start: $y = 2x + 3$ — *Move: divide both sides by 2.*

wrong: ~~$\frac{y}{2} = x + 3$~~ — *only the 2x was divided; the 3 was forgotten*

right: $\frac{y}{2} = \frac{2x+3}{2} = x + \frac{3}{2}$ ✓ *the WHOLE side goes over 2, so the 3 is halved too*

Test with x = 4, y = 11: the wrong version says 5.5 = 7. The right version says 5.5 = 4 + 1.5. ✓ Divide EVERY term.

Undoing in the wrong order.

start: $y = 2x + 3$ — *Move: make x the subject.*

wrong: ~~$x = \frac{y}{2} - 3$~~ — *took off ×2 first, but it's the INNER layer*

right: $x = \frac{y-3}{2}$ ✓ *−3 first (outer layer), then ÷2*

With y = 11 the wrong one gives 2.5, the right one gives 4. ✓ Last on, first off.

Square-rooting term by term.

start: $c^2 = a^2 + b^2$ — *Move: square-root both sides to get c.*

wrong: ~~$c = a + b$~~ — *rooted a^2 and b^2 separately*

right: $c = \sqrt{a^2 + b^2}$ ✓ *the root goes over the WHOLE side*

Test with a = 3, b = 4: the right answer is the square root of 25, which is 5. The wrong one gives 7. (This is also a warning on your Lecture 2 slides.)

The unwrapping routine

1. Circle the letter you want.
2. Work out how it was wrapped: what happens to it, in order? Draw the chain.
3. Messy? Clear fractions first (multiply both sides), collect the letter on one side, and factorise so it appears once.
4. Unwrap: last layer first, using the undo buttons, doing the same to the whole of both sides.
5. Test with numbers.

Coming later: Chapter 15 uses exactly the same moves to find the value of a mystery number, and Chapter 22 adds one more undo button (the logarithm), which undoes powers like e^x. That's how Lecture 7's capacitor formula gets rearranged.

Rearranging rules

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Balance rule	whatever you do to one side, do to the **whole** of the other side
Undo buttons	$+ \leftrightarrow -, \quad \times \leftrightarrow \div, \quad x^2 \leftrightarrow \pm\sqrt{x}, \quad x^3 \leftrightarrow \sqrt[3]{x}$ flip undoes flip
Order	build the chain of layers, then undo them in **reverse** order: last on, first off
Minus in front	$y = 5 - 2x \;\Rightarrow\; 2x = 5 - y$ add the subtracted term to both sides first
Letter on the bottom	$P = \dfrac{V}{R} \;\Rightarrow\; PR = V \;\Rightarrow\; R = \dfrac{V}{P}$ multiply both sides by the bottom first
Flipping	$\dfrac{a}{b} = \dfrac{c}{d} \;\Rightarrow\; \dfrac{b}{a} = \dfrac{d}{c}$ only when each side is ONE fraction
Letter twice	$yx - x = 1 + y \;\Rightarrow\; x(y-1) = 1 + y$ collect, factorise, divide
Roots	$r^2 = k \Rightarrow r = \pm\sqrt{k}, \qquad r^3 = k \Rightarrow r = \sqrt[3]{k}$ keep $+$ only for lengths, speeds and other sizes
Safety	**never divide by something that could be 0**; after squaring both sides, check every answer in the original
Find a value	rearrange first, then substitute, then check in the original formula

Your turn: fill in the chains

For each formula, fill in the wrapping boxes (what happens to the letter, in order), then the unwrapping boxes (reverse order, undo buttons), then write the rearranged formula on the line. Answers to 1 to 5 are on the last page; the later groups have theirs in grey.

1. $y = 5x - 4$ *make x the subject*

wrap: x → [ ] → [ ] → y

unwrap: y → [ ] → [ ] → x *so x =* _____

2. $P = \dfrac{V}{R}$ *make V the subject*

wrap: V → [ ] → P

unwrap: P → [ ] → V *so V =* _____

3. $v = u + at$ *make a the subject*

wrap: a → [ ] → [ ] → v

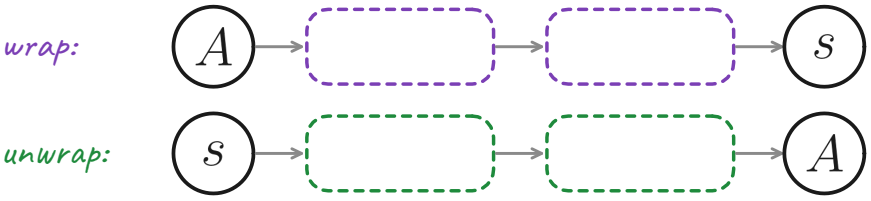
unwrap: v → [ ] → [ ] → a *so a =* _____

4. $E = \frac{1}{2}mv^2$ *make v the subject*

wrap: v → [ ] → [ ] → [ ] → E

unwrap: E → [ ] → [ ] → [ ] → v *so v =* _____

5. $s = \sqrt{\dfrac{A}{6}}$ *make A the subject*



Minus signs, bottoms and cubes

6. $y = 7 - 3x$: make x the subject
7. $I = V/R$: make R the subject
8. $f = 1/(x + 3)$: make x the subject
9. $V = x^3/3$: make x the subject
10. $y = (2x + 1)/(x - 3)$: make x the subject

6. $x = (7 - y)/3$ 7. $R = V/I$ 8. $x = 1/f - 3$ 9. $x = \sqrt[3]{(3V)}$ 10. $x = (3y + 1)/(y - 2)$

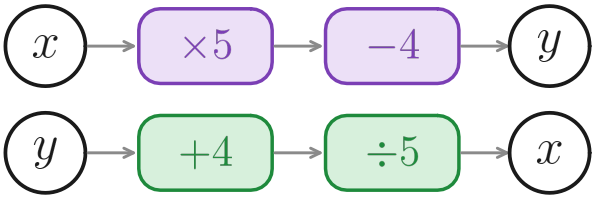
Find the unknown

11. $v^2 = u^2 + 2as$. Find s when v = 20, u = 10, a = 5.
12. $v = u + at$. Find t when v = 25, u = 5, a = 4.
13. $A = \pi r^2$. Find r when A = 154, using $\pi \approx 22/7$.
14. $F = 9C/5 + 32$. Find C when F = 50.
15. $y = (x + 1)/(x - 1)$. Find x when y = 3.
16. $E = \frac{1}{2}mv^2$. Find m when E = 90 and v = 6.

11. s = 30 ($s = (v^2 - u^2)/(2a) = 300/10$) 12. t = 5 ($t = (v - u)/a$) 13. r = 7 ($r^2 = A/\pi = 49$)
14. C = 10 ($C = 5(F - 32)/9$) 15. x = 2 ($x = (y + 1)/(y - 1)$) 16. m = 5 ($m = 2E/v^2 = 180/36$)

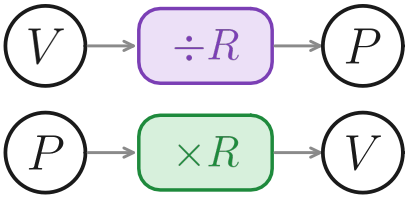
Answers

1.



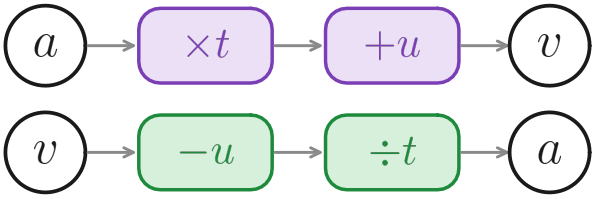
$$x = \frac{y+4}{5}$$

2.



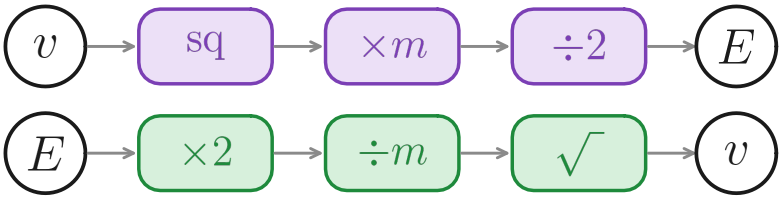
$$V = PR$$

3.



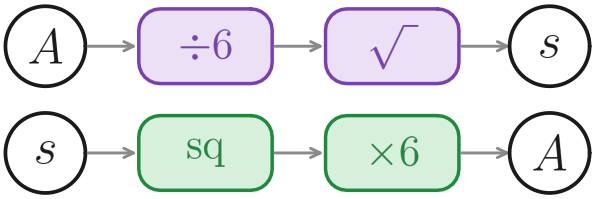
$$a = \frac{v-u}{t}$$

4.



$$v = \sqrt{\frac{2E}{m}}$$

5.



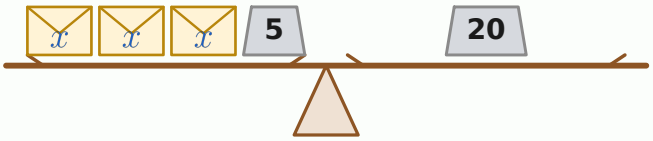
$$A = 6s^2$$

In question 4, $\frac{1}{2}mv^2$ means: square v, times by m, then halve (divide by 2). So unwrapping is: double, divide by m, square root.

Chapter 15

Solving equations

finding the number hiding in the envelope



three sealed envelopes and a 5 g weight balance a 20 g weight. How heavy is one envelope?

$$3x + 5 = 20$$

An **equation** says two things are equal: the scales balance. **Solving** it means finding the number (or numbers) that the letter must be for the balance to hold. That number is called the **solution**.

Every move in this chapter is one of the "undo buttons" from Chapter 14, done to **both sides at once** so the scales stay level. After one letter, you'll meet two letters at once (café receipts) and squared letters, where Chapter 12's pair turns straight into the quadratic formula. Every answer gets put back into the question as a check.

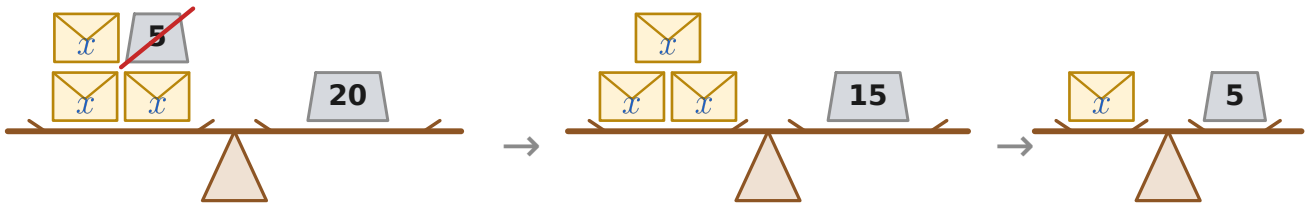
Goes with **Lecture 2** (rearranging), **Lecture 3** (mixing solutions, compounds, simultaneous equations) and **Lecture 14** (solving quadratics).

Balance scales: do the same to both pans

PROCEDURE: solving a linear equation

1. Clear **brackets** (expand) and **fractions** (multiply every term by the LCD).
2. Collect the letter terms on one side (the side with more of them) and the numbers on the other, doing the same to both sides.
3. Divide both sides by the number in front of the letter.
4. **Check:** put the answer back into the original equation.

You could solve $3x + 5 = 20$ by trying numbers until both sides match, but that fails for answers like $x = 1.875$. The no-guessing way: each move keeps the scales level, so whatever you take from one pan, you take from the other:



take 5 from both — 3 envelopes = 15 — one = 5

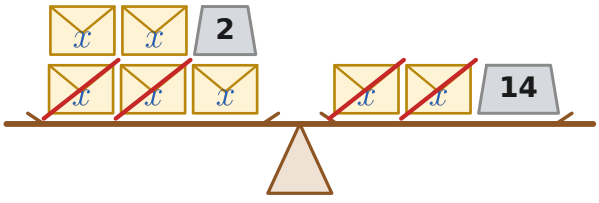
$$3x + 5 = 20$$

$$3x = 15 \quad \longleftarrow \text{ −5 on both sides}$$

$$x = 5 \quad \longleftarrow \text{ ÷3 on both sides (share into 3 equal groups)}$$

Always check: put the answer back. $3(5) + 5 = 20$ ✓.

Letters on both sides: $5x + 2 = 2x + 14$. Take the **same number of envelopes** off each pan, just as you'd take weights off:



take 2 envelopes from each pan

2 envelopes crossed out on EACH pan: what's left still balances

$$5x + 2 = 2x + 14$$

$$3x + 2 = 14 \quad \longleftarrow \text{ −2x on both sides (2 envelopes off each pan)}$$

$$3x = 12 \;\Rightarrow\; x = 4$$ ← −2, then ÷3

Check: left $5(4) + 2 = 22$, right $2(4) + 14 = 22$ ✓. *Tip:* move the letters to whichever side has **more** of them, so you never end up with a negative number of envelopes.

Brackets and fractions: tidy first, then balance

Brackets: expand them (Chapter 11), then it's an ordinary balance. $3(x - 2) = 2(x + 1)$:

$$3x - 6 = 2x + 2$$ ← expand both sides

$$x - 6 = 2$$ ← −2x

$$x = 8$$ ← +6

Check: $3(6) = 18$ and $2(9) = 18$ ✓.

Fractions: multiply **both sides** by something that clears every bottom. For $\frac{x+1}{3} = \frac{x-1}{2}$, use 6 (both 3 and 2 go into it):

$$6 \times \frac{x+1}{3} = 6 \times \frac{x-1}{2}$$ ← × 6 on both sides

$$2(x+1) = 3(x-1)$$ ← 6 ÷ 3 = 2; 6 ÷ 2 = 3

$$2x + 2 = 3x - 3 \;\Rightarrow\; x = 5$$ ← −2x and +3 on both sides

Look at step 2: each top ended up multiplied by the **other** side's bottom. That shortcut has a name.

$$\frac{a}{b} = \frac{c}{d} \;\Rightarrow\; ad = cb$$

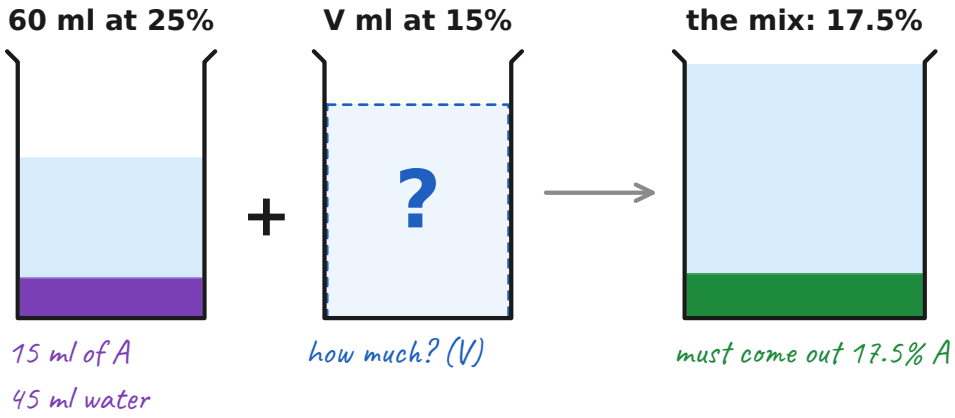
"cross multiplying" is just × b and × d on both sides, done in one go

The Chapter 4 promise: Lecture 3 turns $\frac{5}{V} = \frac{8}{3}$ into $15 = 8V$. That's cross multiplying: $5 \times 3 = 8 \times V$, so $V = \frac{15}{8} = 1.875$ m/s.

Lecture 3, Mixing 2 and 3: an equation from beakers

How much 15% solution must be added to 60 ml of 25% solution to make a 17.5% solution? Before any maths, let's decode what that sentence is asking.

What "a 25% solution" means. A solution is a chemical (call it A) dissolved in water. "25%" says a quarter of the liquid is A and the rest is water. In the pictures below, imagine the A could be poured out separately: the dark part is A, the pale part is water.



In plain English: you have a strong mix. You pour in a weaker one, which dilutes it. Pour a little and it's still strong; pour loads and it's nearly down to 15%. Somewhere in between it is exactly 17.5%. **How much do you pour to stop right there?** That amount is V.

To find it, **follow the chemical A**, not the percentages. Count how much A is in each beaker:

60 ml at 25%	+	**V ml at 15%**	=	**60 + V ml together**
amount of A: $0.25 \times 60 = 15$		amount of A: $0.15V$		amount of A: $15 + 0.15V$

percentage of the mix = (amount of A) ÷ (total volume) × 100, and it must be 17.5

$$\frac{15 + 0.15V}{60 + V} \times 100 = 17.5$$ ← *the equation (the slide's)*

$$(15 + 0.15V) \times 100 = 17.5(60 + V)$$ ← *× (60 + V) on both sides*

$$1500 + 15V = 1050 + 17.5V$$ ← *expand both sides*

$$450 = 2.5V \quad \longleftarrow \text{ } -1050 \text{ and } -15V \text{ on both sides}$$

$$V = \frac{450}{2.5} = 180 \quad \longleftarrow \text{ } \div 2.5$$

Check with the beakers: A = $15 + 0.15 \times 180 = 15 + 27 = 42$ ml in $60 + 180 = 240$ ml. $\frac{42}{240} \times 100 = 17.5\%$ ✓. So add **180 ml** of the 15% solution.

Two unknowns: café receipts

You find two receipts. You don't know the price of a coffee (c) or a cake (k):

CAFÉ	
coffee	
coffee	
cake	
TOTAL	**£7.00**

CAFÉ	
coffee	
cake	
TOTAL	**£4.50**

Put them side by side. The first has exactly ONE more coffee than the second, and costs £2.50 more. So a coffee is £2.50!

That's the whole idea of **simultaneous equations**: two facts about the same two unknowns, used together. Written in algebra:

$$2c + k = 7 \qquad (1)$$

$$c + k = 4.5 \qquad (2)$$

$$c = 2.5 \quad \longleftarrow \text{ } (1) - (2)\text{: the cakes cancel}$$

$$2.5 + k = 4.5 \;\Rightarrow\; k = 2 \quad \longleftarrow \text{ put } c \text{ into } (2)$$

Check in the other one: $2(2.50) + 2 = 7$ ✓. Subtracting whole equations like this is called **elimination**. Here is the rule in full:

PROCEDURE: elimination

1. Label the equations (1) and (2), lined up: x under x, y under y, numbers on the right.
2. Multiply one equation, **or both**, by numbers that make the coefficients of one letter the **same size**.
3. **Same signs: subtract** the equations. **Opposite signs: add** them. That letter cancels.
4. Solve for the letter that's left.
5. Put it back into either original equation to find the other letter. Check in the other one.

Scale one equation (three copies of a receipt cost three times as much):

$$3x + 2y = 16$$

$$x + y = 6$$

$$2x + 2y = 12 \longleftarrow \times 2 \text{ the second equation}$$

$$x = 4 \longleftarrow +2y \text{ in both (same sign): subtract}$$

$$4 + y = 6 \;\Rightarrow\; y = 2 \longleftarrow \text{put } x = 4 \text{ into } x + y = 6$$

Scale both equations: $3x + 2y = 12$ (1) and $4x - 3y = -1$ (2). No single multiplier makes anything match, so aim for $6y$ in both:

$$9x + 6y = 36 \qquad (1) \times 3$$

$$8x - 6y = -2 \qquad (2) \times 2$$

$$17x = 34 \;\Rightarrow\; x = 2 \longleftarrow +6y \text{ and } -6y \text{ (opposite signs): add}$$

$$6 + 2y = 12 \;\Rightarrow\; y = 3 \longleftarrow x = 2 \text{ into (1)}$$

Check in (2): $4(2) - 3(3) = 8 - 9 = -1$ ✓.

The substitution method

PROCEDURE: substitution

1. **Rearrange** one equation to get one letter on its own (pick a letter with no number in front).
2. **Substitute** that expression into the **other** equation, in brackets.
3. Solve the one-letter equation you now have.
4. **Back-substitute**: put the answer into the rearranged equation to find the other letter.
5. Check both answers in **both** original equations.

Example: $2x - y = 1$ (1) and $3x + 2y = 12$ (2).

$$y = 2x - 1$$ ← 1. rearrange (1): y has no number in front

$$3x + 2(2x - 1) = 12$$ ← 2. substitute into (2)

$$3x + 4x - 2 = 12 \;\Rightarrow\; 7x = 14 \;\Rightarrow\; x = 2$$

3. solve

$$y = 2(2) - 1 = 3$$ ← 4. back-substitute

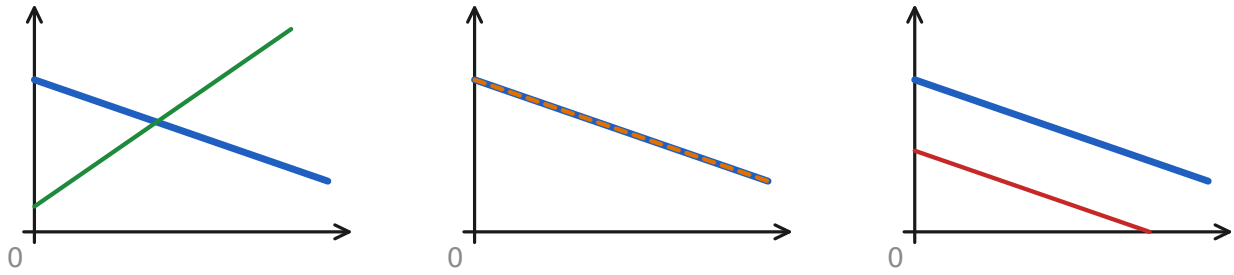
5. Check: (1) $4 - 3 = 1$ ✓, (2) $6 + 6 = 12$ ✓.

Which method? Substitution is quickest when a letter is already on its own (like $y = 2x - 1$ or $x = 1000 - y$). Elimination is quickest when the equations are lined up with whole numbers. Both always give the same answer, and Lecture 12 (matrices) solves the same systems a third way.

The picture: where two lines cross

Each equation on its own has endless answers. $c + k = 4.5$ could be (1, 3.5), (2, 2.5), (3, 1.5)... Plot them all and you get a straight line (Chapter 20). The other receipt gives another line, and the answer is the one point on **both**: where they cross, at (2.5, 2).

Two lines can meet in three ways, and that decides how many answers there are:



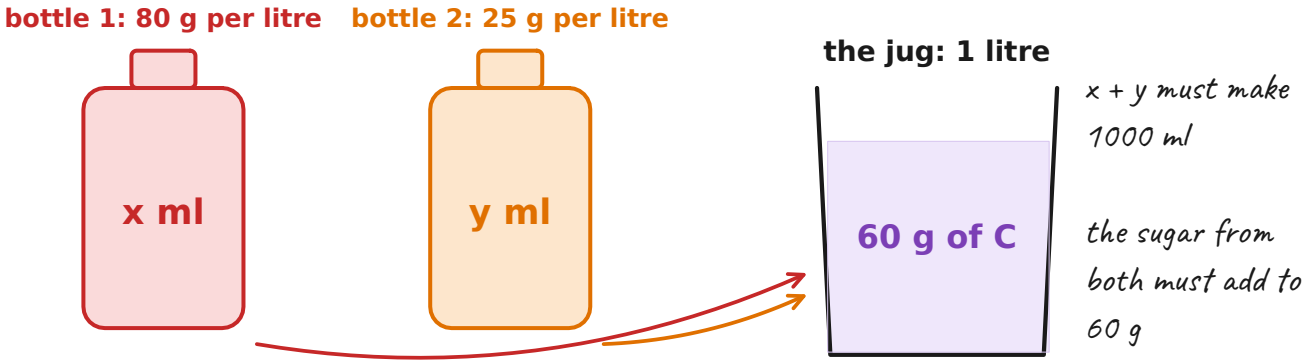
cross once: one answer **same line: endless answers** **parallel: no answer**

Lecture 3 ends with the middle case: $x + 2y = 10$ and $3x + 6y = 30$. The second is just the first multiplied by 3 (the same order bought three times), so it tells you nothing new. Try to solve and everything cancels to $30 = 30$: true, but no letter left to find. These equations are **not independent**. Two equations that contradict each other, like $x + 2y = 10$ and $x + 2y = 12$, are the parallel case: no answer at all.

Lecture 3 compounds: two receipts in disguise

Solution 1 has 80 g of C per litre, solution 2 has 25 g per litre. How much of each makes 1 litre with 60 g per litre?

Decoding it. Think of two bottles of squash, with C as the sugar. "80 g per litre" means every litre of bottle 1 has 80 g of sugar dissolved in it: strong. Bottle 2 has only 25 g per litre: weak. You need to fill a **1-litre jug** that contains exactly **60 g** of sugar. **How many ml do you pour from each bottle?** That's two unknowns.



Every ml of bottle 1 carries $80 \div 1000 = 0.08$ g; every ml of bottle 2 carries 0.025 g. So the two facts are two receipts:

$$x + y = 1000 \qquad (1)$$ ← the volumes add to 1 litre

$$0.08x + 0.025y = 60 \qquad (2)$$ ← the grams of C add to 60

Elimination, exactly as with the café: make the x parts match, then subtract.

$$0.08x + 0.08y = 80$$ ← (1) × 0.08

$$0.055y = 20 \;\Rightarrow\; y = \frac{20}{0.055} \approx 364$$ ← this − (2): the x's cancel

$$x = 1000 - 364 = 636$$ ← back into (1)

Check: $0.08 \times 636 + 0.025 \times 364 = 50.9 + 9.1 = 60$ g ✓. The slide uses **substitution** instead (rearrange (1) to $x = 1000 - y$ and put that into (2): $0.08(1000 - y) + 0.025y = 60$, so $80 - 0.055y = 60$); it gives the same answer. The slide prints $\frac{20}{0.55}$; it should be $\frac{20}{0.055}$, as the answer 364 shows.

Squared letters: two answers, and the zero trick

RULE: the zero product rule

If $A \times B = 0$, then $A = 0$ or $B = 0$. This works **only for 0**.

So **rearrange so one side is 0 BEFORE you factorise**. $(x-1)(x-2) = 6$ does **not** give $x - 1 = 6$: two numbers can multiply to 6 in endless ways.

PROCEDURE: solving a quadratic equation

1. **Rearrange to** $ax^2 + bx + c = 0$: expand any brackets, move every term to one side.
2. **Factorise** (Chapter 12). If it won't factorise, use the **quadratic formula** (next section).
3. Set **each bracket equal to 0** and solve each one.
4. **Check** each answer in the original equation, and reject any that don't fit the question.

$x^2 = 25$ has **two** solutions: $5 \times 5 = 25$ and also $(-5) \times (-5) = 25$ (Chapter 9 warned you). We write $x = \pm 5$, said "plus or minus five".

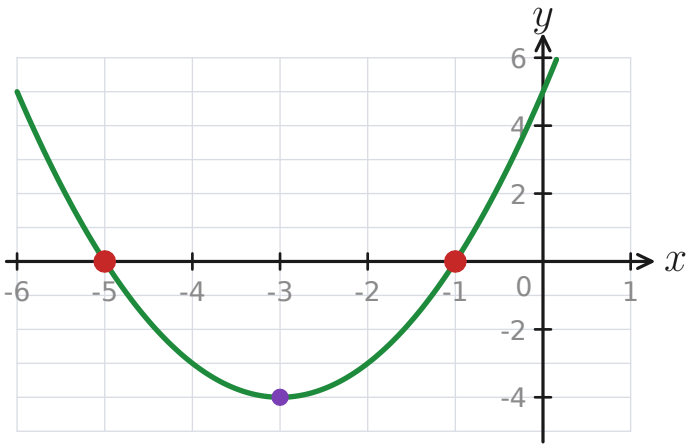
An equation with an x^2 is a **quadratic equation**. Lecture 14's example: $x^2 + 6x + 5 = 0$. The key fact is the **zero trick**: if two numbers multiply to give 0, one of them must be 0 (there's no other way to make 0 by multiplying).

$$x^2 + 6x + 5 = 0$$

$$(x+5)(x+1) = 0 \longleftarrow \text{factorise (Chapter 12: } \Delta = 16\text{, pair 5 and 1)}$$

$$x + 5 = 0 \quad \text{or} \quad x + 1 = 0 \longleftarrow \text{one bracket must be zero}$$

$$x = -5 \quad \text{or} \quad x = -1$$



The graph of $y = x^2 + 6x + 5$ is a U-shaped curve. The solutions are where it crosses the x-axis (where $y = 0$): $x = -5$ and $x = -1$.

The bottom of the U, $(-3, -4)$, is the vertex (two sections on).

Not zero yet: $(x-1)(x-2) = 6$. Expand and rearrange first:

$$x^2 - 3x + 2 = 6$$ ← *expand*

$$x^2 - 3x - 4 = 0$$ ← *−6 on both sides: now one side is 0*

$$(x-4)(x+1) = 0 \Rightarrow x = 4 \text{ or } x = -1$$ ← *factorise; each bracket = 0*

Check: $(3)(2) = 6$ ✓ and $(-2)(-3) = 6$ ✓.

✗ Common mistakes

✗ $(x-1)(x-2) = 6 \Rightarrow x - 1 = 6$

the zero trick only works when the other side is 0

✓ $x^2 - 3x - 4 = 0 \Rightarrow x = 4,\ -1$

✗ $x^2 = 3x \Rightarrow x = 3$

dividing by x loses x = 0: factorise instead

✓ $x(x-3) = 0 \Rightarrow x = 0,\ 3$

✗ $x^2 = 25 \Rightarrow x = 5$

a square root gives two answers

✓ $x = \pm 5$

The quadratic formula: straight from Chapter 12

Chapter 12 factorises **any** quadratic with its pair $m, n = \frac{b \pm \sqrt{\Delta}}{2}$, where $\Delta = b^2 - 4ac$. Chapter 12's grid formula gives the brackets. Now use the zero trick on them:

$$ax^2 + bx + c = \left(x + \frac{n}{a}\right)(ax + m) = 0$$ ← Chapter 12's brackets

$$x = -\frac{n}{a} \quad \text{or} \quad x = -\frac{m}{a}$$ ← one bracket must be zero

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$ ← put in m, n = (b ± √Δ)/2

That's the famous **quadratic formula**. It is Chapter 12's pair with the signs flipped and divided by a: nothing new to learn. Treat it like a recipe card:

RECIPE: solving ax² + bx + c = 0

Ingredients:

a = number in front of x^2
b = number in front of x
c = the plain number

Method:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Lecture 14's example: $x^2 + 3x + 2 = 0$. Ingredients: $a = 1$, $b = 3$, $c = 2$. Work out Δ first, as in Chapter 12:

$$\Delta = 3^2 - 4 \times 1 \times 2 = 1$$ ← a perfect square: it would factorise too

$$x = \frac{-3 \pm 1}{2} = -1 \quad \text{or} \quad -2$$ ← + version, then − version

Where it earns its keep: $2x^2 - 4x - 3 = 0$. $a = 2$, $b = -4$, $c = -3$. **Careful:** $-b$ is $-(-4) = +4$.

$$\Delta = (-4)^2 - 4 \times 2 \times (-3) = 16 + 24 = 40$$

not a square: no neat brackets

$$x = \frac{4 \pm \sqrt{40}}{4} \approx 2.58 \quad \text{or} \quad -0.58 \leftarrow \sqrt{40} \approx 6.32$$

So: Δ a perfect square, factorising and the formula both work. Not a square, the formula is the quick route.

Completing the square: where the U bottoms out

RULE: completing the square

$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c.$

In words: **halve the number in front of x**, put it in a squared bracket, then **subtract that half squared** (the bracket adds it on, so take it off again) and keep c.

If $a \neq 1$: take a out of the x^2 and x terms first, complete the square inside, then multiply the a back in.

The result $a(x+p)^2 + q$ is **vertex form**: the vertex (turning point) is at $(-p,\ q)$.

It's worth knowing because it shows the bottom of the U. Lecture 14's example:

$$x^2 + 6x + 5 = (x+3)^2 - 9 + 5 = (x+3)^2 - 4$$

half of 6 is 3; $3^2 = 9$

When a is not 1: $2x^2 + 12x + 7$ (Lecture 14 uses this vertex form).

$$= 2(x^2 + 6x) + 7 \leftarrow \text{take 2 out of the x terms only}$$

$$= 2\left[(x+3)^2 - 9\right] + 7 \leftarrow \text{complete the square inside}$$

$$= 2(x+3)^2 - 18 + 7 = 2(x+3)^2 - 11 \leftarrow \text{multiply the 2 back in}$$

Vertex $(-3, -11)$. **Check** $x = 0$: 7 and $2(9) - 11 = 7$ ✓.

Solving by completing the square: $x^2 + 4x - 1 = 0$ won't factorise, but:

$$(x+2)^2 - 4 - 1 = 0 \ \Rightarrow\ (x+2)^2 = 5 \leftarrow \text{half of 4 is 2}$$

$$x + 2 = \pm\sqrt{5} \;\Rightarrow\; x = -2 \pm \sqrt{5}$$ ← square root: ±

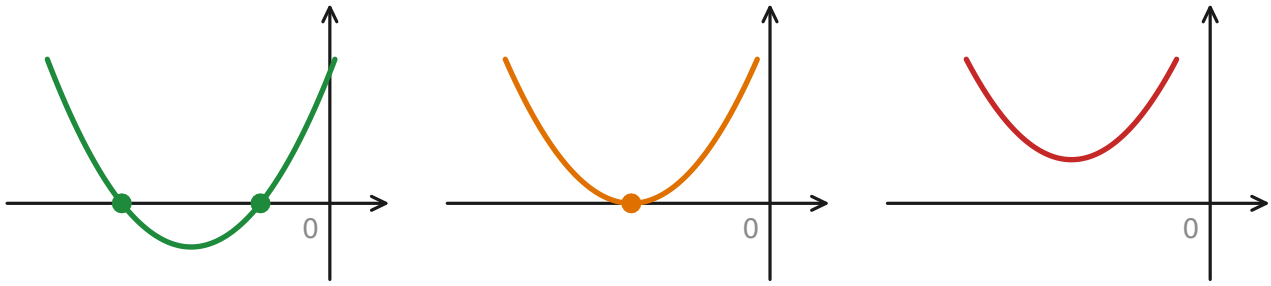
A square is never negative, so $(x+3)^2 - 4$ is smallest when the bracket is 0: at $x = -3$, where it's -4. That point, $(-3, -4)$, is the **vertex**. It also solves the equation by pure undoing: $(x+3)^2 = 4$, so $x + 3 = \pm 2$ and $x = -1$ or -5, the same answers as before.

How many answers? Look at Δ

The part under the root, $\Delta = b^2 - 4ac$, is Chapter 12's test number (the **discriminant**). It decides the brackets there, and the number of answers here. Take three neighbours that differ only in the last number:

✎ YOUR GO (pencil first)

$y = x^2 + 4x + 3$, $y = x^2 + 4x + 4$ and $y = x^2 + 4x + 5$ are three U-shaped graphs. Work out Δ for each, then predict how many times each U crosses the x-axis (where $y = 0$) before you look below.



$x^2 + 4x + 3 :\ \Delta = 4$ — **two crossings**

$x^2 + 4x + 4 :\ \Delta = 0$ — **one touch**

$x^2 + 4x + 5 :\ \Delta = -4$ — **no crossings**

Positive: the formula's $\pm$ gives two different answers, and the U crosses twice. Zero: $+0$ and -0 are the same, so one answer, and the U just touches. Negative: there is no real square root (Chapter 9), so no **real** answers, and the U floats above (or below) the axis. Chapter 12's table of cases is the same fact.

$\Delta > 0$	$\Delta = 0$	$\Delta < 0$
two answers	**one answer**	**no real answers**

Find the unknown: partial information about a quadratic

Exam questions often hide a letter (usually k) in the equation and give you **one fact** about the answers. Each fact turns into an equation for k:

RULE: what each fact means

"**Equal roots**" (or "one repeated root", "touches the x-axis"): $\Delta = b^2 - 4ac = 0$.

"**Two different real roots**": $\Delta > 0$. "**No real roots**": $\Delta < 0$.

"**x = r is a root**": put $x = r$ into the equation; it must come out true. Solve for k.

Then answer the rest (e.g. the other root) by solving the now-complete equation.

Example 1: find k so that $x^2 + kx + 9 = 0$ has equal roots.

$$\Delta = k^2 - 4 \times 1 \times 9 = k^2 - 36$$ ← $a = 1,\ b = k,\ c = 9$

$$k^2 - 36 = 0 \ \Rightarrow\ k^2 = 36 \ \Rightarrow\ k = \pm 6$$ ← equal roots: $\Delta = 0$

Check $k = 6$: $x^2 + 6x + 9 = (x + 3)^2$, one repeated root $x = -3$ ✓. ($k = -6$ gives $(x - 3)^2$.)

Example 2: one root of $x^2 + kx - 10 = 0$ is $x = 2$. Find k and the other root.

$$2^2 + 2k - 10 = 0$$ ← $x = 2$ makes it true

$$2k - 6 = 0 \ \Rightarrow\ k = 3$$ ← solve for k

$$x^2 + 3x - 10 = (x - 2)(x + 5) = 0$$ ← now factorise the full equation

$$\text{other root: } x = -5$$

Check: $(-5)^2 + 3(-5) - 10 = 25 - 15 - 10 = 0$ ✓. (The $(x - 2)$ bracket had to be there, because $x = 2$ is a root.)

Quadratics from words, and x on the bottom

RULE: rejecting answers

After solving, test every answer against the **question**: a length, time, mass or count can't be negative, so reject a negative answer there (say why).

Reject any answer that makes a **denominator 0** in the original equation.

Word problem: a rectangle is 3 cm longer than it is wide, and its area is 40 cm². Find its size.

$$x(x+3) = 40$$ ← width x, length x + 3; area = width × length

$$x^2 + 3x - 40 = 0$$ ← expand, then rearrange to 0

$$(x+8)(x-5) = 0 \;\Rightarrow\; x = -8 \text{ or } x = 5$$ ← pair 8, −5

$$x = 5 : \; 5 \text{ cm by } 8 \text{ cm}$$ ← reject −8: a width can't be negative

Check: $5 \times 8 = 40$ ✓, and 8 is 3 more than 5 ✓.

x on the bottom: $x + \frac{6}{x} = 5$. Banned: $x = 0$. Multiply every term by x (Chapter 13):

$$x^2 + 6 = 5x \;\Rightarrow\; x^2 - 5x + 6 = 0$$ ← × x, then rearrange to 0

$$(x-2)(x-3) = 0 \;\Rightarrow\; x = 2 \text{ or } 3$$ ← neither is banned

Check: $2 + 3 = 5$ ✓ and $3 + 2 = 5$ ✓.

A banned answer: $\frac{x^2}{x-1} = \frac{1}{x-1}$. Banned: $x = 1$.

$$x^2 = 1 \;\Rightarrow\; x = \pm 1$$ ← × (x − 1)

$$x = -1 \text{ only}$$ ← x = 1 makes the bottom 0: reject it

Check: $\frac{1}{-2} = \frac{1}{-2}$ ✓.

Inequalities: solve like an equation, with one twist

RULE: solving an inequality

Use the same moves as for an equation: add or subtract anything on both sides; multiply or divide both sides by a **positive** number. The sign stays the same.

Multiplying or dividing both sides by a NEGATIVE number flips the sign: $<$ becomes $>$, $\leq$ becomes $\geq$ (and the other way round).

The answer is a **range** of values. Show it on a number line: **open circle** for $<$ or $>$ (the end is not included), **filled circle** for $\leq$ or $\geq$ (it is).

Check with one number from inside the range and one from outside.

Why the flip? $2 < 5$ is true. Multiply both sides by -1: -2 and -5. Now -2 is the **bigger** one, so $-2 > -5$. Multiplying by a negative reverses the order of numbers.

Example 1: $3x + 2 < 14$.

$$3x < 12 \;\Rightarrow\; x < 4$$ ← −2, then ÷3 (positive: no flip)

open circle: 4 itself is not included

Example 2: $5 - 2x \geq 11$.

$$-2x \geq 6$$ ← −5 on both sides

$$x \leq -3$$ ← ÷ (−2): FLIP ≥ to ≤

filled circle: −3 is included

Check: $x = -4$ (inside): $5 + 8 = 13 \geq 11$ ✓. $x = 0$ (outside): $5 \geq 11$ is false ✓.

Example 3 (two-sided): $-1 < 2x + 3 \leq 9$. Do each move to **all three** parts:

$$-4 < 2x \leq 6 \;\Rightarrow\; -2 < x \leq 3$$ ← −3, then ÷2

Solving rules

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Balance	do the same to both sides; check the answer in the original
Linear equations	clear brackets and fractions, letters to one side, divide, check
Cross-multiplying	$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$
Elimination	scale one or both equations to match a letter; **same signs subtract, opposite signs add**
Substitution	rearrange one equation for a letter, substitute into the other, solve, back-substitute, check both
Zero product	$AB = 0 \Rightarrow A = 0$ or $B = 0$ **rearrange to = 0 before factorising**
Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ for $ax^2 + bx + c = 0$
Discriminant	$\Delta = b^2 - 4ac$ > 0 two roots, $= 0$ equal roots, < 0 no real roots
Given a root	substitute it; it must make the equation true
Completing the square	$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$ $a \neq 1$: take a out of the x terms first; vertex at $(-p, q)$
Rejecting answers	negative lengths/times; values making a denominator 0
Inequalities	as equations, but **× or ÷ by a negative flips the sign**; open circle $<$, filled $\leq$

Balance scales: your turn

Answers in grey. Put every answer back into the question to check it.

One unknown

1. $4x - 7 = 13$
2. $6x + 1 = 2x + 17$
3. $5(x - 3) = 2x + 3$
4. $(x + 4)/3 = (x - 2)/2$
5. How much 10% solution must be added to 40 ml of 30% solution to make 20%?

1. x = 5 2. x = 4 3. x = 6 4. x = 14 5. 40 ml

Two unknowns

6. $x + y = 10,\ x - y = 4$
7. $2x + 3y = 13,\ x + y = 5$
8. 3 pens and 2 pads cost £7; 1 pen and 2 pads cost £5. Find each price.
9. Is $x + y = 3$, $2x + 2y = 6$ one answer, none, or endless?
10. Compounds 3: how much of the 80 g/L and 25 g/L solutions make 70 ml at 45 g/L?

6. x = 7, y = 3 7. x = 2, y = 3 8. pen £1, pad £2 9. endless (same line)
10. about 25.5 ml and 44.5 ml

Quadratics (Δ first)

11. $x^2 = 49$
12. $x^2 - 5x + 6 = 0$
13. $x^2 + 2x - 8 = 0$
14. $x^2 = 7x$
15. Complete the square: $x^2 + 8x + 10$
16. $x^2 + 4x + 1 = 0$ (2 d.p.)

11. ±7 12. Δ = 1: x = 2 or 3 13. Δ = 36: x = 2 or −4 14. x = 0 or 7 15. $(x + 4)^2 - 6$
16. Δ = 12: −0.27 or −3.73

Methods, words and inequalities

17. Use substitution: $y = 3x - 1$ and $2x + y = 9$
18. Use elimination (scale both): $2x + 3y = 8$, $3x + 2y = 7$
19. Solve $(x + 2)(x - 3) = 6$
20. Complete the square: $3x^2 - 12x + 5$
21. Solve $7 - 3x > 1$ and show it on a number line
22. A rectangle is 2 m longer than it is wide, with area 48 m². Width?
23. Solve $2/x + x = 3$

17. $x = 2$, $y = 5$ 18. $x = 1$, $y = 2$ 19. $x^2 - x - 12 = 0$: $x = 4$ or -3 20. $3(x - 2)^2 - 7$
21. $x < 2$ (open circle) 22. 6 m (reject -8) 23. $x = 1$ or 2

Find the unknown

24. Find k so that $x^2 + kx + 16 = 0$ has equal roots.
25. One root of $x^2 + kx - 12 = 0$ is 3. Find k and the other root.
26. For which k does $x^2 + 6x + k = 0$ have no real roots?
27. $x = -2$ is a root of $2x^2 + 5x + k = 0$. Find k and the other root.
28. $ax + 2y = 11$ and $x - y = 2$ have the solution $x = 3$. Find y and a.
29. $kx^2 + 4x + 1 = 0$ has equal roots. Find k.

24. $k = \pm 8$ ($k^2 - 64 = 0$) 25. $k = 1$, other root -4 ($9 + 3k - 12 = 0$) 26. $k > 9$ ($36 - 4k < 0$)
27. $k = 2$, other root $-1/2$ ($8 - 10 + k = 0$) 28. $y = 1$, $a = 3$ ($3a + 2 = 11$)
29. $k = 4$ ($16 - 4k = 0$)

Chapter 16

Matrices

a spreadsheet you can do sums with

	bread	milk	eggs
North shop	12	30	6
South shop	8	25	10

drop the labels, keep the grid

A **matrix** is a table of numbers in rows and columns, like a spreadsheet with the labels rubbed out. The North shop sold 12 loaves, 30 milks and 6 boxes of eggs; the South shop sold 8, 25 and 10. Keeping the numbers in their grid positions lets you do arithmetic on the whole table at once.

Your module says matrix **methods** are not assessed, but **solving simultaneous equations** is, and Lecture 12 solves them with matrices. So this chapter builds up just enough to follow that: reading a matrix, adding, multiplying (the only odd one), and the inverse, which is an undo button.

Goes with **Lecture 12** (matrices and systems of equations).

Reading a matrix: rows, columns, and seat numbers

Rows run across (like rows of seats in a cinema). **Columns** run up and down (like the columns holding up a building). The **size** is always said **rows first, then columns**:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

2 × 2
2 rows, 2 columns

$$\begin{bmatrix} 12 & 30 & 6 \\ 8 & 25 & 10 \end{bmatrix}$$

row 1 highlighted

2 × 3
2 rows, 3 columns

$$\begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$$

3 × 2
3 rows, 2 columns

Seat numbers. Each number in a matrix is an **entry** (or element). Its address is written with two small numbers below the letter, a **subscript**: a_{ij} means the entry in **row i, column j**, like "row C, seat 7".

$$\begin{bmatrix} 12 & 30 & 6 \\ 8 & 25 & 10 \end{bmatrix}$$

a_{23} = 10: go down to row 2, then across to column 3.

a_{12} = 30 (row 1, column 2)

Memory hook: "RC", like a remote-controlled car: Rows before Columns, for sizes and for addresses.

Adding, subtracting and scaling: cell by cell

RULE: adding, subtracting, scaling and equal matrices

Add or subtract only matrices of the **same size**: combine the entries in the same position. Different sizes: no answer.

Scalar multiplication kA: multiply **every** entry by the number k (a whole number, a negative or a fraction).

Equal matrices: $A = B$ means the same size and **every** pair of matching entries equal. So one matrix equation gives one ordinary equation per entry.

Week 1 and week 2 sales from the two shops. Total sales? Add each cell to the **matching cell**:

$$\begin{bmatrix} 12 & 30 & 6 \\ 8 & 25 & 10 \end{bmatrix} + \begin{bmatrix} 10 & 28 & 4 \\ 9 & 20 & 7 \end{bmatrix} = \begin{bmatrix} 22 & 58 & 10 \\ 17 & 45 & 17 \end{bmatrix}$$

12 + 10 = 22, 30 + 28 = 58, …

Lecture 12's picture of this: $\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} \text{pizza} & \text{kebab} \\ \text{curry} & \text{soup} \end{bmatrix}$ puts a + pizza top-left, and so on. Things only combine with the thing in the **same seat**. So both matrices must be the **same size**; a 2 × 3 plus a 2 × 2 has no answer.

$$\begin{bmatrix} 2 & 3 & -1 \\ 7 & 0 & 4 \end{bmatrix} - \begin{bmatrix} -3 & 4 & 2 \\ 1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 & -1 & -3 \\ 6 & 0 & 2 \end{bmatrix}$$

the slide's subtraction: 2 − (−3) = 5

Scaling (multiplying by one plain number, called a **scalar**): every cell gets multiplied. Four weeks at the same rate:

$$4 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 12 & 16 \end{bmatrix}$$ ← *4 × every entry*

A fraction works the same way. Multiplying by $\frac{1}{3}$ is dividing every entry by 3:

$$\frac{1}{3} \begin{bmatrix} 3 & 6 \\ -9 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -3 & \frac{1}{3} \end{bmatrix}$$ ← *every entry ÷ 3; leave 1/3 as a fraction*

Equal matrices. If $\begin{bmatrix} x+1 & 4 \\ 3 & 2y \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 3 & 10 \end{bmatrix}$, match the entries seat by seat:

$$x+1=5 \Rightarrow x=4, \qquad 2y=10 \Rightarrow y=5$$

top-left and bottom-right seats

Multiplying: row meets column

RULE: multiplying two matrices

The entry in **row i, column j** of AB = **row i of A** paired with **column j of B**: multiply first with first, second with second, ..., then **add**.

Do this for every row of A with every column of B. It is **not** cell-by-cell.

Multiplying two matrices is **not** cell by cell. It comes from a shopping bill. You buy 2 coffees and 3 cakes; a coffee costs £2.50 and a cake £2. Write the order as a **row** and the prices as a **column**:

order $\begin{bmatrix} 2 & 3 \end{bmatrix}$ prices $\begin{bmatrix} 2.5 \\ 2 \end{bmatrix} = 2 \times 2.5 + 3 \times 2 = 11$

first with first, second with second, then add: the bill is £11

That's the whole rule: pair up, multiply, add. For bigger matrices you just do it for every row of the first with every column of the second. Lecture 12's example, traced with two highlighters (yellow row, blue column):

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \to \begin{bmatrix} 19 & \\ & \end{bmatrix}$

row 1 × column 1: 1×5 + 2×7 = 19

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \to \begin{bmatrix} 19 & 22 \\ & \end{bmatrix}$

row 1 × column 2: 1×6 + 2×8 = 22

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \to \begin{bmatrix} 19 & 22 \\ 43 & \end{bmatrix}$

row 2 × column 1: 3×5 + 4×7 = 43

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \to \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$

row 2 × column 2: 3×6 + 4×8 = 50

Slide check: Lecture 12 writes the bottom-left entry as $3 \times 5 + 4 \times 8 = 47$. Row 2 of A is (3, 4) and column 1 of B is (5, 7), so it's $3 \times 5 + 4 \times 7 = 15 + 28 = 43$. The full answer is $\begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$.

Which sizes can multiply? Dominoes

RULE: the size rule

$(m \times n)(n \times p) = (m \times p)$.

AB exists only when the **number of columns of A** equals the **number of rows of B** (the two inside numbers match). The answer has the rows of A and the columns of B (the outside numbers).

Check the sizes **before** multiplying. AB can exist while BA doesn't, and even when both exist, $AB \neq BA$ in general.

Row-meets-column only works if a row of the first is **as long as** a column of the second (every item needs a price). Write each size as a domino and put them side by side: the **touching numbers must match**, and the **outside numbers give the answer's size**.

→ **2 × 1**

→ **1 × 1 (a single number, like the bill)**

→ **3 ≠ 2: can't be done**

Lecture 12's "different size" example: $\begin{bmatrix} 1 & 4 \\ 6 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$ is (2 × 2)(2 × 1), so the answer is 2 × 1:

$$\begin{bmatrix} 1 \times 2 + 4 \times 5 \\ 6 \times 2 + 3 \times 5 \end{bmatrix} = \begin{bmatrix} 22 \\ 27 \end{bmatrix}$$

← *each row meets the one column*

Swap them round and it fails: $\begin{bmatrix}2\\5\end{bmatrix}\begin{bmatrix}1&4\\6&3\end{bmatrix}$ is (2 × 1)(2 × 2); the inside numbers 1 and 2 don't match, so this product **does not exist**. Same two matrices, one order works and the other doesn't.

$$(2\times 3)(3\times 4)=2\times 4, \qquad (3\times 4)(2\times 3):\ 4\neq 2,\ \text{undefined}$$

Order matters: CD is not DC

RULE: order of multiplication

In general $AB\neq BA$: matrix multiplication is **not commutative**. Never swap the order of a product to make it easier.

"Multiply A by B on the left" means BA; "on the right" means AB.

With ordinary numbers $3\times 5=5\times 3$. With matrices, swapping the order usually changes the answer. Lecture 12's "give it a try":

$$C=\begin{bmatrix}7&8\\2&-1\end{bmatrix}, \quad D=\begin{bmatrix}3&2\\-1&4\end{bmatrix}$$

$$CD=\begin{bmatrix}7(3)+8(-1)&7(2)+8(4)\\2(3)+(-1)(-1)&2(2)+(-1)(4)\end{bmatrix}=\begin{bmatrix}13&46\\7&0\end{bmatrix}$$

$$DC=\begin{bmatrix}3(7)+2(2)&3(8)+2(-1)\\-1(7)+4(2)&-1(8)+4(-1)\end{bmatrix}=\begin{bmatrix}25&22\\1&-12\end{bmatrix}$$

Completely different! Think of socks and shoes: socks-then-shoes is not the same as shoes-then-socks. (The slide's heading says "calculate CD" again, but it's working out DC.)

The matrix that acts like 1

RULE: the identity matrix

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (1s on the leading diagonal, 0s elsewhere).

$AI = IA = A$ for any square matrix A of the same size: I behaves like the number 1.

A **diagonal** matrix has numbers only on the diagonal from top-left to bottom-right (zeros elsewhere). The **identity** matrix I is the diagonal matrix with every diagonal entry 1:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$ ← *unchanged!*

Multiplying by I changes nothing, just like multiplying a number by 1. Check the top-left entry: row (1, 2) meets column (1, 0): $1 \times 1 + 2 \times 0 = 1$. The zeros make sure each entry only "sees" itself.

Simultaneous equations as one matrix sum

Here's why matrices are in a course about equations. Multiply a 2 × 2 by a column of letters (Lecture 12):

$$\begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3x + 2y \\ 5x + 3y \end{bmatrix}$$ ← *row meets column, with letters*

Those are the left-hand sides of two equations! So the pair $x + 2y = 13$ and $2x - 5y = 8$ can be written as **one** matrix equation:

$$\begin{bmatrix} 1 & 2 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 13 \\ 8 \end{bmatrix} \qquad AX = B$$

A = numbers in front *X = unknowns* *B = right-hand sides*

It looks just like $3x = 12$, one equation with one unknown. To solve $3x = 12$ you'd undo the "× 3". So we need an **undo button for a matrix**.

Honest note: for two equations, the Chapter 15 methods are quicker by hand. The matrix way wins for computers and for big systems, like Lecture 12's joke equation with 12 unknowns (which needs 12 independent equations).

The inverse: an undo button (2 × 2 recipe)

RULE: determinant and inverse of a 2 × 2 matrix

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ the **determinant** is $\det A = ad - bc$.

If $\det A = 0$ the matrix is **singular**: it has **no inverse**.

If $\det A \neq 0$: $A^{-1} = \dfrac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (swap a and d, change the signs of b and c, divide by the determinant).

The inverse satisfies $AA^{-1} = A^{-1}A = I$. Multiply out to check an inverse.

Only square matrices can have inverses.

For numbers, the undo for "× 3" is "× $\frac{1}{3}$" because $\frac{1}{3} \times 3 = 1$. For a matrix A, the undo is written A^{-1} ("A inverse") and it satisfies $A^{-1}A = I$. Lecture 12 gives the recipe for a 2 × 2:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \longrightarrow A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

1. swap the two diagonal entries (a and d) **2.** change the sign of the other two **3.** divide everything by $ad - bc$

The number $ad - bc$ is called the **determinant**. If it's 0 you'd divide by zero, so there's **no inverse**. That happens exactly when the two equations aren't independent (Chapter 15: the same line, or parallel lines).

PROCEDURE: solving AX = B

1. Write the equations as $AX = B$ (A = coefficients, X = unknowns, B = right-hand sides).
2. Find $\det A$. If it is 0, stop: no unique solution.
3. Find A^{-1} with the recipe.
4. $X = A^{-1}B$, with A^{-1} on the **left**: multiply both sides on the left by A^{-1}, and $A^{-1}AX = IX = X$. (**Not** BA^{-1}: order matters, and here it doesn't even exist.)
5. Check the answers in the original equations.

Solve $AX = B$ with $A = \begin{bmatrix} 1 & 2 \\ 2 & -5 \end{bmatrix}$. Tiny steps:

$$ad - bc = (1)(-5) - (2)(2) = -9$$ ← the determinant (not 0: good)

$$A^{-1} = \frac{1}{-9}\begin{bmatrix} -5 & -2 \\ -2 & 1 \end{bmatrix}$$ ← swap 1 and −5; flip signs of the 2s

$$X = A^{-1}B$$ ← undo A on both sides (inverse × A = I, and IX = X)

$$= \frac{1}{-9}\begin{bmatrix} (-5)(13) + (-2)(8) \\ (-2)(13) + (1)(8) \end{bmatrix} = \frac{1}{-9}\begin{bmatrix} -81 \\ -18 \end{bmatrix}$$ ← row meets column

$$= \begin{bmatrix} 9 \\ 2 \end{bmatrix} \Rightarrow x = 9,\ y = 2$$ ← ÷ (−9)

Check in the original equations: $9 + 2(2) = 13$ ✓ and $2(9) - 5(2) = 8$ ✓.

You can't divide by a matrix. $X = \frac{B}{A}$ means nothing. The "undo" is multiplying by A^{-1}, and because order matters you must say which side. AX has A on the left of X, so A^{-1} goes on the left too. Here B is 2 × 1 and A^{-1} is 2 × 2, so BA^{-1} is (2 × 1)(2 × 2): it doesn't exist.

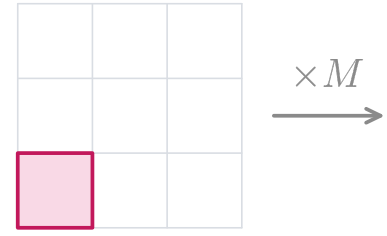
Check an inverse: multiply it by the original and you must get I. Is $\begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix}$ the inverse of $\begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix}$?

$$\begin{bmatrix} 4 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -3 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 4-3 & -12+12 \\ 1-1 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

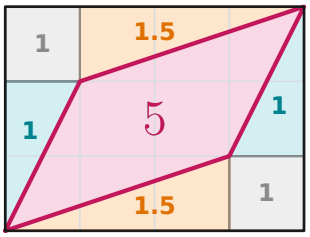
= I, so yes

What the determinant measures: area

$ad - bc$ looks like a random recipe, but it has a shape. Multiply a matrix by the corner points of a 1 by 1 square, $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and they land on the matrix's two **columns**. For $M = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ the square gets pushed over into a slanted shape (a parallelogram):



area 1



the box around it: 4 × 3 = 12

Its area? Take the 4 × 3 box (12) and cut off the corner pieces: 1.5 + 1.5 + 1 + 1 + 1 + 1 = 7.

12 − 7 = 5, and ad − bc = 3×2 − 1×1 = 5 too!

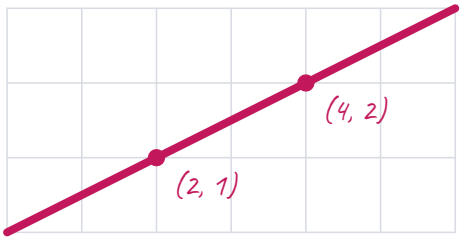
That always happens. With letters the box is $(a+b)$ by $(c+d)$, the orange triangles make one $a \times c$ rectangle, the teal ones one $b \times d$, and there are two grey $b \times c$ rectangles:

$$(a+b)(c+d) - ac - bd - 2bc = ad - bc$$

the box, minus the corners

So the **determinant is how much the matrix stretches area**: every shape comes out 5 times bigger under M. A minus sign means the square also got **flipped over**, like a mirror image: the inverse example's determinant of -9 means "9 times the area, and flipped".

And why 0 means no undo button. Try $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$: $ad - bc = 4 - 4 = 0$. Its columns point the same way, so the square gets squashed completely flat:

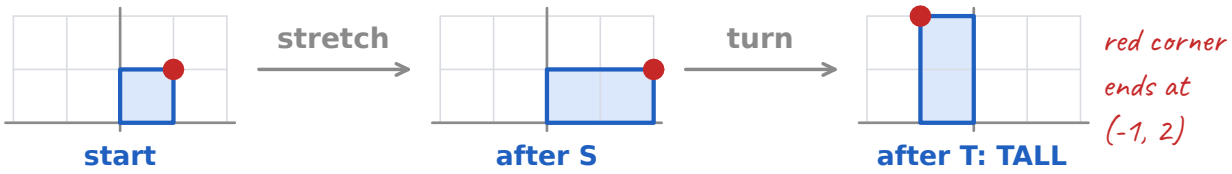


The whole square lands on one line: area 0. Many different points land on the same spot, so there's no way to tell where each came from. You can't un-squash a line back into a square, so there is no inverse. That's the "divide by zero" warning, as a picture.

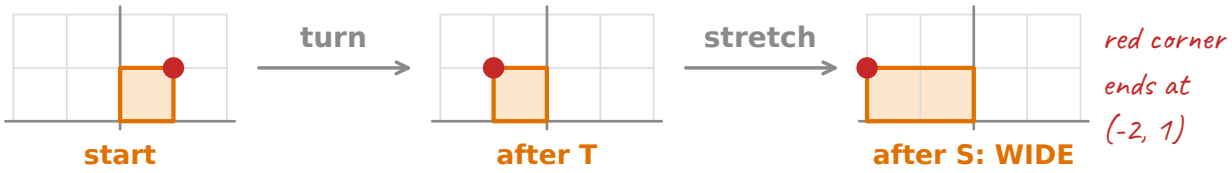
Order matters, seen: stretch then turn, or turn then stretch?

A matrix moves every point, so two matrices in a row are two moves in a row. Take $S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ (**stretch** sideways ×2) and $T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (**turn** a quarter turn anticlockwise). Follow the square and its **red corner**:

stretch first, then turn



turn first, then stretch



Same two moves, different order, different result: one ends up tall, the other wide. **Watch the order of the letters:** "S first, then T" is written TS, because the matrix nearest the point acts first ($TS\,\mathbf{v} = T(S\mathbf{v})$). That's exactly why $CD \neq DC$.

✎ YOUR GO (pencil first)

To UNDO "stretch, then turn" (get the tall shape back to the start square), which move do you undo first: the stretch or the turn?

Answer: undo the **last** move first. Turn back, then un-stretch, just as you take off your shoes before your socks. So the inverse reverses the order:

$$(TS)^{-1} = S^{-1}T^{-1}$$ ← *undo the turn first, then the stretch*

Find the unknown: missing entries

PROCEDURE: partial-information matrix problems

1. Turn each fact into ordinary equations. "No inverse" or "singular" means $ad - bc = 0$. "$AB = C$" means multiply out AB and match it entry by entry with C. "B is the inverse of A" means $AB = I$.
2. Solve those equations for the unknowns.
3. Substitute back and check (work out the determinant or the product again).

Example: find k so that $\begin{bmatrix} k & 4 \\ 3 & 6 \end{bmatrix}$ has no inverse.

$$\det = 6k - 4 \times 3 = 6k - 12 \quad \leftarrow \text{ad – bc}$$

$$6k - 12 = 0 \;\Rightarrow\; k = 2 \quad \leftarrow \text{no inverse: determinant 0}$$

Check: $\begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}$: $12 - 12 = 0$ ✓. Its second row is 1.5 times the first: the two equations it stands for are not independent.

Example: for which values of k is $\begin{bmatrix} k & 2 \\ 8 & k \end{bmatrix}$ singular?

$$k \times k - 2 \times 8 = k^2 - 16 = 0 \quad \leftarrow \text{ad – bc = 0}$$

$$k^2 = 16 \;\Rightarrow\; k = \pm 4 \quad \leftarrow \text{both square roots count}$$

Example: $\begin{bmatrix} a & 1 \\ 2 & b \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix}$. Find a and b.

$$\begin{bmatrix} 3a + 1 \\ 6 + b \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix} \quad \leftarrow \text{multiply out the left side}$$

$$3a + 1 = 7 \Rightarrow a = 2, \qquad 6 + b = 10 \Rightarrow b = 4$$

match the entries

Example: $\begin{bmatrix} 2 & p \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 4 & 1 \\ 1 & q \end{bmatrix} = \begin{bmatrix} 11 & 8 \\ 7 & 7 \end{bmatrix}$. Find p and q.

row 1, col 1: $8 + p = 11 \;\Rightarrow\; p = 3$

row 1, col 2: $2 + pq = 8 \;\Rightarrow\; 2 + 3q = 8 \;\Rightarrow\; q = 2$

use p = 3

row 2: $4 + 3 = 7, \quad 1 + 3q = 1 + 6 = 7$ ← *check the other two entries ✓*

Example: $\begin{bmatrix} 2 & -1 \\ k & 3 \end{bmatrix}$ is the inverse of $\begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$. Find k.

row 2 of A with col 1: $5(2) + 2k = 0$ ← *AB = I, so the bottom-left entry is 0*

$10 + 2k = 0 \;\Rightarrow\; k = -5$

Check: with $k = -5$ the whole product is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ✓.

Mistakes to avoid

✗ Common mistakes

✗ $\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}\begin{bmatrix}5 & 6\\7 & 8\end{bmatrix} = \begin{bmatrix}5 & 12\\21 & 32\end{bmatrix}$ ✓ $= \begin{bmatrix}19 & 22\\43 & 50\end{bmatrix}$

multiplying is row meets column, not cell by cell

✗ $AB = BA$ ✓ $AB \neq BA$

order matters; never swap a product round

✗ $X = \dfrac{B}{A}$ ✓ $X = A^{-1}B$

there is no dividing by a matrix: multiply by the inverse

✗ $X = BA^{-1}$ ✓ $X = A^{-1}B$

the inverse goes on the same side as A (the left)

✗ $\det = ad + bc$ ✓ $\det = ad - bc$

it is a subtraction; and the inverse is not 1 over each entry

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Size	$m \times n$ rows × columns, rows first (also for addresses: a_{ij} is row i, column j)
Add / subtract	same size only; combine entries in the same position
Scalar multiple	$k\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$ every entry, including fractions like $\frac{1}{3}$
Equal matrices	same size and every matching entry equal: one equation per entry
Multiplying	entry (i, j) of AB = row i of A with column j of B: multiply pairs, add
Size rule	$(m \times n)(n \times p) = m \times p$ inside numbers must match, or the product doesn't exist
Order	$AB \neq BA$ in general; AB can exist when BA doesn't
Identity	$AI = IA = A$ I has 1s on the diagonal, 0s elsewhere
Determinant	$\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$ 0 means **singular**: no inverse
Inverse	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ check with $AA^{-1} = A^{-1}A = I$
Solving AX = B	$X = A^{-1}B$ A^{-1} on the left; never divide by a matrix
Undoing two moves	$(TS)^{-1} = S^{-1}T^{-1}$ undo the last move first

Spreadsheets: your turn

Answers in grey. Trace every multiplication with two fingers: one along the row, one down the column.

Reading

1. Size of [1 2 3]?
2. Size of a matrix with 4 rows and 2 columns?
3. In [[5, 1], [7, 9]], what is a_{21}?

1. 1 × 3 2. 4 × 2 3. 7

Adding and scaling

4. [[1, 2], [3, 4]] + [[0, 5], [−1, 2]]
5. 3 × [[2, −1], [0, 4]]
6. [[6, 3], [2, 1]] − [[1, 1], [1, 1]]

4. [[1, 7], [2, 6]] 5. [[6, −3], [0, 12]] 6. [[5, 2], [1, 0]]

Multiplying

7. [1 4] × [[2], [3]]
8. [[2, 1], [0, 3]] × [[1], [4]]
9. [[1, 2], [0, 1]] × [[3, 0], [1, 2]]
10. Can a 3 × 2 multiply a 3 × 2?

7. [14] 8. [[6], [12]] 9. [[5, 4], [1, 2]] 10. No: inner numbers 2 and 3 don't match

Inverse and equations

11. Determinant of [[4, 3], [2, 1]]
12. Inverse of [[2, 1], [5, 3]]
13. Solve 2x + y = 5, 5x + 3y = 13 using your answer to 12

11. 4 − 6 = −2 12. [[3, −1], [−5, 2]] (determinant 1) 13. x = 2, y = 1

Find the unknown

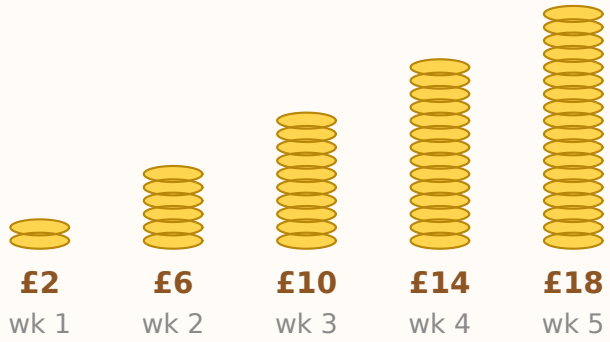
14. Find k so that [[k, 6], [2, 4]] has no inverse.
15. For which values of k is [[k, 3], [3, k]] singular?
16. [[2, a], [b, 1]] × [[1], [3]] = [[11], [7]]. Find a and b.
17. [[x + 2, 6], [1, 3y]] = [[5, 6], [1, −6]]. Find x and y.
18. [[2, −3], [−1, k]] is the inverse of [[k, 3], [1, 2]]. Find k.
19. Is [[1, −2], [−1, 3]] the inverse of [[3, 2], [1, 1]]? Show it.

14. k = 3 (4k − 12 = 0) 15. k = ±3 ($k^2 - 9 = 0$) 16. a = 3, b = 4 (2 + 3a = 11, b + 3 = 7) 17. x = 3, y = −2 (match entries) 18. k = 2 (top-left of the product: 2k − 3 = 1) 19. yes: the product is [[1, 0], [0, 1]] = I

Chapter 17

Sequences and series

patterns that march on, and adding them all up



Each week you save £4 more than the week before. How much after a year?

$$2 + 6 + 10 + \cdots = ?$$

A **sequence** is a list of numbers that follows a rule, like the savings above: 2, 6, 10, 14, ... A **series** is what you get when you **add** the list up. This chapter has two kinds: lists that go up by the **same amount** each time (arithmetic) and lists that get **multiplied** by the same amount each time (geometric).

For each kind you'll learn to find any term without writing out the whole list, and to add up hundreds of terms in one line. Along the way, a trick a ten-year-old found, a Python loop, a bouncing ball, and an endless sum that adds up to exactly 1.

Goes with **Lecture 13** (arithmetic and geometric progressions, summation notation).

Terms live in numbered lockers

Each number in a sequence is a **term**. Terms have positions: 1st, 2nd, 3rd... Picture a row of lockers, each with a number on the door (the position n) and a value inside:

n = 1	n = 2	n = 3	n = 4	n = 5	n = 6	
2	**6**	**10**	**14**	**18**	**22**	...

The value in locker n is often written u_n ("u sub n"). Here $u_1 = 2$, $u_4 = 14$. **Sequence** = the lockers. **Series** = the total of what's inside them: $2 + 6 + 10 + \ldots$

Lecture 13 says "series are lists". Strictly, a list is a sequence and a sum is a series; the slides use the words loosely, so don't let that throw you.

Arithmetic progressions: a staircase of equal steps

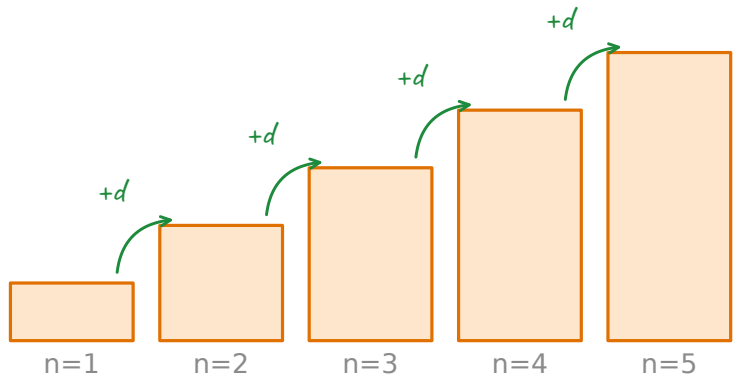
RULE: arithmetic progression (A.P.)

Each term = the previous term **+ d**. The common difference is $d = u_2 - u_1$ (any term minus the one before). a is the first term.

nth term: $u_n = a + (n-1)d$ (to reach term n you make $n-1$ jumps of d).

Any one of u_n, a, n, d can be found from the other three by rearranging.

When each term is the last one **plus a fixed amount**, the sequence is an **arithmetic progression** (A.P.). The fixed amount is the **common difference** d. The savings: $a = 2$ (first term), $d = 4$.



To reach step 5 you make only 4 jumps of d. (Like fence posts: 5 posts, 4 gaps between them.)

So the nth term is the first term plus $(n-1)$ jumps:

$$u_n = a + (n-1)d$$

$$u_8 = 2 + (8-1) \times 4 = 2 + 28 = 30$$ ← 8th week's saving: 7 jumps of 4

Working backwards. The same formula rearranges (Chapter 14) to answer the other questions Lecture 13 asks:

$$n = \frac{l-a}{d} + 1$$ ← how many terms, if the last term is l

$$d = \frac{l-a}{n-1}$$ ← the difference, if you know n

How many terms in 2, 6, ..., 50? The gap from 2 to 50 is 48, that's $48 \div 4 = 12$ jumps, so **13** terms (one more post than gaps). And the slide's check: 2, ..., 30 with 8 terms has $d = \frac{30-2}{8-1} = \frac{28}{7} = 4$ ✓.

Adding them up: Gauss's pairing trick

RULE: sum of an arithmetic progression

First n terms, when you know the last term l: $S_n = \frac{n}{2}(a + l)$.

When you know d instead: $S_n = \frac{n}{2}\big(2a + (n-1)d\big)$.

Both say: (average of the first and last term) × (number of terms). The reason is below.

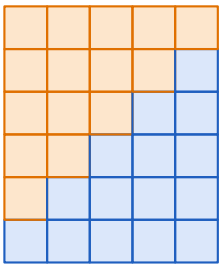
A famous story: a teacher told a class to add 1 + 2 + ... + 100, hoping for a quiet hour. Young Carl Gauss answered in seconds. He wrote the list forwards and backwards and added the columns:

1	**2**	**3**	**…**	**99**	**100**
100	**99**	**98**	**…**	**2**	**1**
101	**101**	**101**	**…**	**101**	**101**

every column makes 101!

100 columns of 101 is 10,100. But that's the list added **twice** (once each way), so the answer is half: **5050**.

The picture: a staircase of blocks, plus the same staircase upside down, makes a rectangle:



5 columns, each 6 tall

blue: 1 + 2 + 3 + 4 + 5 (the staircase)
orange: the same staircase, flipped
together: a 5 × 6 rectangle = 30 blocks
so the staircase alone is 30 ÷ 2 = 15

In general: **(first + last) is the same for every pair, there are n terms, and we halve**. Or in words: the **average term** (halfway between first and last) **times how many terms**:

$$S_n = \frac{a + l}{2} \times n$$

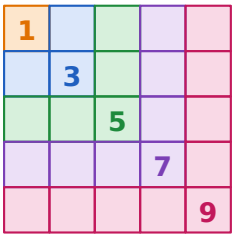
$$2 + 6 + 10 + 14 = \frac{2 + 14}{2} \times 4 = 32$$ ← *Lecture 13, example 1*

$$2 + 6 + \cdots + 50 = \frac{2 + 50}{2} \times 13 = 338$$ ← *example 2 (13 terms, from the last page)*

If you know n but not the last term, swap l for $a + (n-1)d$: $S_n = \frac{n}{2}\big(2a + (n-1)d\big)$. Slide example: 2 + 6 + 10 + …, 8 terms: $\frac{8}{2}(4 + 7 \times 4) = 4 \times 32 = 128$.

The piggy bank, answered: 52 weeks of 2, 6, 10, … is $\frac{52}{2}(4 + 51 \times 4) = 26 \times 208 = 5408$, so £5408.

A special staircase: the odd numbers. Add $1 + 3 + 5 + 7 + 9$. Each odd number is an L-shaped layer that wraps around the previous square and makes the next one:



each colour is one layer; its number sits in its corner

Each new layer adds one row and one column to the square, plus the shared corner: k + k + 1 = the next odd number.

So the first n odd numbers always add up to n^2.

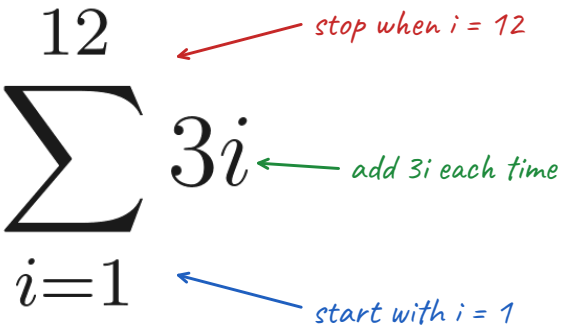
1 + 3 + 5 + 7 + 9 = 5 × 5

$$S_n = \frac{n}{2}\big(2 \times 1 + (n-1) \times 2\big) = \frac{n}{2} \times 2n = n^2$$

the A.P. formula agrees (a = 1, d = 2)

Σ is a for-loop

Writing $3 + 6 + 9 + \cdots + 36$ in full is tedious, so Lecture 13 uses the Greek capital **sigma**, Σ, meaning "add up". If you've written any code, it's a loop:




```
total = 0
for i in range(1, 13):
    total += 3 * i
print(total)   # 234
```

Read it as: "for i from 1 to 12, add $3i$": $3(1) + 3(2) + \cdots + 3(12)$. The letter i is just a counter; it disappears from the answer. Python's range stops **before** its second number, which is why the code says 13.

Lecture 13 writes the same list a second way, counting from 0: $\sum_{i=0}^{11}(3 + 3i) = 3 + 6 + \cdots + 36$. Different counter, same terms, same total. (It's an A.P., so $\frac{3+36}{2} \times 12 = 234$ ✓.)

Geometric progressions: a bouncing ball

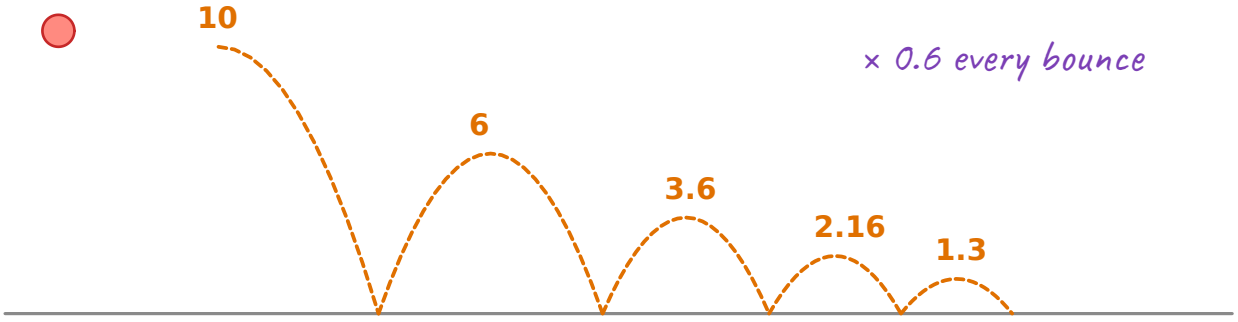
RULE: geometric progression (G.P.)

Each term = the previous term **× r**. The common ratio is $r = \dfrac{u_2}{u_1}$ (any term divided by the one before). a is the first term.

nth term: $u_n = ar^{n-1}$ ($n-1$ multiplications by r).

r can be a fraction (the terms shrink) or negative (the signs alternate).

Drop a ball from 10 m. Each bounce reaches 60% of the height before. The heights are 10, 6, 3.6, 2.16, ... Each term is the last one **multiplied** by the same number, 0.6:



That's a **geometric progression** (G.P.). The multiplier is the **common ratio** r. Find it by dividing any term by the one before: $6 \div 10 = 0.6$.

Term n has been multiplied by r exactly $(n-1)$ times (fence posts again):

$$u_n = ar^{n-1}$$

Lecture 13's example: 3, 6, 12, ..., 3072. Here $a = 3$, $r = 2$. Which term is 3072?

$$3 \times 2^{n-1} = 3072$$

$$2^{n-1} = 1024 = 2^{10}$$ ← ÷3; then keep doubling 1 until 1024: 10 doublings

$$n - 1 = 10 \Rightarrow n = 11$$ ← so 3072 is the 11th term

(The slide's footnote says 3072 is 3×2^{11}; that's 6144. It's 3×2^{10}, and the slide itself then correctly uses $n = 11$.)

Adding a G.P.: shift and subtract

RULE: sum of a geometric progression

First n terms: $S_n = \dfrac{a(1-r^n)}{1-r}$, or the same thing as $\dfrac{a(r^n-1)}{r-1}$ (tidier when $r > 1$).

Only for $r \neq 1$ (otherwise you divide by 0). If $r = 1$ every term is a, so $S_n = na$.

Pairing doesn't work here (the pairs aren't equal). Instead, use this trick: write the sum, then write it again **multiplied by** r. Almost everything lines up:

$$\begin{aligned} S &= a + ar + ar^2 + \ldots + ar^{n-1} \\ rS &= + ar + ar^2 + \ldots + ar^{n-1} + ar^n \end{aligned}$$

the blue parts match: subtracting wipes them out

$$S - rS = a - ar^n \quad \longleftarrow \text{top line minus bottom line}$$

$$S(1-r) = a(1-r^n) \quad \longleftarrow \text{factorise both sides (Chapter 12)}$$

$$S_n = \frac{a(1-r^n)}{1-r} \quad \longleftarrow \div (1-r)$$

Lecture 13: $3 + 6 + 12 + \cdots + 3072$, with $a = 3$, $r = 2$, $n = 11$:

$$S_{11} = \frac{3(1-2^{11})}{1-2} = \frac{3(1-2048)}{-1} = 6141$$

(The slide puts a $\sum$ in front of this formula. It doesn't need one: the formula already **is** the total.)

When $r = 1$ the formula gives $\frac{a \times 0}{0}$, which means nothing. But then the list is just $a, a, a, \ldots$, so add it directly: $4 + 4 + 4 + \ldots$ (10 terms) $= 10 \times 4 = 40$.

Negative ratio: $2 - 6 + 18 - 54 + 162$ has $a = 2$, $r = -3$, $n = 5$:

$$S_5 = \frac{2(1-(-3)^5)}{1-(-3)} = \frac{2(1+243)}{4} = 122$$

$(-3)^5 = -243$: keep the brackets

Walking halfway to the wall, forever

RULE: sum to infinity of a G.P.

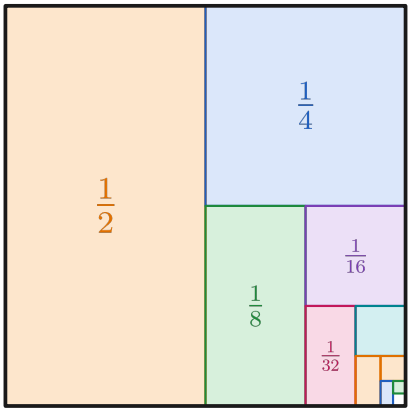
$S_\infty = \dfrac{a}{1-r}$, **only when** $-1 < r < 1$ (written $|r| < 1$). Then $r^n \to 0$ and the sums settle on this value: the series **converges**.

No sum to infinity when $r \geq 1$ or $r \leq -1$: the terms don't shrink, so the total grows without limit or swings about forever.

Stand 1 m from a wall. Walk half the distance, then half what's left, then half again... You walk $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots$ and never stop, yet you never pass the wall:

the pieces fill the gap closer and closer: the total heads to exactly 1 m

The same sum, folded into a square. Let a square have area 1. Shade half of it, then half of what's left, then half of that...



After 1/2 + 1/4 + 1/8 the white part is 1/8.
After 10 pieces it's 1/1024: a speck.

The pieces never spill outside the square, and the white gap halves every time. So the total can't be more than 1, and it gets as close to 1 as you like: it IS 1.

When the terms shrink (the ratio is between −1 and 1), r^n gets closer and closer to 0 as n grows ($0.5^{10} \approx 0.001$). So in the G.P. sum formula the r^n fades away:

$$S_\infty = \frac{a(1-0)}{1-r} = \frac{a}{1-r}$$ ← valid only if $-1 < r < 1$

The slides write the condition as $|r| < 1$: the bars mean "size, ignoring the sign" (the **modulus**). Wall walk: $a = \frac{1}{2}$, $r = \frac{1}{2}$: $\frac{1/2}{1-1/2} = 1$ ✓.

Slide check: Lecture 13 writes $S_\infty = \frac{a}{r-1}$. For the wall walk that gives $\frac{1/2}{-1/2} = -1$: adding up positive distances can't give a negative total! The bottom must be $1 - r$.

The ball, answered: it falls 10 m, then goes up and down each bounce height: $10 + 2(6 + 3.6 + 2.16 + \dots) = 10 + 2 \times \frac{6}{1-0.6} = 10 + 30 = 40$ m in total.

When there is no sum to infinity. If $r \geq 1$ (e.g. 3, 6, 12, ...) the total just keeps growing. If $r \leq -1$ the terms never shrink either: with $r = -1$, $1 - 1 + 1 - 1 + \dots$ flips between 1 and 0 forever; with $r = -2$, $1 - 2 + 4 - 8 + \dots$ swings wider and wider. Either way it never settles, so there is no S_∞. **Always check $-1 < r < 1$ before using $\frac{a}{1-r}$**: blindly using it for $r = 2$ gives $\frac{1}{1-2} = -1$ for $1 + 2 + 4 + \dots$, which is nonsense.

Which kind is it? The gaps test and the ratio test

Given a list, check the **gaps** (subtract neighbours) and the **ratios** (divide neighbours). Whichever stays constant tells you the type:

list	gaps	ratios	verdict
5, 8, 11, 14	3, 3, 3	1.6, 1.375, ...	**A.P., d = 3**
2, 10, 50, 250	8, 40, 200	5, 5, 5	**G.P., r = 5**
80, 40, 20, 10	−40, −20, −10	0.5, 0.5, 0.5	**G.P., r = 0.5**
1, 4, 9, 16	3, 5, 7	4, 2.25, ...	**neither (squares)**

A.P. and G.P. rules side by side

Every formula in this chapter, in one table. The left column adds the same amount each time; the right column multiplies by the same amount.

	arithmetic (A.P.)	geometric (G.P.)
each step	add d	multiply by r
find the step	$d = u_2 - u_1$	$r = \dfrac{u_2}{u_1}$
nth term	$u_n = a + (n-1)d$	$u_n = ar^{n-1}$
sum of n terms	$S_n = \dfrac{n}{2}\big(2a + (n-1)d\big)$ or $S_n = \dfrac{n}{2}(a + l)$	$S_n = \dfrac{a(1-r^n)}{1-r}$ ($r \neq 1$; $r = 1$: $S_n = na$)
sum to infinity	none: the terms don't shrink to 0, so the sum has no limit	$S_\infty = \dfrac{a}{1-r}$ **only if** $-1 < r < 1$
three terms in a row x, y, z	$y - x = z - y$	$\dfrac{y}{x} = \dfrac{z}{y}$

Find the unknown: working backwards

PROCEDURE: given some facts, find the rest

1. Write each fact as an equation using the formulas: "the 3rd term is 11" means $a + 2d = 11$; "the sum is 210" means $S_n = 210$.
2. Two unknowns (a and d, or a and r) need two equations. For an A.P. subtract them; for a G.P. **divide** them, so a cancels.
3. Solve, then check by writing out the first few terms. Reject answers that make no sense (a negative or fractional number of terms).

Example: an A.P. has 3rd term 11 and 8th term 31. Find a and d.

$$a + 2d = 11, \qquad a + 7d = 31$$ ← u_3 and u_8 as equations

$$5d = 20 \;\Rightarrow\; d = 4$$ ← subtract: a cancels

$$a + 2(4) = 11 \;\Rightarrow\; a = 3$$ ← back into the first

Check: 3, 7, **11**, 15, 19, 23, 27, **31** ✓

Example: how many terms of $3 + 7 + 11 + \ldots$ add up to 210?

$$\frac{n}{2}\big(2(3) + (n-1)4\big) = 210$$ ← A.P. sum with a = 3, d = 4

$$\frac{n}{2}(4n + 2) = n(2n + 1) = 210 \;\Rightarrow\; 2n^2 + n - 210 = 0$$

a quadratic in n

$$(2n + 21)(n - 10) = 0 \;\Rightarrow\; n = 10$$ ← n = −10.5 is rejected: n counts terms

Example: a G.P. has 2nd term 6 and 5th term 162. Find a and r.

$$ar = 6, \qquad ar^4 = 162$$ ← u_2 and u_5 as equations

$$\frac{ar^4}{ar} = r^3 = 27 \;\Rightarrow\; r = 3$$ ← divide: a cancels

$$a = \frac{6}{3} = 2$$

Check: 2, **6**, 18, 54, **162** ✓

Example: a G.P. has first term 5 and sum to infinity 20. Find r.

$$\frac{5}{1 - r} = 20 \;\Rightarrow\; 1 - r = \tfrac{1}{4} \;\Rightarrow\; r = \tfrac{3}{4}$$ ← $S_\infty = a/(1 - r)$

$r = \frac{3}{4}$ is between −1 and 1, so a sum to infinity really exists ✓.

Example: what is the smallest number of terms of $1 + 2 + 4 + 8 + \ldots$ that add to more than 1000?

$$S_n = \frac{1(2^n - 1)}{2 - 1} = 2^n - 1 > 1000 \;\Rightarrow\; 2^n > 1001$$

a = 1, r = 2

$$2^9 = 512, \quad 2^{10} = 1024 \;\Rightarrow\; n = 10$$ ← try powers of 2

Trial works because the powers are easy. Chapter 22 solves $2^n > 1001$ directly with logs.

Example: k, $2k+1$ and $4k-2$ are consecutive terms of an A.P. Find k.

$$(2k+1) - k = (4k-2) - (2k+1) \quad \leftarrow \text{equal gaps}$$

$$k + 1 = 2k - 3 \;\Rightarrow\; k = 4 \quad \leftarrow \text{terms 4, 9, 14 ✓}$$

Mistakes to avoid

✗ Common mistakes

✗ $u_n = a + nd$

n − 1 jumps to reach term n

✓ $u_n = a + (n-1)d$

✗ $u_n = ar^n$

the first term has had no multiplications

✓ $u_n = ar^{n-1}$

✗ $r = 2: \; S_\infty = \dfrac{a}{1-2}$

only when −1 < r < 1

✓ $r = 2: \;$ no S_∞

✗ $S_\infty = \dfrac{a}{r-1}$

1 minus r on the bottom

✓ $S_\infty = \dfrac{a}{1-r}$

✗ $-3^5 = 243$

brackets round a negative ratio

✓ $(-3)^5 = -243$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

A.P. term	$u_n = a + (n-1)d$ d = any term minus the one before
A.P. sum	$S_n = \frac{n}{2}\big(2a + (n-1)d\big)$
A.P. sum (last term l)	$S_n = \frac{(a+l)}{2}\,n$
Number of terms	$n = \frac{l-a}{d} + 1$ one more term than gaps
G.P. term	$u_n = ar^{n-1}$ r = any term divided by the one before
G.P. sum	$S_n = \frac{a(1-r^n)}{1-r}$ $r \neq 1$; if $r = 1$, $S_n = na$
Sum to infinity	$S_\infty = \frac{a}{1-r} \quad (\lvert r\rvert < 1)$ none if $r \geq 1$ or $r \leq -1$
Sigma	$\sum_{i=1}^{n} f(i) = f(1) + f(2) + \cdots + f(n)$ a for-loop: add for each counter value
Odd numbers	$1 + 3 + 5 + \cdots + (2n-1) = n^2$
Two terms given	two equations; subtract (A.P.) or divide (G.P.)
Sum given, find n	set S_n equal to the total and solve for n (often a quadratic); n must be a positive whole number

Piggy banks and bouncing balls: your turn

Answers in grey. For sums, try the first few terms by hand to check your formula.

Arithmetic

1. Find the 20th term of 7, 10, 13, ...
2. How many terms in 5, 9, 13, ..., 81?
3. Add 1 + 2 + ... + 50
4. Add 7 + 10 + 13 + ... (20 terms)
5. An A.P. starts at 4 and its 10th term is 31. Find d.

1. 64 2. 20 3. 1275 4. 710 5. 3

Sigma

6. Write out and add: Σ from i = 1 to 5 of (2i + 1)
7. Write 4 + 8 + 12 + ... + 40 using Σ

6. 3 + 5 + 7 + 9 + 11 = 35 7. Σ from i = 1 to 10 of 4i (total 220)

Geometric

8. Find the 8th term of 2, 6, 18, ...
9. Add 5 + 10 + 20 + ... (8 terms)
10. Sum to infinity: 12 + 6 + 3 + ...
11. Sum to infinity: 1 − 1/3 + 1/9 − ...
12. Which is 1, 1.5, 2.25, 3.375: A.P. or G.P.? Does it have a sum to infinity?

8. 4374 9. 1275 10. 24 11. 3/4 12. G.P. with r = 1.5; no (r > 1)

Find the unknown

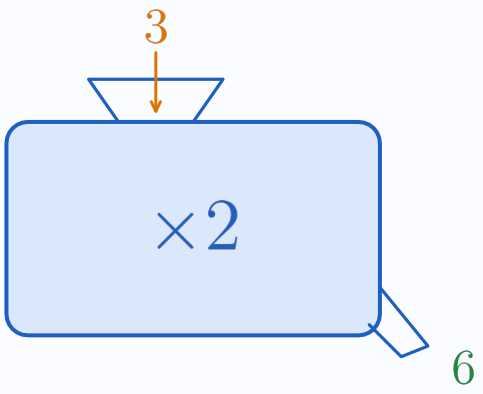
13. An A.P. has 4th term 17 and 10th term 41. Find a and d.
14. How many terms of 4 + 7 + 10 + ... add up to 246?
15. A G.P. has 3rd term 12 and 6th term 96. Find a and r.
16. A G.P. has first term 8 and sum to infinity 12. Find r.
17. Smallest n for which 3 + 6 + 12 + ... (n terms) is more than 1000?
18. x, x + 3 and x + 9 are consecutive terms of a G.P. Find x.

13. d = 4, a = 5 (6d = 24) 14. n = 12 ($3n^2 + 5n - 492 = 0$; reject the negative root) 15. r = 2, a = 3 ($r^3 = 96/12 = 8$) 16. r = 1/3 ($8/(1 - r) = 12$) 17. n = 9 ($S_8 = 3(2^8 - 1) = 765$, $S_9 = 1533$) 18. x = 3 ($(x + 3)^2 = x(x + 9)$, so 3x = 9; terms 3, 6, 12)

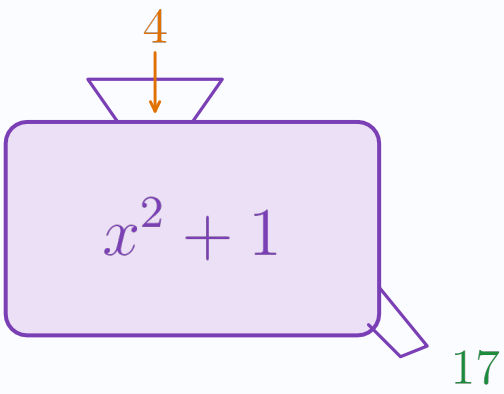
Chapter 18

Functions

a vending machine for numbers



put 3 in, 6 comes out. Every time.



put 4 in: $4^2 + 1 = 17$

A **function** is a rule that turns each input number into exactly **one** output number, like a vending machine: press B3 and you get crisps, every single time. You already know lots: "double it", "square it", "add VAT", and the sin button on your calculator.

This chapter is mostly about **reading the notation**, $f(x)$, which trips everyone up, and about which inputs a machine will accept. Then you'll chain machines together and run them backwards, which is Chapter 14's undo idea again.

Goes with **Lecture 13** (functions, domains, images, piecewise functions).

Three ways to write the same machine

RULE: function and function notation

A **function** is a rule that gives **exactly one output** for each allowed input.

$f(x)$ means "the output of f when the input is x". $f(3)$ means: replace **every** x in the rule by 3 and work it out.

$f(x)$ is **not** $f \times x$. f is the name of the rule; the brackets hold the input.

Lecture 13 writes the doubling machine three ways. They all mean "input a number, output twice it":

$x \mapsto 2x$	$y = 2x$	$f(x) = 2x$
"x maps to 2x"	"output y is 2 times x"	"f of x is 2x"

The third one, **function notation**, is the one to master. The letter f is the **name of the machine** (you could call it g, h or anything). The brackets hold **what you put in**. So:

$$f(3) = 6$$

the input → 3; machine's name → f; the output → 6

Said: "f of 3 is 6".
Meaning: put 3 into machine f, and 6 comes out.

The big trap: $f(x)$ is NOT "f times x". There is no multiplication here. f isn't a number at all, it's a name. Writing $f(3) = 3f$ is like saying "vending-machine(B3)" equals "3 vending machines".

The x is a slot: whatever goes in fills every slot

RULE: substituting into a function

Replace **every** x by the input, **in brackets**: $f(-2)$ for $f(x) = x^2 + 1$ is $(-2)^2 + 1$.

An expression can be the input: $f(a+1)$ means replace each x by $(a+1)$, then simplify.

$f(x+1)$ changes the **input**; $f(x)+1$ changes the **output**. They are different.

Take $f(x) = x^2 + 1$. Think of every x in the rule as an empty slot, $\square^2 + 1$. The input fills **every** slot, and it goes in **with brackets** round it:

$$f(\square) = \square^2 + 1$$ ← the rule with its slots showing

$$f(3) = (3)^2 + 1 = 10$$ ← 3 in the slot

$$f(-2) = (-2)^2 + 1 = 5$$ ← the brackets matter: $(-2)^2 = +4$

$$f(a+1) = (a+1)^2 + 1 = a^2 + 2a + 2$$

a whole expression can go in the slot

Watch out: $f(x+1)$ and $f(x)+1$ are different. The first puts $x+1$ **into** the machine; the second adds 1 to what comes **out**. With $x = 3$: $f(4) = 17$, but $f(3)+1 = 11$.

If you code: a function in maths is exactly a Python function. The letter in the brackets is the parameter; calling it with a number fills the slot.

```
def f(x):
    return x**2 + 1

print(f(3))    # 10
print(f(-2))   # 5
```

def = define the machine

f(3) = use it with input 3

The domain: which coins the machine accepts

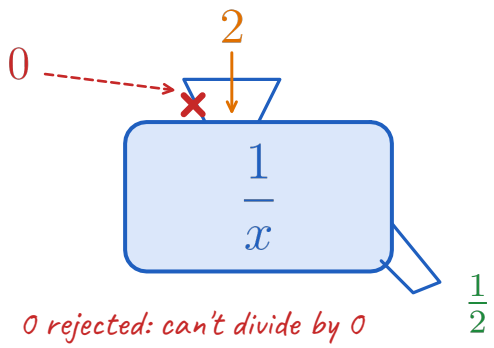
RULE: domain and range

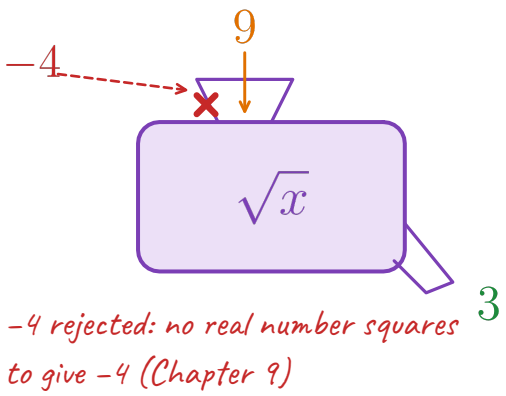
Domain = the set of inputs allowed. **Range** (or image) = the set of outputs that actually come out.

If no domain is given, take every real number **except** those that break the rule: a denominator can't be 0, and you can't square-root a negative number.

A function can be given a **restricted domain** on purpose, e.g. $f(x) = x^2,\ x \geq 0$. The domain is part of the function: change it and you change the range.

A vending machine rejects foreign coins. Some functions reject some inputs, because the rule would break:





The set of inputs a function accepts is its **domain**. The set of outputs that can come out is its **image** (also called its **range**). Lecture 13's examples, in the set notation of Chapter 6:

function	domain	image	why
$f(x) = 2x$	$\mathbb{R}$	$\mathbb{R}$	anything in, anything out
$f(x) = x^2$	$\mathbb{R}$	$[0, \infty)$	squares are never negative
$f(x) = \sin x$	$\mathbb{R}$	$[-1, 1]$	the sin button never gives more than 1
$f(x) = \frac{1}{x}$	$x \neq 0$	$y \neq 0$	no input makes 1/x equal 0

(Lecture 13 writes the squaring function's image as $\mathbb{Q}^+$, but that's the positive **fractions**; the outputs are all the non-negative **real** numbers, $[0, \infty)$, including 0 and numbers like $(\sqrt[3]{2})^2$.)

When no domain is given, assume it's **every number that doesn't break the rule**: usually just avoid dividing by zero and square-rooting negatives.

PROCEDURE: finding the domain and the range

1. **Domain:** set any denominator $\neq 0$ and anything under a square root ≥ 0; solve for x.
2. **Range of a straight line on an interval:** work out the outputs at the two ends of the domain. The range runs between them.
3. **Range of a square:** $(\text{anything})^2 \geq 0$, so $a(x-h)^2 + k \geq k$ when $a > 0$ (Chapter 15's completed square). The smallest output is k.
4. Check with one or two inputs that the outputs really land in your range.

$$g(x) = \sqrt{x-3}: \quad x - 3 \geq 0 \;\Rightarrow\; x \geq 3$$ ← domain: no negatives under the root

$$h(x) = \frac{5}{2x-6}: \quad 2x - 6 \neq 0 \;\Rightarrow\; x \neq 3$$ ← domain: denominator not 0

$$f(x) = 2x + 1,\ 0 \leq x \leq 3: \quad f(0) = 1,\ f(3) = 7$$

range: work out the two ends

$$\Rightarrow\; 1 \leq f(x) \leq 7$$ ← the outputs run between them

$$q(x) = x^2 - 6x + 11 = (x-3)^2 + 2 \geq 2$$ ← range: f(x) ≥ 2 (complete the square)

Is it a function? One button, one snack

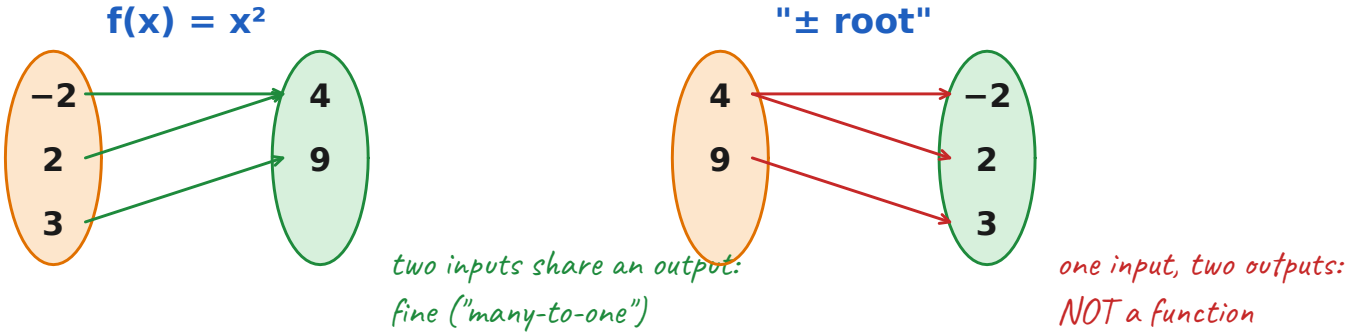
RULE: function tests

A rule is a function only if every input in the domain gives **exactly one** output.

Vertical line test (on a graph, Chapter 19): every vertical line meets the graph **at most once**. Crossing twice means one input has two outputs: not a function.

One-to-one: different inputs always give different outputs (every horizontal line meets the graph at most once). **Many-to-one**: some output comes from two or more inputs, like x^2.

A machine that gave crisps **or** chocolate when you pressed B3 would be broken. A function must give **exactly one output per input**. Draw arrows from inputs to outputs:

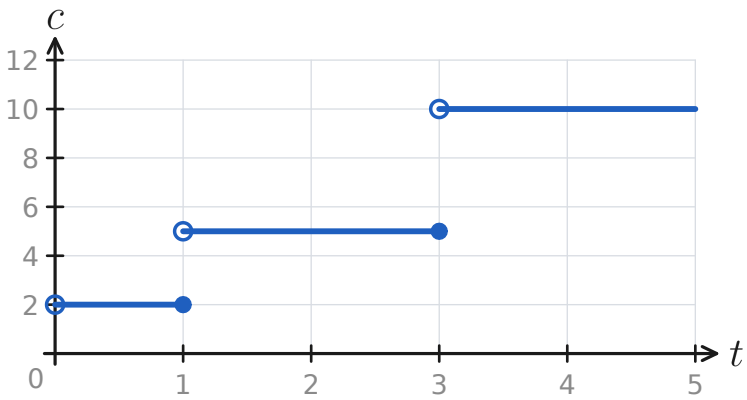


That's why $\sqrt{\ }$ on its own always means the **positive** root (Chapter 9): it keeps the square-root button a proper function.

Piecewise functions: parking charges

Some rules change depending on the input. A car park charges £2 for up to 1 hour, £5 for up to 3 hours, and £10 for longer. One machine, three rules:

$$c(t) = \begin{cases} 2 & 0 < t \le 1 \\ 5 & 1 < t \le 3 \\ 10 & t > 3 \end{cases}$$ ← *t = hours parked, c = cost in £*



filled dot = included
hollow dot = not included

exactly 1 hour costs £2 (filled dot on the lower step); 1 hour 1 minute costs £5

To use it, **first find which piece your input belongs to**, then use that piece's rule: $c(0.5) = 2$, $c(3) = 5$, $c(4.5) = 10$.

Lecture 13's example is the **sign function**: 1 for positive inputs, −1 for negative inputs, 0 for 0. It can't be written as one formula, so it's written in pieces:

$$\operatorname{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$ ← *sgn(7) = 1, sgn(−0.2) = −1*

Machines in a chain: composite functions

RULE: composite functions

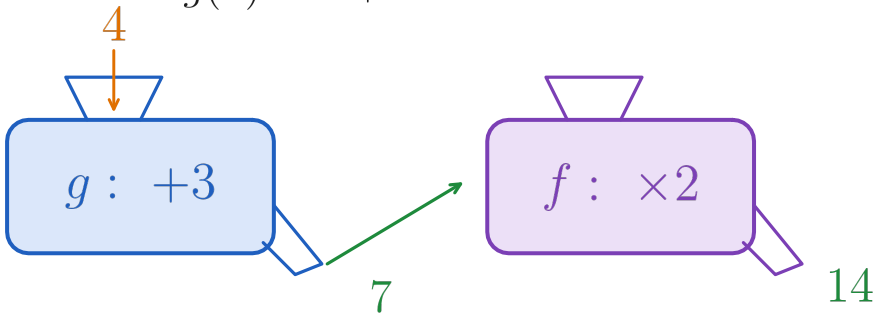
$fg(x)$ means $f(g(x))$: **do g first**, then put its output into f. It is also written $(f \circ g)(x)$, read "f after g".

Work from the **inside out**: the function written nearest to x acts first.

In general $fg(x) \neq gf(x)$: the order matters.

To find $fg(x)$ as a formula: take the rule for f and replace every x by the whole of $g(x)$, in brackets.

The output of one machine can be fed straight into another. With $f(x) = 2x$ and $g(x) = x + 3$:



g first, then f. Written f(g(4)): read from the inside out

$$fg(x) = f(g(x)) = 2(x+3) = 2x + 6$$ ← *g's output fills f's slot*

$$gf(x) = g(f(x)) = 2x + 3$$ ← *other order: different machine!*

Order matters, like socks and shoes (Chapter 14). $f(g(4)) = 14$ but $g(f(4)) = 11$.

With squares: $f(x) = x^2 + 1$ and $g(x) = 3x - 2$.

$$fg(x) = (3x-2)^2 + 1 = 9x^2 - 12x + 5$$ ← *replace x in f by (3x − 2)*

$$gf(x) = 3(x^2+1) - 2 = 3x^2 + 1$$ ← *replace x in g by (x^2 + 1)*

Check with a number: $x = 1$: $g(1) = 1$, $f(1) = 2$; formula: $9 - 12 + 5 = 2$ ✓.

Running a machine backwards: inverse functions

RULE: inverse functions

f^{-1} undoes f: if $f(a) = b$ then $f^{-1}(b) = a$. So $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$.

Only a **one-to-one** function has an inverse. A many-to-one function (like x^2 on all numbers) needs its domain restricted first (e.g. $x \geq 0$).

Domain of f^{-1} = range of f, and range of f^{-1} = domain of f: inputs and outputs swap.

The graph of $y = f^{-1}(x)$ is the graph of $y = f(x)$ **reflected in the line $y = x$**.

$f^{-1}(x)$ is **not** $\dfrac{1}{f(x)}$: the -1 is a label meaning "reverse".

PROCEDURE: finding f⁻¹(x)

1. Write $y = f(x)$.
2. Rearrange to make x the subject (undo each step in reverse order, Chapter 14).
3. Swap the letters: write the answer with x as the input, $f^{-1}(x) = \ldots$
4. State the domain of f^{-1} (the range of f) and check: $f^{-1}(f(a)) = a$ for some number a.

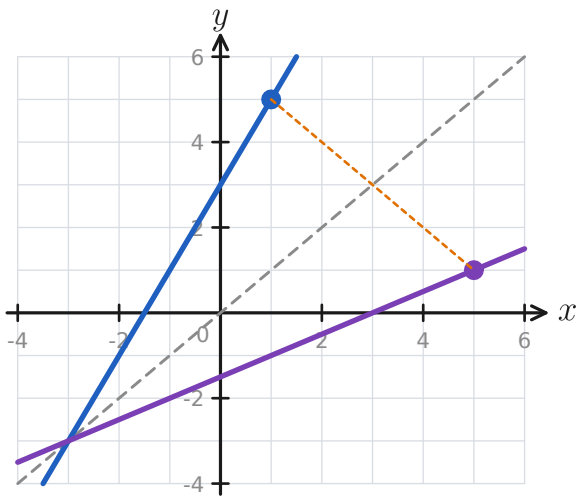
Example: $f(x) = 2x + 3$ (× 2, then + 3).

$$y = 2x + 3$$ ← step 1

$$x = \frac{y - 3}{2}$$ ← step 2: undo +3, then undo ×2

$$f^{-1}(x) = \frac{x - 3}{2}$$ ← step 3: rename the input x

Step 4, check: $f(5) = 13$ and $f^{-1}(13) = \frac{10}{2} = 5$ ✓. Domain of f^{-1}: every number, since f's range is every number.



blue: $y = 2x + 3$
purple: $y = (x - 3)/2$
grey dashed: $y = x$ (the mirror)

(1, 5) on f becomes (5, 1) on the inverse: the coordinates swap, which is a reflection in $y = x$.

Example: $f(x) = \dfrac{x+4}{3}$ with domain $x \geq 2$.

$$y = \frac{x+4}{3} \;\Rightarrow\; 3y = x + 4 \;\Rightarrow\; x = 3y - 4$$

× 3, then − 4

$$f^{-1}(x) = 3x - 4, \quad x \geq 2$$ ← domain of the inverse = range of f

The range of f: the smallest input is 2, giving $f(2) = \frac{6}{3} = 2$, and f goes up, so $f(x) \geq 2$. That becomes the domain of f^{-1}.

Example: $f(x) = x^2$ has no inverse on all numbers: $f(3) = f(-3) = 9$, so $f^{-1}(9)$ couldn't decide. Restrict to $x \geq 0$ and it is one-to-one, with $f^{-1}(x) = \sqrt{x}$, $x \geq 0$.

Find the unknown: working backwards from outputs

PROCEDURE: partial-information function problems

1. "Solve $f(x) = k$": write the rule $= k$ and solve the equation for x (Chapter 15). A square may give two answers; check each is in the domain.
2. "$f(1) = 5$" is an equation: put $x = 1$ into the rule and set it equal to 5. Two unknown letters need two facts: solve them simultaneously.
3. "Solve $fg(x) = k$": first write $fg(x)$ as one formula, then set it equal to k.
4. "Find $f^{-1}(b)$": either find the formula for f^{-1}, or solve $f(x) = b$; it's the same answer.

Example: $f(x) = 3x - 5$. Solve $f(x) = 10$.

$$3x - 5 = 10 \Rightarrow 3x = 15 \Rightarrow x = 5$$ ← check: f(5) = 15 − 5 = 10

Example: $f(x) = x^2 - 2x$. Solve $f(x) = 3$.

$$x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$$

rearrange to = 0, factorise

$$x = 3 \text{ or } x = -1$$ ← two inputs give 3: f is many-to-one

Example: $f(x) = ax + b$, with $f(1) = 5$ and $f(3) = 11$. Find a and b.

$$a + b = 5, \qquad 3a + b = 11$$ ← x = 1 and x = 3 into the rule

$$2a = 6 \Rightarrow a = 3, \quad b = 2$$ ← subtract: b cancels

Check: $f(x) = 3x + 2$: $f(1) = 5$ ✓, $f(3) = 11$ ✓.

Example: $f(x) = 2x + 1$ and $g(x) = x^2$. Solve $fg(x) = 19$.

$$fg(x) = f(x^2) = 2x^2 + 1$$ ← g first, then f

$$2x^2 + 1 = 19 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$$

Example: $f(x) = kx^2 - 3$ and $f(2) = 9$. Find k.

$$4k - 3 = 9 \Rightarrow k = 3$$ ← x = 2 into the rule

Example: $f(x) = 4x - 7$. Find $f^{-1}(5)$ without finding the formula.

$$4x - 7 = 5 \Rightarrow x = 3, \text{ so } f^{-1}(5) = 3$$ ← the input that gives 5

Mistakes to avoid

✗ Common mistakes

✗ $f(3) = 3f$

f is a name, not a number

✓ $f(3) =$ rule with $x = 3$

✗ $f(-2) = -2^2 + 1 = -3$

brackets round the input

✓ $f(-2) = (-2)^2 + 1 = 5$

✗ $fg(x)$: do f first

inside out

✓ $fg(x) = f(g(x))$: g first

✗ $f^{-1}(x) = \dfrac{1}{f(x)}$

−1 means inverse, not a power

✓ f^{-1} undoes f

✗ $f(x+1) = f(x) + 1$

input changed vs output changed

✓ $f(x+1) \neq f(x) + 1$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Function	exactly one output for each input in the domain
Notation	$f(a) =$ the rule with every x replaced by (a)
Domain	allowed inputs: denominator $\neq 0$, nothing negative under a square root
Range	all outputs; for a line on an interval use the end values; $a(x-h)^2 + k \geq k$ $(a > 0)$
Vertical line test	a graph is a function if every vertical line meets it at most once
One-to-one	different inputs give different outputs; needed for an inverse
Composite	$fg(x) = (f \circ g)(x) = f(g(x))$ g first; in general $fg \neq gf$
Inverse	$f(f^{-1}(x)) = f^{-1}(f(x)) = x$ write $y = f(x)$, make x the subject, rename
Inverse domain	domain of f^{-1} = range of f (and range of f^{-1} = domain of f)
Inverse graph	reflect the graph of f in the line $y = x$
Solve f(x) = k	set the rule equal to k and solve; $f(a) = b$ gives an equation

Vending machines: your turn

Answers in grey.

Using the notation

1. $f(x) = 3x - 1$. Find $f(2)$, $f(0)$ and $f(-4)$.
2. $g(x) = x^2 - x$. Find $g(3)$ and $g(-3)$.
3. $h(x) = 10/x$. Find $h(5)$ and $h(0.5)$.
4. $f(x) = 3x - 1$. Find $f(a + 2)$ and simplify.

1. 5, −1, −13 2. 6, 12 3. 2, 20 4. 3a + 5

Domain and image

5. What input does $1/(x - 4)$ reject?
6. What inputs does $\sqrt{(x - 1)}$ accept?
7. What is the image of $f(x) = x^2 + 5$?

5. $x = 4$ 6. $x \geq 1$ 7. $[5, \infty)$: outputs of 5 or more

Pieces, chains and inverses

8. With the car park $c(t)$, find $c(2)$ and $c(3.5)$.
9. $f(x) = x + 4$, $g(x) = 3x$. Find $f(g(2))$ and $g(f(2))$.
10. Find the inverse of $f(x) = 5x - 2$.
11. Find the inverse of $f(x) = (x + 1)/4$.

8. £5, £10 9. 10, 18 10. $f^{-1}(x) = (x + 2)/5$ 11. $f^{-1}(x) = 4x - 1$

Find the unknown

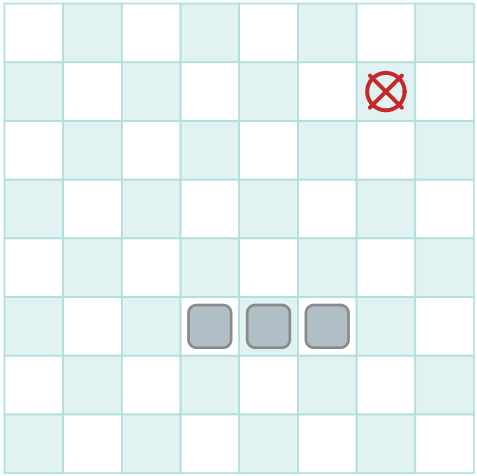
12. $f(x) = 5x + 2$. Solve $f(x) = 27$.
13. $g(x) = x^2 + 4x$. Solve $g(x) = 12$.
14. $f(x) = ax + b$, with $f(2) = 7$ and $f(5) = 16$. Find a and b.
15. $f(x) = 3x - 1$ and $g(x) = x + 2$. Solve $fg(x) = 20$.
16. $h(x) = kx^3 + 1$ and $h(2) = 25$. Find k.
17. $f(x) = (x + 4)/2$. Find the value of a for which $f^{-1}(a) = 6$.

12. $x = 5$ ($5x = 25$) 13. $x = 2$ or $x = -6$ ($(x + 6)(x - 2) = 0$) 14. $a = 3$, $b = 1$ ($3a = 9$) 15. $x = 5$ ($fg(x) = 3(x + 2) - 1 = 3x + 5$) 16. $k = 3$ ($8k + 1 = 25$) 17. $a = 5$ ($f^{-1}(a) = 6$ means $a = f(6)$)

Chapter 19

Graphs of functions

a picture of every input and output at once



"along 7, up 7: hit!"
Every square has an address.
A graph is a grid where every
input gets a dot at
(input, output).

A **graph** turns a function into a picture. Each input x and its output $f(x)$ become one dot, and all the dots together draw a line or curve. You can then **see** things that are hard to spot in a formula: where it's biggest, where it crosses zero, whether it grows fast or slowly.

You'll start with the grid itself (Battleship), plot a curve by hand, learn to read a graph in both directions, and then meet the "family portraits" of the common functions from Lecture 7 so you recognise them on sight.

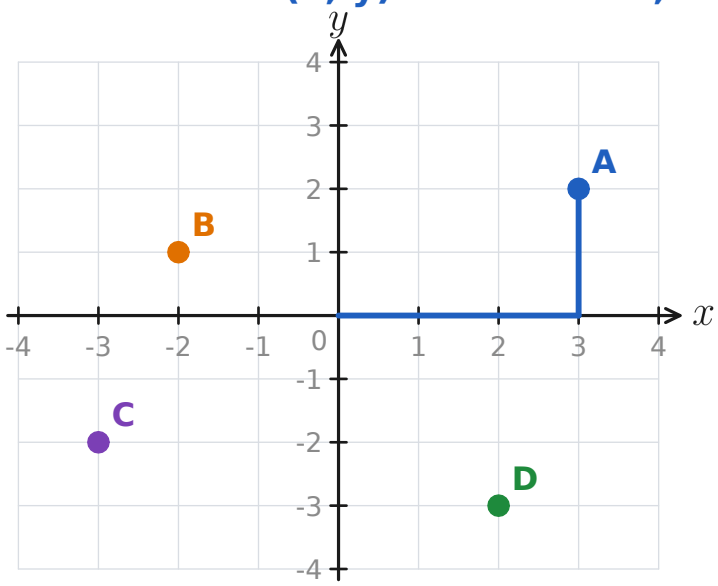
Goes with **Lecture 7** (graphing common functions, dominance), **Lecture 13** (graph of a function) and **Lecture 14** (intercepts and vertex).

Coordinates: Battleship with negative numbers

RULE: coordinates

The point (x, y) is x **across** (right if positive, left if negative) and then y **up** (down if negative), starting from the origin $(0, 0)$. Always x first.

Two number lines cross at right angles. The across one is the **x-axis**, the up one is the **y-axis**, and they meet at the **origin** (0, 0). A point's address is written **(x, y)**: **across first, then up**.



A = (3, 2): along 3, up 2 (blue path: corridor first, then the stairs)

B = (−2, 1): back 2, up 1

C = (−3, −2): back 3, down 2

D = (2, −3): along 2, down 3

Memory hook: "along the corridor, then up the stairs"

The axes split the page into four **quadrants**. Negative x means left of the y-axis; negative y means below the x-axis.

From table to dots to curve

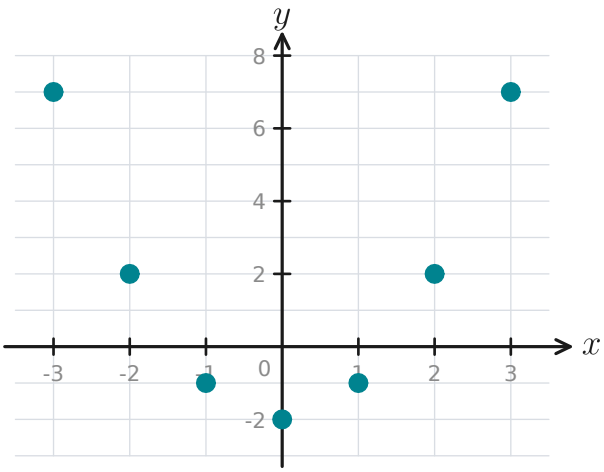
PROCEDURE: plotting y = f(x)

1. Choose inputs x (include negatives and 0) and work out each $y = f(x)$ in a table.
2. Plot each pair (x, y) as a dot.
3. Join the dots with a **smooth** curve (a ruler only if the rule is a straight line).

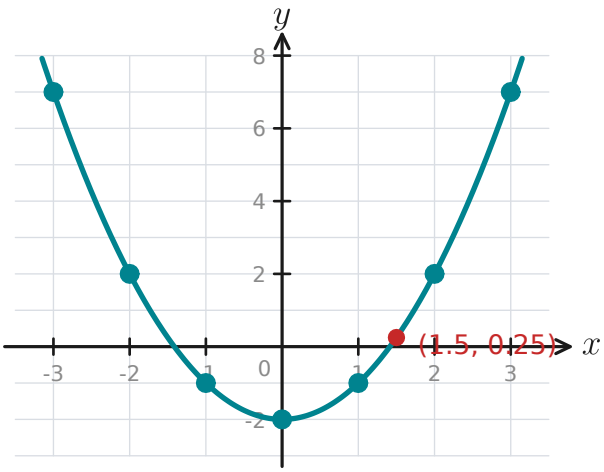
To draw $y = x^2 - 2$: choose some inputs, work out each output (a table), plot each pair as a dot, then join the dots with a smooth curve.

x	−3	−2	−1	0	1	2	3
y = x² − 2	7	2	−1	−2	−1	2	7

Careful with negatives: for $x = -3$, it's $(-3)^2 - 2 = 9 - 2 = 7$, not $-9 - 2$.



1. plot the dots



2. join with a smooth curve

Why smooth, not straight lines between dots? Because in-between inputs have outputs too: $x = 1.5$ gives 0.25, which lands right on the curve (red dot). Lecture 13 puts it neatly: the graph of f is **the set of all points** $(x, f(x))$.

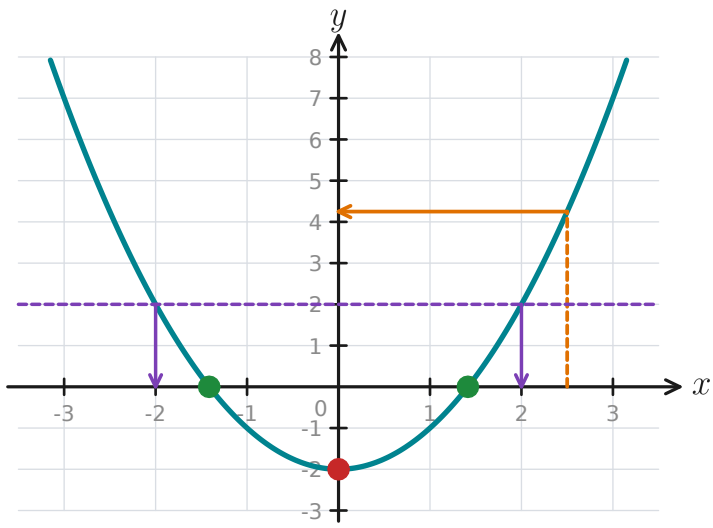
📈 Reading a graph both ways

RULE: intercepts and reading a graph

y-intercept: put $x = 0$; the graph crosses the y-axis at $(0, f(0))$.

x-intercepts (roots): put $y = 0$ and solve $f(x) = 0$.

To solve $f(x) = k$ from a graph: go across from k on the y-axis to the curve, then down to read x.



Input → output: from x = 2.5 go UP to the curve, then ACROSS: y ≈ 4.25

Output → input: which x give y = 2? go ACROSS from 2, then DOWN: x = −2 and x = 2

y-intercept (red): where it crosses the y-axis. Put x = 0: y = −2

x-intercepts (green): where y = 0, the solutions of $x^2 - 2 = 0$: x = ±1.41

That second direction is **solving an equation by picture**: the answers to $x^2 - 2 = 2$ are where the curve meets the horizontal line $y = 2$. (Chapter 15's crossing lines were the same idea.)

Even functions fold, odd functions spin

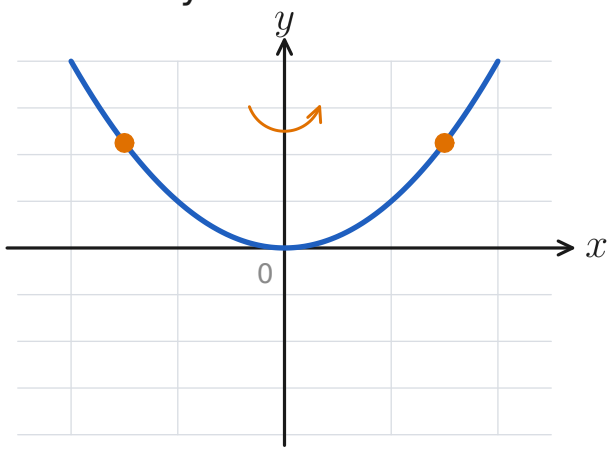
RULE: even and odd functions

Even: $f(-x) = f(x)$ for every x. The graph is symmetric in the y-axis (e.g. x^2, x^4, $x^2 + 3$).

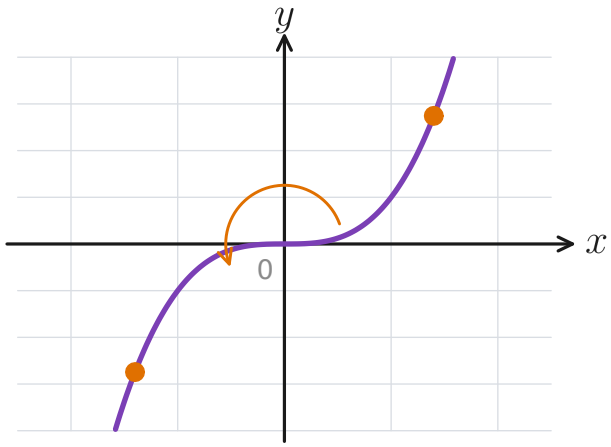
Odd: $f(-x) = -f(x)$ for every x. The graph is unchanged by a half turn about the origin (e.g. x, x^3, $\frac{1}{x}$).

To test: work out $f(-x)$ and simplify. Same as $f(x)$: even. Exactly $-f(x)$: odd. Otherwise: neither.

Lecture 7 sorts the power graphs into **even** and **odd**. You can test by picture or by numbers:



x²: fold along the y-axis



x³: spin half a turn about (0, 0)

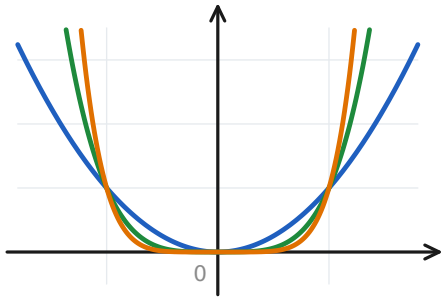
Even: fold the page along the y-axis and the two halves match. In numbers, putting in $-x$ gives the **same** output: $(-1.5)^2 = 1.5^2 = 2.25$. So $f(-x) = f(x)$.

Odd: spin the page half a turn around the origin and the curve lands on itself. In numbers, $-x$ gives the **negative** output: $(-2)^3 = -8$. So $f(-x) = -f(x)$.

The names come from the powers: x^2, x^4, x^6 are even functions; x, x^3, x^5 are odd. Many functions (like $x^2 + x$) are neither.

📈 Family portraits: recognise them on sight

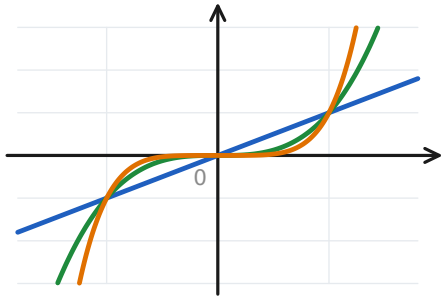
These are the shapes from Lecture 7. Learn them like faces; you'll meet them everywhere.



even powers

x^2 x^4 x^6

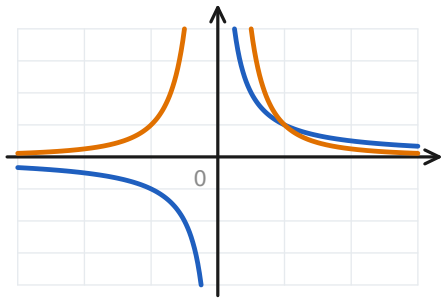
all pass through (−1, 1), (0, 0), (1, 1). Never negative. Higher power: flatter near 0, steeper beyond 1.



odd powers

x x^3 x^5

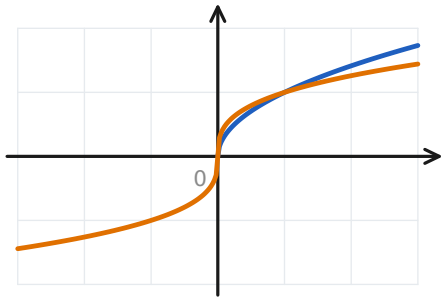
negative on the left, positive on the right; all pass through (−1, −1), (0, 0) and (1, 1).



negative powers

$1/x$ $1/x^2$

blow up near $x = 0$ (can't divide by 0), and creep towards 0 far out, never quite touching the axes.



fractional powers

$\sqrt{x}$ $\sqrt[3]{x}$

$\sqrt{x}$ only exists for $x \geq 0$ (half a curve); $\sqrt[3]{x}$ works for negatives too. Both grow slower and slower.

Those lines that $\frac{1}{x}$ creeps towards but never reaches are called **asymptotes**.

Dominance: what you see depends on the zoom

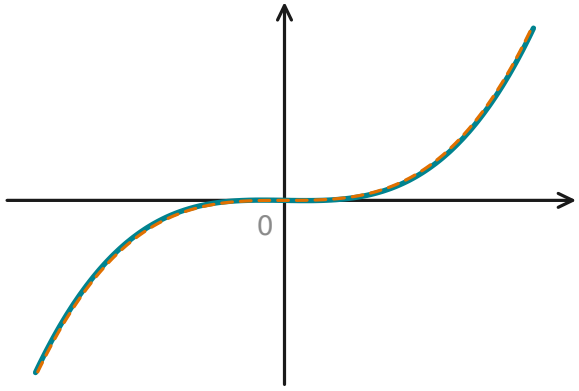
RULE: dominance (Lecture 7)

For $y = ax^m + \cdots + bx^n + c$ (highest power m, lowest power n):

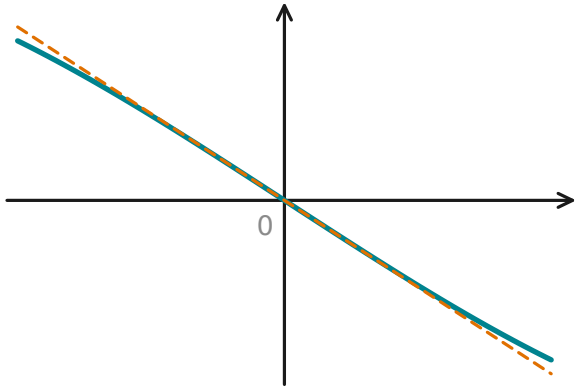
Large x (far from 0, either way): the **highest power** term dominates, so it looks like $y = ax^m$.

Small x (close to 0): the **lowest powers and the constant** dominate, so it looks like $y = bx^n + c$.

Lecture 7's example $y = x^3 - 2x$. Zoom out and the x^3 term swamps everything (the **biggest power dominates** for large x). Zoom in near 0 and x^3 is tiny, so it looks like the line $y = -2x$:



zoomed out: looks like x³ (dashed)



zoomed in: looks like −2x (dashed)

At $x = 10$: $x^3 = 1000$ and $2x = 20$, so the $-2x$ changes the answer by only 2%. At $x = 0.1$: $x^3 = 0.001$ and $2x = 0.2$, so now x^3 hardly matters.

Example: $y = x^4 - 3x + 2$. For large x it looks like $y = x^4$ (a U shape). Near 0 the x^4 is tiny and it looks like the straight line $y = -3x + 2$: at $x = 0.1$, $x^4 = 0.0001$ while $-3x + 2 = 1.7$.

📈 Sliding and flipping graphs

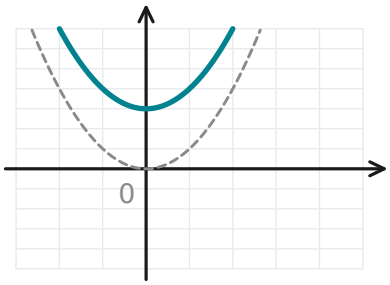
RULE: transformations of y = f(x)

A change **outside** the function (to the output, y) moves the graph **vertically** and does exactly what it says.

A change **inside** the brackets (to the input, x) moves the graph **horizontally** and does the **opposite** of what it looks like: $+a$ moves left, $\times a$ squashes.

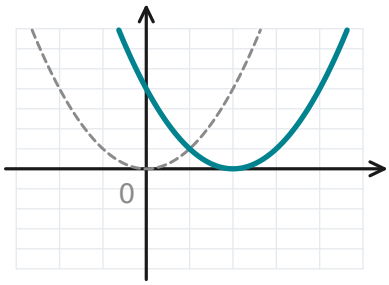
new graph	what happens to y = f(x)	point (x, y) moves to
$y = f(x) + a$	**up** a (down if $a < 0$)	$(x,\ y + a)$
$y = f(x + a)$	**left** a (opposite to the sign!)	$(x - a,\ y)$
$y = f(x - a)$	**right** a	$(x + a,\ y)$
$y = af(x)$	vertical stretch, factor a (from the x-axis)	$(x,\ ay)$
$y = f(ax)$	horizontal squash by factor a (a stretch of factor $\frac{1}{a}$, towards the y-axis)	$(\frac{x}{a},\ y)$
$y = -f(x)$	reflect in the **x-axis** (upside down)	$(x,\ -y)$
$y = f(-x)$	reflect in the **y-axis** (left-right)	$(-x,\ y)$

Change the formula a little and the whole picture moves. Starting from $y = x^2$:



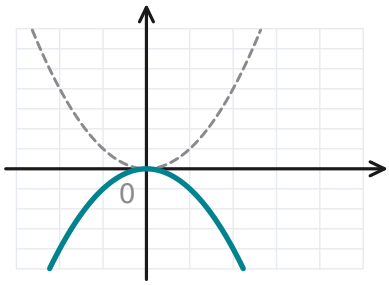
$y = x^2 + 3$

up 3: add 3 to every output



$y = (x - 2)^2$

RIGHT 2 (not left!): x must be 2 bigger to get the same output



$y = -x^2$

flipped upside down: every output changes sign

The one that feels backwards: $(x-2)^2$ moves **right**. Check with numbers: the lowest point of x^2 is at $x=0$; for $(x-2)^2$ the bracket is 0 when $x=2$. So the bottom has moved to $x=2$, to the right.

Chapter 15, explained: $y=x^2+6x+5=(x+3)^2-4$. That's x^2 moved **left 3** and **down 4**, so its lowest point (vertex) is at $(-3,-4)$, just as Lecture 14 said. The completed-square form is the **sat-nav** to the vertex.

RULE: vertex form of a quadratic

$y=a(x-h)^2+k$ has its **vertex** (turning point) at (h,k) and is symmetric about the line $x=h$.

$a>0$: U-shaped, the vertex is the **lowest** point. $a<0$: upside down, the vertex is the **highest** point.

Watch the sign of h: $(x+3)^2$ is $(x-(-3))^2$, so $h=-3$.

PROCEDURE: moving a point

1. Find the **horizontal** change from inside the bracket (opposite sign) and apply it to x.
2. Find the **vertical** change from outside and apply it to y (stretch first, then shift).
3. Check: put the new x into the new formula; it should give the new y.

Example: $(2,5)$ lies on $y=f(x)$. Where does it go on $y=f(x-3)+1$?

$x:\ 2\to 2+3=5$ ← inside: −3 means RIGHT 3

$y:\ 5\to 5+1=6$ ← outside: +1 means up 1

$(2,5)\ \to\ (5,6)$

Check with $f(x)=x^2+1$ (which does pass through (2, 5)): new formula at $x=5$ is $f(2)+1=6$ ✓.

Example: $(4,6)$ lies on $y=f(x)$. Where does it go on $y=f(2x)$, on $y=3f(x)$ and on $y=-f(x)$?

$f(2x):\ (4,6)\to(2,6)$ ← inside ×2 squashes: x ÷ 2

$3f(x):\ (4,6) \to (4,18)$ ← *outside ×3 stretches: y × 3*

$-f(x):\ (4,6) \to (4,-6)$ ← *flip in the x-axis*

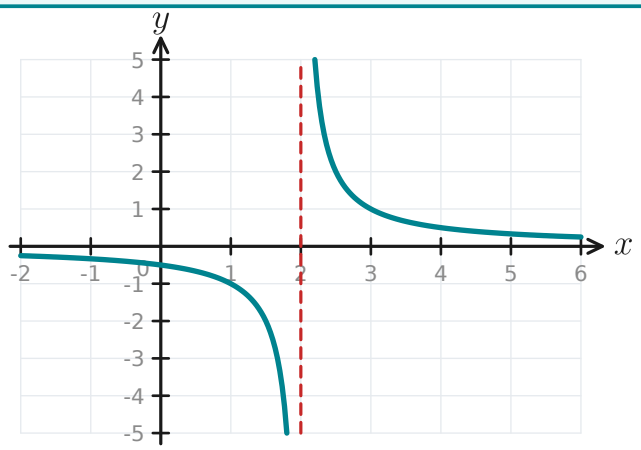
Asymptotes: where a graph can't go

RULE: asymptotes of fractions

A **vertical asymptote** is where the denominator is 0: $y = \dfrac{1}{x-a}$ has the asymptote $x = a$. The graph shoots up or down beside it and never touches it.

A **horizontal asymptote** is the value y approaches as x gets very large: for $\dfrac{1}{x-a}$ it is $y = 0$, because 1 divided by a huge number is nearly 0.

$y = \dfrac{b}{x-h} + k$ is $\frac{1}{x}$ moved: asymptotes $x = h$ and $y = k$.



y = 1/(x − 2)

red dashed: x = 2. Put x = 2 in and you'd divide by 0, so the graph can't be there.

Far left and far right it creeps towards the x-axis (y = 0).

Example: $y = \dfrac{3}{x+1} + 2$. Denominator 0 when $x = -1$: vertical asymptote $x = -1$. For large x the fraction is nearly 0, so y is nearly 2: horizontal asymptote $y = 2$.

Find the unknown: reading a formula from facts

PROCEDURE: partial-information graph problems

1. A point (p, q) on a graph means: put $x = p$ into the formula and you get $y = q$. That's an equation for the unknown.
2. A vertex (h, k) means: write $y = a(x - h)^2 + k$ and fill in h and k.
3. An asymptote $x = a$ means: the denominator is 0 when $x = a$.
4. A moved graph: use the transformation table, on the point or on the formula.

Example: $y = (x - h)^2 + k$ has vertex $(3, -2)$. Find h and k, and write the formula.

$$h = 3,\ k = -2: \quad y = (x - 3)^2 - 2$$ ← vertex (h, k) read straight off

Example: $y = a(x - 1)^2 + 2$ passes through $(3, 10)$. Find a.

$$10 = a(3 - 1)^2 + 2 = 4a + 2 \ \Rightarrow\ a = 2$$

put x = 3, y = 10

Example: the graph of $y = f(x)$ is moved to $y = f(x + 4)$. Where does the point $(6, 1)$ go?

$$(6, 1) \ \rightarrow\ (6 - 4,\ 1) = (2, 1)$$ ← inside +4: LEFT 4

Example: $y = x^2 + bx + 3$ is an even function. Find b.

$$f(-x) = x^2 - bx + 3 = x^2 + bx + 3 \ \Rightarrow\ -b = b \ \Rightarrow\ b = 0$$

even: f(−x) = f(x)

Example: $y = \dfrac{5}{x - k}$ has a vertical asymptote at $x = 4$. Find k.

$$4 - k = 0 \ \Rightarrow\ k = 4$$ ← denominator 0 at the asymptote

Mistakes to avoid

✗ Common mistakes

✗ $y = (x + 2)^2$: right 2 ✓ $y = (x + 2)^2$: left 2

inside the bracket: opposite way

✗ $y = f(2x)$: $(4, 6) \to (8, 6)$ ✓ $(4, 6) \to (2, 6)$

inside ×2 squashes: divide x by 2

✗ $y = (x + 3)^2 - 4$: vertex $(3, -4)$ ✓ vertex $(-3, -4)$

h is the number that makes the bracket 0

✗ $(-3)^2 - 2 = -9 - 2$ ✓ $(-3)^2 - 2 = 9 - 2 = 7$

square the whole negative number

✗ point $(2, 5)$: $y = 2,\ x = 5$ ✓ $x = 2,\ y = 5$

x first, then y

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Coordinates	$(x,\ y)$ across first, then up
Intercepts	y-intercept: put $x = 0$. x-intercepts: solve $f(x) = 0$
Even function	$f(-x) = f(x)$ symmetric in the y-axis
Odd function	$f(-x) = -f(x)$ half-turn symmetry about the origin
Dominance	large x: highest power wins. Near 0: lowest powers and the constant win
Asymptotes	$y = \dfrac{b}{x-h} + k: \quad x = h,\ y = k$ vertical asymptote where the denominator is 0
Vertex form	$y = a(x-h)^2 + k \ \Rightarrow\ \text{vertex}\ (h, k)$ $a > 0$ minimum, $a < 0$ maximum
Shift up / down	$y = f(x) + a$ up a: $(x, y) \to (x, y + a)$
Shift left / right	$y = f(x + a)$ **left** a: $(x, y) \to (x - a, y)$
Vertical stretch	$y = af(x)$ $(x, y) \to (x, ay)$
Horizontal squash	$y = f(ax)$ $(x, y) \to (\frac{x}{a}, y)$
Reflections	$y = -f(x), \quad y = f(-x)$ in the x-axis; in the y-axis

Battleship: your turn

Answers in grey. Sketch every graph on squared paper.

Points and plotting

1. Which quadrant is $(-4, 2)$ in? (top-right is the 1st, go anticlockwise)
2. Make a table for $y = 2x - 1$ with $x = -2$ to 2, and plot it.
3. For $y = x^2 - 4x$, find the y-intercept and the x-intercepts.

1. 2nd (top-left) 2. $y = -5, -3, -1, 1, 3$ (a straight line) 3. y-int 0; $x = 0$ and $x = 4$

Even, odd, neither

4. $y = x^4 + 1$
5. $y = x^3 - x$
6. $y = x^2 + x$
7. $y = 1/x$

4. even 5. odd 6. neither (try $x = 1$ and $x = -1$: 2 and 0) 7. odd

Recognise and move

8. Which family: $y = x^{-2}$?
9. Where is the vertex of $y = (x - 1)^2 + 5$?
10. Describe $y = (x + 4)^2$ compared with $y = x^2$.
11. Which term dominates $y = 5x^2 + x^4$ for large x?

8. negative powers (even: always positive) 9. (1, 5) 10. moved left 4 11. x^4

Find the unknown

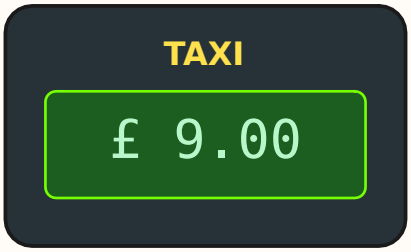
12. $y = (x - h)^2 + k$ has vertex $(3, -2)$. Find h and k.
13. $(6, 1)$ is on $y = f(x)$. Where does it move to on $y = f(x + 4)$?
14. $(2, 5)$ is on $y = f(x)$. Where does it move to on $y = f(x + 1) - 4$?
15. $(4, -3)$ is on $y = f(x)$. Where does it move to on $y = -2f(x)$?
16. $y = a(x + 2)^2 - 1$ passes through $(0, 7)$. Find a.
17. $y = 3/(x - k)$ has a vertical asymptote at $x = -2$. Find k.

12. $h = 3$, $k = -2$ 13. (2, 1): inside +4 moves left 4 14. (1, 1): left 1, down 4 15. (4, 6): $y \times (-2)$ 16. $a = 2$ ($7 = 4a - 1$) 17. $k = -2$ (denominator $-2 - k = 0$)

Chapter 20

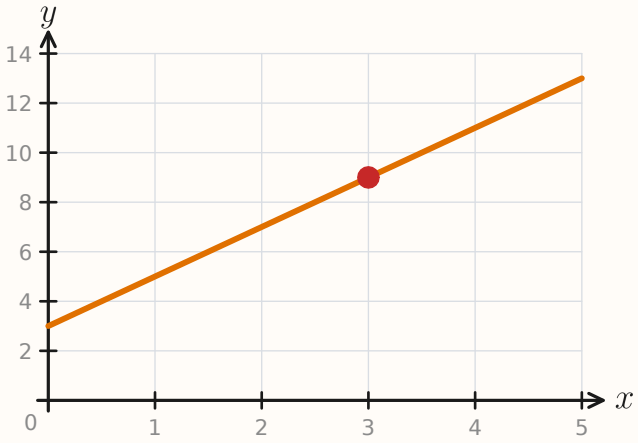
The straight line

y = mx + c, and everything you can be asked about it



£3 to get in, then £2 per mile.

3 miles: 3 + 2 × 3 = £9



Anything that grows by the **same amount for every step** (a taxi fare, a phone bill, a spring) graphs as a **straight line**, and every straight line (except vertical ones) has the formula

$$y = mx + c$$

This chapter states every rule you need about straight lines and then practises each type of question: reading m and c, finding the line through two points, **proving lines are parallel or perpendicular**, midpoints and distances, where lines cross, and **filling in a missing value** when you're only given part of the information.

Goes with **Lecture 7** (y = mx + c, slope and intercept) and **Lecture 14** (line through two points). Chapter 34 reuses this chapter for tangents and normals.

y = mx + c: gradient and intercept

RULE: the equation of a straight line

$y = mx + c$, where **m** is the **gradient** (how much y changes when x goes up by 1) and **c** is the **y-intercept** (the value of y where the line crosses the y-axis, i.e. at $x = 0$).

To read m and c, the equation must start "$y =$" with y on its own. If it doesn't, rearrange first.

A point (a, b) lies on the line exactly when putting $x = a$ into the equation gives $y = b$.

Fare in pounds y for x miles: $y = 2x + 3$. The table (like Lecture 7's for $y = 2x + 1$):

miles x	0		1		2		3		4		5
2x	0		2		4		6		8		10
+ 3	3		3		3		3		3		3
fare y	**3**	+2	**5**	+2	**7**	+2	**9**	+2	**11**	+2	**13**

The fares go up by **2 every time** the miles go up by 1: that's m. At 0 miles you still pay **3**: that's c.

c = 3: where the line crosses the y-axis (purple dot)

m = 2: every 1 step across, the line climbs 2 (the little staircase)

Gradient: rise over run

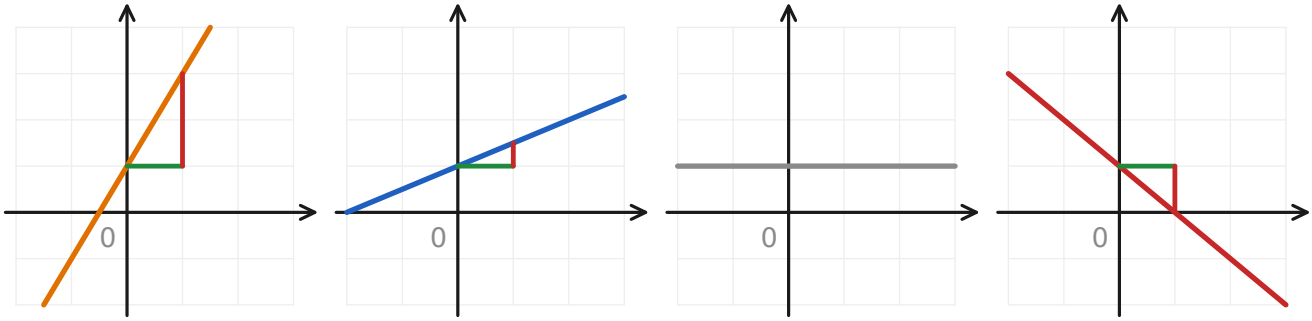
RULE: gradient

$m = \dfrac{\text{rise}}{\text{run}} = \dfrac{\text{change in } y}{\text{change in } x}$, read **left to right**.

$m > 0$: uphill. $m < 0$: downhill. $m = 0$: flat (horizontal).

Steepness depends on the **size** of m ignoring its sign, $|m|$: $y = -5x$ is steeper than $y = 2x$.

Road signs use this. "10%" means rise 10 m for every 100 m across: gradient $\frac{10}{100} = 0.1$. "1 in 4" means rise 1 for every 4 across: 0.25, a steep hill.



m = 2: steep uphill **m = 0.5: gentle uphill** **m = 0: flat (y = c)** **m = −1: downhill**

Lecture 7 says "the greater m, the steeper the line": strictly, it is the greater **size** of m. A negative m tilts the line the other way.

RULE: horizontal and vertical lines

Horizontal line through (a, b): $y = b$. Gradient 0.

Vertical line through (a, b): $x = a$. The run is 0 and you can't divide by 0, so the gradient is **undefined**. A vertical line can't be written as $y = mx + c$.

The x-axis is the line $y = 0$; the y-axis is the line $x = 0$.

The line through two points (Lecture 14)

RULE: gradient through two points

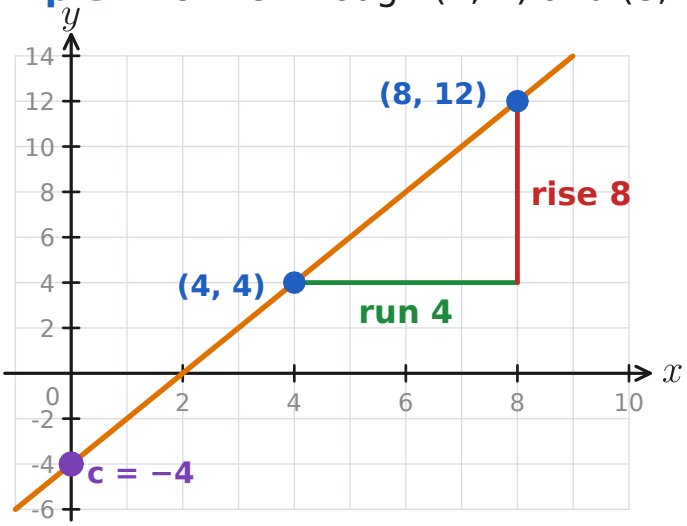
For points (x_1, y_1) and (x_2, y_2): $m = \dfrac{y_2 - y_1}{x_2 - x_1}$.

Same order top and bottom: second minus first in both, or first minus second in both.

PROCEDURE: equation of the line through two points

1. Find m with the gradient formula.
2. Put m into $y = mx + c$.
3. Substitute **either** point for x and y, and solve for c.
4. Write the final equation, and check it with the **other** point.

Example: the line through (4, 4) and (8, 12).



From (4, 4) to (8, 12):
across 8 − 4 = 4
up 12 − 4 = 8

The line crosses the y-axis at −4, below 0.

$$m = \frac{12 - 4}{8 - 4} = \frac{8}{4} = 2$$ ← *step 1: gradient formula*

$$y = 2x + c$$ ← *step 2*

$$4 = 2 \times 4 + c \;\Rightarrow\; c = -4$$ ← *step 3: use (4, 4)*

$$y = 2x - 4$$ ← *step 4. Check (8, 12): 2 × 8 − 4 = 12 ✓*

Point-gradient form: the fastest route

RULE: point-gradient form

The line with gradient m through the point (x_1, y_1) is $y - y_1 = m(x - x_1)$.

Use it whenever you know **one point and the gradient**. Expand the bracket and add y_1 to both sides to get $y = mx + c$.

Watch negative coordinates: through $(-3, -2)$ it is $y - (-2) = m(x - (-3))$, i.e. $y + 2 = m(x + 3)$.

It is the gradient formula rearranged: for any point (x, y) on the line, $\frac{y-y_1}{x-x_1} = m$. Multiply both sides by $(x - x_1)$.

Example: the same line through (4, 4) with gradient 2.

$$y - 4 = 2(x - 4)$$ ← point (4, 4), m = 2

$$y - 4 = 2x - 8$$ ← expand

$$y = 2x - 4$$ ← +4 both sides: same answer as before

Example: gradient $-\frac{1}{2}$ through $(-2, 3)$.

$$y - 3 = -\tfrac{1}{2}(x - (-2)) = -\tfrac{1}{2}(x + 2)$$

$$y = -\tfrac{1}{2}x - 1 + 3 = -\tfrac{1}{2}x + 2$$

A downhill example, in tiny steps

Line through (−1, 5) and (3, −3). The minus signs are where mistakes happen, so every subtraction gets its own bracket:

$$m = \frac{(-3) - (5)}{(3) - (-1)}$$ ← second minus first, top and bottom

$$= \frac{-8}{4} = -2$$ ← −3 − 5 = −8; 3 − (−1) = 3 + 1 = 4

$$5 = -2 \times (-1) + c$$ ← use (−1, 5)

$$5 = 2 + c \;\Rightarrow\; c = 3$$ ← −2 × −1 = +2

$$y = -2x + 3$$

Check: $-2(3) + 3 = -3$ ✓. *Sense check:* from the first point to the second, x went up but y went down, so the gradient must be negative. It is.

Lines in disguise, and the cover-up sketch

RULE: other forms of a line

A line may be written $ax + by = d$ or $ax + by + c = 0$. To find its gradient and intercept, **rearrange to** $y = mx + c$ (Chapter 14). Shortcut: for $ax + by = d$, $m = -\frac{a}{b}$.

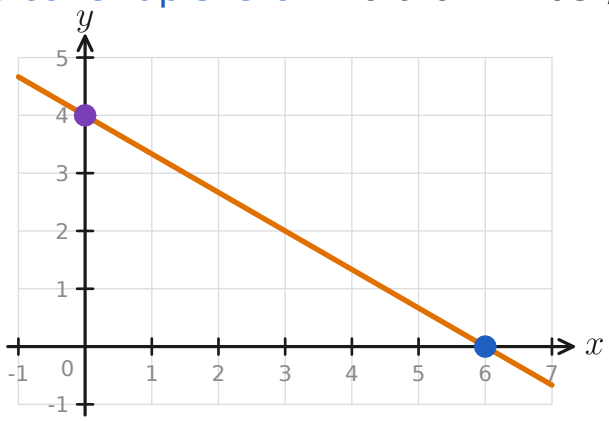
To write $y = mx + c$ as $ax + by + c = 0$ with whole numbers: multiply by the denominators, then move every term to one side.

Axis crossings: the line crosses the y-axis where $x = 0$ and the x-axis where $y = 0$.

$$2x + 3y = 12 \Rightarrow 3y = -2x + 12$$ ← −2x from both sides

$$y = -\frac{2}{3}x + 4$$ ← ÷ 3: so m = −2/3 and c = 4

The cover-up sketch. To draw it fast, find where it crosses each axis:



Cover up the x-term (x = 0):
3y = 12, so y = 4. Point (0, 4)

Cover up the y-term (y = 0):
2x = 12, so x = 6. Point (6, 0)

Join the two dots with a ruler.

Example: write $y = \frac{3}{4}x - 2$ in the form $ax + by + c = 0$.

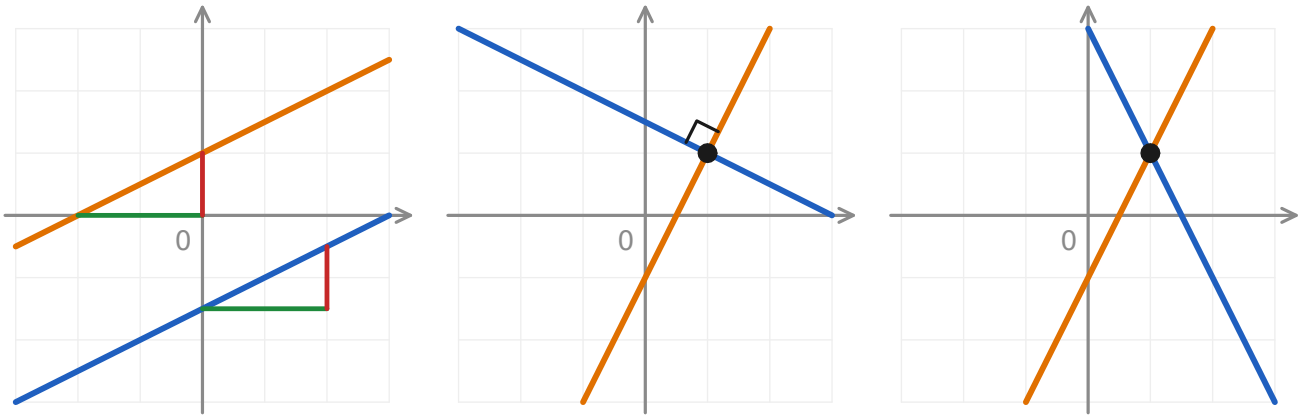
$$4y = 3x - 8$$ ← × 4 to clear the fraction

$$3x - 4y - 8 = 0$$ ← move everything to one side

Two lines at once: only three things can happen

Until now every question had **one** line. When a second line turns up there is only one new question: **how do the two lines sit next to each other?**

Remember what the two numbers do. m is the line's **tilt**. c only **slides** the line up or down without turning it. So to compare how two lines tilt, **ignore the c's and compare the m's**. There are only three possible results:



PARALLEL

same m, different c
same tilt: they never meet
(train tracks)

PERPENDICULAR

m's multiply to −1
meet at a square corner
(a + sign, a crossroads)

NEITHER

any other pair of m's
cross at a slant
(a squashed X)

The words, decoded

Word	Picture it as	What it means for the m's
Parallel	Train tracks, lanes on a motorway, the lines on lined paper. Look at the word itself: para**ll**el has two parallel l's in the middle.	Same tilt: $m_1 = m_2$. The c's are different (same m **and** same c is just one line drawn twice). They never meet.
Perpendicular (symbol ⊥)	The corner of a page, a + sign, a crossroads, a wall standing on the floor. The symbol ⊥ is exactly that: a wall on a floor. The word comes from the Latin for a builder's **plumb line**, a weight on a string that hangs straight down and so meets level ground at a perfect square corner.	They meet at a square corner (90°). Flip one gradient and change its sign to get the other: $m_2 = -\frac{1}{m_1}$.
Neither	Two roads that cross at a slant.	Any other pair of m's. They meet once, but not at a square corner.

Same meaning, different costume: "at right angles to", "normal to" (Chapter 34) and "orthogonal" all mean **perpendicular**.

Name tags. With two lines around, the working needs to say which is which. L_1 and L_2 just mean "line 1" and "line 2"; m_1 is "the gradient of line 1"; m_{AB} is "the gradient of the line from A to B". The small numbers and letters are labels, not something to calculate.

✎ YOUR GO (pencil first)

$y = 3x + 1$ and $y = 3x - 4$. Before any algebra: will these two lines ever meet? Explain using only the numbers in the equations.

Answer: no. Both have m = 3, so they tilt exactly the same way; c = 1 and c = −4 only put one 5 units above the other. They are parallel.

Parallel and perpendicular: the gradient rules

RULE: parallel and perpendicular gradients

Parallel lines have **equal gradients**: $m_1 = m_2$. (Same m and same c means it is the same line.)

Perpendicular lines (meeting at 90°) have gradients that **multiply to −1**: $m_1 \times m_2 = -1$.

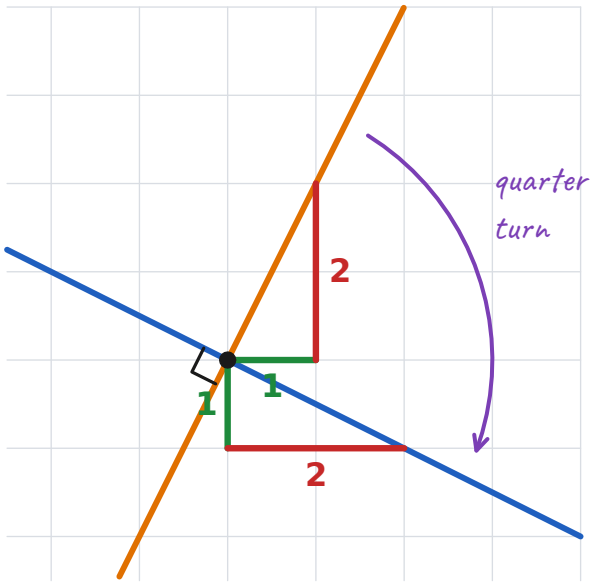
So the perpendicular gradient is the **negative reciprocal**: $m_2 = -\dfrac{1}{m_1}$.

Flip the fraction **and** change the sign: $3 \to -\frac{1}{3}$, $-\frac{2}{5} \to \frac{5}{2}$.

Exception: a horizontal line ($m = 0$) and a vertical line (no gradient) are perpendicular, but the product rule can't be used for them.

Why "flip and change the sign"? Give the staircase a quarter turn.

The orange line's staircase goes across 1, up 2, so $m = 2$. Now turn the whole staircase a quarter turn clockwise, like turning a page on its side, and watch its two sides:



Before: across 1, up 2.
Orange m = 2/1 = 2

After: across 2, down 1.
Blue m = −1/2

The 1 and the 2 swapped places: that's the FLIP.

Uphill turned into downhill: that's the MINUS sign.

So a perpendicular gradient is always the old one **flipped** (rise and run swap places) **with its sign changed** (uphill becomes downhill). Multiply the pair and the numbers cancel, leaving only the minus: $2 \times (-\frac{1}{2}) = -1$. That is all "$m_1 m_2 = -1$" is saying.

SENSE CHECK: does my perpendicular gradient look right?

A perpendicular pair always has **one uphill and one downhill** line (opposite signs), and **if one is steep the other is gentle** (one gradient bigger than 1 in size, the other smaller).

2 and $-\frac{1}{2}$: pass. 2 and -2: fail (both steep). 2 and $\frac{1}{2}$: fail (both uphill). The only pair that is the same size is 1 and -1.

✎ YOUR GO (pencil first)

Hold two pens crossed in a + shape and tilt the whole + slowly. One pen always points uphill while the other points downhill, and as one gets steeper the other gets flatter. That is the minus sign and the flip, in your hands.

PROCEDURE: parallel, perpendicular or neither?

1. Get each gradient: rearrange each equation to $y = mx + c$, or use the gradient formula on two points.
2. If $m_1 = m_2$: **parallel**. If $m_1 \times m_2 = -1$: **perpendicular**. Otherwise: **neither**.
3. Write the conclusion as a sentence that quotes the numbers (this is the "proof" an examiner wants).

Two-line questions are one-line questions in disguise

A question with two lines looks like twice the work. It isn't. Every one is one of **three kinds**, and each kind turns straight back into things you can already do with one line:

Kind	It sounds like	What you actually do
1. Compare	"Show that these lines are parallel." "Parallel, perpendicular or neither?" "Show the triangle has a right angle."	Find each line's m on its own (a one-line job, done twice). Then ask: are they equal? Do they multiply to -1?
2. Build	"Find the line through P parallel (or perpendicular) to L." "Find the perpendicular bisector."	Take m from the given line. Keep it (parallel), or flip it and change its sign (perpendicular). Now you have **one point and one gradient**: $y - y_1 = m(x - x_1)$.
3. Meet	"Where do the lines meet / cross / intersect?"	At the crossing point both equations are true, so set them equal and solve.

The big idea for Build questions: the line you are given is only a **gradient donor**. Take its m and ignore the rest of it, especially its c: your new line goes through a different point, so it gets its own c.

Decoder: what the scary words tell you to do

When the question says...	...it means
"parallel to"	use the **same** m
"perpendicular to", "at right angles to", "normal to", $\perp$	flip m and change its sign: $-\frac{1}{m}$
"passes through (a, b)", "(a, b) lies on"	put $x = a$ and $y = b$ into the equation

When the question says...	...it means
"meet", "cross", "intersect", "point of intersection"	set the two equations equal
"bisect", "bisector"	cut exactly in half: you need the **midpoint**
"show that", "prove"	do the working, then finish with one sentence that quotes the numbers

Example: L_1 has equation $4x + 2y = 5$. L_2 passes through $(1, 4)$ and is perpendicular to L_1. Find the equation of L_2.

Translate it into words first. It's a **Build**. L_1 is only the gradient donor. "Perpendicular" means flip and change the sign. "Passes through (1, 4)" is my point.

$$2y = -4x + 5 \Rightarrow y = -2x + 2.5$$ ← donor: $m_1 = -2$ (its c is not needed)

$$m_2 = +\frac{1}{2}$$ ← flip 2 to ½, and − becomes +

$$y - 4 = \frac{1}{2}(x - 1) \Rightarrow y = \frac{1}{2}x + 3.5$$ ← point (1, 4), gradient ½

Sense check: $-2 \times \frac{1}{2} = -1$ ✓. One line downhill, one uphill; one steep, one gentle ✓. Does (1, 4) fit? $\frac{1}{2} + 3.5 = 4$ ✓

Proving lines are parallel or perpendicular

Example: show that $y = 3x - 2$ and $6x - 2y = 7$ are parallel.

$$6x - 2y = 7 \;\Rightarrow\; 2y = 6x - 7 \;\Rightarrow\; y = 3x - 3.5$$

rearrange the second line

$$m_1 = 3, \quad m_2 = 3 \longleftarrow$$ *read the gradients*

Conclusion: both lines have gradient 3 (and different intercepts), so they are parallel.

Example: are $2x + 3y = 6$ and $3x - 2y = 4$ parallel, perpendicular or neither?

$$3y = -2x + 6 \;\Rightarrow\; y = -\tfrac{2}{3}x + 2 \longleftarrow$$ $m_1 = -2/3$

$$-2y = -3x + 4 \;\Rightarrow\; y = \tfrac{3}{2}x - 2 \longleftarrow$$ $m_2 = 3/2$ (÷ −2 flips every sign)

$$m_1 m_2 = -\tfrac{2}{3} \times \tfrac{3}{2} = -1$$

Conclusion: the product of the gradients is -1, so the lines are perpendicular.

Example: $y = 2x + 1$ and $y = -2x + 5$.

$m_1 = 2$, $m_2 = -2$. Not equal, and $2 \times (-2) = -4 \neq -1$. **Neither**: they cross, but not at a right angle. (**Changing only the sign is not enough** for perpendicular; you must also flip.)

Example: A(1, 2), B(4, 8), C(0, 5), D(6, 2). Show AB is perpendicular to CD.

$$m_{AB} = \frac{8 - 2}{4 - 1} = \frac{6}{3} = 2$$

$$m_{CD} = \frac{2 - 5}{6 - 0} = \frac{-3}{6} = -\tfrac{1}{2}$$

$$m_{AB} \times m_{CD} = 2 \times \left(-\tfrac{1}{2}\right) = -1 \longleftarrow$$ *so AB ⊥ CD*

Example: show the triangle A(1, 1), B(3, 5), C(7, 3) has a right angle at B.

$$m_{AB} = \frac{5-1}{3-1} = 2, \qquad m_{BC} = \frac{3-5}{7-3} = -\tfrac{1}{2}$$

Conclusion: $2 \times (-\frac{1}{2}) = -1$, so AB is perpendicular to BC: the angle at B is 90°.

Parallel or perpendicular lines through a point

PROCEDURE: line parallel / perpendicular to a given line, through a given point

1. Find the gradient of the given line (rearrange to $y = mx + c$ if needed).
2. Parallel: use the **same** gradient. Perpendicular: use $-\frac{1}{m}$.
3. Use $y - y_1 = m(x - x_1)$ with the given point, then tidy into $y = mx + c$.

Example: the line parallel to $y = 3x - 1$ through (2, 9).

$$m = 3: \;\; y - 9 = 3(x - 2) \;\Rightarrow\; y = 3x + 3$$

same gradient

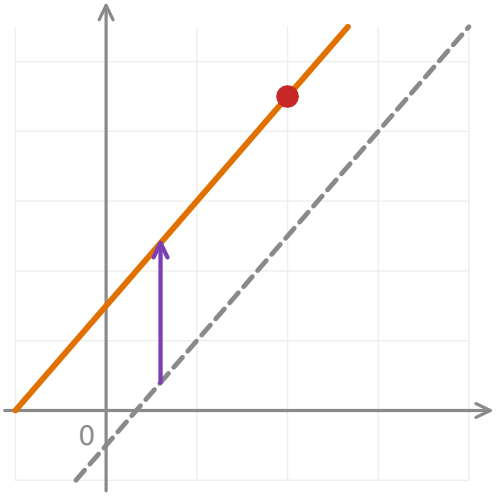
Example: the line perpendicular to $2x + 3y = 6$ through (4, 1).

$$y = -\tfrac{2}{3}x + 2 \;\Rightarrow\; m = -\tfrac{2}{3}$$ ← *rearrange the given line*

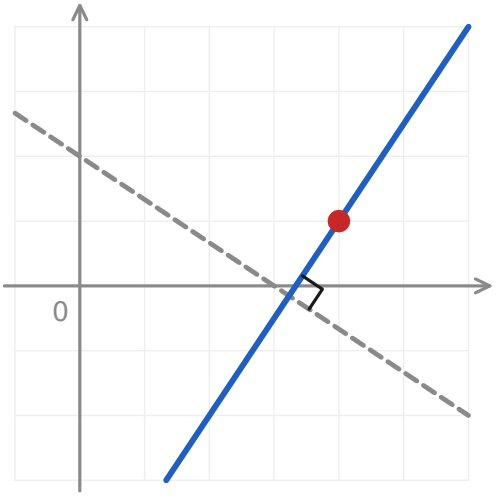
$$m_{\perp} = -\frac{1}{-\frac{2}{3}} = \tfrac{3}{2}$$ ← *negative reciprocal*

$$y - 1 = \tfrac{3}{2}(x - 4) \;\Rightarrow\; y = \tfrac{3}{2}x - 5$$

What those two answers look like (grey dashed = the given line, red dot = the given point):



Parallel: same tilt. Slide it up until it hits (2, 9).



Perpendicular: turn it a quarter turn, then slide it through (4, 1).

RULE: perpendicular bisector of AB

The line that cuts AB in half at a right angle. It passes through the **midpoint** of AB with gradient $-\frac{1}{m_{AB}}$.

Example: the perpendicular bisector of A(2, 1) and B(6, 5).

$$M = \left(\tfrac{2+6}{2}, \tfrac{1+5}{2}\right) = (4, 3)$$ ← *midpoint (next section)*

$$m_{AB} = \frac{5-1}{6-2} = 1 \;\Rightarrow\; m_{\perp} = -1$$

$$y - 3 = -(x - 4) \;\Rightarrow\; y = -x + 7$$

Midpoint and distance

RULE: midpoint and distance between two points

Midpoint of (x_1, y_1) and (x_2, y_2): average the xs and average the ys:

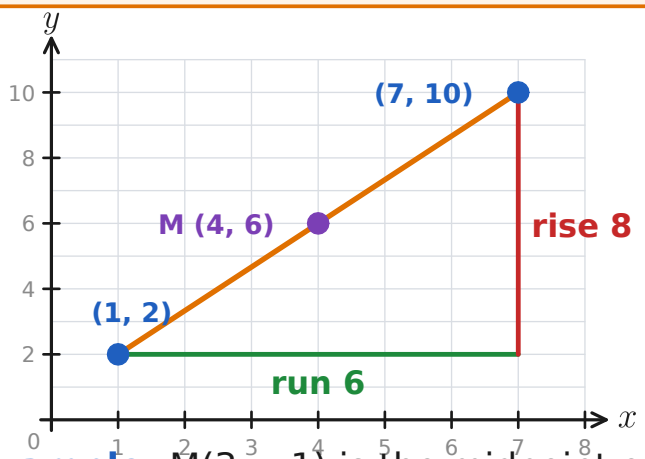
$$M = \left(\frac{x_1 + x_2}{2},\ \frac{y_1 + y_2}{2} \right).$$

Distance: the run and rise are the two short sides of a right-angled triangle, so by Pythagoras ($a^2 + b^2 = c^2$, Chapter 28):

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Missing endpoint: if M is the midpoint of A and B, then

$$B = (2x_M - x_A,\ 2y_M - y_A).$$



midpoint: ((1 + 7)/2, (2 + 10)/2)
= (4, 6)

distance: run 6, rise 8

√(6² + 8²) = √100 = 10

Example: M(3, −1) is the midpoint of A(1, 4) and B. Find B.

$$\frac{1+x_B}{2} = 3 \Rightarrow x_B = 5; \quad \frac{4+y_B}{2} = -1 \Rightarrow y_B = -6$$

So $B = (5, -6)$. Check: the midpoint of (1, 4) and (5, −6) is (3, −1) ✓

Where two lines cross

RULE: intersection of two lines

At the crossing point both equations are true at once, so solve them as **simultaneous equations** (Chapter 15).

If both start $y =$, set the right-hand sides equal and solve for x. Otherwise substitute one into the other.

Put x back into either equation to get y. The answer is a **point** (x, y).

Parallel lines (same m, different c) never cross: the equation has no solution.

Example: where do $y = 2x - 4$ and $y = -x + 5$ meet?

$$2x - 4 = -x + 5$$ ← both equal y, so equal each other

$$3x = 9 \;\Rightarrow\; x = 3$$ ← +x and +4 both sides

$$y = 2(3) - 4 = 2$$ ← back into the first line

They meet at **(3, 2)**. Check in the second line: $-3 + 5 = 2$ ✓

Example: $2x + 3y = 12$ and $y = x - 1$.

$$2x + 3(x - 1) = 12 \;\Rightarrow\; 5x = 15 \;\Rightarrow\; x = 3,\; y = 2$$

substitute y = x − 1

Find the unknown: partial-information problems

Exam questions often give you **part** of the information about a line and a letter (often k, a or p) for something missing. They are all solved the same way:

PROCEDURE: partial-information problems

1. Write down what each piece of information means as an equation: a point on the line means its coordinates satisfy the equation; a given gradient means the gradient formula equals that number; parallel or perpendicular gives you m.
2. Solve the equation for the unknown.
3. Check by substituting back.

Example: the point $(k, 7)$ lies on $y = 2x - 3$. Find k.

$$7 = 2k - 3 \;\Rightarrow\; 2k = 10 \;\Rightarrow\; k = 5$$ ← put x = k, y = 7

Example: does (3, 5) lie on $y = 2x - 1$? Does (4, 6)?

$2(3) - 1 = 5$ ✓ so (3, 5) is on the line. $2(4) - 1 = 7 \neq 6$, so (4, 6) is not.

Example: the gradient from $(2, k)$ to $(5, 11)$ is 3. Find k.

$$\frac{11 - k}{5 - 2} = 3 \;\Rightarrow\; 11 - k = 9 \;\Rightarrow\; k = 2$$ ← gradient formula = 3

Example: $y = kx + 4$ is parallel to $6x - 2y = 1$. Find k. What if it were perpendicular?

$$y = 3x - \tfrac{1}{2} \Rightarrow m = 3$$ ← rearrange the given line

Parallel: $k = 3$. Perpendicular: $k = -\frac{1}{3}$.

Example: a line passes through (1, 3) and has y-intercept 5. Find its gradient.

$$3 = m(1) + 5 \;\Rightarrow\; m = -2$$ ← c = 5; use (1, 3)

Example: the line $ax + 2y = 8$ passes through (2, 1). Find a.

$$a(2) + 2(1) = 8 \;\Rightarrow\; a = 3$$

RULE: collinear points

Three points A, B, C are **collinear** (on one straight line) when $m_{AB} = m_{BC}$: the same gradient, and they share the point B.

Example: are A(1, 1), B(3, 5), C(6, 11) collinear?

$$m_{AB} = \frac{4}{2} = 2, \quad m_{BC} = \frac{6}{3} = 2$$ ← *equal, so yes: collinear*

Example: find k so that (1, 1), (3, 5) and (5, k) are collinear.

$$\frac{k-5}{5-3} = 2 \Rightarrow k = 9$$ ← *gradient must also be 2*

Mistakes to avoid

✗ Common mistakes

✗ $m = \dfrac{x_2 - x_1}{y_2 - y_1}$ ✓ $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

rise over run: the y's go on top

✗ $m = \dfrac{12-4}{4-8}$ ✓ $m = \dfrac{12-4}{8-4}$

same order top and bottom

✗ $2x + 3y = 6 : \ m = 2$ ✓ $y = -\frac{2}{3}x + 2 : \ m = -\frac{2}{3}$

rearrange to y = mx + c before reading m

✗ $m = 3 \Rightarrow m_\perp = -3$ ✓ $m = 3 \Rightarrow m_\perp = -\frac{1}{3}$

flip AND change sign

✗ $(-3, -2) : \ y - 2 = m(x - 3)$ ✓ $(-3, -2) : \ y + 2 = m(x + 3)$

subtracting a negative gives +

Straight lines in real life: temperature

Fahrenheit and Celsius are linked by a straight line: $F = 1.8C + 32$. Water freezes at 0 °C, which is $c = 32$ °F; every 1 °C warmer is 1.8 °F warmer.

$$F = 1.8(100) + 32 = 212$$ ← *boiling point: 100 °C = 212 °F ✓*

$$C = \frac{F - 32}{1.8}$$ ← *undo it for the reverse conversion (Chapter 14)*

Any time a question says "a fixed amount plus so much per...", think $y = mx + c$.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Straight line	$y = mx + c$ m = gradient, c = y-intercept; rearrange to this form first
Gradient	$m = \dfrac{y_2 - y_1}{x_2 - x_1}$ same order top and bottom
Point-gradient form	$y - y_1 = m(x - x_1)$ one point and the gradient
Horizontal / vertical	$y = b \ \ (m = 0), \qquad x = a \ \ (m \text{ undefined})$
Parallel	$m_1 = m_2$
Perpendicular	$m_1 m_2 = -1, \qquad m_2 = -\dfrac{1}{m_1}$ except horizontal with vertical
Perpendicular sense check	one uphill and one downhill; one steep and one gentle
Two-line questions	Compare (find both m's), Build (borrow m, add a point), Meet (set equal)
Midpoint	$\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$
Distance	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Axis crossings	put $x = 0$ for the y-intercept; put $y = 0$ for the x-intercept
Intersection	solve the two equations simultaneously
Point on a line	its coordinates make the equation true
Collinear	$m_{AB} = m_{BC}$
Perpendicular bisector	through the midpoint, gradient $-1/m_{AB}$

Your turn

Answers in grey; worked solutions for 12 onwards on the next page.

Reading y = mx + c

1. Gradient and intercept of $y = 5x - 2$?
2. Gradient and intercept of $y = 7 - 3x$?
3. Gradient and intercept of $4x + 2y = 10$?
4. A plumber charges £40 call-out plus £30 per hour. Write the cost y for x hours.

1. m = 5, c = −2 2. m = −3, c = 7 3. y = −2x + 5: m = −2, c = 5 4. y = 30x + 40

Two points

5. Gradient between (1, 2) and (4, 11)
6. Equation through (1, 2) and (4, 11)
7. Equation through (−2, 7) and (2, −1)
8. Equation through (0, 5) parallel to $y = 3x - 1$

5. 3 6. y = 3x − 1 7. y = −2x + 3 8. y = 3x + 5

Sketch and perpendicular

9. Where does $3x + 5y = 15$ cross the axes?
10. Perpendicular gradient to m = 4?
11. Perpendicular gradient to m = −2/3?

9. (0, 3) and (5, 0) 10. −1/4 11. 3/2

Parallel, perpendicular or neither?

12. $y = 4x + 1$ and $y = 4x - 7$
13. $y = 2x + 3$ and $x + 2y = 6$
14. $3x + y = 2$ and $y = 3x + 1$
15. A(0, 1), B(2, 7), C(1, 5), D(4, 4): AB and CD

12. parallel 13. perpendicular 14. neither 15. perpendicular

Lines through a point

16. Parallel to $y = 2x + 7$ through (3, 1)
17. Perpendicular to $y = 2x + 7$ through (4, 0)
18. Perpendicular bisector of (0, 0) and (4, 8)

16. y = 2x − 5 17. y = −x/2 + 2 18. y = −x/2 + 5

Midpoint, distance, intersection

19. Midpoint of $(-2, 3)$ and $(6, 9)$
20. Distance from $(1, 1)$ to $(4, 5)$
21. $M(2, 5)$ is the midpoint of $A(-1, 3)$ and B. Find B.
22. Where do $y = 3x - 2$ and $y = x + 4$ meet?

19. $(2, 6)$ 20. 5 21. $(5, 7)$ 22. $(3, 7)$

Find the unknown

23. $(k, 11)$ lies on $y = 3x - 4$. Find k.
24. The gradient from $(1, k)$ to $(4, 10)$ is 2. Find k.
25. $y = kx - 3$ is perpendicular to $y = 4x + 1$. Find k.
26. $ax + 3y = 12$ passes through $(3, 2)$. Find a.
27. Are $(0, 2)$, $(2, 8)$, $(5, 17)$ collinear?
28. Gradient -2, through $(3, 1)$: equation, and where it crosses the x-axis?

23. $k = 5$ 24. $k = 4$ 25. $k = -1/4$ 26. $a = 2$ 27. yes 28. $y = -2x + 7$; $(3.5, 0)$

Worked solutions, 12 to 28

12. $m_1 = 4 = m_2$ *parallel*

13. $x + 2y = 6 \Rightarrow y = -\frac{1}{2}x + 3;\quad 2 \times (-\frac{1}{2}) = -1$ *perpendicular*

14. $y = -3x + 2;\quad -3 \times 3 = -9$ *neither: not equal, product not –1*

15. $m_{AB} = \frac{6}{2} = 3,\ m_{CD} = \frac{-1}{3};\quad 3 \times (-\frac{1}{3}) = -1$ *perpendicular*

16. $y - 1 = 2(x - 3) \Rightarrow y = 2x - 5$

17. $m_\perp = -\frac{1}{2};\ y - 0 = -\frac{1}{2}(x - 4) \Rightarrow y = -\frac{1}{2}x + 2$

18. $M = (2, 4),\ m = 2,\ m_\perp = -\frac{1}{2};\ y - 4 = -\frac{1}{2}(x - 2) \Rightarrow y = -\frac{1}{2}x + 5$

19. $\left(\frac{-2+6}{2}, \frac{3+9}{2}\right) = (2, 6)$

20. $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$

21. $B = (2 \times 2 - (-1),\ 2 \times 5 - 3) = (5, 7)$

22. $3x - 2 = x + 4 \Rightarrow x = 3,\ y = 7$

23. $11 = 3k - 4 \Rightarrow k = 5$

24. $\frac{10-k}{3} = 2 \Rightarrow k = 4$

25. $4k = -1 \Rightarrow k = -\frac{1}{4}$ *perpendicular: product –1*

26. $3a + 6 = 12 \Rightarrow a = 2$

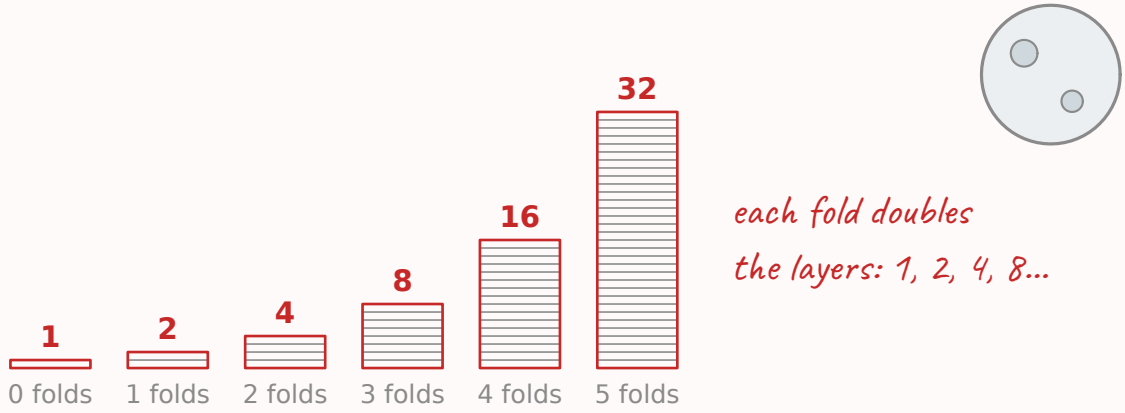
27. $\frac{8-2}{2} = 3,\ \frac{17-8}{3} = 3$ *equal gradients: collinear*

28. $1 = -6 + c \Rightarrow y = -2x + 7;\quad 0 = -2x + 7 \Rightarrow x = 3.5$

Chapter 21

The exponential function

fold a sheet of paper 42 times and reach the Moon



Most growth you know is **adding**: a taxi fare goes up £2 per mile. **Exponential** growth is **multiplying**: the amount doubles (or grows by 5%, or halves) at every step. It starts slowly and then explodes, which is why it surprises everyone.

You'll see it race against x^2, meet its mirror image (decay), and watch the famous number $e = 2.718...$ appear out of a bank account. Then two real decays from your lectures: cooling and the current in a circuit.

Goes with **Lecture 7** (power vs exponential, growth vs decay, the number e).

Folding paper: multiply, multiply, multiply

A sheet of paper is about 0.1 mm thick. Each fold **doubles** the thickness. After n folds there are 2^n layers:

folds n	layers 2^n	thickness	about as tall as
0	1	0.1 mm	a sheet
10	1 024	102 mm	a paperback book
20	1 048 576	105 m	a 30-storey tower
30	1 073 741 824	107 km	into space
42	4 398 046 511 104	440,000 km	past the Moon (384,000 km)

(Nobody can actually fold paper 42 times; about 7 is the limit. But the **maths** is the point.) The thickness is 0.1×2^n mm. Notice where the letter is: **up in the power**. That's what makes a function exponential.

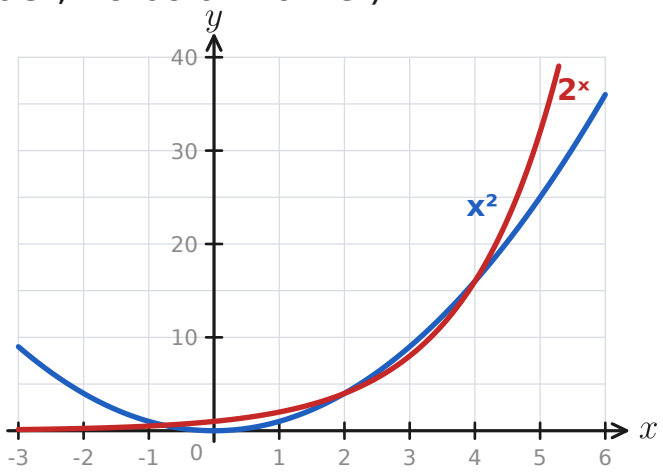
Chapter 17 link: the layers 1, 2, 4, 8, ... are a geometric progression with $r = 2$. An exponential function is a G.P. that's allowed to take in-between inputs too.

The race: x² against 2ˣ

They look similar, but in x^2 the letter is the **base** (a power function, Chapter 19) and in 2^x the letter is the **power** (exponential). Lecture 7 races them:

x	0	1	2	3	4	5	6	8	10	20
x²	0	1	4	**9**	16	25	36	64	100	400
2ˣ	**1**	**2**	4	8	16	**32**	**64**	**256**	**1024**	**1.05M**

They **tie twice**: at $x = 2$ (both are 4) and at $x = 4$ (both are 16). Between those, x^2 is ahead ($3^2 = 9$ beats $2^3 = 8$). After $x = 4$, 2^x runs away: at $x = 20$ it's over a million while x^2 is only 400. (Bold = the leader; no bold = a tie.)



Every step right, x^2 adds a bit more (+5, then +7, then +9…).

2^x DOUBLES every step. Doubling always wins in the end.

On the left, 2^x never goes negative or reaches 0: 2 to the −3 = 1/8.

The "where's the x?" test: letter in the base → power function. Letter in the power → exponential.

The shape: growth and its mirror, decay

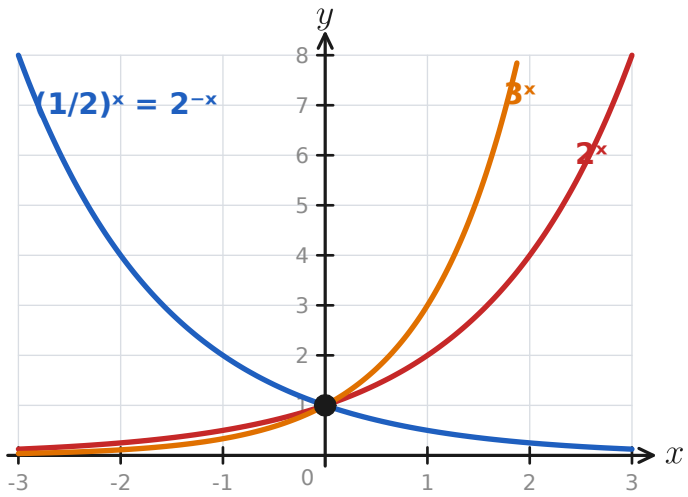
RULE: exponential graphs

$y = A \times b^x$ ($b > 0$): the graph passes through $(0, A)$, because $b^0 = 1$. A is the **starting value**.

$b > 1$: **growth**. $0 < b < 1$: **decay**. A bigger b grows faster.

$y = Ae^{kx}$: $k > 0$ grows, $k < 0$ decays; again A is the value at $x = 0$.

The x-axis is an **asymptote**: $A \times b^x$ (with $A > 0$) is always positive, never 0.



Every one passes through (0, 1), because anything to the power 0 is 1 (Chapter 9).

Base bigger than 1: GROWTH.

Bigger base: steeper.

Base between 0 and 1: DECAY.

$(1/2)^x$ is 2^x reflected in the y-axis.

All of them hug the x-axis on one side without ever touching it: the x-axis is an **asymptote** (Chapter 19). Lecture 7: growth models savings (and student debt!); decay models radioactivity and electric currents dying away.

Start amount × multiplier to the power of steps

RULE: percentage growth and decay

An increase of $r\%$ per step multiplies by $1 + \frac{r}{100}$. A decrease of $r\%$ multiplies by $1 - \frac{r}{100}$ (Chapter 4's multipliers).

After n steps: **amount = start × multiplier**^{n}, i.e. $y = A \times b^n$.

Compound interest, rate r (as a decimal) per year, added n times a year, for t years: $A\left(1 + \frac{r}{n}\right)^{nt}$. Each payment adds $\frac{r}{n}$, and there are nt payments.

Real exponentials are written $y = A \times b^x$: A is the **starting amount**, b is the **multiplier per step** (Chapter 4's multipliers), and x counts the steps.

$$y = A \times b^x$$

Savings: £1000 at 5% interest a year. Each year the money is multiplied by 1.05:

$$1000 \times 1.05 \times 1.05 \times \cdots = 1000 \times 1.05^t$$ ← *t years of × 1.05*

$$1000 \times 1.05^{10} \approx 1628.89$$ ← *after 10 years: £1628.89*

Compare simple interest (£50 added every year, a straight line): £1500 after 10 years. The extra £128.89 is interest earning interest.

Decrease: a £15,000 car loses 15% of its value a year. Multiplier $1 - 0.15 = 0.85$:

$$15000 \times 0.85^3 \approx 9211.88$$ ← *value after 3 years: £9211.88*

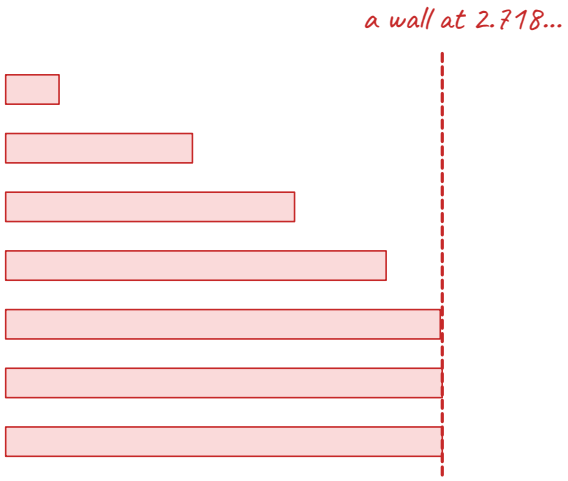
Interest added monthly: £1000 at 6% a year, paid monthly for 2 years: $n = 12$, so each month adds $\frac{0.06}{12} = 0.005$, and there are $12 \times 2 = 24$ payments:

$$1000\left(1 + \frac{0.06}{12}\right)^{24} = 1000 \times 1.005^{24} \approx 1127.16$$

The number e falls out of a bank account

A generous bank offers **100% interest** per year on £1. Paid once at the end: £2. But what if they pay **half** of it (50%) every six months, so the first half starts earning too? Or a quarter every three months? Paying n times a year multiplies by $(1 + \frac{1}{n})$, n times:

paid n times	$(1 + 1/n)^n$
1 (once a year)	£2.00000
2 (every 6 months)	£2.25000
4 (every 3 months)	£2.44141
12 (monthly)	£2.61304
365 (daily)	£2.71457
8 760 (hourly)	£2.71813
1 000 000 (a million times)	£2.71828



More often helps less and less. It never passes a ceiling of about **2.71828**. That number is called e (Lecture 7: $e = (1 + \frac{1}{n})^n$ as $n \to \infty$). Like π, nobody invented it and its decimals never end or repeat.

The function e^x is **the** exponential function; you'll also see it written $\exp(x)$. Your calculator has an e^x button (usually shift + ln). It behaves just like 2^x or 3^x, sitting between them, and it's the one nature and calculus prefer (Chapter 34 shows why).

Real decay: cooling tea and a draining capacitor

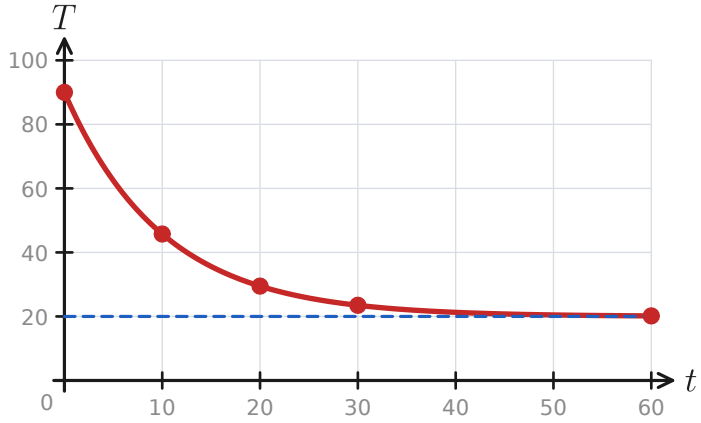
RULE: decay towards a level

$y = A + Be^{-kt}$ (with $k > 0$) starts at $y = A + B$ when $t = 0$ and **levels off at** $y = A$: as t grows, $e^{-kt} \to 0$. The line $y = A$ is a horizontal asymptote.

In $y = Ie^{-t/\tau}$, the **time constant** τ is the time to fall to $e^{-1} \approx 37\%$ of the start. Every further τ multiplies by 0.37 again.

A mug of tea at 90 °C cools in a 20 °C room. The **difference** from room temperature (70 °C to start) decays exponentially:

$$T = 20 + 70e^{-0.1t}$$ ← *t in minutes*



t = 0: 20 + 70 × 1 = 90 °C
t = 10: 20 + 70/e ≈ 45.8 °C
t = 20: ≈ 29.5 °C
t = 30: ≈ 23.5 °C

It creeps towards 20 °C (blue asymptote) but never gets there.

The capacitor (Lecture 7, Question 2). A current starts at I and dies away:

$$i = Ie^{-t/CR}$$ ← *C and R are the circuit's capacitance and resistance*

When $t = CR$ the power is -1, so $i = Ie^{-1} \approx 0.37I$: the current has dropped to 37%. CR is called the **time constant**. After another CR it's 37% of that, and so on: the same fraction lost in every equal stretch of time. That's the fingerprint of every exponential decay.

Chapter 22 answers the reverse question, "how **long** until the current is 10%?", which needs logarithms.

Find the unknown: the model from the data

PROCEDURE: finding A and b from two points

1. Each point (x, y) on $y = A \times b^x$ gives an equation: put in x and y.
2. If one point has $x = 0$, it gives A straight away ($b^0 = 1$).
3. Otherwise **divide** one equation by the other: A cancels and leaves a power of b. Take the root (use the positive one: $b > 0$).
4. Put b back to find A, and check with the other point.

Example: $y = A \times b^x$ passes through $(0, 3)$ and $(2, 12)$. Find A and b.

$$3 = A \times b^0 = A \longleftarrow x = 0 \text{ gives } A$$

$$12 = 3b^2 \;\Rightarrow\; b^2 = 4 \;\Rightarrow\; b = 2 \longleftarrow b > 0, \text{ so not } -2$$

$$y = 3 \times 2^x \longleftarrow \text{check: } 3 \times 2^2 = 12$$

Example: $y = A \times b^x$ passes through $(1, 6)$ and $(3, 54)$. Find A and b.

$$Ab = 6, \qquad Ab^3 = 54 \longleftarrow \text{one equation per point}$$

$$\frac{Ab^3}{Ab} = b^2 = 9 \;\Rightarrow\; b = 3 \longleftarrow \text{divide: } A \text{ cancels}$$

$$A = \frac{6}{3} = 2: \qquad y = 2 \times 3^x \longleftarrow \text{check: } 2 \times 3^3 = 54$$

RULE: half-life and doubling time

Half-life T: the time for the amount to halve. After time t: $y = A\left(\frac{1}{2}\right)^{t/T}$.

Doubling time T: the time for the amount to double. After time t: $y = A \times 2^{t/T}$.

t/T counts the halvings (or doublings). When the answer is a whole number of them, count; otherwise Chapter 22 solves it with logs.

Example: a substance has a half-life of 5 years. How much of 80 g is left after 20 years? When is 10 g left?

$$80 \times \left(\tfrac{1}{2}\right)^{20/5} = 80 \times \tfrac{1}{16} = 5 \text{ g}$$ ← *20 years = 4 half-lives*

$$80 \to 40 \to 20 \to 10 : \ 3 \text{ half-lives} = 15 \text{ years}$$

count the halvings

Example: a population of 500 doubles every 3 hours. When does it reach 8000?

$$\frac{8000}{500} = 16 = 2^4$$ ← *how many doublings?*

$$4 \times 3 = 12 \text{ hours}$$ ← *4 doublings of 3 hours*

Example: £2000 grows to £2420 after 2 years of compound interest. Find the yearly rate.

$$2000\,m^2 = 2420 \ \Rightarrow \ m^2 = 1.21 \ \Rightarrow \ m = 1.1$$

m = the yearly multiplier

$$m = 1 + \tfrac{r}{100} = 1.1 \ \Rightarrow \ r = 10\%$$

Example: a value is multiplied by 0.92 every year. What is the percentage change?

$0.92 = 1 - 0.08$, so it is a **decrease of 8%** a year.

Mistakes to avoid

✗ Common mistakes

✗ 5% rise: $\times 0.05$

the multiplier keeps the original 100%

✓ $\times 1.05$

✗ 20% fall: $\times 0.2$

1 − 0.20 = 0.80

✓ $\times 0.8$

✗ $1000 \times 1.05 \times 10$

multiply by 1.05 ten times: a power

✓ 1000×1.05^{10}

✗ $2 \times 3^x = 6^x$

only the 3 is raised to the power

✓ 2×3^x

✗ $b^2 = 9 \Rightarrow b = \pm 3$

the base of an exponential is positive

✓ $b = 3$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Exponential model	$y = A \times b^x$ A = start value (at $x = 0$); $b > 1$ growth, $0 < b < 1$ decay
e form	$y = Ae^{kx}$ $k > 0$ growth, $k < 0$ decay
Percentage change	$\times \left(1 \pm \dfrac{r}{100}\right)$ per step + for increase, − for decrease
Compound interest	$A\left(1 + \dfrac{r}{n}\right)^{nt}$ rate r as a decimal, n payments a year, t years
The number e	$e = \lim\limits_{n \to \infty} \left(1 + \dfrac{1}{n}\right)^n \approx 2.718$
Levelling off	$y = A + Be^{-kt} \ \rightarrow \ A$ starts at $A + B$; asymptote $y = A$
Time constant	$i = Ie^{-t/\tau}$ after τ, $e^{-1} \approx 37\%$ is left
Half-life	$y = A\left(\frac{1}{2}\right)^{t/T}$ T = half-life
Doubling time	$y = A \times 2^{t/T}$ T = doubling time
Two points	$x = 0$ gives A; otherwise divide the two equations to cancel A

Folding paper: your turn

Answers in grey. Use a calculator for the e questions.

Power or exponential?

1. $y = x^5$
2. $y = 5^x$
3. $y = 3 \times 0.8^x$
4. $y = 1/x^2$

1. power 2. exponential growth 3. exponential decay 4. power

Using $y = A \times b^x$

5. A colony of 200 bacteria doubles every hour. How many after 6 hours?
6. A £15,000 car loses 20% of its value each year. Value after 3 years?
7. £500 at 3% compound interest for 4 years?

5. $200 \times 2^6 = 12{,}800$ 6. $15{,}000 \times 0.8^3 = £7680$ 7. $500 \times 1.03^4 \approx £562.75$

e

8. Work out e^2 and e^{-2} (2 d.p.)
9. Using T = 20 + 70e^(−0.1t), find T when t = 5.
10. A current i = 4e^(−t/2) amps. Find i at t = 0 and t = 2.

8. 7.39 and 0.14 9. about 62.5 °C 10. 4 A and $4e^{-1} \approx 1.47$ A

Find the unknown

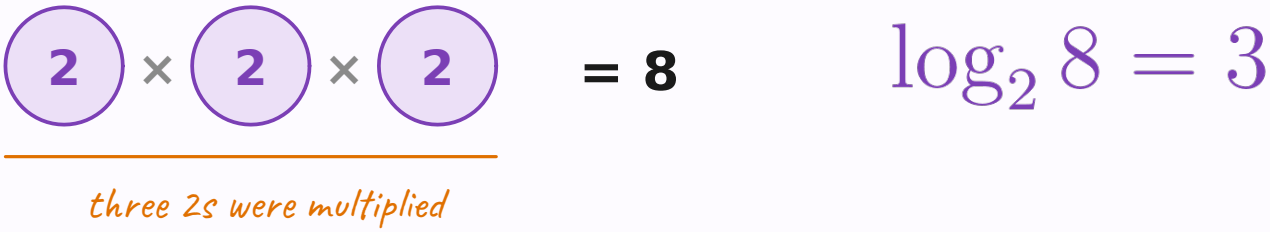
11. $y = A \times b^x$ passes through (0, 5) and (2, 45). Find A and b.
12. $y = A \times b^x$ passes through (1, 10) and (3, 40). Find A and b.
13. A drug has a half-life of 6 hours. How much of a 200 mg dose is left after 24 hours?
14. Bacteria double every 20 minutes. How long until 300 become 4800?
15. £5000 becomes £5618 after 2 years of compound interest. Find the yearly rate.
16. A car's value is multiplied by 0.85 each year. What is the yearly percentage change?

11. A = 5, b = 3 ($b^2 = 45/5 = 9$) 12. b = 2, A = 5 ($b^2 = 40/10 = 4$) 13. 12.5 mg (4 half-lives: $200 \times (1/2)^4$) 14. 80 minutes ($4800/300 = 16 = 2^4$) 15. 6% ($m^2 = 1.1236$, m = 1.06) 16. a 15% decrease (1 − 0.85)

Chapter 22

Logarithms

counting the multiplications



Powers ask: **multiply 2 by itself 3 times, what do you get?** (8). Logarithms ask the question the other way round: **how many 2s must I multiply to get 8?** (3). That's all a log is: **a count of multiplications**, or equivalently, **the missing power**.

Logs are the undo button for exponentials (Chapter 21), so they're how you answer "how long until...?" questions. Each log rule is just an index rule (Chapter 9) wearing a disguise, and this chapter shows them side by side.

Goes with **Lecture 7** (logarithms, laws of logs, natural log, Questions 1-3).

The question a log asks

RULE: definition of a logarithm

$y = \log_b x \iff b^y = x$. A log **is a power**: the power you raise b to, to get x.

The base b is positive and not 1. The number x must be **positive**: there is no log of 0 or of a negative number.

$\log_b b = 1$ (since $b^1 = b$) and $\log_b 1 = 0$ (since $b^0 = 1$).

Read $\log_b N$ as "**log to the base b of N**", and hear it as the question: **what power of b gives N?** The small number b is the **base**: the thing being multiplied.

$\log_2 8$	*how many 2s make 8?*	$2 \times 2 \times 2 = 8$	**3**
$\log_{10} 100$	*how many 10s make 100?*	$10 \times 10 = 100$	**2**
$\log_4 16$	*how many 4s make 16?*	$4 \times 4 = 16$	**2**
$\log_{10} 1000000$	*how many 10s make a million?*	10^6	**6**
$\log_5 5$	*how many 5s make 5?*	5	**1**
$\log_7 1$	*how many 7s make 1?*	$7^0 = 1$	**0**

The same fact in two languages. Every log statement is a power statement in disguise. Lecture 7's translation rule:

$$y = a^x \quad \rightleftharpoons \quad x = \log_a y$$

same fact

A handy chant: "the base stays the base, and the log is the power".

Logs are the undo button for powers

RULE: logs and powers undo each other

$\log_b(b^x) = x$ and $b^{\log_b x} = x$ (for $x > 0$).

Base e: $\ln x$ means $\log_e x$. So $\ln(e^x) = x$, $e^{\ln x} = x$, $\ln e = 1$, $\ln 1 = 0$.

Base 10: $\log x$ (the calculator's log button) means $\log_{10} x$. So $\log 10 = 1$, $\log 1 = 0$.

In Chapter 14 every operation had an undo button: + undoes −, × undoes ÷, squaring undoes square-rooting. **"Raise 10 to the power" and "take $\log_{10}$" undo each other**:

$$\log_{10}(10^x) = x$$ ← *power first, then log: back to x*

$$10^{\log_{10} x} = x$$ ← *log first, then power: back to x*

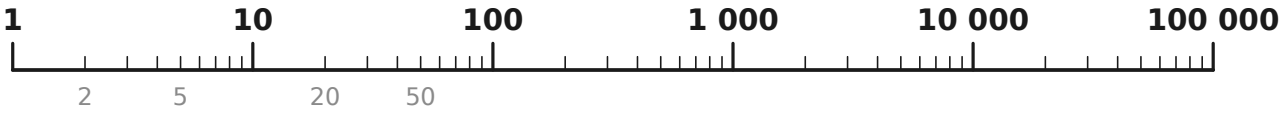
Your calculator has two log buttons. **log** means base 10 (the **common log**); **ln** means base e (the **natural log**, Lecture 7: we write ln rather than $\log_e$).

Logs of in-between numbers are decimals. 50 is between 10 and 100, so $\log_{10} 50$ is between 1 and 2:

What you can't log: $\log 1 = 0$ (since $10^0 = 1$), numbers between 0 and 1 have **negative** logs ($\log 0.01 = -2$), and 0 or negative numbers have **no log at all**: no power of 10 is ever 0 or negative (the curve of 10^x never reaches the x-axis). So the domain of log is positive numbers only.

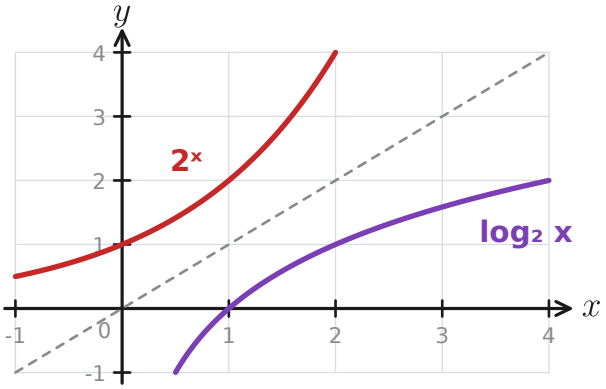
Log scales: a ruler where every step is ×10

On an ordinary ruler every step **adds** the same amount. On a **log scale** every step **multiplies** by the same amount. That squashes enormous ranges onto one page:



equal gaps from 1 to 10, 10 to 100, 100 to 1000… and 2 sits 30% of the way to 10, because log 2 ≈ 0.301

You meet log scales in **decibels** (sound), **pH** (acidity) and the **Richter scale** (earthquakes): one step up means 10 times more intense. They also straighten out exponential curves: plot 10^x on a log axis and you get a straight line.



The graph of log is the graph of the power, reflected in the dashed line y = x (because they undo each other: inputs and outputs swap).

It rises ever more slowly and never exists for x ≤ 0.

The log laws are the index laws in disguise

RULE: the laws of logarithms (same base throughout)

$\log_b(xy) = \log_b x + \log_b y$ $\log_b \dfrac{x}{y} = \log_b x - \log_b y$ $\log_b x^n = n \log_b x$

Special cases: $\log_b \dfrac{1}{x} = -\log_b x$ and $\log_b \sqrt{x} = \frac{1}{2} \log_b x$.

The laws only combine logs with the **same base**: $\log_2 x + \log_3 y$ can't be combined (change base first).

There is **no law** for $\log(x + y)$ or $\log x \times \log y$.

A log **is** a power, so the three index laws of Chapter 9 turn straight into three log laws. Read across each row: the index law on the left, the log law on the right, and a number test.

$$10^m \times 10^n = 10^{m+n} \quad \rightarrow \quad \log(ab) = \log a + \log b$$

$\log(100 \times 1000) = \log 10^5 = 5 = 2 + 3$

multiply inside → add outside

$$\frac{10^m}{10^n} = 10^{m-n} \quad \rightarrow \quad \log \frac{a}{b} = \log a - \log b$$

$\log \dfrac{1000}{10} = \log 100 = 2 = 3 - 1$

divide inside → subtract outside

$$(10^m)^n = 10^{mn} \quad \rightarrow \quad \log a^n = n \log a$$

$\log 1000^2 = \log 10^6 = 6 = 2 \times 3$

power inside → multiply in front

Traps (test with numbers!): $\log(a + b)$ is **not** $\log a + \log b$ (try $\log(10 + 10) = 1.30$, but $1 + 1 = 2$). And $\dfrac{\log a}{\log b}$ is **not** $\log \frac{a}{b}$.

(Lecture 7's example for $\log a^n = n \log a$ is a copy of the first law's example. A real one: $\log_{10} a^3 = 3 \log_{10} a$.)

PROCEDURE: combining several logs into one

1. Check every log has the same base.
2. Move every number in front up into a power: $n \log x = \log x^n$.
3. Every $+$ log goes on top of one fraction, every $-$ log goes underneath: $\log A + \log B - \log C = \log \dfrac{AB}{C}$.
4. Simplify the fraction inside the log.

$$2\log 3 + \log 4 - \log 6 = \log 9 + \log 4 - \log 6$$

step 2: $2 \log 3 = \log 3^2$

$$= \log \frac{9 \times 4}{6} = \log 6$$ ← steps 3 and 4

$$\ln x - 3\ln y + \tfrac{1}{2}\ln z = \ln \frac{x\sqrt{z}}{y^3}$$ ← powers first; minus terms underneath

$$\log \frac{1}{100} = \log 1 - \log 100 = 0 - 2 = -2$$ ← $\log(1/a) = -\log a$

Lecture 7, Question 1: pulling a log apart

Express $\log_{10} \frac{a^2 b^3}{100\sqrt{c}}$ as separate logs of a, b and c. One law per line; do the division first, because it's the "outermost" operation. First rewrite the root as a power (Chapter 9): $\sqrt{c} = c^{1/2}$.

$$\log \frac{a^2 b^3}{100 c^{1/2}} = \log(a^2 b^3) - \log(100 c^{1/2})$$ ← divide → subtract

$$= \log a^2 + \log b^3 - \left(\log 100 + \log c^{1/2}\right)$$ ← multiply → add (keep the bracket!)

$$= 2\log a + 3\log b - (2 + \tfrac{1}{2}\log c)$$ ← powers come to the front; log 100 = 2

$$= 2\log a + 3\log b - 2 - \tfrac{1}{2}\log c$$ ← the minus reaches both terms

Test with numbers: $a = 3$, $b = 5$, $c = 7$: both sides give 0.6286 ✓.

How many doublings? Solving with logs

PROCEDURE: solving an equation with the unknown in the power

1. Get the power on its own on one side: $b^{(\dots)} = N$.
2. Take logs of **both** sides (log or ln; use ln when the base is e).
3. Use the power law to bring the unknown down: $(\dots)\log b = \log N$.
4. Solve the ordinary equation that's left. If the unknown is in two powers, expand, collect the x terms and factorise out x.

When the unknown is **in the power**, no amount of adding or dividing will get it down. Lecture 7: **take logs of both sides** (allowed: do the same to both sides), then use the power law to bring it to the front:

$$2^x = 8$$

$$\log 2^x = \log 8 \quad \longleftarrow \text{log both sides}$$

$$x \log 2 = \log 8 \quad \longleftarrow \text{the power law lifts } x \text{ down}$$

$$x = \frac{\log 8}{\log 2} = \frac{0.903}{0.301} = 3 \quad \longleftarrow \div \log 2 \text{ (the slide's 0.90309)}$$

Now one you couldn't guess: how many years until £1000 at 5% doubles to £2000?

$$1000 \times 1.05^t = 2000 \quad \longleftarrow \text{Chapter 21's formula}$$

$$1.05^t = 2 \quad \longleftarrow \div 1000$$

$$t = \frac{\log 2}{\log 1.05} \approx 14.2 \quad \longleftarrow \text{log both sides; power law; } \div \log 1.05$$

So 15 whole years (after 14 years it's only £1979.93). Change of base (Lecture 7) is the same trick: $\log_5 15 = \frac{\log 15}{\log 5} = \frac{1.1761}{0.6990} = 1.6825$, which lets a base-10 calculator do any base.

RULE: change of base

$\log_a N = \dfrac{\log N}{\log a} = \dfrac{\ln N}{\ln a}$: any base can be worked out with the log or ln button.

Example: solve $3^{2x-1} = 7$ (3 d.p.).

$$(2x-1)\log 3 = \log 7 \quad \leftarrow \text{steps 2 and 3: log both sides, power down}$$

$$2x - 1 = \frac{\log 7}{\log 3} = 1.7712 \quad \leftarrow \div \log 3$$

$$x = \frac{2.7712}{2} = 1.386 \quad \leftarrow +1, \text{ then } \div 2$$

Example: the unknown on both sides: $2^{x+1} = 3^x$ (3 d.p.).

$$(x+1)\log 2 = x \log 3 \quad \leftarrow \text{log both sides}$$

$$x\log 2 + \log 2 = x \log 3 \quad \leftarrow \text{expand}$$

$$\log 2 = x(\log 3 - \log 2) \quad \leftarrow \text{x terms together; factorise out x}$$

$$x = \frac{\log 2}{\log 3 - \log 2} = \frac{0.3010}{0.1761} = 1.710$$

Example: find k if $50 = 100e^{-5k}$ (3 d.p.).

$$e^{-5k} = 0.5 \quad \leftarrow \div 100 \text{ to get the power alone}$$

$$-5k = \ln 0.5 \;\Rightarrow\; k = \frac{\ln 0.5}{-5} = 0.139 \quad \leftarrow \text{ln both sides (base e)}$$

Example: a substance has a half-life of 5 years (Chapter 21). How long until 80 g has decayed to 30 g?

$$80\left(\tfrac{1}{2}\right)^{t/5} = 30 \;\Rightarrow\; 0.5^{t/5} = 0.375 \quad \leftarrow \div 80$$

$$\frac{t}{5} = \frac{\log 0.375}{\log 0.5} = 1.415 \;\Rightarrow\; t \approx 7.08 \text{ years}$$

Example: Chapter 17's question: the smallest n with $2^n > 1001$.

$$n \log 2 > \log 1001 \;\Rightarrow\; n > \frac{\log 1001}{\log 2} = 9.97$$

log 2 > 0, so the > stays

So $n = 10$, the next whole number, as found by trial in Chapter 17.

Solving equations that contain logs

PROCEDURE: solving log equations

1. Use the laws to collect the logs into **one** log on one side (same base): $\log_b(\dots) = c$.
2. Rewrite as a power, by the definition: $\log_b(\dots) = c \;\Rightarrow\; (\dots) = b^c$. (For $\ln$, $b = e$; for $\log$, $b = 10$.)
3. Solve the equation that's left.
4. **Check every answer in the original**: the inside of each log must be positive. Reject any root that makes a log of 0 or of a negative number.

$$\log_2 x = 5 \;\Rightarrow\; x = 2^5 = 32$$ ← *definition: the log is the power*

$$\ln x = 2 \;\Rightarrow\; x = e^2 \approx 7.389$$ ← *base e*

$$\log_3(2x+1) = 2 \;\Rightarrow\; 2x+1 = 9 \;\Rightarrow\; x = 4$$

$3^2 = 9$

Example: solve $\log x + \log(x+3) = 1$ (base 10).

$$\log\big(x(x+3)\big) = 1$$ ← *step 1: add logs = log of the product*

$$x(x+3) = 10^1 = 10$$ ← *step 2: rewrite as a power*

$$x^2 + 3x - 10 = 0 \;\Rightarrow\; (x+5)(x-2) = 0$$

step 3: a quadratic

$$x = 2 \quad (x = -5 \text{ rejected})$$ ← *step 4: log(−5) doesn't exist*

Check: $\log 2 + \log 5 = \log 10 = 1$ ✓. The root $x = -5$ solves the quadratic but not the original, because $\log(-5)$ has no value.

Example: solve $\log_2 x - \log_2(x-2) = 3$.

$$\log_2 \frac{x}{x-2} = 3 \;\Rightarrow\; \frac{x}{x-2} = 2^3 = 8$$ ← subtract logs = log of the quotient

$$x = 8x - 16 \;\Rightarrow\; x = \frac{16}{7}$$ ← × (x − 2); 16/7 > 2, so both logs exist

Lecture 7, Question 2: how long for the current to fade?

The current falls from I to i following $i = Ie^{-t/CR}$. Find an expression for the time t. The unknown is in the power of e, so we'll need **ln**, the undo button for e. Unwrap one layer per line:

$$i = Ie^{-t/CR}$$

$$i = \frac{I}{e^{t/CR}}$$ ← negative power = 1 over (Chapter 9)

$$e^{t/CR} = \frac{I}{i}$$ ← swap i and the e-part (× e^(t/CR), ÷ i)

$$\frac{t}{CR} = \ln \frac{I}{i}$$ ← ln undoes e: ln(e^□) = □

$$t = CR \ln \frac{I}{i}$$ ← × CR

Put numbers in (Chapter 21's question): with $CR = 2$ seconds, how long until the current is 10% of the start? Then $\frac{I}{i} = 10$:

$$t = 2 \ln 10 \approx 2 \times 2.303 = 4.61 \text{ s}$$

Check: $e^{-4.61/2} = e^{-2.303} \approx 0.10$ ✓.

Lecture 7, Question 3: resisted motion

$v = u(1 - e^{-kt})$ gives the velocity v at time t from the starting velocity u. Find t. (The slides do it two ways; this is the shorter, version 3b.)

$$v = u - ue^{-kt}$$ ← expand the bracket

$$v = u - \frac{u}{e^{kt}}$$ ← negative power = 1 over

$$ve^{kt} = ue^{kt} - u$$ ← × e^(kt) every term

$$u = (u - v)e^{kt}$$ ← gather the e-terms: factorise (Chapter 12)

$$e^{kt} = \frac{u}{u - v}$$ ← ÷ (u − v)

$$kt = \ln \frac{u}{u - v}$$ ← ln undoes e

$$t = \frac{1}{k} \ln \frac{u}{u - v}$$ ← ÷ k

Test: $u = 10$, $k = 0.5$, $v = 5$: $t = 2 \ln 2 \approx 1.386$. Back in the original: $10(1 - e^{-0.693}) = 10(1 - 0.5) = 5$ ✓.

Mistakes to avoid

✗ Common mistakes

✗ $\log(x+3) = \log x + \log 3$

the law is for a product, not a sum

✓ $\log(3x) = \log x + \log 3$

✗ $\dfrac{\log a}{\log b} = \log \dfrac{a}{b}$

a quotient of logs isn't a log

✓ $\log \dfrac{a}{b} = \log a - \log b$

✗ $\log_2 x + \log_3 x = \log_6 x$

the laws need the same base

✓ change base first

✗ $(\log x)^2 = 2 \log x$

the power must be inside the log

✓ $\log x^2 = 2 \log x$

✗ $\ln x = 2 \Rightarrow x = \ln 2$

undo ln with e, not with ln

✓ $x = e^2$

✗ $x = -5$ in $\log x$

check answers in the original

✓ reject: no log of a negative

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Definition	$y = \log_b x \iff b^y = x$ $b > 0$, $b \neq 1$; only $x > 0$ has a log
Special values	$\log_b b = 1, \qquad \log_b 1 = 0$
Undo	$\log_b(b^x) = x, \qquad b^{\log_b x} = x$
ln and e	$\ln(e^x) = x, \quad e^{\ln x} = x, \quad \ln e = 1, \quad \ln 1 = 0$ $\ln = \log_e$, $\log = \log_{10}$
Product law	$\log(ab) = \log a + \log b$ same base throughout
Quotient law	$\log \dfrac{a}{b} = \log a - \log b$
Power law	$\log a^n = n \log a$
Reciprocal	$\log \dfrac{1}{a} = -\log a$
Change of base	$\log_a N = \dfrac{\log_b N}{\log_b a}$
Unknown in a power	power alone, log both sides, power law, solve
Log equation	one log each side or log = number; rewrite as a power; reject roots that log a number ≤ 0
No laws for	$\log(a+b)$, $\log a \times \log b$, $\frac{\log a}{\log b}$

Counting multiplications: your turn

Answers in grey.

What is it?

1. $\log_3 9$
2. $\log_2 32$
3. $\log_{10} 0.001$
4. $\log_6 1$
5. Write $5^3 = 125$ as a log
6. Write $\log_2 64 = 6$ as a power

1. 2 2. 5 3. −3 4. 0 5. $\log_5 125 = 3$ 6. $2^6 = 64$

Laws

7. Write as one log: log 4 + log 5
8. Write as one log: 3 log 2 − log 4
9. Expand: log(x^3y/z)

7. log 20 8. log 2 (since 8/4 = 2) 9. 3 log x + log y − log z

Solve

10. $3^x = 81$
11. $5^x = 40$ (2 d.p.)
12. $e^x = 20$ (2 d.p.)
13. How many years for money at 8% to double?
14. Make t the subject: N = N_0e^(−λt)

10. 4 11. 2.29 12. ln 20 ≈ 3.00 13. log 2 / log 1.08 ≈ 9.0 years 14. t = (1/λ) ln(N_0/N)

Find the unknown

15. $\log_2 x = 6$
16. ln x = 3 (2 d.p.)
17. log x + log(x − 3) = 1
18. 5^(2x + 1) = 30 (3 d.p.)
19. 2^x = 5^(x − 1) (3 d.p.)
20. Find k if 40 = 100e^(−2k) (3 d.p.)

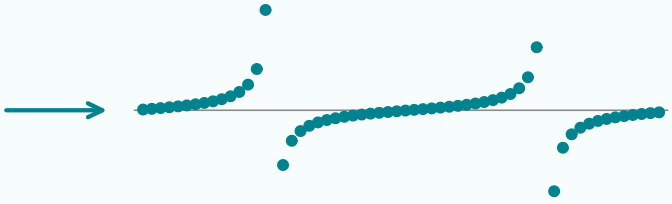
15. x = 64 ($x = 2^6$) 16. x = e^3 ≈ 20.09 17. x = 5 ($x^2 - 3x - 10 = 0$; reject x = −2) 18. x = 0.557 (2x + 1 = log 30 / log 5 = 2.113) 19. x = 1.756 (x = log 5 / (log 5 − log 2)) 20. k = 0.458 (e^(−2k) = 0.4, k = −ln 0.4 / 2)

Chapter 23

Tables and charts

be a graph detective

0.10	0.20	0.31	0.42	0.55	0.68
0.84	1.03	1.26	1.56	1.96	2.57
3.60	5.80	14.10	-34.23	-7.70	-4.29
-2.93	-2.19	-1.71	-1.37	-1.12	-0.92
-0.75	-0.60	-0.47	-0.36	-0.25	-0.14



the same 60 numbers as dots: a repeating pattern of spikes

A list of four numbers you can hold in your head. Lecture 5 points out that research says we can only juggle about **four to seven things** at once, so a list of sixty numbers is just noise. Turn the same numbers into dots (above) and your eyes spot the pattern instantly.

This chapter is about **drawing data honestly** and **reading other people's graphs suspiciously**. Each page is a **case file**: a question, the evidence, and the verdict.

Goes with **Lecture 5** (graphical analysis: basic charts, outliers, bad and good graphs, form / message / criticism).

Two kinds of data

Data just means the facts you've recorded. Before drawing anything, ask one question: **is each fact a word or a number?**

Categorical (words / labels)

favourite drink: tea, coffee, juice

bus route: 7, 36, 58 (numbers used as names!)

eye colour, course, month

Numerical (measured / counted)

height 1.74 m, age 19, wage £16,400

you can sensibly add or average them

(the average bus route is nonsense)

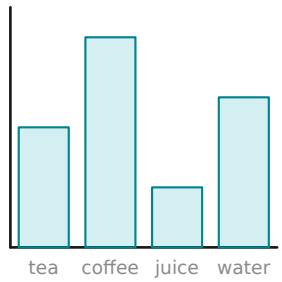
A **frequency** is a count: "how many times did this happen?". Most charts draw frequencies. Lecture 5 asks the difference between a **chart** and a **graph**: loosely, a **graph** plots numbers against numbers on two axes, while **chart** is the wider word (bar charts, pie charts, tables...). People use the words loosely, so don't lose sleep over it.

The line-up: six basic charts

The same small data set drawn six ways (the lecture's Minitab examples). Each chart is good at **one** job.

Bar chart

categories; gaps between bars because tea and coffee are separate things

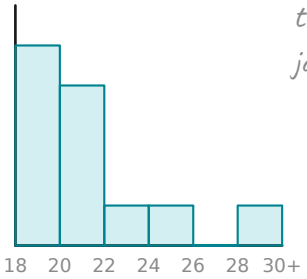


Pie chart

parts of one whole; slice angle = share of 360°

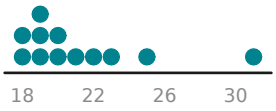
Histogram

numbers in ranges; bars touch because the ranges join up



Dotplot

one dot per value; small data sets, see clusters



Stem and leaf

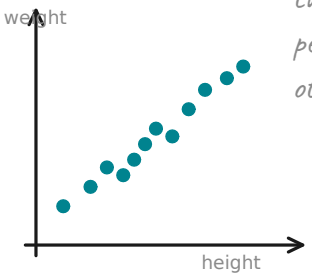
like a dotplot made of the digits themselves

1	8 8 9 9 9
2	0 0 1 2 3 5
3	1

key: 2 | 5 means 25

Scatter plot

two measurements per person: does one go with the other?



Pie angle: coffee is 7 of 18 drinks, so its slice is $\frac{7}{18} \times 360° = 140°$.

Pie charts: the angle rule, both ways

RULE: pie chart angles

The whole circle, $360°$, stands for the **total** frequency.

Frequency to angle: $\text{angle} = \dfrac{\text{frequency}}{\text{total}} \times 360°$.

Angle to frequency: $\text{frequency} = \dfrac{\text{angle}}{360°} \times \text{total}$.

The angles add up to $360°$ and the frequencies add up to the total, so one missing slice is whatever is left over.

Example: a pie chart shows 250 people. One slice is $72°$. How many people is that?

$$\frac{72}{360} \times 250 = \frac{1}{5} \times 250 = 50$$ ← angle ÷ 360 = the fraction of the whole

Example: a $45°$ slice stands for 30 people. How many people does the whole pie show?

$$\frac{45}{360} = \frac{1}{8} \Rightarrow \text{total} = 30 \times 8 = 240$$ ← the slice is 1/8 of the circle

Example: slices of $150°$ and $120°$ and one more slice, for 120 people. The last slice?

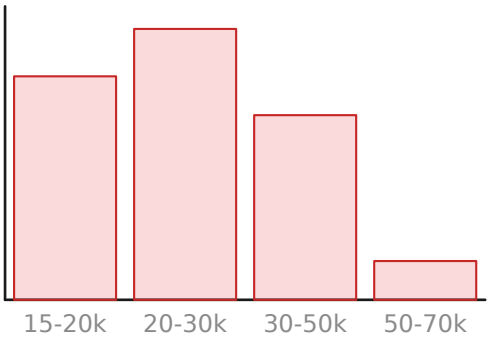
$$360 - 150 - 120 = 90°, \qquad \frac{90}{360} \times 120 = 30 \text{ people}$$

what's left of 360°

Case 1: the HMRC wage brackets

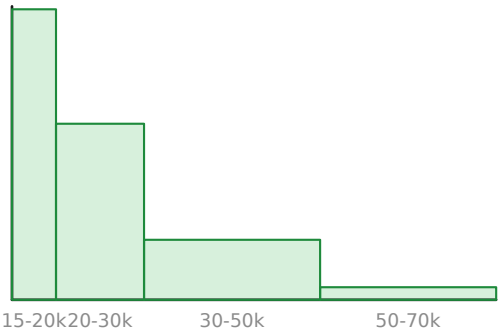
Lecture 4 shows how many UK taxpayers earn in each wage bracket and asks: **"Something is wrong with my choice of graph here. What is it?"** Here are the middle brackets:

SUSPECT: equal-width bars



20-30k looks like the winner

HONEST: width = bracket size



The brackets are **different widths**: £15k to £20k is only £5000 wide, but £30k to £50k is £20,000 wide. A wide bracket collects more people just by being wide. The fix (a proper **histogram**) makes the bar **width** the bracket width and the **area** the number of people, so the height becomes **people per £1000**:

$$\frac{4.95 \text{ million}}{5} = 0.99$$

million per £1000, for £15-20k: the real crowd

$$\frac{6.00 \text{ million}}{10} = 0.60$$

million per £1000, for £20-30k: thinner!

Verdict: the busiest **£1000-slice** is around £15k-£20k, not £20k-£30k. This is why the lecture's mode lands at about £15,000 and the median at £16,400 (Chapter 24).

Histograms with unequal widths: the rule

RULE: frequency density

In a histogram the **area** of a bar is the frequency. The bar's **height** is the **frequency density**, and the vertical axis is labelled "frequency density".

frequency density $= \dfrac{\text{frequency}}{\text{class width}}$, where class width = upper boundary − lower boundary.

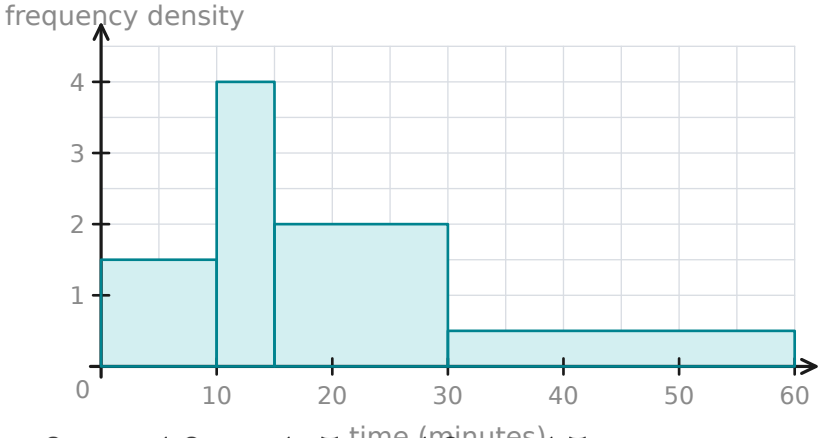
Going back: frequency = frequency density × class width (the area of the bar).

If every class has the same width you may plot plain frequency instead: the shape is the same.

PROCEDURE: reading frequencies off a histogram

1. Read each bar's class boundaries off the horizontal axis and work out its width.
2. Read the bar's height (the frequency density) off the vertical axis.
3. Multiply: frequency = density × width. For "how many between ..." add the bars (or the parts of bars) inside that range.

Example: the commuting times of a group of people. How many took each time, and how many people were asked?



tallest bar (10-15) is not the biggest group: it is only 5 minutes wide

0 to 10: $1.5 \times 10 = 15$ ← *width 10*

10 to 15: $4 \times 5 = 20$ ← *width 5*

15 to 30: $2 \times 15 = 30$ ← *width 15*

30 to 60: $0.5 \times 30 = 15$ ← *width 30*

$$15 + 20 + 30 + 15 = 80 \text{ people}$$ ← *total = sum of the areas*

And forwards again: if 12 more people took 60 to 80 minutes, that bar's height would be $12 \div 20 = 0.6$.

Stem and leaf, one leaf at a time

RULE: stem and leaf diagram

Leaf = the last digit; **stem** = everything in front of it. 34 is stem 3, leaf 4; 5 is stem 0, leaf 5.

Write every stem once, in order, down the left of a vertical line, **including empty ones**. Put each leaf beside its stem, smallest next to the line.

Always give a **key**: "2 | 5 means 25" (or 2.5, or 250: without the key nobody knows).

Computer versions (Lecture 5's Minitab) may split each stem into two rows (leaves 0-4, then 5-9) and add a left-hand column counting the values from each end, with the row holding the median shown in brackets.

Lecture 4's data set, 17 values already in order:

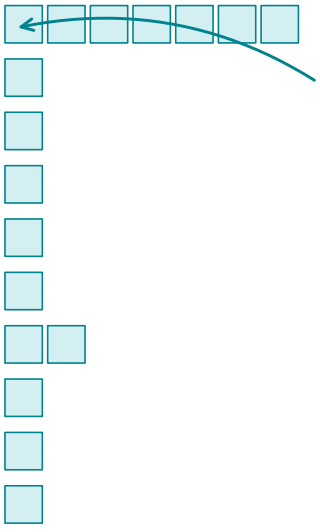
1 2 3 5 5 6 8 18 25 34 47 56 60 69 71 88 94

Split every number into a stem (the tens digit) and a leaf (the units digit). 34 splits into stem 3, leaf 4. Single figures like 5 have stem 0.

34 → 3 | 4 5 → 0 | 5 94 → 9 | 4

0	1 2 3 5 5 6 8
1	8
2	5
3	4
4	7
5	6
6	0 9
7	1
8	8
9	4

key: 2 | 5 means 25



tip it on its side and it's a bar chart of the tens:

7 values under 10, then a long thin tail up to 94

You keep **every original number** (unlike a bar chart) and see the **shape** at a glance. Always give a **key**, or nobody knows if 2 | 5 is 25 or 2.5.

Case 2: the lecturer in the room

Lecture 5: "Consider the distribution of ages in this room." Twelve students and one lecturer:

An **outlier** is a value far from the rest. It drags the average: the mean age is **22.3** with the lecturer and **19.8** without. But the lecturer is **real**: they really are in the room!

RULE: outliers (Lecture 5)

Remove a point only if you think something is **wrong** with it (a typo like age 190, a broken sensor). "Unusual" is not the same as "wrong".

If you do remove one, **always say that you removed it and why**.

Describing a scatter plot: correlation words

RULE: describing correlation

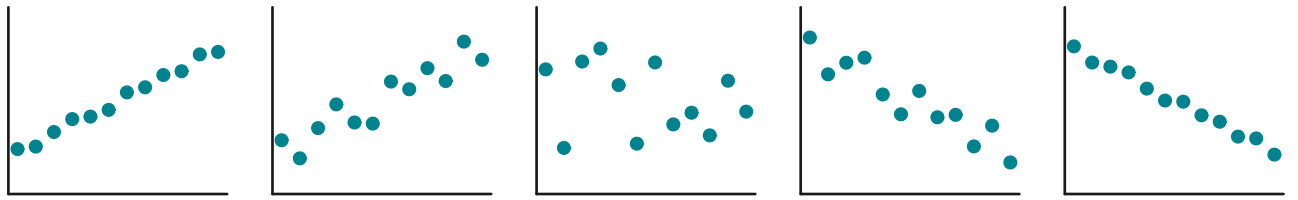
Positive correlation: as one measurement goes up, the other tends to go up (the cloud slopes uphill).

Negative correlation: as one goes up, the other tends to go down (the cloud slopes downhill).

No correlation: no uphill or downhill trend, just a shapeless cloud.

Strong: the points sit close to a straight line. **Weak**: a loose cloud with only a rough trend. Always say both, e.g. "strong negative".

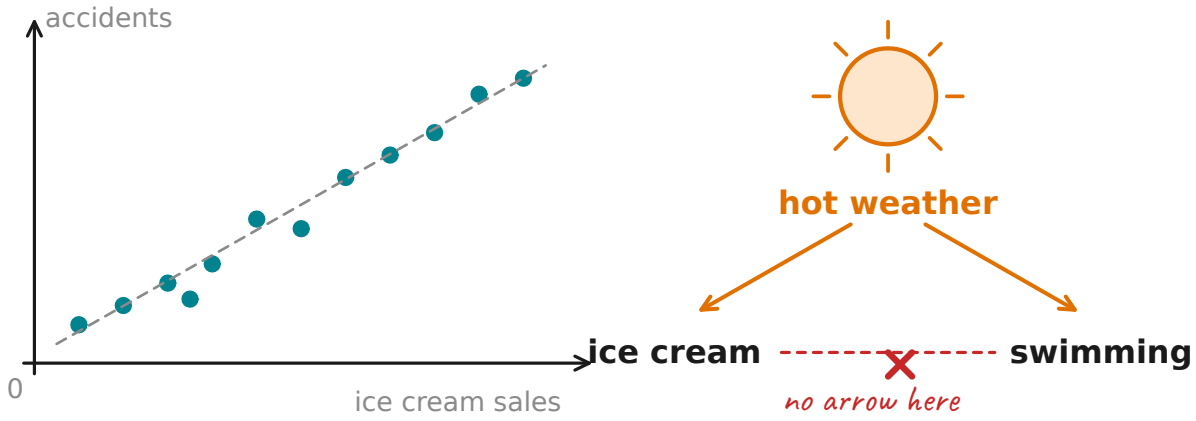
Correlation describes a pattern; it does **not** show that one thing causes the other (next case).



strong positive | weak positive | none | weak negative | strong negative

Case 3: does ice cream cause drowning?

A town records ice cream sales and swimming accidents each month. The scatter plot shows a clear pattern: more ice cream, more accidents. Lock up the ice cream van?



The two **go together** (**correlation**), but neither **makes** the other happen (**causation**). A hidden third thing, hot weather, pushes both up. Lecture 5: **correlation does not equal causation!**

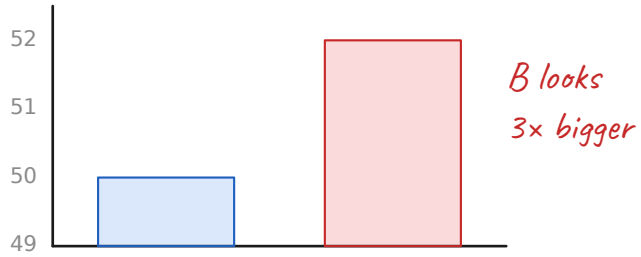
Direction trap: even when there is a real cause, which way does it go? Towns with more police have more crime. Do police cause crime, or does crime bring police?

Joining the dots: only connect the points with a line when there's a reason, like measurements taken one after another in **time**. Here, a line through the months in sales order would be meaningless; a dashed **trend line** is fine.

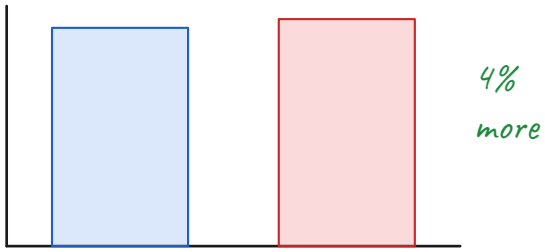
Case 4: crooked graphs and their honest twins

Lecture 5: bad graphs hide or twist information: missing titles or labels, poor use of colour, and a **poor choice of scale**. Each crime on the left, the fix on the right.

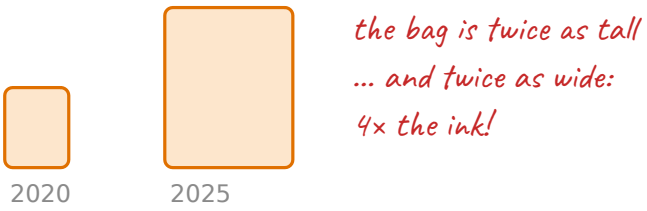
CRIME: axis starts at 49



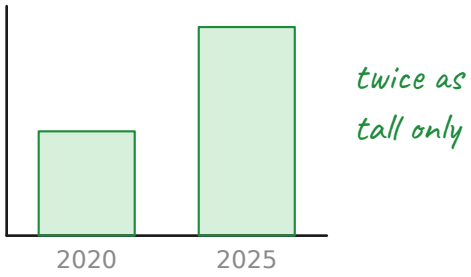
HONEST: axis from 0



CRIME: picture grows both ways



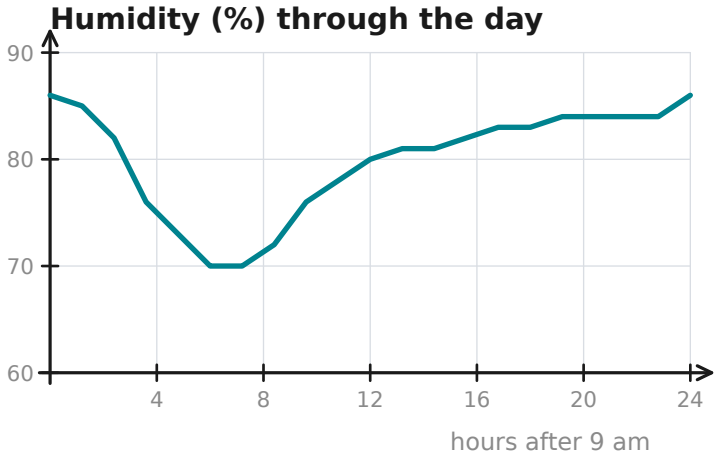
HONEST: plain bars



The checklist of crimes: no title, no axis labels or units, no source, a y-axis that doesn't start at 0 (for bars), uneven scales, 3D effects, pictures that grow in area, and "infographics" (Lecture 5: *very modern, very useful, very easy to lie with!*).

Reading any graph in three passes

Lecture 5's method: **start with how it's built, move on to what it says, finish by being critical.** Try it on this graph:



1 FORM

Title? Yes. Axes? % and hours, but where is the source? Why start at 60?

2 MESSAGE

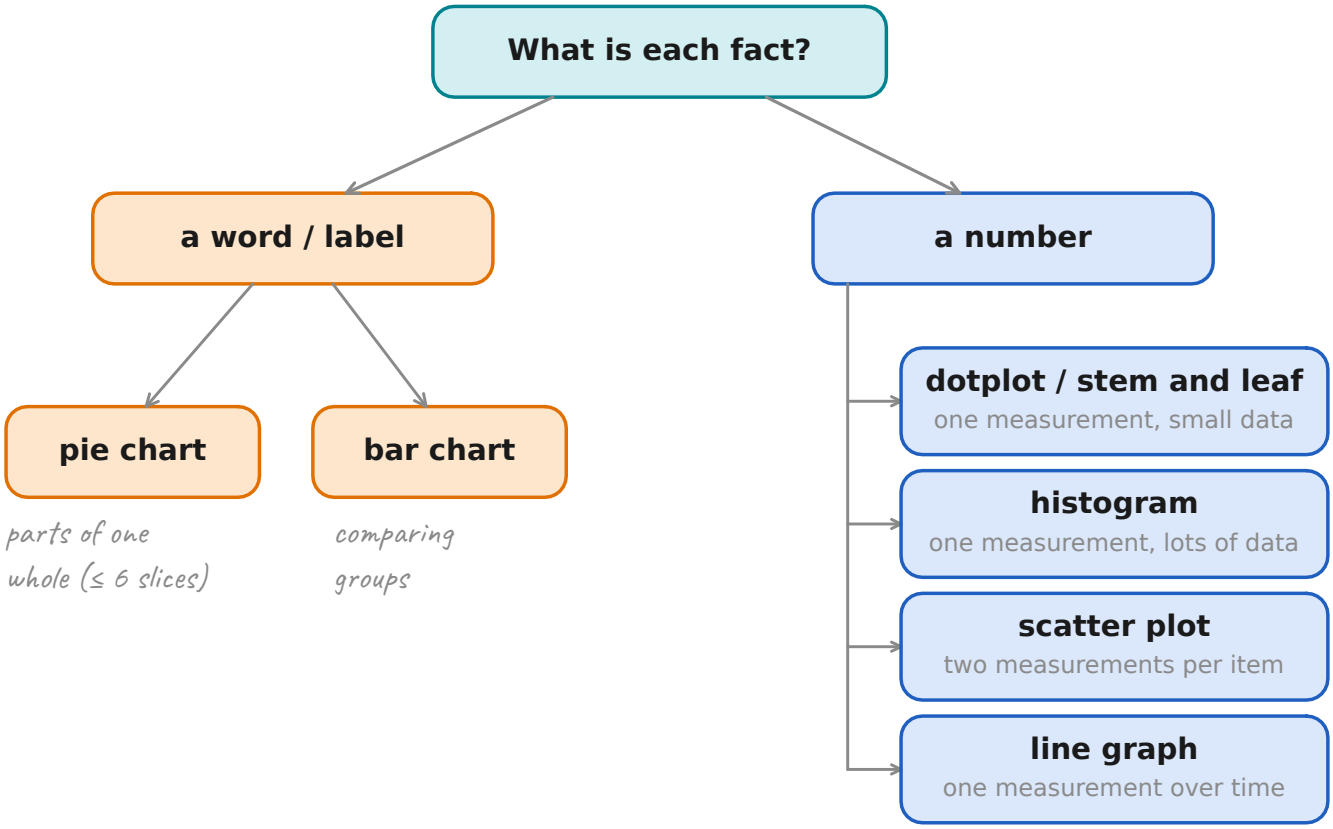
Driest around 3–4 pm (70%), damp again overnight. Implication: sun warms the air and dries it.

3 CRITICISM

One day only, one place. Missing source. Axis from 60 exaggerates the dip.

Form: title and footnotes, axis labels and scale, data source, colours and symbols. **Message:** meaning and implications. **Criticism:** missing data and ease of understanding. (The data are the weather table from the lecture slides.)

Which chart? A flowchart



And when is a table better than any chart? When the reader needs **exact values** to look up, not a shape. Lecture 5's weather forecast is a table because you want to know the temperature at 3 pm, not admire a curve:

time	9:00	12:00	15:00	18:00	21:00	0:00	3:00
temp °C	10	12	12	11	10	10	9
wind	N	NNE	NNE	NNE	NNE	N	N
humidity	86%	76%	70%	78%	81%	83%	84%

Rule of thumb: tables for **looking up**, charts for **seeing patterns**. A table needs the same honesty as a graph: a title, units in the headings, and a source.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Pie chart angle	$\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360°$ all the angles add up to 360°
Pie chart frequency	$\text{frequency} = \frac{\text{angle}}{360°} \times \text{total}$
Frequency density	$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$ the height of a histogram bar; label the axis "frequency density"
Frequency from a histogram	$\text{frequency} = \text{frequency density} \times \text{class width}$ frequency = area of the bar
Bar chart vs histogram	bar chart: categories, gaps between bars. Histogram: numbers in ranges, bars touch, area = frequency
Stem and leaf	leaf = last digit, stem = the rest; leaves in order; always give a key
Correlation	positive (uphill), negative (downhill) or none; strong (close to a line) or weak
Correlation vs causation	a pattern between two things does not show that one causes the other
Outliers	remove a point only if it is wrong (not just unusual), and say that you did and why
Honest graphs	title, labelled axes with units, a source; bar charts start at 0; no pictures that grow in area
Reading a graph	form (how it is built), message (what it says), criticism (what is missing or misleading)

Case files for you

Answers in grey.

Which chart?

1. Share of the budget spent on rent, food, travel, fun
2. Reaction times of 200 people
3. Hours of revision vs exam mark for 30 students
4. Temperature at noon every day in March

1. pie 2. histogram 3. scatter plot 4. line graph

Build it

5. Draw a stem and leaf for 12, 15, 21, 22, 22, 27, 30, 34, 41 (with a key)
6. A pie chart shows 90 people. How many degrees for 25 of them?

5. 1 | 2 5; 2 | 1 2 2 7; 3 | 0 4; 4 | 1; key 2 | 1 means 21 6. 25/90 × 360 = 100°

Detective

7. "Students who sit at the front get better marks, so sitting at the front makes you clever." What's wrong?
8. A bar chart's y-axis runs from 95 to 100. Two bars read 96 and 99. What false impression, and the fix?
9. Data: 4, 5, 5, 6, 7, 480. Should you delete 480?

7. correlation ≠ causation (keen students choose the front) 8. 99 looks 4× bigger than 96; start at 0 9. only if it's an error; check it, and say so if you remove it

Find the unknown

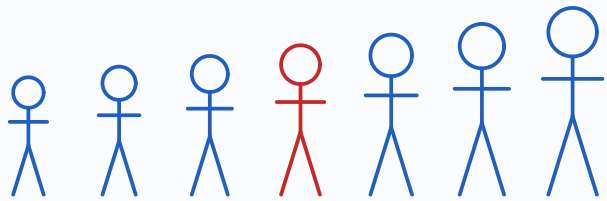
10. A pie chart shows 250 people. One slice is 72°. How many people is that?
11. A 45° slice stands for 30 people. How many people does the whole pie show?
12. A histogram bar for 20-30 cm has frequency density 3.5. What is its frequency?
13. A class 5-8 kg holds 24 parcels. How tall is its histogram bar?
14. Bars: 0-10 at density 2, 10-15 at density 6, 15-30 at density 1. How many values altogether?
15. A pie of 120 people has slices of 150°, 120° and x°. Find x and that slice's frequency.

10. 72/360 × 250 = 50 11. 45° is 1/8 of 360°, so 30 × 8 = 240 12. 3.5 × 10 = 35 (frequency = density × width) 13. 24 ÷ 3 = 8 (density = frequency ÷ width) 14. 2 × 10 + 6 × 5 + 1 × 15 = 20 + 30 + 15 = 65 15. x = 360 − 150 − 120 = 90°; 90/360 × 120 = 30

Chapter 24

Statistics: averages and spread

three numbers that tell the whole story



line them up by height:
the one in the middle is the median

TYPICAL?

HOW FAR?

HOW SPREAD?

Lecture 4 starts from a fact about brains: **we can only work with about four things at once**. A data set of hundreds of values is useless until we squash it into three or four summary numbers:

the average (what number is typical?), the range (what are the outer limits?) and the variance (how spread out are the values?).

Goes with **Lecture 4** (mean, median, mode, quartiles, box plot, variance, standard deviation, sample vs population).

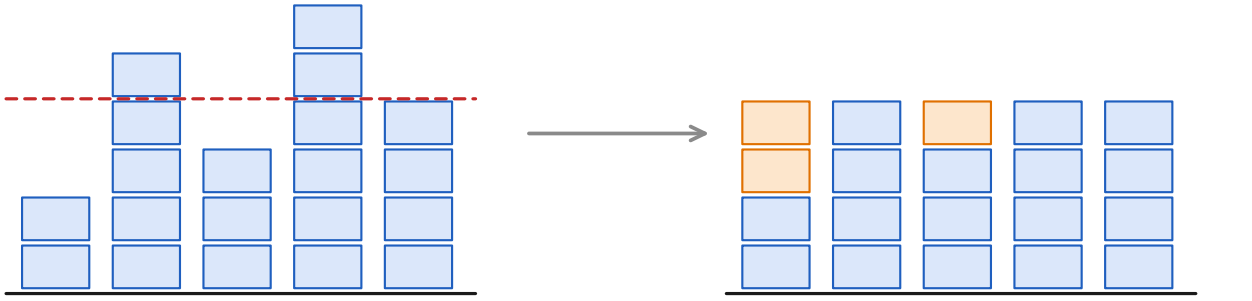
The mean: level the towers

RULE: the mean

mean $= \dfrac{\text{sum of the values}}{\text{number of values}}$, written $\bar{x} = \dfrac{\sum x}{n}$.

Turned round: **total = number of values × mean**. Every "find the missing value" question starts here.

Five friends have 2, 5, 3, 6 and 4 sweets. The **mean** is what each would get if they **shared them out equally**. Picture towers of blocks and move blocks from tall towers to short ones until they're level:



before: 2, 5, 3, 6, 4 *after: every tower is 4 (orange = moved)*

Sharing out equally is the same as: **add them all up, divide by how many there are.**

$$\bar{x} = \frac{2+5+3+6+4}{5} = \frac{20}{5} = 4$$

Lecture 4 writes it with the sigma (adding-up) sign of Chapter 17:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$\bar{x}$ ("x bar") = the mean

$\sum x_i$ = add up every value $x_1, x_2, \ldots x_n$

n = how many values there are

The lecture's giant example: UK GDP is about 2.5 trillion dollars and there are about 65 million people, so the mean income is $2.5 \times 10^{12} \div (6.5 \times 10^{7}) \approx 38\,500$ dollars, about £24,000.

The median: the middle of the queue

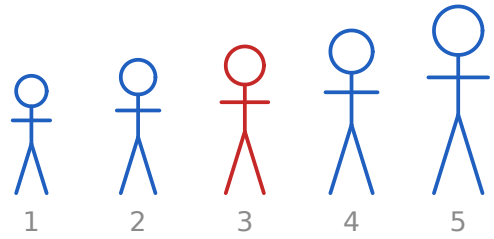
RULE: median and mode

Median: put the values **in order**; the median is the value in position $\frac{n+1}{2}$.

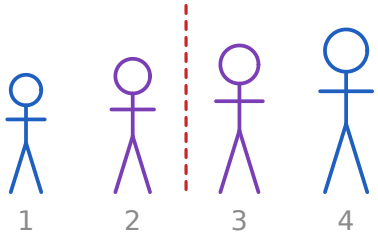
If n is even the position ends in .5: the median is halfway between the two middle values (add them and halve).

Mode: the value that occurs most often. There can be more than one mode, or none.

Line everyone up in order of size. The **median** is the person standing in the middle.



5 people: position 3 is the middle



4 people: nobody is in the middle, so go halfway between 2 and 3

The position of the middle is $\frac{n+1}{2}$ (Lecture 4 writes $M = x_{(n+1)/2}$). For $[1, 2, 3, 4, 5]$ that's position 3, so $M = 3$. For $[1, 2, 3, 4]$ it's position 2.5, so halfway between the 2nd and 3rd values: $M = \frac{2+3}{2} = 2.5$.

The mode is the **most common** value: the tallest stack in a dotplot. In 3, 5, 5, 6, 8, 5, 2 the mode is 5. No formula, just counting. For measurements like wages, where nobody earns exactly the same, you first group them into brackets (Chapter 23), then the mode is the busiest bracket.

Missing values: work with the total

PROCEDURE: missing-value problems

1. Turn every mean you are given into a total: total = number of values × mean.
2. Subtract the values you know from the total: what's left is the missing value (or their sum).
3. If a value is added or removed, change the total **and** the count, then divide again: new mean = new total ÷ new count.
4. Check: add the values up and divide.

Example: the mean of five numbers is 8. Four of them are 6, 7, 9 and 10. Find the fifth.

$$\text{total} = 5 \times 8 = 40$$ ← *total = n × mean*

$$40 - (6 + 7 + 9 + 10) = 40 - 32 = 8$$ ← *take away the four you know*

Example: four values have mean 15. A fifth value, 25, is added. What is the new mean?

$$4 \times 15 = 60, \qquad \frac{60 + 25}{5} = \frac{85}{5} = 17$$ ← *new total ÷ new count*

Example: six values have mean 10. The value 20 is removed. What is the new mean?

$$6 \times 10 = 60, \qquad \frac{60 - 20}{5} = \frac{40}{5} = 8$$ ← *one fewer value*

Example: the mean of x, $2x$ and $3x$ is 12. Find x.

$$\frac{x + 2x + 3x}{3} = 12 \;\Rightarrow\; 6x = 36 \;\Rightarrow\; x = 6$$

total = 3 × 12 = 36

Example: four tests average 66. What must the fifth score be to bring the average up to 70?

$$5 \times 70 - 4 \times 66 = 350 - 264 = 86$$ ← *total needed − total so far*

✗ Common mistake: averaging two means

✗ $\dfrac{20+40}{2} = 30$

group A: 10 people, mean 20; group B: 30 people, mean 40. Combine the totals, not the means

✓ $\dfrac{10 \times 20 + 30 \times 40}{40} = 35$

Averages from a frequency table

RULE: frequency tables

f = how many times the value x occurs. The number of values is $n = \sum f$ (not the number of rows).

Mean $= \dfrac{\sum fx}{\sum f}$: multiply each value by its frequency, add these up, divide by the total frequency.

Median: position $\frac{n+1}{2}$ with $n = \sum f$. Keep a running total of f to see which value sits in that position. Mode: the value with the biggest f.

Grouped data (classes like 10-20): use each class's **midpoint** as x. The mean is then only an **estimate**, because the exact values are lost.

Example: the number of brothers and sisters of 20 students.

siblings x	frequency f	fx	running total of f
0	5	0 × 5 = 0	5
1	8	1 × 8 = 8	13
2	4	2 × 4 = 8	17
3	3	3 × 3 = 9	20
total	**Σf = 20**	**Σfx = 25**	

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{25}{20} = 1.25$$

$$\text{median position} = \frac{20+1}{2} = 10.5$$ ← between the 10th and 11th values

The running total says values 1 to 5 are 0s and values 6 to 13 are 1s, so the 10th and 11th are both 1: **median = 1**. The mode is also 1 (frequency 8).

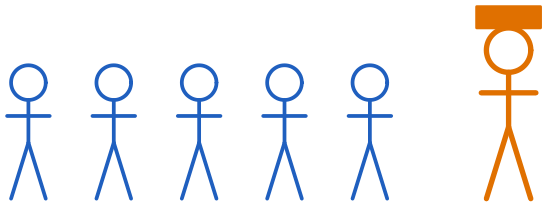
Example: grouped data: minutes spent on homework by 20 students.

minutes	frequency f	midpoint x	fx
0-10	4	5	5 × 4 = 20
10-20	10	15	15 × 10 = 150
20-40	6	30	30 × 6 = 180
total	**20**		**350**

$$\bar{x} \approx \frac{350}{20} = 17.5 \text{ minutes}$$ ← an estimate: midpoints stand in for the real values

A billionaire walks into a pub

Five people in a pub earn £20k, £22k, £25k, £28k and £30k. Mean £25k, median £25k. Then a billionaire earning £1,000 000k walks in:

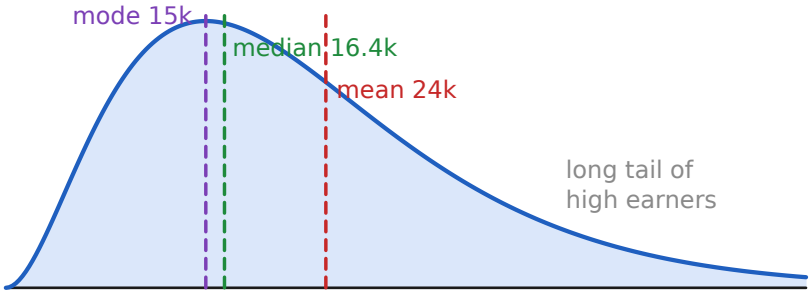


mean **£25k → £166 687k**

median **£25k → £26.5k**

the mean now describes nobody in the room

One extreme value (an outlier) drags the mean but barely moves the median. Data with a long tail on one side are called **skewed**, and UK wages are exactly like that. Lecture 4's three answers to "what is the average UK wage?":



The mega-rich few in the tail pull the mean right.

Median: usually the most honest for skewed data. Mean: best for symmetric (Normal) data. Mode: rarely used.

Lecture 4 also lists what's wrong with the wage data itself (part-time work not pro-rata, self-reporting, tax avoidance...), and hides a puzzle on each slide: a wrong title, a bad choice of graph (Chapter 23's case 1) and overlapping brackets like "£6000 to £7000" then "£7000 to £8000": which bracket does exactly £7000 go in?

Range, quartiles and the box plot

RULE: range, quartiles and IQR (Lecture 4's method)

Put the data in order. Median position $= \dfrac{n+1}{2}$.

Quartile position $= \dfrac{\text{median position} + 1}{2}$. Count that far **up from the smallest** for the lower quartile (LQ) and **down from the largest** for the upper quartile (UQ).

Range = largest − smallest. IQR = UQ − LQ. Semi-interquartile range = IQR ÷ 2.

RULE: fractional positions

Position 3.5 means **halfway** between the 3rd and 4th values: add them and halve.

In general, position $k + d$ (whole part k, decimal part d) means: start at the kth value and go the fraction d of the gap to the next one: $x_k + d\,(x_{k+1} - x_k)$.

So position 1.75 is **¾ of the way** from the 1st value to the 2nd, and 2.25 is ¼ of the way from the 2nd to the 3rd. Counting down from the top for the UQ, move the fraction **down** towards the next value.

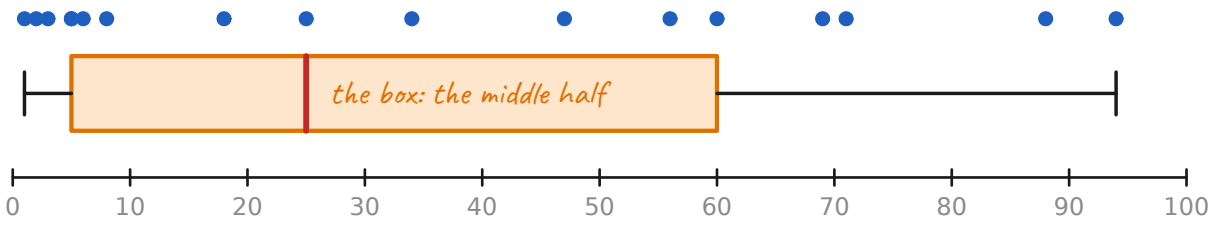
Calculators and software use slightly different quartile rules, so their answers can differ a little. Say which method you used.

Lecture 4's 17 values, in order. We'll find five landmarks and hang a box on them.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1	2	3	5	**5**	6	8	18	**25**	34	47	56	**60**	69	71	88	**94**
min				LQ				median				UQ				max

Median: position $\frac{17+1}{2} = 9$, so the 9th value: **25**.

Quartiles cut the data into quarters. Lecture 4: the quartile position is $\frac{\text{median position}+1}{2} = \frac{9+1}{2} = 5$. Count 5 from the bottom for the lower quartile (LQ = 5) and 5 from the top for the upper quartile (UQ = 60).



range	largest − smallest	**94 − 1 = 93**
interquartile range (IQR)	UQ − LQ: the width of the box	**60 − 5 = 55**
semi-interquartile range	half the IQR	**55 ÷ 2 = 27.5**

The range uses only the two most extreme values, so one outlier wrecks it. The IQR ignores the top and bottom quarters, so it's robust, like the median.

Example: quartiles with fractional positions: 2, 4, 6, 8, 10, 12, 14, 16 ($n = 8$).

$$\text{median position} = \frac{8+1}{2} = 4.5, \quad \text{median} = \frac{8+10}{2} = 9$$

halfway between the 4th and 5th

$$\text{quartile position} = \frac{4.5+1}{2} = 2.75$$

$$\text{LQ} = 4 + 0.75 \times (6 - 4) = 5.5$$ ← ¾ of the way from the 2nd value (4) to the 3rd (6)

$$\text{UQ} = 14 - 0.75 \times (14 - 12) = 12.5$$ ← 2.75th from the top: ¾ of the way from 14 down to 12

$$\text{IQR} = 12.5 - 5.5 = 7$$

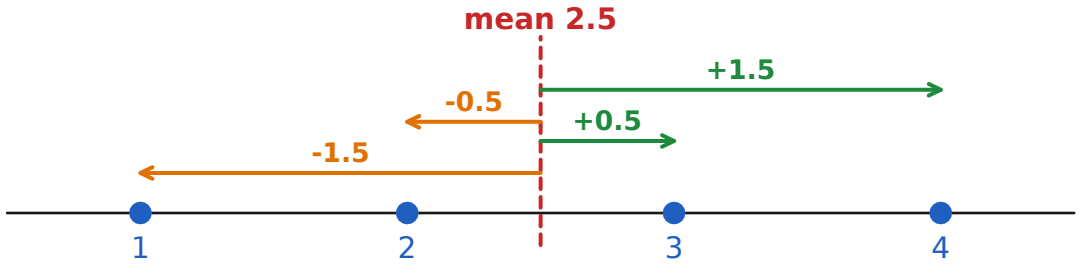
How spread out? A tug of war that always draws

RULE: deviation

The deviation of a value is its distance from the mean: $x - \bar{x}$ (negative below the mean).

The deviations of any data set **always add up to 0**, so their plain average is useless as a measure of spread.

Take $[1, 2, 3, 4]$: mean 2.5. But so does $[-1000, 608, 400, 2]$! The mean can't tell them apart; we need to measure how far values sit from the mean. That distance is the **deviation**: $x - \bar{x}$.



left pulls: −1.5, −0.5. Right pulls: +0.5, +1.5. Total: 0. Always!

Averaging the deviations is useless: the mean is the balance point, so the pulls to the left and right **always cancel to zero**. We need to get rid of the minus signs. Lecture 4's trick: **square them**.

Variance: the area of the average square

RULE: variance and standard deviation

Variance = the **mean of the squared deviations**. Standard deviation = $\sqrt{\text{variance}}$, back in the units of the data.

Whole **population** (N values, mean μ): $\sigma^2 = \dfrac{\sum(x-\mu)^2}{N}$, $\sigma = \sqrt{\sigma^2}$.

Sample (n values, mean $\bar{x}$): $s^2 = \dfrac{\sum(x-\bar{x})^2}{n-1}$, $s = \sqrt{s^2}$ (why $n-1$: three sections on).

Squaring a deviation means drawing a **real square** whose side is the deviation. Minus signs vanish (area is never negative), and far-away values get **much** bigger squares:

(-1.5)² = 2.25 (-0.5)² = 0.25 (+0.5)² = 0.25 (+1.5)² = 2.25

sum of squares = 2.25 + 0.25 + 0.25 + 2.25 = 5

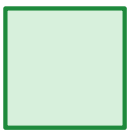
Problem: $[1, 1, 2, 2, 3, 3, 4, 4]$ is just as spread out, but has 8 squares, so its sum is 10. Bigger groups get bigger sums. **Fix: divide by the number of values.** That's the average square:

$$\frac{5}{4} = 1.25 \qquad \frac{10}{8} = 1.25$$ ← *same spread, same answer*

This average-square area is the **variance**, σ^2. Its side length (the square root) is the **standard deviation** σ, which is back in the same units as the data. Treating the values as a whole population, Lecture 4 writes the mean as μ and the count as N:

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

$$\sigma = \sqrt{1.25} \approx 1.118$$



area 1.25 = variance

side 1.118 = standard deviation

Units check (Lecture 4): $[10, 20, 30, 40]$ has variance 125, a hundred times bigger than 1.25, because squares of 10× longer sides have 100× the area. The standard deviation, $\sqrt{125} = 11.18$, is exactly 10× bigger: sensible. That's why we usually quote σ.

Wider range means more spread? Not always

Lecture 4's "range fallacy": $[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 5]$ has range 4, wider than $[1, 2, 3, 4]$ (range 3). Yet its variance is 1.06, **slightly less** than 1.25: thirteen values sit almost on top of the mean, and only one lonely square is big. **Range looks at two values; variance looks at all of them.**

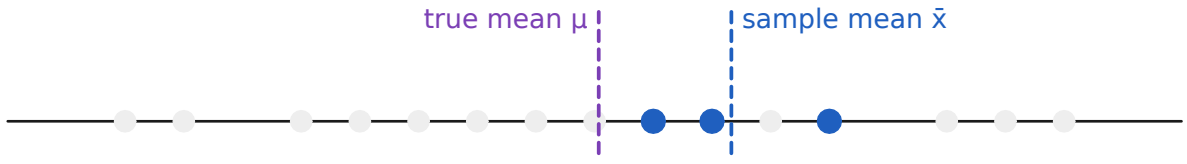
Sample or population, and the mysterious n − 1

A **population** is everyone you care about (all NTU students). A **sample** is the handful you actually measured (this class). Lecture 4 uses different letters so you always know which you mean:

	population (Greek)	sample (English)
mean	μ ("mew")	x̄ ("x bar")
standard deviation	σ ("sigma")	s
how many	N	n

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}} \qquad s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Why $n - 1$ (Bessel's correction)? A sample huddles closer to **its own** mean than to the true population mean, because its mean was calculated from those very values. So its squares come out **too small**. Dividing by a slightly smaller number ($n - 1$) nudges the answer back up:



the 3 blue values are closer to x̄ than to μ, so their spread looks smaller than it really is

The fix matters most for small samples: dividing by 2 instead of 3 is a big change; by 999 instead of 1000 is tiny.

The recipe, one step per line

Lecture 4's "break it down" for the sample standard deviation of $[1, 2, 3, 4]$:

x	x − x̄	(x − x̄)²
1	-1.5	2.25
2	-0.5	0.25
3	+0.5	0.25
4	+1.5	2.25
		total 5

1 find the mean: 10 ÷ 4 = 2.5
2 subtract the mean from each x
3 square each one
4 add the squares: 5
5 divide by n − 1 = 3: 1.667
6 square root: s ≈ 1.291

Compare: dividing by $n = 4$ gave $\sigma = 1.118$ (treating the four values as the whole population). Calculators have both buttons: σ_n (or σx) and σ_{n-1} (or sx).

Shifting and stretching the data

RULE: adding or multiplying every value

Add a constant c to every value: the mean, median, mode and quartiles all go up by c. The spread does **not** change: range, IQR, variance and standard deviation stay the same.

Multiply every value by k: the mean, median, mode and quartiles are multiplied by k; the range, IQR and standard deviation by $|k|$; the variance by k^2.

Both at once ($y = kx + c$): new mean $= k\bar{x} + c$, new standard deviation $= |k| \times$ old.

Sliding every value along by 10 moves the whole data set, but the gaps between the values (and so the spread) are untouched. Tripling every value triples every gap.

Example: 2, 4, 6 has mean 4 and $\sigma \approx 1.633$.

$$\text{add } 10\text{: } 12, 14, 16 \;\Rightarrow\; \text{mean } 14,\ \sigma \approx 1.633$$

mean + 10, spread unchanged

$$\text{times } 3\text{: } 6, 12, 18 \;\Rightarrow\; \text{mean } 12,\ \sigma \approx 4.899$$ ← *both × 3*

Example: temperatures have mean 20 °C and standard deviation 5 °C. In Fahrenheit, $F = 1.8C + 32$.

$$\text{mean} = 1.8 \times 20 + 32 = 68\,^\circ\text{F}$$ ← *multiply and add*

$$\text{SD} = 1.8 \times 5 = 9\,^\circ\text{F}$$ ← *multiply only: the +32 doesn't spread anything*

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Mean	$\bar{x} = \dfrac{\sum x}{n}$ add them up, divide by how many
Total from a mean	$\text{total} = n \times \bar{x}$ the first step of every missing-value problem
Mean of a frequency table	$\bar{x} = \dfrac{\sum fx}{\sum f}$ grouped data: x = class midpoint (an estimate)
Median position	$\dfrac{n+1}{2}$ data in order; a .5 position means halfway between two values
Quartile position	$\dfrac{\text{median position} + 1}{2}$ count up from the bottom for LQ, down from the top for UQ
Fractional position	$k + d \;\rightarrow\; x_k + d\,(x_{k+1} - x_k)$ e.g. 1.75 = ¾ of the way from the 1st value to the 2nd
Mode	the most common value (or busiest class)
Range	$\text{largest} - \text{smallest}$
Interquartile range	$\text{IQR} = \text{UQ} - \text{LQ}$ semi-interquartile range = IQR ÷ 2
Population variance and SD	$\sigma^2 = \dfrac{\sum(x-\mu)^2}{N}, \qquad \sigma = \sqrt{\sigma^2}$ the whole population: mean μ, N values
Sample SD	$s = \sqrt{\dfrac{\sum(x-\bar{x})^2}{n-1}}$ a sample: mean $\bar{x}$, divide by $n-1$ (Bessel)
Add c to every value	averages + c; range, IQR, SD, variance unchanged
Multiply every value by k	averages × k; range, IQR, SD × |k|; variance × k²
Which average?	skewed data or outliers: median. Symmetric data: mean

Your turn

Data A: 3, 7, 7, 8, 10, 12, 15. Answers in grey.

Averages

1. Mean of A
2. Median of A
3. Mode of A
4. Add 100 to data A. Which average changes most?

1. 62 ÷ 7 ≈ 8.86 2. 8 3. 7 4. the mean (it becomes 20.25; median 9)

Spread

5. Range of A
6. Median position for 11 values? Quartile position?
7. Population variance and σ of 2, 4, 6
8. Sample s of 2, 4, 6

5. 12 6. 6th; (6 + 1)/2 = 3.5th from each end (halfway between the 3rd and 4th) 7. 8/3 ≈ 2.67, σ ≈ 1.63 8. √(8/2) = 2

Think

9. Why is median house price reported, not the mean?
10. Two classes both average 60%. What else would you want to know?

9. a few mansions skew the mean upward 10. the spread (standard deviation / IQR)

Find the unknown

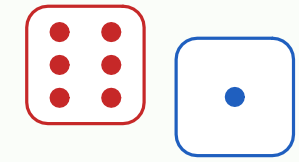
11. The mean of 5 numbers is 8. Four of them are 6, 7, 9, 10. Find the fifth.
12. The mean of x, 2x and 3x is 12. Find x.
13. 4 values have mean 15. The value 25 is added. New mean?
14. 6 values have mean 10. The value 20 is removed. New mean?
15. Ten values have mean 7 and σ = 2. New mean and σ if every value is (a) increased by 5, (b) tripled?
16. x = 1, 2, 3 have frequencies 4, k, 2 and the mean is 1.8. Find k.

11. 8 (total 5 × 8 = 40; 40 − 32 = 8) 12. x = 6 (6x = 3 × 12 = 36) 13. 17 (60 + 25 = 85; 85 ÷ 5) 14. 8 (60 − 20 = 40; 40 ÷ 5) 15. (a) mean 12, σ = 2 (b) mean 21, σ = 6 16. k = 4 (Σfx = 10 + 2k = 1.8 × (6 + k))

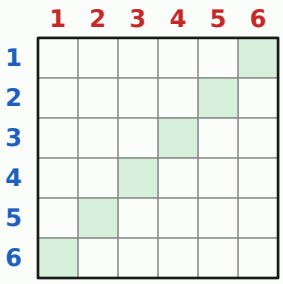
Chapter 25

Probability and Bayes

count the squares



two dice = 36 possible results, all equally likely



the 6 ways to roll a total of 7: 6 out of 36

Probability sounds like guesswork, but most of it is **counting**. List every possible result, shade the ones you care about, and the probability is **shaded ÷ total**.

This chapter uses one picture for almost everything: the **6 × 6 grid of two dice**. With it you can **see** what "not", "and", "or" and "given that" mean, and then the lecture's famous puzzle (only 15% of doctors got it right) becomes a matter of counting little people.

Goes with **Lecture 8** (sample space, complements, joint, exclusive and conditional probability, independence, the urn, Bayes' theorem, the mammogram).

Probability = shaded squares ÷ all squares

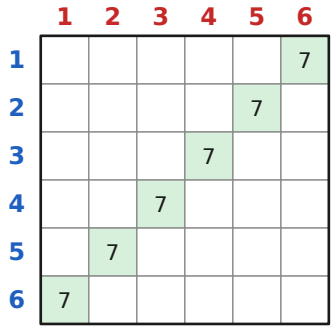
RULE: probability

For equally likely outcomes: $p(E) = \dfrac{\text{number of outcomes in } E}{\text{total number of outcomes}}$.

$0 \le p(E) \le 1$: 0 means impossible, 1 means certain. The probabilities of all the possible outcomes add up to 1.

Complement: $p(\text{not } E) = p(E') = 1 - p(E)$.

Roll a red die and a blue die. Every square below is one possible result: column = red die, row = blue die. The whole grid is the **sample space** S (every possible outcome). A set of outcomes you're interested in is an **event** E, a subset of S (Chapter 6).

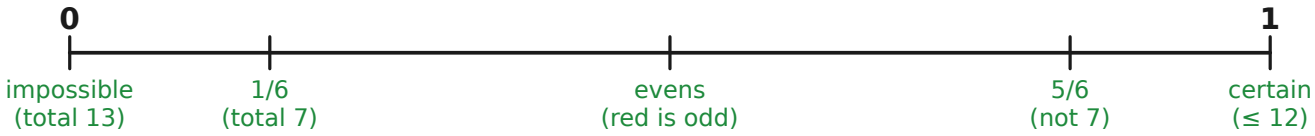


E = "the total is 7"

$$p(E) = \frac{\text{squares in } E}{\text{squares in } S} = \frac{6}{36} = \frac{1}{6}$$

read p(E) as "the probability of E"

Probabilities always sit between 0 and 1 (or 0% and 100%):



Rule 1, complements: the squares **not** in E are everything else, so

$$p(\text{not } E) = 1 - p(E)$$ ← *not a total of 7: 1 − 1/6 = 5/6*

Handy when "not" is easier to count:
$p(\text{at least one six}) = 1 - p(\text{no sixes}) = 1 - \frac{25}{36} = \frac{11}{36}$.

AND is the overlap, OR is both stripes

RULE: AND and OR

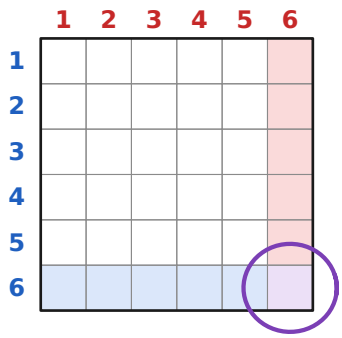
$p(A \cap B)$ = the probability that A **and** B both happen (the overlap).

$p(A \cup B)$ = the probability that A **or** B happens (either one, or both).

General addition rule, always true:
$p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Mutually exclusive events can't both happen: $p(A \cap B) = 0$, so $p(A \cup B) = p(A) + p(B)$.

Let E = "red die shows 6" (a column) and F = "blue die shows 6" (a row):



E (red column): 6 squares
F (blue row): 6 squares
both (purple corner): 1 square

$p(E \cap F) = \frac{1}{36}$ $\quad$ $p(E \cup F) = \frac{6+6-1}{36} = \frac{11}{36}$

AND (both happen, the **joint** probability) is the **overlap**: $\cap$, the intersection from Chapter 6. OR (at least one happens) is **everything shaded**: $\cup$, the union. Counting the column (6) and the row (6) counts the corner **twice**, so subtract it once:

$$p(E \cup F) = p(E) + p(F) - p(E \cap F)$$ ← *Lecture 8's Rule 3*

Mutually exclusive events can't happen together: "total is 2" and "total is 12" have no squares in common, so $p(E \cap F) = 0$ (Rule 2) and the OR rule becomes simply $p(E) + p(F) = \frac{1}{36} + \frac{1}{36}$.

"Given that": zoom in, and the stripe becomes the world

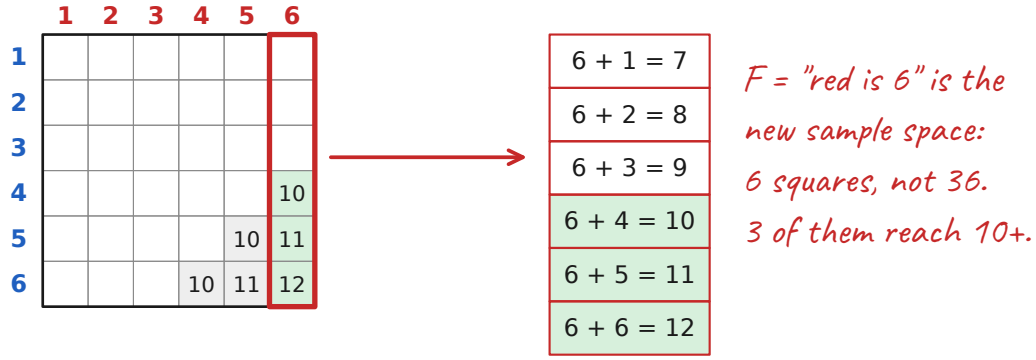
RULE: conditional probability

$p(A \mid B)$ = the probability of A **given that** B has happened. B becomes the whole sample space.

$p(A \mid B) = \dfrac{p(A \cap B)}{p(B)}$ (as long as $p(B) \neq 0$).

Multiplication rule (the same rule turned round):
$p(A \cap B) = p(B) \times p(A \mid B) = p(A) \times p(B \mid A)$.

Your friend rolls, peeks, and says "the red die is a 6". What's the chance the total is 10 or more **now**? The bar $\mid$ is read "**given**":
$p(E \mid F)$ = probability of E given that F has happened.



Without the peek, $p(10+) = \frac{6}{36} = \frac{1}{6}$. With it, $p(10+ \mid \text{red } 6) = \frac{3}{6} = \frac{1}{2}$. New information **changes** the probability. Lecture 8: **"F is the new S!"** As a formula, count the overlap inside F, as a fraction of F:

$$p(E \mid F) = \frac{p(E \cap F)}{p(F)} = \frac{3/36}{6/36} = \frac{1}{2} \quad \leftarrow \text{Rule 4}$$

Rearranged: $p(E \cap F) = p(F) \times p(E \mid F)$: the chance of both = the chance of the first × the chance of the second **once the first has happened**.

Independent: when the peek tells you nothing

RULE: independent events

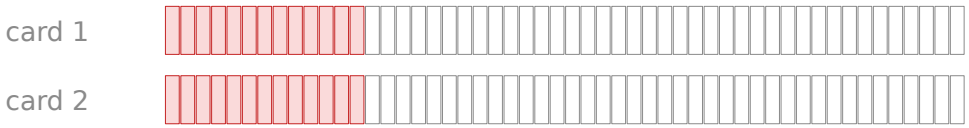
A and B are **independent** when knowing that one happened doesn't change the probability of the other: $p(A \mid B) = p(A)$.

Then, and only then: $p(A \cap B) = p(A) \times p(B)$.

Test: work out $p(A) \times p(B)$ and compare it with $p(A \cap B)$. Equal: independent. Not equal: dependent.

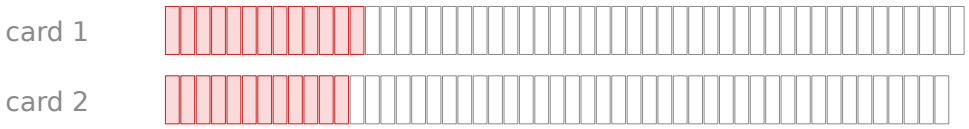
Two events are **independent** if knowing one doesn't change the other: $p(E \mid F) = p(E)$. Then the multiplication rule becomes simply $p(E \cap F) = p(E) \times p(F)$. Lecture 8's cards: **draw two cards; what's the chance both are hearts?**

put the first card back and shuffle



$= \frac{1}{16}$

keep the first card out



$= \frac{1}{17}$

With replacement the deck is identical for card 2: **independent**, $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16} = 0.0625$. Without replacement one heart has gone: **dependent**, $\frac{1}{4} \times \frac{12}{51} = \frac{3}{51} = \frac{1}{17} \approx 0.0588$.

Sun and red cars (independent): $p(\text{sun}) = 0.4$, $p(\text{red car}) = 0.2$, so $p(\text{both}) = 0.08$. Sun and umbrellas (dependent): the weather obviously changes your chance of seeing an umbrella, so you **can't** just multiply.

Exclusive or independent? The rules side by side

These two words sound alike and get mixed up constantly. They mean very different things:

	Mutually exclusive	Independent
meaning	A and B can't both happen	one happening doesn't change the chance of the other
AND	$p(A \cap B) = 0$	$p(A \cap B) = p(A)\,p(B)$
OR	$p(A \cup B) = p(A) + p(B)$	$p(A \cup B) = p(A) + p(B) - p(A)\,p(B)$
GIVEN	$p(A \mid B) = 0$	$p(A \mid B) = p(A)$
dice example	total is 2 / total is 12	red shows 6 / blue shows 6

RULE: always true, for any two events

General addition rule: $p(A \cup B) = p(A) + p(B) - p(A \cap B)$.

Conditional probability: $p(A \mid B) = \dfrac{p(A \cap B)}{p(B)}$, so $p(A \cap B) = p(B)\,p(A \mid B)$.

The two special cases in the table are these general rules with $p(A \cap B)$ replaced by 0 or by $p(A)\,p(B)$.

RULE: exclusive events are never independent

If A and B are mutually exclusive and both have non-zero probability, they **cannot** be independent: $p(A \cap B) = 0$ but $p(A)\,p(B) > 0$.

Knowing A happened tells you B definitely **didn't**: the biggest change of all.

Check on the dice: "red shows 1" and "red shows 2" are exclusive. $p(\text{both}) = 0$, but $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$. Not equal, so they are dependent.

✗ Common mistakes

✗ $p(A \cup B) = p(A) + p(B)$

only drop the overlap when A and B are mutually exclusive

✓ $p(A \cup B) = p(A) + p(B) - p(A \cap B)$

✗ $p(A \cap B) = p(A)\,p(B)$ ✓ $p(A \cap B) = p(B)\,p(A \mid B)$

multiply the plain probabilities only when A and B are independent

✗ $p(A \mid B) = p(B \mid A)$ ✓ $p(A \mid B) = \dfrac{p(B \mid A)\,p(A)}{p(B)}$

a conditional can't simply be turned round (Bayes, later in the chapter)

The urn: a table does the counting

RULE: reading a two-way table

Joint $p(A \cap B)$: one cell ÷ the grand total.

Marginal $p(A)$: a row or column total ÷ the grand total.

Conditional $p(A \mid B)$: the cell ÷ the total of the row or column for B (zoom into B).

Any missing cell = its row (or column) total minus the other cells in that row (or column).

Lecture 8's urn holds 100 balls, each Red, Green or Blue and Large or Small. A **two-way table** counts every combination; the totals in the **margins** give the **marginal** probabilities:

	Red	Green	Blue	total
Large	5	30	15	**50**
Small	15	20	15	**50**
	20	**50**	**30**	**100**

Joint: $p(R \cap L) = \frac{5}{100} = 0.05$ (one cell). Marginal: $p(R) = \frac{20}{100} = 0.20$ (a margin total). Conditional: told the ball is red, **zoom into the red column** (20 balls) and it's the new world:

$$p(L \mid R) = \frac{5}{20} = 0.25 \qquad p(S \mid R) = \frac{15}{20} = 0.75$$

$$p(L \mid G) = \frac{30}{50} = 0.60$$ ← a different answer for green

Before the peek, large was 50-50. Knowing the colour changes the odds for size, so colour and size are **dependent**.

Find the missing probability

PROCEDURE: partial-information problems

1. Write down every probability you are given, in symbols: $p(A) = 0.5$, $p(A \cup B) = 0.7$, ...
2. Pick the rule that links them: the addition rule (OR), the conditional / multiplication rule (GIVEN), or $p(A \cap B) = p(A)\,p(B)$ if you're told they're independent.
3. Substitute and solve for the unknown. A Venn diagram or a two-way table helps you see it.
4. Sense-check: every probability must be between 0 and 1.

Example: $p(A) = 0.5$, $p(B) = 0.4$, $p(A \cup B) = 0.7$. Find $p(A \cap B)$. Are A and B independent?

$$0.7 = 0.5 + 0.4 - p(A \cap B) \;\Rightarrow\; p(A \cap B) = 0.2$$

addition rule

$$p(A)\,p(B) = 0.5 \times 0.4 = 0.2 = p(A \cap B)$$ ← equal, so independent ✓

Example: $p(A \mid B) = 0.3$ and $p(B) = 0.6$. Find $p(A \cap B)$.

$$p(A \cap B) = p(B)\,p(A \mid B) = 0.6 \times 0.3 = 0.18$$

multiplication rule

Example: A and B are independent, $p(A) = 0.3$ and $p(A \cup B) = 0.65$. Find $p(B)$.

$$0.65 = 0.3 + p - 0.3p$$ ← addition rule with $p(A \cap B) = 0.3p$

$$0.35 = 0.7p \;\Rightarrow\; p(B) = 0.5$$

Example: 200 people: 40% wear glasses, 15% are left-handed, and 30% of the glasses-wearers are left-handed. Complete the table. Are glasses and left-handedness independent?

Turn each percentage into a count, **of the right group**: glasses $0.4 \times 200 = 80$; left-handed $0.15 \times 200 = 30$; left-handed glasses-wearers $0.3 \times 80 = 24$ (30% **of the 80**). Every other cell is a subtraction from a total:

	left-handed	right-handed	total
glasses	**24**	80 − 24 = 56	**80**
no glasses	30 − 24 = 6	120 − 6 = 114	200 − 80 = 120
total	**30**	200 − 30 = 170	**200**

$$p(G \cap L) = \frac{24}{200} = 0.12, \qquad p(G)\,p(L) = 0.4 \times 0.15 = 0.06$$

$0.12 \neq 0.06$, so they are **dependent** (in this made-up group, glasses-wearers are more likely to be left-handed). Zoom in, as before: $p(G \mid L) = \frac{24}{30} = 0.8$, much more than $p(G) = 0.4$.

Bayes: turning a conditional round

RULE: Bayes' theorem

$p(A \mid B)$ and $p(B \mid A)$ are **different** numbers: same overlap on top, different crowd underneath.

To turn one into the other: $p(A \mid B) = \dfrac{p(B \mid A)\, p(A)}{p(B)}$.

If $p(B)$ isn't given, build it from the law of total probability (two sections on).

Lecture 8's school: 100 children. Two questions that **sound** the same but aren't:

	boy	girl	total
dark hair	30	36	66
light hair	25	9	34
total	55	45	100

light hair: zoom into the row of 34

girl: zoom into the column of 45

If you see a light-haired child, how likely is it a girl? Zoom into the light row: $p(\text{girl} \mid \text{light}) = \frac{9}{34} = 0.26$.

If you see a girl, how likely is light hair? Zoom into the girl column: $p(\text{light} \mid \text{girl}) = \frac{9}{45} = 0.20$.

Same 9 children on top, **different crowd underneath**. **Bayes' theorem** converts one into the other using the marginal probabilities, when you don't have the whole table:

$$p(a \mid b) = \frac{p(b \mid a)\, p(a)}{p(b)}$$

swap what's known and what's asked, then correct by the ratio of the crowd sizes

$$p(\ell \mid g) = \frac{p(g \mid \ell)\, p(\ell)}{p(g)} = \frac{0.26 \times 0.34}{0.45} = 0.20$$

ℓ = light, g = girl ✓

Tree diagrams and the law of total probability

RULE: tree diagrams

Split first by one event (A or A′), then split each branch by the second event (B or B′). The second-level branches carry **conditional** probabilities such as $p(B \mid A)$.

Multiply along a path to get the probability of that whole path: $p(A \cap B) = p(A)\,p(B \mid A)$.

Add between the paths that end in the outcome you want. The branches leaving any point add up to 1.

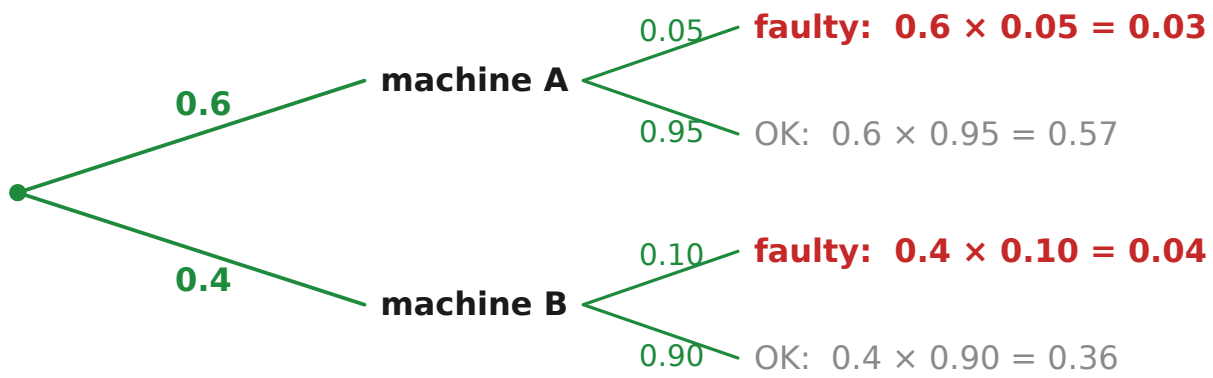
RULE: law of total probability

$p(B) = p(B \mid A)\,p(A) + p(B \mid A')\,p(A')$: every way B can happen, added up.

Bayes' theorem uses it for the bottom line:

$$p(A \mid B) = \frac{p(B \mid A)\,p(A)}{p(B \mid A)\,p(A) + p(B \mid A')\,p(A')}.$$

Example: machine A makes 60% of a factory's bolts and machine B the other 40%. 5% of A's bolts are faulty and 10% of B's. What fraction of all bolts is faulty? A bolt is faulty: what's the chance machine A made it?



$$p(F) = 0.05 \times 0.6 + 0.10 \times 0.4 = 0.03 + 0.04 = 0.07$$

add the two red paths

$$p(A \mid F) = \frac{p(F \mid A)\, p(A)}{p(F)} = \frac{0.03}{0.07} = \frac{3}{7} \approx 0.43$$

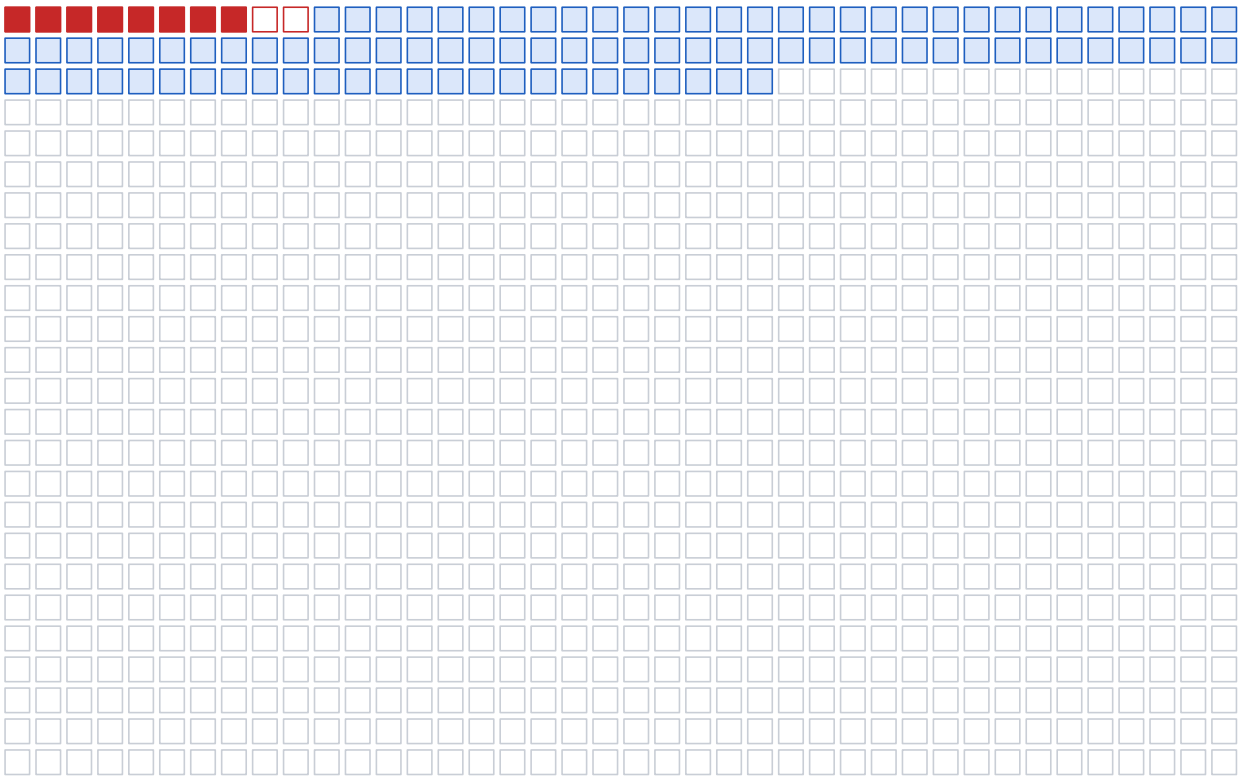
Bayes: the A path as a share of all faulty paths

Machine A makes most of the bolts, yet a faulty bolt is more likely to have come from B ($\frac{4}{7}$): B's higher fault rate wins. The mammogram puzzle next is the same calculation.

⚄ The mammogram puzzle, as 1000 people

1% of women aged 40 who are screened have breast cancer. 80% of those get a positive mammogram. 9.6% of women without cancer also get a positive. A woman tests positive. What's the chance she has cancer?

Guess first! Almost certain? Fairly likely? About half? (Lecture 8: only 15% of doctors got it right.) Now picture **1000 women**, one square each:



Step by step: 1% of 1000 = **10 have cancer**; 80% of them = **8 test positive**. The other 990 are healthy, but 9.6% of 990 = 95 test positive anyway. So **103 positives, only 8 of whom are ill**:

$$p(\text{cancer} \mid +) = \frac{8}{103} \approx 0.078 = 7.8\% \leftarrow \textit{fairly unlikely!}$$

The healthy group's chance of a false alarm is small, but **there are so many more of them** that their false alarms swamp the real cases. Bayes' theorem gives the same answer, with $p(+)$ built from both groups by the law of total probability (Lecture 8's alternative formula):

$$p(c \mid m) = \frac{p(m \mid c)\, p(c)}{p(m \mid c)p(c) + p(m \mid c')p(c')} = \frac{0.8 \times 0.01}{0.8 \times 0.01 + 0.096 \times 0.99}$$

$$= \frac{0.008}{0.10304} \approx 0.078$$

Vocabulary: $p(\text{cancer}) = 1\%$ is the **prior** (belief before the evidence); $p(\text{cancer} \mid +) = 7.8\%$ is the **posterior** (belief after). The positive test raised the risk almost eightfold, but it's still far from certain. That's the scientific method in miniature: update your beliefs as evidence arrives.

Lecture 8's extension question asks whether $p(a_1 \mid b) = \frac{p(a_1 \cap b)}{p(a_1 \cap b) + p(a_2 \cap b)}$ when a_1 and a_2 can't both happen but one of them must. **Yes:** the bottom is just every way b can happen, split into the a_1 part and the a_2 part. It's exactly the counting we did: $\frac{8}{8+95}$, with a_1 = cancer, a_2 = no cancer, b = positive.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Probability	$p(E) = \dfrac{\text{outcomes in } E}{\text{all outcomes}}$ equally likely outcomes; always between 0 and 1
Complement	$p(E') = 1 - p(E)$
Addition rule (OR)	$p(A \cup B) = p(A) + p(B) - p(A \cap B)$ always true
Mutually exclusive	$p(A \cap B) = 0, \qquad p(A \cup B) = p(A) + p(B)$ can't both happen
Conditional	$p(A \mid B) = \dfrac{p(A \cap B)}{p(B)}$ B becomes the new sample space
Multiplication rule (AND)	$p(A \cap B) = p(B)\,p(A \mid B) = p(A)\,p(B \mid A)$ always true
Independent	$p(A \cap B) = p(A)\,p(B), \qquad p(A \mid B) = p(A)$ test: compare $p(A)p(B)$ with $p(A \cap B)$
Exclusive vs independent	exclusive events with non-zero probabilities are never independent
Tree diagram	multiply along the branches, add between the paths; branches from a point add to 1
Total probability	$p(B) = p(B \mid A)\,p(A) + p(B \mid A')\,p(A')$
Bayes' theorem	$p(A \mid B) = \dfrac{p(B \mid A)\,p(A)}{p(B)}$ bottom line from the law of total probability
Two-way table	joint = one cell ÷ grand total; marginal = a total ÷ grand total; conditional = one cell ÷ its row or column total

Your turn

Use the dice grid where it helps. Answers in grey.

Two dice

1. p(a double)
2. p(total is 5)
3. p(not a double)
4. p(red is even OR blue is 1)
5. p(total is 8 | red is 3)

1. 6/36 = 1/6 2. 4/36 = 1/9 3. 5/6 4. (18 + 6 − 3)/36 = 21/36 = 7/12 5. 1/6 (only blue 5)

Cards and tables

6. Two cards without replacement: p(both aces)?
7. Urn table: p(Small | Blue)?
8. Are 'coin shows heads' and 'die shows 6' independent? p(both)?

6. 4/52 × 3/51 = 1/221 7. 15/30 = 0.5 8. yes; 1/2 × 1/6 = 1/12

Bayes

9. A disease affects 2% of people. A test catches 90% of cases but also gives 5% false positives. Using 1000 people, find p(disease | positive).

9. 20 ill, 18 positive; 980 well, 49 positive; 18/67 ≈ 0.27

Find the unknown

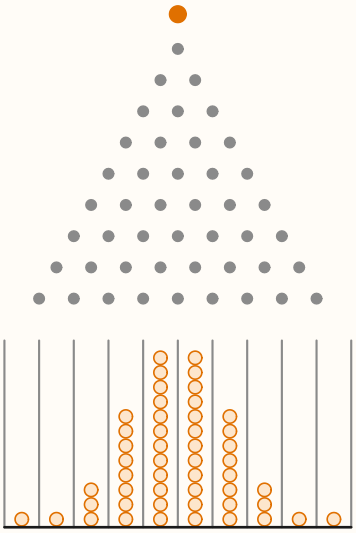
10. p(A) = 0.6, p(B) = 0.5, p(A ∪ B) = 0.8. Find p(A ∩ B). Are A and B independent?
11. p(A | B) = 0.25 and p(B) = 0.4. Find p(A ∩ B).
12. p(A ∩ B) = 0.12 and p(A) = 0.3. Find p(B | A).
13. A and B are mutually exclusive, p(A) = 0.35, p(A ∪ B) = 0.8. Find p(B).
14. A and B are independent, p(A) = 0.4, p(A ∪ B) = 0.7. Find p(B).
15. 300 students: 40% come by bus; 25% of bus users are late, and 10% of the others. How many are late? p(bus | late)?

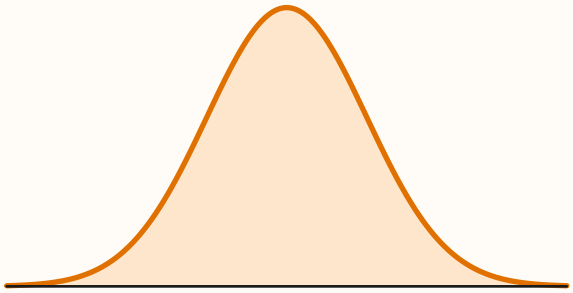
10. 0.6 + 0.5 − 0.8 = 0.3; 0.6 × 0.5 = 0.3, so independent 11. 0.4 × 0.25 = 0.1 12. 0.12 ÷ 0.3 = 0.4 13. 0.8 − 0.35 = 0.45 (no overlap) 14. 0.7 = 0.4 + p − 0.4p, so p = 0.5 15. bus 120, late 30; others 180, late 18; 48 late; 30/48 = 0.625

Chapter 26

The Normal distribution

why so many things pile up into a bell





the shape the balls are aiming for

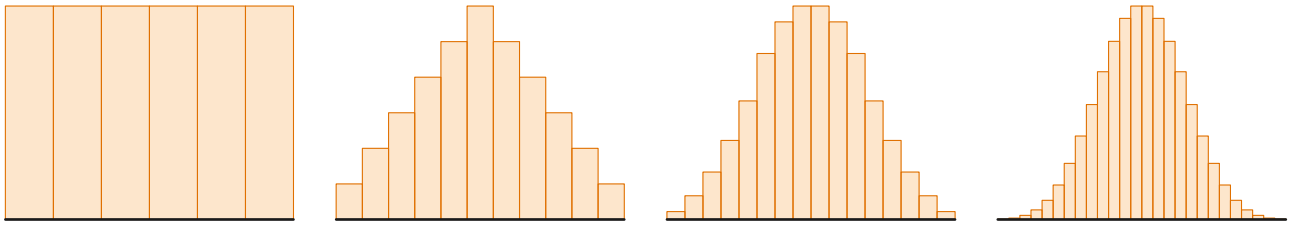
Drop balls into a board of pegs. At each peg a ball bounces left or right at random, yet the pile at the bottom always makes the same shape: a **bell**, high in the middle and fading at the edges. Heights, test scores and measurement errors make the same bell.

This chapter shows **why** the bell appears, how to read it (it's all about **area**), and how the **standard deviation** of Chapter 24 becomes a ruler for answering "what fraction of people...?" questions. You will never need to integrate the scary formula by hand.

Goes with **Lecture 9** (uniform vs Normal, sums of distributions, the bell curve equation, IQ, skewness and kurtosis, probability density, the seat-width problem).

/\ From flat to bell: add things together

Roll one die many times and every face turns up about equally: a **flat** histogram. Lecture 9 calls this **uniformly random** (coins, dice, roulette and lottery balls are all flat). A **distribution** is just the pattern of how often each value turns up.

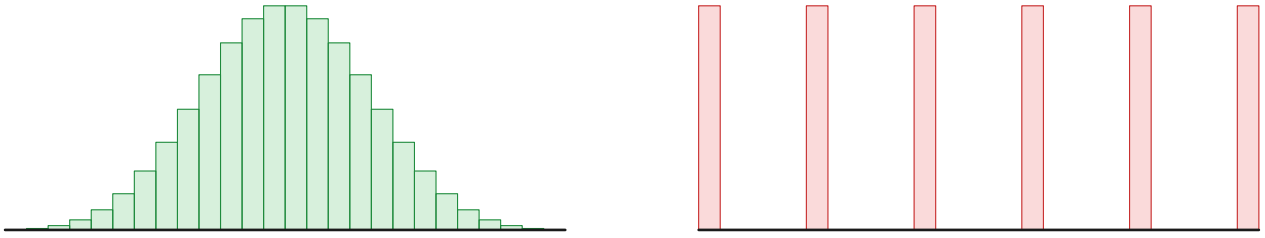


total of 1 die **total of 2 dice** **total of 3 dice** **total of 5 dice**

Now add two dice. A total of 7 can happen 6 ways but 2 only one way (the diagonals of Chapter 25's grid), so the histogram becomes a **triangle**. Three dice: a hump. Five: already a bell. Lecture 9 does the same with coins: one coin is flat; the number of heads in 100 coins is a beautiful bell (mean 50, standard deviation 5).

That's the secret: whenever a quantity is the **sum (or average) of many small random bits**, it comes out bell-shaped, whatever the bits looked like. Lecture 9: *"Any sum (or average) of multiple identically distributed distributions"* is Normal. Your height is the sum of thousands of small genetic and dietary nudges; that's why it's (roughly) Normal.

But not just any sum. The bits must be **separate** (independent, or only weakly linked). Both pictures below add up five dice numbers, but only one makes a bell:



5 separate rolls, added

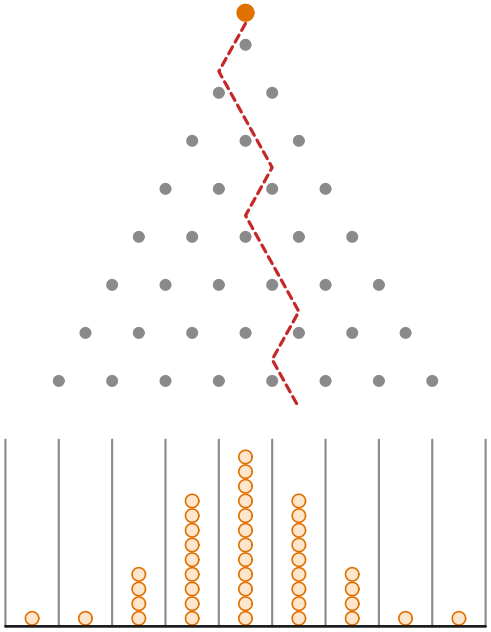
bell ✓

1 roll, copied 5 times, added

still flat: only 5, 10, ..., 30

Copying one roll adds no new randomness, so the shape is just the single die, stretched. The honest rule: a sum of **many** contributions is close to Normal when they're **independent or only weakly linked**, each has a **finite spread**, and **no single one dominates** the total. (If one huge nudge outweighs all the rest, the total just looks like that one nudge.)

The Galton board: coin flips you can watch



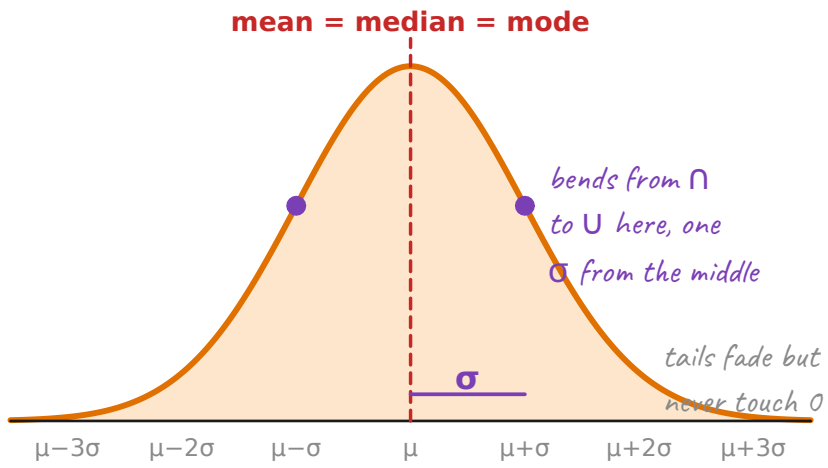
Each peg is a coin flip: left or right.
8 rows = 8 flips. The bin a ball lands in
= the number of rights.

The red path went L R R L R R L R:
5 rights, so bin 5.

Ending in the middle needs a mix of lefts
and rights, and there are many such paths.
The edges need all-left or all-right:
only one path each.

With 8 rows there are $2^8 = 256$ possible paths: 70 of them end in the middle bin but only 1 reaches each edge. Count paths, and the bell appears. (The counts 1, 8, 28, 56, 70, 56, 28, 8, 1 are the same numbers as expanding $(a+b)^8$.)

∧ Anatomy of the bell



Symmetric: the left half mirrors the right.

One peak in the middle.

Two numbers pin it down completely: the mean μ (where) and the standard deviation σ (how wide).

Lecture 9's equation of the curve looks terrifying, but read it piece by piece:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

(x − μ)/σ: how many σ's from the middle (the z-score, two pages on)

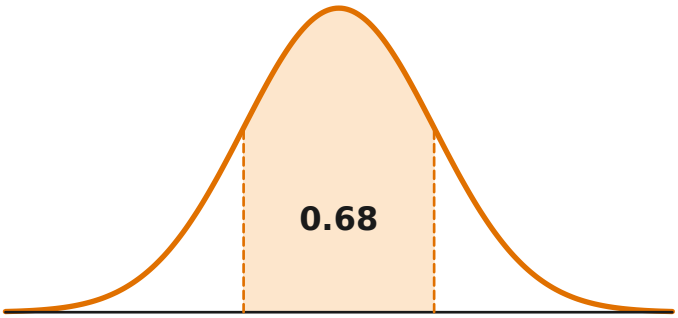
e to minus a square: 1 in the middle, shrinking fast either side (Chapter 21)

the front fraction just makes the total area exactly 1

With $\mu = 0$ and $\sigma = 1$ it's the **standard Normal**: $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. Every Normal curve is this one, slid along and stretched.

Probability is area

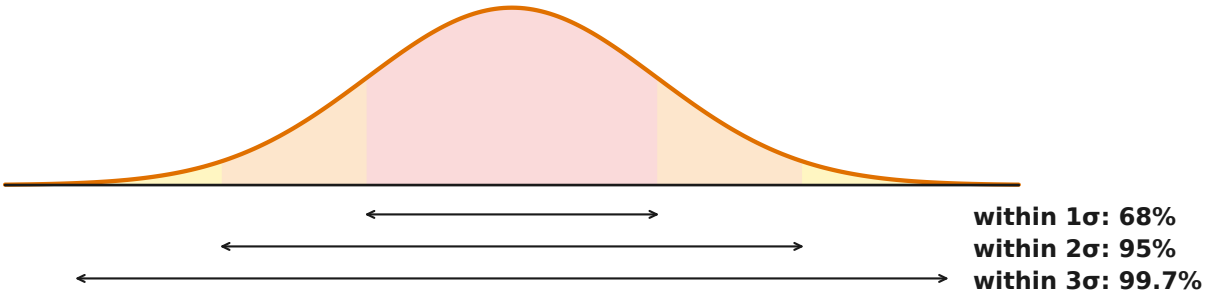
Imagine a histogram of 100,000 people's heights with ever thinner bars: the tops of the bars melt into the smooth curve. The height of the curve is called **probability density**. On its own a single height means little; **areas** are what count:



the whole area under the curve = 1 (everybody, 100%)

the shaded slice = the fraction of people between those values = the probability a random person lands there

The 68-95-99.7 rule. For **every** Normal distribution:



So about 2 in 3 people are within one standard deviation of the mean, 19 in 20 within two, and almost everyone within three.

Λ z-scores: a ruler marked in sigmas

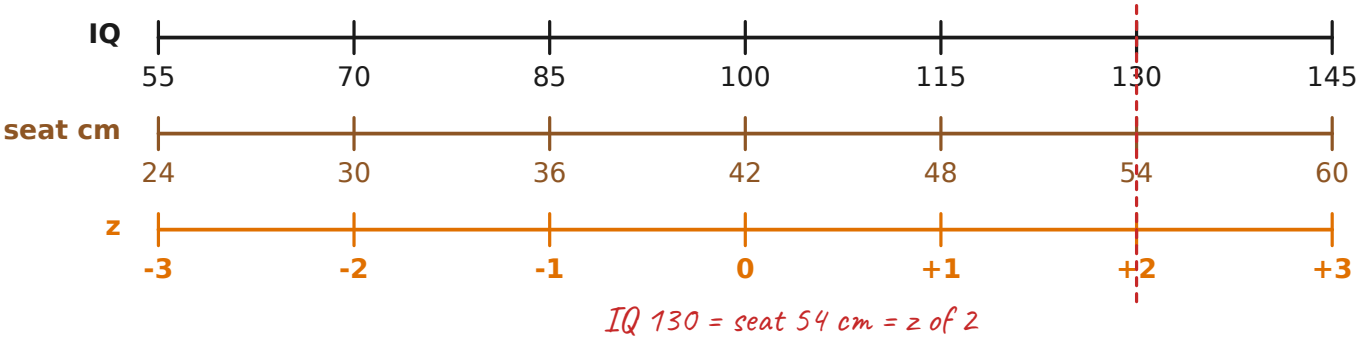
RULE: z-scores

$z = \dfrac{x - \mu}{\sigma}$: how many standard deviations x is above (+) or below (−) the mean.

Turned round: $x = \mu + z\sigma$.

Whatever μ and σ are, the z-scores follow the **standard Normal** (mean 0, standard deviation 1), so one table serves every Normal distribution.

IQ scores are Normal with mean 100 and standard deviation 15 (Lecture 9). Lay a **second ruler** under the IQ ruler, marked in steps of one σ. The reading on that ruler is the **z-score**: **how many standard deviations above (+) or below (−) the mean**.



$$z = \frac{x - \mu}{\sigma}$$ ← distance from the mean ÷ the size of one step

$$\text{IQ } 75: \quad z = \frac{75 - 100}{15} = \frac{-25}{15} \approx -1.67$$ ← 1.67 σ's below

Once everything is in z-units, **one** table (or calculator button) works for every Normal distribution in the world. That table gives $\Phi(z)$ ("phi of z"): the area to the **left** of z, i.e. the fraction of people **below** that point.

z = 0	z = 0.5	z = 1	z = 1.28	z = 1.33	z = 1.645	z = 1.96	z = 2	z = 2.326	z = 3
0.5000	0.6915	0.8413	0.8997	0.9082	0.9500	0.9750	0.9772	0.9900	0.9987

Areas from the table: the four rules

RULE: areas under a Normal curve

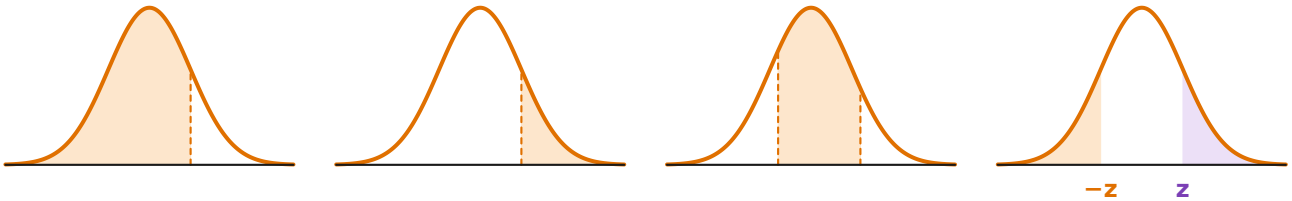
$\Phi(z)$ = the area to the **left** of z. So $p(X < x) = \Phi(z)$.

Above a value: $p(X > x) = 1 - \Phi(z)$, because the whole area is 1.

Between two values: $p(a < X < b) = \Phi(z_b) - \Phi(z_a)$.

Negative z: printed tables only give positive z. By symmetry $\Phi(-z) = 1 - \Phi(z)$. Also $\Phi(0) = 0.5$.

For a smooth curve, $<$ and $\leq$ give the same area: the chance of hitting one exact value is 0.



$p(X < x) = \Phi(z)$ $\quad$ $p(X > x) = 1 - \Phi(z)$ $\quad$ $\Phi(z_b) - \Phi(z_a)$ $\quad$ $\Phi(-z) = 1 - \Phi(z)$

PROCEDURE: finding a Normal probability

1. **Standardise**: turn each x into $z = \frac{x-\mu}{\sigma}$.
2. **Sketch** the bell, mark the mean and your value(s), and shade the area the question asks for.
3. **Look up** Φ for each z (for a negative z use $1 - \Phi(|z|)$).
4. **Finish**: left area, done; right area, $1 - \Phi$; between, larger Φ minus smaller Φ. Check against the sketch: should the answer be more or less than 0.5?

Example: IQ is Normal with $\mu = 100$, $\sigma = 15$. What fraction score **above** 120?

$$z = \frac{120 - 100}{15} = 1.33$$ ← standardise (2 d.p.)

$$p(X > 120) = 1 - \Phi(1.33) = 1 - 0.9082 = 0.0918$$

right of z: 1 − Φ

Example: what fraction score **below** 85?

$$z = \frac{85 - 100}{15} = -1$$ ← standardise

$$\Phi(-1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$$

negative z: symmetry

Example: what fraction score **between** 85 and 120?

$$\Phi(1.33) - \Phi(-1) = 0.9082 - 0.1587 = 0.7495$$

larger minus smaller: about 75%

✗ Common mistakes

✗ $p(X > x) = \Phi(z)$ ✓ $p(X > x) = 1 - \Phi(z)$

Φ always gives the area to the LEFT

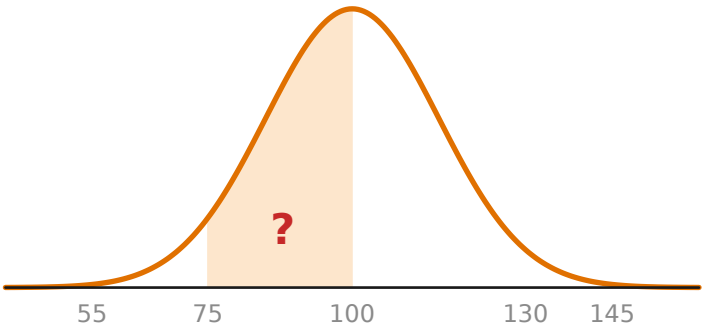
✗ $z = \frac{x - \mu}{\sigma^2}$ ✓ $z = \frac{x - \mu}{\sigma}$

divide by the standard deviation, not the variance

✗ $\Phi(-1) = -0.8413$ ✓ $\Phi(-1) = 1 - 0.8413 = 0.1587$

an area is never negative: use the symmetry rule

Lecture 9's IQ range: what fraction scores 75 to 100?



Area between 75 and 100
= (area left of 100)
− (area left of 75)

$$z_{100} = 0, \quad z_{75} = -1.667$$ ← *convert both ends to z*

$$\Phi(-1.667) = 1 - \Phi(1.667) = 1 - 0.9522 = 0.0478$$

negative z: use the symmetry rule

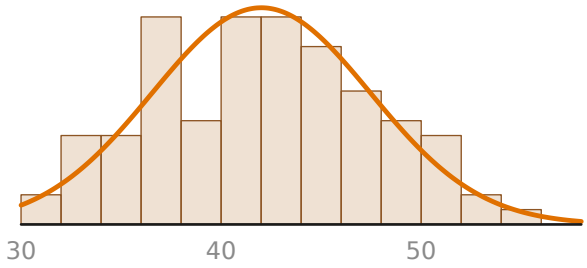
$$\Phi(0) - \Phi(-1.667) = 0.5000 - 0.0478$$ ← *half the area is left of the middle*

$$= 0.4522 \approx 45\%$$ ← *exactly the lecture's figure*

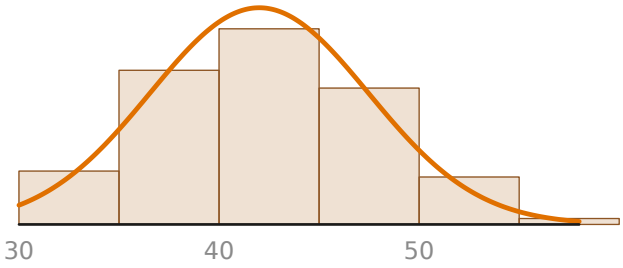
(Lecture 9 uses this to show how a far-right group targeted its propaganda at people with IQ 75-100. The maths is neutral; what people do with it isn't.)

The aeroplane seat problem

Lecture 9: design seats wide enough for **90% of people**; the widest 10% buy two tickets. First, measure a sample. The slides list 100 seat widths (cm); here they are as histograms, binned two ways:



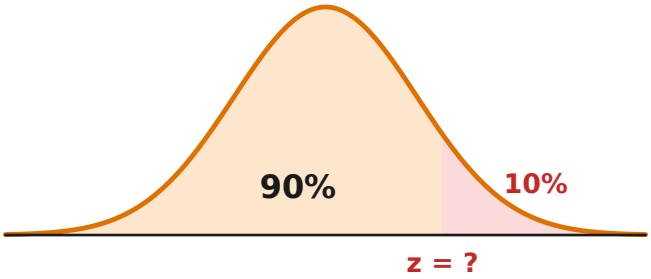
2 cm bins



5 cm bins

Same data, different binning, different look (the lecture's point). Either way, the orange Normal curve fits well. The data give mean **42.0 cm** and standard deviation **5.46 cm**; the slides round this to **6 cm**, so we'll use 6.

Now work backwards. Before, we had x and wanted the area. Now we have the area (90% to the left) and want x:



Find the z with Φ(z) = 0.90.
The table (previous page):
Φ(1.28) ≈ 0.90, so z ≈ 1.28.
Two tickets beyond here.

$$z = \frac{x - \mu}{\sigma} \Rightarrow x = \mu + z\sigma$$ ← *transpose (Chapter 14)*

$$x = 42 + 1.28 \times 6 = 49.7 \text{ cm}$$

Seats 49.7 cm wide fit 90% of passengers. (With the unrounded 5.46 cm you'd get 49.0 cm; rounding σ costs airlines 7 mm per seat.)

Working backwards: cut-offs, μ and σ

PROCEDURE: inverse Normal problems

1. Turn the percentage into an area to the **left**: bottom 10% is 0.10; top 10% is 0.90; the middle 90% leaves 5% in each tail, so 0.05 and 0.95.
2. Find the z with that $\Phi(z)$: read the table backwards, or use a calculator's inverse Normal. A left-tail area below 0.5 gives a **negative** z.
3. Put z into $z = \frac{x-\mu}{\sigma}$ and solve for the missing quantity: $x = \mu + z\sigma$, $\mu = x - z\sigma$, or $\sigma = \dfrac{x-\mu}{z}$.
4. If μ and σ are **both** unknown you need two facts: write $x = \mu + z\sigma$ for each and solve them as simultaneous equations (Chapter 15).

area to the left, Φ(z)	z	use it for
0.90	1.2816	top 10% (or bottom 90%)
0.95	1.6449	top 5%; middle 90% is ±1.645
0.975	1.96	top 2.5%; middle 95% is ±1.96
0.99	2.3263	top 1%; middle 98% is ±2.326
0.10	−1.2816	bottom 10%: the mirror image of 0.90

Example: IQ ($\mu = 100$, $\sigma = 15$): what score cuts off the **bottom** 10%?

$$\Phi(z) = 0.10 \;\Rightarrow\; z = -1.2816$$ ← a bottom tail: negative z

$$x = 100 + (-1.2816) \times 15 = 80.8$$ ← $x = \mu + z\sigma$

Example: between which two IQ scores do the **middle** 90% lie?

$$z = \pm 1.645, \qquad 100 \pm 1.645 \times 15 = 100 \pm 24.7$$

5% in each tail

So from **75.3** to **124.7**.

Example: a machine fills bags with $\sigma = 5$ g. Only 2.5% of bags may weigh under 500 g. What mean μ should it be set to?

$$\Phi(z) = 0.025 \;\Rightarrow\; z = -1.96$$

$$500 = \mu - 1.96 \times 5 \;\Rightarrow\; \mu = 509.8 \text{ g}$$ ← $\mu = x - z\sigma$

Example: exam marks have mean 60 and 10% of students score over 80. Find σ.

$$\Phi(z) = 0.90 \;\Rightarrow\; z = 1.2816$$ ← top 10% means 90% to the left

$$\sigma = \frac{80 - 60}{1.2816} = 15.6$$ ← $\sigma = (x - \mu)/z$

✗ Common mistakes

✗ $\text{top } 10\% : \; z = -1.28$

the top tail is above the mean

✓ $\text{top } 10\% : \; z = +1.28$

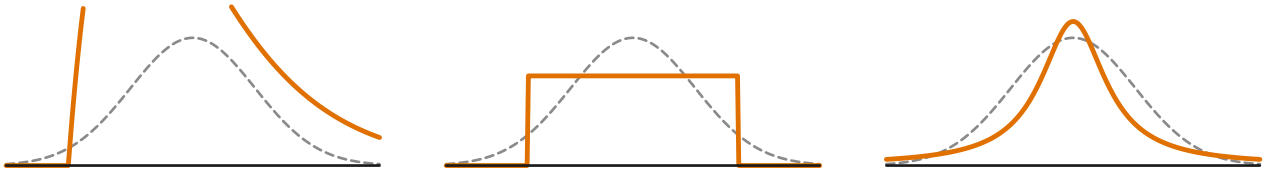
✗ $\text{middle } 90\% : \; \Phi(z) = 0.90$

the 10% left over is split between two tails

✓ $\text{middle } 90\% : \; \Phi(z) = 0.95$

Is it Normal? And stay critical

Probably Normal (Lecture 9): averaged qualities like IQ, test scores, rainfall, molecular speeds, football crowds, women's heights. **Definitely not:** anything **lopsided** or with a **hard limit on one side**, like wages (skewed, Chapter 24). Two numbers measure the damage:



skewness ≠ 0
lopsided

kurtosis < 0
too flat (e.g. uniform)

kurtosis > 0
too peaky, heavy tails

A Normal distribution has skewness 0 and kurtosis 0. Lecture 9's table: one uniform variable has kurtosis −1.18 (flat), the sum of four has −0.32: **adding makes it more Normal**, the lesson of page 2.

Be critical. Lecture 9 tells the story of *The Bell Curve* (1994) and a world IQ map built from biased samples: one country's "national IQ" came from a group of children in a Spanish care home; inconvenient high-scoring samples were dropped. Chapter 23's detective questions (who was sampled? what's missing?) matter more than the maths. The lecture's last word: **scientists can be lazy, prejudiced and wrong but still get published, so develop your own analytical skills**.

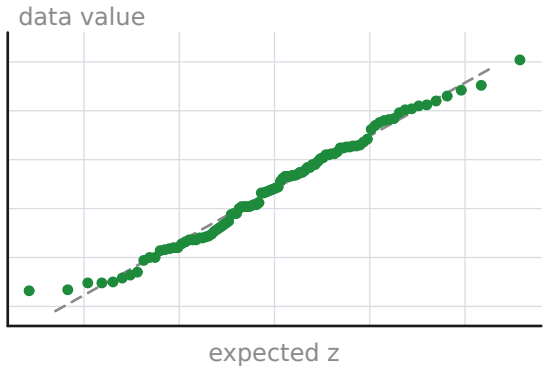
Checking for Normality: the probability plot

RULE: normal probability plot

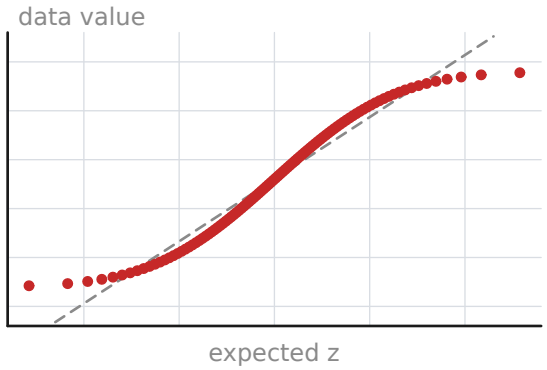
Sort the data. Plot each value against the z-score it **would** have if the data were exactly Normal (the kth of n values goes with the z that has $\Phi(z) = \frac{k-0.5}{n}$).

Normal data lie close to a **straight line**. Ends that bend away from the line mean skew, or tails that are too light or too heavy.

Software (Minitab, in Lecture 9) draws it for you: your job is to judge whether the points are straight. Lecture 9: the difference from a uniform data set is much clearer here than in a histogram.



the 100 seat widths: straight, so Normal



uniform data: bent ends, not Normal

Is human height Normal? Lecture 9 lists **women's** heights as roughly Normal, but **people's** heights (men and women together) as not: the mix is two overlapping bells with different means, which is wider and flatter on top than one Normal. Even one group's heights are only **approximately** Normal: the curve's tails go on for ever, but nobody is 50 cm or 3 m tall. A model can be very useful without being exactly true.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

z-score	$z = \dfrac{x - \mu}{\sigma}$ how many standard deviations from the mean
Back from z	$x = \mu + z\sigma, \qquad \mu = x - z\sigma, \qquad \sigma = \dfrac{x - \mu}{z}$
Below a value	$p(X < x) = \Phi(z)$ Φ is the area to the left
Above a value	$p(X > x) = 1 - \Phi(z)$
Between two values	$p(a < X < b) = \Phi(z_b) - \Phi(z_a)$
Negative z	$\Phi(-z) = 1 - \Phi(z), \qquad \Phi(0) = 0.5$ tables only give positive z
68-95-99.7 rule	within 1σ: 68%; within 2σ: 95%; within 3σ: 99.7%
Normal probability	standardise, sketch and shade, look up, then complement or subtract
Inverse Normal	percentage → area to the left → z (table backwards) → x, μ or σ
Useful z values	Φ = 0.90: z = 1.2816; 0.95: 1.645; 0.975: 1.96; 0.99: 2.326. Middle 90%: ±1.645; middle 95%: ±1.96
Normal curve	$f(x) = \dfrac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ total area 1; symmetric about μ; mean = median = mode
When is it Normal?	sums or averages of many small independent effects; check with a probability plot (straight line)

Λ Your turn

Use the Φ table on the z-score page. Answers in grey.

z-scores

1. IQ 115: z = ?
2. IQ 70: z = ?
3. A seat width of 36 cm (μ = 42, σ = 6): z = ?
4. What IQ has z = 1.645?

1. 1 2. −2 3. −1 4. 100 + 1.645 × 15 ≈ 124.7

Areas

5. Fraction of people with IQ below 130
6. Fraction with IQ above 115
7. Fraction between IQ 85 and 115
8. Fraction with IQ between 100 and 124.7

5. Φ(2) = 0.977 6. 1 − Φ(1) = 0.159 7. 0.683 (the 68% rule) 8. 0.95 − 0.5 = 0.45

Backwards

9. An exam is Normal, μ = 60, σ = 10. The top 10% get a distinction. What mark is needed?
10. Which is more Normal: one die, or the average of 20 dice? Why?

9. 60 + 1.28 × 10 ≈ 72.8 10. the average of 20: sums of many independent bits become bell-shaped

Find the unknown

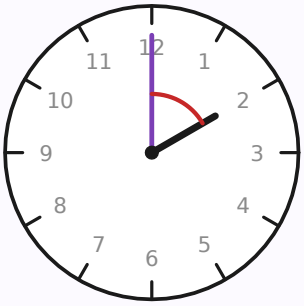
11. IQ (μ = 100, σ = 15): which score cuts off the bottom 10%?
12. Between which two IQ scores do the middle 95% lie?
13. Bags have σ = 4 g, and 5% weigh under 250 g. Find μ.
14. Exam marks: μ = 55, and 10% score above 75. Find σ.
15. A value of 70 has z = 2 and a value of 40 has z = −1. Find μ and σ.
16. Heights: μ = 170 cm and p(X > 185) = 0.0668. Find σ.

11. Φ(z) = 0.10, z = −1.2816; 100 − 1.2816 × 15 ≈ 80.8 12. z = ±1.96; 100 ± 29.4: 70.6 to 129.4 13. z = −1.645; μ = 250 + 1.645 × 4 ≈ 256.6 g 14. z = 1.2816; σ = (75 − 55)/1.2816 ≈ 15.6 15. 70 = μ + 2σ and 40 = μ − σ, so σ = 10, μ = 50 16. Φ(z) = 0.9332, z = 1.5; σ = 15/1.5 = 10

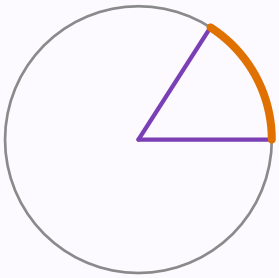
Chapter 27

Angles: degrees and radians

how much have you turned?



2 o'clock: the hands are 60° apart



string one radius long, laid on the rim: 1 radian

An angle isn't a length or a shape: it's **an amount of turning**. Open a door, turn a key, spin round on the spot: each is an angle.

We measure turning two ways: **degrees** (360 for a full turn, the everyday way) and **radians** (about 6.28 for a full turn, the way calculus and your lecturer prefer). This chapter shows where both come from and how to swap between them.

Goes with **Lecture 6** (radians vs degrees, notation and conversion).

Turning on the spot

RULE: degrees and the names of angles

A full turn is $360°$. A half turn (a straight line) is $180°$. A quarter turn (a **right angle**) is $90°$.

Acute: less than $90°$. **Obtuse**: between $90°$ and $180°$. **Reflex**: more than $180°$.

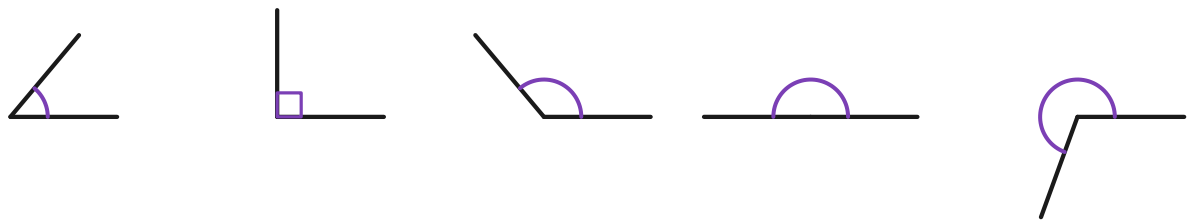
Angles on a straight line add to $180°$; angles round a point add to $360°$.

Stand facing north and turn. A **full turn** brings you back to where you started. We cut a full turn into **360 equal steps** called **degrees**, written with a little circle: 360°.

quarter turn **90°** | **half turn** **180°** | **three quarters** **270°** | **full turn** **360°**

Why 360? Nobody really knows: the Babylonians liked it (a year has about 360 days, and 360 divides by 2, 3, 4, 5, 6, 8, 9, 10, 12...). Lecture 6 calls degrees **arbitrary**, and that's fair.

Names for sizes of angle:



acute less than 90° | **right angle** exactly 90° | **obtuse** 90° to 180° | **straight** exactly 180° | **reflex** more than 180°

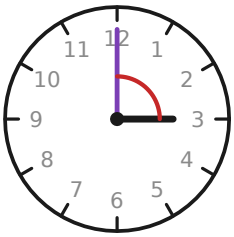
∠ Clock hands: angles you already read every day

RULE: angle between clock hands

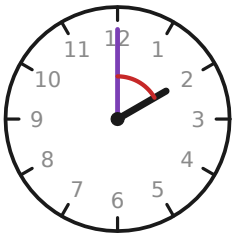
Measure both hands clockwise from 12. Minute hand: $6°$ per minute. Hour hand: $30°$ per hour **plus** $0.5°$ per minute past the hour.

The angle between them is the difference. If it is over $180°$, the smaller angle is $360°$ minus it.

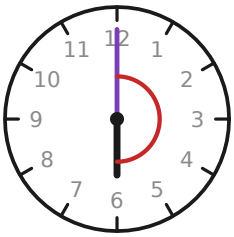
A clock face is a full turn split into 12 hours, so **each hour mark is 360 ÷ 12 = 30°** apart. The minute hand does a full turn in 60 minutes: **6° per minute**.



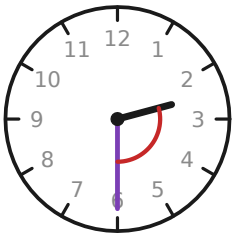
3:00
90°



2:00
60°



6:00
180°



2:30
105°

2 o'clock: the hour hand is 2 marks round, so $2 \times 30° = 60°$ (Lecture 6's cosine-rule clock, Chapter 31). The trap at 2:30: the hour hand has moved **halfway** towards 3, so it's at 75° from 12, while the minute hand is at 180°. Gap: $180° - 75° = 105°$, not 90°.

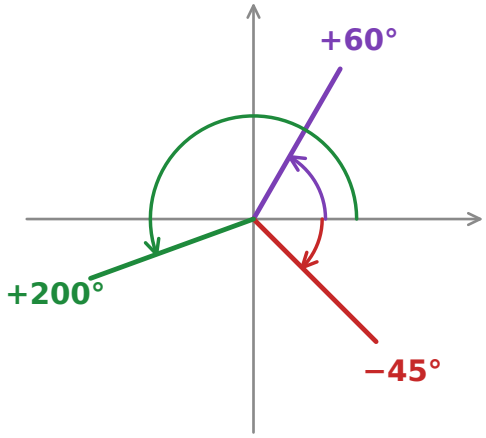
Which way is positive? Anticlockwise, from the east

RULE: positive and negative angles

Maths angles are measured from the **positive x-axis** (east). **Anticlockwise is positive**, clockwise is negative.

Adding or subtracting $360°$ (2π) gives the **same direction**: $-45°$, $315°$ and $675°$ all point the same way.

Clocks go clockwise from 12. Mathematicians do the opposite: **start pointing east (along the positive x-axis) and turn anticlockwise**. Lecture 6 measures angles this way on the unit circle. Turning clockwise gives a **negative** angle.



+60°: a sixth of a turn anticlockwise
−45°: an eighth of a turn clockwise
+200°: past the half-turn

More than 360° just means more than one full turn: 550° = 360° + 190°. (Lecture 6 uses 550° in an example.)

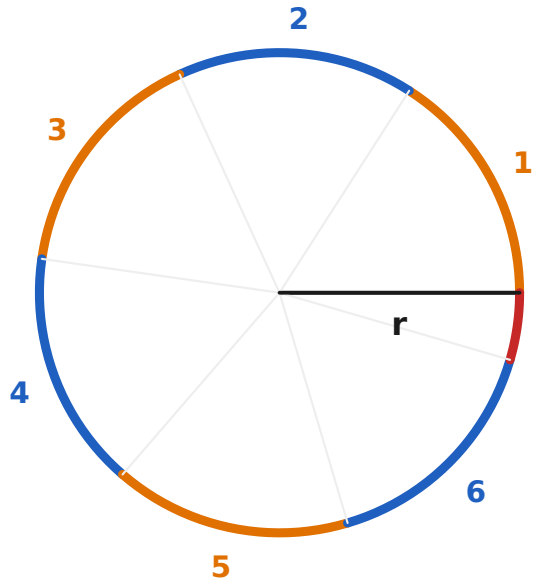
∠ The radian: a string one radius long

RULE: the radian

1 radian is the angle at the centre when the arc is exactly as long as the radius.

$2\pi \text{ rad} = 360°$, so $\pi \text{ rad} = 180°$ and $1 \text{ rad} \approx 57.3°$.

Take a piece of string **exactly as long as the circle's radius** and lay it along the rim. The angle it covers at the centre is **one radian**. (Lecture 6: "1 radian is the angle subtended by an arc of length R"; *subtended* just means "made at the centre by".)



Lay radius-long strings round the rim: 6 whole strings fit, plus a little red bit (0.28 of a string).

The circumference is 2πr (Chapter 5), so exactly 2π ≈ 6.28 strings fit.

So a full turn is 2π radians:

$$360° = 2\pi \text{ rad} \approx 6.28 \text{ rad}$$

$$1 \text{ rad} = \frac{360°}{2\pi} \approx 57.3°$$ ← *a bit less than 60°*

Why bother? A radian is **arc length ÷ radius**, a length divided by a length, so it has **no units**: it's a pure number. That makes formulas simpler (next pages) and calculus (Chapter 34) only works cleanly in radians.

∠ Converting: the unit-conversion chain again

RULE: converting and reducing angles

Degrees → radians: multiply by $\frac{\pi}{180}$. **Radians → degrees**: multiply by $\frac{180}{\pi}$.

Leave π in the answer when asked for an **exact** value ($60^\circ = \frac{\pi}{3}$); otherwise use $\pi \approx 3.1416$.

Reducing: an angle over 360° or below 0° points the same way as one between 0° and 360°. Subtract or add 360° (in radians, 2π) until it lands in range.

Chapter 5 converted units by multiplying by a fraction equal to 1 (like $\frac{100\text{ cm}}{1\text{ m}}$). Here the "1" is $\frac{\pi\text{ rad}}{180^\circ}$, because π rad and 180° are the same half-turn:

degrees → radians

$$\times \frac{\pi}{180}$$

radians → degrees

$$\times \frac{180}{\pi}$$

$$60^\circ \times \frac{\pi}{180} = \frac{60\pi}{180} = \frac{\pi}{3}$$ ← *cancel 60/180 = 1/3*

$$\frac{3\pi}{4} \times \frac{180}{\pi} = \frac{3 \times 180}{4} = 135^\circ$$ ← *the π's cancel*

Reducing angles into range (0° to 360°, or 0 to 2π):

$$550^\circ - 360^\circ = 190^\circ$$ ← *over 360°: subtract a whole turn*

$$-45^\circ + 360^\circ = 315^\circ$$ ← *negative: add a whole turn*

$$\frac{9\pi}{4} - 2\pi = \frac{9\pi}{4} - \frac{8\pi}{4} = \frac{\pi}{4}$$ ← *in radians a whole turn is 2π*

$$-\frac{\pi}{3} + 2\pi = \frac{5\pi}{3}$$

Pizza shortcut: think in fractions of a turn. 90° is a quarter turn, and a quarter of 2π is $\frac{\pi}{2}$. The ones to know by heart:

30°	45°	60°	90°	180°	270°	360°
$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

Notation warning (Lecture 6): degrees always carry the ° sign. Radians are written "rad", a small c (2^c), or **nothing at all**. So an angle with no ° is probably radians, but many sources are lazy: always check which one you're in.

∡ The calculator trap: DEG or RAD?

RULE: calculator mode

Angle written with ° → **DEG** mode. Angle with "rad", a π, or no unit at all → **RAD** mode.

Check the mode **before** every trig calculation. The calculator never warns you.

Your calculator has a mode: **DEG** or **RAD**. Type the same keys in the wrong mode and you get a confident, wrong answer:

DEG
```
sin(30)
                0.5000
```
✓

RAD
```
sin(30)
               -0.9880
```
✗

In RAD mode, "30" means 30 radians (almost 5 full turns), so sin gives -0.988 instead of 0.5. If a trig answer looks mad, check the mode first.

∠ Pizza maths: crust length and slice area

RULE: arc length and sector area

A slice is a **fraction of the whole circle**: arc = fraction × circumference $2\pi r$; sector = fraction × area πr^2.

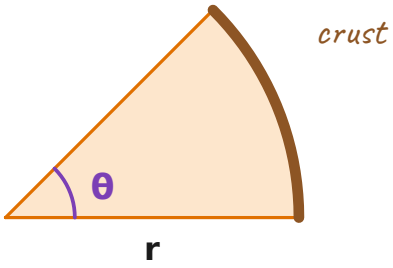
The fraction is $\dfrac{\theta}{360}$ when θ is in degrees, or $\dfrac{\theta}{2\pi}$ when θ is in radians.

That gives the two sets of formulas in the table.

Only use $s = r\theta$ and $A = \frac{1}{2}r^2\theta$ with θ in **radians**.

	θ in radians	θ in degrees
Arc length s	$s = r\theta$	$s = \dfrac{\theta}{360} \times 2\pi r$
Sector area A	$A = \frac{1}{2}r^2\theta$	$A = \dfrac{\theta}{360} \times \pi r^2$

With the angle θ in **radians**, the formulas for a slice of circle are beautifully simple (that's the payoff for radians):



crust

$$s = r\theta$$ arc (crust) length

$$A = \frac{1}{2}r^2\theta$$ sector (slice) area

Why they work: a full turn ($\theta = 2\pi$) gives $s = 2\pi r$ (the whole circumference) and $A = \frac{1}{2}r^2 \cdot 2\pi = \pi r^2$ (the whole area). A slice is just that fraction of the whole.

Example: a 30 cm pizza (radius 15 cm) cut into 8 slices. Each slice angle is $\frac{2\pi}{8} = \frac{\pi}{4}$:

$$s = 15 \times \frac{\pi}{4} \approx 11.8 \text{ cm}$$ ← crust per slice

$$A = \tfrac{1}{2} \times 15^2 \times \frac{\pi}{4} \approx 88.4 \text{ cm}^2$$ ← *cheese per slice*

The same in degrees: radius 10 cm, angle $72°$ (a fifth of a turn):

$$s = \frac{72}{360} \times 2\pi \times 10 = 4\pi \approx 12.57 \text{ cm}$$ ← *fraction × circumference*

$$A = \frac{72}{360} \times \pi \times 10^2 = 20\pi \approx 62.83 \text{ cm}^2$$ ← *fraction × area*

PROCEDURE: finding r or θ from an arc or a sector

1. Write down what you know: r, θ, s or A. Decide degrees or radians.
2. Write the matching formula from the table with the known numbers put in.
3. Rearrange for the unknown (Chapter 14). For r from an area you get r^2 first: take the square root at the end.
4. Check by putting your answer back into the formula.

Find r: a sector has angle 1.5 rad and area 48 cm². Find the radius.

$$48 = \tfrac{1}{2} r^2 (1.5) = 0.75\, r^2 \;\Rightarrow\; r^2 = 64,\; r = 8 \text{ cm}$$

÷ 0.75, then √

Find θ: an arc of 14 cm on a circle of radius 8 cm. Find the angle in radians and degrees.

$$14 = 8\theta \;\Rightarrow\; \theta = 1.75 \text{ rad} = 1.75 \times \tfrac{180}{\pi} = 100.3°$$

s = rθ

Find r in degrees: a $40°$ sector has an arc of 4π cm. Find r.

$$\frac{40}{360} \times 2\pi r = 4\pi \;\Rightarrow\; \frac{2\pi r}{9} = 4\pi \;\Rightarrow\; r = 18 \text{ cm}$$

✗ Common mistakes

✗ $s = r\theta = 5 \times 60 = 300$ ✓ $s = 5 \times \tfrac{\pi}{3} = 5.24$

s = rθ needs θ in radians: convert 60° first

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Full turn	$360^\circ = 2\pi$ rad half turn $180^\circ = \pi$; right angle $90^\circ = \frac{\pi}{2}$
Angle types	acute $< 90^\circ$; obtuse 90°–180°; reflex $> 180^\circ$
Clock hands	minute hand 6°/min; hour hand 30°/hour + 0.5°/min
Direction	from the positive x-axis; anticlockwise $+$, clockwise $-$
Degrees → radians	$\theta_{\text{rad}} = \theta_{\text{deg}} \times \dfrac{\pi}{180}$
Radians → degrees	$\theta_{\text{deg}} = \theta_{\text{rad}} \times \dfrac{180}{\pi}$
Reduce an angle	add or subtract 360° (2π) until it is in range
Arc length	$s = r\theta \quad \text{or} \quad s = \dfrac{\theta}{360} \times 2\pi r$ first with θ in radians, second in degrees
Sector area	$A = \frac{1}{2}r^2\theta \quad \text{or} \quad A = \dfrac{\theta}{360} \times \pi r^2$ first with θ in radians, second in degrees
Calculator	° → DEG mode; rad, π or no unit → RAD mode

📐 Your turn

Answers in grey.

Degrees

1. Angle between clock hands at 4:00
2. … at 9:30
3. Is 170° acute, obtuse or reflex?
4. Describe −90° as a turn

1. 120° 2. 105° (hour hand at 285°, minute at 180°) 3. obtuse 4. a quarter turn clockwise

Converting

5. 120° in radians
6. π/6 in degrees
7. 2.5 rad in degrees (1 d.p.)
8. 225° in radians

5. 2π/3 6. 30° 7. 143.2° 8. 5π/4

Arcs

9. Arc length: r = 10 cm, θ = 1.2 rad
10. Sector area: r = 4 m, θ = π/2
11. A slice has a crust of 6 cm on a 12 cm-radius pizza. Its angle in radians?

9. 12 cm 10. $4\pi \approx 12.6$ m² 11. 6 ÷ 12 = 0.5 rad

Find the unknown

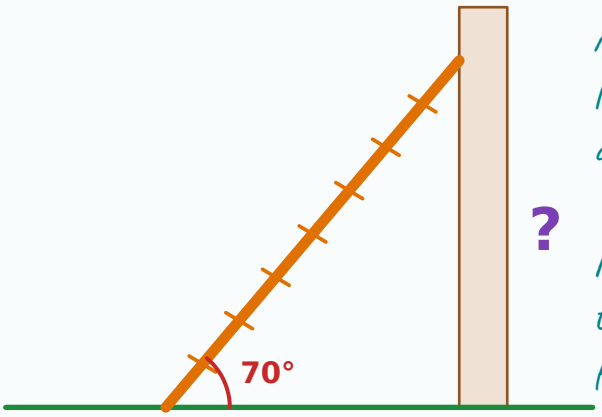
12. A sector has angle 1.5 rad and area 27 cm². Find the radius.
13. An arc is 15 cm long on a circle of radius 6 cm. Find the angle in radians and in degrees.
14. A 40° sector has an arc of 6π cm. Find the radius.
15. A sector of radius 9 cm has area 27π cm². Find its angle in degrees.
16. Write 750° as an angle from 0° to 360°, and −5π/3 as an angle from 0 to 2π.

12. $r = 6$ cm ($\frac{1}{2}r^2 \times 1.5 = 27$, so $r^2 = 36$) 13. $\theta = 15 \div 6 = 2.5$ rad ≈ 143.2° 14. $r = 27$ cm ($(40/360) \times 2\pi r = 6\pi$, so $2\pi r/9 = 6\pi$) 15. 120° ($(\theta/360) \times 81\pi = 27\pi$, so $\theta/360 = 1/3$) 16. 750° − 720° = 30°; $-5\pi/3 + 2\pi = \pi/3$

Chapter 28

Introduction to trigonometry

triangles that measure what you can't reach



A 5 m ladder leans at 70°. How high up the wall does it reach?

No tape measure needed: trigonometry answers it from the angle alone.

Trigonometry means "triangle measuring". Surveyors use it to find the height of mountains, sat-navs to find where you are, engineers to find forces. It all rests on one fact about right-angled triangles: **the angle decides the shape**.

We'll name the sides properly, meet Pythagoras (with real squares), see **why** sin, cos and tan exist, and work Lecture 6's examples one tiny step at a time.

Goes with **Lecture 6** (right-angled triangles, Pythagoras in 2D and 3D, the circle, SOHCAHTOA, inverse functions, Examples 1-3).

Naming the sides: stand at the angle and look

RULE: naming the sides of a right-angled triangle

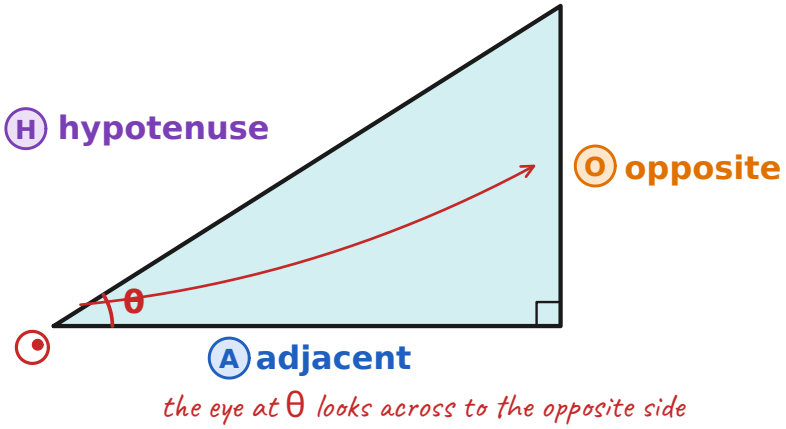
Hypotenuse (H): the longest side, opposite the right angle. It never changes.

Opposite (O): the side across from the angle θ you are using.

Adjacent (A): the side next to θ that is not the hypotenuse.

O and A depend on which angle is θ: change the angle and they swap.

A right-angled triangle has one 90° corner (the little square). Its three sides get names, and two of the names depend on **which angle you're standing at**, called θ ("theta", a Greek letter used for angles).

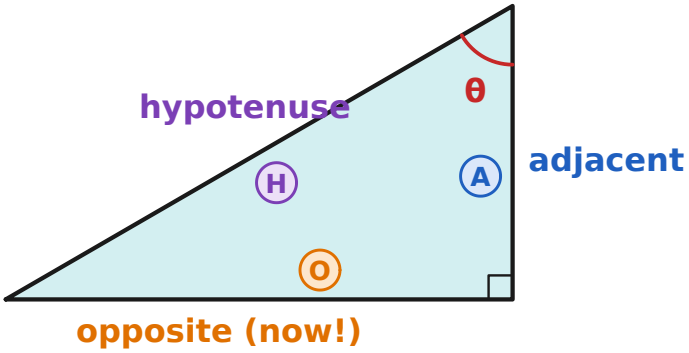


HYPOTENUSE: the longest side, opposite the right angle. It never changes name.

OPPOSITE: across from θ, the side θ is "looking at".

ADJACENT: next to θ ("adjacent" means next to), but not the hypotenuse.

Stand at the other corner and opposite and adjacent **swap**; the hypotenuse stays put:



Always label from θ's point of view, one side at a time:
1. hypotenuse (easy, longest)
2. opposite (across from θ)
3. adjacent (the one left over)

Lecture 6 writes them h, o and a.

The labelling habit. From here on, every triangle in this book's trig chapters wears three badges: **H** (hypotenuse, purple), **O** (opposite, orange) and **A** (adjacent, blue), always in the same colours. Do the same on paper: **mark θ, then badge all three sides, before anything else**, even when a question only needs two of them. After a week it becomes automatic, and choosing SOH, CAH or TOA becomes a matter of reading your own badges.

◿ Pythagoras: squares on the sides

RULE: Pythagoras' theorem

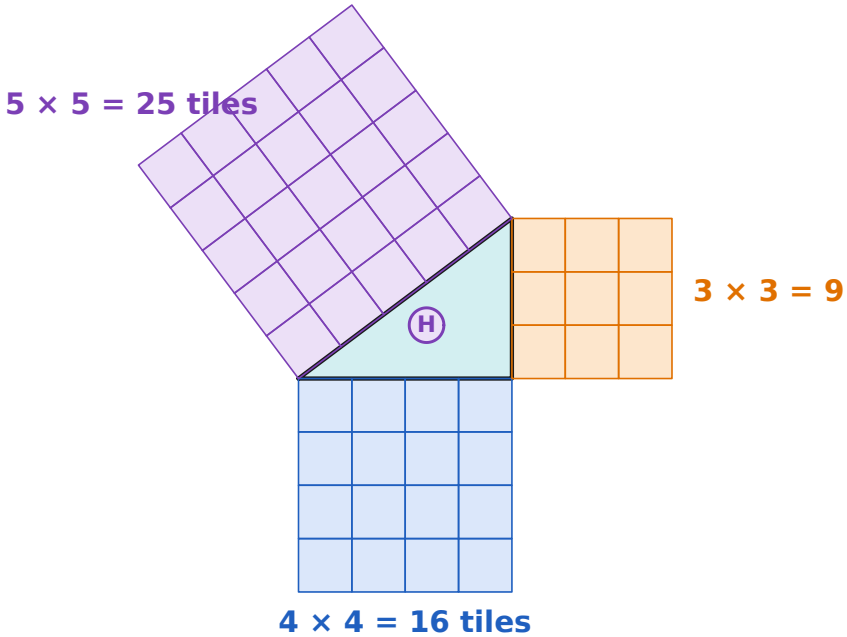
In a **right-angled** triangle with hypotenuse c and shorter sides a and b: $a^2 + b^2 = c^2$.

Finding the hypotenuse: square the two short sides, **add**, square-root: $c = \sqrt{a^2 + b^2}$.

Finding a short side: square both, **subtract** (big minus small), square-root: $a = \sqrt{c^2 - b^2}$.

Only for right-angled triangles. Other triangles need Chapter 31's cosine rule.

Build a real square on each side of a 3-4-5 triangle out of 1 × 1 tiles and count them:



9 + 16 = 25. The two small squares exactly fill the big one.

True for every right-angled triangle.

$$h^2 = o^2 + a^2$$ ← *Pythagoras: hypotenuse squared = the other two squared, added*

Finding a short side means subtracting instead. Lecture 6, Example 3: $h = 5$, $o = 2.5$:

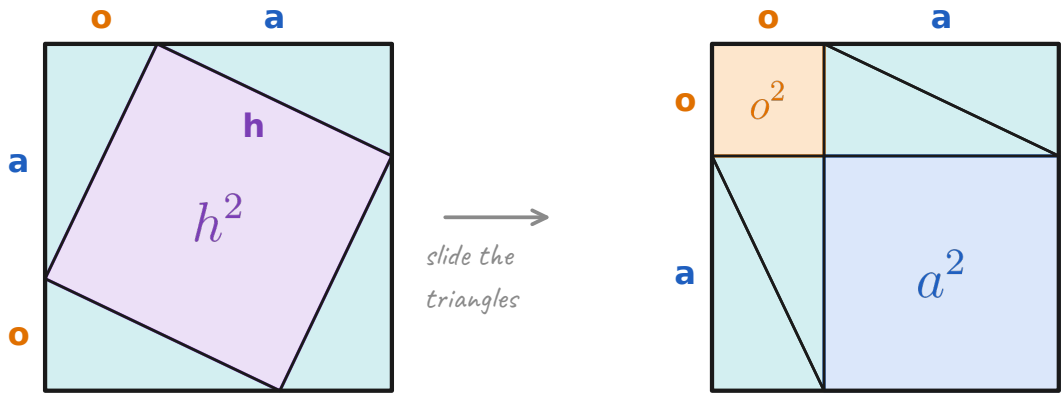
$$a^2 = 5^2 - 2.5^2 = 25 - 6.25 = 18.75$$

$$a = \sqrt{18.75} \approx 4.33$$ ← don't forget the square root!

(The slide writes "$5^2 - 2.5^2 = a = 4.33$", dropping the square root: $5^2 - 2.5^2$ is 18.75, not 4.33.)

Why it's always true: four triangles in a box

Counting tiles only checks one triangle. Here's why it works for **every** right-angled triangle. Take four copies of any one, with short sides o and a and hypotenuse h, and a square box with sides $o + a$. Pack the triangles in two different ways:



4 triangles, and the tilted square on the hypotenuse

the same 4 triangles, paired into 2 rectangles, and two squares

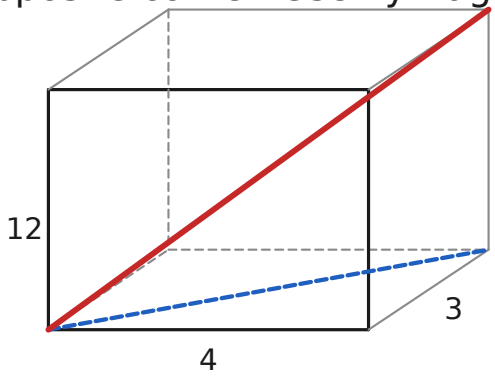
Both boxes are the same size and hold the same four triangles. So the space the triangles leave must be the same too: on the left that's h^2, on the right it's $o^2 + a^2$. Nothing here depended on the lengths, so it's true for **every** right-angled triangle:

$$h^2 = o^2 + a^2$$ ← the leftover space, counted two ways

(Chapter 11's perfect square is hiding here: the whole box is $(o + a)^2 = o^2 + 2oa + a^2$, and the four triangles are the two oa rectangles.)

◿ Pythagoras in 3D, and the circle

A box: the longest stick that fits in a 3 × 4 × 12 box goes corner to opposite corner. Use Pythagoras twice: across the floor, then up:



floor: $\sqrt{4^2 + 3^2} = 5$

stick: $\sqrt{5^2 + 12^2} = 13$

$$d^2 = a^2 + b^2 + c^2$$

A circle is every point at distance r from the centre (a, b). The distance from the centre to a point (x, y) is the hypotenuse of a triangle with sides $x - a$ and $y - b$, so Pythagoras gives Lecture 6's circle equation:

$$(x - a)^2 + (y - b)^2 = r^2$$ ← centred at the origin: $x^2 + y^2 = r^2$

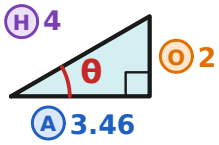
Why sin, cos and tan exist: zoomed photos

RULE: SOHCAHTOA

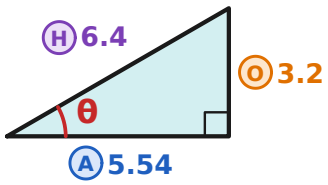
$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} \text{ (SOH) } \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} \text{ (CAH) } \tan\theta = \frac{\text{opposite}}{\text{adjacent}} \text{ (TOA)}$$

Each ratio depends **only on the angle**, not on the size of the triangle. Right-angled triangles only.

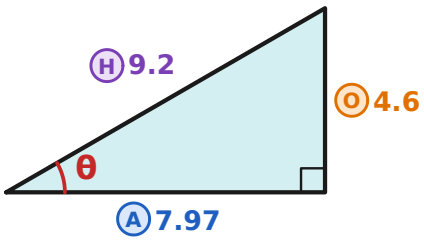
Draw a right-angled triangle with a 30° angle. Now photograph it and zoom in: the triangle gets bigger but its **shape** can't change, because the angles are the same. Measure the sides of each copy:



O ÷ H: 2 ÷ 4 = 0.5



O ÷ H: 3.2 ÷ 6.4 = 0.5



O ÷ H: 4.6 ÷ 9.2 = 0.5

Every 30° triangle has opposite ÷ hypotenuse = 0.5, however big. The ratio depends **only on the angle**. So we give each ratio a name and let the calculator remember them:

S O H

$$\sin\theta = \frac{\text{opp}}{\text{hyp}}$$

(sine)

C A H

$$\cos\theta = \frac{\text{adj}}{\text{hyp}}$$

(cosine)

T O A

$$\tan\theta = \frac{\text{opp}}{\text{adj}}$$

(tangent)

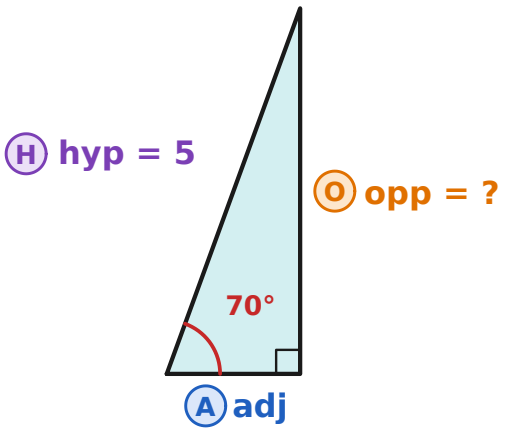
SOHCAHTOA glues the codes together. Lecture 6 admits it's a nightmare to pronounce and suggests your own sentence, e.g. *"Some Old Horses Can Always Hear Their Owners Approach"* . Degrees or radians doesn't matter here: like measuring in feet or inches.

◿ The ladder, in five tiny steps

PROCEDURE: which ratio? (sin, cos or tan)

1. Mark the angle θ you know (or want) and the right angle.
2. Label the sides H, O, A from θ's point of view.
3. Tick the side you **know** and the side you **want**.
4. Pick the ratio that contains exactly those two: O and H → **sin**; A and H → **cos**; O and A → **tan**.
5. Write the equation and solve it. Unknown on top: **multiply** ($\mathrm{O} = \mathrm{H}\sin\theta$). Unknown on the bottom: **divide** ($\mathrm{H} = \mathrm{O} \div \sin\theta$). Unknown angle: use the **inverse** ($\theta = \sin^{-1}\frac{\mathrm{O}}{\mathrm{H}}$).

A 5 m ladder makes 70° with the ground. How high up the wall does it reach?



1 Label the sides from θ = 70°: the ladder is the hypotenuse, the wall height is opposite.

2 What do I know, what do I want? Know hyp, want opp: O and H, so SOH, sine.

3 Write it: sin 70° = opp ÷ 5.

4 Transpose (Chapter 14): opp = 5 × sin 70°.

5 Calculate: 5 × 0.9397 = 4.70 m.

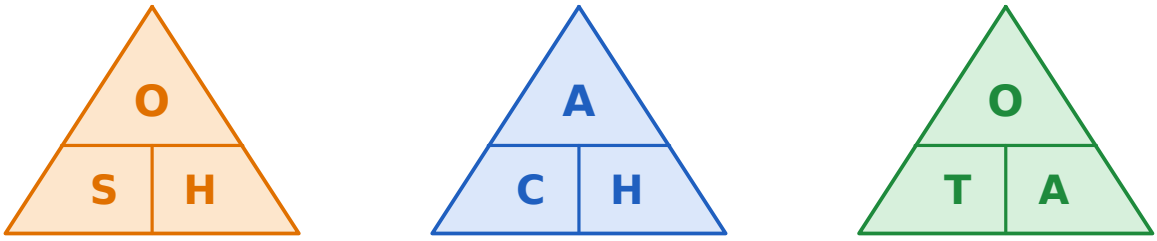
How far is its foot from the wall? Now we want the adjacent side: A and H, so CAH, cosine:

$$\cos 70^\circ = \frac{\text{adj}}{5}$$

$$\text{adj} = 5\cos 70^\circ = 5 \times 0.342 = 1.71 \text{ m}$$

Check with Pythagoras: $4.70^2 + 1.71^2 = 22.09 + 2.92 = 25.0 = 5^2$ ✓.

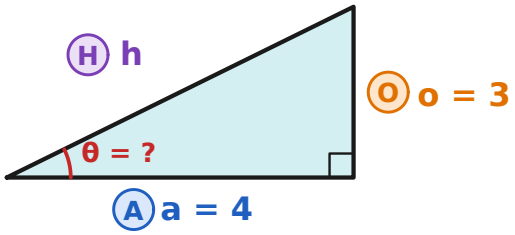
Cover-up triangles save the transposing. Put the top letter over the other two; cover what you want:



Cover O in the first: O = S × H. Cover H: H = O ÷ S. (Same idea as the speed-distance-time triangle.)

Lecture 6's examples, and the undo button

Example 1: opposite = 3, adjacent = 4. Find θ.

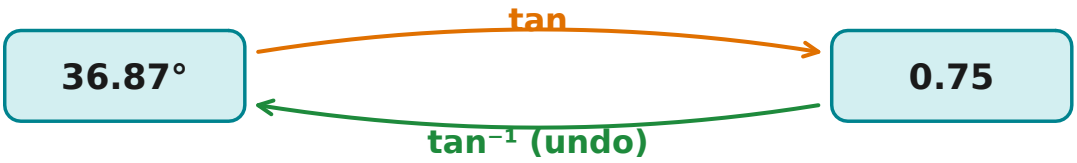


badges: O and A are known, so TOA (tan)

$$\tan\theta = \frac{3}{4} = 0.75$$

Now θ is **stuck inside** the tan. We need an **undo button** for tan, just as in Chapter 14 (÷ undoes ×) and Chapter 22 (log undoes power). It's called **inverse tan**, written $\tan^{-1}$ (or arctan):

$$\theta = \tan^{-1}(0.75) = 36.87^\circ$$ ← *SHIFT then tan on the calculator*



Warning: the -1 here means "undo", **not** "1 over" (Chapter 9's negative powers). $\sin^{-1} x$ is not $\frac{1}{\sin x}$. Sorry, mathematicians reused the symbol.

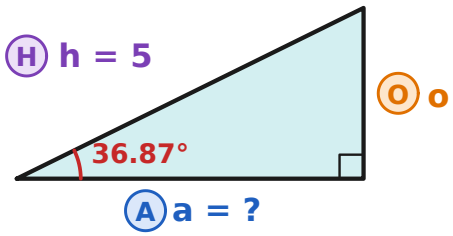
RULE: inverse trig gives ONE angle (the principal value)

$\sin^{-1}$ always answers between $-90°$ and $90°$. $\cos^{-1}$ between $0°$ and $180°$. $\tan^{-1}$ between $-90°$ and $90°$.

In a right-angled triangle that is the angle you want: it is always between $0°$ and $90°$.

But other angles share the same ratio: $\sin 150° = \sin 30° = 0.5$, yet $\sin^{-1}(0.5)$ only says $30°$. The other answers come from the quadrant rules (Chapter 29) and equation solving (Chapter 30).

Example 2: $h = 5$, $\theta = 36.87°$. Find a.

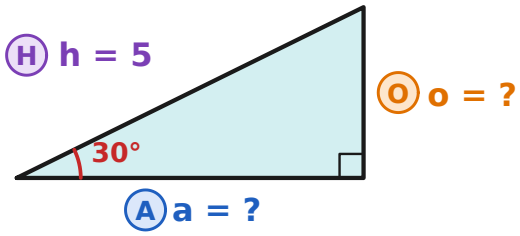


badges: know H, want A, so CAH (cos)

$$a = 5 \times \cos 36.87° = 4$$

(Lecture 6 asks "what do we think about the notation here?": it writes 36.870° with a superscript o. Use a proper ° sign.)

Example 3: $h = 5$, $\theta = 30°$. Find all the sides.



badges: know H, want O first, so SOH (sin); then Pythagoras for A

$$o = 5 \sin 30° = 2.5, \qquad a = \sqrt{5^2 - 2.5^2} = 4.33$$

Elevation, depression, and two linked triangles

RULE: angles of elevation and depression

Angle of elevation: measured **up** from the horizontal to the line of sight (looking up).

Angle of depression: measured **down** from the horizontal to the line of sight (looking down).

The angle of depression from A down to B equals the angle of elevation from B up to A (the two horizontal lines are parallel).

Both are measured from the **horizontal**, never from the vertical.

PROCEDURE: multi-step problems (two linked triangles)

1. Draw the picture and split it into right-angled triangles. Mark every known length and angle.
2. Find the side the two triangles **share**, using the triangle you know enough about.
3. Use that shared side in the second triangle to get the answer.
4. Keep full calculator accuracy in the middle; round only at the end.

Example 1 (elevation): from 40 m away, the angle of elevation of the top of a building is 35°. How tall is the building?

$$\tan 35^\circ = \frac{h}{40} \Rightarrow h = 40 \tan 35^\circ = 28.01 \text{ m}$$

know A, want O: TOA

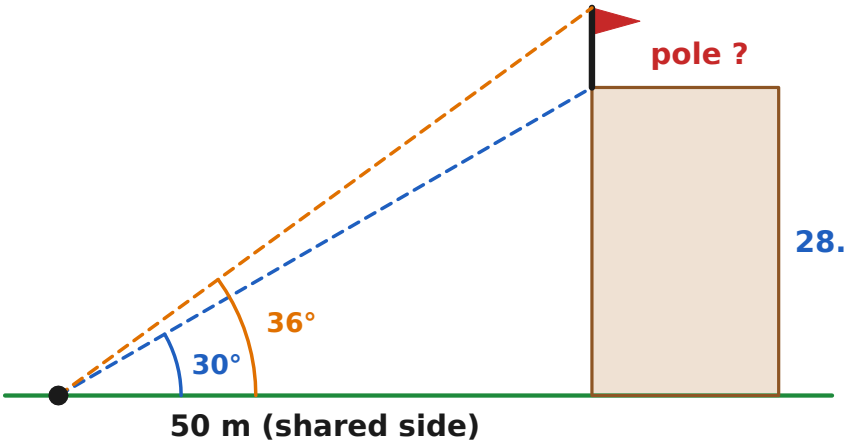
Example 2 (depression): from the top of a 60 m cliff, the angle of depression of a boat is 25°. How far is the boat from the foot of the cliff?

$$\tan 25^\circ = \frac{60}{d} \Rightarrow d = \frac{60}{\tan 25^\circ} = 128.67 \text{ m}$$

← d on the bottom: divide

(The 25° at the cliff top is also the angle of elevation from the boat, so it sits at the boat's corner of the triangle.)

Example 3 (two triangles): a flagpole stands on top of a building. From a point 50 m from the base, the angle of elevation of the top of the building is 30°, and of the top of the flagpole 36°. How tall is the flagpole?



Two right triangles share the 50 m side. Solve each with TOA, then subtract.

$$\text{building} = 50\tan 30^\circ = 28.87 \text{ m}$$ ← *smaller triangle*

$$\text{building} + \text{pole} = 50\tan 36^\circ = 36.33 \text{ m}$$ ← *bigger triangle, same 50 m*

$$\text{pole} = 36.33 - 28.87 = 7.46 \text{ m}$$ ← *subtract*

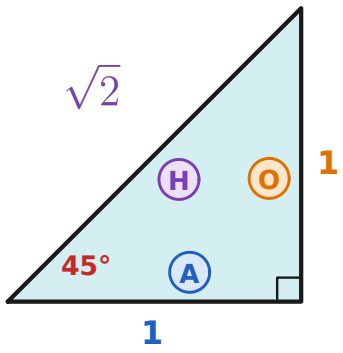
◿ Two special triangles give exact values

RULE: exact values

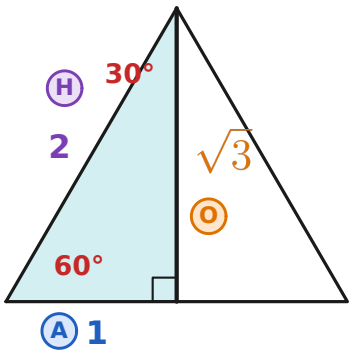
The $45°$ triangle has sides $1, 1, \sqrt{2}$. The $30°$–$60°$ triangle has sides $1, \sqrt{3}, 2$. Read sin, cos and tan off them with SOHCAHTOA (table below).

Useful patterns: $\sin 30° = \cos 60° = \frac{1}{2}$, $\sin 60° = \cos 30° = \frac{\sqrt{3}}{2}$, $\tan 45° = 1$.

A few angles have sin, cos and tan you can write exactly, with no calculator. They come from two triangles:



half a square



badges from the 60° corner.
From the 30° corner, O and A swap:

opp = 1, adj = √3

half an equilateral triangle (all sides 2)

Pythagoras gives the missing sides ($\sqrt{1^2 + 1^2} = \sqrt{2}$ and $\sqrt{2^2 - 1^2} = \sqrt{3}$). Now read off SOHCAHTOA:

	30°	45°	60°
sin	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
tan	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$

Lecture 6 uses the 45° one: $(\cos 45°, \sin 45°) = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Side names	H opposite the right angle (longest); O across from θ; A next to θ
Pythagoras	$a^2 + b^2 = c^2$ c = hypotenuse. Hypotenuse: add then $\sqrt{}$. Short side: subtract then $\sqrt{}$
Pythagoras in 3D	$d^2 = a^2 + b^2 + c^2$ diagonal of a box
Circle	$(x-a)^2 + (y-b)^2 = r^2$ centre (a, b), radius r
SOHCAHTOA	$\sin\theta = \frac{O}{H}, \quad \cos\theta = \frac{A}{H}, \quad \tan\theta = \frac{O}{A}$ right-angled triangles only
Which ratio?	the one containing the side you know and the side you want
Rearranging	unknown on top: multiply; unknown on the bottom: divide; unknown angle: inverse
Inverse trig	$\theta = \sin^{-1}\frac{O}{H}, \ \cos^{-1}\frac{A}{H}, \ \tan^{-1}\frac{O}{A}$ one answer only: $\sin^{-1}, \tan^{-1}$ in $-90°$ to $90°$; $\cos^{-1}$ in $0°$ to $180°$
Elevation / depression	measured up / down from the **horizontal**; they are equal (parallel lines)
Two triangles	solve for the shared side first, then use it in the other triangle
Exact values	$\sin 30° = \frac{1}{2}, \ \sin 45° = \frac{1}{\sqrt{2}}, \ \sin 60° = \frac{\sqrt{3}}{2}$ cos: the same list backwards; $\tan 45° = 1$

◿ Your turn

Draw and label the triangle first (Lecture 6's advice). Answers in grey.

Pythagoras

1. Short sides 5 and 12: hypotenuse?
2. Hypotenuse 10, one side 6: the other side?
3. Diagonal of a 2 × 3 × 6 box

1. 13 2. 8 3. $\sqrt{(4 + 9 + 36)} = 7$

Sides

4. hyp 8, $\theta = 40°$: opposite?
5. adj 6, $\theta = 25°$: opposite?
6. opp 4, $\theta = 55°$: hypotenuse?

4. $8 \sin 40° \approx 5.14$ 5. $6 \tan 25° \approx 2.80$ 6. $4 \div \sin 55° \approx 4.88$

Angles

7. opp 5, hyp 13: θ?
8. adj 7, hyp 10: θ?
9. A ramp rises 1 m over 12 m of ground. Its angle?

7. $\sin^{-1}(5/13) \approx 22.6°$ 8. $\cos^{-1}(0.7) \approx 45.6°$ 9. $\tan^{-1}(1/12) \approx 4.8°$

Find the unknown

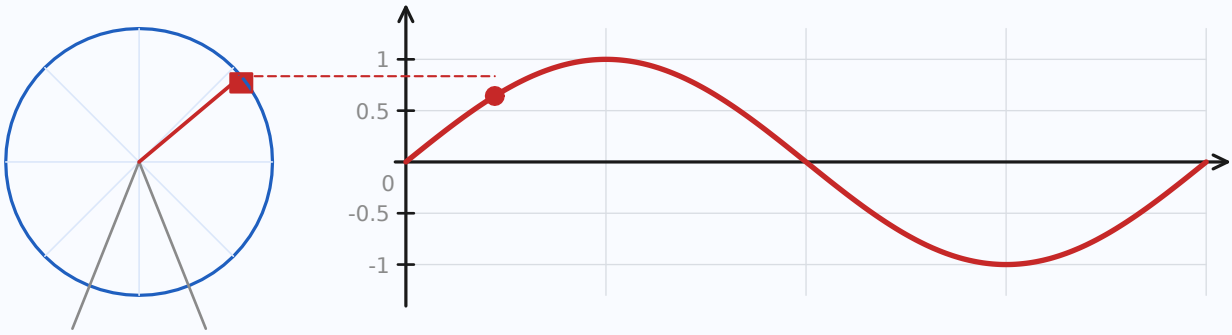
10. A 6 m ladder reaches 5 m up a wall. What angle does it make with the ground?
11. From 30 m away, the angle of elevation of a tree top is 28°. How tall is the tree?
12. From the top of a 45 m cliff, the angle of depression of a boat is 18°. How far is the boat from the foot of the cliff?
13. A ramp must rise 0.8 m at an angle of 5°. How long is the sloping surface?
14. $\tan \theta = 0.6$ and the adjacent side is 15. Find the opposite side and the hypotenuse.
15. From 40 m away, the elevation of the top of a building is 25° and of the top of a mast on its roof 31°. How tall is the mast?

10. $\sin \theta = 5/6$, $\theta \approx 56.4°$ 11. $30 \tan 28° \approx 15.95$ m 12. $45 \div \tan 18° \approx 138.50$ m 13. $\sin 5° = 0.8/H$, $H = 0.8 \div \sin 5° \approx 9.18$ m 14. $O = 0.6 \times 15 = 9$; $H = \sqrt{(81 + 225)} \approx 17.49$ 15. $40 \tan 31° - 40 \tan 25° \approx 24.03 - 18.65 = 5.38$ m

Chapter 29

Trig functions and their graphs

ride the Ferris wheel



your seat's height, turn by turn, draws a wave

In Chapter 28, sin, cos and tan lived inside right-angled triangles, so the angle had to be less than 90°. But angles can be anything: 200°, −45°, ten full turns. This chapter frees sin, cos and tan from the triangle by putting them on a **circle**, and then watches them move.

The star of the show is a Ferris wheel. Ride it, record your height, and you've drawn the sine graph. Everything about the graphs (the repeats, the heights, the shifts) is something about the wheel.

Goes with **Lecture 6** (the unit circle, sin, cos and tan graphs, period, amplitude, power series, odd and even functions).

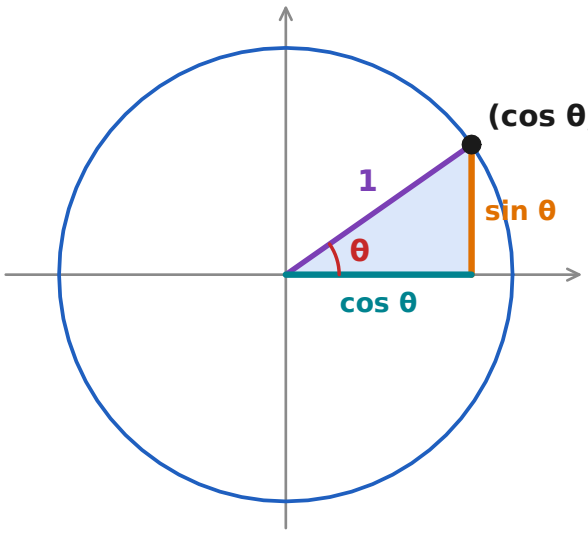
∿ The unit circle: sin and cos become coordinates

RULE: trig of any angle

On the unit circle (radius 1), the point at angle θ is $(\cos\theta,\ \sin\theta)$: **cos is the x-coordinate, sin is the y-coordinate**, and $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$.

This defines sin, cos and tan for **every** angle, not just those under $90°$.

Draw a circle of radius 1 (the **unit circle**, $x^2 + y^2 = 1$ from Chapter 28). Walk anticlockwise from the east by an angle θ and drop a line down to the x-axis. You've made a right-angled triangle with hypotenuse 1:



SOH with hyp = 1: opp = sin θ.
CAH with hyp = 1: adj = cos θ.

So the point on the circle is (cos θ, sin θ) (Lecture 6).

Now let θ keep growing: the point keeps going round, and its x and y define cos and sin for ANY angle.

Signs come free: in the second quarter (90° to 180°) the point is left of the y-axis, so cos is **negative** while sin is still positive. E.g. $\cos 120° = -0.5$, $\sin 120° = 0.866$. And since the point is on $x^2 + y^2 = 1$:

$$\cos^2\theta + \sin^2\theta = 1$$ ⟵ *Pythagoras on the circle (Chapter 30 uses it)*

∿ Any angle: quadrant signs and the reference angle

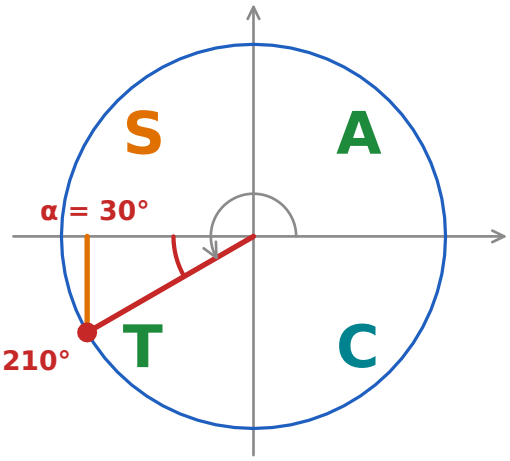
RULE: the quadrant sign rule (CAST)

The four quadrants are numbered anticlockwise from east: Q1 is $0°$–$90°$, Q2 $90°$–$180°$, Q3 $180°$–$270°$, Q4 $270°$–$360°$.

Which ratios are **positive**: Q1 **All**; Q2 only **Sin**; Q3 only **Tan**; Q4 only **Cos**. Memory aid, going round from Q1: "*All Students Take Calculus* ". The rest are negative.

The **reference angle** α is the acute angle between the radius and the **x-axis**. sin, cos and tan of θ are those of α, with the sign from CAST.

Quadrant	θ is between	Positive	Ref. angle α	θ from α
Q1	0° and 90°	all	$\alpha = \theta$	$\theta = \alpha$
Q2	90° and 180°	sin	$\alpha = 180° - \theta$	$180° - \alpha$ $(\pi - \alpha)$
Q3	180° and 270°	tan	$\alpha = \theta - 180°$	$180° + \alpha$ $(\pi + \alpha)$
Q4	270° and 360°	cos	$\alpha = 360° - \theta$	$360° - \alpha$ $(2\pi - \alpha)$



S, A, T, C: the ratios that are positive in each quadrant.

210° is in Q3: the point is below the axis (sin < 0) and left of it (cos < 0). Its reference angle is 210° − 180° = 30°, so sin 210° = −sin 30° = −0.5.

PROCEDURE: sin, cos or tan of any angle

1. Reduce the angle to between $0°$ and $360°$ (add or subtract $360°$).
2. Find its quadrant.
3. Find the reference angle α (table above).
4. Work out sin, cos or tan of α (exact value or calculator).
5. Attach the sign from CAST: $+$ if the ratio is positive in that quadrant, $-$ if not.

$$\sin 210^\circ = -\sin 30^\circ = -\frac{1}{2}$$ ← Q3, α = 30°; sin is −

$$\cos 300^\circ = +\cos 60^\circ = \frac{1}{2}$$ ← Q4, α = 60°; cos is +

$$\tan 135^\circ = -\tan 45^\circ = -1$$ ← Q2, α = 45°; tan is −

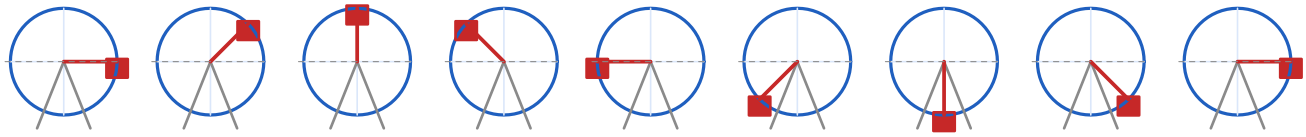
$$\cos(-120^\circ) = \cos 240^\circ = -\cos 60^\circ = -\frac{1}{2}$$

+360°, then Q3, α = 60°

Check on a calculator in DEG mode: $\sin 210^\circ = -0.5$, $\cos 300^\circ = 0.5$, $\tan 135^\circ = -1$. The method matters for exact answers and, in Chapter 30, for running the rule backwards to find every angle with a given sine.

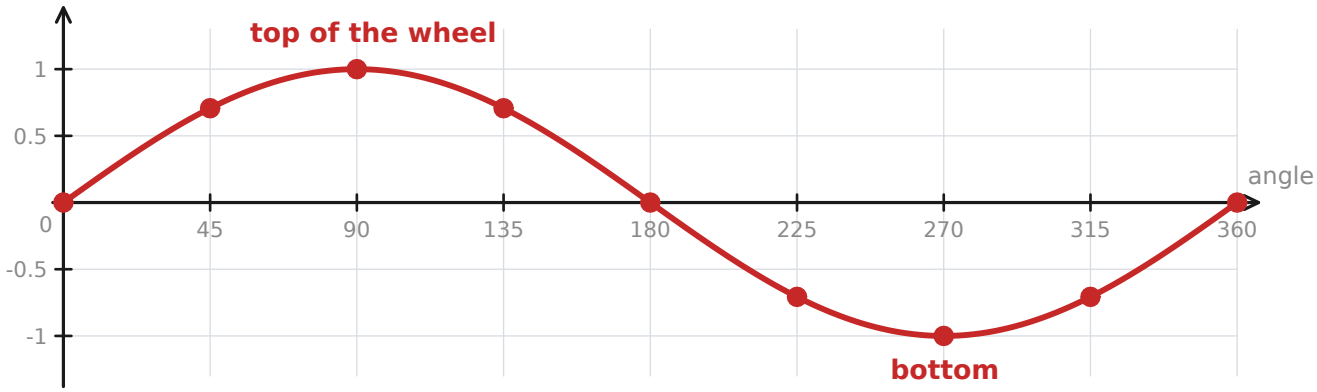
∿ The comic strip: unrolling the wheel

A wheel of radius 1. Start level with the hub (on the right) and turn anticlockwise. In each frame, note how high the seat is **above the hub**:



0°	45°	90°	135°	180°	225°	270°	315°	360°
0	+0.71	+1.00	+0.71	0	-0.71	-1.00	-0.71	0

Plot each height above its angle, then join the dots smoothly:



That wave is the **graph of** $y = \sin\theta$. Keep riding and it repeats every turn, forever (and backwards for negative angles).

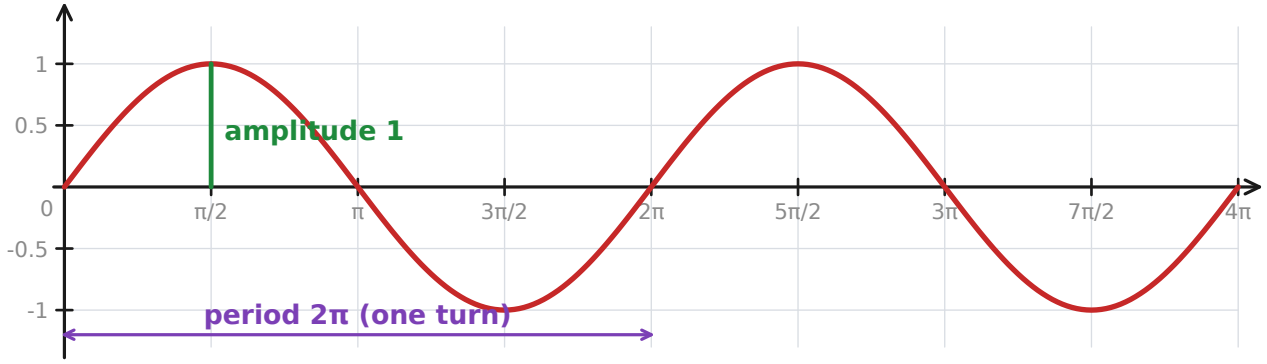
∿ Two numbers describe a wave: period and amplitude

RULE: the sine and cosine graphs

$y = \sin x$ and $y = \cos x$ both have **period** $360°$ (2π) and **amplitude** 1: they stay between -1 and 1.

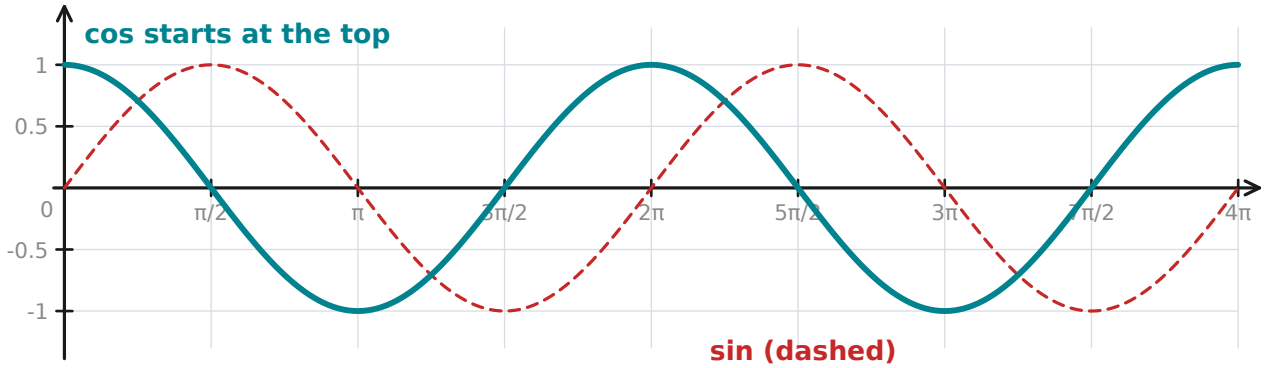
sin starts at 0 going up; cos starts at its peak 1. cos is sin shifted $90°$ left: $\cos x = \sin(x + 90°)$.

cos is even: $\cos(-x) = \cos x$. sin is odd: $\sin(-x) = -\sin x$.



The **period** is how long before it repeats: one full turn, 2π (360°). The **amplitude** is how high it rises above the middle: the wheel's radius, 1. (Lecture 6: sin has period 2π and amplitude 1.)

Cosine is the same ride seen from the side. The seat's **sideways** position (its x-coordinate) starts at 1 (far right), so cos is the sine wave **shifted a quarter turn left**:



Same period (2π) and amplitude (1). Even and odd (Chapter 19): cos is **even**, a mirror image either side of the y-axis ($\cos(-x) = \cos x$); sin is **odd**, it spins onto itself ($\sin(-x) = -\sin x$).

∿ Tan: the wave that explodes

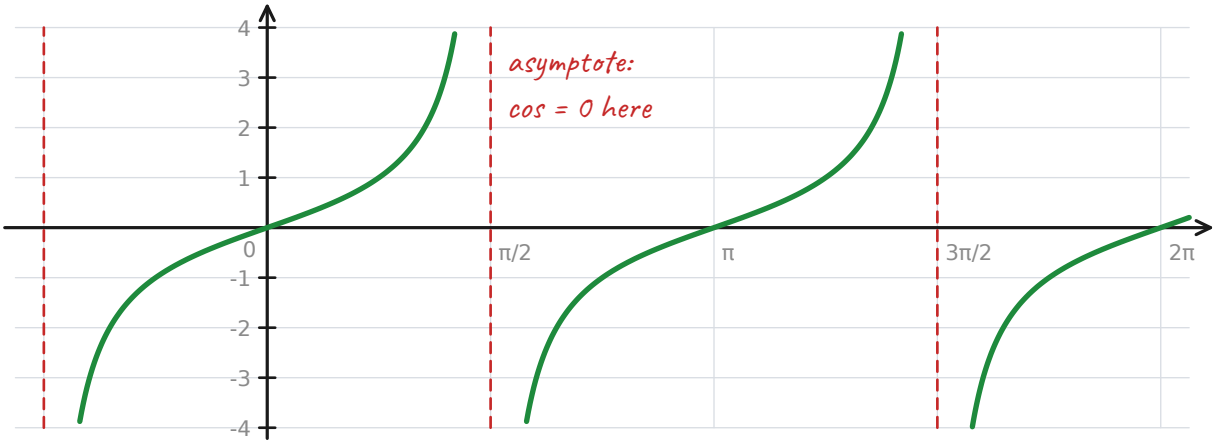
RULE: the tangent graph

$\tan x = \dfrac{\sin x}{\cos x}$. It has **period** $180°$ (π) and no amplitude (no top or bottom).

Vertical **asymptotes** wherever $\cos x = 0$: $x = 90°, 270°, \ldots$ ($\frac{\pi}{2} + k\pi$).

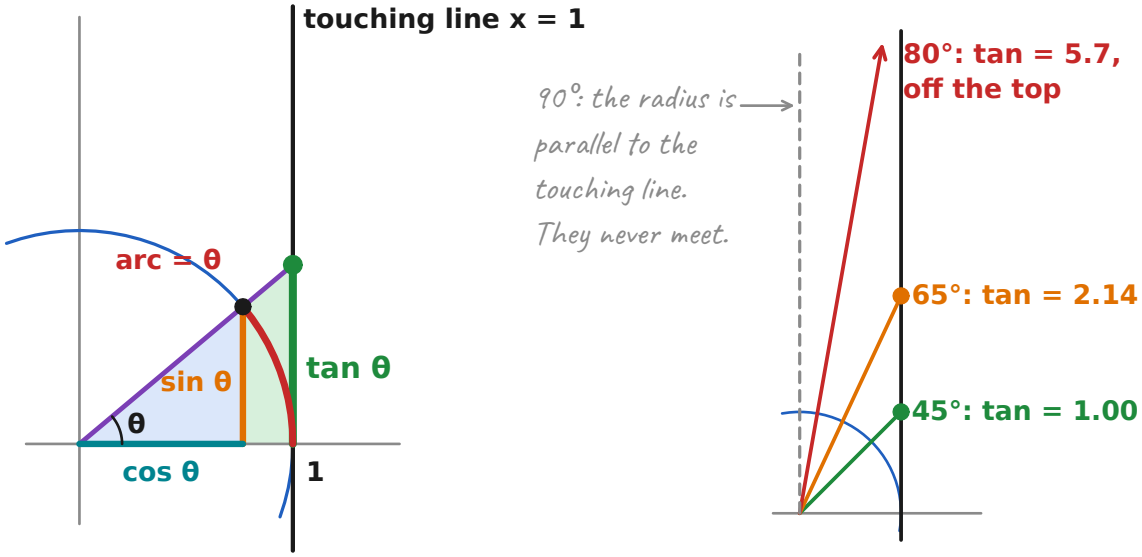
$y = \tan(Bx)$ has period $\dfrac{180°}{B}$, i.e. $\dfrac{\pi}{B}$.

$\tan\theta = \frac{\sin\theta}{\cos\theta}$ (opp/adj = (opp/hyp) ÷ (adj/hyp)). Whenever cos hits 0, at 90° and 270°, we'd be dividing by zero: the graph shoots off to infinity at **vertical asymptotes** (Chapter 19):



Tan repeats every π (180°), not 2π: Lecture 6 says period π and, jokingly, "amplitude of ∞" (it has no top). Tan is odd, like sin.

Why is it called tan? A picture. Draw the line that just **touches** the circle at $(1, 0)$: the upright line $x = 1$. A line that touches a circle is a **tangent** (Latin *tangere*, to touch). Now extend the radius until it hits that line:



The two shaded triangles are the same shape (same angle θ), one scaled up so its bottom is 1. So $\frac{\tan\theta}{1} = \frac{\sin\theta}{\cos\theta}$: **tan θ is the height where the radius line meets the touching line**, which is also the **slope** of the radius (rise ÷ run, Chapter 20). As the angle nears 90° the meeting point races up the line, and at 90° there is no meeting point: that's the asymptote. Past 90°, extend the radius backwards through the centre: it hits the touching line again, which is why tan repeats every half turn.

Notice three heights for the same angle: **sin θ** (shortest), the red **arc θ** (middle) and **tan θ** (tallest). Keep that order in mind; the recipes below will show it.

sin

angle in (the distance walked round the unit circle) → the point's HEIGHT out

cos

angle in → the point's SIDEWAYS position out

tan

angle in → where the radius line hits the touching line x = 1, i.e. the radius's SLOPE (sin ÷ cos)

∿ Bigger wheels, faster wheels

RULE: amplitude, period and shifts

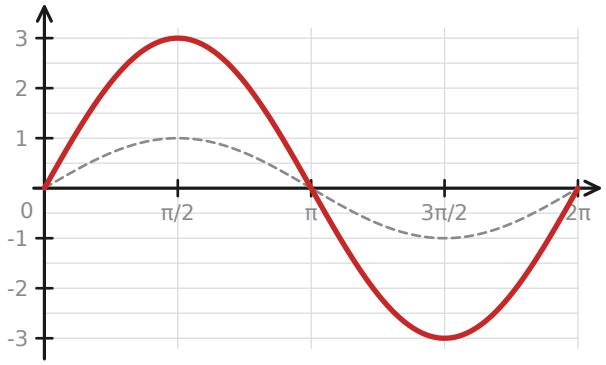
For $y = A\sin\big(B(x - C)\big) + D$ (the same for cos):

Amplitude $= |A|$. **Period** $= \dfrac{360^\circ}{B} = \dfrac{2\pi}{B}$. **Shift** C to the **right** (left if C is negative). **Midline** $y = D$.

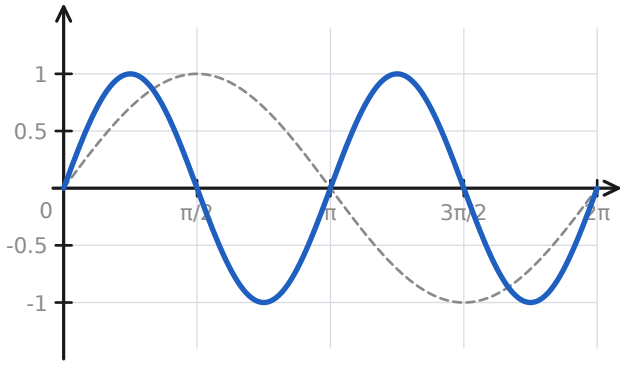
Maximum $= D + |A|$, minimum $= D - |A|$. A negative A flips the wave upside down.

For $y = \tan(Bx)$ the period is $\dfrac{180^\circ}{B} = \dfrac{\pi}{B}$ (tan has no amplitude).

Change the wheel, change the wave. For $y = A\sin(Bx)$:



y = 3 sin x: radius 3, amplitude 3



y = sin 2x: twice as fast, period π

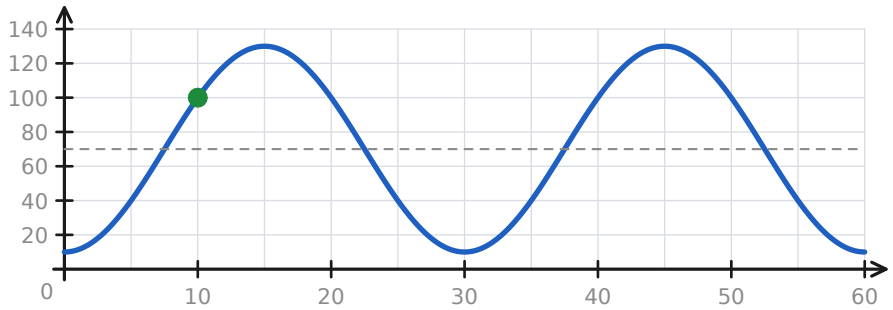
$$\text{amplitude} = A, \qquad \text{period} = \frac{2\pi}{B}$$

Add a number D on the end and the whole wave lifts up by D: the hub is higher off the ground.

The London Eye: radius 60 m, hub 70 m up, one turn every 30 minutes. You board at the **bottom**, so your height starts low: a flipped cosine.

$$h(t) = 70 - 60\cos\left(\frac{2\pi t}{30}\right)$$ ← *t in minutes*

$$h(10) = 70 - 60\cos\left(\tfrac{2\pi}{3}\right) = 70 - 60(-0.5) = 100 \text{ m}$$



minutes along the bottom, metres up the side. Highest (130 m) at 15 min.

Reading a formula: $y = 2\sin\big(3(x - 20°)\big) + 1$.

$$\text{amplitude } 2, \quad \text{period } \frac{360°}{3} = 120° \leftarrow A = 2,\ B = 3$$

$$\text{shift } 20° \text{ right}, \quad \text{midline } y = 1 \leftarrow C = 20°,\ D = 1$$

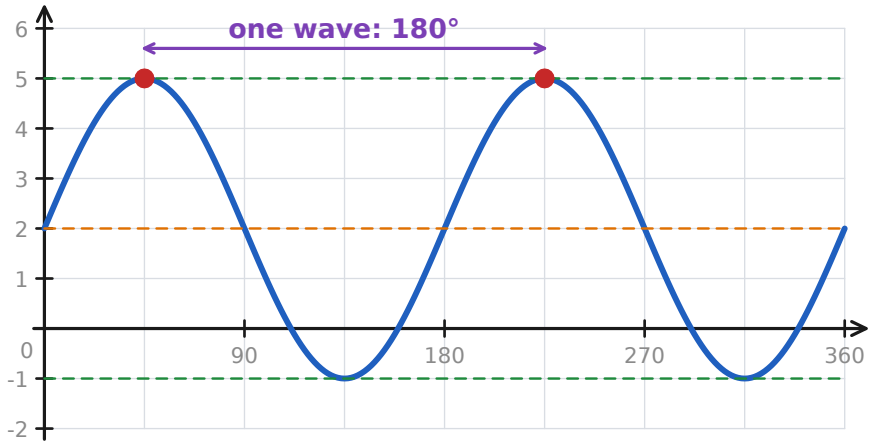
$$\max = 1 + 2 = 3, \quad \min = 1 - 2 = -1 \leftarrow D \pm |A|$$

And $y = \tan 3x$ repeats every $\frac{180°}{3} = 60°$ ($\frac{\pi}{3}$), with asymptotes $60°$ apart.

∿ Reading A, B and D off a graph

PROCEDURE: find A, B and D from a graph of y = A sin(Bx) + D

1. Read the **maximum** and **minimum** heights.
2. Midline $D = \dfrac{\max + \min}{2}$ (halfway). Amplitude $A = \dfrac{\max - \min}{2}$ (half the height).
3. Read the **period**: the distance from one peak to the next (or one full wave).
4. $B = \dfrac{360°}{\text{period}}$ (or $\dfrac{2\pi}{\text{period}}$ in radians).
5. Check one point: at $x = 0$, $y = A\sin 0 + D = D$.



max 5, min −1 (green lines)

orange line: halfway

peaks at 45° and 225°

$$D = \frac{5 + (-1)}{2} = 2, \qquad A = \frac{5 - (-1)}{2} = 3$$

midline, then half the height

$$\text{period} = 225° - 45° = 180° \;\Rightarrow\; B = \frac{360°}{180°} = 2$$

$$y = 3\sin 2x + 2$$ ← check x = 0: y = 2, on the midline ✓

✗ Common mistakes

✗ period of $\sin 2x = 720°$ ✓ period $= \frac{360°}{2} = 180°$

a bigger B squashes the wave: divide by B, don't multiply

✗ $A = \max = 5$ ✓ $A = \frac{5-(-1)}{2} = 3$

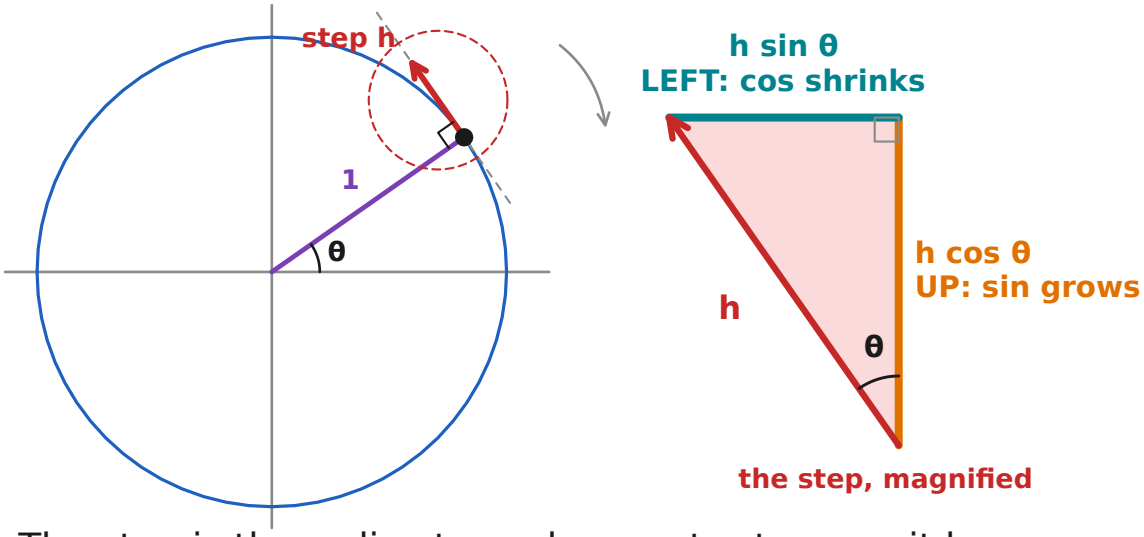
amplitude is measured from the midline, not from zero

∿ The engine inside sin and cos: one tiny step

A calculator can only add, subtract, multiply and divide. So how does it find sin 1? With a recipe made of powers (the **power series**, coming up). The recipe isn't magic: it grows out of one picture of the point walking round the unit circle. Two facts set it up.

1. In radians, the angle IS the distance walked. A radian is one radius of string laid on the rim (Chapter 27). On a circle of radius 1, turning by θ radians moves the point exactly θ along the rim. So a tiny extra turn h moves it a distance h. (In degrees this fails, which is why the recipes need radians.)

2. The point always moves at right angles to the radius, along the line that touches the circle. Zoom in on one tiny step:



The step is the radius turned a quarter turn, so it leans away from upright by the same θ that the radius leans away from flat. Read the little triangle with SOHCAHTOA (hypotenuse h): the point rises by $h\cos\theta$ and slides left by $h\sin\theta$.

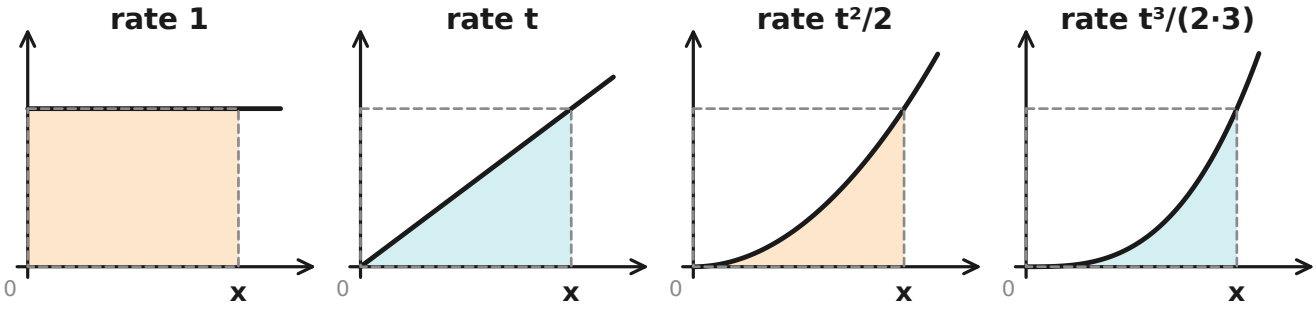
$$\sin(\theta + h) \approx \sin\theta + h\cos\theta \quad \longleftarrow \text{sin grows at the rate cos says}$$

$$\cos(\theta + h) \approx \cos\theta - h\sin\theta \quad \longleftarrow \text{cos shrinks at the rate sin says}$$

That's the whole engine: sin grows at cos's rate, cos shrinks at sin's rate. Each one drives the other. Everything else (the waves, the recipes, even Chapter 35's derivatives) comes out of this little triangle.

∿ Building the recipe: leapfrog

Start at angle 0: the point is at $(1, 0)$, so $\sin 0 = 0$ and $\cos 0 = 1$. Now let the angle grow to x and use the engine again and again, each round correcting the last. One tool: when something grows at a known **rate**, its total growth is the **area under the rate graph** (like a speed–time graph, where area = distance). The areas we'll need are all a fraction of a box:



box ÷ 1
$\text{area} = x$

box ÷ 2
$\text{area} = \frac{x^2}{2}$

box ÷ 3
$\text{area} = \frac{x^3}{2 \cdot 3}$

box ÷ 4
$\text{area} = \frac{x^4}{2 \cdot 3 \cdot 4}$

The triangle is obviously half its box. The next two (a third, a quarter) are Chapter 36's pyramid facts; for now, trust the pattern: **each new area is the box ÷ the next whole number**.

Now leapfrog. sin feeds cos, cos feeds sin, and each round uses the latest guess:

cos | **sin**

1

sin grows by the area under cos ≈ 1

x

cos shrinks by the area under sin ≈ x

$1 - \frac{x^2}{2}$

sin grows by the area under 1 − x²/2

$x - \frac{x^3}{2 \cdot 3}$

cos shrinks by the area under x − x³/6

$1 - \frac{x^2}{2} + \frac{x^4}{2 \cdot 3 \cdot 4}$

Why the signs flip: cos **shrinks** by its area, so every correction to cos is subtracted, and that minus is carried into sin on the next round. Why only odd powers in sin and even ones in cos: each round raises the power by one and swaps sides.

The bottoms are a pattern too: 1, 2, 2×3, 2×3×4, ... Each round divides by the next whole number. That product has a name: $n!$ ("n factorial") means multiply the whole numbers from n down to 1, so $3! = 3 \times 2 \times 1 = 6$ and $5! = 120$. Carry on leapfrogging forever and you get:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$ ← x in radians!

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

And there's the order from the tan picture: $\sin x = x -$ (something), so sin is a little **less** than the arc x.

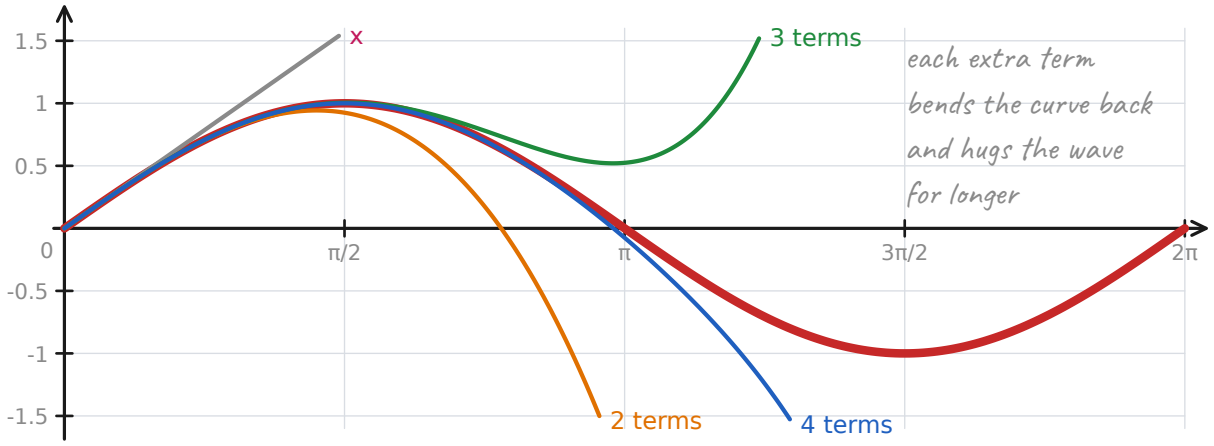
∿ How does a calculator know sin 1?

It follows the recipe. Put $x = 1$ and watch the running total home in on the true value:

add 1	1.000000
add − 1/6	0.833333
add + 1/120	0.841667
add − 1/5040	0.841468

Four terms already agree to 5 decimal places: Lecture 6's $\sin(1) \approx 0.841$. Only **odd** powers appear in sin (an odd function) and **even** ones in cos (an even function): that's where the names odd and even come from.

Watch each term earn its place. Here's the real sine wave (thick red) against the recipe cut off after 1, 2, 3 and 4 terms:



x alone rises forever. The $-\frac{x^3}{3!}$ term pulls it back down; $+\frac{x^5}{5!}$ stops it falling too far; and so on. Add enough terms and the recipe matches the wave as far along as you like: the factorials on the bottom grow so fast that they always win in the end.

∿ Tan: one recipe divided by another

$\tan x = \frac{\sin x}{\cos x}$, so divide the recipes. Dividing by $1 - (\text{something small})$ is Chapter 17's sum to infinity read backwards: $\frac{1}{1-r} = 1 + r + r^2 + \dots$

$$\tan x = \frac{x - \frac{x^3}{6} + \dots}{1 - \frac{x^2}{2} + \dots}$$ ← sin recipe ÷ cos recipe

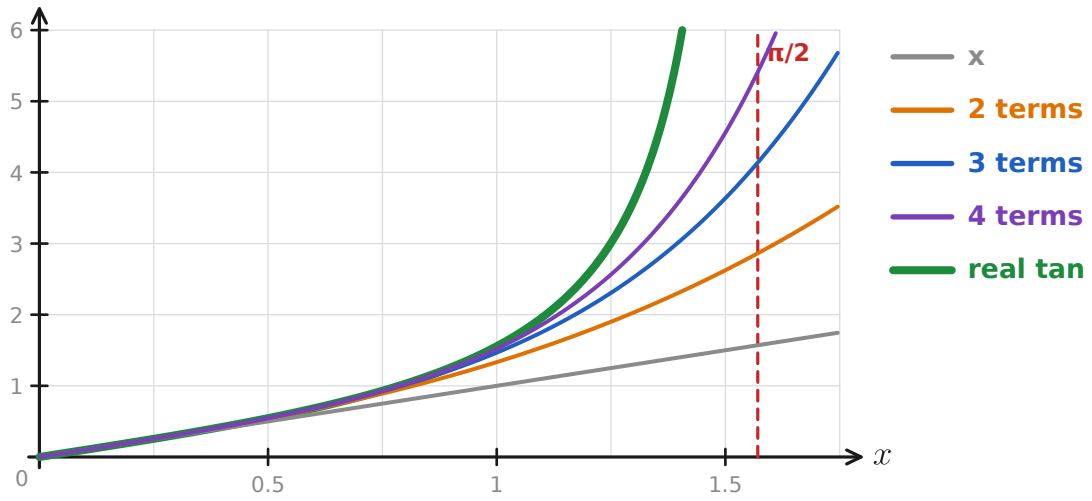
$$\frac{1}{1 - \frac{x^2}{2}} \approx 1 + \frac{x^2}{2}$$ ← Chapter 17 with r = x²/2

$$\tan x \approx \left(x - \frac{x^3}{6}\right)\left(1 + \frac{x^2}{2}\right) = x + \frac{x^3}{2} - \frac{x^3}{6} + \dots$$

expand; keep powers up to x^3

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \dots$$ ← the rest: by computer

Compare the starts: $\sin x = x - \frac{x^3}{6}$ and $\tan x = x + \frac{x^3}{3}$. sin dips just **below** the arc x, tan climbs **above** it: exactly the order of the three heights in the touching-line picture.



Every recipe is a smooth polynomial, and a polynomial can never shoot off to infinity at a fixed x. So the tan recipe can only work **between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$**. That's Chapter 17's warning in disguise: $\frac{1}{1-r}$ needs r below 1, and here $r = 1 - \cos x$ reaches 1 exactly when $\cos x = 0$, at the asymptote. The division breaks exactly where the picture's radius stops meeting the touching line.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Unit circle	$(\cos\theta,\ \sin\theta)$, $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ the point at angle θ
Pythagorean identity	$\cos^2\theta + \sin^2\theta = 1$
CAST	positive: Q1 **All**, Q2 **Sin**, Q3 **Tan**, Q4 **Cos**
Reference angle	Q2: $180° - \theta$; Q3: $\theta - 180°$; Q4: $360° - \theta$
Any angle	reduce to 0–$360°$, find the quadrant, use α, attach the CAST sign
sin and cos graphs	period $360°$ (2π), amplitude 1; cos is even, sin is odd
tan graph	period $180°$ (π); asymptotes where $\cos x = 0$
General wave	$y = A\sin(B(x - C)) + D$ amplitude $\lvert A\rvert$, period $\frac{360°}{B}$, shift C right, midline D; max/min $D \pm \lvert A\rvert$
tan(Bx)	$\text{period} = \dfrac{180°}{B} = \dfrac{\pi}{B}$
Read off a graph	$D = \dfrac{\max + \min}{2}$, $A = \dfrac{\max - \min}{2}$, $B = \dfrac{360°}{\text{period}}$
Power series	$\sin x = x - \dfrac{x^3}{3!} + \dfrac{x^5}{5!} - \dots$, $\cos x = 1 - \dfrac{x^2}{2!} + \dfrac{x^4}{4!} - \dots$ x in radians

∿ Your turn

Answers in grey.

Circle

1. Coordinates of the point at 90° on the unit circle
2. Is cos 150° positive or negative?
3. sin 210° = ? (use the 30° triangle and the sign)

1. (0, 1) 2. negative (left half) 3. −0.5

Graphs

4. Period and amplitude of $y = 4 \sin x$
5. Period of $y = \sin 3x$
6. Period of $y = \cos(x/2)$
7. Where does $y = \tan x$ have asymptotes between 0 and 2π?

4. 2π, 4 5. 2π/3 6. 4π 7. x = π/2 and 3π/2

Wheels

8. A wheel of radius 10 m, hub 12 m up, turning once a minute: write h(t) if you board at the bottom.
9. Use the sine series with 3 terms to estimate sin 0.5

8. h = 12 − 10 cos(2πt), t in minutes 9. 0.5 − 0.125/6 + 0.03125/120 ≈ 0.479427

Recipes

10. Use the cosine series with 3 terms to estimate cos 0.5
11. Estimate tan 0.3 with the start of the tan recipe, $x + x^3/3$
12. At 60° (x ≈ 1.047 radians) put sin x, the arc x and tan x in order, with values
13. In the one-tiny-step triangle, which side tells you how fast sin grows?

10. 1 − 0.25/2 + 0.0625/24 ≈ 0.877604 (true 0.877583) 11. 0.3 + 0.027/3 = 0.309 (true 0.3093) 12. sin 0.866 < arc 1.047 < tan 1.732 13. the upright side, h cos θ: sin grows at cos's rate

Find the unknown

14. Exact values of cos 225° and tan 300°

15. θ is in quadrant 2 and sin θ = 0.6. Find θ, cos θ and tan θ.

16. y = A sin(Bx) + D has maximum 7, minimum 1 and repeats every 90°. Find A, B and D.

17. y = 5 sin(Bx) has period 72°. Find B.

18. y = tan(Bx) has asymptotes 45° apart. Find B.

14. $\cos 225° = -\cos 45° = -1/\sqrt{2}$ (Q3); $\tan 300° = -\tan 60° = -\sqrt{3}$ (Q4) 15. $\alpha = \sin^{-1} 0.6 = 36.87°$, $\theta = 180° - 36.87° = 143.13°$; $\cos\theta = -0.8$, $\tan\theta = -0.75$ 16. $D = (7 + 1)/2 = 4$, $A = (7 - 1)/2 = 3$, $B = 360/90 = 4$ 17. $B = 360/72 = 5$ 18. period $45° = 180°/B$, so $B = 4$

Chapter 30

Trig identities and equations

always true, or only sometimes?

ALWAYS TRUE

$$\sin^2\theta + \cos^2\theta = 1$$

try any angle you like

SOMETIMES TRUE

$$\sin\theta = 0.5$$

only at 30°, 150°, 390°...

Some trig statements hold for **every** angle: these are **identities**, tools you can swap in whenever convenient. Others hold only for **particular** angles: these are **equations**, puzzles to solve.

The big idea for equations: trig waves repeat, so an equation usually has **many** answers, and your calculator only tells you one of them. We'll catch them all with a picture.

Goes with **Lecture 6** ($\cos^2 + \sin^2 = 1$, multiple values, $r\sin(\theta + \alpha)$, the advanced question).

sin² + cos² = 1 is Pythagoras on the circle

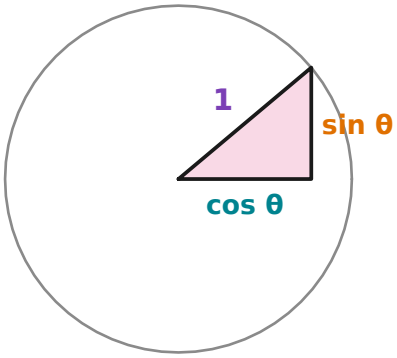
RULE: the Pythagorean identity

For **every** angle: $\sin^2\theta + \cos^2\theta = 1$. Rearranged: $\cos^2\theta = 1 - \sin^2\theta$ and $\sin^2\theta = 1 - \cos^2\theta$.

Also $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$, and dividing by $\cos^2\theta$: $\tan^2\theta + 1 = \dfrac{1}{\cos^2\theta}$.

$\sin^2\theta$ means $(\sin\theta)^2$, not $\sin(\theta^2)$.

On the unit circle (Chapter 29) the point at angle θ is $(\cos\theta, \sin\theta)$, and every point on it obeys $x^2 + y^2 = 1$. Swap in $x = \cos\theta$, $y = \sin\theta$:



$$(\cos\theta)^2 + (\sin\theta)^2 = 1^2$$

$$\cos^2\theta + \sin^2\theta = 1$$

short sides squared, added = hypotenuse squared

Notation: $\sin^2\theta$ means $(\sin\theta)^2$: work out the sine, then square it. It does **not** mean $\sin(\theta^2)$. (Lecture 6 writes "cos 25² + sin 25² = 1", which you should read as $\cos^2 25° + \sin^2 25°$.)

θ = 25°

(0.9063)² + (0.4226)²

0.8214 + 0.1786 = 1 ✓

θ = 550°

(-0.9848)² + (-0.1736)²

0.9698 + 0.0302 = 1 ✓

θ = -73°

(0.2924)² + (-0.9563)²

0.0855 + 0.9145 = 1 ✓

Using it: if $\sin\theta = \frac{3}{5}$ (and θ is acute), then $\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$, so $\cos\theta = \frac{4}{5}$. No angle needed. (It's the 3-4-5 triangle in disguise.)

Dividing the identity by $\cos^2\theta$ also shows $\tan\theta = \frac{\sin\theta}{\cos\theta}$ fits in: $\tan^2\theta + 1 = \frac{1}{\cos^2\theta}$.

PROCEDURE: given one ratio, find the others

1. Use the identity: $\cos^2\theta = 1 - \sin^2\theta$ (or $\sin^2\theta = 1 - \cos^2\theta$).
2. Square-root. This gives **two** possible values, $\pm$.
3. Choose the sign from the **quadrant** θ is in (CAST, Chapter 29): in Q2 only sin is positive, in Q3 only tan, in Q4 only cos.
4. Then $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$, keeping the signs.

Example: $\sin\theta = \frac{3}{5}$ and θ is **obtuse** (between $90°$ and $180°$). Find $\cos\theta$ and $\tan\theta$.

$$\cos^2\theta = 1 - \left(\tfrac{3}{5}\right)^2 = \tfrac{16}{25}$$ ← step 1

$$\cos\theta = \pm\tfrac{4}{5}$$ ← step 2: two candidates

$$\cos\theta = -\tfrac{4}{5}$$ ← step 3: obtuse means Q2, where cos is negative

$$\tan\theta = \frac{3/5}{-4/5} = -\tfrac{3}{4}$$ ← step 4

Check: $\theta = 180° - 36.87° = 143.13°$, and a calculator gives $\cos 143.13° = -0.8$ ✓.

Solving: catch every crossing

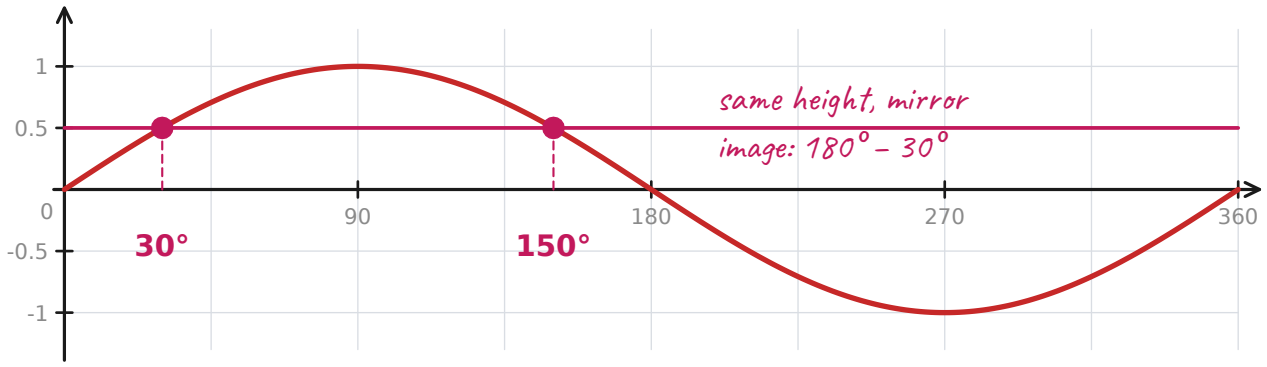
RULE: trig equations have many solutions

The calculator ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) gives **one** answer, the principal value.

Each function has a **partner rule** giving a second answer in the same turn (table below).

Every answer repeats every whole turn: add or subtract $360°$ (2π) to list all the solutions in the range asked for.

Solve $\sin\theta = 0.5$ for $0° \le \theta \le 360°$. The calculator's $\sin^{-1}(0.5)$ says 30°. But the Ferris-wheel seat is at height 0.5 **twice** per turn: once going up, once coming down. Draw the line $y = 0.5$ across the wave:



The second answer comes from the wave's symmetry. Each function has its own partner rule. If the calculator's answer is θ_1:

Equation	Calculator gives θ₁ in	Partner (degrees)	Partner (radians)	Why
$\sin\theta = k$	$-90°$ to $90°$	$180° - \theta_1$	$\pi - \theta_1$	wave symmetric about $90°$
$\cos\theta = k$	$0°$ to $180°$	$360° - \theta_1$	$2\pi - \theta_1$	wave symmetric about $180°$
$\tan\theta = k$	$-90°$ to $90°$	$\theta_1 + 180°$	$\theta_1 + \pi$	tan repeats every $180°$

Then add or subtract **whole turns** (360°) to reach any other range. So $\sin\theta = 0.5$ also has $390°, 510°, -210°, \ldots$

Lecture 6's example: $\cos^{-1}(-1)$. The calculator says 180° (π). But cos touches -1 at the bottom of **every** dip: $\theta = \pi, 3\pi, -\pi, -3\pi, \ldots$, i.e. $180°, 540°, -180°, -540°$.

A harder one, step by step

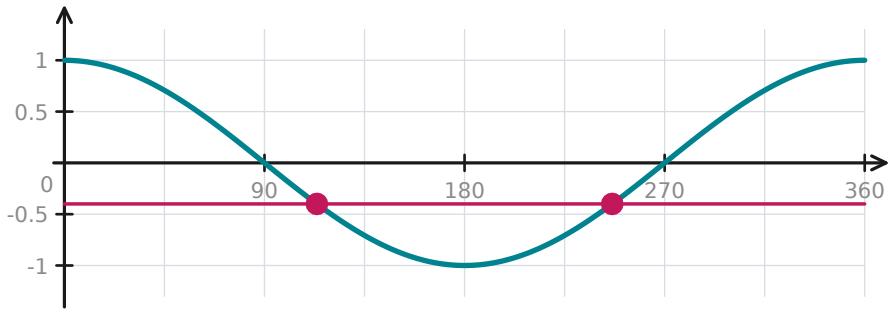
Solve $5\cos\theta + 2 = 0$ **for** $0° \le \theta \le 360°$**.**

$$5\cos\theta = -2$$ ← − 2 both sides

$$\cos\theta = -0.4$$ ← ÷ 5: get the cos on its own first

$$\theta = \cos^{-1}(-0.4) = 113.58°$$ ← calculator (one answer)

$$\theta = 360° - 113.58° = 246.42°$$ ← cos partner rule



two crossings, mirror images about 180°

Check: $5\cos 246.42° + 2 = 5(-0.4) + 2 = 0$ ✓.

When the calculator gives a negative angle

PROCEDURE: solving sin θ = k, cos θ = k or tan θ = k in a range

1. Rearrange so the trig function is on its own: $\sin\theta = k$.
2. Get the principal value $\theta_1 = \sin^{-1} k$ (or $\cos^{-1} k$, $\tan^{-1} k$). It may be **negative** or out of range: keep it anyway.
3. Get the partner: $180° - \theta_1$ for sin, $360° - \theta_1$ for cos, $\theta_1 + 180°$ for tan (radians: $\pi - \theta_1$, $2\pi - \theta_1$, $\theta_1 + \pi$).
4. Add or subtract $360°$ (2π) to each answer until it lands in the range. Keep every one that does.
5. Check each answer on the calculator.

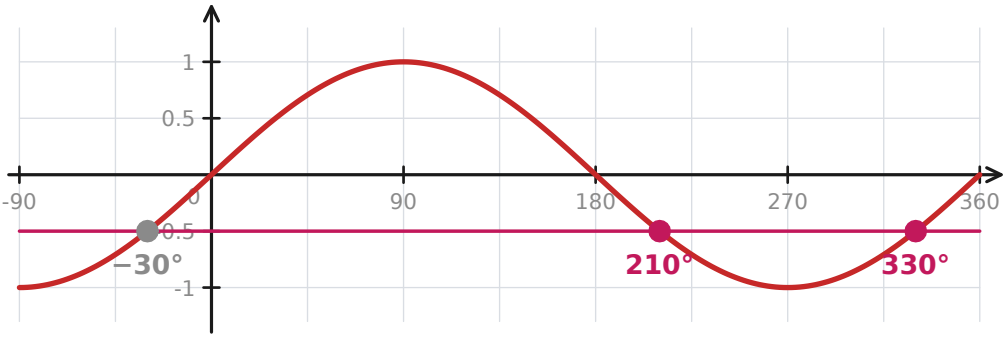
Example 1: solve $\sin\theta = -0.5$ for $0° \le \theta < 360°$.

$$\theta_1 = \sin^{-1}(-0.5) = -30°$$ ← step 2: out of range, but keep it

$$180° - (-30°) = 210°$$ ← step 3: sin partner

$$-30° + 360° = 330°$$ ← step 4: bring θ_1 into range

$$\theta = 210°,\ 330°$$



calculator: −30° (grey), outside the range.

In range: 210° and 330°

Example 2: solve $\tan\theta = -2$ for $0° \le \theta < 360°$.

$$\theta_1 = \tan^{-1}(-2) = -63.43°$$

$$-63.43° + 180° = 116.57°, \quad -63.43° + 360° = 296.57°$$

tan: add 180° each time

Example 3 (radians): solve $\sin\theta = -0.4$ for $0 \le \theta < 2\pi$.

$$\theta_1 = \sin^{-1}(-0.4) = -0.412$$ ← RAD mode

$$\pi - (-0.412) = 3.553, \qquad -0.412 + 2\pi = 5.872$$

partner π − θ_1; add 2π

Multiple angles: sin 2θ = 0.5

PROCEDURE: equations like sin(nθ) = k

1. Call the whole angle $\phi = n\theta$. Multiply the range by n too: if $0° \le \theta < 360°$ then $0° \le \phi < n \times 360°$.
2. Solve for ϕ in the stretched range: principal value, partner, then add $360°$ to each until you pass the top of the range.
3. Divide every ϕ by n to get θ. Expect n times as many answers.

Example: solve $\sin 2\theta = 0.5$ for $0° \le \theta < 360°$.

$$\phi = 2\theta, \quad 0° \le \phi < 720°$$ ← *step 1: double the range*

$$\phi = 30°,\ 150°,\ 390°,\ 510°$$ ← *30°, its partner 150°, then + 360°*

$$\theta = 15°,\ 75°,\ 195°,\ 255°$$ ← *step 3: halve*

Four answers, not two: $\sin 2\theta$ has period $180°$ (Chapter 29), so it goes through two full waves between $0°$ and $360°$.

✗ Common mistakes

✗ $\sin\theta = -0.5 \Rightarrow \theta = -30°$ ✓ $\theta = 210°,\ 330°$

−30° is outside 0° to 360°: use the partner and add 360°

✗ $\sin 2\theta = 0.5 \Rightarrow \theta = 15°, 75°$ ✓ $\theta = 15°, 75°, 195°, 255°$

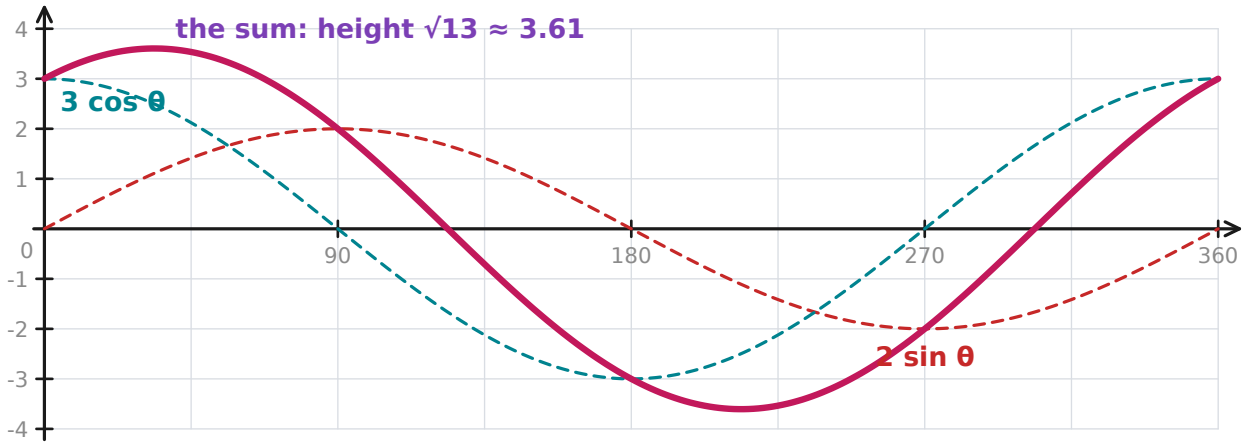
stretch the range to 0°–720° before halving

Two waves make one: r sin(θ + α)

PROCEDURE: a sin θ + b cos θ = r sin(θ + α)

1. $r = \sqrt{a^2 + b^2}$.
2. $\alpha = \tan^{-1}\dfrac{b}{a}$, then the quadrant fix: sketch the point (a, b); if $a < 0$, add $180°$.
3. The largest value is r, the smallest $-r$.
4. To solve $a\sin\theta + b\cos\theta = c$: write $\sin(\theta + \alpha) = \frac{c}{r}$, solve for the whole bracket $\phi = \theta + \alpha$ (shift the range by α), then subtract α.

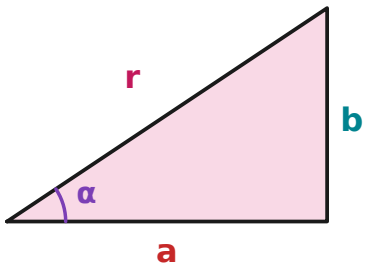
Add a sine wave and a cosine wave, $2\sin\theta + 3\cos\theta$, and something lovely happens: the result is **a single sine wave**, just taller and shifted:



Lecture 6 writes $a\sin\theta + b\cos\theta = r\sin(\theta + \alpha)$, with

$$r = \sqrt{a^2 + b^2}$$

$$\alpha = \tan^{-1}\frac{b}{a}$$



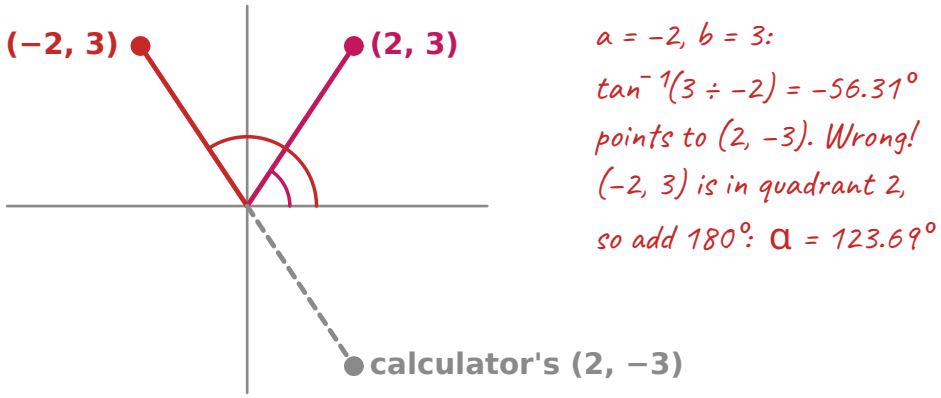
a and b as the sides of a right triangle

Why? There's a standard identity for the sine of a sum (the **compound angle formula**):

$$\sin(\theta + \alpha) = \sin\theta\cos\alpha + \cos\theta\sin\alpha$$

Multiply by r and match with $a\sin\theta + b\cos\theta$: we need $r\cos\alpha = a$ and $r\sin\alpha = b$. Those are exactly the sides of the triangle above (CAH and SOH with hypotenuse r), and Pythagoras gives r. Lecture 6 notes this is the same move as going from (x, y) to polar coordinates (Chapter 33).

Watch the signs. $r\cos\alpha = a$ and $r\sin\alpha = b$ say that α is the angle of the **point** (a, b). When a and b are both positive that point is in quadrant 1 and $\tan^{-1}\frac{b}{a}$ is right. When a is **negative**, the calculator still answers in the right-hand half, so it points the wrong way:



So: **sketch the point** (a, b) **first**. If $a < 0$ (quadrants 2 and 3), add $180°$ to the calculator's answer. Chapter 33 builds this **quadrant fix** properly for polar coordinates; it's exactly the same job.

Lecture 6's advanced question, both answers

Solve $2\sin\theta + 3\cos\theta = 2$ **for** $0° \le \theta \le 360°$**.**

$$r = \sqrt{2^2 + 3^2} = \sqrt{13}, \quad \alpha = \tan^{-1}\tfrac{3}{2} = 56.31°$$

a = 2, b = 3

$$\sqrt{13}\,\sin(\theta + 56.31°) = 2 \leftarrow \text{one wave now}$$

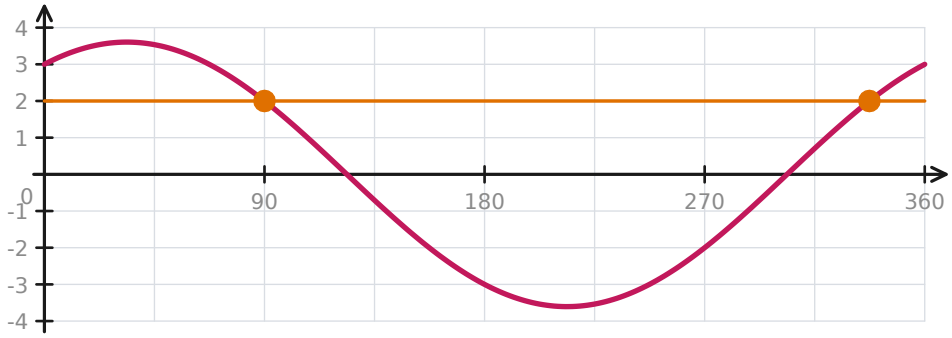
$$\sin(\theta + 56.31°) = \frac{2}{\sqrt{13}} = 0.5547 \leftarrow \div\sqrt{13}$$

Treat the whole bracket as one angle, call it ϕ ("fie"). The calculator gives $\phi = 33.69°$; the sin partner is $180° - 33.69° = 146.31°$:

$$\theta + 56.31 = 33.69 \;\Rightarrow\; \theta = -22.62° \to 337.38°$$

+ 360° to get into range

$$\theta + 56.31 = 146.31 \;\Rightarrow\; \theta = 90° \leftarrow \text{the one the slide asks you to find}$$



the line y = 2 crosses the wave at 90° and 337.38°

Check 90°: $2\sin 90° + 3\cos 90° = 2(1) + 3(0) = 2$ ✓. Easy to miss, which is exactly why the lecture asks.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Pythagorean identity	$\sin^2\theta + \cos^2\theta = 1$ also $\cos^2\theta = 1 - \sin^2\theta$, $\sin^2\theta = 1 - \cos^2\theta$
tan	$\tan\theta = \dfrac{\sin\theta}{\cos\theta}$, $\qquad \tan^2\theta + 1 = \dfrac{1}{\cos^2\theta}$
Choosing ±	after a square root, take the sign from the quadrant (CAST)
Partner: sin	$\theta_1, \quad 180° - \theta_1 \quad (\pi - \theta_1)$ $\sin^{-1}$ gives $-90°$ to $90°$
Partner: cos	$\theta_1, \quad 360° - \theta_1 \quad (2\pi - \theta_1)$ $\cos^{-1}$ gives $0°$ to $180°$
Partner: tan	$\theta_1, \quad \theta_1 + 180° \quad (\theta_1 + \pi)$ $\tan^{-1}$ gives $-90°$ to $90°$
All solutions	add or subtract $360°$ (2π) to each; keep those in range, even if θ_1 itself is negative
Multiple angles	for $\sin n\theta = k$: multiply the range by n, solve, divide by n
r sin form	$a\sin\theta + b\cos\theta = r\sin(\theta + \alpha)$ $r = \sqrt{a^2 + b^2}$, $\alpha = \tan^{-1}\frac{b}{a}$ (+180° if $a < 0$); max r, min $-r$
Compound angle	$\sin(A + B) = \sin A\cos B + \cos A\sin B$

Your turn

Range 0° to 360° unless stated. Answers in grey.

Identities

1. If $\cos\theta = 5/13$ (θ acute), find $\sin\theta$
2. Simplify $3\sin^2 x + 3\cos^2 x$
3. Is $\sin(2\theta) = 2\sin\theta$ an identity? Test $\theta = 90°$.

1. 12/13 2. 3 3. no: sin 180° = 0 but 2 sin 90° = 2

Equations

4. $\sin\theta = 0.8$
5. $\cos\theta = 0.5$
6. $\tan\theta = 1$
7. $2\sin\theta + 1 = 0$

4. 53.13°, 126.87° 5. 60°, 300° 6. 45°, 225° 7. 210°, 330°

r sin form

8. Write $3\sin\theta + 4\cos\theta$ as $r\sin(\theta + \alpha)$
9. Hence the largest value of $3\sin\theta + 4\cos\theta$
10. Solve $3\sin\theta + 4\cos\theta = 5$

8. $5\sin(\theta + 53.13°)$ 9. 5 10. $\theta + 53.13 = 90$, so $\theta = 36.87°$

Find the unknown

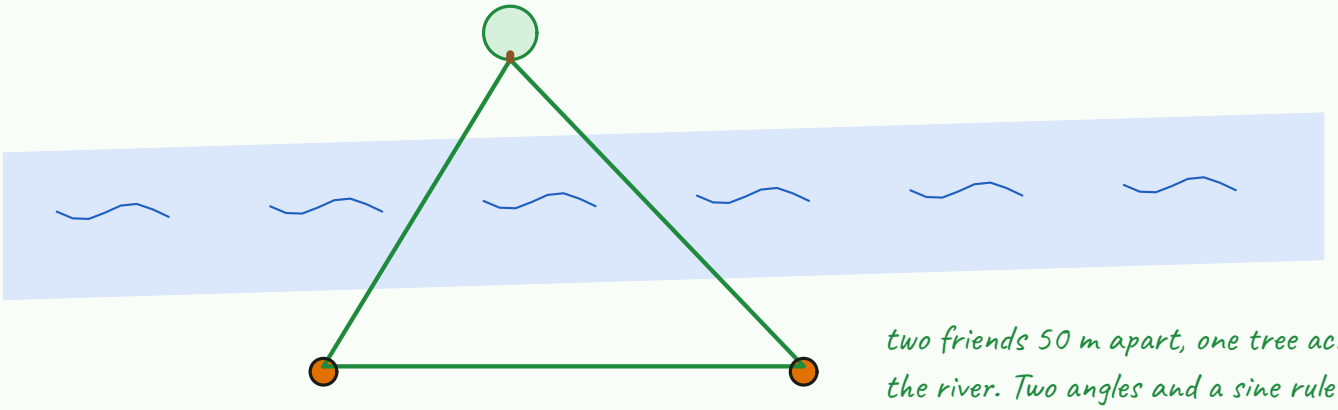
11. $\sin\theta = -0.6$
12. $\cos\theta = -0.25$ for $0 \le \theta < 2\pi$ (radians, 3 d.p.)
13. $\tan\theta = -1.5$
14. $\cos 2\theta = 0.5$
15. $\cos\theta = -5/13$ and θ is in quadrant 3. Find $\sin\theta$ and $\tan\theta$.

11. $\theta_1 = -36.87°$; $180° - (-36.87°) = 216.87°$, $-36.87° + 360° = 323.13°$ 12. $\theta_1 = \cos^{-1}(-0.25) = 1.823$; $2\pi - 1.823 = 4.460$ 13. $\theta_1 = -56.31°$; + 180° = 123.69°, + 360° = 303.69° 14. $2\theta = 60°, 300°, 420°, 660°$ (range 0°–720°), so $\theta = 30°, 150°, 210°, 330°$ 15. $\sin^2\theta = 1 - 25/169 = 144/169$, $\sin\theta = -12/13$ (Q3); $\tan\theta = 12/5$

Chapter 31

Solving any triangle

measure a river without getting wet



two friends 50 m apart, one tree across the river. Two angles and a sine rule give the width.

SOHCAHTOA only works when there's a right angle. Real triangles (plots of land, roof trusses, navigation) usually don't have one. This chapter gives you two tools that work on **any** triangle: the **sine rule** and the **cosine rule**, plus a map for choosing between them.

Goes with **Lecture 6** (triangle types and notation, sine rule, cosine rule, the clock example, answering questions).

△ Names, and the 180° rule

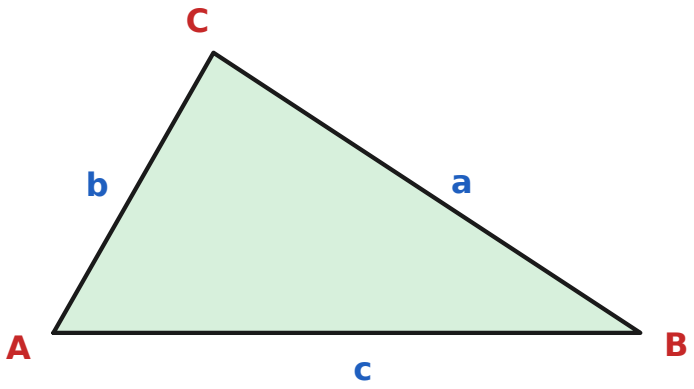
RULE: triangle notation and the angle sum

Angles are capital letters A, B, C; the side **opposite** each angle is the same letter in lower case: side a faces angle A.

The three angles add up to $180°$: $A + B + C = 180°$. Two angles always give the third.

The biggest angle faces the longest side; the smallest angle faces the shortest side.

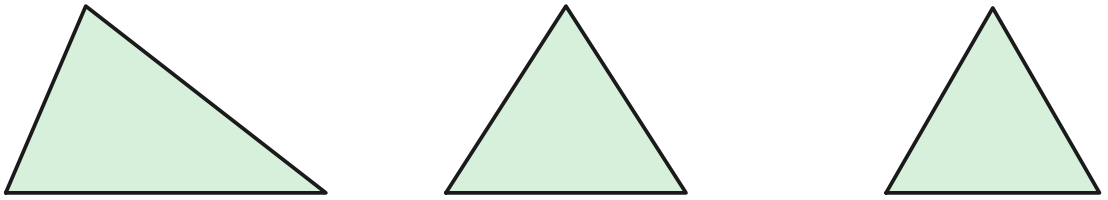
Lecture 6's labelling: **capital letters for the angles** (at the corners) and the **same letter in lower case for the side opposite**. Side a faces angle A:



a is opposite A,
b is opposite B,
c is opposite C.

Bigger angle ↔ longer side opposite it.

Three kinds (Lecture 6): **scalene** (all sides different, from Greek "unequal"), **isosceles** (two equal sides: "equal legs") and **equilateral** (all three equal, so all angles 60°).



scalene **isosceles** **equilateral**

Angles in a triangle add up to 180°. Tear the three corners off a paper triangle and line them up: they always make a straight line. So two angles always give you the third.

△ The sine rule: drop a height

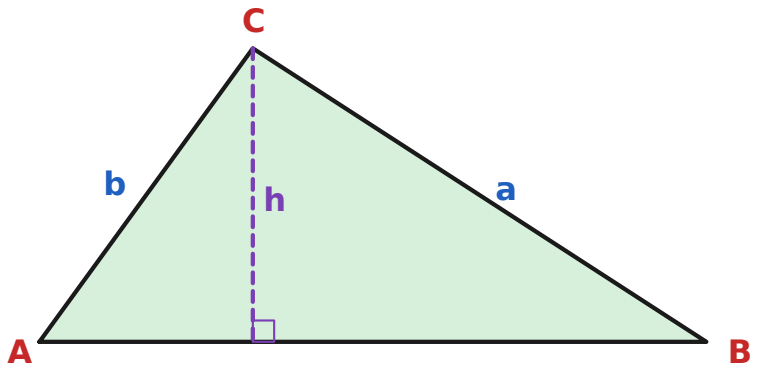
RULE: the sine rule

In **any** triangle: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$. Each side over the sine of the angle **opposite** it.

To find an **angle**, flip it: $\frac{\sin A}{a} = \frac{\sin B}{b}$ (the unknown goes on top).

Use it when you know a **matching pair** (a side and the angle opposite it) plus one more side or angle. Use only the two fractions you need.

Where does it come from? Drop a line straight down from C to the base, height h. That splits the triangle into **two right-angled triangles**, where SOHCAHTOA works again:



left triangle: sin A = h / b
so h = b sin A

right triangle: sin B = h / a
so h = a sin B

Both expressions are the same height, so

$$b \sin A = a \sin B$$

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$ ← *÷ both sides by sin A sin B*

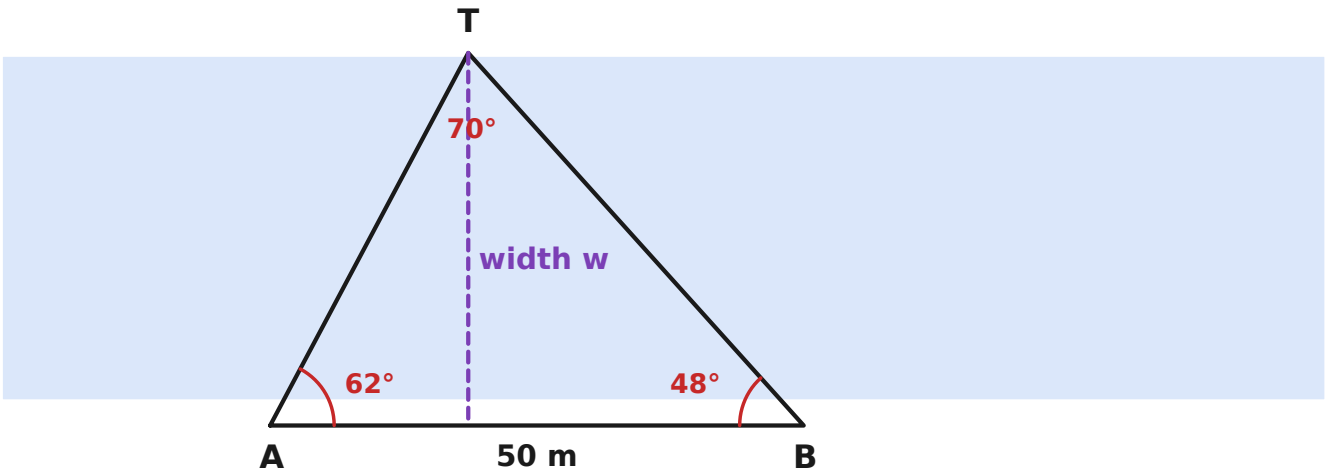
Dropping the height from a different corner brings in c the same way. The **sine rule**:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$ ← *each side ÷ the sine of its opposite angle*

When to use it: you know a **matching pair** (a side and the angle opposite it) plus one more fact. Use only the two fractions you need.

△ Measuring the river

Ana (A) and Ben (B) stand 50 m apart on one bank. Each measures the angle between the bank and a tree T on the far side: 62° at A, 48° at B. How wide is the river?



$$T = 180^\circ - 62^\circ - 48^\circ = 70^\circ$$ ← angle sum: now 50 m faces a known angle

$$\frac{AT}{\sin 48^\circ} = \frac{50}{\sin 70^\circ}$$ ← sine rule: AT faces B, 50 faces T

$$AT = \frac{50 \sin 48^\circ}{\sin 70^\circ} = 39.54 \text{ m}$$ ← × sin 48°

$$w = AT \sin 62^\circ = 34.9 \text{ m}$$ ← SOH in the right triangle at A

Lecture 6's example: $a = 17$, $b = 2$, $A = 24^\circ$. Find B:

$$\frac{17}{\sin 24^\circ} = \frac{2}{\sin B} \Rightarrow \sin B = \frac{2 \sin 24^\circ}{17}$$

$$B = \sin^{-1}\left(\frac{2 \sin 24^\circ}{17}\right) = 2.743^\circ$$

△ The cosine rule: Pythagoras plus a correction

RULE: the cosine rule (all three forms)

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$b^2 = a^2 + c^2 - 2ac\cos B$$

$$c^2 = a^2 + b^2 - 2ab\cos C$$

Pattern: the side on the left is **opposite the angle** in the cosine; the other two sides appear squared and in the $2 \times \ldots$ term.

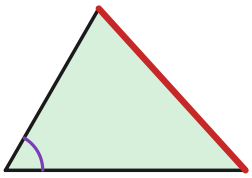
Angle form (rearranged): $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$, $\cos B = \dfrac{a^2 + c^2 - b^2}{2ac}$, $\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$.

Use it with **two sides and the angle between them** (find the third side) or **all three sides** (find an angle).

If you know **two sides and the angle between them**, there's no matching pair, so the sine rule is stuck. The **cosine rule** takes over:

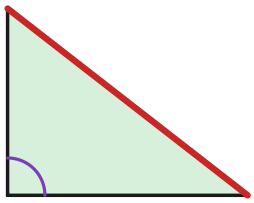
$$a^2 = b^2 + c^2 - 2bc\cos A$$

Look closely: it's **Pythagoras** ($a^2 = b^2 + c^2$) with a correction term. The correction depends on how far A is from 90°:



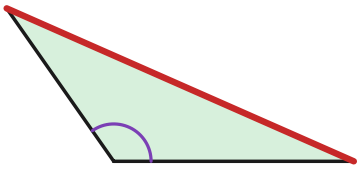
A < 90°

cos A > 0: subtract, a is shorter



A = 90°

cos A = 0: plain Pythagoras



A > 90°

cos A < 0: add, a is longer

Lecture 6's clock: hands 10 cm and 7 cm long. How far apart are the tips at 2 o'clock?

$$A = \frac{2}{12} \times 360^\circ = 60^\circ$$ ← *the angle between the hands (Chapter 27)*

$$a^2 = 10^2 + 7^2 - 2 \times 10 \times 7\cos 60^\circ$$

$$a^2 = 100 + 49 - 140 \times 0.5 = 79$$

$$a = \sqrt{79} = 8.89 \text{ cm}$$

Same rule, other letters: $a = 8$, $c = 5$, $B = 110°$. Find b (it faces B, so use the b^2 form):

$$b^2 = 8^2 + 5^2 - 2 \times 8 \times 5 \cos 110° = 89 + 27.36 = 116.36$$

cos 110° < 0: the term is added

$$b = \sqrt{116.36} = 10.79$$

Finding an angle from three sides: rearrange the rule (Chapter 14) to

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\text{sides } 5, 6, 7: \quad \cos C = \frac{25 + 36 - 49}{2 \times 5 \times 6} = 0.2, \quad C = 78.46°$$

C faces the 7

△ Which rule? Ask "what do I know?"

RULE: sine rule or cosine rule?

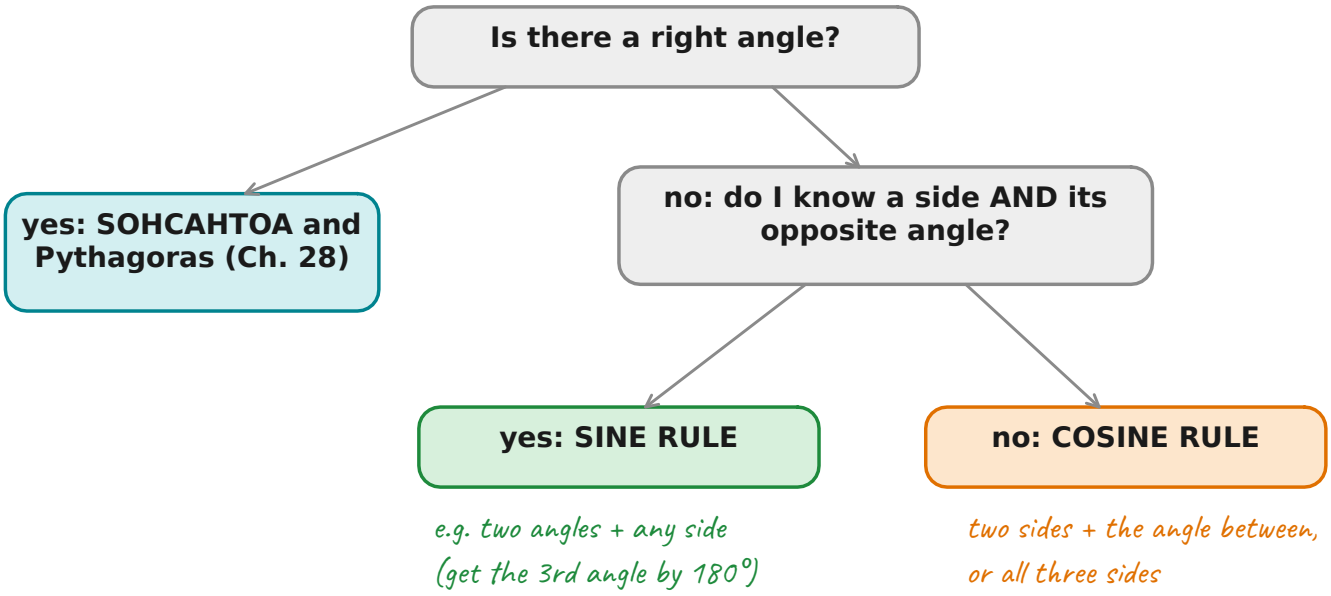
Right angle in the triangle → SOHCAHTOA and Pythagoras (Chapter 28).

Know a side **and the angle opposite it** (a matching pair) → **sine rule**.

Know two sides and the angle **between** them (SAS), or **all three sides** (SSS) → **cosine rule**.

You know	Use	Then
two angles + any side (AAS / ASA)	sine rule	third angle $= 180° -$ the other two, first
two sides + an angle opposite one of them (SSA)	sine rule (for an angle)	check the **ambiguous case** (below)
two sides + the angle between them (SAS)	cosine rule	third side; then the sine rule for an angle
three sides (SSS)	cosine rule, angle form	find the **largest** angle first
three angles only	can't be solved	only the shape is fixed, not the size

Lecture 6's advice: **first draw the triangle and fill in everything you know**. Then:



RULE: the ambiguous case (two possible triangles)

It can only happen with **two sides and a non-included angle** (SSA): you know a, b and A.

It happens only when the given angle A is **acute** and the side opposite it is the **shorter** one ($a < b$), but still longer than the height $b \sin A$.

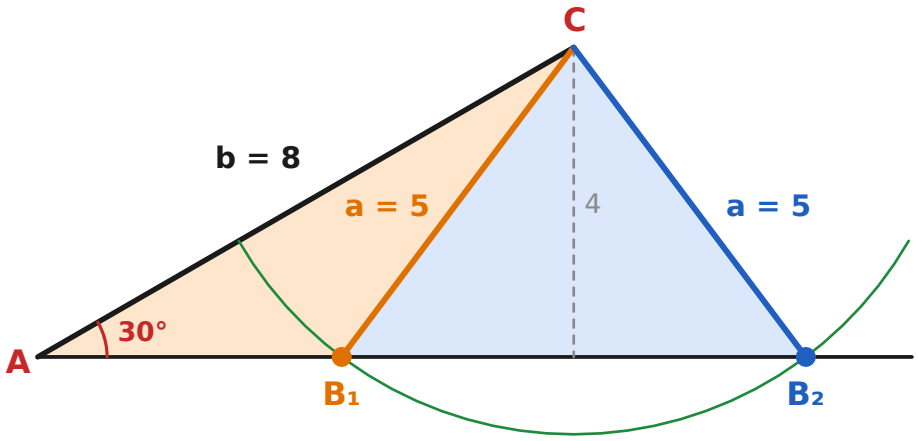
Then $\sin B$ gives two angles: $B_1 = \sin^{-1}(\dots)$ (acute) and $B_2 = 180° - B_1$ (obtuse). Keep B_2 only if $A + B_2 < 180°$.

If A is **obtuse** (or $90°$), or $a \geq b$: only **one** triangle. If $a < b \sin A$: no triangle at all (the calculator gives an error).

See it with a hinge. $A = 30°$, $b = 8$, $a = 5$. Draw the 30° angle at A and the 8-long side up to C. Side a (5 long) hangs from C on a hinge, and must reach the bottom line to close the triangle.

✎ **YOUR GO (pencil first)**

Swing the 5-long side from C down to the bottom line (a compass set to 5 does it). How many places can it land?



Green arc: everywhere the hinged side can reach. It crosses the line TWICE.

Orange triangle: A, B_1, C. Blue triangle: A, B_2, C (the orange one sits inside it).

Two different triangles, both with $A = 30°$, $b = 8$, $a = 5$. The sine rule can't tell them apart:

$$\sin B = \frac{8 \sin 30°}{5} = 0.8$$ ← *sine rule, rearranged*

$$B = 53.13° \text{ (calculator)} \quad \text{or} \quad B = 180° - 53.13° = 126.87°$$

$\sin^{-1}$ **only ever gives the acute one** (the blue triangle, at B_2). The obtuse partner (orange, at B_1) has exactly the same sine, and it fits too: $30° + 126.87° = 156.87°$, under 180°. When the hinge is **shorter than b but longer than the height** ($4 < a < 8$ here), expect two answers. Longer than b: only one. Shorter than the height: it can't reach, so no triangle.

△ Area of any triangle

RULE: area from two sides and the included angle

$\text{Area} = \frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A = \frac{1}{2}ac\sin B$: half × two sides × the sine of the angle **between** them.

Find an angle from the area: $\sin C = \dfrac{2 \times \text{Area}}{ab}$. Two angles have that sine, C and $180° - C$; both can be valid.

Find a side from the area: $a = \dfrac{2 \times \text{Area}}{b\sin C}$.

"Half base × height" needs the height. But the height is $b\sin A$ (from the sine-rule picture), so with two sides and the angle between them:

$$\text{Area} = \frac{1}{2}\,bc\sin A$$

$$\text{clock triangle: } \frac{1}{2} \times 10 \times 7 \times \sin 60° = 30.3 \text{ cm}^2$$

And the **perimeter** is just the three sides added: for the clock triangle $10 + 7 + 8.89 = 25.89$ cm.

△ Find the unknown: working backwards and chaining

PROCEDURE: partial-information triangle problems

1. Sketch the triangle and label **everything** you know, including the area if given.
2. Write the equation that links what you know to what you want (sine rule, cosine rule, area formula, or $A + B + C = 180°$).
3. Solve for the unknown. For an angle from a sine, check whether $180° -$ it also fits.
4. If the rule you need has a gap, fill it first with another rule (a chain: angle → side → area).

Area → side: a triangle has area 20 cm², $b = 8$ cm and $C = 30°$. Find a.

$$20 = \tfrac{1}{2} \times a \times 8 \times \sin 30° = 2a \;\Rightarrow\; a = 10 \text{ cm}$$

½ × 8 × 0.5 = 2

Area → angle: area 30 cm², sides 8 cm and 10 cm. Find the angle C between them.

$$30 = \tfrac{1}{2} \times 8 \times 10 \sin C \;\Rightarrow\; \sin C = \frac{30}{40} = 0.75$$

$$C = 48.59° \quad \text{or} \quad C = 180° - 48.59° = 131.41°$$

both fit: two triangles

Chain (angle → side → area): $A = 50°$, $B = 60°$, $c = 12$. Find the area.

$$C = 180° - 50° - 60° = 70°$$ ← angle sum: now c = 12 has its matching angle

$$a = \frac{12 \sin 50°}{\sin 70°} = 9.78$$ ← sine rule

$$\text{Area} = \tfrac{1}{2} ac \sin B = \tfrac{1}{2} \times 9.78 \times 12 \times \sin 60° = 50.83$$

B is between a and c

✗ Common mistakes

✗ $\text{Area} = \frac{1}{2}ab\sin A$ ✓ $\text{Area} = \frac{1}{2}ab\sin C$

the angle must be the one between the two sides you use

✗ $c^2 = \ldots - 2ab\cos A$ ✓ $c^2 = \ldots - 2ab\cos C$

the side on the left faces the angle in the cosine

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Notation	side a is opposite angle A, and so on
Angle sum	$A + B + C = 180^\circ$
Sine rule	$\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$ for an angle: $\frac{\sin A}{a} = \frac{\sin B}{b}$
Cosine rule	$a^2 = b^2 + c^2 - 2bc\cos A$ also $b^2 = a^2 + c^2 - 2ac\cos B$, $c^2 = a^2 + b^2 - 2ab\cos C$
Cosine rule (angle)	$\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$ SSS: find the largest angle first
Which rule?	matching side + angle → sine rule; SAS or SSS → cosine rule
Ambiguous case	SSA with the given angle acute and the side opposite it shorter: try B and $180^\circ - B$. Obtuse given angle: one triangle only
Area	$\text{Area} = \frac{1}{2}ab\sin C$ two sides and the angle between them
Area backwards	$\sin C = \dfrac{2 \times \text{Area}}{ab}$, $\quad a = \dfrac{2 \times \text{Area}}{b\sin C}$ an angle from a sine may have two answers

△ Your turn

Draw each triangle first. Answers in grey.

Sine rule

1. A = 40°, B = 75°, a = 8. Find b.
2. a = 12, A = 110°, b = 9. Find B.
3. Two angles are 35° and 65°; the side between them is 20. Find the side opposite 65°.

1. $8 \sin 75°/\sin 40° \approx 12.02$ 2. $\sin^{-1}(9 \sin 110°/12) \approx 44.8°$ 3. third angle 80°; $20 \sin 65°/\sin 80° \approx 18.41$

Cosine rule

4. b = 5, c = 8, A = 50°. Find a.
5. Sides 4, 7, 9: find the largest angle.
6. Clock hands 10 cm and 7 cm at 4 o'clock: distance between the tips?

4. ≈ 6.13 5. $\cos^{-1}((16 + 49 - 81)/56) \approx 106.6°$ 6. 120°: $\sqrt{(149 + 70)} \approx 14.80$ cm

Choose

7. You know all three angles. Can you find the sides?
8. Area with sides 6, 9 and 30° between

7. no, only their ratios (the triangle could be any size) 8. 13.5

Find the unknown

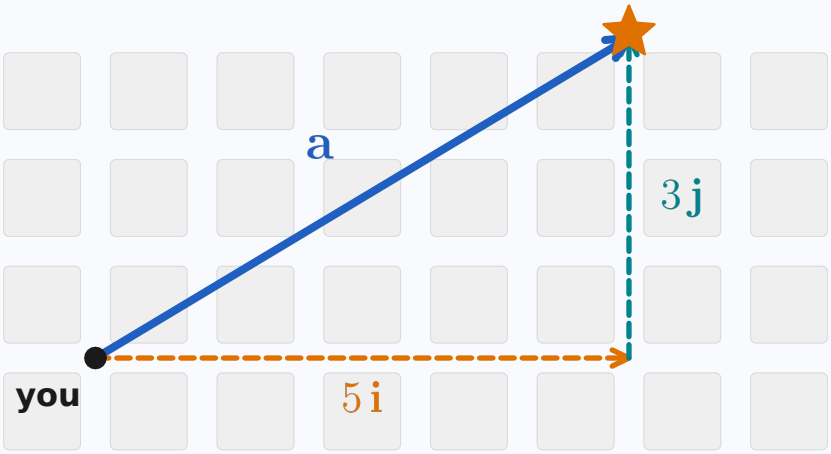
9. A triangle has area 24 cm² and sides 8 cm and 10 cm. Find the angle between them (two answers).
10. Area 35 cm², C = 150°, a = 10 cm. Find b.
11. A = 40°, B = 70°, c = 15. Find a, then the area.
12. Sides 7 and 9 with 60° between them: find the third side, then the area.
13. A = 120°, a = 9, b = 5. How many triangles? Find B.

9. $\sin C = 48/80 = 0.6$, C = 36.87° or 143.13° 10. $35 = ½ \times 10 \times b \times 0.5$, so b = 14 cm 11. C = 70°; $a = 15 \sin 40°/\sin 70° \approx 10.26$; area = $½ \times 10.26 \times 15 \times \sin 70° \approx 72.31$ 12. $c^2 = 49 + 81 - 63 = 67$, $c \approx 8.19$; area = $½ \times 7 \times 9 \times \sin 60° \approx 27.28$ 13. A is obtuse, so one triangle: $\sin B = 5 \sin 120°/9 \approx 0.481$, B ≈ 28.76°

Chapter 32

Vectors

a vector is a trip: how far, and which way



5 blocks east, then 3 north.
The blue arrow is the whole
trip in one go: that's the vector a = 5i + 3j.

Some things are just a size: 20 °C, 3 kg, 10 miles. Others only make sense with a **direction** too: a push, a wind, a trip across town. Those are **vectors**, and this chapter treats every one of them as a little trip on a map: how many blocks east, how many north, and (in 3D) how many floors up.

You'll add trips, stretch them, measure them, and multiply them in two very different ways (the dot product and the cross product), each with its own picture.

Goes with **Lecture 10** (scalars and vectors, notation, unit vectors, adding and subtracting, scalar multiples, dot product, cross product, determinants).

↗ Size only, or size and direction?

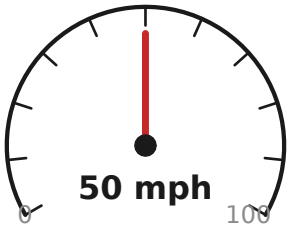
RULE: scalars and vectors

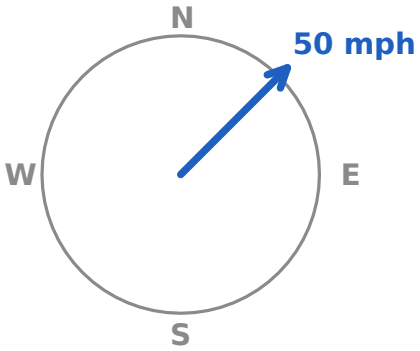
A **scalar** has size only (distance, speed, mass, temperature).

A **vector** has size **and** direction (displacement, velocity, force).

Distance is the total length travelled; **displacement** is where you end up relative to the start.

A **scalar** is one piece of information: its size. A **vector** needs two: its size **and** its direction. Same speed, two different stories:

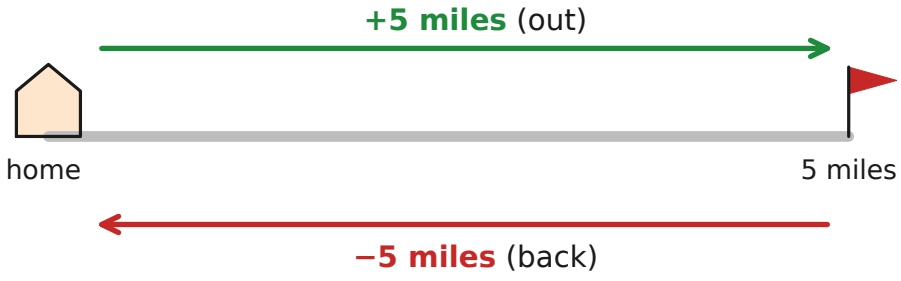




scalar: speed = 50 mph

vector: velocity = 50 mph north-east

Lecture 10's runner jogs 5 miles away from home, then 5 miles back:



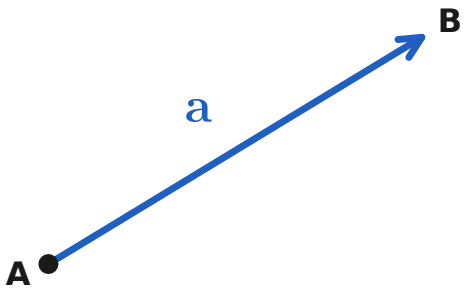
distance (scalar):
5 + 5 = 10 miles

displacement (vector):
+5 + (−5) = 0
he ends where he began

Same with north then south, or any direction and its opposite: the two trips cancel.

↗ One arrow, four spellings

Lecture 10 admits the notation is "quite confusing". All four of these mean the same arrow, the trip from point A to point B, named $\mathbf{a}$:



$\vec{a}$	*arrow over the letter*
$\overrightarrow{AB}$	*the trip from A to B*
$\mathbf{a}$	*bold: how books print it*
$\underline{a}$	*underlined: how you write it by hand (you can't write bold with a pen)*

↗ i, j and k: one block east, one north, one floor up

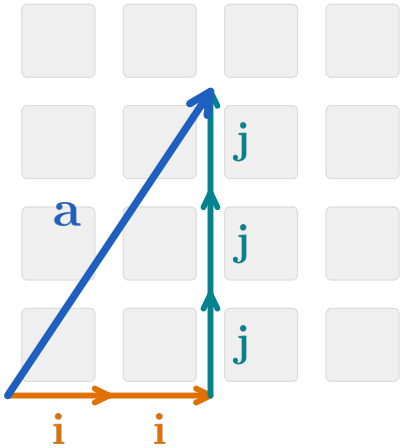
RULE: components

$\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ are vectors of length 1 along the x, y and z axes.

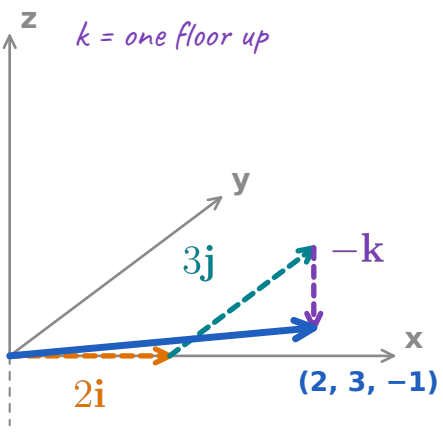
Any vector can be written $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, or as a column $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$, or as (x, y, z). The numbers are its **components**.

In a city you give directions in blocks: "2 east, 3 north". Vectors do exactly this using three standard mini-trips, each exactly **1 long** (a **unit vector**):

$\mathbf{i}$ = one step along x (east), $\mathbf{j}$ = one step along y (north), $\mathbf{k}$ = one step along z (one floor up). Lecture 10 writes them with hats: $\hat{\imath}, \hat{\jmath}, \hat{k}$.



a = 2i + 3j



On the right is Lecture 10's 3D example, $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$: 2 east, 3 north, then one floor **down** (that's the $-\mathbf{k}$). So $x = 2$, $y = 3$, $z = -1$. These are **Cartesian** coordinates; Chapter 33 shows the other way (polar).

The same trip can be written as a column (or a row), the numbers in order x, y, z:

$$2\mathbf{i} + 3\mathbf{j} - \mathbf{k} = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = (2,\ 3,\ -1) \quad \longleftarrow \textit{three ways, one trip}$$

↗ Adding: one trip, then the next (tip to tail)

RULE: adding and subtracting vectors

Add **component by component**: $(a_x, a_y) + (b_x, b_y) = (a_x + b_x,\ a_y + b_y)$. Never mix $\mathbf{i}$ and $\mathbf{j}$ parts.

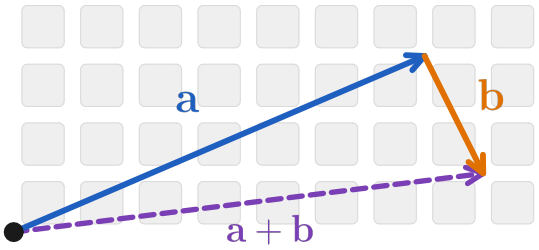
Subtract the same way: $\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$, and $-\mathbf{b}$ has every sign flipped.

Order doesn't matter for adding: $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ (tip to tail either way).

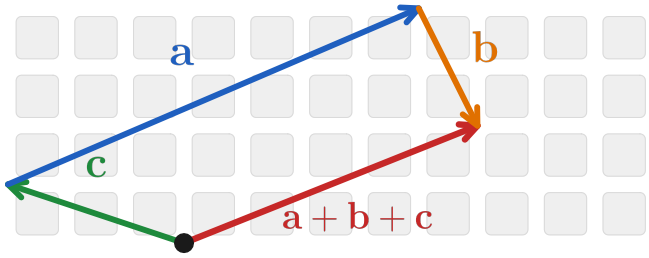
To add vectors, do the first trip, then start the second one **where the first one ended**. The answer is the single arrow from the very start to the very end. That's why "1 up + 2 right" isn't 3 of anything: it's one diagonal trip.

Lecture 10's example: $\mathbf{a} = 7\mathbf{i} + 3\mathbf{j}$, $\mathbf{b} = \mathbf{i} - 2\mathbf{j}$, $\mathbf{c} = -3\mathbf{i} + \mathbf{j}$.

a + b



c + a + b: a different order, same finish



With numbers it's even easier: keep $\mathbf{i}$'s with $\mathbf{i}$'s and $\mathbf{j}$'s with $\mathbf{j}$'s, like two separate tally columns:

add down the i column, then down the j column. Never mix them.

	$\mathbf{i}$	$\mathbf{j}$
$\mathbf{a}$	7	3
$\mathbf{b}$	1	-2
$\mathbf{c}$	-3	1
	5	2

$\mathbf{a} + \mathbf{b} + \mathbf{c} = 5\mathbf{i} + 2\mathbf{j}$

$$\mathbf{a} + \mathbf{b} = 7\mathbf{i} + 3\mathbf{j} + \mathbf{i} - 2\mathbf{j} = 8\mathbf{i} + \mathbf{j}$$ ← *i's: 7 + 1, j's: 3 − 2*

Subtracting is adding the **reverse**. $-\mathbf{c}$ is $\mathbf{c}$ turned round: same length, opposite way, so every sign flips:

$$\mathbf{c} = -3\mathbf{i} + \mathbf{j}$$

$$-\mathbf{c} = 3\mathbf{i} - \mathbf{j}$$

$$\mathbf{a} - \mathbf{c} = \mathbf{a} + (-\mathbf{c}) = 7\mathbf{i} + 3\mathbf{j} + 3\mathbf{i} - \mathbf{j} = 10\mathbf{i} + 2\mathbf{j}$$

↗ Stretching, and measuring the length

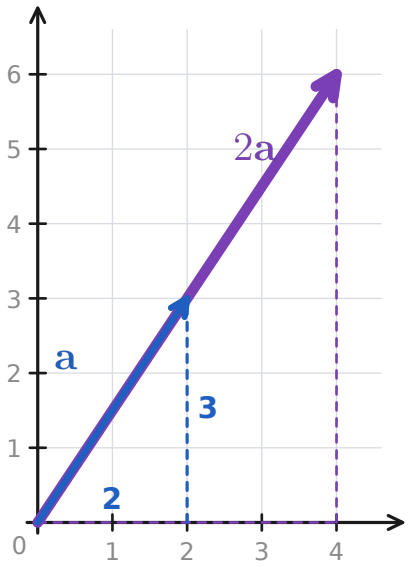
RULE: scalar multiples and magnitude

$k\mathbf{a}$: multiply **every** component by k. Same direction if $k > 0$, reversed if $k < 0$; $|k|$ times as long.

Magnitude (length): $|\mathbf{a}| = \sqrt{x^2 + y^2}$ in 2D, $\sqrt{x^2 + y^2 + z^2}$ in 3D (Pythagoras).

Multiply a vector by an ordinary number (a scalar) and you do the trip that many times: $2\mathbf{a}$ is the same direction, **twice as long**. A negative number also turns it round.

How long is the arrow? The $\mathbf{i}$ part and the $\mathbf{j}$ part meet at a right angle, so it's **Pythagoras** (Chapter 28). The length is called the **magnitude**, written $|\mathbf{a}|$.



$$|\mathbf{a}| = \sqrt{2^2 + 3^2} = \sqrt{13} \approx 3.61$$

$$|2\mathbf{a}| = \sqrt{4^2 + 6^2} = \sqrt{52} \approx 7.21$$

= 2 × 3.61: twice the trip, twice the length

$$|\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$$

in 3D add the k part too (Chapter 28's box)

↗ A unit vector in any direction

RULE: the unit vector of a

The **unit vector** in the direction of $\mathbf{a}$ is $\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{|\mathbf{a}|}$: divide every component by the length.

It points the same way as $\mathbf{a}$ and has length exactly 1.

A vector of length L in the direction of $\mathbf{a}$ is $L\,\hat{\mathbf{a}}$.

Example: the unit vector in the direction of $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}$.

$$|\mathbf{a}| = \sqrt{3^2 + 4^2} = 5 \quad \longleftarrow \textit{length first}$$

$$\hat{\mathbf{a}} = \tfrac{1}{5}(3\mathbf{i} + 4\mathbf{j}) = 0.6\,\mathbf{i} + 0.8\,\mathbf{j} \quad \longleftarrow \div 5$$

$$\text{check: } \sqrt{0.6^2 + 0.8^2} = \sqrt{0.36 + 0.64} = 1$$

A vector of length 10 in that direction: $10\,\hat{\mathbf{a}} = 6\mathbf{i} + 8\mathbf{j}$.

In 3D: $\mathbf{b} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

$$|\mathbf{b}| = \sqrt{4 + 1 + 4} = 3, \qquad \hat{\mathbf{b}} = \tfrac{2}{3}\mathbf{i} - \tfrac{1}{3}\mathbf{j} + \tfrac{2}{3}\mathbf{k}$$

↗ Position vectors: from point to point

RULE: position vectors, distance and midpoint

The **position vector** of a point A is the trip from the origin O to A: if A is $(2, -1)$ then $\mathbf{a} = \overrightarrow{OA} = 2\mathbf{i} - \mathbf{j}$.

The trip from A to B: $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ (**end minus start**).

The distance from A to B is its length, $|\overrightarrow{AB}| = |\mathbf{b} - \mathbf{a}|$.

The midpoint M of AB has position vector $\mathbf{m} = \dfrac{\mathbf{a} + \mathbf{b}}{2}$.

Example: A is $(2, -1)$ and B is $(5, 3)$.

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = (5-2)\mathbf{i} + (3-(-1))\mathbf{j} = 3\mathbf{i} + 4\mathbf{j}$$

end minus start

$$|\overrightarrow{AB}| = \sqrt{3^2 + 4^2} = 5$$ ← *distance AB*

$$\mathbf{m} = \tfrac{1}{2}\big((2,-1) + (5,3)\big) = \tfrac{1}{2}(7,2) = (3.5,\ 1)$$

midpoint

In 3D: A$(1,0,2)$, B$(3,-2,3)$: $\overrightarrow{AB} = (2,-2,1)$, length $\sqrt{4+4+1} = 3$, midpoint $(2,-1,2.5)$.

RULE: direction angle of a 2D vector

The angle $\mathbf{a} = x\mathbf{i} + y\mathbf{j}$ makes with the positive x-axis: $\theta = \tan^{-1}\dfrac{y}{x}$, then the **quadrant fix**.

Sketch it first. If $x < 0$ (pointing left, quadrants 2 and 3) add $180°$; if it points into quadrant 4, add $360°$ to get an angle from $0°$ to $360°$ (quadrants: Chapter 29; Chapter 33 reuses this fix for polar coordinates).

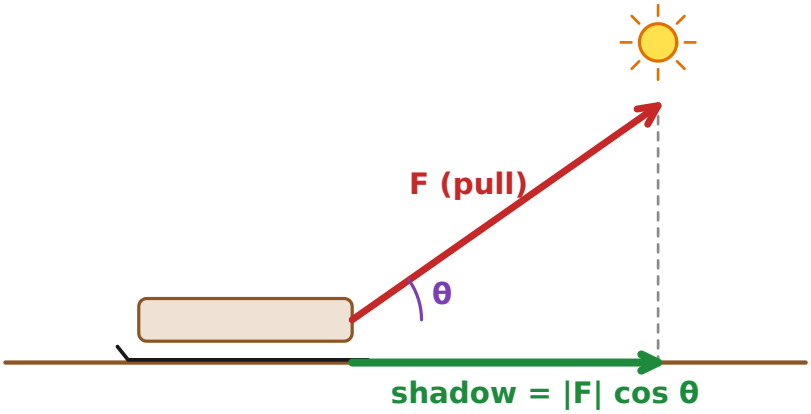
$$\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}: \quad \theta = \tan^{-1}\tfrac{4}{3} = 53.13°$$ ← *quadrant 1: as it is*

$$\mathbf{c} = -3\mathbf{i} + 4\mathbf{j}: \quad \tan^{-1}\tfrac{4}{-3} = -53.13° \to -53.13° + 180° = 126.87°$$

x < 0: add 180°

↗ The dot product: the rope's shadow

Pull a sled with a rope at an angle. Only the part of your pull **along the ground** moves the sled; the upward part is wasted. The **dot product** measures exactly that: how much one vector pulls in the direction of another.



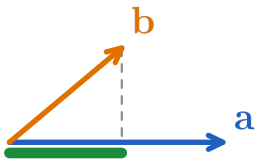
the sun is straight overhead. The rope's shadow on the ground is the useful part.

work done = |F| cos θ × distance

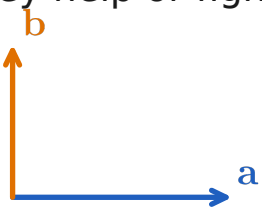
Put the two vectors tail to tail with angle θ between them. The dot product is the length of one times the length of the other's shadow on it:

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta$$ ← *say "a dot b"*

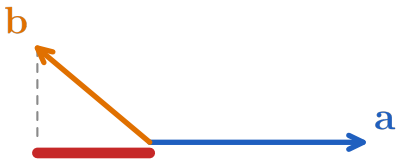
The sign tells you whether they help or fight:



positive: b helps a (angle under 90°)



zero: at right angles, no shadow at all



negative: the shadow points backwards

↗ Dot product with numbers

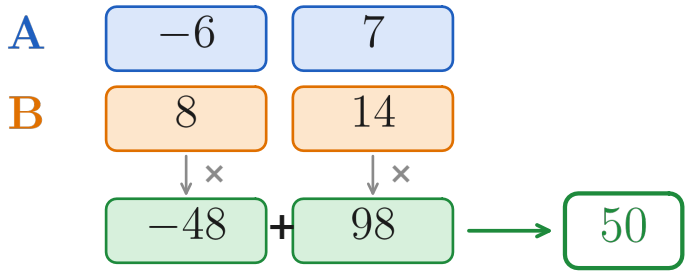
RULE: the dot product

Components: $\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y \,(+ a_z b_z)$. The answer is a **number**.

Angle form: $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta$, so $\cos\theta = \dfrac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}|\,|\mathbf{b}|}$.

$\mathbf{a} \cdot \mathbf{b} = 0$ means **perpendicular**; positive means the angle is under 90°; negative means over 90°.

With components you don't need the angle at all: **pair up** the matching parts, **multiply** each pair, **add**. The answer is a single number (a scalar), not an arrow.



Lecture 10: A = (−6, 7), B = (8, 14)

pair up → multiply → add = one plain number

$$\mathbf{A} \cdot \mathbf{B} = (-6)(8) + (7)(14) = -48 + 98 = 50$$

Now find the angle. Put the 50 into the angle form and undo the cosine (Chapter 28's $\cos^{-1}$):

$$|\mathbf{A}| = \sqrt{36 + 49} = 9.22, \quad |\mathbf{B}| = \sqrt{64 + 196} = 16.12$$

$$\cos\theta = \frac{50}{9.22 \times 16.12} = 0.336 \quad \leftarrow \textit{a·b ÷ both lengths}$$

$$\theta = \cos^{-1}(0.336) = 70.3°$$

Slide slip: Lecture 10 uses $\theta = 60°$ here, but $9.22 \times 16.12 \times \cos 60° = 74.3$, not 50. The true angle is 70.3°.

Lecture 10's second example: $\mathbf{A} = (2, 3)$, $\mathbf{B} = (-1, 5)$.

$$\mathbf{A} \cdot \mathbf{B} = 2 \times (-1) + 3 \times 5 = -2 + 15 = 13$$

$$\theta = \cos^{-1} \frac{13}{\sqrt{13}\sqrt{26}} = \cos^{-1} \frac{13}{3.6 \times 5.1} = \cos^{-1}(0.707) = 45°$$

The right-angle test. At 90° the shadow vanishes ($\cos 90^\circ = 0$), so **perpendicular vectors have dot product 0**. Lecture 10: $\mathbf{a} = 2\mathbf{i} + 0\mathbf{j}$ (along x), $\mathbf{b} = 0\mathbf{i} + 1\mathbf{j}$ (along y): $2 \times 0 + 0 \times 1 = 0$.

Where it turns up: work = force · displacement (the sled), power = force · velocity, and a solar panel's energy depends on the dot product of the sun's rays with the panel's direction.

↗ Find k: perpendicular or parallel

RULE: perpendicular and parallel vectors

Perpendicular: $\mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0$.

Parallel: $\mathbf{a} \parallel \mathbf{b} \iff \mathbf{b} = t\,\mathbf{a}$ for some number t: every component is multiplied by the **same** t (negative t: opposite directions).

PROCEDURE: finding an unknown k

1. Write both vectors with k in place.
2. Perpendicular: work out the dot product in terms of k, set it **= 0**, solve.
3. Parallel: write $\mathbf{b} = t\,\mathbf{a}$, find t from a component with no k, then use that t in the component with k.
4. Check by putting k back in.

Example 1: find k so that $3\mathbf{i} + k\mathbf{j}$ is perpendicular to $2\mathbf{i} - 6\mathbf{j}$.

$$(3)(2) + (k)(-6) = 0 \quad \longleftarrow \text{dot product = 0}$$

$$6 - 6k = 0 \;\Rightarrow\; k = 1$$

Example 2 (3D): $(k, 2, -1)$ and $(3, k, 4)$ are perpendicular.

$$3k + 2k - 4 = 0 \;\Rightarrow\; 5k = 4 \;\Rightarrow\; k = 0.8$$

Example 3: find k so that $2\mathbf{i} + k\mathbf{j}$ is parallel to $6\mathbf{i} + 9\mathbf{j}$.

$$6 = t \times 2 \;\Rightarrow\; t = 3 \quad \longleftarrow \text{x-parts give t}$$

$$9 = 3k \;\Rightarrow\; k = 3 \quad \longleftarrow \text{same t for y}$$

Example 4 (3D): $(4, -2, k)$ is parallel to $(-6, 3, 9)$. From the x-parts, $t = -6/4 = -1.5$; check y: $-2 \times (-1.5) = 3$ ✓; so $k \times (-1.5) = 9$ and $k = -6$.

✗ Common mistakes

✗ $\mathbf{a} \parallel \mathbf{b} \Rightarrow \mathbf{a} \cdot \mathbf{b} = 0$ ✓ $\mathbf{a} \parallel \mathbf{b} \Rightarrow \mathbf{b} = t\,\mathbf{a}$

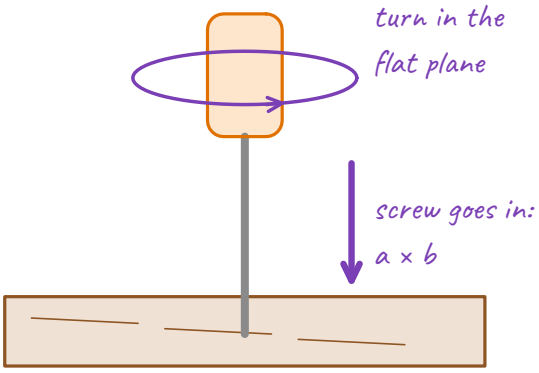
dot product 0 means perpendicular; parallel vectors have cos θ = ±1

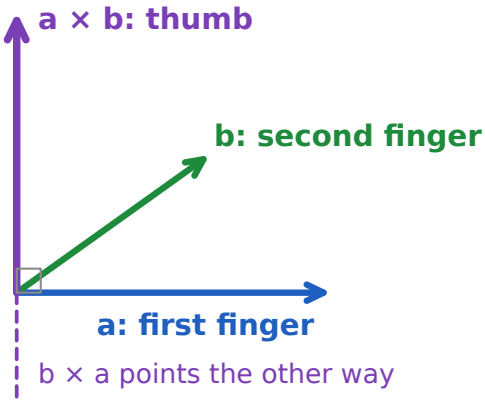
✗ $(2, k) \parallel (6, 9) :\ k = 5$ ✓ $t = 3, \quad k = 9 \div 3 = 3$

6 − 2 = 4 so k = 9 − 4? No: parallel means multiplied by the same t, not shifted

↗ The cross product: a screwdriver

The **cross product** $\mathbf{a} \times \mathbf{b}$ multiplies two vectors and gives a new **vector**, pointing at right angles to both of them. Think of a screwdriver: you turn the handle in a flat circle (that's where $\mathbf{a}$ and $\mathbf{b}$ live), and the screw moves straight in or out of the wood (that's the answer).





Right-hand rule (Lecture 10): point your right hand's first finger along $\mathbf{a}$ and your second finger along $\mathbf{b}$; your thumb points along $\mathbf{a} \times \mathbf{b}$. Swap the order and the thumb flips: $\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$. (Left-handers: use your right hand anyway!)

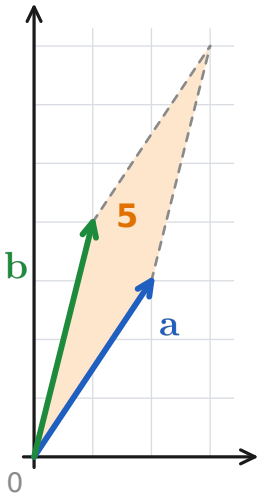
RULE: the cross product

$\mathbf{a} \times \mathbf{b}$ is a **vector** at right angles to both $\mathbf{a}$ and $\mathbf{b}$ (direction: right-hand rule). Order matters: $\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$.

Size: $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}| \sin\theta$ = area of the parallelogram.

2D shortcut: for (a, b) and (c, d) the area is $|ad - bc|$ (take the size; area is never negative).

How long is it? $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}|\,|\mathbf{b}| \sin\theta$, which is exactly the **area of the parallelogram** the two arrows make. Lecture 10: $\mathbf{a} = (2, 3)$, $\mathbf{b} = (1, 4)$.



$$|\mathbf{a}|\,|\mathbf{b}| = \sqrt{13}\sqrt{17} = 14.87$$

$$\mathbf{a}\cdot\mathbf{b} = 2\times 1 + 3\times 4 = 14$$

$$\theta = \cos^{-1}\frac{14}{14.87} = 19.65^\circ$$

$$|\mathbf{a}\times\mathbf{b}| = 14.87\sin 19.65^\circ = 5.00$$

shortcut in 2D: 2×4 − 3×1 = 5 (Chapter 16's ad − bc)

Slide slip: Lecture 10's answer is 11.18, but $14.87\sin 19.7^\circ = 5.0$. The parallelogram really has area 5. Where did 11.18 come from? $14.87\times\sin(19.7) = 11.18$ with the calculator in **radians**: Chapter 27's DEG/RAD trap.

Order and sign: $(1,4)$ and $(2,3)$ give $ad - bc = 1\times 3 - 4\times 2 = -5$. The minus sign only says $\mathbf{b}$ is clockwise from $\mathbf{a}$; the area is $|-5| = 5$.

Where it turns up: turning a screwdriver or spanner (turning force in the flat plane, movement along the axis), and the force on a wire carrying current in a magnetic field.

↗ The 3 × 3 determinant: cover up and work out

In 3D, the cross product has a recipe using a 3 × 3 **determinant** (Chapter 16 did the 2 × 2 kind: $ad - bc$). Put $\mathbf{i}, \mathbf{j}, \mathbf{k}$ along the top, then the two vectors. Lecture 10: $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}$, $\mathbf{b} = 4\mathbf{i} + 9\mathbf{j} + 2\mathbf{k}$.

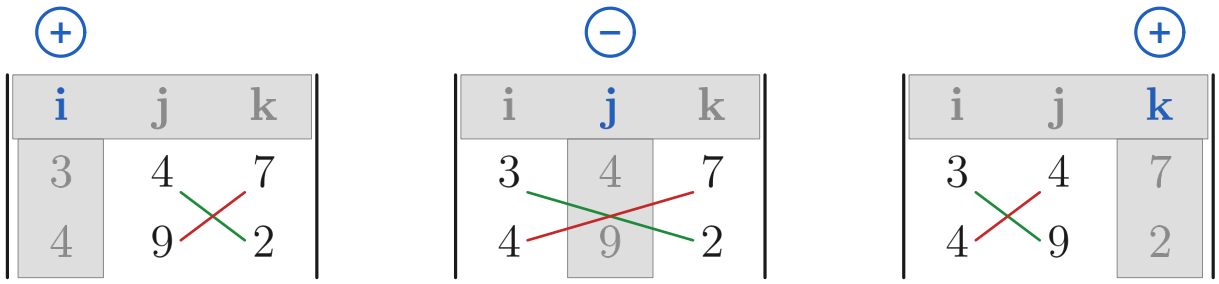
PROCEDURE: 3D cross product by determinant

1. Top row $\mathbf{i}, \mathbf{j}, \mathbf{k}$; row 2 is $\mathbf{a}$; row 3 is $\mathbf{b}$.
2. For each top-row letter, cover its row and column; work out the 2 × 2 left (**down-right minus up-right**).
3. Combine with signs **+ − +**: $+\mathbf{i}(\ldots) - \mathbf{j}(\ldots) + \mathbf{k}(\ldots)$.
4. Check: the answer dotted with $\mathbf{a}$ and with $\mathbf{b}$ must both give 0.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 4 & 7 \\ 4 & 9 & 2 \end{vmatrix}$$

← *top row: letters*
row 2: a
row 3: b

The cover-up method. For each letter on the top row, cover its row and its column with your fingers, and work out the little 2 × 2 that's left (down-right diagonal minus up-right diagonal). The signs go **+ − +**.



$+\mathbf{i}\,(4 \times 2 - 7 \times 9)$ $\quad -\mathbf{j}\,(3 \times 2 - 7 \times 4)$ $\quad +\mathbf{k}\,(3 \times 9 - 4 \times 4)$

$= -55\,\mathbf{i}$ $\quad = -(-22)\,\mathbf{j} = +22\,\mathbf{j}$ $\quad = +11\,\mathbf{k}$

$$\mathbf{a} \times \mathbf{b} = -55\,\mathbf{i} + 22\,\mathbf{j} + 11\,\mathbf{k}$$

Check it! The answer must be at right angles to **both** $\mathbf{a}$ and $\mathbf{b}$, so both dot products must be 0:

$$(-55)(3) + (22)(4) + (11)(7) = -165 + 88 + 77 = 0$$

✓ *perpendicular to a*

$$(-55)(4) + (22)(9) + (11)(2) = -220 + 198 + 22 = 0$$

✓ *perpendicular to b*

Slide slip: Lecture 10's final answer says $-11\mathbf{k}$, but the $\mathbf{k}$ part is $+(3 \times 9 - 4 \times 4) = +11$. The check catches it: with $-11\mathbf{k}$ the first dot product comes out as -154, not 0.

The general recipe, if you prefer letters:

$$\mathbf{a} \times \mathbf{b} = \mathbf{i}(a_y b_z - a_z b_y) - \mathbf{j}(a_x b_z - a_z b_x) + \mathbf{k}(a_x b_y - a_y b_x)$$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Components	$\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$
Add / scale	component by component; $k\mathbf{a}$ multiplies every part by k
Magnitude	$\lvert\mathbf{a}\rvert = \sqrt{x^2 + y^2 + z^2}$
Unit vector	$\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{\lvert\mathbf{a}\rvert}$ length L in that direction: $L\hat{\mathbf{a}}$
Position vectors	$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}, \qquad \mathbf{m} = \dfrac{\mathbf{a} + \mathbf{b}}{2}$ distance $AB = \lvert\mathbf{b} - \mathbf{a}\rvert$
Direction angle	$\theta = \tan^{-1}\dfrac{y}{x}$ then add 180° if $x < 0$, 360° if in quadrant 4
Dot product	$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \cos\theta$
Angle between	$\cos\theta = \dfrac{\mathbf{a} \cdot \mathbf{b}}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$
Perpendicular / parallel	$\mathbf{a} \cdot \mathbf{b} = 0$ / $\mathbf{b} = t\mathbf{a}$
Cross product size	$\lvert\mathbf{a} \times \mathbf{b}\rvert = \lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert \sin\theta$ parallelogram area; 2D: $\lvert ad - bc\rvert$
3D cross product	determinant with $\mathbf{i}, \mathbf{j}, \mathbf{k}$ on top, signs + − +; check both dot products are 0

↗ Your turn

Sketch every vector on squared paper first. Answers in grey.

Trips, adding, lengths

1. $p = 3i - 2j$ and $q = -i + 5j$. Find $p + q$ and $p - q$.
2. Find $|p|$, exactly and to 2 d.p.
3. Find $3q$ and its length.

1. $2i + 3j$; $4i - 7j$ 2. $\sqrt{13} \approx 3.61$ 3. $-3i + 15j$; $\sqrt{234} \approx 15.30$

Dot product

4. $(4, 1) \cdot (2, -3)$
5. The angle between $(1, 2)$ and $(3, 1)$
6. Are $2i + 3j$ and $6i - 4j$ perpendicular?

4. $8 - 3 = 5$ 5. $\cos^{-1}(5 \div (\sqrt{5} \times \sqrt{10})) = 45°$ 6. yes: $12 - 12 = 0$

Cross product

7. Area of the parallelogram made by $(3, 1)$ and $(1, 2)$
8. $(1, 2, 0) \times (0, 1, 3)$ by the cover-up method
9. Check your answer to 8 with two dot products.

7. $3 \times 2 - 1 \times 1 = 5$ 8. $6i - 3j + k$ 9. $6 - 6 + 0 = 0$ and $0 - 3 + 3 = 0$

Distance or displacement?

10. A bus drives 3 km east then 4 km north. Distance travelled? Size of the displacement?

10. 7 km; $\sqrt{(9 + 16)} = 5$ km

Find the unknown

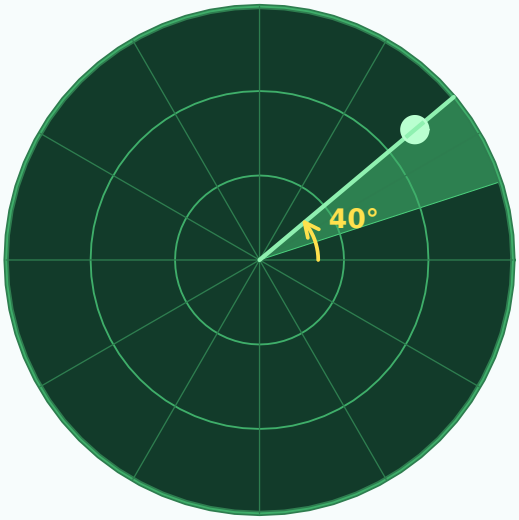
11. Find k so that $ki + 4j$ is perpendicular to $2i - j$.
12. Find k so that $ki + 6j$ is parallel to $2i + 3j$.
13. Find the unit vector in the direction of $5i - 12j$.
14. A is $(1, 3)$ and B is $(7, -5)$. Find AB, the distance AB and the midpoint.
15. Find the vector of length 15 in the direction of $3i + 4j$.
16. Find the direction angle of $-2i - 2j$ ($0°$ to $360°$).

11. $2k - 4 = 0$, $k = 2$ 12. $t = 2$ (from $3 \to 6$), so $k = 4$ 13. length 13: $(5/13)i - (12/13)j$ 14. $b - a = 6i - 8j$; $\sqrt{(36 + 64)} = 10$; $((1 + 7)/2, (3 - 5)/2) = (4, -1)$ 15. $15 \times (0.6i + 0.8j) = 9i + 12j$ 16. $\tan^{-1}(1) = 45°$; $x < 0$ so $+ 180°$: $225°$

Chapter 33

Polar coordinates

turn, then walk



The radar operator doesn't say "3 km east, 2.5 km north". She says "40° round from east, 4 km out": turn to face the angle, then walk the distance. (Maths angles go anticlockwise from east. A navigator's bearing goes clockwise from north.)

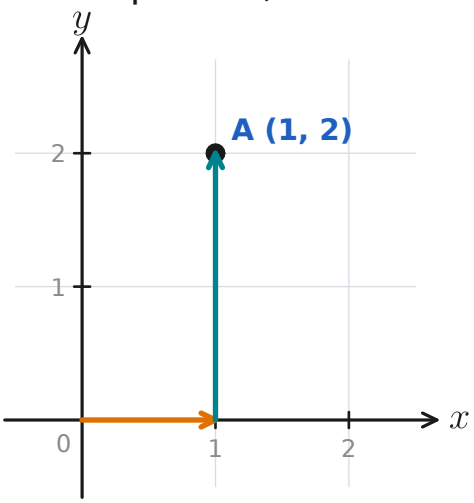
You've always said where things are with (x, y): so far across, so far up. **Polar coordinates** use a different pair: **how far from the centre** (r) and **which direction** (θ). Anything that spins or radiates (wheels, radar, waves, sound from a speaker) is easier this way.

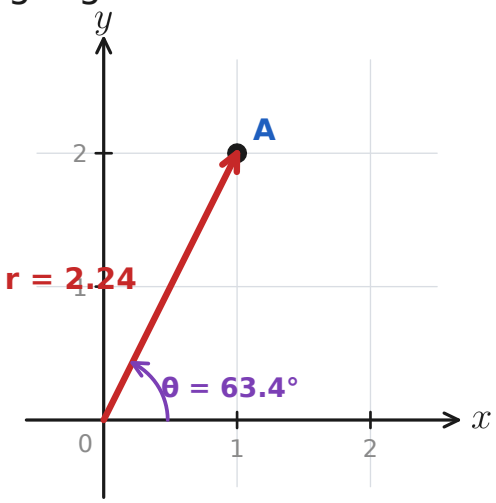
You'll translate between the two languages, dodge the calculator's blind spot, and draw some curves that would be horrible in x and y.

Goes with **Lecture 11** (Cartesian vs polar, converting, degrees vs radians, finding θ in each quadrant, polar curves, converting equations, the extended trig list, the cardioid).

Two ways to say where

The same point A, described in both languages:





Cartesian: walk 1 east, then 2 north

Polar: turn to 63.4°, then walk 2.24

Cartesian writes A as $(x, y) = (1, 2)$. Polar writes it as $(r, \theta) = (2.24, 63.4°)$. r is the **radius** (distance from the centre, the origin) and θ is the angle turned **anticlockwise from the positive x axis** (Chapter 27's convention).

Lecture 11: Cartesian is what graphs, slopes and vectors are built on; polar is how you describe circular motion and navigate on a round planet.

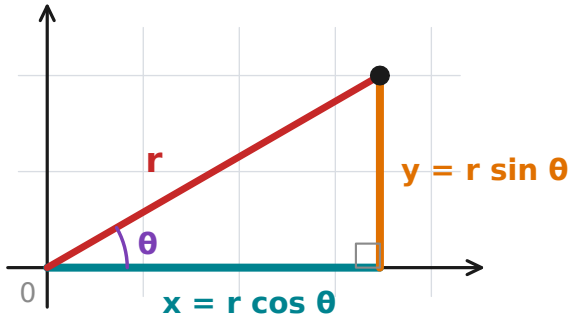
Polar to Cartesian: the triangle hiding inside

RULE: polar to Cartesian

$x = r\cos\theta$ and $y = r\sin\theta$.

This works for **any** angle: the signs of $\cos\theta$ and $\sin\theta$ put the point in the right quadrant automatically. No fix needed in this direction.

Drop a line from the point straight down to the x axis. You get a right-angled triangle with hypotenuse r and angle θ at the origin, so SOHCAHTOA (Chapter 28) does the rest:



$$x = r\cos\theta$$

CAH: adjacent = hyp × cos

$$y = r\sin\theta$$

SOH: opposite = hyp × sin

Example: a ship at $(r, \theta) = (4, 30°)$.

$$x = 4\cos 30° = 4 \times 0.866 = 3.46$$

$$y = 4\sin 30° = 4 \times 0.5 = 2$$

$$(r, \theta) = (4, 30°) \longrightarrow (x, y) = (3.46,\ 2)$$

Slide slip: Lecture 11's summary slide writes $y = x\sin\theta$. It should be $y = r\sin\theta$ (the hypotenuse is r, not x).

Cartesian to polar, and the calculator's blind spot

RULE: Cartesian to polar

$r = \sqrt{x^2 + y^2}$ (Pythagoras; r is never negative here).

$\theta = \tan^{-1}\dfrac{y}{x}$ **plus a correction** that depends on the quadrant: the calculator only answers between $-90°$ and $90°$.

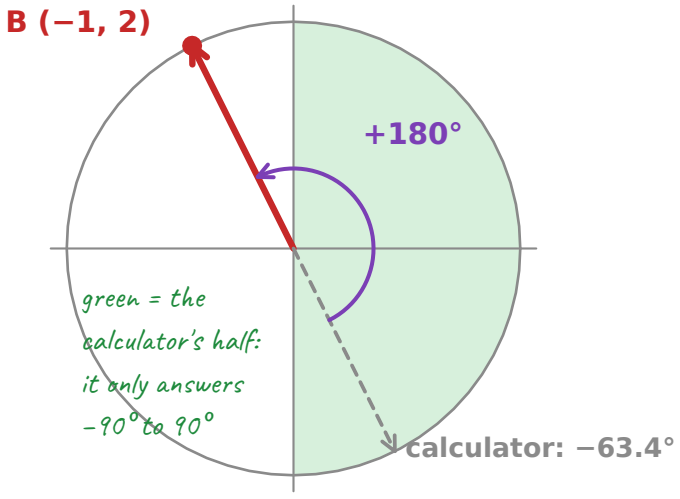
Always sketch the point first so you know which quadrant the answer must be in.

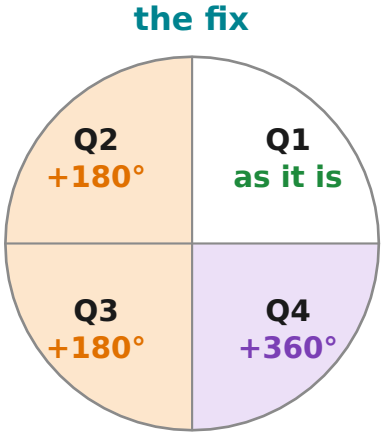
Going the other way, Pythagoras gives the distance and TOA gives the angle:

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \tan^{-1}\frac{y}{x}$$

Lecture 11: $A = (1, 2)$ in quadrant 1. $r = \sqrt{1+4} = \sqrt{5} = 2.24$, $\theta = \tan^{-1}\frac{2}{1} = 63.43°$ (1.107 rad). Perfect.

Now $B = (-1, 2)$, in quadrant 2. The calculator says $\tan^{-1}\frac{2}{-1} = \tan^{-1}(-2) = -63.43°$. That points down into quadrant 4: **the wrong way!**





Why +180°? The calculator can't tell $\frac{2}{-1}$ from $\frac{-2}{1}$: both are -2. So it always answers in the right-hand half. If the point is really on the left (quadrants 2 or 3), its direction is exactly opposite: turn another half-turn.

$$\theta_B = -63.43^\circ + 180^\circ = 116.57^\circ \quad (-1.107 + \pi = 2.03 \text{ rad})$$

The quadrant fix in full. Work out $\tan^{-1}\frac{y}{x}$ on the calculator, then:

Point is in	Signs	Add (degrees)	Add (radians)	θ ends up in
Quadrant 1	x > 0, y > 0	nothing	nothing	0° to 90° (0 to π/2)
Quadrant 2	x < 0, y > 0	180°	π	90° to 180° (π/2 to π)
Quadrant 3	x < 0, y < 0	180°	π	180° to 270° (π to 3π/2)
Quadrant 4	x > 0, y < 0	360°	2π	270° to 360° (3π/2 to 2π)

RULE: points on the axes

No calculator needed (and $\tan^{-1}\frac{y}{0}$ gives an error anyway). Read the angle off the sketch:
positive x axis 0; positive y axis $90^\circ = \frac{\pi}{2}$; negative x axis $180^\circ = \pi$; negative y axis $270^\circ = \frac{3\pi}{2}$. And r is just the distance from O.

$$(0, -3): \quad r = 3, \quad \theta = 270^\circ = \frac{3\pi}{2}$$ ← straight down

$$(-4, 0): \quad r = 4, \quad \tan^{-1}\frac{0}{-4} = 0^\circ \to 0^\circ + 180^\circ = 180^\circ = \pi$$

x < 0: the fix still works

Lecture 11's four test points. Watch the calculator column repeat itself:

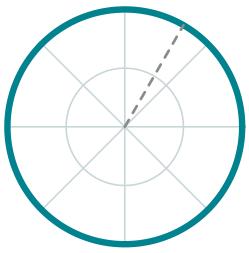
point	y ÷ x	calculator	quadrant fix	true angle
(1, 1)	1	45°	Q1: as it is	**45° = π/4**
(−1, 1)	−1	−45°	Q2: +180°	**135° = 3π/4**
(−1, −1)	1	45°	Q3: +180°	**225° = 5π/4**
(1, −1)	−1	−45°	Q4: +360°	**315° = 7π/4**

Slide slip: Lecture 11 labels 225° as $\frac{4\pi}{3}$; it's $\frac{5\pi}{4}$ ($\frac{225}{180} = \frac{5}{4}$). $\frac{4\pi}{3}$ would be 240°.

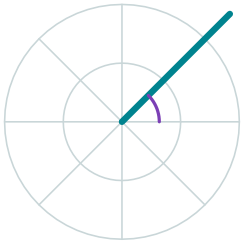
Degrees or radians? Lecture 11: **calculate in radians**, and present in degrees only when the context wants it (navigation). If a question names a format, use it; if not, either is fine. Check the calculator's DEG/RAD mode first (Chapter 27).

Drawing with a spinning pen

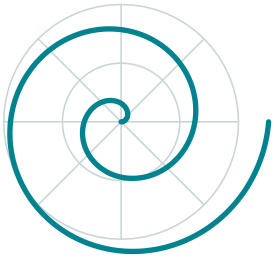
Picture a pen on a spinning arm, like the radar sweep. As the arm turns through θ, you slide the pen in or out to distance r. A **polar equation** is the rule that says how far out the pen is at each angle.



r = 2, any θ:
a circle

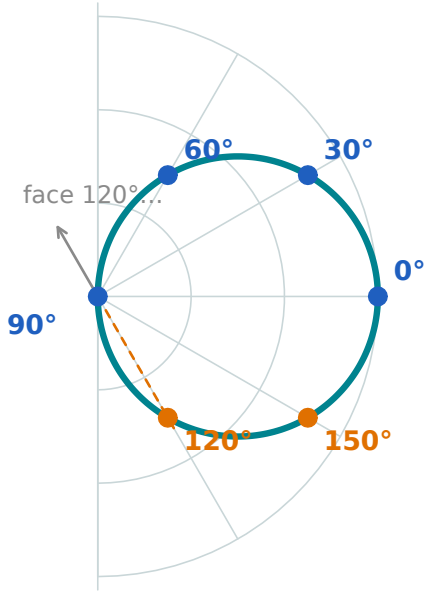


θ = 45°, any r:
a ray from the centre



r grows with θ
(r = 0.2θ): a spiral

Lecture 11's example: $r = 1.5\cos\theta$. Make a table, then plot each point by turning to θ and walking r:



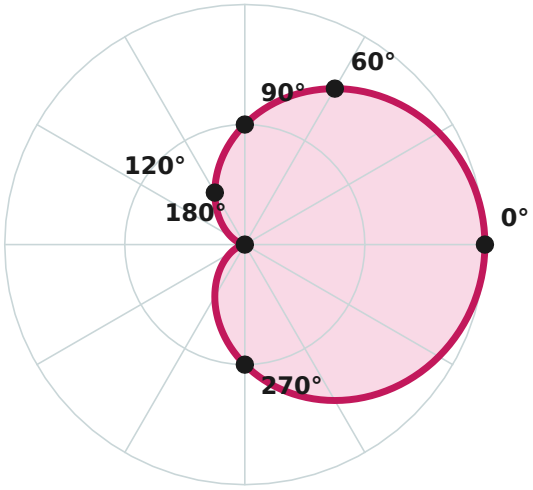
θ	cos θ	r = 1.5 cos θ
0°	1.00	**1.50**
30°	0.87	**1.30**
60°	0.50	**0.75**
90°	0.00	**0.00**
120°	−0.50	**−0.75**
150°	−0.87	**−1.30**
180°	−1.00	**−1.50**

Negative r: face θ, then walk BACKWARDS. So 120° lands where 300° would: the pen goes round the same circle again.

Surprise: it's a **circle** of diameter 1.5 sitting on the origin. In x and y that's $(x - 0.75)^2 + y^2 = 0.75^2$, which is much uglier to plot.

The cardioid: a heart from one line of maths

Change the rule to $r = 1 + \cos\theta$. The pen starts 2 out, comes in to 1 when facing up, and reaches 0 (the centre) facing left:



θ = 0°: r = 1 + 1 = 2 (far right)
θ = 90°: r = 1 + 0 = 1 (straight up)
θ = 120°: r = 1 − 0.5 = 0.5
θ = 180°: r = 1 − 1 = 0 (the dent)
θ = 270°: r = 1 (straight down)

"Cardioid" means heart-shaped. Microphones that hear mostly from the front have this pick-up pattern, and it turns up in roller and cam design (Lecture 11).

Converting equations: swap x and y out

RULE: converting equations, both ways

Cartesian → polar: replace x by $r\cos\theta$ and y by $r\sin\theta$ (and $x^2 + y^2$ by r^2), then make r the subject.

Polar → Cartesian: look for $r\cos\theta$, $r\sin\theta$ and r^2 and swap them for x, y and $x^2 + y^2$. If there is a lone $\cos\theta$ or $\sin\theta$, **multiply both sides by r** first.

A fixed angle $\theta = \alpha$ is the line through O: $\dfrac{y}{x} = \tan\alpha$.

To turn a Cartesian equation into a polar one: **replace every x by $r\cos\theta$ and every y by $r\sin\theta$**, collect the r terms, then make r the subject (Chapter 14).

Lecture 11: the straight line $y = 2x + 2$.

$$r\sin\theta = 2r\cos\theta + 2 \quad \longleftarrow \text{swap y and x}$$

$$r\sin\theta - 2r\cos\theta = 2 \quad \longleftarrow \text{r terms together}$$

$$r(\sin\theta - 2\cos\theta) = 2 \quad \longleftarrow \text{factor out r (Chapter 12)}$$

$$r = \frac{2}{\sin\theta - 2\cos\theta}$$ ← ÷ the bracket

Check one point: at $\theta = 90°$, $r = \frac{2}{1-0} = 2$: straight up 2, the point $(0, 2)$. And $y = 2x + 2$ does cross the y axis at 2. ✓

Lecture 11: the oval $x^2 + 2y^2 = 9$.

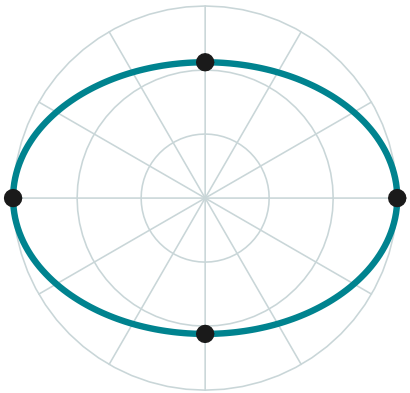
$$(r\cos\theta)^2 + 2(r\sin\theta)^2 = 9$$ ← swap

$$r^2(\cos^2\theta + 2\sin^2\theta) = 9$$ ← factor out r^2

$$r^2(\underbrace{\cos^2\theta + \sin^2\theta}_{=1} + \sin^2\theta) = 9$$ ← split 2 $\sin^2$ into $\sin^2 + \sin^2$, then Chapter 30's identity

$$r^2 = \frac{9}{1 + \sin^2\theta} \quad \Rightarrow \quad r = \frac{3}{\sqrt{1 + \sin^2\theta}}$$

Check: $\theta = 0$ gives $r = 3$, the point $(3, 0)$: $9 + 0 = 9$ ✓. $\theta = 90°$ gives $r = \frac{3}{\sqrt{2}} = 2.12$, the point $(0, 2.12)$: $2 \times 2.12^2 = 9$ ✓.



Lecture 11's "Polar curves 3" slide is this oval. As the pen turns, r changes smoothly: largest (r = 3) facing 0 and π, smallest (r = 2.12) facing π/2 and 3π/2. So the oval is wider than it is tall.

Converting back, and quick wins

Circle centred on O: $x^2 + y^2 = 9$.

$$r^2 = 9 \;\Rightarrow\; r = 3$$ ← $x^2 + y^2$ is r^2

Line through O: $y = x$.

$$r\sin\theta = r\cos\theta \;\Rightarrow\; \tan\theta = 1 \;\Rightarrow\; \theta = \tfrac{\pi}{4}$$ ← ÷ r cos θ

(With negative r allowed, $\theta = \frac{\pi}{4}$ covers the whole line, including the part in quadrant 3.)

Polar → Cartesian: $r = 4\cos\theta$. There is a lone $\cos\theta$, so multiply by r:

$$r^2 = 4r\cos\theta$$ ← × r

$$x^2 + y^2 = 4x$$ ← $r^2 \to x^2 + y^2$, r cos θ → x

$$x^2 - 4x + y^2 = 0 \;\Rightarrow\; (x-2)^2 + y^2 = 4$$ ← complete the square

A circle of radius 2 centred at $(2, 0)$, passing through O, just like $r = 1.5\cos\theta$ earlier.

✗ Common mistakes

✗ $r = 4\cos\theta \;\to\; r = 4x$ — x is r cos θ, not cos θ: multiply by r first

✓ $r^2 = 4r\cos\theta = 4x$

✗ $x^2 + y^2 = 9 \;\to\; r = 9$ — $x^2 + y^2$ is r^2, not r

✓ $r^2 = 9,\; r = 3$

Lecture 11's extended trig list

Converting equations often needs these. The first four are the compound angle rules (Chapter 30 met the first). Notice the sign **flips** in the cosine ones:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

Now just **set** $B = A$ in the "+" versions, and the **double angle** rules drop out:

$$\sin 2A = 2 \sin A \cos A \longleftarrow \sin A \cos A + \cos A \sin A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

The last two come from swapping $\sin^2 A = 1 - \cos^2 A$ (or the other way round) into the first one. Quick test with $A = 30°$: $\sin 60° = 0.866$ and $2 \times 0.5 \times 0.866 = 0.866$ ✓.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Polar → Cartesian	$x = r\cos\theta, \qquad y = r\sin\theta$ works for any angle; no fix needed
Distance	$r = \sqrt{x^2 + y^2}$
Angle	$\theta = \tan^{-1}\dfrac{y}{x}$ then the quadrant fix
Quadrant fix	Q1: nothing; Q2 and Q3: + 180° (+ π); Q4: + 360° (+ 2π)
Points on the axes	read off the sketch: 0, 90° (π/2), 180° (π), 270° (3π/2)
Links	$x^2 + y^2 = r^2, \qquad \tan\theta = \dfrac{y}{x}$
Cartesian → polar equation	swap $x \to r\cos\theta$, $y \to r\sin\theta$, make r the subject
Polar → Cartesian equation	multiply by r if needed, then swap $r\cos\theta \to x$, $r\sin\theta \to y$, $r^2 \to x^2 + y^2$
Simple curves	$r = a$: circle radius a; $\theta = \alpha$: line through O; $r = a\cos\theta$: circle through O
Double angle	$\sin 2A = 2\sin A\cos A, \quad \cos 2A = \cos^2 A - \sin^2 A$

Your turn

Sketch the point or curve first. For θ from $\tan^{-1}$, always ask "which quadrant?". Answers in grey.

Polar to Cartesian

1. $(r, \theta) = (6, 60°)$
2. $(r, \theta) = (2, 3\pi/4)$

1. (3, 5.20) 2. (−1.41, 1.41)

Cartesian to polar (mind the quadrant!)

3. (3, 4)
4. (−3, 4)
5. (−2, −2)
6. (5, −5)

3. (5, 53.13°) 4. (5, 126.87°) 5. (2.83, 225°) 6. (7.07, 315°)

Curves and equations

7. Sketch $r = 3$. What shape is it?
8. Where is the pen for $r = 1 + \cos\theta$ at $\theta = 90°$? Give x and y.
9. Write the line $x = 4$ in polar form.
10. Write the circle $x^2 + y^2 = 25$ in polar form.

7. a circle of radius 3 8. r = 1, the point (0, 1) 9. $r\cos\theta = 4$, so $r = 4/\cos\theta$ 10. $r^2 = 25$, so r = 5

Double angle

11. Use $\sin 2A = 2\sin A\cos A$ to find sin 90° from A = 45°.

11. $2 \times 0.7071 \times 0.7071 = 1$

Find the unknown

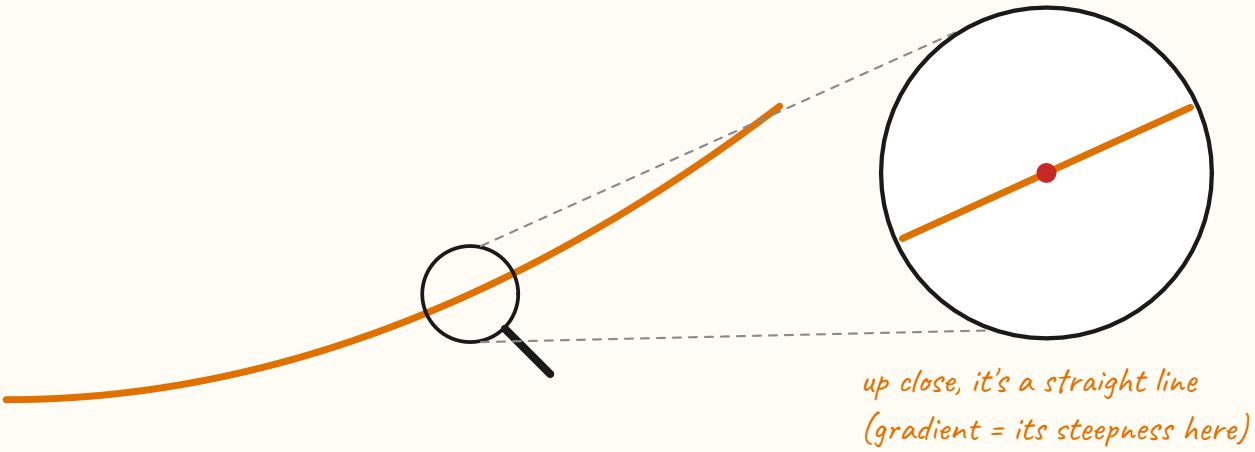
12. Write $r = 6\sin\theta$ in Cartesian form and describe the curve.
13. Write the line $y = \sqrt{3}\,x$ in polar form.
14. Write the point (0, −5) in polar form, in degrees and radians.
15. A point has r = 8, y = 4 and is left of the y axis. Find θ and x.
16. A point has x = −3 and θ = 135°. Find y and r.

12. × r: $r^2 = 6r\sin\theta$, $x^2 + y^2 = 6y$, $x^2 + (y - 3)^2 = 9$: circle, centre (0, 3), radius 3 13. $\tan\theta = \sqrt{3}$, θ = π/3 (60°) 14. on the negative y axis: (5, 270°) = (5, 3π/2) 15. $\sin\theta = 4/8 = 0.5$ and x < 0, so θ = 150°; $x = 8\cos 150° \approx -6.93$ 16. $y = x\tan 135° = (-3)(-1) = 3$; $r = \sqrt{(9 + 9)} \approx 4.24$

Chapter 34

Gradients of curves

zoom in until it's straight



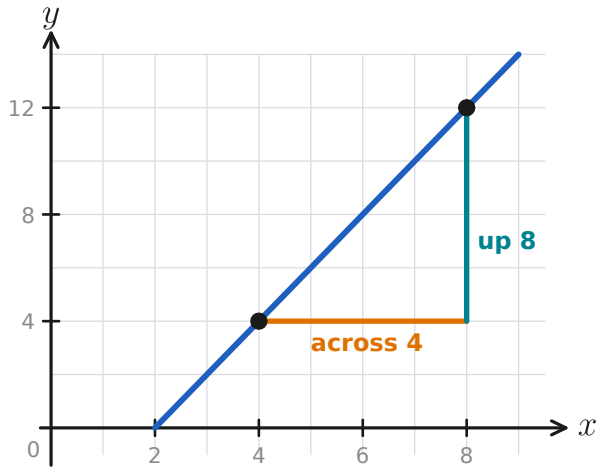
A straight line has the same steepness everywhere (Chapter 20). A curve doesn't: a road over a hill gets steeper, levels off at the top, then tips downwards. This chapter answers **"how steep is it right here?"**

The trick is in the picture: zoom in on any smooth curve and it looks straight, so it has a gradient at that spot. **Differentiation** is the method that finds that gradient for every point at once. It's the first half of calculus.

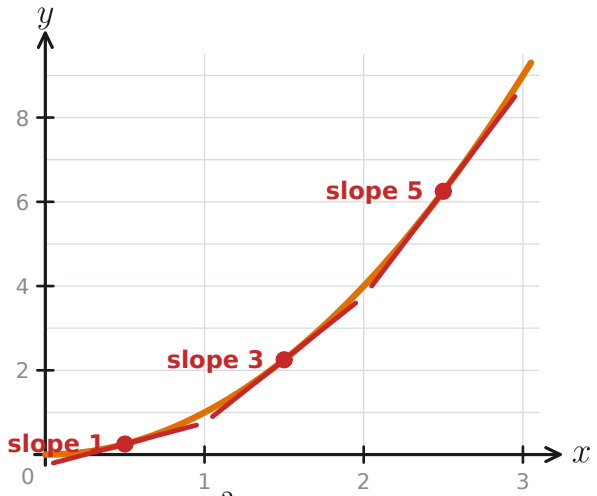
Goes with **Lecture 14** (straight-line recap, tangents, approximating the gradient, differentiation and notation, standard derivatives, stationary points, the second derivative).

A line has one gradient; a curve has many

Recap (Lecture 14): the line through $(4, 4)$ and $(8, 12)$ goes up 8 while it goes across 4, so $m = \frac{8}{4} = 2$. Then $4 = 2 \times 4 + c$ gives $c = -4$: the line is $y = 2x - 4$.



a line: **one** gradient everywhere

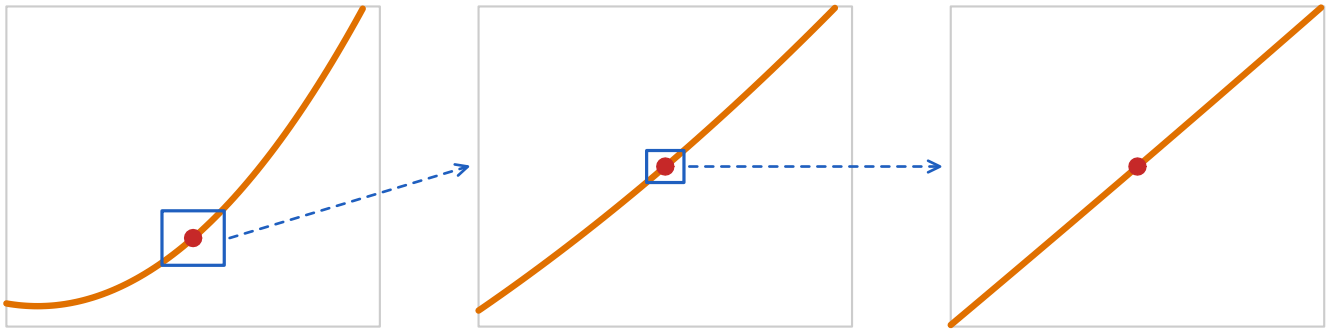


the curve $y = x^2$: a **different** gradient at every point

The gradient of a curve at a point is the gradient of its **tangent** there: the straight line that just touches the curve at that point, heading the same way (the red lines).

Zoom in until it's straight

Lecture 14 zooms in on $y = x^2$. Here's the same idea around the point $(1, 1)$:



×1: clearly curved | **×10: less curved** | **×100: a straight line, gradient 2**

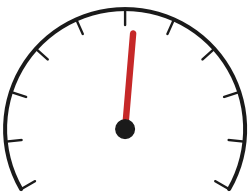
Lecture 14: "The slope that we see is 'roughly' correct, but only for that location!" Zoom in somewhere else and you'd see a line with a different slope.

Average speed vs the speedometer

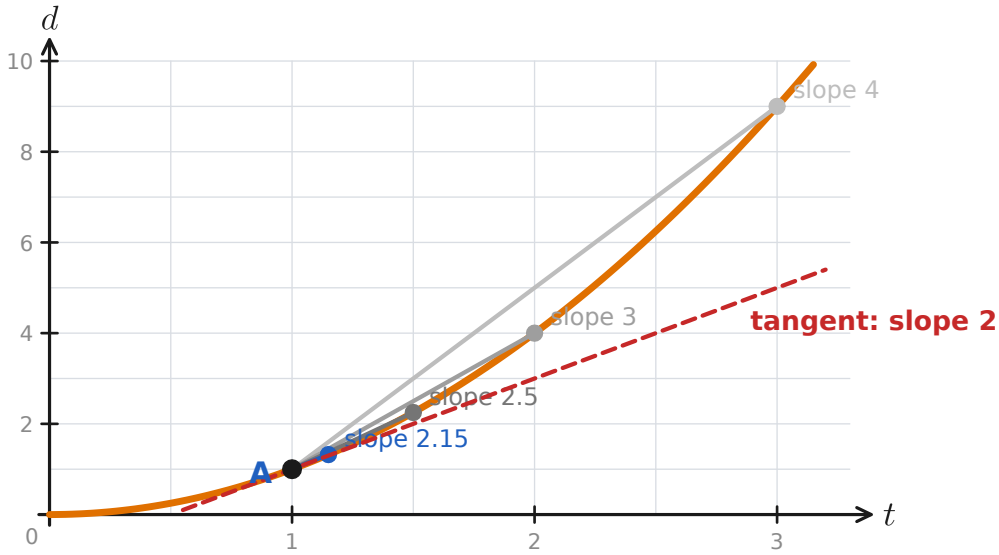
A trip computer says you averaged 40 mph (60 miles in 1.5 hours). The speedometer says what you're doing **right now**. On a distance–time graph:

• the **average** is the gradient of a **chord**: a straight line joining two points of the curve;
• **right now** is the gradient of the **tangent**.

To turn one into the other, slide the second point closer and closer to the first:



right now: 52 mph



The chord slopes 4, 3, 2.5, 2.15 close in on 2: the speed right now at A.

Lecture 14's shrinking step

With numbers: take a small step h along the x axis. The chord from $x = a$ to $x = a + h$ has gradient rise ÷ run:

$$f'(a) \approx \frac{f(a+h) - f(a)}{h}$$ ← *rise ÷ run, and the run is h*

Lecture 14: the gradient of $f(x) = 2x^3 - 3x^2$ at the point $(2, 4)$. (Check the point: $f(2) = 16 - 12 = 4$ ✓.) Shrink h and watch the last column:

step h	f(2 + h)	rise = f(2 + h) − 4	rise ÷ h
1	27	23	**23**
0.1	5.292	1.292	**12.92**
0.01	4.120902	0.120902	**12.0902**
0.001	4.012009	0.012009	**12.009**
0.00001	4.00012	0.000120001	**12.0001**

→ 12

The answers home in on **12**. Now imagine h shrinking **all the way** to nothing. That's **differentiation**, and Lecture 14 lists its two advantages over the table:

1. it gives the gradient at **any** point, not just at $x = 2$;
2. it's **exact** (12, not 12.0001).

The phrasebook: how to read the symbols

$\frac{\mathrm{d}y}{\mathrm{d}x}$ — *"dee y by dee x": the gradient of y as x changes (the derivative)*

$f'(x)$ — *"f dash of x" (or "f prime of x"): the same thing for a function f*

$\delta x,\ \delta y$ — *"delta x": a small step in x (δ is small Greek delta). δy is the matching small step in y*

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ — *"dee two y by dee x squared": differentiate twice (later in this chapter)*

$\frac{\partial y}{\partial x}$ — *"partial y by partial x": when y depends on more than one letter (Chapter 38)*

$\dot{f}(x),\ \dot{y}$ — *"f dot": Newton's dot notation. Lecture 14 writes f(x) for the derivative; most books keep the dot for a rate over time t: ẏ = dy/dt*

Careful: $\frac{\mathrm{d}y}{\mathrm{d}x}$ is **one symbol**, not a fraction of $d \times y$ over $d \times x$. You can't cancel the d's.

The power rule

RULE: the power rule (and constants)

$\frac{\mathrm{d}}{\mathrm{d}x}(ax^n) = anx^{n-1}$: the power comes down and multiplies the number in front, then the power drops by 1. It works for **any** number n: whole, negative or a fraction.

A constant on its own differentiates to **0**: $\frac{\mathrm{d}}{\mathrm{d}x}(7) = 0$.

A term kx differentiates to k: $\frac{\mathrm{d}}{\mathrm{d}x}(5x) = 5$.

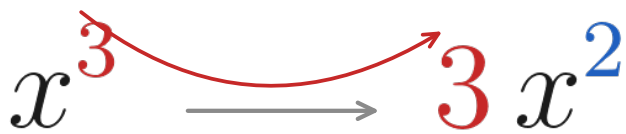
Several terms added or subtracted: differentiate each term on its own and keep the + and − signs.

Every term must look like ax^n **before** you use the rule: rewrite roots and fractions first (box later in this section).

Where the rule comes from. Doing the shrinking-step table every time would be exhausting. So let's measure some gradients (with a tiny h, exactly as above) and hunt for a pattern:

curve	at x = 1	at x = 2	at x = 3	pattern?
$y = x^2$	2	4	6	$2x$
$y = x^3$	3	12	27	$3x^2$
$y = x^4$	4	32	108	$4x^3$

Look at x^3: the gradients 3, 12, 27 are 3×1^2, 3×2^2, 3×3^2. The power **dives to the front**, and what's left has power **one smaller**:



the power jumps down in front...

...and the power goes down by one: 3 − 1 = 2

If a number is already in front, the diving power multiplies it: $5x^4 \to 5 \times 4\,x^3 = 20x^3$. In general:

$$y = ax^n \quad \Rightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = a\,n\,x^{n-1}$$

← *a and n are constants*

A constant has gradient 0: $y = 2$ is a flat line. The rule agrees: $y = 2x^0$ gives $0 \times 2x^{-1} = 0$ (Lecture 14).

Several terms? Differentiate each one separately and keep the + and − signs.

$$y = x \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = 1x^0 = 1$$ ← x is x^1

$$y = 2x + 3 \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = 2 + 0 = 2$$ ← the +3 just lifts the line

$$f(x) = x^3 + 3x \;\Rightarrow\; f'(x) = 3x^2 + 3$$

$$f(z) = 3z^3 - 2z^2 + z \;\Rightarrow\; f'(z) = 9z^2 - 4z + 1$$

any letter works

RULE: rewrite as powers of x before differentiating (Chapter 9)

$\frac{4}{x^3} = 4x^{-3}$: only the x^3 moves up and its power turns negative; the 4 stays a multiplier. It is **not** $(4x)^{-3}$.

$\sqrt{x} = x^{\frac{1}{2}}$, $\sqrt[3]{x} = x^{\frac{1}{3}}$, $x\sqrt{x} = x^1 \times x^{\frac{1}{2}} = x^{\frac{3}{2}}$ (add the powers).

$\frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$, $\frac{3}{2x^2} = \frac{3}{2}x^{-2}$ (numbers on the bottom stay as a fraction in front).

After differentiating you may turn negative and fractional powers back into fractions and roots.

$$y = \frac{4}{x^3} = 4x^{-3} \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = 4(-3)x^{-4} = -\frac{12}{x^4}$$

−3 − 1 = −4

$$y = \frac{1}{x^2} = x^{-2} \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = -2x^{-3} = -\frac{2}{x^3}$$

$$y = x\sqrt{x} = x^{\frac{3}{2}} \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3}{2}x^{\frac{1}{2}} = \frac{3}{2}\sqrt{x}$$ ← 3/2 − 1 = 1/2

$$y = \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{2}x^{-\frac{3}{2}}$$ ← $-\frac{1}{2} - 1 = -\frac{3}{2}$

$$y = \sqrt[4]{x} = x^{\frac{1}{4}} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{4}\,x^{-\frac{3}{4}} = \frac{1}{4x^{3/4}}$$ ← $\frac{1}{4} - 1 = -\frac{3}{4}$

✗ Common mistakes

✗ $\frac{4}{x^3} = (4x)^{-3}$

only the x^3 moves; the 4 stays in front

✓ $\frac{4}{x^3} = 4x^{-3}$

✗ $\frac{\mathrm{d}}{\mathrm{d}x}(5x) = 0$

5x is a line of gradient 5; only a constant on its own gives 0

✓ $\frac{\mathrm{d}}{\mathrm{d}x}(5x) = 5$

✗ $\sqrt{x} \to \frac{1}{2}\sqrt{x}$

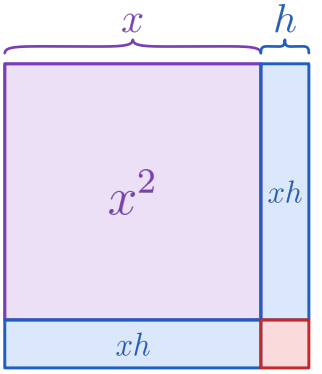
rewrite as a power first, then take 1 off the power

✓ $x^{\frac{1}{2}} \to \frac{1}{2}x^{-\frac{1}{2}}$

Lecture 14's sneaky one: $x = \frac{1}{3}(y+2)$. Make y the subject first (Chapter 14): $y = 3x - 2$, so $\frac{\mathrm{d}y}{\mathrm{d}x} = 3$.

Why the power rule works: grow a square, grow a cube

The table found the pattern; a picture shows **why**. Think of $y = x^2$ as the **area of a square** with side x. Nudge the side up by a tiny amount h. How much new area appears?



(h drawn far bigger than it really is)

New area = two thin strips (x by h each) plus a tiny corner (h by h).

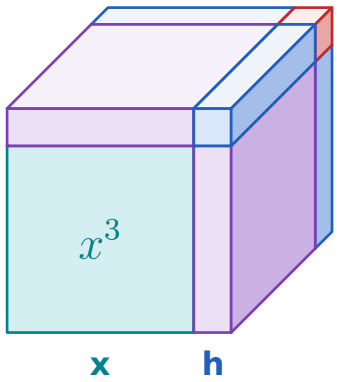
Gradient = new area ÷ nudge:

$$\frac{2xh + h^2}{h} = 2x + h$$

shrink h to nothing and only 2x is left

So the **2 in $2x$ is the two strips**: the square grows along two edges. (It's Chapter 11's $(x+1)^2$ picture with the 1 shrunk down to a tiny h.)

Now $y = x^3$: the volume of a cube. Nudge every edge by h. The cube grows the same way in 3D: three thin **slabs** (x by x by h, one on the top, one on the right, one hidden on the back), three thin **rods** along the edges where slabs meet, and a speck of a cube in the corner:



3 slabs: 3 × x²h

3 rods: 3 × xh²

1 speck: h³

$$\frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2$$

shrink h: the rods and the speck vanish, the slabs stay. Gradient $3x^2$

The **3 in $3x^2$ is the three slabs**: a cube grows on three faces, and each slab has area x^2. You can't draw x^4, but the pattern carries on: n "slabs" of size x^{n-1}, which is exactly the power diving down to the front and dropping by one.

The standard derivatives

Lecture 14's list. Learn these like vocabulary: every harder derivative is built from them. (Angles in radians, Chapter 27.)

function y	derivative dy/dx	
c	0	flat line
ax^n	anx^{n-1}	the power rule
$\ln x$	$\frac{1}{x}$	natural log (Chapter 22)
ae^{kx}	ake^{kx}	e is its own gradient; the k comes from the inside kx
$a\sin(bx)$	$ab\cos(bx)$	the b comes from the inside bx
$a\cos(bx)$	$-ab\sin(bx)$	note the minus
$\tan x$	$\sec^2 x$	sec x = 1/cos x

Where the extra b and k come from. In $\sin(bx)$ the angle bx changes b times as fast as x does, so every gradient is b times bigger: $\frac{\mathrm{d}}{\mathrm{d}x}\sin(bx) = b\cos(bx)$. The same happens with $e^{kx} \to ke^{kx}$. This is the **chain rule** (Chapter 35): differentiate the outside function, then multiply by the derivative of the inside ($bx \to b$, $kx \to k$). A number a in front just multiplies the answer.

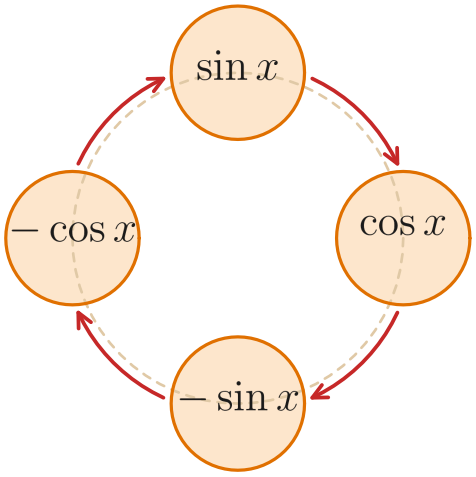
$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(3x) = 3\cos(3x)$$ ← inside 3x → 3

$$\frac{\mathrm{d}}{\mathrm{d}x}5e^{-2x} = 5 \times (-2)e^{-2x} = -10e^{-2x}$$ ← k = −2

$$\frac{\mathrm{d}}{\mathrm{d}x}4\cos\left(\frac{x}{2}\right) = -4 \times \frac{1}{2}\sin\left(\frac{x}{2}\right) = -2\sin\left(\frac{x}{2}\right)$$

b = ½; cos gives a minus

The sin–cos wheel. Differentiating sine and cosine goes round in a circle. Each step clockwise is one differentiation:



sin → cos → −sin → −cos → back to sin.

Four differentiations bring you home. (Powers vanish instead: $x^3 \to 3x^2 \to 6x \to 6 \to 0$.)

Gradients are rates of change

A gradient is "how fast one thing changes as another changes". With time t instead of x you write $\frac{\mathrm{d}}{\mathrm{d}t}$. Lecture 14: velocity $v = 3t^2 + 2$.

$$\frac{\mathrm{d}v}{\mathrm{d}t} = 6t$$ ← *power rule; the +2 vanishes*

$$t = 3: \quad \frac{\mathrm{d}v}{\mathrm{d}t} = 6 \times 3 = 18$$

How fast velocity changes is **acceleration**, so at $t = 3$ the acceleration is 18 (m/s² if v is in m/s). Likewise, the gradient of distance against time is velocity: the speedometer.

The gradient at a point, and finding unknowns

RULE: gradient at a point

The gradient of $y = f(x)$ at $x = a$ is $f'(a)$: differentiate **first**, then put $x = a$ into $\frac{dy}{dx}$.

Never substitute into y first: y at one point is just a number, and a number differentiates to 0.

Given a point (a, b), only the x-value a goes into $\frac{dy}{dx}$.

Example: the gradient of $y = x^3 - 2x^2 + 5$ at the point $(2, 5)$.

$$\frac{dy}{dx} = 3x^2 - 4x$$ ← differentiate first

$$x = 2: \quad 3(2)^2 - 4(2) = 12 - 8 = 4$$ ← then substitute x = 2

PROCEDURE: given the gradient, find the unknown

1. Differentiate, treating any unknown letter (a, k, p...) as an ordinary number.
2. Write "$\frac{dy}{dx}$ at the given x = the given gradient" as an equation.
3. Solve it. If the question asks for a **point**, put each x into the **original** y equation to get its y-coordinate.
4. Check by substituting back.

Example: $y = ax^2 + 3x$ has gradient 7 at $x = 1$. Find a.

$$\frac{dy}{dx} = 2ax + 3$$ ← a is just a number

$$2a(1) + 3 = 7 \Rightarrow 2a = 4 \Rightarrow a = 2$$ ← x = 1, gradient 7

Example: where on $y = x^2 - 4x + 1$ is the gradient 2?

$$2x - 4 = 2 \Rightarrow x = 3$$ ← set dy/dx equal to 2

$$y = 3^2 - 4(3) + 1 = -2$$ ← y from the ORIGINAL equation

The point is **(3, −2)**.

Example: find the points on $y = x^3 - 3x^2$ where the gradient is 9.

$$3x^2 - 6x = 9 \;\Rightarrow\; x^2 - 2x - 3 = 0$$ ← ÷ 3 and rearrange

$$(x-3)(x+1) = 0 \;\Rightarrow\; x = 3 \text{ or } x = -1$$ ← factorise (Chapter 12)

$$y(3) = 27 - 27 = 0, \qquad y(-1) = -1 - 3 = -4$$

original equation

Two points: **(3, 0)** and **(−1, −4)**. A quadratic for x often gives two answers; keep both.

Tangents and normals

RULE: tangent and normal

The **tangent** at a point touches the curve there and has the curve's gradient, $m = f'(a)$.

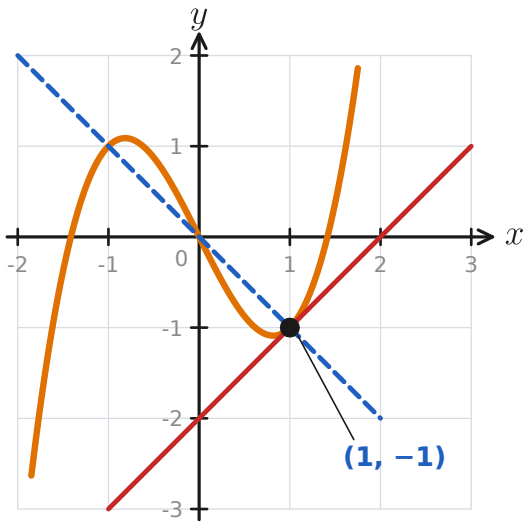
The **normal** is the line through the same point at right angles to the tangent. Its gradient is $-\frac{1}{m}$ (Chapter 20: perpendicular gradients multiply to -1).

Both are straight lines: write them with Chapter 20's point-gradient form $y - y_1 = m(x - x_1)$.

PROCEDURE: equation of the tangent or normal at x = a

1. If only $x = a$ is given, find y_1 by putting a into the **original** equation. The point is (a, y_1).
2. Differentiate.
3. Tangent gradient: $m = f'(a)$ (put a into $\frac{\mathrm{d}y}{\mathrm{d}x}$).
4. Tangent: $y - y_1 = m(x - x_1)$, then tidy into $y = mx + c$ (or $ax + by + c = 0$ if asked).
5. Normal: same point, gradient $-\frac{1}{m}$. (If $m = 0$ the tangent is flat and the normal is the vertical line $x = a$.)

Example: the tangent and normal to $y = x^3 - 2x$ at $x = 1$.



orange: the curve $y = x^3 - 2x$
red: the tangent, gradient 1
blue dashes: the normal, gradient −1

The normal crosses the tangent at a right angle (the grid squares are true squares here).

$$y_1 = 1^3 - 2(1) = -1$$ ← *step 1: the point is (1, −1)*

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 2$$ ← *step 2*

$$m = 3(1)^2 - 2 = 1$$ ← *step 3*

$$y - (-1) = 1(x - 1) \;\Rightarrow\; y = x - 2$$ ← *step 4: tangent*

$$m_N = -\tfrac{1}{1} = -1 : \quad y + 1 = -(x - 1) \;\Rightarrow\; y = -x$$

step 5: normal

Example: the tangent and normal to $y = \frac{2}{x}$ at $(1, 2)$.

$$y = 2x^{-1} \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = -2x^{-2} = -\frac{2}{x^2}$$ ← *rewrite first*

$$m = -\frac{2}{1^2} = -2$$ ← *point given, so skip step 1*

$$y - 2 = -2(x - 1) \;\Rightarrow\; y = -2x + 4$$ ← *tangent*

$$m_N = \tfrac{1}{2} : \quad y - 2 = \tfrac{1}{2}(x - 1) \;\Rightarrow\; y = \tfrac{1}{2}x + \tfrac{3}{2}$$

normal: flip and change sign

As $ax + by + c = 0$ the normal is $x - 2y + 3 = 0$ (multiply by 2 and collect).

✗ Common mistakes

✗ $m = f(a)$ ✓ $m = f'(a)$

the gradient comes from dy/dx, not from y

✗ $m_N = -m$ ✓ $m_N = -\frac{1}{m}$

flip the fraction AND change the sign

✗ y_1 from $\frac{\mathrm{d}y}{\mathrm{d}x}$ ✓ y_1 from $y = f(x)$

the point lies on the curve, so use the curve's equation

Stationary points: hilltops, valleys and ledges

RULE: stationary points

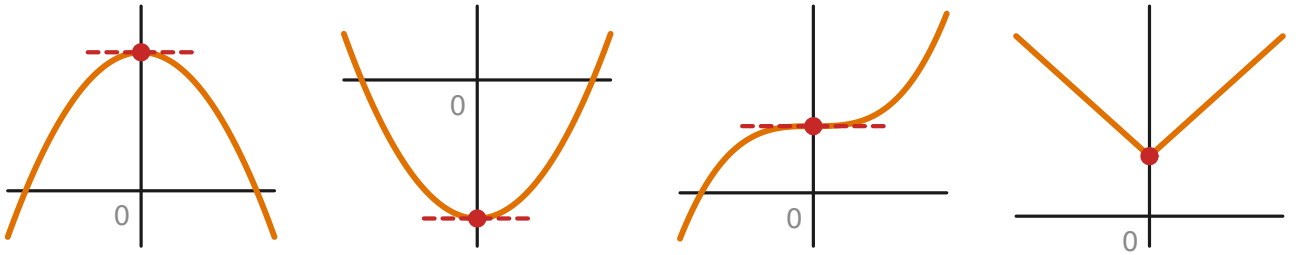
A **stationary point** is where $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$: the tangent is horizontal.

It is a **local maximum** (hilltop), a **local minimum** (valley) or a **stationary point of inflection** (a ledge, where the curve carries on the same way).

PROCEDURE: finding and classifying stationary points

1. Differentiate to get $\frac{\mathrm{d}y}{\mathrm{d}x}$.
2. Set $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$.
3. Solve for x (factorise or use the quadratic formula; there may be more than one answer).
4. Put each x into the **original** y equation to get the y-coordinate. (Not into $\frac{\mathrm{d}y}{\mathrm{d}x}$: that is 0 there by definition.)
5. Classify each one: second-derivative test (next section), or the sign of the gradient just either side: $+, 0, -$ hilltop; $-, 0, +$ valley; same sign both sides: inflection.

Where the gradient is **exactly 0** the curve has stopped rising or falling, for a moment. Lecture 14's gallery:



local maximum (0, 3): stationary and turning

local minimum (0, −3): stationary and turning

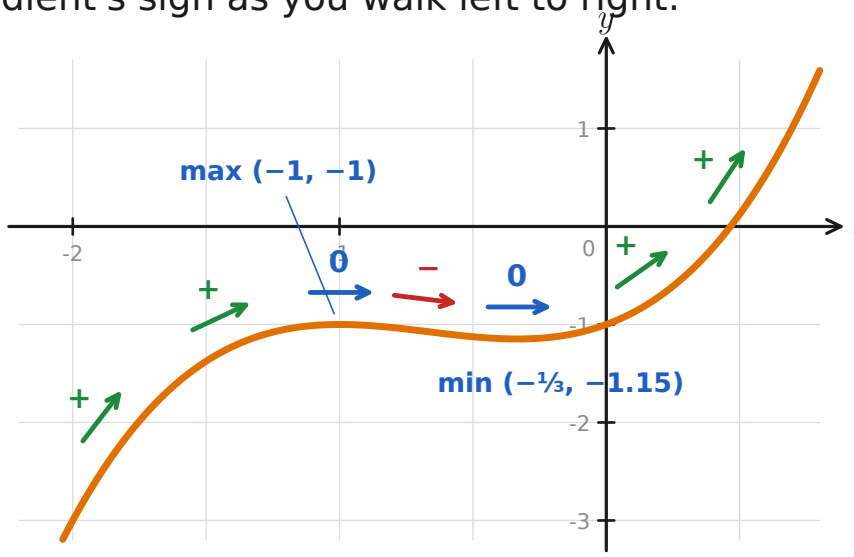
inflection (0, 2): stationary, NOT turning

sharp corner (0, 1): turning, NOT stationary

Lecture 14, by the procedure: $y = -x^2 + 3$. Step 1: $\frac{\mathrm{d}y}{\mathrm{d}x} = -2x$. Steps 2–3: $-2x = 0$ only when $x = 0$. Step 4: $y = -0^2 + 3 = 3$. The stationary point is $(0, 3)$.

Careful with words: the slide says differentiating finds the curve's "zeros". It finds where the **gradient** is zero (stationary points), not where the curve crosses the x axis.

Lecture 14's big example: $y = x^3 + 2x^2 + x - 1$. The arrows show the gradient's sign as you walk left to right:



green +: uphill
blue 0: flat
red –: downhill

+ then 0 then –:
a hilltop

– then 0 then +:
a valley

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 + 4x + 1 = 0$$ ← set the gradient to zero

$$x = \frac{-4 \pm \sqrt{16 - 12}}{6} = \frac{-4 \pm 2}{6}$$ ← quadratic formula, a = 3, b = 4, c = 1 (Chapter 15)

$$x = -\frac{1}{3} \quad \text{or} \quad x = -1$$

Which kind is $x = -1$? Test the gradient just either side (Lecture 14):

$$x = -1.1 : \; 3(-1.1)^2 + 4(-1.1) + 1 = 3.63 - 4.4 + 1 = 0.23$$

\+ uphill

$$x = -0.9 : \; 3(-0.9)^2 + 4(-0.9) + 1 = 2.43 - 3.6 + 1 = -0.17$$

– downhill

Uphill, flat, downhill: $x = -1$ is a **maximum**, at $y = -1 + 2 - 1 - 1 = -1$.

The second derivative: smile or frown?

RULE: the second-derivative test

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ means: differentiate $\frac{\mathrm{d}y}{\mathrm{d}x}$ again. Also written $f''(x)$.

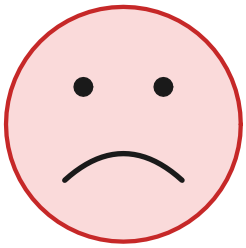
At a stationary point $x = a$, work out $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$ at $x = a$:

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} > 0$: **minimum**. $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} < 0$: **maximum**.

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$: the test is **inconclusive** (it could be a maximum, a minimum or an inflection). Use the gradient-sign test either side instead.

Differentiate the derivative and you get the **second derivative** $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$: how fast the **gradient itself** is changing (Lecture 14).

positive: gradient growing
(−, 0, +) → a valley: minimum



negative: gradient shrinking
(+, 0, −) → a hilltop: maximum

Lecture 14 finishes the big example:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 + 4x + 1 \quad \Rightarrow \quad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x + 4 \leftarrow \textit{differentiate again}$$

$$x = -1: \;\; 6(-1) + 4 = -2 < 0 \leftarrow \textit{frown: maximum} \checkmark$$

$$x = -\tfrac{1}{3}: \;\; 6(-\tfrac{1}{3}) + 4 = 2 > 0 \leftarrow \textit{smile: minimum}$$

Why zero decides nothing. $y = x^3$ and $y = x^4$ both have $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ and $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$ at $x = 0$. Yet x^3 has a ledge there (gradient $+, 0, +$) and x^4 has a minimum (gradient $-, 0, +$). Only the either-side test can tell them apart.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Notation

$$\frac{dy}{dx} = f'(x) = \dot{f}(x)$$

the gradient of the tangent; the dot is mostly used for rates over time

Power rule

$$\frac{d}{dx}(ax^n) = anx^{n-1}$$

any n: whole, negative or fraction

Constants, kx, sums

$$\frac{d}{dx}(c) = 0, \qquad \frac{d}{dx}(kx) = k$$

differentiate a sum term by term, keeping the signs

Rewrite first

$$\frac{a}{x^n} = ax^{-n}, \quad \sqrt{x} = x^{\frac{1}{2}}, \quad \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$$

$\frac{4}{x^3} = 4x^{-3}$, not $(4x)^{-3}$

ln and e

$$\frac{d}{dx}\ln x = \frac{1}{x}, \qquad \frac{d}{dx}e^{kx} = ke^{kx}$$

Trig (radians)

$$\frac{d}{dx}\sin bx = b\cos bx, \quad \frac{d}{dx}\cos bx = -b\sin bx, \quad \frac{d}{dx}\tan x = \sec^2 x$$

Gradient at x = a

$$m = f'(a)$$

differentiate first, then substitute. Unknown constant or x: set $f'(a)$ equal to the given gradient and solve

Tangent at x = a

$$y - y_1 = f'(a)(x - x_1)$$

point $(a, f(a))$

Normal at x = a

$$m_N = -\frac{1}{f'(a)}$$

same point, perpendicular gradient

Stationary points

$$\frac{dy}{dx} = 0$$

solve for x; get y from the original equation

Second-derivative test

$$\frac{d^2y}{dx^2} > 0 \text{ min}, \quad < 0 \text{ max}, \quad = 0 \text{ test fails}$$

if it fails, check the gradient's sign either side

Your turn

Answers in grey. For stationary points, sketch the curve afterwards: does your answer match the picture?

Gradient from scratch

1. Estimate the gradient of $y = x^2$ at $x = 3$ using $h = 0.01$.

1. $(9.0601 - 9) \div 0.01 = 6.01$ (the exact answer is 6)

Power rule

2. $y = 5x^4$
3. $y = 3x^2 - 7x + 2$
4. $y = \sqrt{x}$
5. $y = 4/x^3$

2. $20x^3$ 3. $6x - 7$ 4. $\frac{1}{2}x^{(-\frac{1}{2})} = 1/(2\sqrt{x})$ 5. $-12x^{-4} = -12/x^4$

Standard derivatives

6. $y = 3e^{(2x)}$
7. $y = 2 \sin 5x$
8. the gradient of $y = \ln x$ at $x = 4$

6. $6e^{(2x)}$ 7. $10 \cos 5x$ 8. $1/4 = 0.25$

Stationary points

9. $y = x^2 - 6x + 1$: find the stationary point and say what kind it is.
10. $y = x^3 - 3x$: find both stationary points and classify them with the second derivative.

9. $x = 3$, $y = -8$: minimum (second derivative $2 > 0$) 10. $(1, -2)$ minimum ($6 > 0$); $(-1, 2)$ maximum ($-6 < 0$)

Tangents and normals

11. Find the tangent to $y = x^2 + 2x$ at $x = 1$.
12. Find the normal to $y = x^2 + 2x$ at $x = 1$.
13. Find the tangent to $y = 6/x$ at $(2, 3)$.

11. point $(1, 3)$, $m = 4$: $y = 4x - 1$ 12. $m = -1/4$: $y = -x/4 + 13/4$ 13. $dy/dx = -6/x^2 = -3/2$: $y = -3x/2 + 6$

Find the unknown

14. $y = ax^2 + 3x$ has gradient 7 at $x = 1$. Find a.

15. Find the point on $y = x^2 - 5x + 2$ where the gradient is 3.

16. $y = x^3 + kx$ has a stationary point at $x = 2$. Find k.

17. Find the points on $y = x^3 - 12x + 1$ where the gradient is 15.

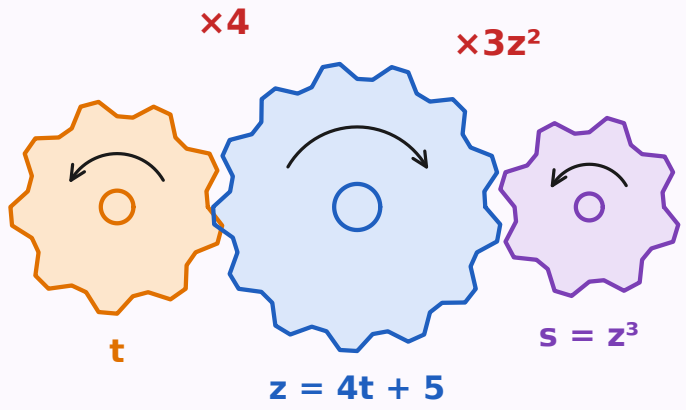
18. The tangent to $y = x^2 + bx$ at $x = 2$ has gradient 1. Find b, then the tangent.

14. $a = 2$ ($2a + 3 = 7$) 15. $(4, -2)$ ($2x - 5 = 3$, so $x = 4$) 16. $k = -12$ ($3(2)^2 + k = 0$) 17. $(3, -8)$ and $(-3, 10)$ ($3x^2 - 12 = 15$, so $x^2 = 9$) 18. $b = -3$ ($4 + b = 1$); point $(2, -2)$, tangent $y = x - 4$

Chapter 35

Chain, product and quotient rules

rates multiply, one layer at a time



Turn the first gear once: the middle one turns 4 times as fast, the last one $3z^2$ times as fast as the middle. Altogether $4 \times 3z^2$. Rates multiply.

Chapter 34's rules handle one simple term at a time. Real formulas are built from pieces: one thing **inside** another, two things **multiplied**, one thing **divided** by another. Each way of building gets its own rule and its own picture:

- **chain rule**: gears turning gears (and nesting dolls)
- **product rule**: a rectangle growing on two sides
- **quotient rule**: a product wearing a disguise
- **implicit differentiation**: every y wears a luggage tag

Goes with **Lecture 15** (higher-order derivatives, functions of functions, chain rule, product rule, quotient rule, options, implicit differentiation).

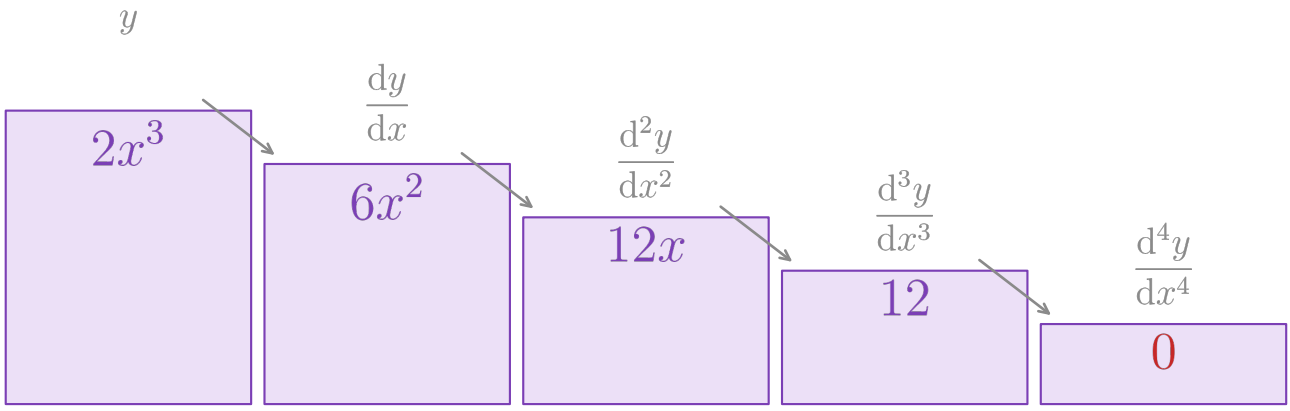
Differentiate again, and again

RULE: higher derivatives

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$: differentiate $\frac{\mathrm{d}y}{\mathrm{d}x}$ once more. $\frac{\mathrm{d}^3y}{\mathrm{d}x^3}$: once more again, and so on. Also written $f''(x)$, $f'''(x)$.

Use the same rules every time; rewrite $\frac{1}{x}$ as x^{-1} first and watch the signs of negative powers.

You can differentiate a derivative. Lecture 15: velocity differentiated is acceleration; acceleration differentiated says how the acceleration changes. For $y = 2x^3$ each differentiation walks one step down the stairs:



Powers of x eventually vanish; sine and cosine cycle for ever (Chapter 34's wheel). Lecture 15's examples:

$$y = 4x^3 - 2x^2 + \frac{1}{x} + 2 \longleftarrow 1/x \text{ is } x^{-1}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 12x^2 - 4x - \frac{1}{x^2} \longleftarrow x^{-1} \to -1x^{-2}$$

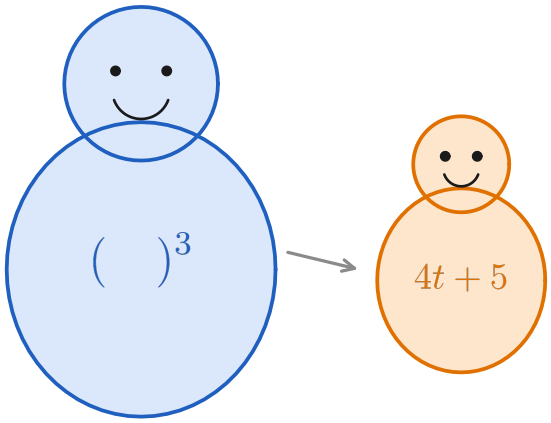
$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 24x - 4 + \frac{2}{x^3} \longleftarrow -x^{-2} \to +2x^{-3}$$

Slide slip: Lecture 15 writes $+\frac{1}{x^3}$ in the second derivative. The power rule on $-x^{-2}$ gives $(-1)(-2)x^{-3} = +2x^{-3}$, so it's $\frac{2}{x^3}$.

$$y = \sin x + 2x^3 \;\Rightarrow\; \frac{\mathrm{d}y}{\mathrm{d}x} = \cos x + 6x^2 \;\Rightarrow\; \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\sin x + 12x$$

A function inside a function

Lecture 15: $s = (4t + 5)^3$. It's one function (**cube it**) wrapped around another ($4t + 5$), like nesting dolls:



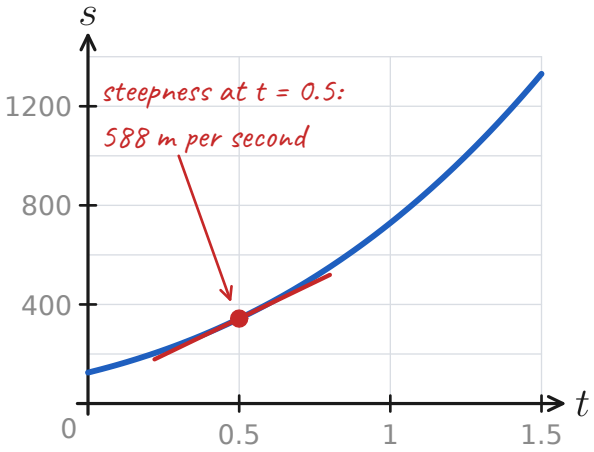
outside doll: "cube whatever is inside"
inside doll: 4t + 5

(Chapter 18 called this a composite function.)

So what are we trying to find? **Nothing here needs solving.** $s = (4t + 5)^3$ is a finished recipe: pick a time t and it tells you the distance s. Think of t as seconds and s as metres travelled. Lecture 15's question is a different one: **how fast is s changing?** That speed is $\frac{ds}{dt}$, Chapter 34's gradient (how steep the graph is).

t (s)	s (m)	jump
0	125	
		+218
0.5	343	
		+386
1	729	
		+602
1.5	1,331	

the jumps keep growing: s speeds up, so the speed depends on t



So the goal is a **new formula**, $\frac{ds}{dt}$: a speedometer. Feed in any t and it gives the speed at that moment. We'll get $\frac{ds}{dt} = 12(4t + 5)^2$, so at $t = 0.5$ the speed is $12 \times 7^2 = 588$, the slope of the red line.

The long way (what Lecture 15 does first): multiply it all out, then differentiate.

$$(4t + 5)^3 = (4t + 5)(16t^2 + 40t + 25) \quad \longleftarrow \textit{square first}$$

$$= 64t^3 + 240t^2 + 300t + 125$$

$$\frac{\mathrm{d}s}{\mathrm{d}t} = 192t^2 + 480t + 300 = 12(16t^2 + 40t + 25) = 12(4t+5)^2$$

take out 12, spot the square

Six lines, and imagine a power of 10! Look at the answer though: $12(4t+5)^2 = 3(4t+5)^2 \times 4$. That's the outside doll differentiated ($3(\ )^2$) times the inside doll differentiated (4). That's the shortcut.

The chain rule: gears

RULE: the chain rule

Use it for a **function of a function**: something with an inside chunk, like $(4t+5)^3$, e^{x^2}, $\ln(3x+1)$, $\sqrt{1+x^3}$, $\sin(2x)$.

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x}$, where u is the inside: **derivative of the outside** (inside left alone) **× derivative of the inside**.

PROCEDURE: chain rule step by step

1. Name the inside chunk u (any spare letter: z, u...).
2. Write y in terms of u alone.
3. Differentiate the outside with respect to u, and the inside with respect to x.
4. Multiply the two.
5. Put the inside back in place of u and tidy.

Name the inside z. Then t drives z, and z drives s, like the gears on the title page. How fast s changes with t is the two rates **multiplied**:

$$\frac{\mathrm{d}s}{\mathrm{d}t} = \frac{\mathrm{d}s}{\mathrm{d}z} \times \frac{\mathrm{d}z}{\mathrm{d}t}$$ ← *outside rate × inside rate*

(It looks as if the $\mathrm{d}z$'s cancel. They don't really, since these aren't fractions, but it's a handy way to remember it.)

Five small steps for $s = (4t+5)^3$:

$z = 4t + 5$ ← *1. name the inside doll*

$$s = z^3 \quad \longleftarrow \text{2. the outside doll}$$

$$\frac{\mathrm{d}s}{\mathrm{d}z} = 3z^2, \qquad \frac{\mathrm{d}z}{\mathrm{d}t} = 4 \quad \longleftarrow \text{3. differentiate each doll}$$

$$\frac{\mathrm{d}s}{\mathrm{d}t} = 3z^2 \times 4 = 12z^2 \quad \longleftarrow \text{4. multiply the rates}$$

$$\frac{\mathrm{d}s}{\mathrm{d}t} = 12(4t+5)^2 \quad \longleftarrow \text{5. put the inside back}$$

Same answer as the six-line slog. Lecture 15's second example: $z = 2(3t^3 + 2t^2)^4$.

$$u = 3t^3 + 2t^2, \qquad z = 2u^4 \quad \longleftarrow \text{name the inside}$$

$$\frac{\mathrm{d}z}{\mathrm{d}u} = 8u^3, \qquad \frac{\mathrm{d}u}{\mathrm{d}t} = 9t^2 + 4t \quad \longleftarrow \text{each doll}$$

$$\frac{\mathrm{d}z}{\mathrm{d}t} = 8(9t^2 + 4t)(3t^3 + 2t^2)^3 \quad \longleftarrow \text{multiply, put u back}$$

Lecture 15's quick version for brackets: bring the power down, knock one off the power, then multiply by the derivative of what's in the bracket. And the table in Chapter 34 was secretly using the chain rule all along: the k in $e^{kx} \to ke^{kx}$ is the inside's rate.

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x^2) = \cos(x^2) \times 2x \quad \longleftarrow \text{outside: sin} \to \text{cos; inside: } x^2 \to 2x$$

The chain rule with e, ln and roots

These patterns come up constantly. Each is the chain rule with $g(x)$ as the inside and $g'(x)$ as its derivative:

outside	derivative	example
$(g(x))^n$	$n(g(x))^{n-1} \times g'(x)$	$(2x+1)^5 \to 5(2x+1)^4 \times 2$
$e^{g(x)}$	$g'(x)\,e^{g(x)}$: the exponential copies itself, times the inside's derivative	$e^{x^2} \to 2x\,e^{x^2}$
$\ln g(x)$	$\dfrac{g'(x)}{g(x)}$: inside's derivative over the inside	$\ln(3x+1) \to \dfrac{3}{3x+1}$
$\sqrt{g(x)} = g(x)^{\frac{1}{2}}$	$\frac{1}{2}g(x)^{-\frac{1}{2}}g'(x) = \dfrac{g'(x)}{2\sqrt{g(x)}}$	$\sqrt{x^2+1} \to \dfrac{2x}{2\sqrt{x^2+1}}$
$\sin g(x)$, $\cos g(x)$	$g'(x)\cos g(x)$, $-g'(x)\sin g(x)$	$\cos(x^2) \to -2x\sin(x^2)$

Worked examples. Name g, find g', then use the pattern:

$$y = e^{x^2+3x}: \quad g' = 2x+3, \quad \frac{\mathrm{d}y}{\mathrm{d}x} = (2x+3)\,e^{x^2+3x}$$

e copies itself × g'

$$y = \ln(x^2+4): \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2x}{x^2+4}$$ ← *g' over g*

$$y = \ln(5x): \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{5}{5x} = \frac{1}{x}$$ ← *same as ln x! (ln 5x = ln 5 + ln x)*

$$y = \sqrt{4x+1} = (4x+1)^{\frac{1}{2}}$$ ← *√ is the power ½*

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \tfrac{1}{2}(4x+1)^{-\frac{1}{2}} \times 4 = \frac{2}{\sqrt{4x+1}}$$ ← *× 4 from the inside*

$$y = \frac{3}{(2x-1)^2} = 3(2x-1)^{-2}$$ ← rewrite as a power first

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -6(2x-1)^{-3} \times 2 = -\frac{12}{(2x-1)^3}$$ ← then chain: inside 2x − 1 → 2

✗ Common mistakes

✗ $\frac{\mathrm{d}}{\mathrm{d}x} e^{x^2} = e^{x^2}$

multiply by the inside's derivative

✓ $\frac{\mathrm{d}}{\mathrm{d}x} e^{x^2} = 2x\, e^{x^2}$

✗ $\frac{\mathrm{d}}{\mathrm{d}x} \ln(x^2+4) = \frac{1}{x^2+4}$

the top is g', not 1

✓ $\frac{\mathrm{d}}{\mathrm{d}x} \ln(x^2+4) = \frac{2x}{x^2+4}$

✗ $\frac{\mathrm{d}}{\mathrm{d}x}(2x+1)^5 = 5(2x+1)^4$

× 2 from the inside

✓ $\frac{\mathrm{d}}{\mathrm{d}x}(2x+1)^5 = 10(2x+1)^4$

The product rule: a rectangle growing on two sides

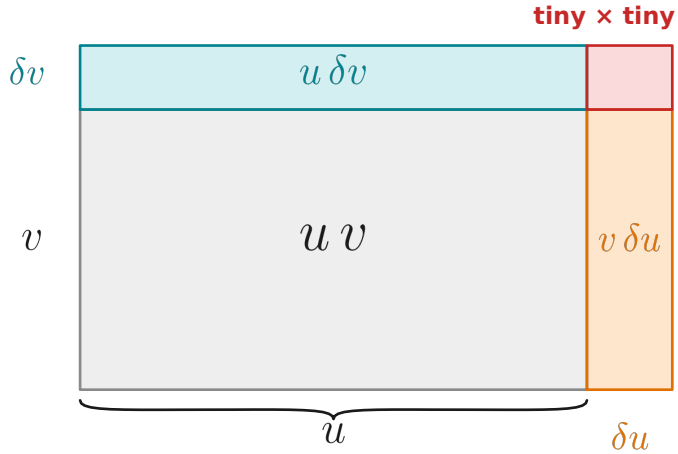
RULE: the product rule

For $y = uv$, where **both** u and v contain x: $\frac{dy}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$. In words: (second × derivative of first) + (first × derivative of second).

Steps: write down u, v, u', v' (use the chain rule for these if needed), substitute, then simplify, often by taking out a common factor.

A number times a function is not a product: $\frac{d}{dx}(5\sin x) = 5\cos x$. The number just waits in front.

$y = u \times v$, where u and v both depend on x. Picture y as the **area** of a rectangle with sides u and v. Nudge x a little: both sides grow.



extra area =
v δu (orange) + u δv (teal)
+ δu δv (red corner)

Shrink the nudge: the red corner is tiny × tiny, so it vanishes. What's left:

$$\frac{dy}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$$ ← *each side's growth × the other side*

Test it where you can check (Lecture 15): $y = (3x+1)(5x-2)$.

expand first: $y = 15x^2 - x - 2$, $\frac{dy}{dx} = 30x - 1$

the old way

$u = 3x+1,\ v = 5x-2,\ \frac{du}{dx} = 3,\ \frac{dv}{dx} = 5$

$$\frac{dy}{dx} = (5x-2)\times 3 + (3x+1)\times 5 = 15x - 6 + 15x + 5 = 30x - 1$$

✓ *the same*

Where you have no choice: $y = x^3 \ln x$ can't be expanded.

$$u = x^3, \quad v = \ln x, \quad \frac{\mathrm{d}u}{\mathrm{d}x} = 3x^2, \quad \frac{\mathrm{d}v}{\mathrm{d}x} = \frac{1}{x}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \ln x \times 3x^2 + x^3 \times \frac{1}{x} = 3x^2 \ln x + x^2 = x^2(1 + 3\ln x)$$

Product rule: Lecture 15's harder ones

Your turn slide: $y = x^{-2}(4 + 3x^{-3})$.

$$u = x^{-2}, \; v = 4 + 3x^{-3}, \; \frac{\mathrm{d}u}{\mathrm{d}x} = -2x^{-3}, \; \frac{\mathrm{d}v}{\mathrm{d}x} = -9x^{-4}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = (4 + 3x^{-3})(-2x^{-3}) + x^{-2}(-9x^{-4}) \longleftarrow v\,u' + u\,v'$$

$$= -8x^{-3} - 6x^{-6} - 9x^{-6} = -8x^{-3} - 15x^{-6} = -\frac{8x^3 + 15}{x^6}$$

add powers (Chapter 9)

Combined example: $y = (x + 3)^2(4x + 1)^3$. Each factor is itself a nesting doll, so the chain rule works **inside** the product rule:

$$u = (x + 3)^2 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2(x + 3) \times 1 \longleftarrow \text{chain}$$

$$v = (4x + 1)^3 \Rightarrow \frac{\mathrm{d}v}{\mathrm{d}x} = 3(4x + 1)^2 \times 4 = 12(4x + 1)^2$$

chain

$$\frac{\mathrm{d}y}{\mathrm{d}x} = (4x + 1)^3 \cdot 2(x + 3) + (x + 3)^2 \cdot 12(4x + 1)^2$$

product rule

$$= 320x^4 + 1728x^3 + 2628x^2 + 1010x + 114 \longleftarrow \text{multiplied out}$$

Slide slip: Lecture 15's multiplied-out answer starts $340x^4$; it's $320x^4$ ($2 \times 64 + 12 \times 16 = 128 + 192 = 320$).

Tidier: take out the common factors $2(x+3)(4x+1)^2$, leaving $\frac{dy}{dx} = 2(x+3)(4x+1)^2(10x+19)$.

The quotient rule: a product in disguise

RULE: the quotient rule

For $y = \frac{u}{v}$: $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{v\frac{\mathrm{d}u}{\mathrm{d}x} - u\frac{\mathrm{d}v}{\mathrm{d}x}}{v^2}$. The order on top matters because of the minus: **bottom × top′ − top × bottom′**.

Do not expand v^2 on the bottom; simplify the top only.

If the bottom is just a power of x (like $\frac{x^2+1}{x}$), split the fraction and use the power rule instead. If the bottom is a number, it is just a constant multiplier.

A fraction $\frac{u}{v}$ is really $u \times v^{-1}$. So you don't need anything new: use the product rule, with the chain rule for v^{-1}:

$$y = u\,v^{-1}$$ ← dividing = multiplying by the reciprocal

$$\frac{\mathrm{d}y}{\mathrm{d}x} = v^{-1}\frac{\mathrm{d}u}{\mathrm{d}x} + u\cdot(-1)v^{-2}\frac{\mathrm{d}v}{\mathrm{d}x}$$ ← product rule; chain on v^{-1}

$$= \frac{v\frac{\mathrm{d}u}{\mathrm{d}x} - u\frac{\mathrm{d}v}{\mathrm{d}x}}{v^2}$$ ← over a common denominator v^2

Memory rhyme: **"low d-high minus high d-low, over the square of what's below"**. The order matters because of the minus.

Lecture 15: $y = \frac{x-1}{x+2}$, so $u = x-1$, $v = x+2$, and both derivatives are 1.

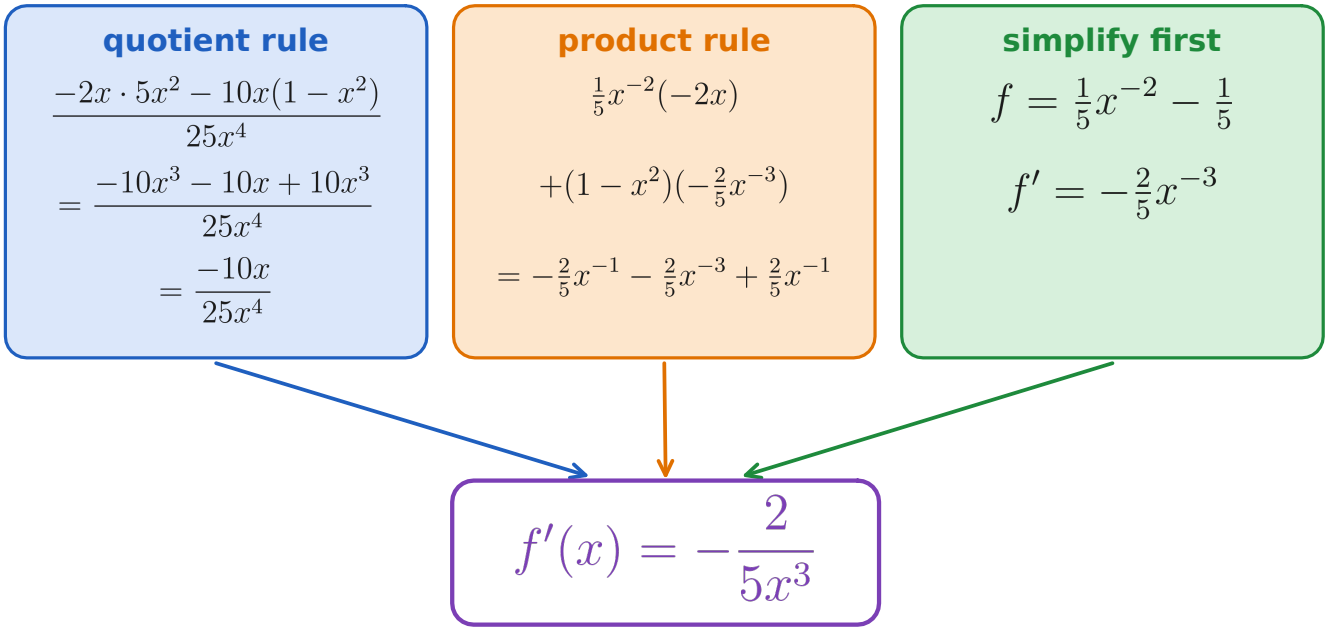
$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(x+2)(1) - (x-1)(1)}{(x+2)^2} = \frac{3}{(x+2)^2}$$

Harder: $y = \frac{e^{-3t}}{t^2+1}$, with $u = e^{-3t}$, $u' = -3e^{-3t}$, $v = t^2+1$, $v' = 2t$.

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{(t^2+1)(-3e^{-3t}) - e^{-3t}(2t)}{(t^2+1)^2} = \frac{-(3t^2+2t+3)\,e^{-3t}}{(t^2+1)^2}$$

Three roads to the same answer

Lecture 15's "options": $f(x) = \frac{1-x^2}{5x^2}$. The quotient rule works, the product rule works, and a third road (simplify first) is the shortest:



Moral: **tidy up before you differentiate**. Splitting the fraction ($\frac{1}{5x^2} - \frac{x^2}{5x^2}$) turned a quotient into two plain powers.

Implicit differentiation: every y wears a tag

PROCEDURE: implicit differentiation

1. Differentiate **every term** of the equation with respect to x, both sides.
2. An x term differentiates as usual. A y term: differentiate as if y were x, then multiply by $\frac{\mathrm{d}y}{\mathrm{d}x}$ (chain rule). A number on its own gives 0.
3. Collect every $\frac{\mathrm{d}y}{\mathrm{d}x}$ term on one side and everything else on the other.
4. Factorise out $\frac{\mathrm{d}y}{\mathrm{d}x}$ and divide.
5. For a gradient at a point you need **both** x and y. If only x is given, put it into the **original** equation and solve for y first.

Explicit: $y =$ (something with x). **Implicit**: x and y are tangled together, like $x^3 + 3y^2 - 4x + y = 11$, and you can't easily get y on its own.

The trick: y is secretly a function of x (an inside doll!). So when you differentiate a y term with respect to x, the chain rule sticks a **tag** on it: differentiate as if y were x, then multiply by $\frac{\mathrm{d}y}{\mathrm{d}x}$.

$$\underset{\downarrow}{x^3} \quad \underset{\downarrow}{+3y^2} \quad \underset{\downarrow}{-4x} \quad \underset{\downarrow}{+y} \quad = 11$$
$$3x^2 \qquad +6y \cdot \tfrac{\mathrm{d}y}{\mathrm{d}x} \qquad -4 \qquad +1 \cdot \tfrac{\mathrm{d}y}{\mathrm{d}x} \qquad = 0$$

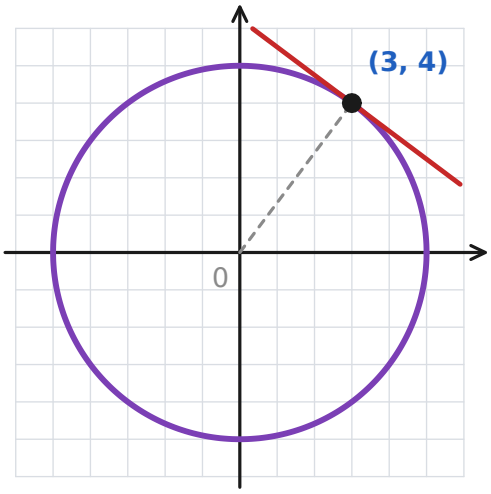
$$3x^2 + 6y\frac{\mathrm{d}y}{\mathrm{d}x} - 4 + \frac{\mathrm{d}y}{\mathrm{d}x} = 0 \quad \leftarrow \text{the tagged terms hold } dy/dx$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}(6y + 1) = 4 - 3x^2 \quad \leftarrow \text{tagged terms on one side; factor (Chapter 12)}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{4 - 3x^2}{6y + 1} \quad \leftarrow \div (6y + 1)$$

Notice the answer has y in it: the gradient depends on **where** you are on the curve, which needs both coordinates.

A picture you can check: the circle $x^2 + y^2 = 25$ (Chapter 28).



$$2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 0$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y}$$

$$\text{at } (3,4): \ -\tfrac{3}{4}$$

The radius has gradient 4/3, the tangent −3/4: perpendicular, as Chapter 20 promised (they multiply to −1).

When only x is given. Find the gradient of $y^3 + x^2 = 9$ at the point where $x = 1$.

$$y^3 + 1^2 = 9 \ \Rightarrow \ y^3 = 8 \ \Rightarrow \ y = 2$$ ← *step 5 first: the point is (1, 2)*

$$3y^2\frac{\mathrm{d}y}{\mathrm{d}x} + 2x = 0$$ ← *steps 1–2: y^3 gets a tag*

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{2x}{3y^2}$$ ← *steps 3–4*

$$\text{at } (1,2): \ -\frac{2(1)}{3(2)^2} = -\frac{2}{12} = -\frac{1}{6}$$

Which rule? Sort by how it's built

PROCEDURE: choosing the rule

1. **Simplify first** if you can: split a fraction over a single power of x, expand a small bracket, use log laws ($\ln x^3 = 3\ln x$). Often no special rule is needed afterwards.
2. A **number times a function** ($5\sin x$, $3e^{2x}$, $\frac{x^2}{4}$): keep the number in front. **Not** a product-rule case.
3. A **function of a function** (an inside chunk): **chain rule**.
4. Two functions of x **multiplied**: **product rule**.
5. One function of x **divided** by another: **quotient rule** (or rewrite as $u \times v^{-1}$ and use the product rule).
6. x and y mixed in one equation: **implicit differentiation**.
7. Rules can nest: inside a product, each of u and v may need the chain rule.

Ask: what is the **last** thing you'd do if you worked it out on a calculator? That's the outer structure, and it picks the rule.

something inside something: CHAIN	two x-things multiplied: PRODUCT	one divided by another: QUOTIENT	x and y tangled: IMPLICIT
$(2x+1)^5$	$x^2 \sin x$	$\frac{x+1}{x-1}$	$x^2 + y^2 = 25$
e^{3x^2}	xe^x	$\frac{\sin x}{x}$	$x^3 + 3y^2 = 11$
$\sin(x^2)$	$(x+1)^3(x-2)$	$\frac{e^x}{x^2+1}$	$y^3 + x = y$
$\sqrt{1+x^3}$	$x \ln x$		

Sums and differences are never a problem: split them up and do each term with its own rule.

✗ Common mistakes

✗ $\frac{\mathrm{d}}{\mathrm{d}x}\ln(5x) = \frac{1}{5x}$

ln of an inside chunk: g' over g

✓ $\frac{\mathrm{d}}{\mathrm{d}x}\ln(5x) = \frac{5}{5x} = \frac{1}{x}$

✗ $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(x^2 \sin x\right) = 2x \cos x$ ✓ $2x \sin x + x^2 \cos x$

a product is NOT the product of the derivatives

✗ $\dfrac{\mathrm{d}}{\mathrm{d}x}\dfrac{u}{v} = \dfrac{u'}{v'}$ ✓ $\dfrac{vu' - uv'}{v^2}$

a quotient is NOT the quotient of the derivatives

Exam combinations

Exam questions mix this chapter's rules with Chapter 34's jobs: gradient at a point, stationary points, tangents. Pick the rule, differentiate, then carry on exactly as in Chapter 34.

Stationary point of $y = xe^{-x}$. Product rule, with the chain rule for e^{-x}:

$$u = x, \quad v = e^{-x}, \quad u' = 1, \quad v' = -e^{-x}$$ ← chain: inside −x → −1

$$\frac{\mathrm{d}y}{\mathrm{d}x} = e^{-x} \cdot 1 + x \cdot \left(-e^{-x}\right) = e^{-x}(1 - x)$$ ← v u′ + u v′, take out e^{-x}

$$e^{-x}(1 - x) = 0 \;\Rightarrow\; x = 1$$ ← e^{-x} is never 0, so 1 − x = 0

$$y = 1 \cdot e^{-1} = \frac{1}{e} \approx 0.368$$ ← y from the ORIGINAL equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = e^{-x}(x - 2); \quad x = 1 : \; -e^{-1} < 0$$ ← product rule again: maximum

So $\left(1, \frac{1}{e}\right)$ is a **maximum**.

Gradient at a point after the quotient rule: $y = \frac{2x}{x+1}$ at $x = 1$.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(x+1) \cdot 2 - 2x \cdot 1}{(x+1)^2} = \frac{2}{(x+1)^2}$$ ← simplify the top

$$x = 1 : \quad \frac{2}{2^2} = \frac{1}{2}$$ ← then substitute

Gradient at a point after the chain rule: $y = \sqrt{2x+5}$ at $x = 2$.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2}{2\sqrt{2x+5}} = \frac{1}{\sqrt{2x+5}}; \quad x = 2: \ \frac{1}{\sqrt{9}} = \frac{1}{3}$$

Find the unknown: $y = e^{kx}$ has gradient 6 at $x = 0$. Find k.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = ke^{kx}; \quad x = 0: \ ke^{0} = k = 6 \longleftarrow e^{0} = 1$$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Higher derivatives	$\dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left(\dfrac{dy}{dx}\right)$ differentiate again
Chain rule	$\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$ outside's derivative × inside's derivative
Chain: powers	$\dfrac{d}{dx}\big(g(x)\big)^n = n\big(g(x)\big)^{n-1} g'(x)$
Chain: e and ln	$\dfrac{d}{dx}e^{g(x)} = g'(x)e^{g(x)}$, $\qquad \dfrac{d}{dx}\ln g(x) = \dfrac{g'(x)}{g(x)}$
Chain: roots	$\dfrac{d}{dx}\sqrt{g(x)} = \dfrac{g'(x)}{2\sqrt{g(x)}}$
Product rule	$\dfrac{d(uv)}{dx} = v\dfrac{du}{dx} + u\dfrac{dv}{dx}$ only when both factors contain x
Quotient rule	$\dfrac{d}{dx}\left(\dfrac{u}{v}\right) = \dfrac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$ bottom × top′ − top × bottom′, over bottom²
Constant multiple	$\dfrac{d}{dx}\big(kf(x)\big) = kf'(x)$ not a product
Implicit	differentiate every term; each y term gets $\times\frac{dy}{dx}$; collect and factorise. Need y? Use the original equation
Which rule?	simplify first; inside chunk: chain; multiplied: product; divided: quotient; x and y mixed: implicit

Your turn

Name the rule before you start. Answers in grey.

Higher derivatives

1. Find d^2y/dx^2 for $y = x^4 - 3x^2 + 5$.
2. Find d^3y/dx^3 for $y = \cos x$.

1. $12x^2 - 6$ 2. $\sin x$

Chain rule

3. $y = (2x + 1)^5$
4. $y = e^{(3x^2)}$
5. $y = \sqrt{(1 + x^3)}$

3. $10(2x + 1)^4$ 4. $6x\, e^{(3x^2)}$ 5. $3x^2 \div (2\sqrt{(1 + x^3)})$

Product rule

6. $y = x^2 \sin x$
7. $y = x\, e^x$

6. $2x \sin x + x^2 \cos x$ 7. $e^x + x\, e^x = (1 + x)e^x$

Quotient rule (or rewrite!)

8. $y = (x + 1)/(x - 1)$
9. $y = (x^2 + 1)/x$

8. $-2/(x - 1)^2$ 9. $1 - 1/x^2$ (easiest: $x + x^{-1}$ first)

Implicit

10. Find dy/dx for $x^2 + y^2 = 100$, and its value at $(6, 8)$.
11. Find dy/dx for $y^3 + x = y$.

10. $-x/y$; $-3/4$ 11. $1/(1 - 3y^2)$

Exam combinations

12. Find the stationary point of $y = x\, e^{(-2x)}$ and say what kind it is.
13. Find the gradient of $y = x^2/(x + 1)$ at $x = 1$.
14. $y = \ln(x^2 + 1)$: find dy/dx and its value at $x = 1$.

12. $dy/dx = e^{(-2x)}(1 - 2x) = 0$, $x = ½$; $(½, 1/(2e))$, maximum 13. $(x^2 + 2x)/(x + 1)^2 = 3/4$
14. $2x/(x^2 + 1)$; 1

Find the unknown

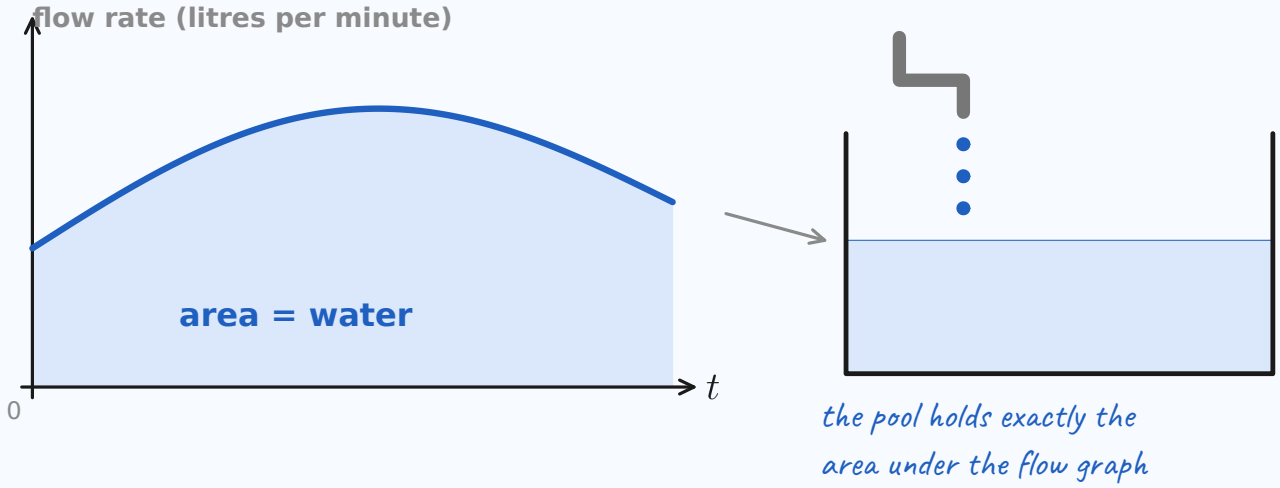
15. y = e^(ax) has gradient 6 at x = 0. Find a.
16. y = (ax + 1)⁴ has gradient 24 at x = 0. Find a.
17. y = ln(x² + k) has gradient ½ at x = 1. Find k.
18. y = x/(x + c) has gradient ¼ at x = 0 (c > 0). Find c.
19. (2, b) lies on x² + y³ = 12. Find b, then dy/dx there.

15. a = 6 (a e⁰ = 6) 16. a = 6 (4a(1)³ = 24) 17. k = 3 (2/(1 + k) = ½) 18. c = 4 (c/c² = 1/c = ¼) 19. b = 2 (4 + b³ = 12); dy/dx = −2x/(3y²) = −1/3

Chapter 36

Integration and areas

add up the tiny pieces



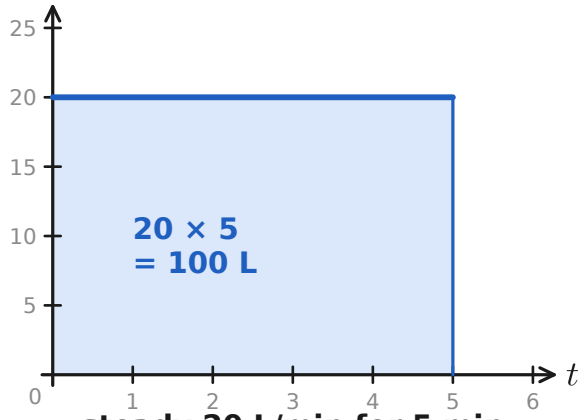
If water pours in at 20 litres a minute for 5 minutes, the pool gains $20 \times 5 = 100$ litres. On a graph of flow rate against time that's a rectangle's **area**. When the rate keeps changing, the water is still the **area under the curve**, and finding it is called **integration**.

It's the second half of calculus, and it turns out to be differentiation run backwards.

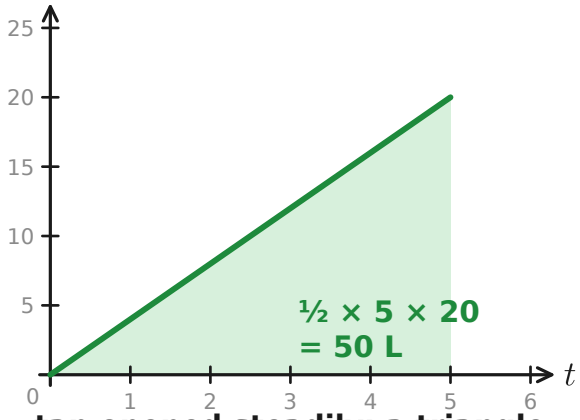
Goes with **Lecture 16** (area under a curve, the trapezium rule and its accuracy, the integral sign, the general formula, +c, 1/x, standard integrals, adding and multiples, definite integrals, negative areas).

Area under a graph = a total

Rate × time = amount. On a rate–time graph, rate is the height and time is the width, so the amount is an **area**:



steady 20 L/min for 5 min



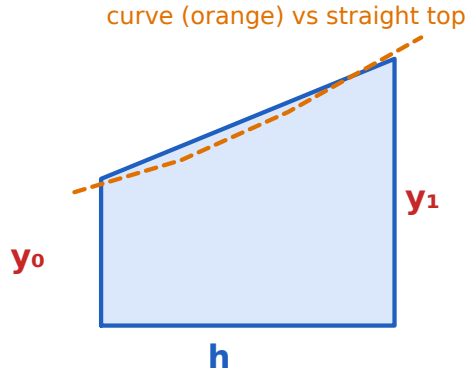
tap opened steadily: a triangle

Rectangles and triangles are easy. A **curved** top isn't, which is what the rest of this chapter is for. Lecture 16's examples of "area under a curve" in real life:

- a **probability**: the area under the Normal curve (Chapter 26)
- **displacement**: the area under a velocity–time graph
- **drug absorption** (pharmacokinetics): the area under a concentration curve

Strips with sloping tops: the trapezium rule

Chop the area into vertical strips of width h and give each strip a straight sloping top. Each strip is a **trapezium** (four sides, two of them parallel). Its area is the **average height × width**:



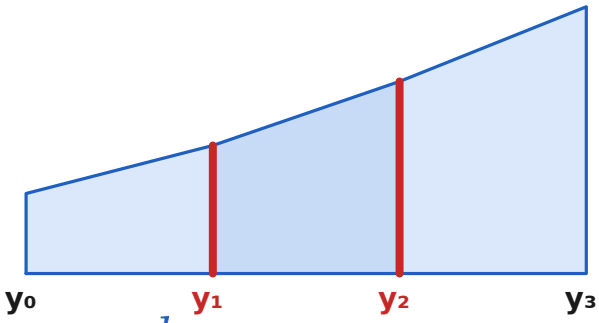
$$\text{area} \approx \frac{y_0 + y_1}{2} \times h = \frac{h}{2}(y_0 + y_1)$$

the straight top misses the curve a little, so this over- or under-estimates (Lecture 16's "imperfections")

Lecture 16, one strip: area under $y = x^2$ from $x = 1$ to $x = 4$, so $h = 3$:

$$\text{area} \approx \tfrac{3}{2}\left(1^2 + 4^2\right) = \tfrac{3}{2} \times 17 = 25.5$$

More strips. With several strips side by side, every **inside wall** belongs to two strips (the right side of one, the left side of the next), so it gets counted **twice**:



red walls y_1 and y_2 are shared by two strips, so they appear twice. The two end walls y_0 and y_3 only once.

$$\text{area} \approx \frac{h}{2}\left(y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n\right)$$

$$h = \frac{b - a}{n}$$ ← *n strips between x = a and x = b*

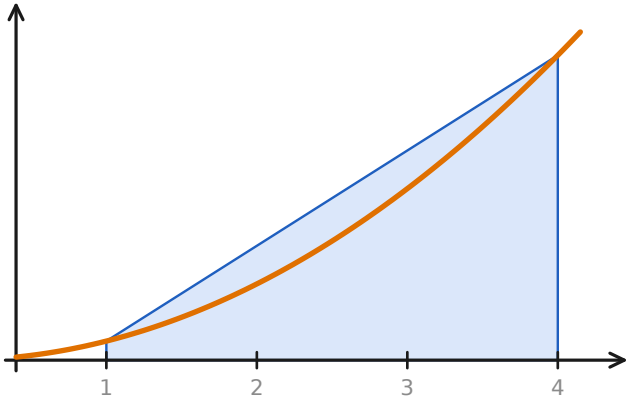
Slide slip: Lecture 16's general formula leaves out the 2 in front of the middle heights. Its own two-strip example (next page) does use the 2.

1, 2, 4, 8 strips: watch the error shrink

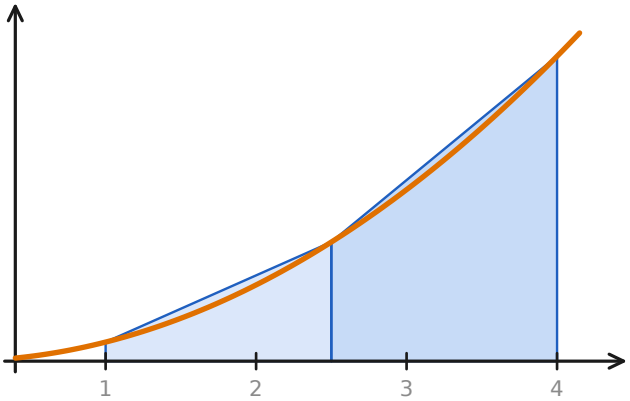
Lecture 16's two-strip version: $h = 1.5$, middle height $2.5^2 = 6.25$:

$$\frac{1.5}{2}\left(1 + 2 \times 6.25 + 16\right) = 0.75 \times 29.5 = 22.125$$

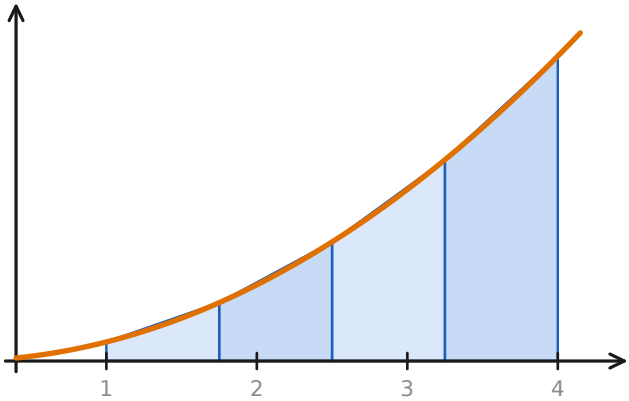
The true area is 21 (you'll prove it later in this chapter). Keep doubling:



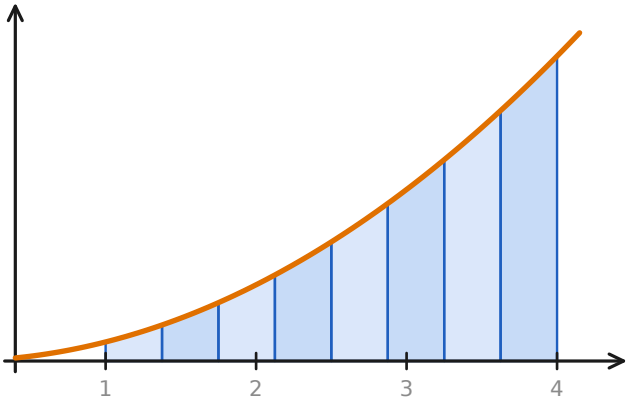
n = 1: 25.5 **error 4.5**



n = 2: 22.125 **error 1.125**



n = 4: 21.2812 **error 0.28125**



n = 8: 21.0703 **error 0.0703125**

Look at the errors: 4.5, 1.125, 0.28, 0.07. For this curve, **halving the strip width cuts the error to exactly a quarter.** For other smooth curves it is only **roughly** a quarter, and only once the strips are fairly narrow; for some curves the pattern is different. What always holds: as the number of strips heads to infinity, the error heads to 0 (Lecture 16: as $n \to \infty$, $\varepsilon \to 0$).

Infinitely many strips: the integral sign

With a finite number of strips you'd write a sum with Σ (Chapter 17). With infinitely many, infinitely thin strips, the S gets stretched into a new symbol, $\int$ (a tall S for "sum"):

$$\int_a^b f(x)\,\mathrm{d}x$$

$\int$ *"the integral of": add up infinitely many thin strips*

a and b: the limits (where the area starts and stops)

f(x): the integrand, the height of each strip

dx: the strip width, shrunk to nothing. It also says which letter you're integrating with.

Integration is the undo button for differentiation

RULE: the reverse power rule

$\int ax^n \, dx = \frac{a}{n+1}x^{n+1} + c$: add 1 to the power, then divide by the **new** power.

It works for every n **except** $n = -1$, which would divide by 0. That case is $\int \frac{1}{x} \, dx = \ln|x| + c$ (later in this chapter).

A constant: $\int k \, dx = kx + c$ (it is kx^0, so the power goes up to 1). For example $\int 5 \, dx = 5x + c$.

Every indefinite integral ends with $+c$. Check by differentiating your answer: you must get back what you started with.

Chapter 14's **undo button**: every operation has an inverse that takes you back. Integration is the undo button for differentiation (Lecture 16): it asks **"what did I differentiate to get this?"**

$$y = 5x^3 + 3 \xrightarrow{\text{differentiate}} 15x^2 \xrightarrow{\text{integrate}} 5x^3 + ?$$

The "+3" vanished when we differentiated, so undoing can't know what it was. More on that in a moment. First the rule. Differentiating made the power **dive down and drop by one**. Undo it in reverse order:

$$x^3 \longrightarrow \frac{x^4}{4}$$

1. the power goes UP by one: 3 → 4

2. divide by the NEW power

$$\int ax^n \, dx = \frac{a}{n+1}x^{n+1} + c \qquad (n \neq -1)$$

Lecture 16: $\int 10x^4\,\mathrm{d}x$.

$$\int 10x^4\,\mathrm{d}x = \frac{10}{4+1}x^{4+1} + c = 2x^5 + c$$

check: $\dfrac{\mathrm{d}}{\mathrm{d}x}(2x^5 + c) = 10x^4$ ✓ ← always check by differentiating back

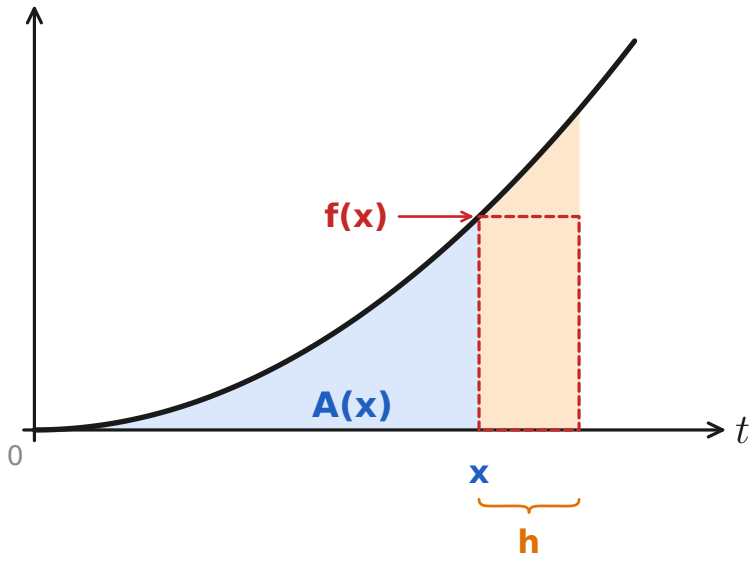
Why area undoes a gradient: slide the right edge

Why should **area** (adding strips) be the undo of **gradient** (Chapter 34)? Fix the left edge at 0 and let the right edge sit at x. Call the shaded area $A(x)$: a number that changes as you slide the edge.

Take $f(t) = t^2$:

✎ **YOUR GO (pencil first)**

Slide the right edge a tiny distance h further. What shape is the extra sliver of area, roughly? How tall is it, and so what's its area?



The new orange sliver is almost a rectangle (red dashes): height f(x), width h.

So extra area ≈ f(x) × h.

The thinner the sliver, the smaller the curved error on top.

Turn that into a gradient. The **rise** of A is the sliver, the **run** is h:

$$A(x+h) - A(x) \approx f(x) \times h \quad \leftarrow \text{extra area = the sliver}$$

$$\frac{A(x+h) - A(x)}{h} \approx f(x) \quad \leftarrow \div h\text{: rise over run}$$

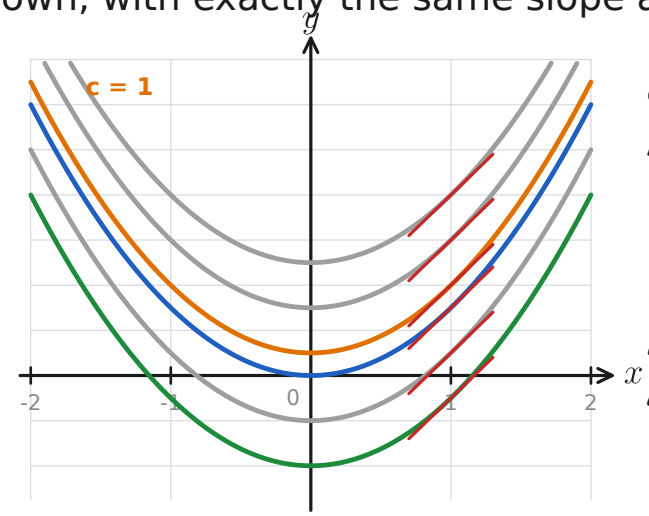
$$A'(x) = f(x) \quad \leftarrow \text{let h shrink to 0 (Chapter 34)}$$

The rate the area grows is the height of the curve at the moving edge. So A is a function whose derivative is f: finding area **is** undoing differentiation. Check with $f(t) = t^2$: the reverse power rule says $A(x) = \frac{x^3}{3}$, and at $x = 2$, with a sliver of width 0.01:

$$\frac{A(2.01) - A(2)}{0.01} = 4.02003 \approx f(2) = 4 \checkmark$$

+ c: a whole family of answers

$3x^2$, $3x^2 + 1$ and $3x^2 - 4$ all differentiate to $6x$ (Lecture 16 calls this **degenerate**: many answers). They're the same curve lifted up or down, with exactly the same slope at every x:



every curve $y = 3x^2 + c$.
Red: the tangents at $x = 1$ are parallel (all have slope 6).

So every integral gets a "+ c": the constant of integration, meaning "some number we don't know yet".

PROCEDURE: finding c from a point on the curve

1. Integrate $\frac{dy}{dx}$ to get $y = F(x) + c$.
2. Substitute the given point: its x into F, its y on the left.
3. Solve for c.
4. Write the final equation with that number in place of c.

Lecture 16: $\frac{dy}{dx} = 12x^3$, and $y = 5$ when $x = 1$. The extra fact picks out one member of the family:

$$y = \int 12x^3 \, dx = 3x^4 + c \quad \longleftarrow \text{step 1}$$

$$5 = 3(1)^4 + c \;\Rightarrow\; c = 2$$ ← steps 2–3: x = 1, y = 5

$$y = 3x^4 + 2$$ ← step 4: the specific solution

Another: a curve has gradient $3x^2 - 4$ and passes through $(2, 5)$.

$$y = x^3 - 4x + c$$ ← integrate each term

$$5 = 2^3 - 4(2) + c = 0 + c \;\Rightarrow\; c = 5$$ ← x = 2, y = 5

$$y = x^3 - 4x + 5$$

The one exception: 1/x

Try the rule on $\int \frac{1}{x}\,\mathrm{d}x = \int x^{-1}\,\mathrm{d}x$: the power goes up to 0 and you'd divide by 0. **Impossible.** But Chapter 34 showed $\ln x$ differentiates to $\frac{1}{x}$, so:

$$\int \frac{1}{x}\,\mathrm{d}x = \ln|x| + c$$ ← |x|: the size of x, sign removed

Slide slip: Lecture 16 writes $\ln x + c$. The modulus matters: $\frac{1}{x}$ exists for negative x too, but $\ln$ only accepts positive numbers, so we feed it $|x|$ (Chapter 17's modulus).

Negative and fractional powers: rewrite first

PROCEDURE: integrating anything built from powers of x

1. Rewrite every term as ax^n: $\frac{3}{x^2} = 3x^{-2}$, $\sqrt{x} = x^{\frac{1}{2}}$, $\frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$, $\frac{5}{x} = 5x^{-1}$. **Expand** brackets; **split** a fraction with a single term on the bottom ($\frac{x^2+1}{x} = x + \frac{1}{x}$).
2. Integrate term by term: for $n \neq -1$, add 1 to the power and divide by the new power; a term ax^{-1} gives $a\ln|x|$; a constant k gives kx.
3. Add $+c$ and, if you like, turn the powers back into fractions and roots.
4. Check by differentiating.

$$\int \frac{3}{x^2}\,\mathrm{d}x = \int 3x^{-2}\,\mathrm{d}x = \frac{3x^{-1}}{-1} + c = -\frac{3}{x} + c$$

$-2 + 1 = -1$, $\div (-1)$

$$\int \sqrt{x}\,\mathrm{d}x = \int x^{\frac{1}{2}}\,\mathrm{d}x = \frac{x^{3/2}}{3/2} + c = \tfrac{2}{3}x^{\frac{3}{2}} + c$$

$\div \frac{3}{2}$ is $\times \frac{2}{3}$

$$\int \frac{1}{\sqrt{x}}\,\mathrm{d}x = \int x^{-\frac{1}{2}}\,\mathrm{d}x = \frac{x^{1/2}}{1/2} + c = 2\sqrt{x} + c$$

$-\frac{1}{2} + 1 = \frac{1}{2}$

$$\int (x+1)^2\,\mathrm{d}x = \int (x^2 + 2x + 1)\,\mathrm{d}x = \frac{x^3}{3} + x^2 + x + c$$

expand first

$$\int \frac{x^2+1}{x}\,\mathrm{d}x = \int \left(x + \frac{1}{x}\right)\mathrm{d}x = \frac{x^2}{2} + \ln|x| + c$$

split, then $1/x \to \ln$

✗ Common mistakes

✗ $\displaystyle\int \frac{3}{x^2}\,\mathrm{d}x = \frac{3}{x^3/3}$

rewrite as a power; never integrate the bottom on its own

✓ $\displaystyle\int 3x^{-2}\,\mathrm{d}x = -3x^{-1} + c$

✗ $\displaystyle\int (2x+1)^2\,\mathrm{d}x = \frac{(2x+1)^3}{3}$

expand first: the answer is $\frac{4}{3}x^3 + 2x^2 + x + c$ (Chapter 37 has a shortcut)

✓ $\displaystyle\int (4x^2 + 4x + 1)\,\mathrm{d}x$

✗ $\displaystyle\int x^{-1}\,\mathrm{d}x = \frac{x^0}{0}$

$n = -1$ is the one exception

✓ $\displaystyle\int x^{-1}\,\mathrm{d}x = \ln|x| + c$

Standard integrals and the easy rules

RULE: standard integrals

Each one is a derivative from Chapter 34 read backwards. Add $+c$ to every indefinite integral.

Sums and differences: integrate term by term. **A number in front** waits outside: $\int kf(x)\,dx = k\int f(x)\,dx$.

Products and quotients have no such rule: expand or split them first (previous section), or use Chapter 37's methods.

Lecture 16's list (angles in radians):

function	integral	function	integral
k	kx	$\sin x$	$-\cos x$
x	$\frac{1}{2}x^2$	$\cos x$	$\sin x$
x^n	$\frac{1}{n+1}x^{n+1}$	$\sin kx$	$-\frac{1}{k}\cos kx$
$\frac{1}{x}$	$\ln\lvert x\rvert$	$\cos kx$	$\frac{1}{k}\sin kx$
e^x	e^x	$\tan x$	$\ln\lvert\sec x\rvert$
e^{kx}	$\frac{1}{k}e^{kx}$		

About $\tan x$. Lectures 16 and 17 list $\int \tan x\,dx = \ln(\sec x) + c$, where $\sec x = \frac{1}{\cos x}$. As with $\frac{1}{x}$, write $\ln\lvert\sec x\rvert$ so that it also works where $\cos x$ is negative. (Check: by the chain rule, $\ln(\sec x)$ differentiates to $\frac{\sec x \tan x}{\sec x} = \tan x$.)

The easy rules (Lecture 16). Integrate a sum or difference term by term, and pull constant multiples out front. All the separate c's merge into one:

$$\int (x^2 + x^3)\,dx = \frac{x^3}{3} + \frac{x^4}{4} + c \quad \longleftarrow \text{term by term}$$

$$\int 3x^4\,dx = 3\int x^4\,dx = \frac{3x^5}{5} + c \quad \longleftarrow \text{the 3 waits outside}$$

Lecture 16's example: $\int (4x - 2x^2 + 8)\,\mathrm{d}x$.

$$= \frac{4x^2}{2} - \frac{2x^3}{3} + 8x + c = 2x^2 - \frac{2x^3}{3} + 8x + c$$

each term: power up, ÷ new power

Definite integrals: top minus bottom

PROCEDURE: evaluating a definite integral

1. Integrate (rewrite first if needed) and write the result in square brackets with the limits: $\big[F(x)\big]_a^b$. Leave out the $+c$: it would cancel.
2. Substitute the **top** limit b: $F(b)$.
3. Substitute the **bottom** limit a: $F(a)$. Use brackets; negative limits cause most mistakes.
4. Subtract: $F(b) - F(a)$. The answer is a number (a signed area), not a formula.

The square brackets mean "this, evaluated at the top minus at the bottom":

$$\int_a^b f(x)\,\mathrm{d}x = \Big[F(x)\Big]_a^b = F(b) - F(a)$$

F is any integral of f

The pool's exact answer: the area under x^2 from 1 to 4, which the strips were chasing:

$$\int_1^4 x^2\,\mathrm{d}x = \Big[\frac{x^3}{3}\Big]_1^4 = \frac{64}{3} - \frac{1}{3} = \frac{63}{3} = 21$$

← ✓ the strips' target

Lecture 16: $\int_1^3 2x^5\,\mathrm{d}x$.

$$= 2\Big[\frac{x^6}{6}\Big]_1^3 = \Big[\frac{x^6}{3}\Big]_1^3 = \frac{729}{3} - \frac{1}{3} = 243 - \tfrac{1}{3} = 242.\dot{6}$$

Lecture 16's your turn: $\int_{-1}^1 (3x^2 + 7)\,\mathrm{d}x$.

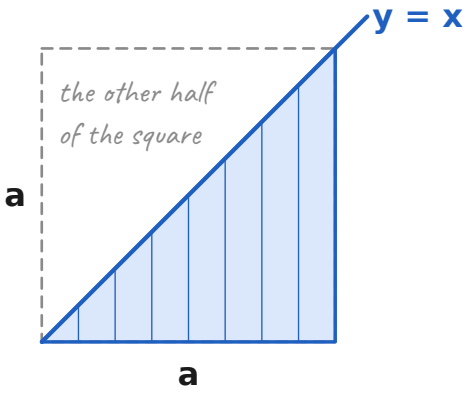
$$= \Big[x^3 + 7x\Big]_{-1}^{1} = (1+7) - (-1-7) = 8 + 8 = 16$$

careful with the minus signs

Slide slip: Lecture 16's general limits formula ends with $\frac{a}{n+1}(b^{n+1} - c^{n+1})$ for limits b (bottom) to c (top). It's the other way round: top minus bottom, $\frac{a}{n+1}(c^{n+1} - b^{n+1})$, as its own line above it says.

Where the ÷2 and ÷3 come from: a triangle and a pyramid

The rule "power up by one, divide by the new power" can be checked with shapes you already know. Area under $y = x$ from 0 to a: the strips build a triangle, and a triangle is **half** of the a by a square:

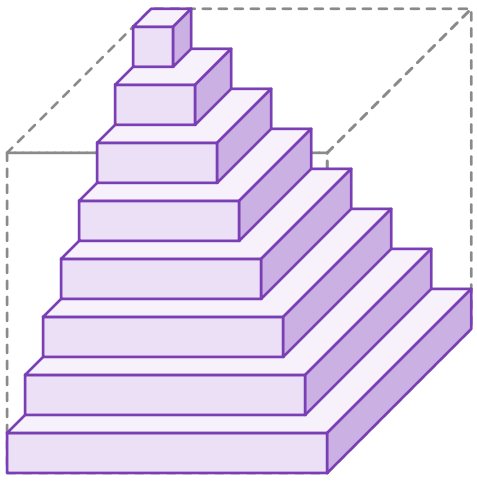


triangle = half of a × a:

$$\int_0^a x\,\mathrm{d}x = \left[\frac{x^2}{2}\right]_0^a = \frac{a^2}{2}$$

the ÷2 IS the half

Area under $y = x^2$ from 0 to a. Each strip is x^2 tall and $\mathrm{d}x$ wide. Now turn every strip into a **square slab**: x by x, and $\mathrm{d}x$ thick (same number, $x^2\,\mathrm{d}x$). Stack the slabs, biggest at the bottom, and they build a **pyramid** with an a by a base and height a:



dashed: the whole a × a × a cube

Three of these pyramids fit together to fill the whole cube (a famous fact, known to Euclid). So the pyramid is a THIRD of a^3:

$$\int_0^a x^2\,\mathrm{d}x = \left[\frac{x^3}{3}\right]_0^a = \frac{a^3}{3}$$

the ÷3 IS the third

With 8 chunky slabs the staircase is a bit more than a third (0.398 of the cube); with 100 thin ones it's 0.338; with infinitely thin ones, exactly $\frac{1}{3}$. That's the integral doing its job. The pattern continues past what we can draw: $\int x^3 \, \mathrm{d}x$ is a 4D pyramid, a quarter of a 4D cube, $\frac{x^4}{4}$.

Area below the axis counts as negative

RULE: areas that cross the axis

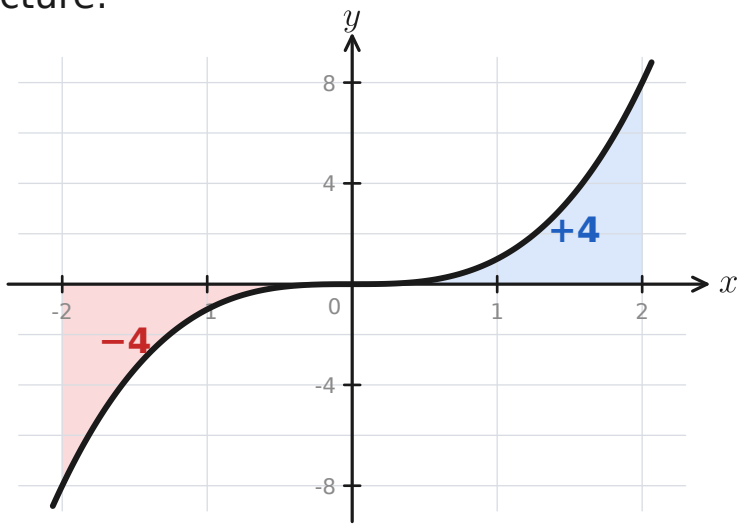
A definite integral counts area **above** the x-axis as positive and area **below** it as negative, so pieces on opposite sides cancel.

To find the **total area** between a curve and the axis: solve $f(x) = 0$ to find where the curve crosses the axis between the limits.

Integrate over each piece separately (limit to crossing, crossing to crossing, crossing to limit).

Add the **sizes** of the pieces (ignore their minus signs).

Lecture 16: $\int_{-2}^{2} x^3 \, \mathrm{d}x = \left[\frac{x^4}{4}\right]_{-2}^{2} = \frac{16}{4} - \frac{16}{4} = 0$. Zero area? Look at the picture:



Area above the axis counts +, area below counts −. Here they cancel exactly.

If you want the total shaded area, split at the crossing and add the sizes: 4 + 4 = 8.

Take care interpreting them (Lecture 16): a velocity graph dipping below the axis means going backwards, and the integral gives your net displacement (Chapter 32), not the distance run.

Worked example: the crossing point isn't given. Find the total area between $y = x^2 - x - 2$ and the x-axis from $x = 0$ to $x = 3$.

$$x^2 - x - 2 = 0 \;\Rightarrow\; (x-2)(x+1) = 0 \;\Rightarrow\; x = 2 \text{ or } x = -1$$

solve f(x) = 0

Only $x = 2$ lies between 0 and 3, so split there. Let $F(x) = \frac{x^3}{3} - \frac{x^2}{2} - 2x$:

$$\int_0^2 f(x)\,\mathrm{d}x = F(2) - F(0) = \left(\tfrac{8}{3} - 2 - 4\right) - 0 = -\tfrac{10}{3}$$

below the axis

$$\int_2^3 f(x)\,\mathrm{d}x = F(3) - F(2) \quad \longleftarrow \textit{above the axis}$$

$$= \left(9 - \tfrac{9}{2} - 6\right) - \left(-\tfrac{10}{3}\right) = -\tfrac{3}{2} + \tfrac{10}{3} = \tfrac{11}{6}$$

$$\text{total area} = \tfrac{10}{3} + \tfrac{11}{6} = \tfrac{31}{6} \approx 5.17 \quad \longleftarrow \textit{add the sizes}$$

Integrating straight from 0 to 3 would give $-\frac{10}{3} + \frac{11}{6} = -\frac{3}{2}$: not an area at all.

Find the unknown

PROCEDURE: an integral with a missing value

1. Do the integral with the unknown letter (k, a…) treated as a number, exactly as usual.
2. Set the result equal to the value you are given. This gives an equation in the unknown.
3. Solve it. If you get two answers, keep only the one that fits the question (e.g. $k > 1$).
4. Check by putting your answer back into the integral.

Unknown limit: $\int_1^k 2x\,\mathrm{d}x = 8$ with $k > 1$. Find k.

$$\left[x^2\right]_1^k = k^2 - 1$$ ← integrate, top minus bottom

$$k^2 - 1 = 8 \;\Rightarrow\; k^2 = 9 \;\Rightarrow\; k = 3$$ ← $k = -3$ is ruled out by $k > 1$

Unknown constant: $\int_0^2 (ax + 1)\,\mathrm{d}x = 10$. Find a.

$$\left[\tfrac{a}{2}x^2 + x\right]_0^2 = 2a + 2$$ ← a is just a number

$$2a + 2 = 10 \;\Rightarrow\; a = 4$$

Unknown c with a negative power: $\frac{\mathrm{d}y}{\mathrm{d}x} = 4x - \frac{3}{x^2}$, and the curve passes through $(1, 6)$.

$$y = \int \left(4x - 3x^{-2}\right)\mathrm{d}x = 2x^2 + 3x^{-1} + c = 2x^2 + \frac{3}{x} + c$$

rewrite first

$$6 = 2 + 3 + c \;\Rightarrow\; c = 1$$ ← $x = 1$, $y = 6$

$$y = 2x^2 + \frac{3}{x} + 1$$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Reverse power rule

$$\int ax^n\,dx = \frac{a}{n+1}x^{n+1} + c \quad (n \neq -1)$$

power up by 1, ÷ new power

Constant

$$\int k\,dx = kx + c$$

The exception

$$\int \frac{1}{x}\,dx = \ln|x| + c$$

Rewrite first

$$\frac{3}{x^2} = 3x^{-2}, \qquad \sqrt{x} = x^{\frac{1}{2}}, \qquad \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$$

expand brackets; split fractions over a single term

Sums and multiples

$$\int (f \pm g)\,dx = \int f\,dx \pm \int g\,dx, \qquad \int kf\,dx = k\int f\,dx$$

no such rule for products or quotients

e and trig

$$\int e^{kx}dx = \tfrac{1}{k}e^{kx}, \quad \int \sin kx\,dx = -\tfrac{1}{k}\cos kx, \quad \int \cos kx\,dx = \tfrac{1}{k}\sin kx$$

tan

$$\int \tan x\,dx = \ln|\sec x| + c$$

the slides write $\ln(\sec x)$

Finding c

integrate, substitute the known point, solve for c

Definite integral

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

top minus bottom; no $+c$

Total area

solve $f(x) = 0$, integrate each piece, add the sizes

Find the unknown

integrate with the letter as a number, set equal to the given value, solve

Trapezium rule

$$\frac{h}{2}\big(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\big), \quad h = \frac{b-a}{n}$$

Your turn

Check every indefinite answer by differentiating it. Answers in grey.

Trapezium rule

1. Estimate the area under $y = x^2$ from 0 to 2 with 2 strips, then 4 strips. (True: 8/3.)

1. 3 and 2.75

Indefinite integrals

2. $\int 6x^2\,dx$
3. $\int (3x^2 - 4x + 5)\,dx$
4. $\int (2/x)\,dx$
5. $\int e^{(3x)}\,dx$
6. $\int 4\cos 2x\,dx$

2. $2x^3 + c$ 3. $x^3 - 2x^2 + 5x + c$ 4. $2\ln|x| + c$ 5. $\frac{1}{3}e^{(3x)} + c$ 6. $2\sin 2x + c$

Find the curve

7. $dy/dx = 6x + 2$ and the curve passes through (1, 10).

7. $y = 3x^2 + 2x + 5$

Definite integrals

8. $\int$ from 0 to 3 of $x^2\,dx$
9. $\int$ from 1 to 2 of $(4x^3 - 1)\,dx$
10. The shaded area between $y = x^3$ and the x axis from -1 to 1

8. 9 9. 14 10. ½ (each side is ¼; the integral itself is 0)

Rewrite first

11. $\int 4/x^3\,dx$
12. $\int 6\sqrt{x}\,dx$
13. $\int (x + 2)^2\,dx$
14. $\int (x^3 - 5)/x\,dx$
15. Total area between $y = x^2 - 1$ and the x axis from $x = 0$ to $x = 2$

11. $-2/x^2 + c$ ($4x^{-3} \to 4x^{-2}/(-2)$) 12. $4x^{(3/2)} + c$ 13. $x^3/3 + 2x^2 + 4x + c$ 14. $x^3/3 - 5\ln|x| + c$ 15. 2 (crosses at $x = 1$: pieces $-\frac{2}{3}$ and 4/3)

Find the unknown

16. $dy/dx = 4x + 3$ and the curve passes through $(1, 2)$. Find y.

17. $\int$ from 1 to k of $2x\,dx = 15$, with $k > 1$. Find k.

18. $\int$ from 0 to 3 of $(ax + 2)\,dx = 15$. Find a.

19. $\int$ from 0 to k of $3x^2\,dx = 64$. Find k.

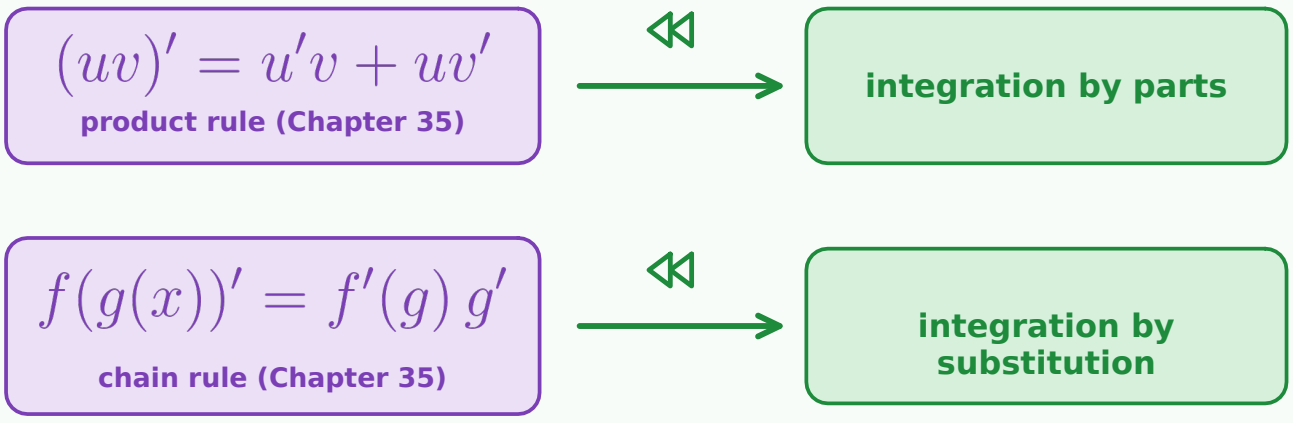
20. $f'(x) = 6x^2 - 2/x^2$ and $f(1) = 5$. Find $f(x)$.

16. $y = 2x^2 + 3x - 3$ ($2 + 3 + c = 2$) 17. $k = 4$ ($k^2 - 1 = 15$) 18. $a = 2$ ($9a/2 + 6 = 15$) 19. $k = 4$ ($k^3 = 64$) 20. $f(x) = 2x^3 + 2/x + 1$ ($2 + 2 + c = 5$)

Chapter 37

Integration by parts and substitution

rewind the rules



each technique is a differentiation rule played in reverse

Chapter 36 could integrate single pieces: x^n, e^{kx}, $\sin kx$. It could also add pieces. But what about pieces **multiplied** together, like $x \sin x$, or a function **inside** another, like $(3x+1)^4$?

There's no simple "multiply the integrals" rule. Instead there are two techniques, and each one is a differentiation rule from Chapter 35 run backwards, like pressing rewind.

Goes with **Lecture 17** (recap, products of functions, integration by parts and its derivation, repeated parts, ln x, choosing u, definite integrals by parts, integration by substitution, substitution with limits, the general solution).

⏪ Warm-up: top minus bottom

Lecture 17 opens with a reminder of Chapter 36. Put the top limit in, put the bottom limit in, subtract. A scoreboard helps keep the two apart:

$$\int_0^2 3x^3 \, \mathrm{d}x = \left[\tfrac{3}{4}x^4\right]_0^2$$ ← *power up, ÷ new power*

TOP (x = 2)	BOTTOM (x = 0)
$\frac{3}{4} \times 2^4 = 12$	$\frac{3}{4} \times 0^4 = 0$

$$12 - 0 = 12$$

Lecture 17's your turn: $\int_1^2 \left(2x^{1/2} - 18x\right) \mathrm{d}x$. The half power is just another power (Chapter 9): up by one to $\frac{3}{2}$, then divide by $\frac{3}{2}$, which is the same as multiplying by $\frac{2}{3}$.

$$= \left[\tfrac{4}{3}x^{3/2} - 9x^2\right]_1^2$$ ← *2 × ⅔ = 4/3, 18 ÷ 2 = 9*

TOP (x = 2)	BOTTOM (x = 1)
$\frac{4}{3}(2)^{3/2} - 36 \approx -32.23$	$\frac{4}{3} - 9 \approx -7.67$

$$-32.23 - (-7.67) = -32.23 + 7.67 = -24.56$$

minus a minus is a plus

⏪ Multiplied pieces: the trap

Adding is easy: $\int (f+g) = \int f + \int g$. So it's tempting to guess that multiplying works the same way. **It doesn't.** Test it on the simplest product there is, $x \times x$:

$$\int x \cdot x \, \mathrm{d}x = \int x^2 \, \mathrm{d}x = \frac{x^3}{3} \quad \longleftarrow \text{the true answer}$$

$$\left(\int x \, \mathrm{d}x\right)\left(\int x \, \mathrm{d}x\right) = \frac{x^2}{2} \cdot \frac{x^2}{2} = \frac{x^4}{4} \quad \times \quad \longleftarrow \text{the guess: wrong power, wrong number}$$

Lecture 17's examples of products: $\int (x^2+7)\sin x \, \mathrm{d}x$, $\int x e^{2x} \, \mathrm{d}x$, $\int f(x)\, g(x) \, \mathrm{d}x$. For these we need **integration by parts**.

Where parts comes from: one rectangle, two pieces

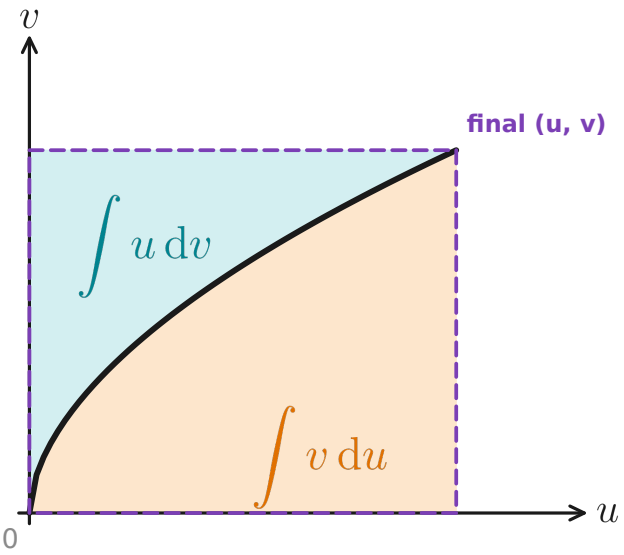
RULE: integration by parts

$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$. Use it for a **product of two different kinds of function** ($x\sin x$, xe^{2x}, $x^2\ln x$).

One factor is u (you will **differentiate** it); the other is $\frac{\mathrm{d}v}{\mathrm{d}x}$ (you will **integrate** it to get v).

Choosing u: LIATE. Whichever factor comes first in Logs, Inverse trig, Algebra (powers of x), Trig, Exponentials is u. A good choice makes the leftover integral $\int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$ simpler than the original.

Where the formula comes from. Chapter 35 pictured $u \times v$ as the area of a rectangle. Now let u and v change together, so that as u grows from 0 to its final size, v grows along a curve. The curve cuts the rectangle into two pieces:



orange: the area under the curve, strips standing on the u axis: ∫ v du

teal: the area beside the curve, strips lying against the v axis: ∫ u dv

together: the whole rectangle, uv

$$\int u\,\mathrm{d}v + \int v\,\mathrm{d}u = uv \quad \longleftarrow \text{the two pieces fill the rectangle}$$

$$\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u \quad \longleftarrow \text{move the orange piece across}$$

So if you want the teal piece but it's hard, you can get it from the rectangle minus the orange piece. That swap is the whole trick. Writing $\mathrm{d}v$ as $\frac{\mathrm{d}v}{\mathrm{d}x}\mathrm{d}x$ gives Lecture 17's version:

$$\int u\,\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = u\,v - \int v\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

Lecture 17's algebra route (the same thing without the picture): start from the product rule for $y = uv$.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = v\frac{\mathrm{d}u}{\mathrm{d}x} + u\frac{\mathrm{d}v}{\mathrm{d}x}$$ ← Chapter 35

$$u\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}x} - v\frac{\mathrm{d}u}{\mathrm{d}x}$$ ← subtract to isolate one term

$$\int u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = \int \frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x - \int v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$ ← integrate every term

$$\int \frac{\mathrm{d}y}{\mathrm{d}x}\,\mathrm{d}x = y = uv$$ ← integrating undoes differentiating

⏪ The workings box: x sin x

PROCEDURE: integration by parts, step by step

1. Choose u by LIATE; the other factor is $\frac{dv}{dx}$.
2. Differentiate u to get $\frac{du}{dx}$. Integrate $\frac{dv}{dx}$ to get v (no $+C$ yet).
3. Substitute into $uv - \int v\frac{du}{dx}\,dx$.
4. Do the leftover integral. If it is still a product, use parts again.
5. Add a single $+C$ at the end, and check by differentiating.

The formula has four ingredients. Lecture 17 lays them out in a box. You **choose** the top row; the bottom row follows. u gets **differentiated** and $\frac{dv}{dx}$ gets **integrated**:

$u = x$

↓ differentiate

$\frac{du}{dx} = 1$

$\frac{dv}{dx} = \sin x$

↓ integrate

$v = -\cos x$

u = x because it turns into a plain 1 when differentiated.

Now feed the four boxes into the formula, colour for colour:

$$\int u\,\frac{dv}{dx}\,dx = u\,v - \int v\,\frac{du}{dx}\,dx$$

$$\int x\sin x\,dx = x\,(-\cos x) - \int (-\cos x)\,(1)\,dx$$

$$= -x\cos x + \int \cos x\,dx \quad \leftarrow \text{minus times minus}$$

$$= \sin x - x\cos x + C \quad \leftarrow \int \cos x = \sin x \text{ (Chapter 36)}$$

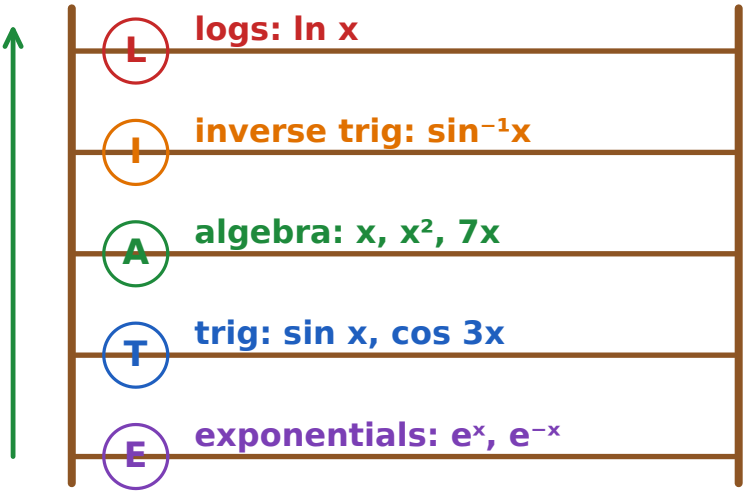
check: $\frac{d}{dx}(\sin x - x\cos x) = \cos x - (\cos x - x\sin x) = x\sin x$ ✓

product rule on x cos x

About the +c. Lecture 17 carries a $+c$ on v and it cancels out ($+cx$ then $-cx$). So leave v without one and add a single $+C$ at the very end. (The slide ends with "$-d$": any constant, plus or minus, means the same thing.)

⏪ Which one is u? The ladder

Lecture 17: you must be able to **differentiate u easily** and **integrate dv/dx easily**, and the leftover integral must be no harder than the one you started with. A handy ordering: climb the ladder, and whichever piece sits **higher** becomes u.



higher rung = make it u

single powers of x are great u's (Lecture 17): differentiate them and they shrink to a number

What happens if you pick the wrong one? Try $x \sin x$ both ways:

u = x (A beats T)

$$\frac{du}{dx} = 1, \quad v = -\cos x$$

$$\text{left over: } \int -\cos x \, dx$$

power of x fell: $x \to 1$

✓ **easier**

u = sin x (wrong way)

$$\frac{du}{dx} = \cos x, \quad v = \frac{x^2}{2}$$

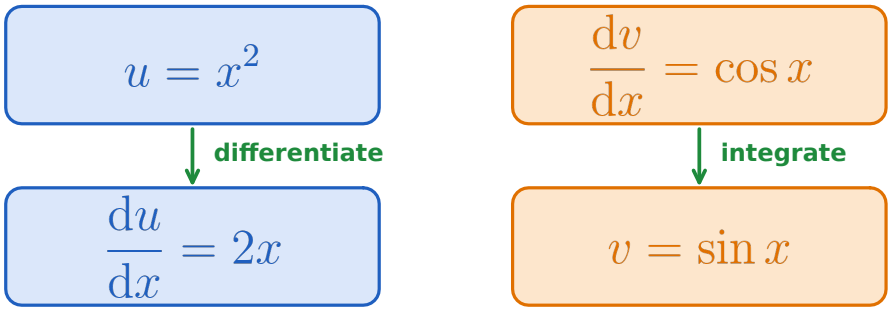
$$\text{left over: } \int \frac{x^2}{2} \cos x \, dx$$

power of x rose: $x \to x^2$

✗ **harder**

⏪ Parts twice: x² cos x

Lecture 17's harder example: $\int x^2 \cos x \, dx$. By the ladder, $u = x^2$ (A beats T).



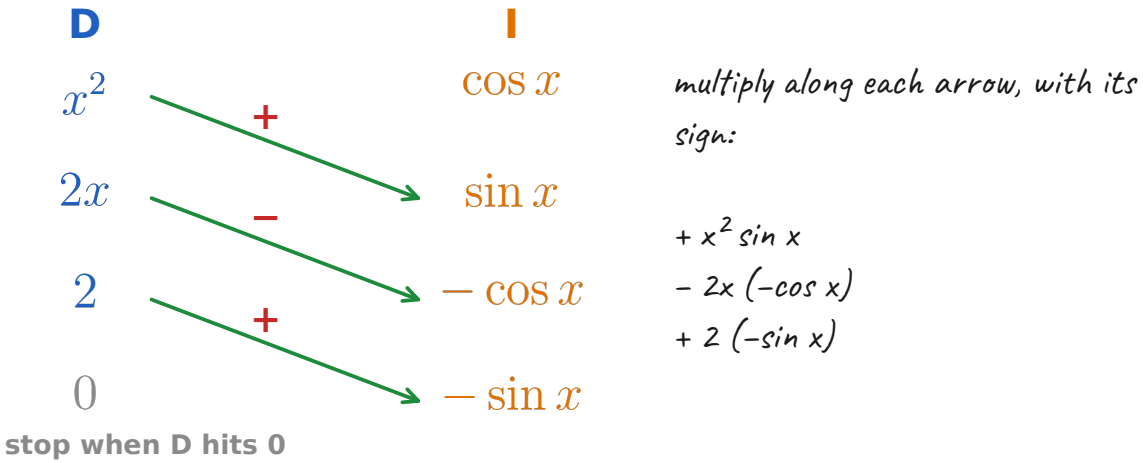
$$\int x^2 \cos x \, dx = x^2 \sin x - \int \sin x \cdot 2x \, dx$$

The leftover $\int 2x \sin x \, dx$ is **still a product**. But it's $2\times$ the example we just did, so reuse it:

$$= x^2 \sin x - 2(\sin x - x \cos x) + C \quad \longleftarrow \textit{Lecture 17's answer}$$

$$= x^2 \sin x + 2x \cos x - 2 \sin x + C \quad \longleftarrow \textit{brackets opened}$$

A shortcut table. When u is a power of x, parts keeps going until u is differentiated down to 0. Make two columns: **D** (keep differentiating u) and **I** (keep integrating). Join each D to the **next row's** I, with signs $+, -, +, \ldots$ taking turns:



$$= x^2 \sin x + 2x \cos x - 2 \sin x + C \quad \longleftarrow \checkmark \textit{ the same}$$

⏪ The invisible 1: integrating ln x

$\ln x$ isn't in Chapter 36's table, and it isn't a product... unless you spot the **invisible 1** that's always there:

$$\int \ln x \, \mathrm{d}x \longrightarrow \int 1 \times \ln x \, \mathrm{d}x$$

ln is top of the ladder, so u = ln x and the 1 is dv/dx

$u = \ln x$	$\dfrac{\mathrm{d}v}{\mathrm{d}x} = 1$
↓ differentiate	↓ integrate
$\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1}{x}$	$v = x$

$$\int \ln x \, \mathrm{d}x = x \ln x - \int x \cdot \frac{1}{x} \, \mathrm{d}x$$

$$= x \ln x - \int 1 \, \mathrm{d}x = x \ln x - x + C$$ ← *x × 1/x = 1*

⏪ Parts with limits

RULE: definite integration by parts (Lecture 17)

$$\int_a^b u\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = \Big[uv\Big]_a^b - \int_a^b v\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

The uv part gets square brackets and its own top-minus-bottom; the leftover integral keeps the **same limits**. No $+C$ anywhere.

Tidy approach: find the whole indefinite answer first, then put it in one set of square brackets and do top minus bottom once.

Lecture 17: $\int_0^2 xe^{-x}\,\mathrm{d}x$. By the ladder, $u = x$ (A beats E).

$u = x$ → differentiate → $\dfrac{\mathrm{d}u}{\mathrm{d}x} = 1$

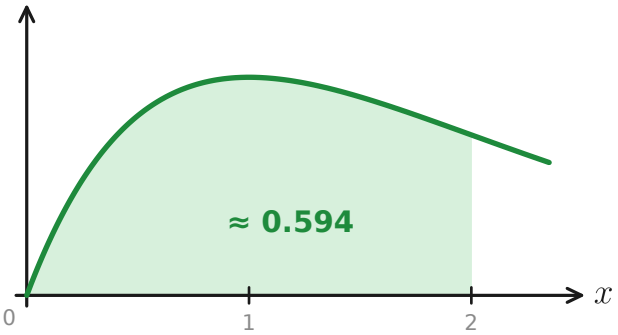
$\dfrac{\mathrm{d}v}{\mathrm{d}x} = e^{-x}$ → integrate → $v = -e^{-x}$

∫e^(−x) = −e^(−x): the k = −1 case of ∫e^(kx) = (1/k)e^(kx)

$$= \Big[-xe^{-x}\Big]_0^2 + \int_0^2 e^{-x}\,\mathrm{d}x = \Big[-xe^{-x} - e^{-x}\Big]_0^2$$

$$= (-2e^{-2} - e^{-2}) - (0 - 1) = 1 - 3e^{-2} \approx 0.594$$

top minus bottom



sanity check: the shape is under 0.37 tall and 2 wide, so an area a bit over half makes sense

Slide slip: Lecture 17's working writes $-xe^{x}$ where it should be $-xe^{-x}$, and e^{2} where it should be e^{-2}. Its final answer $1 - 3e^{-2}$ is right.

⏪ A trig one with limits in radians

Lecture 17: $\int_{\pi/4}^{\pi/2} 7x \sin 3x \, dx$. The limits are in radians (Chapter 27).

$u = 7x$ → differentiate → $\frac{du}{dx} = 7$

$\frac{dv}{dx} = \sin 3x$ → integrate → $v = -\frac{1}{3}\cos 3x$

∫ sin kx = −(1/k)cos kx with k = 3

$$= \left[-\frac{7x}{3}\cos 3x \right] + \int \frac{7}{3}\cos 3x \, dx = \left[-\frac{7x}{3}\cos 3x + \frac{7}{9}\sin 3x \right]_{\pi/4}^{\pi/2}$$

TOP x = π/2

$\cos\frac{3\pi}{2} = 0, \ \sin\frac{3\pi}{2} = -1$

$0 + \frac{7}{9}(-1) = -\frac{7}{9}$

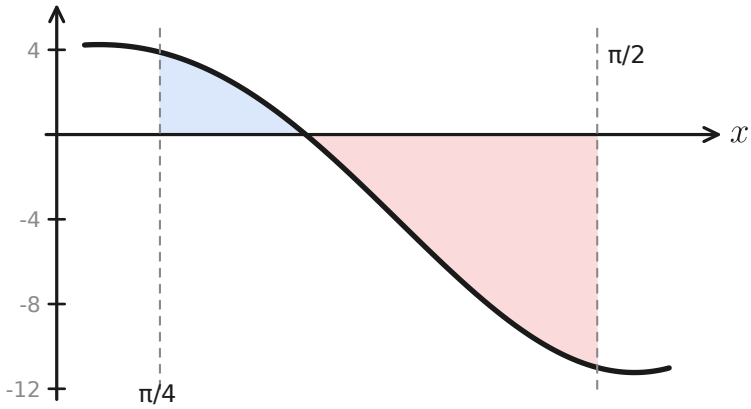
BOTTOM x = π/4

$\cos\frac{3\pi}{4} = -\frac{1}{\sqrt{2}}, \ \sin\frac{3\pi}{4} = \frac{1}{\sqrt{2}}$

$\frac{7\pi}{12\sqrt{2}} + \frac{7}{9\sqrt{2}} \approx 1.846$

$$-\frac{7}{9} - 1.846 \approx -0.778 - 1.846 \approx -2.62$$ ← top minus bottom

A negative answer? Chapter 36: area below the axis counts negative. The graph shows why:



a small blue piece above the axis (π/4 to π/3), a much bigger red piece below (π/3 to π/2). Net: negative.

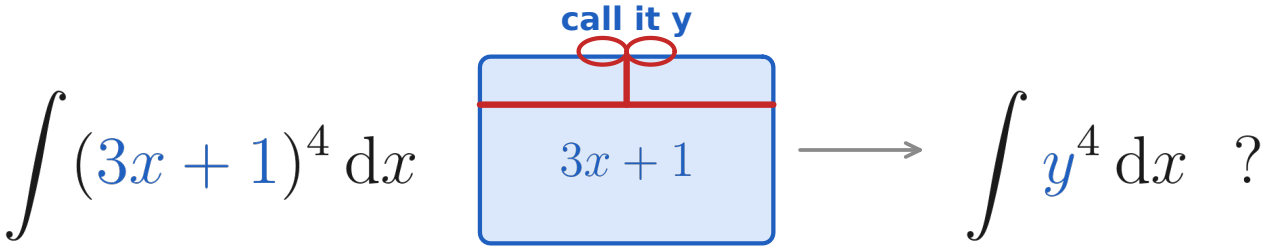
⏪ Substitution: rename the messy chunk

PROCEDURE: integration by substitution

1. **Choose the new letter**: let $y =$ the inside of a bracket, power or function, ideally one whose derivative (give or take a number) also appears in the integral.
2. Differentiate: find $\frac{\mathrm{d}y}{\mathrm{d}x}$, then rearrange to $\mathrm{d}x = \dfrac{\mathrm{d}y}{\mathrm{d}y/\mathrm{d}x}$.
3. Substitute both, so that **every** x disappears (matching x factors cancel). If an x is left over, this choice of y doesn't work.
4. Integrate with respect to y.
5. Indefinite: substitute back, $y =$ (the chunk), and add $+C$. Definite: instead **change the limits** into y-values and never go back to x.

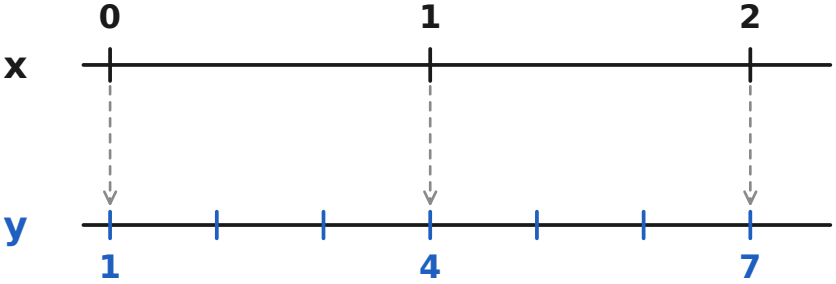
Lecture 17 calls the new letter y, because u is already busy in the parts formula. Many books call it u; the letter doesn't matter as long as it's not x.

Lecture 17: $\int (3x+1)^4 \,\mathrm{d}x$. You could multiply out the fourth power, but that's a pain. Instead, wrap the messy chunk in a parcel and give it a one-letter name:



$$\int (3x+1)^4 \,\mathrm{d}x \longrightarrow \int y^4 \,\mathrm{d}x \quad ?$$

y^4 is easy, but there's a problem: the $\mathrm{d}x$ still says "x". The strips are measured in the wrong units. How do x-steps convert into y-steps? Put two rulers side by side:



one step along x is three steps along y. So dy = 3 dx, which means dx = dy ÷ 3

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3 \;\Rightarrow\; \mathrm{d}x = \frac{\mathrm{d}y}{3}$$ ← Lecture 17

$$\int (3x+1)^4 \,\mathrm{d}x = \int y^4 \,\frac{\mathrm{d}y}{3} = \frac{1}{3} \cdot \frac{y^5}{5} + C$$ ← now everything is y

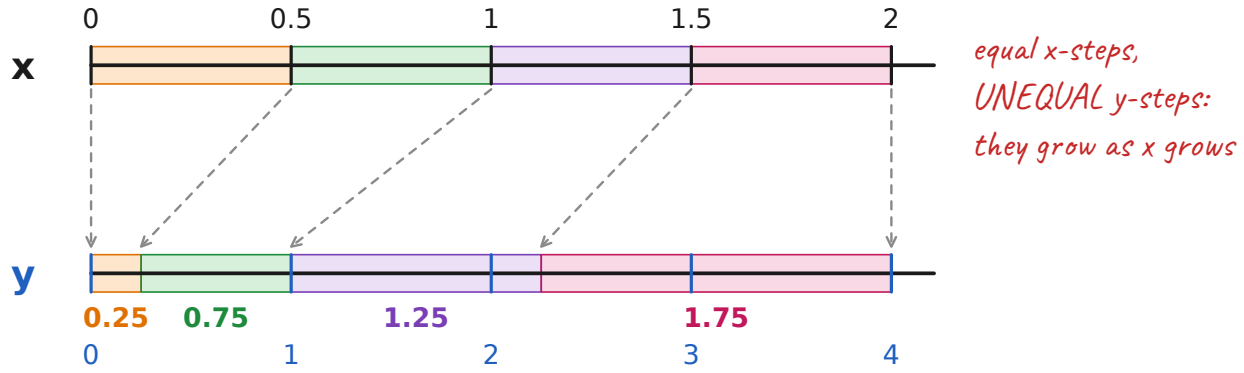
$$= \frac{(3x+1)^5}{15} + C$$ ← unwrap the parcel: y = 3x + 1

check: $\dfrac{\mathrm{d}}{\mathrm{d}x}\dfrac{(3x+1)^5}{15} = \dfrac{5(3x+1)^4 \times 3}{15} = (3x+1)^4$ ✓

chain rule

⏪ The partner outside the bracket

First, why a partner is needed at all. With $y = 3x + 1$ the rulers matched evenly: every x-step was 3 y-steps. Try $y = x^2$. Mark equal x-steps of 0.5 and see where they land on a y-ruler:



Each y-step is $2x \times 0.5$, where x is the middle of that strip: $2(0.25)(0.5) = 0.25$, $2(0.75)(0.5) = 0.75$, and so on. So $\mathrm{d}y = 2x\,\mathrm{d}x$: the exchange rate between the rulers **changes as you go**, and it's $\frac{\mathrm{d}y}{\mathrm{d}x}$. When you swap $\mathrm{d}x$ for $\mathrm{d}y$ you must bring that rate along, and it only disappears if a matching x is sitting outside to cancel it. That's the partner.

Lecture 17: $\int x \sin(2x^2)\,\mathrm{d}x$. The messy chunk is $2x^2$, so let $y = 2x^2$. Its derivative is $4x$... and look: an x is already sitting outside, waiting. Colour them:

$$\int x\,\sin(2x^2)\,\mathrm{d}x, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = 4x \;\Rightarrow\; \mathrm{d}x = \frac{\mathrm{d}y}{4x}$$

$$= \int \cancel{x}\,\sin y\,\frac{\mathrm{d}y}{4\cancel{x}} = \frac{1}{4}\int \sin y\,\mathrm{d}y \quad \leftarrow \text{the partners cancel: only } y \text{ is left}$$

$$= -\frac{1}{4}\cos y + C = -\frac{\cos(2x^2)}{4} + C \quad \leftarrow \int \sin = -\cos$$

Slide slip: Lecture 17 writes $+\frac{\cos(2x^2)}{4}$. Differentiate it and you get $-x\sin(2x^2)$, the wrong sign, because $\int \sin y\,\mathrm{d}y = -\cos y$.

Lecture 17's your turn: $\int t\sqrt{t^2+1}\,\mathrm{d}t$. Chunk t^2+1, derivative $2t$, partner t outside:

$$y = t^2+1, \quad \frac{\mathrm{d}y}{\mathrm{d}t} = 2t, \quad \mathrm{d}t = \frac{\mathrm{d}y}{2t}$$

$$\int \cancel{t}\sqrt{y}\,\frac{\mathrm{d}y}{2\cancel{t}} = \frac{1}{2}\int y^{1/2}\,\mathrm{d}y \leftarrow \sqrt{}\text{ is the power } \tfrac{1}{2}$$

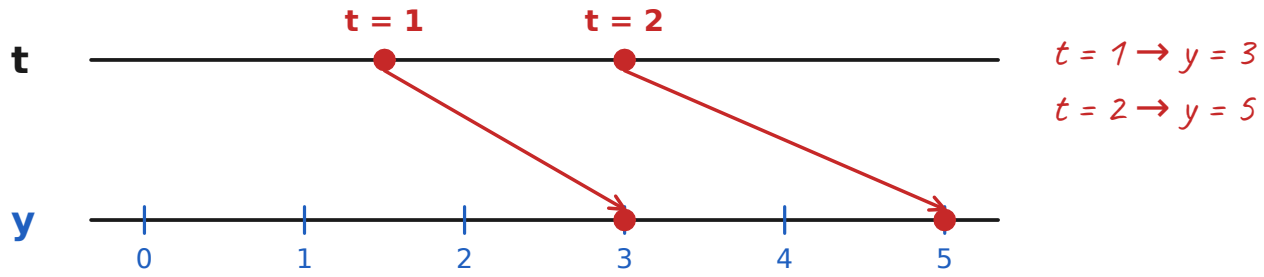
$$= \frac{1}{2}\cdot\frac{2}{3}y^{3/2} + C = \frac{1}{3}(t^2+1)^{3/2} + C$$

The spotting test. Substitution works when the derivative of the chunk (give or take a number) is sitting outside as a partner. No partner, no cancelling: $\int \sin(2x^2)\,\mathrm{d}x$ can't be done this way, because $\frac{\mathrm{d}x}{4x}$ would leave an x stranded among the y's.

⏪ Substitution with limits: change the limits too

Lecture 17: $\int_1^2 \frac{3}{(2t+1)^2}\,\mathrm{d}t$. Let $y = 2t+1$, so $\frac{\mathrm{d}y}{\mathrm{d}t} = 2$ and $\mathrm{d}t = \frac{\mathrm{d}y}{2}$.

The limits 1 and 2 are values of t. Once everything is in y, the limits must be y values too. Convert them with the same rule, like converting pounds into pence:



$$\int_1^2 \frac{3}{(2t+1)^2}\,\mathrm{d}t = \int_3^5 \frac{3}{y^2}\cdot\frac{\mathrm{d}y}{2} = \frac{3}{2}\int_3^5 y^{-2}\,\mathrm{d}y$$

new limits, new letter

$$= \frac{3}{2}\Big[-\frac{1}{y}\Big]_3^5 = \frac{3}{2}\Big(-\frac{1}{5}+\frac{1}{3}\Big) = \frac{3}{2}\cdot\frac{2}{15} = 0.2$$

no need to swap back to t

The general pattern (Lecture 17). Every substitution example had a chunk $g(x)$ **and its derivative** $g'(x)$ multiplied together:

$$\int_a^b f\big(g(x)\big)\,g'(x)\,\mathrm{d}x = \int_{g(a)}^{g(b)} f(y)\,\mathrm{d}y \longleftarrow y = g(x)$$

That's the **chain rule backwards**: Chapter 35's chain rule makes exactly this shape (outer derivative × inner derivative). In $x\sin(2x^2)$, x is a quarter of the derivative of $2x^2$; in $t\sqrt{t^2+1}$, t is half the derivative of t^2.

⏩ Shortcut: a linear inside, ax + b

RULE: integrating f(ax + b)

If $\int f(x)\,dx = F(x) + C$, then $\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C$.

In words: integrate as if the bracket were plain x, then **divide by** a, the number in front of x.

Only when the inside is linear ($ax + b$). For an inside like $x^2 + 1$ you can't divide by its derivative $2x$: use substitution with a partner, or expand.

It is substitution done in your head: $y = ax + b$ gives $dx = \frac{dy}{a}$, and that $\frac{1}{a}$ comes straight out front. Lecture 17's first example agrees: $\int (3x+1)^4\,dx = \frac{1}{3}\cdot\frac{(3x+1)^5}{5} = \frac{(3x+1)^5}{15}$.

$$\int (3x+1)^5\,dx = \frac{1}{3}\cdot\frac{(3x+1)^6}{6} = \frac{(3x+1)^6}{18} + C$$

$a = 3$

$$\int \cos(2x)\,dx = \tfrac{1}{2}\sin(2x) + C \quad \leftarrow a = 2$$

$$\int e^{4x}\,dx = \tfrac{1}{4}e^{4x} + C \quad \leftarrow a = 4$$

$$\int \frac{1}{2x+3}\,dx = \tfrac{1}{2}\ln|2x+3| + C \quad \leftarrow 1/x \to \ln|x|,\ a = 2$$

$$\int \sqrt{4x-1}\,dx = \frac{1}{4}\cdot\frac{(4x-1)^{3/2}}{3/2} = \frac{(4x-1)^{3/2}}{6} + C$$

$a = 4$

✗ Common mistakes

✗ $\int (3x+1)^5 \mathrm{d}x = \frac{(3x+1)^6}{6}$ ✓ $\frac{(3x+1)^6}{18} + C$

divide by the 3 in front of x as well

✗ $\int e^{4x}\,\mathrm{d}x = 4e^{4x}$ ✓ $\int e^{4x}\,\mathrm{d}x = \frac{1}{4}e^{4x} + C$

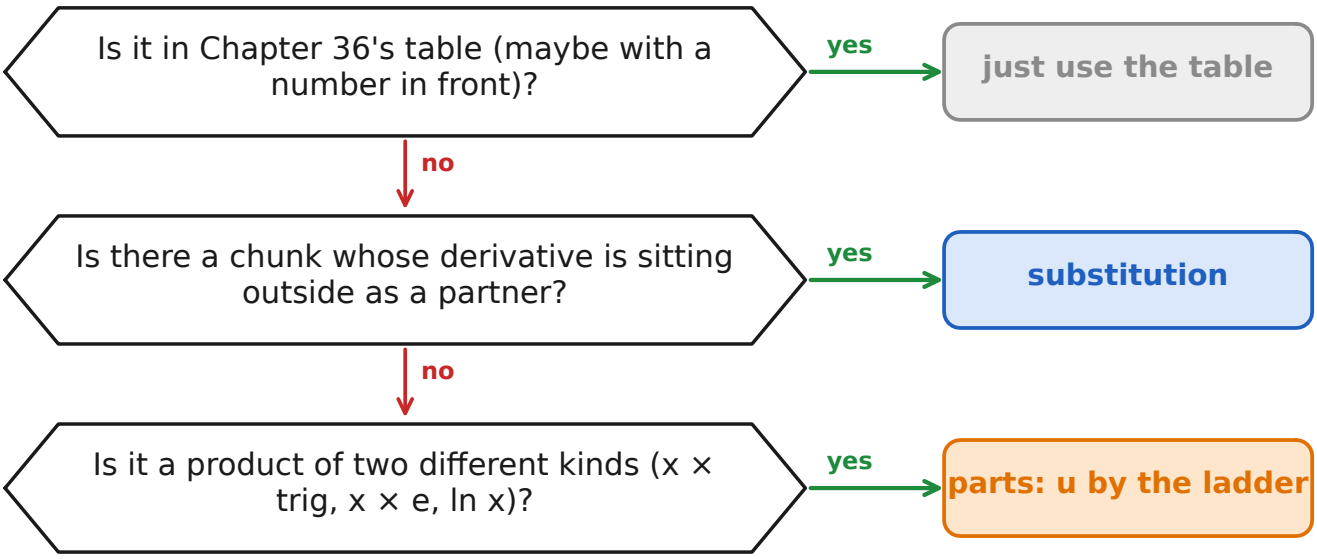
integrating divides by a; differentiating multiplies

✗ $\int (x^2+1)^5\,\mathrm{d}x = \frac{(x^2+1)^6}{6 \cdot 2x}$ ✓ expand first

the shortcut needs a linear inside

⏪ Parts, substitution, or neither? A sorting machine

Ask the questions in this order. Stop at the first yes.



no to all three: try rewriting it first (expand, split, use an identity)

Sort these (answers checked by differentiating):

$\int x\,e^{x^2}\,\mathrm{d}x$	**substitution: y = x², partner x**	$\frac{1}{2}e^{x^2}+C$
$\int x\,e^{x}\,\mathrm{d}x$	**parts: u = x**	xe^x-e^x+C
$\int e^{3x}\,\mathrm{d}x$	**table**	$\frac{1}{3}e^{3x}+C$
$\int \cos x\,\sin^4 x\,\mathrm{d}x$	**substitution: y = sin x, partner cos x**	$\frac{1}{5}\sin^5 x+C$
$\int x\ln x\,\mathrm{d}x$	**parts: u = ln x (L beats A)**	$\frac{x^2}{2}\ln x-\frac{x^2}{4}+C$

And the big picture: **integration by parts undoes the product rule; substitution undoes the chain rule.** Both are the undo button (Chapter 36) for a Chapter 35 rule.

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Integration by parts	$\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$ for a product of two different kinds of function
Choosing u	LIATE: Logs, Inverse trig, Algebra, Trig, Exponentials; the earlier one is u
Parts with limits	$\int_a^b u\frac{dv}{dx}\,dx = \Big[uv\Big]_a^b - \int_a^b v\frac{du}{dx}\,dx$ same limits on both parts
Parts twice	if the leftover is still a product, use parts again (or the D/I table)
ln x	$\int \ln x\,dx = x\ln x - x + C$ parts with $u = \ln x$, $\frac{dv}{dx} = 1$
Substitution	let y = inside chunk; $dx = \frac{dy}{dy/dx}$; every x must disappear; integrate; substitute back
Substitution with limits	$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(y)\,dy$ change the limits; don't go back to x
Linear inside	$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C$ only for $ax+b$
Which method?	in the table: use it. Chunk with its derivative outside: substitution. Product of different kinds: parts. Otherwise rewrite first

⏪ Your turn

Check every answer by differentiating it. Answers in grey.

Integration by parts

1. ∫ x cos x dx
2. ∫ x e^(2x) dx
3. ∫ x² e^x dx (use the D/I table)
4. ∫ from 1 to e of ln x dx

1. x sin x + cos x + C 2. ½x e^(2x) − ¼e^(2x) + C 3. e^x(x² − 2x + 2) + C 4. 1

Integration by substitution

5. ∫ (2x − 5)⁶ dx
6. ∫ 6x (x² + 4)⁵ dx
7. ∫ cos x · e^(sin x) dx
8. ∫ from 0 to 1 of 2x (x² + 1)³ dx (change the limits)

5. (2x − 5)⁷/14 + C 6. (x² + 4)⁶/2 + C 7. e^(sin x) + C 8. 15/4 (limits become 1 and 2)

Which technique?

9. ∫ x sin(x²) dx
10. ∫ x sin x dx
11. ∫ sin 5x dx

9. substitution: −½cos(x²) + C 10. parts: sin x − x cos x + C 11. table: −⅕cos 5x + C

Linear inside shortcut

12. ∫ (5x − 2)³ dx
13. ∫ sin(3x + 1) dx
14. ∫ e^(−2x) dx
15. ∫ 1/(4x + 1) dx

12. (5x − 2)⁴/20 + C 13. −⅓cos(3x + 1) + C 14. −½e^(−2x) + C 15. ¼ ln|4x + 1| + C

Find the unknown

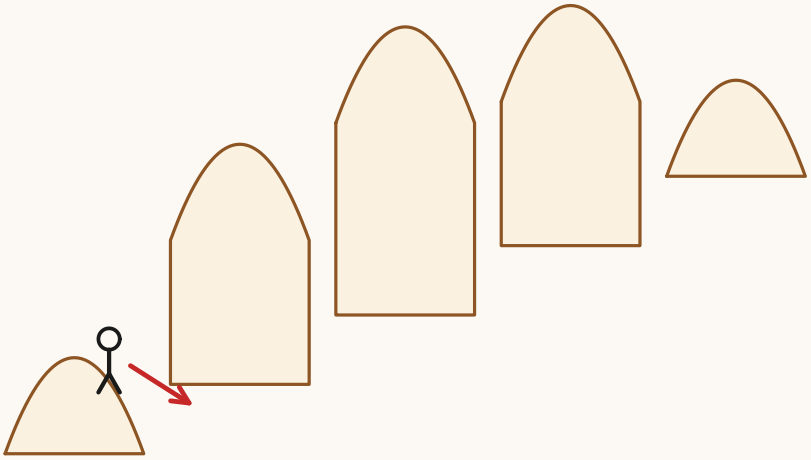
16. ∫ from 0 to k of x e^x dx = 1, with k > 0. Find k.
17. ∫ (ax + 1)⁴ dx = (ax + 1)⁵/10 + C. Find a.
18. ∫ from 0 to k of 2x(x² + 1)³ dx = 255/4, with k > 0. Find k.
19. dy/dx = x e^x and y = 3 when x = 0. Find y.
20. Find the smallest k > 0 with ∫ from 0 to k of cos 2x dx = ½.

16. k = 1 ([x e^x − e^x] gives (k − 1)e^k + 1 = 1) 17. a = 2 (1/(5a) = 1/10) 18. k = √3 (((k² + 1)⁴ − 1)/4 = 255/4, so k² + 1 = 4) 19. y = (x − 1)e^x + 4 (−1 + C = 3) 20. k = π/4 (½ sin 2k = ½, so 2k = π/2)

Chapter 38

Partial differentiation

one direction at a time



a hill is a height that depends on two things: how far east and how far north you are

Every function so far had one input: put in x, get out y. Plenty of real things depend on **two** inputs. The height of a hill depends on how far east and how far north you stand; the volume of a can depends on its radius and its height.

You can't ask "what's the slope?" of a hill: it depends which way you face. So we ask one direction at a time, keeping everything else still. That is **partial differentiation**.

Goes with **Lecture 14** (the notation slide: "∂y/∂x is used when the function is made up of more than one variable"). Not a lecture of its own, but it builds straight on Chapter 34.

Two inputs, one output: a height table

Take the hill $h = 12 - x^2 - 2y^2$, where x is metres east and y metres north of the summit. Every pair (x, y) gives one height, so a spreadsheet of heights describes the hill. Darker means higher:

	x = -2	x = -1	x = 0	x = 1	x = 2
y = 2	0	3	4	3	0
y = 1	6	9	10	9	6
y = 0	8	11	12	11	8
y = -1	6	9	10	9	6
y = -2	0	3	4	3	0

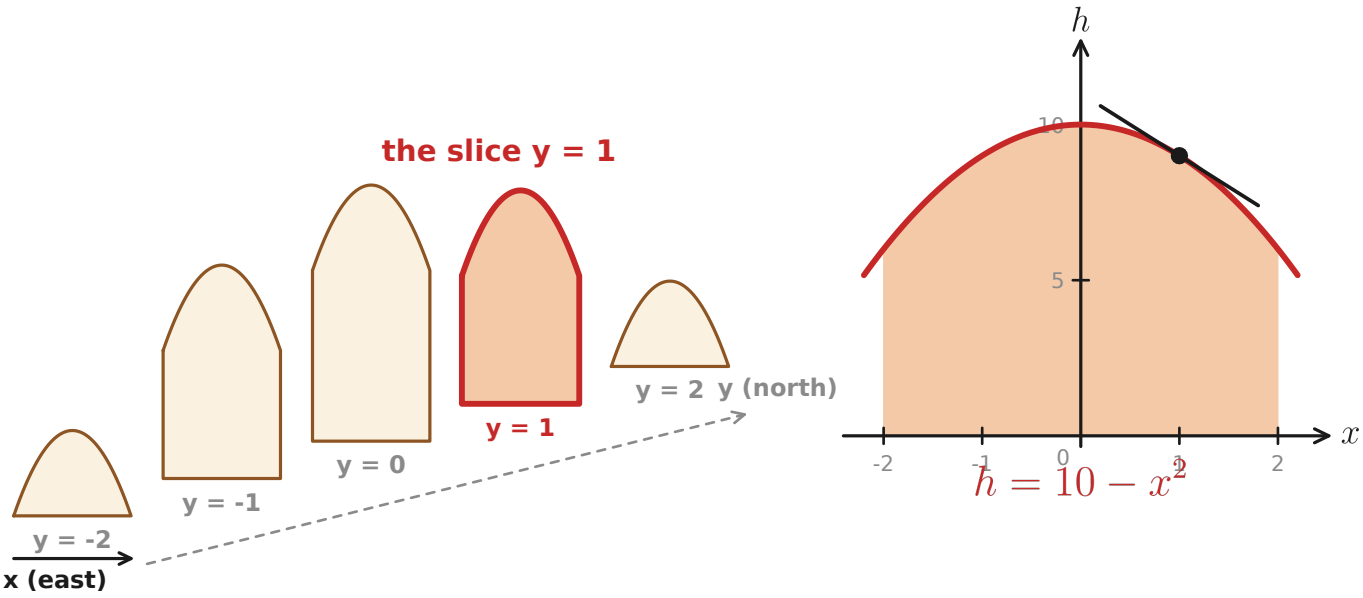
the red cell: 1 m east, 1 m north.
h = 12 − 1 − 2 = 9

walk along a row (east): the heights go 9, 10, 9, 6 … gently.
walk up a column (north): 3, 9, 11, 9, 3: steeper.

We write $h(x, y)$: "h of x and y". The question for the rest of the chapter: **how steep is it** at the red cell? Answer: it depends which way you walk.

Slice the loaf

Draw the hill in 3D and it looks like a loaf of bread. Cut it into slices, each at a fixed y. **Every slice is an ordinary curve**, with only x changing, and Chapter 34 already knows how to find the slope of those.



With y stuck at 1 the formula becomes $h = 12 - x^2 - 2 = 10 - x^2$, a plain one-letter curve. Its slope is $-2x$, so at $x = 1$ it's -2: walking east there, you drop 2 m for every metre.

Freeze the other letter

RULE: partial differentiation

$\frac{\partial f}{\partial x}$: differentiate with respect to x while **holding every other letter constant**. Treat y exactly like a number such as 7.

So a term with no x in it (like $3y^2$) differentiates to **0**, and a frozen letter multiplying x stays as a multiplier: $\frac{\partial}{\partial x}(yx^2) = 2xy$.

All the Chapter 34–35 rules (power, chain, product, quotient) still apply, to the moving letter only.

Subscript notation: f_x means $\frac{\partial f}{\partial x}$ and f_y means $\frac{\partial f}{\partial y}$.

Slicing at $y = 1$ worked, but we'd like a formula for **any** slice. So don't put in a number: just **pretend** y is a number. Picture it frozen in an ice cube. It can't move, so it behaves exactly like the 7 in $7x^2$:

$7x^2 \longrightarrow 14x$

a number in front stays in front

$y\,x^2 \longrightarrow 2x\,y$

so does a frozen y

$7 \longrightarrow 0$

a number alone vanishes

$y^2 \longrightarrow 0$

so does a frozen y on its own

The new notation: a curly ∂ (say "partial" or "del") instead of d, so nobody forgets that another letter was frozen:

$$\frac{\partial h}{\partial x}$$

"partial h by partial x": slope walking east, y frozen

The hill, both ways. $h = 12 - x^2 - 2y^2$.

$$\frac{\partial h}{\partial x} = 0 - 2x - \underbrace{2\,y^2}_{\text{frozen} \to 0} = -2x$$

← 12 → 0 as usual

$$\frac{\partial h}{\partial y} = 0 - \underbrace{x^2}_{\text{frozen} \to 0} - 4y = -4y$$

← now freeze x instead

$$\text{at } (1,1): \quad \frac{\partial h}{\partial x} = -2, \qquad \frac{\partial h}{\partial y} = -4$$

← north is twice as steep

That matches the height table: the column (north) changed faster than the row (east).

Term by term, ice cube by ice cube

Example. $f = x^2y + 3y^2 + 5x$. Go through one term at a time, asking "does this term contain the moving letter?"

term	∂/∂x (freeze y)	∂/∂y (freeze x)
x^2y	$2x\ y$	x^2
$3y^2$	0	$6y$
$5x$	5	0

$$\frac{\partial f}{\partial x} = 2xy + 5, \qquad \frac{\partial f}{\partial y} = x^2 + 6y$$ ← add the columns

Lecture 14's example, printed as $y = 2x^2x + 3z$. Here y is the output and the inputs are x and z. As printed, $2x^2x = 2x^3$:

$$\frac{\partial y}{\partial x} = 6x^2, \qquad \frac{\partial y}{\partial z} = 3$$

If it was meant to be $2x^2z + 3z$ (a more typical example), freeze z for the first and x for the second:

$$\frac{\partial y}{\partial x} = 4x\,z, \qquad \frac{\partial y}{\partial z} = 2x^2 + 3$$

With Chapter 35's rules. A frozen letter changes nothing about the chain rule. It's just a number riding along:

$$z = e^{2x}\sin y: \quad \frac{\partial z}{\partial x} = 2e^{2x}\sin y, \quad \frac{\partial z}{\partial y} = e^{2x}\cos y$$

sin y or e^{2x} is a constant factor

$$z = \ln(x^2 + y^2): \quad \frac{\partial z}{\partial x} = \frac{2x}{x^2 + y^2}$$ ← chain rule: inside's slope ÷ inside

RULE: when the product rule is needed

Look at the factors of a product. If only **one** factor contains the moving letter, the other is a constant multiplier: no product rule.

If **both** factors contain the moving letter, use the product rule $vu' + uv'$, with $'$ meaning $\frac{\partial}{\partial x}$ (or $\frac{\partial}{\partial y}$).

Example. $z = xe^{xy}$. Both factors, x and e^{xy}, contain x:

$$u = x, \quad v = e^{xy}, \quad \frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial x} = y\,e^{xy} \longleftarrow \text{chain: inside } xy \to y$$

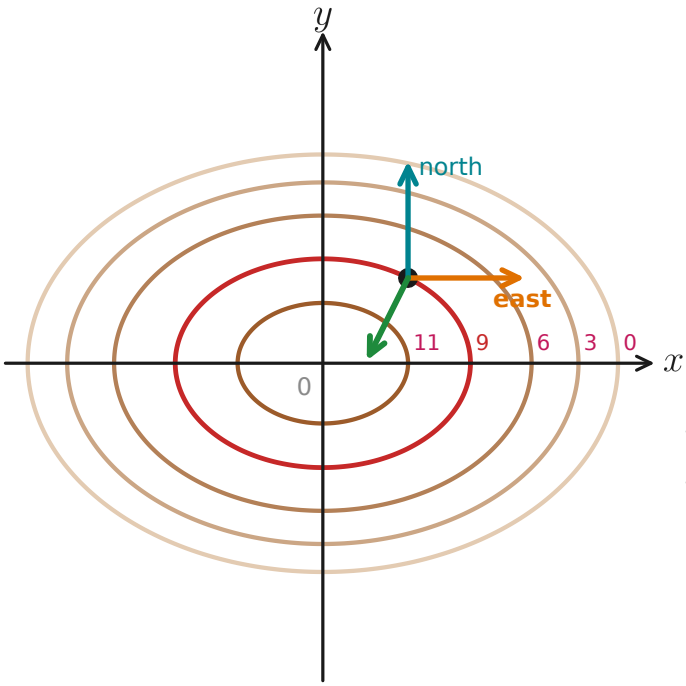
$$\frac{\partial z}{\partial x} = e^{xy} \cdot 1 + x \cdot ye^{xy} = (1 + xy)\,e^{xy} \longleftarrow \text{product rule}$$

Now with respect to y: the first factor x is frozen, so it is just a multiplier:

$$\frac{\partial z}{\partial y} = x \cdot x\,e^{xy} = x^2 e^{xy} \longleftarrow \text{no product rule; chain: } xy \to x$$

The walker's map: contour lines

Hikers don't carry 3D models. They carry a **contour map**: lines joining points of equal height, seen from above. For our hill the contours are ovals. Stand at $(1, 1)$, height 9:



orange: walk east. Slope ∂h/∂x = −2.
teal: walk north. Slope ∂h/∂y = −4 (the contours are more tightly packed that way: closer lines = steeper).

the red oval is height 9, the one you're standing on (12 − 1² − 2 × 1² = 9). Walk along it and the height doesn't change at all.

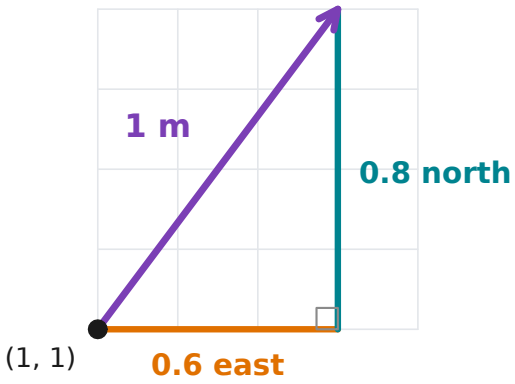
The uphill arrow. Put the two partial slopes together as a vector (Chapter 32):

$$\nabla h = \begin{pmatrix} \partial h/\partial x \\ \partial h/\partial y \end{pmatrix} = \begin{pmatrix} -2 \\ -4 \end{pmatrix} \text{ at } (1, 1)$$ ← *say "grad h"*

This arrow (green, scaled down, on the map) points the **steepest way uphill** and cuts across the contour at a right angle. It points south-west, roughly at the summit, but not exactly: it leans **south** (its y-part, -4, is bigger than its x-part, -2), because the hill is steeper in the north–south direction. Its length, $\sqrt{(-2)^2 + (-4)^2} = \sqrt{20} \approx 4.47$, is how steep that steepest way is.

Walking any other way. Most walks aren't due east or due north. Walk diagonally, and each metre you go splits into an **east part** and a north part:

> ✎ **YOUR GO (pencil first)**
>
> *From (1, 1) you walk 1 m in a direction that takes you 0.6 m east and 0.8 m north. East drops you 2 per metre, north drops you 4 per metre. Roughly how much height do you lose?*



east part: 0.6 × (−2) = −1.2
north part: 0.8 × (−4) = −3.2

total: about 4.4 lost per metre of diagonal walking.

That's the **directional slope**: take each partial slope, weight it by how much of your step goes that way, and add. It's the dot product (Chapter 32) of the uphill arrow with your walking direction: $(-2)(0.6) + (-4)(0.8) = -4.4$. Walk straight along ∇h and you get the steepest possible climb, 4.47; walk along the contour and you get 0.

Second partials: the mixed ones agree

RULE: second partial derivatives and their notation

$\frac{\partial^2 f}{\partial x^2} = f_{xx}$: differentiate by x twice. $\frac{\partial^2 f}{\partial y^2} = f_{yy}$: by y twice.

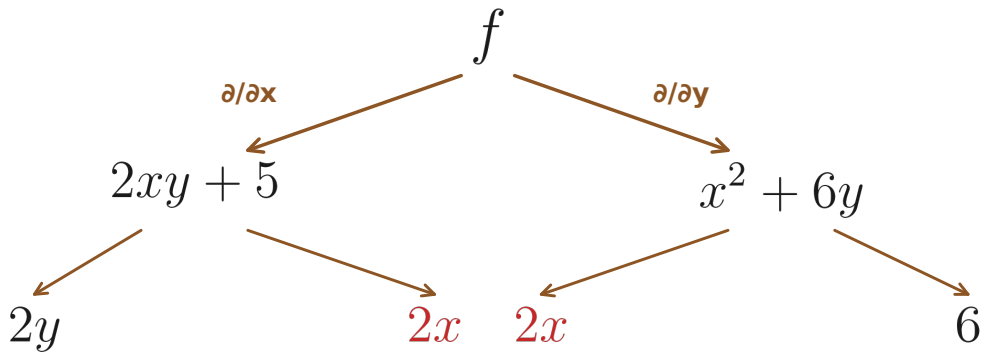
$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$: differentiate by x **first**, then by y. Read the bottom **right to left**.

In subscripts the order reads **left to right**: $f_{xy} = (f_x)_y$, x first then y.

So $f_{xy} = \frac{\partial^2 f}{\partial y\,\partial x}$.

For the smooth functions in this course the two mixed partials are equal, so the order never changes the answer; it is still worth reading the notation correctly.

You can differentiate again, in either letter. With $f = x^2y + 3y^2 + 5x$ from before:

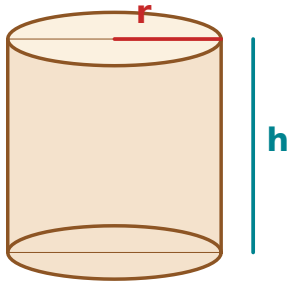


the two middle routes (x then y, y then x) land on the same 2x. They always do for the smooth functions you'll meet.

$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y} \longleftarrow \text{mixed partials are equal}$$

What partials are for: small changes

A can's volume is its circle's area times its height: $V = \pi r^2 h$. A factory makes cans with $r = 2$ cm and $h = 5$ cm, but the machine drifts: r grows by 0.1 and h shrinks by 0.2. Roughly how much does the volume change?



Each partial says how sensitive V is to one measurement while the other stays put. Multiply each by its own nudge and add (the δ's are Lecture 14's "small increments"):

$$\delta V \approx \frac{\partial V}{\partial r}\,\delta r + \frac{\partial V}{\partial h}\,\delta h$$

$$\frac{\partial V}{\partial r} = 2\pi r h = 20\pi, \qquad \frac{\partial V}{\partial h} = \pi r^2 = 4\pi \quad \longleftarrow \text{at } r = 2,\ h = 5$$

$$\delta V \approx 20\pi(0.1) + 4\pi(-0.2) = 2\pi - 0.8\pi = 1.2\pi \approx 3.77 \text{ cm}^3$$

Check with the real volumes: $\pi(2.1^2)(4.8) - \pi(2^2)(5) \approx 3.67$. Close, and the smaller the nudges, the closer it gets.

Find the unknown

PROCEDURE: a partial derivative with a missing value

1. Differentiate with respect to the stated letter, treating the unknown constant (a, k...) and the other letters as numbers.
2. Substitute the given point (x, y).
3. Set the result equal to the given value and solve for the unknown. Check by substituting back.

Example. $f = kxy^2 + x^3$ has $\frac{\partial f}{\partial y} = 8$ at $(1, 2)$. Find k.

$$\frac{\partial f}{\partial y} = 2kxy$$ ← x and k frozen; $x^3 \to 0$

$$2k(1)(2) = 8 \;\Rightarrow\; 4k = 8 \;\Rightarrow\; k = 2$$ ← at (1, 2)

Example. Where are both slopes of $z = x^2 + y^2 - 4x + 6y$ zero (the bottom of the bowl)?

$$\frac{\partial z}{\partial x} = 2x - 4 = 0 \Rightarrow x = 2, \qquad \frac{\partial z}{\partial y} = 2y + 6 = 0 \Rightarrow y = -3$$

Both are zero at $(2, -3)$: the surface is flat in every direction there.

✗ Common mistakes

✗ $\frac{\partial}{\partial x}(3y^2) = 6y$

no x in the term: it is a constant

✓ $\frac{\partial}{\partial x}(3y^2) = 0$

✗ $\frac{\partial}{\partial x}(x^2 y) = 2x + x^2$

frozen y is a multiplier, not a product

✓ $\frac{\partial}{\partial x}(x^2 y) = 2xy$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Partial derivative	$\dfrac{\partial f}{\partial x} = f_x$ differentiate by x, holding every other letter constant
Frozen letters	a term with no moving letter → 0; a frozen factor stays as a multiplier
Product rule	only when both factors contain the moving letter
Second partials	$f_{xx} = \dfrac{\partial^2 f}{\partial x^2}, \qquad f_{xy} = \dfrac{\partial^2 f}{\partial y\,\partial x}$ f_{xy}: x first, then y
Mixed partials	$\dfrac{\partial^2 f}{\partial y\,\partial x} = \dfrac{\partial^2 f}{\partial x\,\partial y}$ for smooth functions
Gradient vector	$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix}$ points steepest uphill
Small changes	$\delta z \approx \dfrac{\partial z}{\partial x}\delta x + \dfrac{\partial z}{\partial y}\delta y$
Find the unknown	differentiate, substitute the point, set equal to the given value, solve

Your turn

Freeze, then differentiate. Answers in grey.

Freeze the other letter

1. $f = x^3 + 4xy + y^2$: find $\partial f/\partial x$ and $\partial f/\partial y$
2. $f = 5x^2y^3$: find $\partial f/\partial x$ and $\partial f/\partial y$
3. $z = 7y - 3x$ (careful: very short answers)

1. $3x^2 + 4y$, $4x + 2y$ 2. $10xy^3$, $15x^2y^2$ 3. $\partial z/\partial x = -3$, $\partial z/\partial y = 7$

With the chain rule

4. $z = \sin(3x + 2y)$
5. $z = e^{(xy)}$

4. $3\cos(3x + 2y)$, $2\cos(3x + 2y)$ 5. $y\, e^{(xy)}$, $x\, e^{(xy)}$

The hill $h = 12 - x^2 - 2y^2$

6. Find both slopes at $(2, -1)$.
7. Which way is steeper there, east or north?

6. $\partial h/\partial x = -4$, $\partial h/\partial y = 4$ 7. equally steep (both size 4); east goes down, north goes up

Second partials

8. $f = x^2y^3$: check that the two mixed partials agree.

8. both are $6xy^2$

Small changes

9. A rectangle is 10 by 5. Each side grows by 0.1. Estimate the change in area $A = xy$.

9. $\delta A \approx 5(0.1) + 10(0.1) = 1.5$ (true: 1.51)

Product rule and notation

10. $z = x\, e^{(xy)}$: find $\partial z/\partial x$ and $\partial z/\partial y$.
11. $z = x^2 \sin(xy)$: find $\partial z/\partial x$.
12. $f = x^3y^2$: find f_x, f_y and f_{xy}.

10. $(1 + xy)e^{(xy)}$, $x^2e^{(xy)}$ 11. $2x \sin(xy) + x^2y \cos(xy)$ 12. $3x^2y^2$, $2x^3y$, $6x^2y$

Find the unknown

13. $f = ax^2y + 3y^2$ and $\partial f/\partial x = 12$ at $(1, 2)$. Find a.
14. $f = x^3 + bxy$ and $\partial f/\partial x = 0$ at $(1, 3)$. Find b.
15. f = e^(kxy) and $\partial f/\partial x = 6$ at $(0, 3)$. Find k.
16. $z = x^2 + y^2 - 6x + 2y$: find the point where both partials are zero.
17. $V = \pi r^2 h$ with $r = 2$. Find h so that $\partial V/\partial r = \partial V/\partial h$.

13. $a = 3$ ($2axy = 4a = 12$) 14. $b = -1$ ($3 + 3b = 0$) 15. $k = 2$ ($ky\,e^0 = 3k = 6$) 16. $(3, -1)$ ($2x - 6 = 0$, $2y + 2 = 0$) 17. $h = 1$ ($2\pi rh = \pi r^2$, so $4\pi h = 4\pi$)

Chapter 39

Exam pocket cards

the handful of facts that unlock each topic — like y = mx + c for lines

You said it yourself: once **y = mx + c**, the gradient formula, and "parallel / perpendicular" are in your head, every straight-line question is rinse and repeat. These cards do that for every exam-critical topic.

How to use them

- Read the matching chapter once for the why (pictures, derivations).
- Memorise the card until you can write it from blank paper.
- Drill past-paper questions of that type until the card fires automatically.
- Tier 1 cards first (pass → 70%). Tier 2 after those are automatic.

Past papers reviewed: Goldsmiths Foundation Maths IS50002 (2010–2019), NCC FDM May 2016, Bolton EEE4011, plus your NTU lecture sequence L01–L17. **IS51002** (graph theory, relations, induction) is a **different** module — not the focus of this guide.

Four-week priority map

Your NTU module end-goals (graphs, descriptive stats, rearrange equations, calculus on curves, geometry & probability, vectors) sit mostly in Tier 1. Goldsmiths-style foundation papers match Tier 1 almost question-for-question.

Week 1 — number fluency

Ch 1–5, 9–10: fractions, decimals, %, ratio, indices, bases. No calculator habits.

Week 2 — algebra engine

Ch 8, 11–15: expand, factorise, transpose, linear / simultaneous / quadratic.

Week 3 — structures + chance

Ch 6–7, 16–18, 25: sets, logic, matrices, sequences, functions, probability trees.

Week 4 — graphs, geo, calc core

Ch 19–20, 24, 28, 32, 34, 36: lines, stats, trig, vectors, differentiate, integrate.

Daily drill: **one pocket card from memory**, then **two past-paper parts** of that type. Aim for speed with full working shown — foundation papers often ban calculators.

What the papers actually ask

Goldsmiths IS50002 Foundation Maths is remarkably stable across years. Ten questions, ~10 marks each, no calculator. Typical map → guide chapters:

Q-type (every year-ish)	Guide chapters
Fractions, %, order of ops, d.p./s.f., standard form	**2, 3, 4, 9**
Algebra: expand, substitute, indices, factors	**8, 9, 11, 12**
Number bases (bin/hex) ± matrices sometimes	**10, 16**
Sets, Venn, truth tables (tautology / contingency / contradiction)	**6, 7**
Linear, simultaneous, quadratic	**15**
Sequences and series (AP/GP, Σ)	**17**
Functions: evaluate, compose, inverse	**18**
Graphs / straight lines / intersection	**19, 20**
Trig: Pythagoras, sin/cos/tan, sketch sin	**27, 28, 29**
Matrices 2×2: +, ×, I, inverse	**16**
Probability: dice/coins, trees, with/without replacement	**25**

NTU lectures add descriptive stats (L04), charts (L05), Bayes/Normal (L08–09), vectors (L10), polar (L11), and calculus (L14–17). Those are Tier 1 for your course even when Goldsmiths papers stop before them.

Skip for now (unless your paper is IS51002-shaped): graph theory, relations / orders, proof by induction, homogeneous coordinates, complex numbers, ODEs.

Card index

Tier 1 first. Tick each card when you can reproduce it blank.

○	Ch 02	Fractions	T1
○	Ch 03	Decimals, rounding, standard form	T1
○	Ch 04	Percentage and ratio	T1
○	Ch 09	Indices	T1
○	Ch 10	Number bases	T1
○	Ch 12	Expand and factorise	T1
○	Ch 14	Transposing	T1
○	Ch 15	Solving equations	T1
○	Ch 06	Sets and Venn	T1
○	Ch 07	Logic / truth tables	T1
○	Ch 17	Sequences and series	T1
○	Ch 18	Functions	T1
○	Ch 20	Straight line	T1
○	Ch 16	Matrices (2×2)	T1
○	Ch 25	Probability	T1
○	Ch 28	Trigonometry (right-angled)	T1
○	Ch 24	Statistics (descriptive)	T1
○	Ch 32	Vectors	T1
○	Ch 34	Differentiation (core)	T1
○	Ch 36	Integration (core)	T1
○	Ch 22	Logs and exponentials	T2
○	Ch 26	Bayes and Normal	T2
○	Ch 35	Further differentiation	T2
○	Ch 33	Trig identities / polar	T2

TIER 1 · exam core

FRACTIONS

full explanation → Chapter 2: Fractions

Add / subtract

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

common denom is already in the formula

Multiply

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Divide

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

flip the second, then multiply

Inputs

whole = n/1; mixed → improper first; simplify only if asked

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

DECIMALS, ROUNDING, STANDARD FORM

full explanation → Chapter 3: Decimals

d.p.

look at the next digit: ≥5 rounds up

s.f.

start counting from the first non-zero digit

Standard form

$$a \times 10^n,\ 1 \leq a < 10$$

scientific notation is the same idea

Recurring → fraction

x = 0.ȧḃ…; multiply by 10^k; subtract; solve for x

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

PERCENTAGE AND RATIO

full explanation → Chapter 4: Percentage and ratio

Percent of

$$\frac{x}{100} \times y$$

Multiplier

$$\text{new} = \text{old} \times \frac{100 \pm r}{100}$$

\+ for increase, − for decrease

Reverse %

$$\text{original} = \frac{\text{new}}{1 \pm r/100}$$

undo a completed % change

Share in ratio a:b:c

parts = a+b+c; one part = total ÷ parts; each = parts × one

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

INDICES

full explanation → Chapter 9: Indices (powers)

Laws

$$a^m a^n = a^{m+n},\ \frac{a^m}{a^n} = a^{m-n},\ (a^m)^n = a^{mn}$$

Zero / negative

$$a^0 = 1,\ a^{-n} = \frac{1}{a^n}$$

$a \neq 0$

Fractional

$$a^{1/n} = \sqrt[n]{a},\ a^{m/n} = (\sqrt[n]{a})^m$$

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

NUMBER BASES

full explanation → Chapter 10: Number bases

Binary → decimal

add place values: … 16 8 4 2 1

Decimal → binary

repeated ÷2; remainders bottom→top

Hex

groups of 4 bits; A=10 … F=15

Binary +/−

same as decimal but place values are powers of 2; show working

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

EXPAND AND FACTORISE

full explanation → Chapter 12: Factorisation

Expand

$$(a+b)(c+d) = ac + ad + bc + bd$$

every left term hits every right term

Common factor

$$ab + ac = a(b+c)$$

pull out the HCF of the terms

Quadratic

find two numbers that multiply to ac and add to b; split middle

Difference of squares

$$a^2 - b^2 = (a-b)(a+b)$$

TIER 1 · exam core

TRANSPOSING

full explanation → Chapter 14: Transposing formulae

Balance

do the same inverse op to both sides

Undo order

unwrap outside-in (opposite of BIDMAS)

Goal

get the wanted letter alone on one side

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

SOLVING EQUATIONS

full explanation → Chapter 15: Solving equations

Linear

expand → collect like terms → divide

Simultaneous

elimination (align one letter) or substitution

Quadratic factorise

$$(x-p)(x-q)=0 \Rightarrow x=p \text{ or } x=q$$

zero-product rule

Quadratic formula

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

when factorising fails

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

SETS AND VENN

full explanation → Chapter 6: Sets

Symbols

∪ union (OR), ∩ intersection (AND), ′ complement, − difference

Count union

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

don't double-count the overlap

Power set

$$|P(S)| = 2^n$$

all subsets of an n-element set

Venn habit

fill from the middle out; unknown region = x

TIER 1 · exam core

LOGIC / TRUTH TABLES

full explanation → Chapter 7: Logic

Rows

$$2^n$$

n letters → that many rows

AND / OR / NOT

∧ both T; ∨ either T; ¬ flips

Implication

$$P \Rightarrow Q$$

false only when P is T and Q is F

Classify

final column: all T = tautology; all F = contradiction; mixed = contingency

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

SEQUENCES AND SERIES

full explanation → Chapter 17: Sequences and series

AP nth term

$$a_n = a + (n-1)d$$

a = first term, d = common difference

AP sum

$$S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l)$$

l = last term

GP nth term

$$a_n = ar^{n-1}$$

r = common ratio

GP sum

$$S_n = \frac{a(1-r^n)}{1-r} \ (r \neq 1)$$

|r|<1:\ S_\infty=\frac{a}{1-r}

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

FUNCTIONS

full explanation → Chapter 18: Functions

Evaluate

replace every x with the given number; brackets help

Compose

$$(f \circ g)(x) = f(g(x))$$

inside first

Inverse

swap x and y (or set y = f(x)), then make x the subject; rename

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

STRAIGHT LINE

full explanation → Chapter 20: The straight line

Core

$$y = mx + c$$

m = gradient, c = y-intercept

Gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

rise over run

Point-gradient

$$y - y_1 = m(x - x_1)$$

when you know a point and m

Parallel / perp

$$m_1 = m_2; \quad m_1 m_2 = -1$$

perp: negative reciprocal

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

MATRICES (2×2)

full explanation → Chapter 16: Matrices

Multiply

row of left · column of right; sizes must match

Determinant

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

Inverse

$$\frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

swap diagonal, flip off-diagonal signs

Solve Ax = b

x = $A^{-1}b$ when det ≠ 0

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

PROBABILITY

full explanation → Chapter 25: Probability and Bayes

Basic

$$p = \frac{\text{favourable}}{\text{total}}$$

equally likely outcomes

AND / OR

independent: multiply; mutually exclusive: add

Conditional

$$p(E|F) = \frac{p(E \cap F)}{p(F)}$$

Tree

multiply along a path; add across paths; watch with/without replacement

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

TRIGONOMETRY (RIGHT-ANGLED)

full explanation → Chapter 28: Intro to trigonometry

SOH CAH TOA

$$\sin = \frac{o}{h},\ \cos = \frac{a}{h},\ \tan = \frac{o}{a}$$

Pythagoras

$$h^2 = o^2 + a^2$$

Exact values

0°, 30°, 45°, 60°, 90°: memorise sin/cos/tan table

CAST / graph

find all solutions in the given range from the principal angle

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

STATISTICS (DESCRIPTIVE)

full explanation → Chapter 24: Statistics

Mean

$$\bar{x} = \frac{\sum x}{n}$$

or Σfx / Σf from a frequency table

Median / mode

middle when ordered; most frequent

IQR

Q3 − Q1; resistant to outliers

SD

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \text{ or } n - 1$$

spread around the mean

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

VECTORS

Full explanation → Chapter 32: Vectors

Add / scale

component-wise; k⟨x,y⟩ = ⟨kx, ky⟩

Magnitude

$$|\mathbf{a}| = \sqrt{x^2 + y^2 (+z^2)}$$

Dot product

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta = x_1 x_2 + y_1 y_2$$

perp ⇒ dot = 0

Unit vector

$$\hat{\mathbf{a}} = \mathbf{a}/|\mathbf{a}|$$

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

DIFFERENTIATION (CORE)

full explanation → Chapter 34: Gradients of curves

Power rule

$$\frac{d}{dx}x^n = nx^{n-1}$$

constants differentiate to 0

Sum / multiple

term by term; constant factors stay

Turning points

dy/dx = 0; 2nd derivative (or sign check) for max/min

Tangent

m = dy/dx at the point; then $y - y_1 = m(x - x_1)$

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

INTEGRATION (CORE)

full explanation → Chapter 36: Integration and areas

Reverse power

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \ (n \neq -1)$$

Definite

$$\int_a^b f = F(b) - F(a)$$

area under curve (watch sign below axis)

1/x and e^x

$$\int \frac{1}{x} = \ln|x| + C, \quad \int e^x = e^x + C$$

cover the card · write from memory · check · then do 2 paper parts

TIER 2 · stretch / later lectures

LOGS AND EXPONENTIALS

full explanation → Chapter 22: Logarithms

Definition

$$y = \log_b x \iff b^y = x$$

Laws

$\log(ab) = \log a + \log b,\ \log(a/b) = \log a - \log b,\ \log(a^k) = k \log a$

Solve a^x = b

take logs both sides, or rewrite as log

e and ln

ln = log base e; inverse of exp

cover the card · write from memory · check · then do 2 paper parts

TIER 2 · stretch / later lectures

BAYES AND NORMAL

full explanation → Chapter 26: The Normal distribution

Bayes

$$p(A|B) = \frac{p(B|A)\,p(A)}{p(B)}$$

prior → posterior after evidence

Total probability

partition B via A and A′ (or a tree)

z-score

$$z = \frac{x - \mu}{\sigma}$$

then use the Normal table

cover the card · write from memory · check · then do 2 paper parts

TIER 2 · stretch / later lectures

FURTHER DIFFERENTIATION

full explanation → Chapter 35: Chain, product, quotient

Product

$$(uv)' = u'v + uv'$$

Quotient

$$(u/v)' = (u'v - uv')/v^2$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

function of a function

cover the card · write from memory · check · then do 2 paper parts

TIER 2 · stretch / later lectures

TRIG IDENTITIES / POLAR

full explanation → Chapter 33: Polar coordinates

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Polar ↔ Cartesian

$$x = r\cos\theta,\ \ y = r\sin\theta,\ \ r = \sqrt{x^2 + y^2}$$

fix θ for quadrant

cover the card · write from memory · check · then do 2 paper parts

Appendices

The back of the book

look it up, say it, check it

Word Finder — 402 words, A to Z, with meanings — page 2

Symbol Phrasebook — 116 symbols: see it, say it, mean it — page 14

Formula Pages — 361 formulas, Part by Part — page 22

Answer Key and Tracker — 783 practice questions to tick off — page 46

Slide Errata — 31 slips on the lecture slides, corrected — page 61

Everything here was collected automatically while the chapters were being built, so every page reference points to the place where the idea is **properly explained**, with pictures. References look like 24·p3: Chapter 24, page 3 of that chapter's file.

Use the drawers above like a filing cabinet. In Xournal++ you can also tick, circle and scribble on every page here.

Word Finder

Every maths word in the book, A to Z, with a one-line meaning and where it is explained. Numbers like 36·p3 mean Chapter 36, page 3. The thumb tabs on the right edge work like a dictionary's.

A

acute / obtuse / reflex 27·p2
less than 90° / between 90° and 180° / more than 180°

adding vectors (tip to tail) 32·p5
start each vector where the last one ended; the answer runs from the first start to the last end

addition rule 25·p3
p(E or F) = p(E) + p(F) − p(E and F)

algebraic fraction 13·p2
a fraction with letters in the top, bottom or both, e.g. 3/x or (x + 1)/(x − 2)

all solutions 30·p4
trig equations have many answers: find the calculator's one, its partner, then add or subtract whole turns

ambiguous case 31·p8
two sides and a non-included angle can give two different triangles, since sin B = sin(180° − B)

amplitude 29·p5
the height of a wave's peak above its middle line

angle 27·p2
an amount of turning between two directions

angle sum of a triangle 31·p2
the three angles of any triangle add up to 180°

anticlockwise positive 27·p4
maths angles start at the positive x-axis and turn anticlockwise; clockwise is negative

arc length 27·p8
the distance along the curved edge of a slice of circle: s = rθ (θ in radians)

area 5·p5
the amount of flat surface inside a shape, counted in unit squares

area function 36·p8
A(x) = area under f from a fixed start up to x; its derivative is f(x)

area of a triangle (trig) 31·p9
½ × two sides × sine of the angle between them

area under a curve 36·p2
the region between a graph and the x axis; it measures a total (amount, distance, probability)

arithmetic progression (A.P.) 17·p2
a sequence where you add the same number d each time

asymptote 19·p6
a line a graph gets closer and closer to but never reaches, e.g. the axes for y = 1/x

average 24·p2
a single typical value for a data set: mean, median or mode

axes 19·p2
the two number lines of a graph: the x-axis across and the y-axis up (singular: axis)

B

bar chart 23·p3
separate bars (with gaps) whose heights show the frequency of each category

base 9·p2
in a^n, the number a that gets multiplied by itself

base (of a log) 22·p2
the number being raised to a power, written small: the 2 in $\log_2 8$

Bayes' theorem 25·p10
p(a|b) = p(b|a) p(a) / p(b): turns one conditional probability round

bell curve 26·p4
the Normal distribution's graph: symmetric, one central peak, tails fading to zero

Bessel's correction 24·p12
dividing by n − 1 instead of n for a sample, to undo the underestimate of spread

binary 10·p3
base 2: counting with only the digits 0 and 1

bit 10·p3
a binary digit, 0 or 1

BODMAS 1·p10
the order to do operations: Brackets, Orders (powers), Division/Multiplication, Addition/Subtraction

bounds 3·p13
the smallest and largest values a rounded number could really have been

box plot 24·p9
a box from LQ to UQ with a line at the median and whiskers out to the min and max

byte 10·p11
8 bits; can store $2^8 = 256$ different values (0-255)

C

cancel 13·p5
divide the top and bottom of a fraction by the same factor (something multiplying the whole top and the whole bottom)

cardioid 33·p7
the heart-shaped polar curve $r = a(1 + \cos\theta)$

CAST rule 29·p3
which trig ratios are positive in each quadrant: All (Q1), Sin (Q2), Tan (Q3), Cos (Q4)

categorical data 23·p2
data that are labels or groups (tea / coffee), not quantities

causation 23·p10
one thing actually makes the other happen

chain rule 35·p4
for a function inside a function: dy/dx = dy/dz × dz/dx (outside rate × inside rate)

chord 34·p3
a straight line joining two points on a curve; its gradient is an average rate

circumference 5·p4
the perimeter of a circle

class width 23·p7
the size of a group in a histogram: upper boundary minus lower boundary

coefficient 8·p4
the number multiplying a letter, e.g. 5 in 5x

collinear 20·p19
lying on the same straight line

common difference 17·p2
d, the fixed amount added each time in an arithmetic progression

common factor 12·p3
something that divides every term; taking it out means writing it once outside a bracket

common log 22·p3
log to base 10, the log button: log 1000 = 3

common ratio 17·p7
r, the fixed multiplier in a geometric progression (any term ÷ the one before)

commutative 1·p7, 16·p6
order doesn't matter: a + b = b + a and a × b = b × a (not true for − or ÷)

compare decimals 3·p3
line up the dots, pad with zeros, compare digits from the left

complement 6·p3
everything in the universal set that is not in the given set

complement rule 25·p2
p(E doesn't happen) = 1 − p(E)

completing the square 15·p13
rewriting $x^2 + bx + c$ as $(x + b/2)^2 - (b/2)^2 + c$; for $a \neq 1$ take a out of the x terms first. Gives vertex form $a(x + p)^2 + q$

complex fraction 13·p12
a fraction with fractions in its top or bottom, e.g. (1 + 1/x)/(1 − 1/x); clear it by multiplying top and bottom by the LCD of the small fractions

composite function 18·p8
one function applied to the output of another, written f(g(x)), fg(x) or (f∘g)(x)

compound angle formula 30·p8
sin(A + B) = sin A cos B + cos A sin B

compound interest 21·p5
interest added each period and then earning interest itself: amount = $A \times (1 + \text{rate})^t$

concentration 4·p17
how much of a substance is in each amount of mixture, e.g. 20% or 60 g per litre

conditional probability 25·p4
p(E | F): the probability of E once you know F happened; F becomes the new sample space

constant 8·p4
a term with no letter: a fixed number

constant of integration (+c) 36·p8
the unknown constant added to every indefinite integral

contingency 7·p7
a statement that is sometimes true and sometimes false in its truth table

contour map 38·p8
a map with lines joining points of equal height; lines close together = steep

contradiction 7·p7
a statement that is false in every row of its truth table, e.g. A ∧ ¬A

contrapositive 7·p9
not Q ⇒ not P; always has the same truth as P ⇒ Q

converge 17·p10
a never-ending sum converges if its total settles towards a fixed number

converse 7·p9
the 'if' statement with its two halves swapped: Q ⇒ P

converting to polar 33·p8
replace x by r cos θ and y by r sin θ, then make r the subject

coordinates 19·p2
the pair (x, y) giving a point's position: x across, then y up

correlation 23·p10
two measurements tend to rise (or fall) together; seen as a sloping cloud on a scatter plot

cosine graph 29·p5
the wave y = cos x: the sine wave shifted π/2 left; starts at its peak, 1

cosine rule 31·p5
$a^2 = b^2 + c^2 - 2bc \cos A$: use it with two sides and the angle between, or with all three sides

cross product 32·p13
a×b: a vector at right angles to a and b, of length |a||b| sin θ

cross-multiply 15·p3
from a/b = c/d go straight to ad = cb; it's multiplying both sides by b and by d

D

data 23·p2
recorded facts: measurements, counts or labels

De Morgan's laws 7·p6
not(A and B) = (not A) or (not B); not(A or B) = (not A) and (not B)

De Morgan's laws (sets) 6·p9
$(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$

decimal places (d.p.) 3·p12
the number of digits after the decimal point

decimal point 3·p2
the dot that separates whole units (left) from parts of a unit (right)

definite integral 36·p13
∫ from a to b: a number, F(b) − F(a), the (signed) area between x = a and x = b

degree 27·p2
1/360 of a full turn, written °

degree-radian conversion 27·p6
× π/180 turns degrees into radians; × 180/π turns radians into degrees

denominator 2·p2
the bottom number of a fraction: how many equal parts make one whole

density 5·p12
mass per unit volume: mass ÷ volume

derivative 34·p4
the gradient function dy/dx: the exact gradient of the curve at every x

derivative of a constant 34·p6
zero: a flat line has no slope

determinant 16·p8
for a 2 × 2 matrix, ad − bc; if it is 0 the matrix has no inverse

deviation 24·p10
how far a value is from the mean: $x - \bar{x}$ (negative if below)

diagonal matrix 16·p7
a square matrix whose only non-zero entries are on the top-left to bottom-right diagonal

diameter 5·p4
distance straight across a circle through the centre; twice the radius

difference of two squares 12·p11
something squared minus something else squared; it always factorises as (a − b)(a + b)

differentiation 34·p4
the process of finding the gradient function (derivative) of a function

digit 1·p2
one of the ten symbols 0 1 2 3 4 5 6 7 8 9

direction angle 32·p8
the angle a vector makes with the positive x-axis: $\tan^{-1}(y/x)$, then the quadrant fix

directional derivative 38·p9
the slope in any direction: $(\partial f/\partial x)$(east part) + $(\partial f/\partial y)$(north part), a dot product with ∇f

discriminant 12·p8
$\Delta = b^2 - 4ac$ for $ax^2 + bx + c$. A perfect square: whole-number brackets. Not a square: square-root brackets. Negative: no real brackets (and no real solutions)

displacement 32·p2
where you ended up compared with the start (how far and which way): a vector

distance 32·p2
how far you travelled in total: a scalar

distance between two points 20·p16
$\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$: Pythagoras on the run and rise

distribution 26·p2
the pattern of how often each possible value occurs

distributive law 1·p7
a(b + c) = ab + ac: multiply each part of the bracket

doesn't factorise (with whole numbers) 12·p9
a quadratic like $x^2 + 4x + 2$ whose Δ is not a perfect square: no whole-number brackets, only square-root ones. Exam questions that say "factorise" expect whole numbers

domain 18·p4
the set of inputs a function accepts

dominant term 19·p7
the term that decides a graph's shape in some range: the highest power for large x

dot product 32·p9
a·b = |a||b| cos θ = axbx + ayby (+ azbz): a scalar measuring how much two vectors point the same way

dotplot 23·p3
a number line with one dot stacked up for every data value

double angle formulas 33·p10
sin 2A = 2 sin A cos A; cos 2A = cos²A − sin²A = 2cos²A − 1 = 1 − 2sin²A

doubling time 21·p8
the time it takes for a growing quantity to double

dy/dx notation 34·p4
dy/dx is one symbol meaning 'the gradient of y as x changes'; the d's never cancel

E

e 21·p6
the number 2.71828..., the limit of $(1 + 1/n)^n$; the natural base for growth and decay

element (member) 6·p2
one of the things in a set

elimination 15·p5
adding or subtracting whole equations so one unknown cancels out

empty set 6·p10
the set with no elements, written {} or ∅

entry (element) 16·p2
one number inside a matrix; a_{23} is the one in row 2, column 3

equal matrices 16·p4
matrices of the same size whose matching entries are all equal

equals sign (=) Part 0·p4
the two sides have exactly the same value, like a balanced see-saw

equation Part 0·p2, 8·p5
a maths sentence saying two things have the same value (it has an = sign)

equation of a circle 28·p5
$(x − a)^2 + (y − b)^2 = r^2$: centre (a, b), radius r

equivalent fractions 2·p9
fractions with the same value, like 1/2 and 2/4: top and bottom multiplied by the same number

even function 19·p5
f(−x) = f(x): the graph is a mirror image across the y-axis (e.g. x^2, x^4)

event 25·p2
a set of outcomes you are interested in; a subset of the sample space

exact trig values 28·p12
sin, cos, tan of 30°, 45°, 60° written with roots, from the two special triangles

exclusive or (XOR) 7·p5
true when exactly one input is true, not both

explicit / implicit function 35·p12
explicit: y = f(x); implicit: x and y mixed in one equation

exponent 9·p2
another name for the power or index

exponential decay 21·p4
decrease by the same multiplier (between 0 and 1) each step

exponential function 21·p2
a function with the variable in the power, like 2^x or e^x

exponential growth 21·p4
increase by the same multiplier (bigger than 1) each step

expression 8·p4
a combination of numbers, letters and operations with no equals sign

F

factor 1·p13
a number that divides exactly into another: the factors of 12 are 1, 2, 3, 4, 6, 12

factorial 29·p13
n! = n × (n − 1) × ... × 2 × 1; e.g. 4! = 24

factorise 12·p2
rewrite an expression as a product (things multiplied together); the reverse of expanding

factorise by grouping 12·p4
split four terms into two pairs, factorise each pair, then take out the bracket they share

form / message / criticism 23·p12
Lecture 5's three passes for reading a graph: how it's built, what it says, what's wrong or missing

formula Part 0·p9, 8·p5
a shorthand recipe connecting quantities, written with letters

fraction 2·p2
part of a whole, written as top number over bottom number; also means top ÷ bottom

fraction of an amount 2·p8
a/b of N = (N ÷ b) × a

frequency 23·p2
how many times a value (or group of values) occurs

frequency density 23·p7
frequency ÷ class width: the height of a histogram bar, so that area = frequency

frequency table 24·p6
a table listing each value (or class) with how many times it occurs

frozen variable 38·p4
in a partial derivative, every letter except the one you differentiate by acts like a number

function 18·p2
a rule that gives exactly one output for each allowed input

function of two variables 38·p2
one output from two inputs, like h(x, y): a height for each map position

G

geometric progression (G.P.) 17·p7
a sequence where you multiply by the same number r each time

gradient (slope) 20·p2
how steep a line is: how much y goes up for each 1 that x goes across; m in y = mx + c

gradient at a point 34·p12
the value of dy/dx when the point's x-coordinate is substituted in

gradient vector 38·p8
∇f = (∂f/∂x, ∂f/∂y): points steepest uphill, at right angles to the contours

graph of a function 19·p3
all the points (x, f(x)) drawn on axes

Greek letters 8·p12
letters like θ, π, σ used as variables or constants with traditional meanings

grid method (factorising) 12·p8
work out $\Delta = b^2 - 4ac$ and the pair m, n = $(b \pm \sqrt{\Delta})/2$; the grid's edges are then x, n/a on the side and ax, m on top

H

half-life 21·p8
the time it takes for a decaying quantity to halve

HCF (highest common factor) 1·p15
the biggest number that divides exactly into both numbers

hexadecimal (hex) 10·p6
base 16: digits 0-9 then A-F for 10-15

higher derivatives 35·p2
differentiate repeatedly: d^2y/dx^2, d^3y/dx^3, …, d^ny/dx^n

histogram 23·p3
touching bars for numerical data grouped into ranges; area shows frequency

hundredth 3·p2
one of a hundred equal pieces of 1; written 0.01 or 1/100

hypotenuse 28·p2
the longest side of a right-angled triangle, opposite the right angle

I

identity 8·p5
an equation true for every value of the letters, written with ≡

identity matrix 16·p7
the square matrix I with 1s down the diagonal and 0s elsewhere; AI = IA = A

if and only if (iff) 7·p10
P ⇔ Q: P and Q are always both true or both false

image (range) 18·p4
the set of all outputs a function can produce

implication (if... then) 7·p8
P ⇒ Q: false only when P is true but Q is false

implicit differentiation 35·p12
differentiate every term; each y term gets an extra × dy/dx (chain rule); then solve for dy/dx

improper fraction 2·p2
a fraction whose top is bigger than its bottom (more than one whole)

indefinite integral 36·p8
∫ f(x) dx with no limits: a family of functions, + c

independent equations 15·p8
equations that each give genuinely new information; you need as many as there are unknowns

independent events 25·p5
knowing one happened doesn't change the other's probability; then p(E and F) = p(E) × p(F)

inequality 15·p17
a statement that one side is less than (<), greater than (>), at most (≤) or at least (≥) the other; its solution is a range of values

integer 1·p4
a whole number that may be positive, negative or zero: ..., -2, -1, 0, 1, 2, ...

integers 6·p12
whole numbers including 0 and negatives (ℤ)

integral of 1/x 36·p9
ln|x| + c: the one power the reverse power rule can't do

integral sign ∫ 36·p5
a stretched S for 'sum': adds up infinitely many infinitely thin strips

integrand 36·p5
the function being integrated (the strip height)

integration 36·p5
finding the area under a curve; also the reverse of differentiation

integration by parts 37·p5
∫u (dv/dx) dx = uv − ∫v (du/dx) dx: the product rule run backwards

integration by substitution 37·p12
rename an inner chunk as a new letter, swap dx for the new letter's d, integrate, swap back

intercept 19·p4
where a graph crosses an axis: the y-intercept at x = 0, x-intercepts where y = 0

interquartile range 24·p9
UQ − LQ: the spread of the middle half of the data

intersection 6·p3, 20·p17
the elements that are in both sets

interval 6·p11
all the numbers between two end points

inverse function 18·p9
f^{-1}, the function that reverses f: if f(3) = 9 then $f^{-1}(9) = 3$

inverse matrix 16·p8
A^{-1}, the matrix with $A^{-1}A = I$; it undoes multiplying by A

inverse Normal 26·p12
working backwards from an area (a percentage) to the z-score or x-value that gives it

inverse operation 14·p2
the operation that undoes another one: minus undoes plus, divide undoes times

inverse proportion 4·p16
two amounts whose product stays the same: when one doubles, the other halves

inverse trig functions 28·p8
$\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ (arcsin, arccos, arctan): given the ratio, they give back the angle

irrational number 6·p12
a real number that is not a fraction; its decimals never end or repeat

J

joint probability 25·p3
the probability that two events both happen, p(E ∩ F)

K

kurtosis 26·p13
how peaky or flat a distribution is compared with the Normal (0 for Normal)

L

law of total probability 25·p11
p(B) = p(B|A)p(A) + p(B|A′)p(A′): add up every way B can happen

laws of logs 22·p5
log(ab) = log a + log b, log(a/b) = log a − log b, log a^n = n log a (any base)

LCM (lowest common multiple) 1·p14
the smallest number that is in both times tables

LIATE 37·p7
order for choosing u in parts: Logs, Inverse trig, Algebra, Trig, Exponentials

like terms 11·p2
terms with exactly the same letters raised to the same powers, e.g. $3x^2$ and $-5x^2$

limits of integration 36·p5
the start and end values of x for a definite integral

linear equation 15·p3
an equation where the letter appears only to the power 1 (no x^2, no 1/x)

linearity of integration 36·p12
$\int(f \pm g) = \int f \pm \int g$ and $\int k\,f = k \int f$

local maximum / minimum 34·p15
a hilltop / valley bottom: higher / lower than the points just around it

log scale 22·p4
a scale where equal steps mean multiplying by the same amount (e.g. ×10)

logarithm 22·p2
$\log_b N$ is the power you must raise b to, to get N; it counts how many b's multiply to N

logically equivalent 7·p7
two statements with the same truth value in every row (≡)

lowest common denominator (LCD) 13·p9
the simplest bottom that every fraction's bottom divides into: each different factor of the bottoms, taken the most times it appears in any one bottom

M

magnitude 32·p6
the length of a vector, found by Pythagoras on its components

many-to-one 18·p6
a function where different inputs can give the same output, like x^2 (2 and −2 both give 4)

marginal probability 25·p8
the overall probability of one characteristic, from a margin total

matrix 16·p2
a rectangular table of numbers in rows and columns (plural: matrices)

matrix equation 16·p8
AX = B: a set of simultaneous equations written with one matrix of coefficients A

matrix multiplication 16·p5
entry (i, j) of AB = row i of A paired with column j of B: multiply the pairs and add

mean 24·p2
add all the values and divide by how many there are (the fair-share average)

median 24·p3
the middle value when the data are in order; position (n+1)/2

midpoint 20·p16
the point halfway between two points: average the x-coordinates and the y-coordinates

midpoint (of a class) 24·p6
halfway between a class's boundaries; stands in for its values when estimating a mean

misleading graph 23·p11
a graph whose scale, shape or missing labels give a false impression

mixed number 2·p2
a whole number and a fraction together, such as 1 3/4

mixed partial derivatives 38·p10
$\partial^2 f/\partial x \partial y$: differentiate by one letter then the other; the order doesn't matter

mode 24·p3
the most common value (or busiest group)

modulus (absolute value) 17·p10
the size of a number ignoring its sign, written $|x|$; $|-0.5| = 0.5$

multiple 1·p13
a number in another number's times table: 12 is a multiple of 4

multiplier 4·p6
the number you multiply by to apply a percentage change, e.g. ×1.1 for +10%

mutually exclusive 25·p3
events that cannot both happen: p(E and F) = 0

N

natural log 22·p3
log to base e (2.718...), written ln: ln e = 1

natural numbers 6·p12
the counting numbers 1, 2, 3, ... ($\mathbb{N}$)

necessary condition 7·p11
Q is necessary for P when $P \Rightarrow Q$: P cannot happen without Q

negative area 36·p15
area below the x axis comes out negative in a definite integral

negative correlation 23·p9
one measurement tends to fall as the other rises: the scatter slopes downhill

negative fraction 2·p7
a fraction below zero; -a/b = (-a)/b = a/(-b)

negative of a vector 32·p6
the same arrow pointing the opposite way: every component changes sign

normal (to a curve) 34·p12
the straight line through a point of a curve at right angles to the tangent there

normal distribution 26·p2
the symmetric bell-shaped distribution you get when many small random effects add up

normal probability plot 26·p14
sorted data plotted against the z-scores Normal data would have; straight means Normal

Normal table 26·p6
a table of $\Phi(z)$, the area to the left of z under the standard Normal curve

number base 10·p3
how many digits a counting system uses; each column is worth base × the one to its right

numerator 2·p2
the top number of a fraction: how many parts you have

numerical data 23·p2
data that are measured or counted quantities (heights, ages)

numerical gradient 34·p3
$(f(a + h) - f(a)) \div h$ for a small h: an estimate of the gradient at x = a

O

octal 10·p6
base 8: digits 0-7

odd function 19·p5
$f(-x) = -f(x)$: the graph looks the same after a half turn about the origin (e.g. x, x^3)

one-to-one 18·p6
a function where different inputs always give different outputs, like 2x + 3

opposite / adjacent 28·p2
the side across from the angle θ / the side next to θ that isn't the hypotenuse

origin 19·p2
the point (0, 0) where the axes cross

outlier 23·p8
a value far away from the rest of the data

P

parallel lines 20·p3
lines with the same gradient; they never meet

parallel vectors 32·p11
vectors where one is a scalar multiple of the other: b = t a

partial derivative 38·p4
the derivative with respect to one letter, all the others treated as constants

parts formula 37·p5
$\int u\,dv = uv - \int v\,du$

percentage (%) 4·p2
a number of pieces out of 100 equal pieces of the whole

perfect square (expression) 12·p12
an expression that is a bracket times itself, $a^2 \pm 2ab + b^2 = (a \pm b)^2$, e.g. $x^2 + 6x + 9 = (x + 3)^2$

perimeter 5·p4
the total length of the boundary of a shape

period 29·p5
the length of one complete cycle before a graph repeats

perpendicular 20·p9
meeting at a right angle (90°); for lines, gradients multiply to −1

perpendicular bisector 20·p15
the line through the midpoint of AB at right angles to AB

perpendicular vectors 32·p11
vectors at right angles; their dot product is 0

pie chart 23·p3
a circle cut into slices; each slice's angle is its share of 360°

piecewise function 18·p7
a function defined by different formulas on different parts of its domain

place value 1·p2
the position of a digit tells you its size: ones, tens, hundreds...

point of inflection (stationary) 34·p15
a flat 'ledge' where the curve keeps going the same way

polar coordinates 33·p2
(r, θ): distance r from the origin and angle θ anticlockwise from the positive x axis

polar curve 33·p6
the path traced by r = f(θ) as θ turns: the pen is r from the centre at angle θ

polar to Cartesian 33·p3
$x = r\cos\theta$, $y = r\sin\theta$

polar to Cartesian equation 33·p9
swap $r\cos\theta \to x$, $r\sin\theta \to y$, $r^2 \to x^2 + y^2$; multiply by r first if needed

population 24·p12
every member of the group you want to know about

position vector 32·p7
the vector from the origin to a point; for A(x, y) it is xi + yj

positive correlation 23·p9
both measurements tend to rise together: the scatter slopes uphill

posterior probability 25·p14
the updated probability after the evidence, e.g. p(cancer | positive) = 7.8%

power (index, exponent) 8·p4
the small raised number: how many times to multiply by itself

power rule 34·p5
the derivative of a x^n is a $n\,x^{n-1}$: the power comes down in front and drops by one

power series 29·p13
a function written as an endless sum of powers of x, like $\sin x = x - x^3/3! + \ldots$

power set 6·p10
P(S): the set of all subsets of S, including ∅ and S; |P(S)| = 2^n

prefix (metric) 5·p2
a word-start like kilo- or milli- that multiplies a unit by 1000, 1/1000 ...

prime factorisation 1·p14
a number written as primes multiplied together: 60 = 2 x 2 x 3 x 5

prime number 1·p13
a number bigger than 1 whose only factors are 1 and itself

prior probability 25·p14
the probability before new evidence, e.g. p(cancer) = 1%

probability 25·p2
how likely an event is, from 0 (impossible) to 1 (certain); shaded ÷ total when outcomes are equally likely

probability density 26·p5
the height of a smooth distribution curve; areas under it are probabilities

product rule 35·p8
$d(uv)/dx = v\,du/dx + u\,dv/dx$

proper fraction 2·p2
a fraction less than one whole: top smaller than bottom

proper subset 6·p10
a subset that is not the whole set: $A \subset B$ means $A \subseteq B$ and $A \neq B$

proportion 4·p14
two amounts that always stay in the same ratio

Pythagoras in 3D 28·p5
the diagonal of a box: $d^2 = a^2 + b^2 + c^2$

Pythagoras' theorem 28·p3
in a right-angled triangle, hypotenuse² = (one short side)² + (other short side)²

P

Pythagorean identity 30·p2
$\sin^2\theta + \cos^2\theta = 1$: Pythagoras on the unit circle

Q

quadrant 19·p2
one of the four regions the axes divide the plane into

quadrant correction 33·p5
after $\theta = \tan^{-1}(y/x)$: add 180° (π) in quadrants 2 and 3, add 360° (2π) in quadrant 4; points on the axes are read off the sketch

quadratic 12·p8
an expression whose highest power is x^2, like $x^2 + 4x + 3$

quadratic equation 15·p10
an equation with x^2 as the highest power, e.g. $x^2 + 6x + 5 = 0$

quadratic formula 15·p12
$x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$, the solutions of $ax^2 + bx + c = 0$; the roots are Chapter 12's pair with the sign flipped, divided by a

quantifier 7·p12
'for all' (∀) or 'there exists' (∃): says how many things a statement covers

quartiles 24·p9
the values a quarter (LQ) and three quarters (UQ) of the way through the ordered data

quotient rule 35·p10
$d(u/v)/dx = (v\,du/dx - u\,dv/dx) \div v^2$

R

r sin(θ + α) form 30·p8
$a \sin\theta + b \cos\theta$ rewritten as one wave $r \sin(\theta + \alpha)$, $r = \sqrt{(a^2 + b^2)}$, $\tan\alpha = b/a$

radian 27·p5
the angle made at the centre by an arc as long as the radius; 2π radians = 360°

radius 5·p4
distance from the centre of a circle to its edge

range 24·p9
largest value minus smallest value

rate of change 34·p10
how fast one quantity changes as another does: the derivative

ratio 4·p9
how many parts of one thing for every so many parts of another, written a : b

rational number 6·p12
a number that can be written as a fraction of two integers (ℚ)

real numbers 6·p12
all numbers on the number line (ℝ)

reciprocal 2·p8
a fraction flipped upside down: the reciprocal of 4/5 is 5/4, of 4 (= 4/1) is 1/4

recurring decimal 3·p10
a decimal whose digits repeat forever, e.g. 0.333...; shown with dots

reference angle 29·p3
the acute angle between the radius and the x-axis; trig of θ = ± trig of it

remainder 1·p9
what is left over when a number doesn't share out exactly

restricted domain 18·p5
a domain deliberately limited, e.g. $x \geq 0$, often to make a function one-to-one

reverse chain rule 37·p16
$\int f(g(x))\,g'(x)\,dx = F(g(x)) + C$: substitution undoes the chain rule

reverse power rule 36·p6
$\int a\,x^n\,dx = a\,x^{n+1} \div (n + 1) + c$, for any n except −1

right angle 27·p2
a quarter turn, 90°, marked with a little square

right-hand rule 32·p13
first finger along a, second finger along b, thumb gives the direction of a×b

rounding 3·p12
replacing a number with the nearest one on a coarser ruler

S

sample 24·p12
the members you actually measured, used to estimate the population

sample space 25·p2
the set S of every possible outcome

scalar 16·p4, 32·p2
an ordinary single number (as opposed to a matrix or vector)

scalar multiple 32·p6
k times a vector: k times as long, same direction (reversed if k is negative)

scalene / isosceles / equilateral 31·p2
no equal sides / two equal sides / all three sides equal

scatter plot 23·p3
one dot per item, placed by two measurements (x, y), to see whether they are related

second derivative 34·p17
d^2y/dx^2: the derivative of the derivative; positive at a minimum, negative at a maximum

sector area 27·p8
the area of a slice of circle: $A = \frac{1}{2}r^2\theta$ (θ in radians)

sequence 17·p2
an ordered list of numbers following a rule, e.g. 2, 6, 10, 14, ...

series 17·p2
the sum of the terms of a sequence, e.g. 2 + 6 + 10 + 14

set 6·p2
a collection of distinct things, written inside curly brackets { }

set-builder notation 6·p11
{x | rule}: the set of all x that obey the rule

sigma notation 17·p6
Σ with a start value below and a stop value above: add up the expression for every counter value

significant figures (s.f.) 3·p16
digits counted from the first non-zero digit; how precise a number is

similar triangles 28·p6
triangles with the same angles: one is a zoomed copy of the other, so side ratios match

simplify a fraction 2·p10
divide top and bottom by the same number (ideally the HCF) so the numbers are as small as possible

simultaneous equations 15·p5
two (or more) equations about the same unknowns, which must all be true at once

sine graph 29·p4
the wave y = sin x: starts at 0, peaks at 1 (90°), back to 0 (180°), down to −1 (270°), 0 at 360°

sine rule 31·p3
a/sin A = b/sin B = c/sin C: use it when you know a side and its opposite angle

sine, cosine, tangent 28·p6
the ratios opp/hyp, adj/hyp and opp/adj of a right-angled triangle, fixed by the angle

singular matrix 16·p8
a square matrix with determinant 0; it has no inverse

size of a matrix 16·p2
(number of rows) × (number of columns), said "2 by 3"

skewed data 24·p7
data with a long tail on one side; the mean is dragged towards the tail

small changes formula 38·p11
δz ≈ (∂z/∂x)δx + (∂z/∂y)δy: add up each input's effect

SOHCAHTOA 28·p6
memory aid: Sin = Opp/Hyp, Cos = Adj/Hyp, Tan = Opp/Adj

solution 15·p2
a value of the letter that makes the equation true (also called a root)

solve 15·p2
find the value(s) of the letter that make an equation true

solving exponential equations 22·p7
take logs of both sides and use log a^n = n log a to bring the unknown down

speed 5·p12
distance travelled per unit of time: distance ÷ time

split the middle 12·p6
for $ax^2 + bx + c$: find two numbers that multiply to ac and add to b, use them to split bx into two terms, then factorise in pairs

square root Part 0·p3, 9·p3
the number which, multiplied by itself, gives the number you started with

squared Part 0·p3
multiplied by itself: 5 squared is 5 x 5

standard derivatives 34·p9
the basic list: constants, powers, ln x, e^(kx), sin, cos, tan

standard deviation 24·p11
the square root of the variance; a typical distance from the mean, in the data's units

standard form 9·p15
a number written as a × 10^n with 1 ≤ a < 10

standard integrals 36·p12
the basic list: k, powers, 1/x, e^(kx), sin kx, cos kx, tan x

standard Normal distribution 26·p4
the Normal distribution with mean 0 and standard deviation 1

statement 7·p2
a sentence that is either true or false

stationary point 34·p15
a point where the gradient is zero (the tangent is flat)

stem and leaf diagram 23·p7
each value split into stem (leading digit) and leaf (last digit); leaves listed beside their stem

strip width h 36·p3
h = (b − a) ÷ n: the width of each strip

subject of a formula 14·p2
the letter on its own on one side of the = sign

subscript 16·p2
a small number or letter written low after a symbol, used as a label: a_1, x_2, a_{23}

subset 6·p10
a set whose elements are all inside another set

substitute Part 0·p7
replace a letter with a number and work out the value

S

substitution 8·p6
replacing letters by given numbers, then working out the value

substitution limits 37·p16
when substituting in a definite integral, convert the limits into the new letter

substitution method 15·p7
rearrange one equation for one letter and put it into the other equation

sufficient condition 7·p11
P is sufficient for Q when P ⇒ Q: P on its own guarantees Q

sum of squares 24·p11
add up the squared deviations $\Sigma(x - \bar{x})^2$

sum rule (differentiation) 34·p6
differentiate a sum or difference term by term

sum-product rule 12·p6
to factorise $x^2 + bx + c$, find two numbers that add to b and multiply to c; they go in the brackets $(x + p)(x + q)$

surface area 5·p10
the total area of all the faces of a solid

T

tabular integration 37·p8
D/I table for repeated parts: differentiate u down to 0, integrate the other, alternate signs

tangent (to a curve) 34·p2
the straight line that just touches a curve at one point, going the same way as the curve there

tangent equation 34·p14
$y - y_1 = f'(a)(x - a)$: the line through (a, f(a)) with the curve's gradient there

tangent graph 29·p6
y = tan x: repeats every π, with vertical asymptotes where cos x = 0 (90°, 270°, ...)

tautology 7·p7
a statement that is true in every row of its truth table, e.g. A ∨ ¬A

tenth 3·p2
one of ten equal pieces of 1; written 0.1 or 1/10

term 8·p4, 17·p2
one piece of an expression, separated from the others by + or − signs

time constant 21·p7
for e^(−t/τ), the time τ in which the quantity drops to 1/e ≈ 37% of its value

transformation 19·p9
a change to a formula that moves, stretches or flips its graph

transpose 14·p1
rearrange a formula so that a different letter is on its own

trapezium rule 36·p3
estimate an area with n sloping-topped strips: h/2 × (first + 2 × middles + last)

tree diagram 25·p11
branches for each outcome of each stage; multiply along a path, add between paths

triangle notation 31·p2
angles A, B, C at the corners; sides a, b, c opposite them

trig equation 30·p4
a statement like sin θ = 0.5 that is true only for particular angles

trig identity 30·p2
a trig statement that is true for every angle, e.g. $\sin^2\theta + \cos^2\theta = 1$

trig of any angle 29·p2
cos θ and sin θ are the x- and y-coordinates of the point at angle θ on the unit circle

trig transformations 29·p8
y = A sin(Bx) + D: A stretches the height, B squashes the period to 2π/B, D lifts it

truncate 3·p15
chop off digits after a point without rounding

truth table 7·p3
a table listing every combination of T/F inputs and the result

truth value 7·p2
whether a statement is true (T, 1) or false (F, 0)

turning point 34·p15
where a curve changes from going up to going down, or down to up

two-way table 25·p8
a table counting every combination of two characteristics, with totals in the margins

U

undefined 13·p2
has no value; a fraction is undefined when its bottom equals 0

uniform distribution 26·p2
every value equally likely: a flat histogram

union 6·p3
all elements in either set (or both), listed once

unit 5·p2
the fixed amount you count with: a metre, a gram, a second...

unit circle 29·p2
the circle of radius 1 centred at the origin; the point at angle θ is (cos θ, sin θ)

unit vector 32·p3
a vector of length exactly 1; i, j, k point along the x, y, z axes

unit vector of a 32·p7
â = a ÷ |a|: the vector of length 1 pointing the same way as a

unitary method 4·p14
scale a known pair down to 1 unit, then up to the amount you want

universal set 6·p3
the set of everything under discussion, written ξ or U

V

variable Part 0·p6, 8·p2
a letter standing for a number that can change

variance 24·p11
the mean of the squared deviations; σ² for a population

vector 32·p2
a quantity with size AND direction (velocity, displacement, force)

vector notation 32·p3
an arrow over the letter, bold type, or an underline all mark a vector

Venn diagram 6·p4
overlapping circles inside a rectangle, showing how sets share elements

vertex 19·p9
the turning point of a parabola: (h, k) for y = a(x − h)² + k

vertex form 15·p13
a quadratic written a(x + p)² + q; its vertex (turning point) is (−p, q)

vertical line test 18·p6
a graph is a function if no vertical line crosses it more than once

volume 5·p9
the amount of 3-D space inside a solid, counted in unit cubes

Y

y-intercept 20·p2
c in y = mx + c: the value of y where the line crosses the y-axis (x = 0)

Z

z-score 26·p6
how many standard deviations a value is above (+) or below (−) the mean: z = (x − μ)/σ

zero product rule 15·p10
if A × B = 0 then A = 0 or B = 0

#

3 × 3 determinant 32·p16
cover each top-row entry's row and column, take the 2 × 2 determinant left over, and combine with signs + − +

Symbol Phrasebook

See it, say it, mean it. The middle column is how you'd read the symbol out loud; saying it is half of understanding it (Chapter 0).

Start here: The language of maths

π	*pie*	a fixed number, about 3.14159, from circles **Part 0·p10**
$=$	*equals*	has the same value as **Part 0·p4**
$\neq$	*is not equal to*	different value **Part 0·p4**
$<$	*is less than*	smaller than **Part 0·p4**
$>$	*is greater than*	bigger than **Part 0·p4**
$\leq$	*less than or equal to*	smaller or the same **Part 0·p4**
$\geq$	*greater than or equal to*	bigger or the same **Part 0·p4**
$\approx$	*approximately equal to*	roughly the same **Part 0·p4**
$\equiv$	*identically equal to*	true for every possible number **Part 0·p5**
$\gg$	*much greater than*	far bigger **Part 0·p5**
$\sqrt{}$	*square root of*	which number times itself gives this **Part 0·p3**
x^2	*x squared*	x times x **Part 0·p3**
x_1	*x one*	a name tag: the first x **Part 0·p10**

Part 1: Numbers

$0.\dot{3}$	*nought point three recurring*	the 3 repeats forever **3·p10**
$\approx$	*is approximately equal to*	close to, but not exactly **3·p10**

$\%$	*per cent*	out of 100, so 25% means 25/100 **4·p2**
$a : b$	*a to b*	a parts of one thing for every b parts of the other **4·p9**
$::$	*is in the same ratio as*	a:b :: c:d means a/b = c/d **4·p10**
cm^2	*square centimetres*	a count of 1 cm x 1 cm squares **5·p5**
cm^3	*cubic centimetres*	a count of 1 cm cubes **5·p9**
$\mathrm{m\,s}^{-1}$	*metres per second*	same as m/s **5·p12**
$\{\ \}$	*the set of / braces*	curly brackets that hold the members of a set **6·p2**
$\in$	*is an element of / is in*	x ∈ A means x belongs to set A **6·p2**
$\notin$	*is not an element of*	x ∉ A means x is not in A **6·p2**
$n(A)$	*n of A / the size of A*	how many elements are in set A **6·p2**
ξ	*ksee (or 'the universal set')*	everything being talked about **6·p3**
$\cap$	*intersection / and*	elements in both sets **6·p3**
$\cup$	*union / or*	elements in either set (or both) **6·p3**
A'	*A complement / not A*	everything in ξ that is not in A **6·p3**
$\subset$	*is a subset of*	every element of the left set is in the right set **6·p10**
$\varnothing$	*the empty set*	a set with nothing in it **6·p10**
$\subseteq$	*is a subset of (or equal to)*	every element of the left set is in the right set **6·p10**
$\mid$	*such that*	separates the variable from the rule it must obey **6·p11**

$[a, b]$	the closed interval from a to b	all numbers from a to b, ends included **6·p11**
(a, b)	the open interval from a to b	all numbers between a and b, ends excluded **6·p11**
$[a, \infty)$	a to infinity	all real numbers greater than or equal to a **6·p11**
$\mathbb{N}$	N, the natural numbers	counting numbers 1, 2, 3, ... **6·p12**
$\mathbb{Z}$	Z, the integers	whole numbers including negatives and 0 **6·p12**
$\mathbb{Q}$	Q, the rational numbers	anything that can be written as a fraction **6·p12**
$\mathbb{R}$	R, the real numbers	every point on the number line **6·p12**
$\neg P$	not P	true exactly when P is false **7·p2**
$A \wedge B$	A and B	true only when both A and B are true **7·p3**
$A \vee B$	A or B	true when at least one of A, B is true **7·p4**
$A \oplus B$	A x-or B (exclusive or)	true when exactly one of A, B is true **7·p5**
$\equiv$	is logically equivalent to	same truth value in every row of the table **7·p7**
$P \Rightarrow Q$	P implies Q / if P then Q	false only when P is true and Q is false **7·p8**
$P \Leftrightarrow Q$	P if and only if Q (iff)	P and Q are both true or both false **7·p10**
$\forall$	for all	the statement holds for every value **7·p12**
$\exists$	there exists	at least one value makes the statement true **7·p12**

Part 2: Algebra

$\equiv$	*is identically equal to*	true for all values **8·p5**
α	*alpha (AL-fa)*	angles; radiation (α particles) **8·p12**
β	*beta (BEE-ta)*	angles; radiation **8·p12**
γ	*gamma (GAM-a)*	angles; gamma rays **8·p12**
$\delta,\ \Delta$	*delta (DEL-ta)*	a small change (Δx: change in x) (Ch. 20, 34) **8·p12**
ε	*epsilon (EP-si-lon)*	a tiny amount; errors **8·p12**
θ	*theta (THEE-ta)*	the most common angle letter (Ch. 27-31) **8·p12**
ϕ	*phi (FYE (or fee))*	another angle **8·p12**
λ	*lambda (LAM-da)*	wavelength; eigenvalues **8·p12**
μ	*mu (MEW)*	the mean in statistics (Ch. 24, 26) **8·p12**
$\sigma,\ \Sigma$	*sigma (SIG-ma)*	σ: spread (Ch. 24); Σ: add up (Ch. 17) **8·p12**
π	*pi (PIE)*	3.14159..., the circle number **8·p12**
ω	*omega (oh-MEE-ga)*	angular speed; spinning **8·p12**
χ	*chi (KYE)*	the chi-squared test in statistics **8·p12**
a^n	*a to the power n*	n copies of a multiplied together **9·p2**
$\sqrt{x}$	*the square root of x*	the positive number whose square is x **9·p3**
$\sqrt[n]{x}$	*the n-th root of x*	the number whose n-th power is x **9·p3**
$\pm$	*plus or minus*	both the + and − versions **9·p3**

10110_2	*one-zero-one-one-zero in base two*	the subscript gives the base **10·p4**
$\pm$	*plus or minus*	both the + version and the − version are answers **15·p10**
a_{ij}	*a sub i j*	the entry in row i, column j of matrix A **16·p2**
A^{-1}	*A inverse*	the matrix that undoes A: $A^{-1}A = I$ **16·p8**
u_n	*u sub n*	the nth term of a sequence **17·p2**
S_n	*S sub n*	the sum of the first n terms **17·p4**
$\sum$	*sigma (sum)*	add up the expression for each value of the counter from the bottom number to the top **17·p6**
S_∞	*S infinity*	the sum of all the terms of a never-ending series **17·p10**
$\lvert r \rvert$	*mod r (the modulus of r)*	the size of r ignoring its sign: $\lvert -3 \rvert = 3$ **17·p10**

Part 3: Functions and graphs

$\mapsto$	*maps to*	the input on the left is turned into the output on the right **18·p2**
$f(x)$	*f of x*	the output of function f when the input is x **18·p2**
$f \circ g$	*f after g (f compose g)*	the composite function f(g(x)): g first, then f **18·p8**
f^{-1}	*f inverse*	the function that undoes f **18·p9**
e	*e (Euler's number)*	2.71828..., the base of natural growth **21·p6**
$\to \infty$	*tends to infinity*	gets bigger and bigger without end **21·p6**
$\exp(x)$	*exp of x*	another way to write e^x **21·p6**

$\log_b N$	*log to the base b of N*	the power of b that gives N **22·p2**
$\ln$	*L-N, natural log*	log to base e **22·p3**

Part 4: Data and chance

$\bar{x}$	*x bar*	the mean of a sample of x-values **24·p2**
σ	*sigma (lower case)*	standard deviation of a population **24·p11**
μ	*mew*	the mean of a whole population **24·p12**
$p(E)$	*p of E*	the probability that event E happens **25·p2**
$p(E \mid F)$	*p of E given F*	probability of E, knowing that F has happened **25·p4**
$\Phi(z)$	*fie of z (capital phi)*	area under the standard Normal curve to the left of z **26·p6**

Part 5: Geometry and trigonometry

$\circ$	*degrees*	360° is one full turn **27·p2**
θ	*theta*	a common letter for an unknown angle **28·p2**
$\sin\theta$	*sine theta (say 'sign')*	opp ÷ hyp for the angle θ **28·p6**
$\sin^{-1}$	*inverse sine, or arcsine*	the angle whose sine is ... **28·p8**
$n!$	*n factorial*	n × (n−1) × ... × 1 **29·p13**
$\sin^2\theta$	*sine squared theta*	$(\sin\theta)^2$, the sine worked out first, then squared **30·p2**
$\vec{a}$	*vector a*	a vector named a (also bold a, or a underlined) **32·p3**
$\overrightarrow{AB}$	*vector A B*	the vector that takes you from point A to point B **32·p3**

$\mathbf{i},\ \mathbf{j},\ \mathbf{k}$	*i, j, k*	unit vectors along x, y and z (also written with hats) **32·p3**
$\lvert\mathbf{a}\rvert$	*the magnitude of a*	the length of vector a **32·p6**
$\hat{\mathbf{a}}$	*a hat*	the unit vector (length 1) in the direction of a **32·p7**
$\mathbf{a} \cdot \mathbf{b}$	*a dot b*	the dot (scalar) product: a plain number **32·p9**
$\mathbf{a} \times \mathbf{b}$	*a cross b*	the cross (vector) product **32·p13**
(r, θ)	*r, theta*	a point in polar coordinates: radius then angle **33·p2**

Part 6: Calculus

h	*h*	a small step along the x axis **34·p3**
$\dfrac{dy}{dx}$	*dee y by dee x*	the derivative: gradient of y with respect to x **34·p4**
$f'(x)$	*f dash of x*	the derivative of the function f **34·p4**
δx	*delta x*	a small change in x **34·p4**
$\dot{y}$	*y dot*	Newton's notation for the derivative of y with respect to time, dy/dt **34·p4**
$\sec x$	*seck x (secant)*	1 ÷ cos x **34·p9**
$\dfrac{d^2y}{dx^2}$	*dee two y by dee x squared*	the second derivative **34·p17**
$\displaystyle\int_a^b f(x)\,dx$	*the integral from a to b of f of x, dee x*	the exact area under y = f(x) between x = a and x = b **36·p5**
$\dfrac{\partial f}{\partial x}$	*partial f by partial x (or del f by del x)*	the slope of f when only x changes; every other letter is frozen as a constant **38·p4**
$f_x,\ f_y$	*f sub x, f sub y*	short for ∂f/∂x and ∂f/∂y **38·p4**

∇f	*grad f (or del f)*	the vector of partial derivatives; points the steepest way uphill **38·p8**

Formula Pages

Every boxed formula from the book in one place, Part by Part. The label says what it's for; the reference says where it's explained and practised. The standard derivative and integral tables come last.

Part 1: Numbers

Adding a negative
Whole numbers · 1·p18
$$a + (-b) = a - b$$

Subtracting a negative
Whole numbers · 1·p18
$$a - (-b) = a + b$$

× and ÷ signs
Whole numbers · 1·p18
$$(-a) \times (-b) = ab, \quad (-a) \times b = -ab$$

Fraction bar
Whole numbers · 1·p18
$$\frac{a+b}{c+d} = (a+b) \div (c+d)$$

Powers of negatives
Whole numbers · 1·p18
$$(-3)^2 = 9, \qquad -3^2 = -9$$

Commutative
Whole numbers · 1·p18
$$a + b = b + a, \quad a \times b = b \times a$$

Distributive law
Whole numbers · 1·p18
$$a(b+c) = ab + ac$$

Division
Whole numbers · 1·p18
$$a \times b = c \iff c \div b = a$$

HCF and LCM
Whole numbers · 1·p18
$$\mathrm{HCF}(a,b) \times \mathrm{LCM}(a,b) = a \times b$$

Fraction = division
Fractions · 2·p15
$$\frac{a}{b} = a \div b, \quad n = \frac{n}{1}$$

Equivalent fractions
Fractions · 2·p15
$$\frac{a}{b} = \frac{a \times k}{b \times k} = \frac{a \div k}{b \div k}$$

Mixed to improper
Fractions · 2·p15
$$p\tfrac{a}{b} = \frac{pb + a}{b}$$

Add / subtract
Fractions · 2·p15
$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

Multiply
Fractions · 2·p15
$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Divide
Fractions · 2·p15

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

Reciprocal
Fractions · 2·p15

$$\frac{a}{b} \to \frac{b}{a}, \qquad \frac{a}{b} \times \frac{b}{a} = 1$$

Compare
Fractions · 2·p15

$$\frac{a}{b} > \frac{c}{d} \iff ad > bc$$

Negative fractions
Fractions · 2·p15

$$-\frac{a}{b} = \frac{-a}{b} = \frac{a}{-b}$$

Fraction of an amount
Fractions · 2·p15

$$\frac{a}{b} \text{ of } N = \frac{N}{b} \times a$$

Decimal to fraction
Decimals · 3·p19

$$0.abc = \frac{abc}{1000}$$

Fraction to decimal
Decimals · 3·p19

$$\frac{a}{b} = a \div b$$

Divide
Decimals · 3·p19

$$\frac{a}{b} = \frac{10a}{10b}$$

Recurring to fraction
Decimals · 3·p19

$$x = 0.\dot{2}\dot{7} : \;\; 100x - x = 27 \Rightarrow x = \frac{27}{99}$$

Bounds
Decimals · 3·p19

$$\text{value} - \tfrac{1}{2}\text{step} \leq x < \text{value} + \tfrac{1}{2}\text{step}$$

Bounds of a ÷
Decimals · 3·p19

$$\max \frac{a}{b} = \frac{a_{\max}}{b_{\min}}, \qquad \min \frac{a}{b} = \frac{a_{\min}}{b_{\max}}$$

percentage of an amount
Percentage and ratio · 4·p4

$$x\% \text{ of } y = \frac{x}{100} \times y$$

percentage increase / decrease
Percentage and ratio · 4·p6

$$\text{new} = \text{old} \times \frac{100 \pm r}{100}$$

percentage change
Percentage and ratio · 4·p7

$$\frac{\text{new} - \text{old}}{\text{old}} \times 100$$

reverse percentage
Percentage and ratio · 4·p8

$$\text{original} = \frac{\text{new}}{1 \pm \frac{r}{100}}$$

cross-multiplication
Percentage and ratio · 4·p15

$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

Per cent
Percentage and ratio · 4·p19

$$x\% = \frac{x}{100}$$

Whole from a part
Percentage and ratio · 4·p19

$$\text{whole} = \text{part} \div \frac{p}{100}$$

Increase / decrease by r%
Percentage and ratio · 4·p19

$$\text{new} = \text{old} \times \left(1 \pm \frac{r}{100}\right)$$

Ratio as fractions
Percentage and ratio · 4·p19

$$a : b \Rightarrow \frac{a}{a+b} \text{ and } \frac{b}{a+b} \text{ of the total}$$

Ratio to percentage
Percentage and ratio · 4·p19

$$\frac{\text{part}}{\text{sum of parts}} \times 100$$

Inverse proportion
Percentage and ratio · 4·p19

$$x \times y = \text{constant}$$

Mixing
Percentage and ratio · 4·p19

$$\frac{\text{total A}}{\text{total volume}} \times 100$$

perimeter of a rectangle
Measurement · 5·p4

$$P = 2(l + w)$$

circumference of a circle
Measurement · 5·p4

$$C = \pi d = 2\pi r$$

area of a rectangle
Measurement · 5·p5

$$A = lw$$

area of a triangle
Measurement · 5·p5

$$A = \tfrac{1}{2}bh$$

area of a circle
Measurement · 5·p5

$$A = \pi r^2$$

area of a trapezium
Measurement · 5·p6

$$A = \tfrac{1}{2}(a + b)h$$

volume of a cuboid
Measurement · 5·p9

$$V = lwh$$

volume of a cylinder
Measurement · 5·p9

$$V = \pi r^2 h$$

surface area of a cuboid
Measurement · 5·p10

$$S = 2(lw + lh + wh)$$

speed
Measurement · 5·p12

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Rectangle, triangle
Measurement · 5·p15

$$A = lw, \qquad A = \tfrac{1}{2}bh$$

Parallelogram, trapezium
Measurement · 5·p15

$$A = bh, \qquad A = \tfrac{1}{2}(a+b)h$$

Volume
Measurement · 5·p15

$$V = lwh, \qquad V = \pi r^2 h$$

Speed
Measurement · 5·p15

$$s = \frac{d}{t}, \quad d = st, \quad t = \frac{d}{s}$$

Density
Measurement · 5·p15

$$\rho = \frac{m}{V}, \quad m = \rho V, \quad V = \frac{m}{\rho}$$

counting a complement
Sets · 6·p3

$$n(A') = n(\xi) - n(A)$$

counting a union (don't count the overlap twice)
Sets · 6·p5

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

De Morgan's law for sets
Sets · 6·p9

$$(A \cup B)' = A' \cap B'$$

De Morgan's law for sets
Sets · 6·p9

$$(A \cap B)' = A' \cup B'$$

subsets of a set with n elements
Sets · 6·p10

$$\text{number of subsets} = 2^n$$

size of the power set
Sets · 6·p10

$$|P(S)| = 2^{|S|}$$

Three sets
Sets · 6·p13

$$n(A \cup B \cup C) = \text{singles} - \text{pairs} + n(A \cap B \cap C)$$

rows in a truth table with n letters
Logic · 7·p5

$$\text{rows} = 2^n$$

De Morgan's law
Logic · 7·p7

$$\neg(A \wedge B) = \neg A \vee \neg B$$

De Morgan's law
Logic · 7·p7

$$\neg(A \vee B) = \neg A \wedge \neg B$$

implication as OR
Logic · 7·p8

$$P \Rightarrow Q \equiv \neg P \vee Q$$

negating an implication
Logic · 7·p8

$$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$$

if and only if
Logic · 7·p10

$$P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P)$$

negating 'for all'
Logic · 7·p12

$$\neg\, \forall x\, P(x) \equiv \exists x\, \neg P(x)$$

negating 'there exists'
Logic · 7·p12

$$\neg\, \exists x\, P(x) \equiv \forall x\, \neg P(x)$$

Part 2: Algebra

Hidden ×
Algebra · 8·p13

$$3x = 3 \times x, \quad xy = x \times y$$

Division
Algebra · 8·p13

$$\frac{x}{2} = x \div 2$$

Powers
Algebra · 8·p13

$$3x^2 = 3 \times x \times x, \quad (3x)^2 = 9x^2$$

Formula / identity
Algebra · 8·p13

$$2(x+3) \equiv 2x + 6$$

Sign rules
Algebra · 8·p13

$$(-) \times (-) = +, \quad (+) \times (-) = -$$

Powers of negative numbers
Algebra · 8·p13

$$(-3)^2 = 9, \quad -3^2 = -9, \quad (-2)^3 = -8$$

multiplying powers of the same base
Indices (powers) · 9·p5

$$a^m \times a^n = a^{m+n}$$

dividing powers of the same base
Indices (powers) · 9·p6

$$\frac{a^m}{a^n} = a^{m-n}$$

power of a power
Indices (powers) · 9·p7

$$(a^m)^n = a^{mn}$$

power of a product
Indices (powers) · 9·p7
$$(ab)^n = a^n b^n$$

zero power
Indices (powers) · 9·p8
$$a^0 = 1$$

negative power
Indices (powers) · 9·p8
$$a^{-n} = \frac{1}{a^n}$$

fractional power
Indices (powers) · 9·p10
$$a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$$

Power
Indices (powers) · 9·p18
$$a^n = a \times a \times \cdots \times a$$

Brackets
Indices (powers) · 9·p18
$$(ab)^n = a^n b^n, \quad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

Zero power
Indices (powers) · 9·p18
$$a^0 = 1 \quad (a \neq 0)$$

Negative power
Indices (powers) · 9·p18
$$a^{-n} = \frac{1}{a^n}, \quad \frac{1}{a^{-n}} = a^n$$

Negative power of a fraction
Indices (powers) · 9·p18
$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$$

Fraction power
Indices (powers) · 9·p18
$$a^{\frac{1}{n}} = \sqrt[n]{a}, \quad a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m$$

Negative base
Indices (powers) · 9·p18
$$(-2)^4 = 16, \ (-2)^3 = -8, \ -3^2 = -9$$

Roots of negatives
Indices (powers) · 9·p18
$$\sqrt[3]{-8} = -2$$

Unknown power
Indices (powers) · 9·p18
$$a^p = a^q \Rightarrow p = q$$

Unknown base
Indices (powers) · 9·p18
$$x^{\frac{m}{n}} = c \Rightarrow x = c^{\frac{n}{m}}$$

Standard form
Indices (powers) · 9·p18
$$a \times 10^n, \quad 1 \leq a < 10$$

Place value in base b
Number bases · 10·p14
$$(d_2\, d_1\, d_0)_b = d_2 b^2 + d_1 b + d_0$$

Binary addition
Number bases · 10·p14
$$1 + 1 = 10_2, \quad 1 + 1 + 1 = 11_2$$

Binary subtraction
Number bases · 10·p14
$$10_2 - 1 = 1$$

Bits and bytes
Number bases · 10·p14

$$n \text{ bits} \to 2^n \text{ values}$$

expanding a bracket
Simplifying · 11·p5

$$a(b+c) = ab + ac$$

expanding two brackets
Simplifying · 11·p6

$$(a+b)(c+d) = ac + ad + bc + bd$$

perfect square
Simplifying · 11·p8

$$(a+b)^2 = a^2 + 2ab + b^2$$

perfect square (minus)
Simplifying · 11·p8

$$(a-b)^2 = a^2 - 2ab + b^2$$

difference of two squares
Simplifying · 11·p8

$$(a-b)(a+b) = a^2 - b^2$$

Collecting
Simplifying · 11·p12

$$3x^2 - 5x^2 = -2x^2$$

Unlike terms
Simplifying · 11·p12

$$x + x^2, \quad 7a + 3b$$

Multiplying terms
Simplifying · 11·p12

$$-4x \times 3x^2 = -12x^3$$

Sign rules
Simplifying · 11·p12

$$(+)(+) = +, \ (-)(-) = +, \ (+)(-) = -$$

Minus outside
Simplifying · 11·p12

$$-(x-3) = -x + 3$$

Perfect squares
Simplifying · 11·p12

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Difference of two squares
Simplifying · 11·p12

$$(a+b)(a-b) = a^2 - b^2$$

Dividing by x
Simplifying · 11·p12

$$\frac{2x+3}{x} = 2 + \frac{3}{x}$$

factorising x² + bx + c
Factorisation · 12·p5

$$x^2 + bx + c = (x+p)(x+q), \quad p+q = b, \ pq = c$$

the pair for factorising ax² + bx + c
Factorisation · 12·p7

$$m, \ n = \frac{b \pm \sqrt{b^2 - 4ac}}{2}$$

the grid formula (m, n the pair)
Factorisation · 12·p7

$$ax^2 + bx + c = \left(x + \frac{n}{a}\right)(ax + m)$$

difference of two squares (factorising)
Factorisation · 12·p11

$$a^2 - b^2 = (a - b)(a + b)$$

perfect squares (factorising)
Factorisation · 12·p12

$$a^2 \pm 2ab + b^2 = (a \pm b)^2$$

difference of cubes
Factorisation · 12·p13

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

sum of cubes
Factorisation · 12·p13

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

x² + bx + c
Factorisation · 12·p15

$$(x + p)(x + q): \quad p + q = b, \quad pq = c$$

ax² + bx + c (split the middle)
Factorisation · 12·p15

$$pq = ac, \quad p + q = b: \quad ax^2 + px + qx + c$$

Grid formula (alternative)
Factorisation · 12·p15

$$m,\ n = \frac{b \pm \sqrt{b^2 - 4ac}}{2}, \qquad \left(x + \frac{n}{a}\right)(ax + m)$$

Does it factorise?
Factorisation · 12·p15

$$\Delta = b^2 - 4ac$$

Grouping (4 terms)
Factorisation · 12·p15

$$ax + ay + bx + by = (a + b)(x + y)$$

adding / subtracting fractions
Algebraic fractions · 13·p9

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

Simplify
Algebraic fractions · 13·p15

$$\frac{x^2 + 3x}{x + 3} = \frac{\cancel{x(x + 3)}}{\cancel{(x + 3)}} = x$$

Back-to-front brackets
Algebraic fractions · 13·p15

$$a - b = -(b - a), \qquad \frac{a - b}{b - a} = -1$$

Subtracting
Algebraic fractions · 13·p15

$$\frac{x + 3}{2} - \frac{x - 1}{3} = \frac{3(x + 3) - 2(x - 1)}{6}$$

Multiply
Algebraic fractions · 13·p15

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Divide
Algebraic fractions · 13·p15

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

Splitting
Algebraic fractions · 13·p15

$$\frac{a+b}{c} = \frac{a}{c} + \frac{b}{c}, \qquad \frac{a}{b+c} \neq \frac{a}{b} + \frac{a}{c}$$

Whole number with a fraction
Algebraic fractions · 13·p15

$$1 - \frac{3}{x+1} = \frac{x+1-3}{x+1}$$

Undo buttons
Transposing formulae · 14·p21

$$+ \leftrightarrow -, \quad \times \leftrightarrow \div, \quad x^2 \leftrightarrow \pm\sqrt{x}, \quad x^3 \leftrightarrow \sqrt[3]{x}$$

Minus in front
Transposing formulae · 14·p21

$$y = 5 - 2x \;\Rightarrow\; 2x = 5 - y$$

Letter on the bottom
Transposing formulae · 14·p21

$$P = \frac{V}{R} \;\Rightarrow\; PR = V \;\Rightarrow\; R = \frac{V}{P}$$

Flipping
Transposing formulae · 14·p21

$$\frac{a}{b} = \frac{c}{d} \;\Rightarrow\; \frac{b}{a} = \frac{d}{c}$$

Letter twice
Transposing formulae · 14·p21

$$yx - x = 1 + y \;\Rightarrow\; x(y-1) = 1 + y$$

Roots
Transposing formulae · 14·p21

$$r^2 = k \Rightarrow r = \pm\sqrt{k}, \qquad r^3 = k \Rightarrow r = \sqrt[3]{k}$$

quadratic formula (for $ax^2 + bx + c = 0$)
Solving equations · 15·p12

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

completing the square
Solving equations · 15·p13

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

Cross-multiplying
Solving equations · 15·p18

$$\frac{a}{b} = \frac{c}{d} \;\Rightarrow\; ad = bc$$

Zero product
Solving equations · 15·p18

$$AB = 0 \;\Rightarrow\; A = 0 \text{ or } B = 0$$

Discriminant
Solving equations · 15·p18

$$\Delta = b^2 - 4ac$$

inverse of a 2 × 2 matrix
Matrices · 16·p8

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Size
Matrices · 16·p16

$$m \times n$$

Scalar multiple
Matrices · 16·p16

$$k \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

Size rule
Matrices · 16·p16

$$(m \times n)(n \times p) = m \times p$$

Order
Matrices · 16·p16

$$AB \neq BA$$

Identity
Matrices · 16·p16

$$AI = IA = A$$

Determinant
Matrices · 16·p16

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

Solving AX = B
Matrices · 16·p16

$$X = A^{-1}B$$

Undoing two moves
Matrices · 16·p16

$$(TS)^{-1} = S^{-1}T^{-1}$$

nth term of an arithmetic progression
Sequences and series · 17·p3

$$u_n = a + (n-1)d$$

sum of an arithmetic progression (first a, last l, n terms)
Sequences and series · 17·p4

$$S_n = \frac{(a+l)}{2}n$$

sum of an arithmetic progression (n terms, difference d)
Sequences and series · 17·p5

$$S_n = \frac{n}{2}\big(2a + (n-1)d\big)$$

nth term of a geometric progression
Sequences and series · 17·p7

$$u_n = ar^{n-1}$$

sum of the first n terms of a geometric progression
Sequences and series · 17·p8

$$S_n = \frac{a(1-r^n)}{1-r}$$

sum to infinity of a geometric progression
Sequences and series · 17·p10

$$S_\infty = \frac{a}{1-r} \quad (|r| < 1)$$

Number of terms
Sequences and series · 17·p14

$$n = \frac{l-a}{d} + 1$$

Sigma
Sequences and series · 17·p14

$$\sum_{i=1}^{n} f(i) = f(1) + f(2) + \cdots + f(n)$$

Odd numbers
Sequences and series · 17·p14

$$1 + 3 + 5 + \cdots + (2n-1) = n^2$$

Part 3: Functions and graphs

Notation
Functions · 18·p13
$f(a) = \text{the rule with every } x \text{ replaced by } (a)$

Composite
Functions · 18·p13
$fg(x) = (f \circ g)(x) = f(g(x))$

Inverse
Functions · 18·p13
$f(f^{-1}(x)) = f^{-1}(f(x)) = x$

Coordinates
Graphs of functions · 19·p13
$(x,\ y)$

Even function
Graphs of functions · 19·p13
$f(-x) = f(x)$

Odd function
Graphs of functions · 19·p13
$f(-x) = -f(x)$

Asymptotes
Graphs of functions · 19·p13
$y = \dfrac{b}{x-h} + k: \quad x = h,\ y = k$

Vertex form
Graphs of functions · 19·p13
$y = a(x-h)^2 + k \ \Rightarrow\ \text{vertex } (h, k)$

Shift up / down
Graphs of functions · 19·p13
$y = f(x) + a$

Shift left / right
Graphs of functions · 19·p13
$y = f(x + a)$

Vertical stretch
Graphs of functions · 19·p13
$y = af(x)$

Horizontal squash
Graphs of functions · 19·p13
$y = f(ax)$

Reflections
Graphs of functions · 19·p13
$y = -f(x), \quad y = f(-x)$

gradient through two points
The straight line · 20·p4
$m = \dfrac{y_2 - y_1}{x_2 - x_1}$

line with gradient m through (x_1, y_1)
The straight line · 20·p5
$y - y_1 = m(x - x_1)$

parallel lines
The straight line · 20·p9
$m_1 = m_2$

perpendicular lines
The straight line · 20·p9
$m_1 m_2 = -1 \quad \Leftrightarrow \quad m_2 = -\dfrac{1}{m_1}$

midpoint
The straight line · 20·p16

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

distance between two points
The straight line · 20·p16

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Straight line
The straight line · 20·p21

$$y = mx + c$$

Horizontal / vertical
The straight line · 20·p21

$$y = b \ \ (m = 0), \qquad x = a \ \ (m \text{ undefined})$$

Perpendicular
The straight line · 20·p21

$$m_1 m_2 = -1, \qquad m_2 = -\frac{1}{m_1}$$

Midpoint
The straight line · 20·p21

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Distance
The straight line · 20·p21

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Collinear
The straight line · 20·p21

$$m_{AB} = m_{BC}$$

Exponential model
The exponential function · 21·p11

$$y = A \times b^x$$

e form
The exponential function · 21·p11

$$y = Ae^{kx}$$

Percentage change
The exponential function · 21·p11

$$\times \left(1 \pm \frac{r}{100}\right) \text{ per step}$$

Compound interest
The exponential function · 21·p11

$$A\left(1 + \frac{r}{n}\right)^{nt}$$

The number e
The exponential function · 21·p11

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.718$$

Levelling off
The exponential function · 21·p11

$$y = A + Be^{-kt} \ \rightarrow \ A$$

Time constant
The exponential function · 21·p11

$$i = Ie^{-t/\tau}$$

Half-life
The exponential function · 21·p11

$$y = A\left(\tfrac{1}{2}\right)^{t/T}$$

Doubling time
The exponential function · 21·p11

$$y = A \times 2^{t/T}$$

log law: product
Logarithms · 22·p5

$$\log(ab) = \log a + \log b$$

log law: quotient
Logarithms · 22·p5

$$\log \frac{a}{b} = \log a - \log b$$

log law: power
Logarithms · 22·p5

$$\log a^n = n \log a$$

change of base
Logarithms · 22·p7

$$\log_a N = \frac{\log_b N}{\log_b a}$$

Definition
Logarithms · 22·p13

$$y = \log_b x \iff b^y = x$$

Special values
Logarithms · 22·p13

$$\log_b b = 1, \qquad \log_b 1 = 0$$

Undo
Logarithms · 22·p13

$$\log_b(b^x) = x, \qquad b^{\log_b x} = x$$

ln and e
Logarithms · 22·p13

$$\ln(e^x) = x, \quad e^{\ln x} = x, \quad \ln e = 1, \quad \ln 1 = 0$$

Reciprocal
Logarithms · 22·p13

$$\log \frac{1}{a} = -\log a$$

Part 4: Data and chance

Pie chart angle
Tables and charts · 23·p14

$$\text{angle} = \frac{\text{frequency}}{\text{total}} \times 360°$$

Pie chart frequency
Tables and charts · 23·p14

$$\text{frequency} = \frac{\text{angle}}{360°} \times \text{total}$$

Frequency density
Tables and charts · 23·p14

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

Frequency from a histogram
Tables and charts · 23·p14

$$\text{frequency} = \text{frequency density} \times \text{class width}$$

mean
Statistics · 24·p2

$$\bar{x} = \frac{\sum x_i}{n}$$

population variance
Statistics · 24·p11

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

sample standard deviation
Statistics · 24·p12

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Mean
Statistics · 24·p15

$$\bar{x} = \frac{\sum x}{n}$$

Total from a mean
Statistics · 24·p15

$$\text{total} = n \times \bar{x}$$

Mean of a frequency table
Statistics · 24·p15

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Median position
Statistics · 24·p15

$$\frac{n + 1}{2}$$

Quartile position
Statistics · 24·p15

$$\frac{\text{median position} + 1}{2}$$

Fractional position
Statistics · 24·p15

$$k + d \;\rightarrow\; x_k + d\,(x_{k+1} - x_k)$$

Range
Statistics · 24·p15

$$\text{largest} - \text{smallest}$$

Interquartile range
Statistics · 24·p15

$$\text{IQR} = \text{UQ} - \text{LQ}$$

Population variance and SD
Statistics · 24·p15

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}, \qquad \sigma = \sqrt{\sigma^2}$$

Sample SD
Statistics · 24·p15

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

addition rule (OR)
Probability and Bayes · 25·p3

$$p(E \cup F) = p(E) + p(F) - p(E \cap F)$$

conditional probability
Probability and Bayes · 25·p4

$$p(E \mid F) = \frac{p(E \cap F)}{p(F)}$$

Bayes' theorem
Probability and Bayes · 25·p10

$$p(a \mid b) = \frac{p(b \mid a)\, p(a)}{p(b)}$$

law of total probability
Probability and Bayes · 25·p11

$$p(B) = p(B \mid A)\, p(A) + p(B \mid A')\, p(A')$$

Probability
Probability and Bayes · 25·p15

$$p(E) = \frac{\text{outcomes in } E}{\text{all outcomes}}$$

Complement
Probability and Bayes · 25·p15

$$p(E') = 1 - p(E)$$

Addition rule (OR)
Probability and Bayes · 25·p15

$$p(A \cup B) = p(A) + p(B) - p(A \cap B)$$

Mutually exclusive
Probability and Bayes · 25·p15

$$p(A \cap B) = 0, \qquad p(A \cup B) = p(A) + p(B)$$

Conditional
Probability and Bayes · 25·p15

$$p(A \mid B) = \frac{p(A \cap B)}{p(B)}$$

Multiplication rule (AND)
Probability and Bayes · 25·p15

$$p(A \cap B) = p(B)\, p(A \mid B) = p(A)\, p(B \mid A)$$

Independent
Probability and Bayes · 25·p15

$$p(A \cap B) = p(A)\, p(B), \qquad p(A \mid B) = p(A)$$

Bayes' theorem
Probability and Bayes · 25·p15

$$p(A \mid B) = \frac{p(B \mid A)\, p(A)}{p(B)}$$

Normal curve
The Normal distribution · 26·p4

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

z-score
The Normal distribution · 26·p6

$$z = \frac{x - \mu}{\sigma}$$

Back from z
The Normal distribution · 26·p15

$$x = \mu + z\sigma, \qquad \mu = x - z\sigma, \qquad \sigma = \frac{x - \mu}{z}$$

Below a value
The Normal distribution · 26·p15

$$p(X < x) = \Phi(z)$$

Above a value
The Normal distribution · 26·p15

$$p(X > x) = 1 - \Phi(z)$$

Between two values
The Normal distribution · 26·p15

$$p(a < X < b) = \Phi(z_b) - \Phi(z_a)$$

Negative z
The Normal distribution · 26·p15

$$\Phi(-z) = 1 - \Phi(z), \qquad \Phi(0) = 0.5$$

Part 5: Geometry and trigonometry

degrees and radians
Angles · 27·p5
$$360^\circ = 2\pi \text{ rad}$$

degrees to radians
Angles · 27·p6
$$\text{rad} = \text{deg} \times \frac{\pi}{180}$$

arc length (θ in degrees)
Angles · 27·p8
$$s = \frac{\theta}{360} \times 2\pi r$$

sector area (θ in degrees)
Angles · 27·p8
$$A = \frac{\theta}{360} \times \pi r^2$$

arc length (θ in radians)
Angles · 27·p8
$$s = r\theta$$

sector area (θ in radians)
Angles · 27·p8
$$A = \tfrac{1}{2} r^2 \theta$$

Degrees → radians
Angles · 27·p10
$$\theta_{\text{rad}} = \theta_{\text{deg}} \times \frac{\pi}{180}$$

Radians → degrees
Angles · 27·p10
$$\theta_{\text{deg}} = \theta_{\text{rad}} \times \frac{180}{\pi}$$

Arc length
Angles · 27·p10
$$s = r\theta \quad \text{or} \quad s = \frac{\theta}{360} \times 2\pi r$$

Sector area
Angles · 27·p10
$$A = \tfrac{1}{2} r^2 \theta \quad \text{or} \quad A = \frac{\theta}{360} \times \pi r^2$$

Pythagoras
Intro to trigonometry · 28·p3
$$h^2 = o^2 + a^2$$

circle, centre (a, b), radius r
Intro to trigonometry · 28·p5
$$(x - a)^2 + (y - b)^2 = r^2$$

sine
Intro to trigonometry · 28·p6
$$\sin\theta = \frac{o}{h}$$

cosine
Intro to trigonometry · 28·p6
$$\cos\theta = \frac{a}{h}$$

tangent
Intro to trigonometry · 28·p6
$$\tan\theta = \frac{o}{a}$$

Pythagoras
Intro to trigonometry · 28·p13

$$a^2 + b^2 = c^2$$

Pythagoras in 3D
Intro to trigonometry · 28·p13

$$d^2 = a^2 + b^2 + c^2$$

SOHCAHTOA
Intro to trigonometry · 28·p13

$$\sin\theta = \frac{O}{H}, \quad \cos\theta = \frac{A}{H}, \quad \tan\theta = \frac{O}{A}$$

Inverse trig
Intro to trigonometry · 28·p13

$$\theta = \sin^{-1}\frac{O}{H},\ \cos^{-1}\frac{A}{H},\ \tan^{-1}\frac{O}{A}$$

Exact values
Intro to trigonometry · 28·p13

$$\sin 30^\circ = \tfrac{1}{2},\ \sin 45^\circ = \tfrac{1}{\sqrt{2}},\ \sin 60^\circ = \tfrac{\sqrt{3}}{2}$$

general sine wave
Trig graphs · 29·p8

$$y = A\sin(B(x-C)) + D:\ \text{amplitude } |A|,\ \text{period } \frac{2\pi}{B},\ \text{shift } C,\ \text{midline } D$$

stretching trig graphs
Trig graphs · 29·p8

$$y = A\sin(Bx):\ \text{amplitude } A,\ \text{period } \frac{2\pi}{B}$$

sine series
Trig graphs · 29·p13

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

cosine series
Trig graphs · 29·p13

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

tangent series
Trig graphs · 29·p15

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots \quad (|x| < \tfrac{\pi}{2})$$

Unit circle
Trig graphs · 29·p17

$$(\cos\theta,\ \sin\theta), \qquad \tan\theta = \frac{\sin\theta}{\cos\theta}$$

Pythagorean identity
Trig graphs · 29·p17

$$\cos^2\theta + \sin^2\theta = 1$$

General wave
Trig graphs · 29·p17

$$y = A\sin(B(x-C)) + D$$

tan(Bx)
Trig graphs · 29·p17

$$\text{period} = \frac{180^\circ}{B} = \frac{\pi}{B}$$

Read off a graph
Trig graphs · 29·p17

$$D = \frac{\max + \min}{2}, \quad A = \frac{\max - \min}{2}, \quad B = \frac{360^\circ}{\text{period}}$$

Power series
Trig graphs · 29·p17

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots, \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

Pythagorean identity
Trig identities and equations · 30·p2

$$\sin^2\theta + \cos^2\theta = 1$$

r sin form
Trig identities and equations · 30·p8

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha),\ r = \sqrt{a^2+b^2},\ \alpha = \tan^{-1}\frac{b}{a}$$

tan
Trig identities and equations · 30·p11

$$\tan\theta = \frac{\sin\theta}{\cos\theta}, \qquad \tan^2\theta + 1 = \frac{1}{\cos^2\theta}$$

Partner: sin
Trig identities and equations · 30·p11

$$\theta_1, \quad 180^\circ - \theta_1 \quad (\pi - \theta_1)$$

Partner: cos
Trig identities and equations · 30·p11

$$\theta_1, \quad 360^\circ - \theta_1 \quad (2\pi - \theta_1)$$

Partner: tan
Trig identities and equations · 30·p11

$$\theta_1, \quad \theta_1 + 180^\circ \quad (\theta_1 + \pi)$$

r sin form
Trig identities and equations · 30·p11

$$a\sin\theta + b\cos\theta = r\sin(\theta+\alpha)$$

Compound angle
Trig identities and equations · 30·p11

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$

sine rule
Solving triangles · 31·p3

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

cosine rule (angle form)
Solving triangles · 31·p5

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

cosine rule
Solving triangles · 31·p5

$$a^2 = b^2 + c^2 - 2bc\cos A$$

area of a triangle
Solving triangles · 31·p9

$$\text{Area} = \tfrac{1}{2}bc\sin A$$

Angle sum
Solving triangles · 31·p12

$$A + B + C = 180^\circ$$

Area
Solving triangles · 31·p12

$$\text{Area} = \tfrac{1}{2}ab\sin C$$

Area backwards
Solving triangles · 31·p12

$$\sin C = \frac{2\times\text{Area}}{ab}, \qquad a = \frac{2\times\text{Area}}{b\sin C}$$

magnitude (length) of a vector
Vectors · 32·p6

$$|\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$$

unit vector in the direction of a
Vectors · 32·p7

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

position vectors: A to B, and the midpoint
Vectors · 32·p7
$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}, \qquad \mathbf{m} = \frac{\mathbf{a} + \mathbf{b}}{2}$$

dot product (angle form)
Vectors · 32·p9
$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$$

angle between two vectors
Vectors · 32·p10
$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$$

dot product (components)
Vectors · 32·p10
$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$$

size of the cross product
Vectors · 32·p13
$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \sin\theta$$

cross product (components)
Vectors · 32·p16
$$\mathbf{a} \times \mathbf{b} = \mathbf{i}(a_y b_z - a_z b_y) - \mathbf{j}(a_x b_z - a_z b_x) + \mathbf{k}(a_x b_y - a_y b_x)$$

Components
Vectors · 32·p17
$$\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

Direction angle
Vectors · 32·p17
$$\theta = \tan^{-1}\frac{y}{x}$$

Dot product
Vectors · 32·p17
$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z = |\mathbf{a}||\mathbf{b}| \cos\theta$$

polar to Cartesian
Polar coordinates · 33·p3
$$x = r\cos\theta, \quad y = r\sin\theta$$

Cartesian to polar
Polar coordinates · 33·p4
$$r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}\frac{y}{x} \ (+\text{quadrant fix})$$

polar and Cartesian links
Polar coordinates · 33·p7
$$x^2 + y^2 = r^2, \qquad \tan\theta = \frac{y}{x}$$

double angle
Polar coordinates · 33·p10
$$\sin 2A = 2\sin A\cos A, \quad \cos 2A = \cos^2 A - \sin^2 A$$

Polar → Cartesian
Polar coordinates · 33·p11
$$x = r\cos\theta, \qquad y = r\sin\theta$$

Distance
Polar coordinates · 33·p11
$$r = \sqrt{x^2 + y^2}$$

Angle
Polar coordinates · 33·p11
$$\theta = \tan^{-1}\frac{y}{x}$$

Double angle
Polar coordinates · 33·p11

$$\sin 2A = 2\sin A\cos A, \quad \cos 2A = \cos^2 A - \sin^2 A$$

Part 6: Calculus

power rule
Gradients of curves · 34·p5

$$\frac{d}{dx}\left(ax^n\right) = anx^{n-1}$$

Notation
Gradients of curves · 34·p18

$$\frac{dy}{dx} = f'(x) = \dot{f}(x)$$

Constants, kx, sums
Gradients of curves · 34·p18

$$\frac{d}{dx}(c) = 0, \qquad \frac{d}{dx}(kx) = k$$

Rewrite first
Gradients of curves · 34·p18

$$\frac{a}{x^n} = ax^{-n}, \quad \sqrt{x} = x^{\frac{1}{2}}, \quad \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$$

ln and e
Gradients of curves · 34·p18

$$\frac{d}{dx}\ln x = \frac{1}{x}, \qquad \frac{d}{dx}e^{kx} = ke^{kx}$$

Trig (radians)
Gradients of curves · 34·p18

$$\frac{d}{dx}\sin bx = b\cos bx, \quad \frac{d}{dx}\cos bx = -b\sin bx, \quad \frac{d}{dx}\tan x = \sec^2 x$$

Gradient at x = a
Gradients of curves · 34·p18

$$m = f'(a)$$

Tangent at x = a
Gradients of curves · 34·p18

$$y - y_1 = f'(a)(x - x_1)$$

Normal at x = a
Gradients of curves · 34·p18

$$m_N = -\frac{1}{f'(a)}$$

Stationary points
Gradients of curves · 34·p18

$$\frac{dy}{dx} = 0$$

Second-derivative test
Gradients of curves · 34·p18

$$\frac{d^2y}{dx^2} > 0 \text{ min}, \quad < 0 \text{ max}, \quad = 0 \text{ test fails}$$

chain rule
Chain, product, quotient · 35·p4

$$\frac{dy}{dx} = \frac{dy}{dz} \times \frac{dz}{dx}$$

product rule
Chain, product, quotient · 35·p8

$$\frac{d(uv)}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$$

quotient rule
Chain, product, quotient · 35·p10

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Higher derivatives
Chain, product, quotient · 35·p17

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

Chain rule
Chain, product, quotient · 35·p17

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Chain: powers
Chain, product, quotient · 35·p17

$$\frac{d}{dx}\big(g(x)\big)^n = n\big(g(x)\big)^{n-1} g'(x)$$

Chain: e and ln
Chain, product, quotient · 35·p17

$$\frac{d}{dx}e^{g(x)} = g'(x)e^{g(x)}, \qquad \frac{d}{dx}\ln g(x) = \frac{g'(x)}{g(x)}$$

Chain: roots
Chain, product, quotient · 35·p17

$$\frac{d}{dx}\sqrt{g(x)} = \frac{g'(x)}{2\sqrt{g(x)}}$$

Constant multiple
Chain, product, quotient · 35·p17

$$\frac{d}{dx}\big(kf(x)\big) = kf'(x)$$

trapezium rule
Integration and areas · 36·p3

$$\int_a^b y\,dx \approx \frac{h}{2}\big(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\big),\ h = \frac{b-a}{n}$$

power rule for integration
Integration and areas · 36·p6

$$\int ax^n\,dx = \frac{a}{n+1}x^{n+1} + c \quad (n \neq -1)$$

integral of 1/x
Integration and areas · 36·p9

$$\int \frac{1}{x}\,dx = \ln|x| + c$$

definite integral
Integration and areas · 36·p13

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Constant
Integration and areas · 36·p18

$$\int k\,dx = kx + c$$

Rewrite first
Integration and areas · 36·p18

$$\frac{3}{x^2} = 3x^{-2}, \quad \sqrt{x} = x^{\frac{1}{2}}, \quad \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$$

Sums and multiples
Integration and areas · 36·p18

$$\int (f \pm g)\,dx = \int f\,dx \pm \int g\,dx, \qquad \int kf\,dx = k\int f\,dx$$

e and trig
Integration and areas · 36·p18

$$\int e^{kx}dx = \tfrac{1}{k}e^{kx}, \quad \int \sin kx\,dx = -\tfrac{1}{k}\cos kx, \quad \int \cos kx\,dx = \tfrac{1}{k}\sin kx$$

tan
Integration and areas · 36·p18

$$\int \tan x \, dx = \ln|\sec x| + c$$

Trapezium rule
Integration and areas · 36·p18

$$\frac{h}{2}\big(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\big), \quad h = \frac{b-a}{n}$$

integration by parts
Parts and substitution · 37·p5

$$\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$$

definite integration by parts
Parts and substitution · 37·p10

$$\int_a^b u\frac{dv}{dx}\,dx = \Big[uv\Big]_a^b - \int_a^b v\frac{du}{dx}\,dx$$

integration by substitution (y = g(x))
Parts and substitution · 37·p16

$$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(y)\,dy$$

linear inside shortcut
Parts and substitution · 37·p17

$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b) + C$$

ln x
Parts and substitution · 37·p20

$$\int \ln x \, dx = x\ln x - x + C$$

small changes (two variables)
Partial differentiation · 38·p11

$$\delta z \approx \frac{\partial z}{\partial x}\delta x + \frac{\partial z}{\partial y}\delta y$$

Partial derivative
Partial differentiation · 38·p13

$$\frac{\partial f}{\partial x} = f_x$$

Second partials
Partial differentiation · 38·p13

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \qquad f_{xy} = \frac{\partial^2 f}{\partial y\,\partial x}$$

Mixed partials
Partial differentiation · 38·p13

$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}$$

Gradient vector
Partial differentiation · 38·p13

$$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix}$$

Standard derivatives and integrals

Chapters 34–37

Read across: differentiate a function to get the middle column; integrate it to get the right-hand column. Angles are in **radians** whenever you differentiate or integrate trig functions.

Function $f(x)$	Derivative $f'(x)$	Integral $\int f(x)\,dx$
k (a constant)	0	$kx + C$
x^n	nx^{n-1}	$\dfrac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
$\dfrac{1}{x} = x^{-1}$	$-\dfrac{1}{x^2}$	$\ln\lvert x\rvert + C$
e^x	e^x	$e^x + C$
e^{kx}	$k\,e^{kx}$	$\dfrac{1}{k}e^{kx} + C$
a^x	$a^x \ln a$	$\dfrac{a^x}{\ln a} + C$
$\ln x$	$\dfrac{1}{x}$	$x\ln x - x + C$
$\sin x$	$\cos x$	$-\cos x + C$
$\cos x$	$-\sin x$	$\sin x + C$
$\tan x$	$\sec^2 x$	$\ln\lvert\sec x\rvert + C$
$\sin kx$	$k\cos kx$	$-\dfrac{1}{k}\cos kx + C$
$\cos kx$	$-k\sin kx$	$\dfrac{1}{k}\sin kx + C$

Calculus rules

Chapters 34–37

Sum rule: do each term separately

$$(f \pm g)' = f' \pm g', \qquad \int (f \pm g) = \int f \pm \int g$$

A constant multiple stays put (not a product rule case)

$$(kf)' = kf', \qquad \int kf = k\int f$$

Product rule

$$\frac{\mathrm{d}}{\mathrm{d}x}(uv) = u\frac{\mathrm{d}v}{\mathrm{d}x} + v\frac{\mathrm{d}u}{\mathrm{d}x}$$

Quotient rule

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u}{v}\right) = \frac{v\,u' - u\,v'}{v^2}$$

Chain rule (function of a function)

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \cdot \frac{\mathrm{d}u}{\mathrm{d}x}$$

Linear inside: integrate, then divide by a

$$\int f(ax+b)\,\mathrm{d}x = \frac{1}{a}F(ax+b) + C$$

Integration by parts

$$\int u\,\frac{\mathrm{d}v}{\mathrm{d}x}\,\mathrm{d}x = uv - \int v\,\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x$$

Definite integral

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a)$$

Answer Key and Tracker

The answers to every "Your turn" page, without the questions, so you can check a whole chapter at a glance. Each small circle is one question: tick it in Xournal++ when you get it right, and watch the book fill up.

1 Whole numbers

○○○ Bundles

1. 6 hundreds (600) 2. 125 (ones: 8+7 = 15, carry 1) 3. 45 (open a bundle: 11 - 6, 7 - 3)

○○○○ The lift

4. 5 5. -5 6. 3 (taking away a debt) 7. 20 (same signs: positive)

○○○ Getting dressed

8. 4 (multiply first) 9. 24 (brackets first) 10. 11 (power first: 9, then + 2)

○○○○ Dots and lights

11. 1, 2, 3, 6, 9, 18 12. No: 21 = 3 x 7 13. 24 14. 12

○○○○○○ Signs, order and primes

15. -15 (both negative: add sizes) 16. -8 (left to right: -4 × 2) 17. 25 and -25 18. 2 (inner bracket 3, then 10 - 9 = 1) 19. 3 (15 ÷ 5) 20. $18 = 2 \times 3^2$, $24 = 2^3 \times 3$: HCF = $2 \times 3 = 6$, LCM = $2^3 \times 3^2 = 72$

○○○○○○ Find the unknown

21. 4 (? = -3 + 7) 22. -7 (? = 28 ÷ (-4)) 23. -30 (? = 5 × (-6)) 24. -6 (? = 9 - 15) 25. 20 or 28 (multiples of 4; 24 is out because HCF(12, 24) = 12) 26. 5 (30 = 2 × 3 × 5 and 6 = 2 × 3, so n must supply the 5; 10 is too big)

2 Fractions

○○○○ Simplify and convert

1. 3/4 2. 4 3/5 3. 23/7 4. 7/11 (5×11 = 55 < 56 = 7×8)

○○○○○ Add and subtract

5. 7/9 6. 7/20 7. 29/24 = 1 5/24 8. −5/12 9. 23/15 = 1 8/15

○○○○○ Multiply and divide

10. 1/6 11. 7/2 = 3 1/2 12. 3/4 13. 1/8 14. 1/2

○○○○ Mixed operations

15. 1 16. 7/8 17. 27 18. −1/4

○○○○○ Find the unknown

19. 25 20. 5/8 21. 2/3 22. 24 23. 1/2

3 Decimals

○○○○ Reading and comparing

1. 6 hundredths 2. 0.7 3. 0.3, 0.303, 0.33 4. 7/20

○○○○○ Working with them

5. 16.15 6. 6.38 7. 0.42 8. 4.2 9. 90

○○○ Fraction to decimal

10. 0.625 11. 0.6 recurring 12. 0.35

○○○○○ Rounding

13. 6.7 14. 6.75 15. 0.030 16. 73,000 17. $10 \times 50 = 500$

○○○○○○ Significant figures, recurring, bounds

18. 3, 2, 3 19. 0.0410 (4, 0, 9; decider 6 bumps the 9) 20. 4/9 ($10x - x = 4$) 21. yes ($40 = 2^3 \times 5$); no (6 has a 3) 22. $12.55 \leq x < 12.65$ 23. $8.5 \div 1.5 = 5.67$ (upper ÷ lower)

○○○○○ Find the unknown

24. 0.9 ($0.36 \div 0.4 = 3.6 \div 4$) 25. 4.45 (2.65 + 1.8) 26. 3 (15×0.2) 27. $4.25 \leq x < 4.35$ ($\pm$ 0.05) 28. $0.555 \leq x < 0.565$ ($\pm$ 0.005)

4 Percentage and ratio

○○○○○ Percentages

1. 0.35 = 7/20 2. 7% 3. 13 4. 14 5. 40

○○○○ Up, down and change

6. 280 7. 68 8. -25% 9. $1.2 \times 0.8 = 0.96$ (a 4% fall)

○○○○ Ratio

10. 3 : 4 11. £36 and £48 12. 100 ml 13. 1 km

○ Mixtures

14. (60 + 15)/500 = 15%

○○○○○ Ratio rules

15. 1 : 4 (50 : 200) 16. 60%, 20%, 20% (9/15, 3/15, 3/15) 17. 2 : 3 (40 : 60) 18. 15 days (inverse: $6 \times 10 = 60$, ÷ 4) 19. A £49, B £14 (5 parts = £35, 1 part = £7)

○○○○○○ Find the unknown

20. £50 ($55 \div 1.1$) 21. £60 ($45 \div 0.75$) 22. 36 (5 parts = 20, 1 part = 4, 9 parts) 23. 10 ($6x = 15 \times 4 = 60$) 24. 4 ($5x = 2.5 \times 8 = 20$) 25. 150 ($18 \div 0.12$)

5 Measurement

○○○○○ Changing units

1. 3200 m 2. 0.45 m 3. 2500 g 4. 0.75 l 5. 4 ms

○○○ Shapes

6. 30 cm, 54 cm^2 7. 35 cm^2 8. 31.4 cm, 78.5 cm^2

○○○○ Area and volume units

9. 30,000 cm^2 10. 60 cm^3 11. 2000 cm^3 12. 500,000 cm^3

○○○ Time and per

13. 2 h 24 min 14. 60 km/h 15. 72 km/h

○○○○○○ More shapes, units and time

16. 28 cm² (½ × 14 × 4) 17. 88 m² (108 − 20) 18. 72 cm² (2(18 + 12 + 6)) 19. 6.5 cm³ (÷ 1000) 20. 3.67 h (3 + 40/60) 21. 10:25 (9 h 85 min = 10 h 25 min)

○○○○○○ Find the unknown

22. 8 cm (√64) 23. 9 m (54 ÷ 6) 24. 10.0 cm (62.8 ÷ 2π) 25. 7.0 cm (√(154 ÷ π)) 26. 2.5 h = 2 h 30 min (210 ÷ 84) 27. 5 cm (120 ÷ (6 × 4))

6 Sets

○○○○○○ Reading sets

1. no 2. {2, 4} 3. {1, 2, 3, 4, 5, 6, 8, 10} 4. {1, 3, 5} 5. {1, 3, 5, 7, 9} 6. 8

○○ Venn counting

7. 40 - (13 + 12 + 8) = 7 8. 13

○○○○ Decoding

9. {-2, -1, 0, 1, 2} 10. ℤ 11. yes (3/4); no 12. filled dot at 2, hollow at 5

○○○○○○ Set rules

13. {7, 9} 14. {6, 8, 10} 15. both {1, 3, 5, 6, 7, 8, 9, 10}; yes (De Morgan) 16. $2^4 = 16$ 17. yes; no (a proper subset must be smaller) 18. [4, ∞)

○○○○○○ Find the unknown

19. (28 − x) + x + (22 − x) + 9 = 50, so 59 − x = 50, x = 9 20. 25 + 21 − 40 = 6 21. 35 = 20 + n(B) − 8, n(B) = 23 22. 60 − 42 = 18 23. 39 − x = 32, x = 7 24. $2^n = 64$, n = 6

○○ Three sets

25. 45 + 40 + 35 − 15 − 12 − 10 + 5 = 88 like at least one, so 100 − 88 = 12 26. 45 − (15 − 5) − (12 − 5) − 5 = 23

7 Logic

○○○ Statements

1. only once x has a value (true if x = 3) 2. T 3. F

○○○ Tables

4. T only in the row A = F, B = T 5. always T (called a tautology) 6. the columns match

○○ If... then

7. no (row 4) 8. converse: if even then divisible by 4 (false: 6); contrapositive: if odd then not divisible by 4 (true)

○○ Python

9. True 10. False

○○○○○○ Rules and bigger tables

11. $2^4 = 16$ 12. T F T F T F T F 13. (¬P) ∨ (Q ∧ R) 14. ¬P ∨ Q 15. some bus is not late 16. drive ⇒ licence (a licence is necessary, not sufficient)

○○○○○○ Find the unknown

17. P = T, Q = F (the only F row) 18. both F (the only F row of OR) 19. F (P = T would make T ⇒ F, which is F) 20. F (⇔ needs them to match) 21. F (P = T, Q = T, so T ⇒ F) 22. $2^n = 32$, n = 5

8 Algebra

○○○ Reading algebra

1. 4 × a × b 2. no: 9 vs 6 3. 7, −1, 2

○○○○○○○○ Substitution (x = 3, y = −2)

4. 11 5. 11 6. 12 7. −10 8. 1 9. −8 10. 1/5 11. 5/(−4) = −5/4

○○○○ Words to algebra

12. 4n − 7 13. p/3 14. 5q + 90 15. 2n + 7

○ Types

16. (a) expression (b) equation (c) formula (d) identity

○○○○○○ Find the unknown

17. a = 5 (20 = 5 + 3a) 18. p = 5 (3p = 15) 19. w = 6 (30 = 18 + 2w) 20. b = 8 (24 = 3b) 21. 7 (4n − 7 = 21, 4n = 28) 22. k = 9 (6 + k = 15)

9 Indices (powers)

○○○○○○○○ Values

1. 64 2. 125 3. 1 4. 1/4 5. 1/9 6. 5 7. 9 8. 7

○○○○○○ Simplify

9. x^{9} 10. y^{6} 11. a^{12} 12. $9x^{2}$ 13. x^{3} 14. $x^{4}y^{6}$

○○○ Standard form

15. 4.5×10^{4} 16. 6.1×10^{-4} 17. 8×10^{-2}

○ Spot the error

18. negative power means one over: 1/9

○○○○○○○○ Coefficients and negatives

19. $3x^{3}$ 20. $4x^{6}$ 21. $3a^{4}b$ 22. −32 23. −16 24. −3 25. x^{3} 26. 16/9

○○○○ Calculating in standard form

27. 3×10^{9} 28. 3×10^{10} 29. 4.2×10^{3} 30. 2.5×10^{5}

○○○○○○ Find the unknown

31. n = −2 ($1/9 = 3^{-2}$) 32. x = 5 ($32 = 2^{5}$) 33. x = 3/2 ($2^{2x} = 2^{3}$) 34. x = 4 (x = 8^(2/3)) 35. k = 7 (k + 3 = 10) 36. x = 2 ($125 = 5^{3}$, x + 1 = 3)

10 Number bases

○○○ Binary to ordinary

1. 13 2. 32 3. 255

○○○ Ordinary to binary

4. 1001 5. 11001 6. 1100100

○○○○ Hexadecimal

7. 26 8. C8 9. pure blue 10. B7

○○○ Arithmetic

11. 1000_{2} (5 + 3 = 8) 12. 10000_{2} 13. $2^{4} = 16$

○○ Other bases

14. 17 15. octal has no digit 8

○○○○ More conversions

16. 144_8 (100 ÷ 8 = 12 r 4, 12 ÷ 8 = 1 r 4, 1 ÷ 8 = 0 r 1) 17. 1001_2 (14 − 5 = 9) 18. 111100_2 (0011 1100) 19. 153_8 (001 101 011)

○○○○○○ Find the unknown

20. x = 1 (8 + 2 + x = 11) 21. base 3 (2b + 1 = 7) 22. base 6 ($b^2 = 36$) 23. x = B (48 + x = 59, x = 11) 24. b = 6 (2b + 5 = 17) 25. x = 1 (4 + 2x + 1 = 7)

11 Simplifying

○○○ Collect like terms

1. 3a + 4b 2. $7x^2 + 4x - 2$ 3. $-2p - 2q$

○○○ Multiply

4. 12xy 5. $-10a^2$ 6. $6x^5$

○○○○ One bracket

7. 10x − 15 8. −3x + 12 9. $2x^2 + 7x$ 10. −x + 13

○○○○○ Two brackets

11. $x^2 + 7x + 10$ 12. $x^2 + 4x - 21$ 13. $2x^2 - 7x - 4$ 14. $x^2 - 12x + 36$ 15. $9x^2 - 4$

○○○ Minus signs outside

16. −2x + 7 17. $-3x^2 + 4x$ 18. −2x + 14

○○○○○○ Find the unknown

19. a = 4 (a + 3 = 7; check 3a = 12) 20. k = 4 (2k − 3 = 5) 21. p = 5 (2p = 10; p^2 = 25) 22. a = 4 (a + 3 = 7; −2a = −8) 23. b = 3 (constants: b = 3; check 2b + 1 = 7) 24. c = 4 (3c = 12)

12 Factorisation

○○○○ Common factor

1. 4(x + 3) 2. 3x(2x − 3) 3. 5a(b + 2) 4. $2x(x^2 + 2x + 4)$

○○○○○○○ Three terms: the pair adds to b and multiplies to ac (Δ shows whether it exists)

5. Δ = 1, pair 4, 3: (x + 3)(x + 4) 6. Δ = 81, pair 4, −5: (x − 5)(x + 4) 7. take out 3 first; Δ = 36, pair 4, −2: 3(x + 4)(x − 2) 8. Δ = 81, pair 10, 1: (2x + 1)(x + 5) 9. Δ = 1, pair 4, 3: (2x + 1)(3x + 2) 10. Δ = 64, pair 2, −6: (2x − 3)(2x + 1) 11. Δ = 289, pair 15, −2: (5x − 2)(x + 3)

○○○ Whole, square roots, or nothing? (Δ first)

12. $\Delta = 4 = 2^2$: pair 4, 2, (x + 2)(x + 4) 13. Δ = 8: pair $3 \pm \sqrt{2}$, $(x + 3 - \sqrt{2})(x + 3 + \sqrt{2})$ 14. Δ = −8: no brackets

○○○○ Difference of two squares

15. (x − 5)(x + 5) 16. (2x − 7)(2x + 7) 17. 2(x − 3)(x + 3) 18. $100^2 - 3^2 = 9991$

○○○ Grouping and chunks

19. (a + b)(x + y) 20. (x − 1)(x + 2) 21. x = b/(a − c)

○○ Cubes

22. $(x - 2)(x^2 + 2x + 4)$ 23. $(3x + 1)(9x^2 - 3x + 1)$

○○○○○○ Find the unknown

24. $p = 4$, $k = 7$ ($3p = 12$, $k = 3 + p$) 25. $k = 9$ ($(6 \div 2)^2 = 9$) 26. $q = 3$, $c = 6$ ($2 + q = 5$, $c = 2q$) 27. $k = -12$, $(x - 3)$ ($4 + p = 1$, so $p = -3$; $k = 4p$) 28. $k = 12$ ($(2x + 3)^2$, middle $2 \times 2x \times 3$) 29. $b = 5$ (expand: $6x - x = 5x$)

13 Algebraic fractions

○○ When is it undefined?

1. $x = 5$ 2. $x = 2$ or $x = -2$

○○○○○ Cancel (factorise first!)

3. $2x/3$ 4. $x + 2$ 5. $x - 1$ 6. $x + 3$ 7. can't cancel: it's $x/5 + 1$

○○○ Multiply and divide

8. $6/x$ 9. 5 10. x

○○○○○ Add and subtract

11. $3/x$ 12. $3/(2x)$ (LCD $2x$) 13. $(3y - 2x)/(xy)$ 14. $2x/((x + 1)(x - 1))$ 15. $(x + 5)/4$

○○○○○ Signs, nested fractions, equations

16. -1 17. $(x + 1)/(x + 3)$ 18. $x/(x - 2)$ ($\times x^2$, then cancel $x + 2$) 19. $x(x + 2)$ 20. $x = 3$ ($\times 3x$: $12 - x = 3x$)

○○○○○ Find the unknown

21. $A = 2$ ($3(x + 1) + A(x - 1) = 5x + 1$) 22. $k = 5$ ($(x + 2)(x + 3) = x^2 + 5x + 6$) 23. $k = 4$ ($\times x$: $1 + 1/k = 5/4$) 24. undefined at $x = 3$ (bottom 0), zero at $x = -2$ (top 0) 25. $a = 2$ ($ax - (x - 2) = x + 2$)

14 Transposing formulae

○○○○○ Fill in the chains

1. $x = (y + 4)/5$ 2. $V = PR$ 3. $a = (v - u)/t$ 4. $v = \sqrt{(2E/m)}$ 5. $A = 6s^2$

○○○○○ Minus signs, bottoms and cubes

6. $x = (7 - y)/3$ 7. $R = V/I$ 8. $x = 1/f - 3$ 9. $x = \sqrt[3]{(3V)}$ 10. $x = (3y + 1)/(y - 2)$

○○○○○○ Find the unknown

11. $s = 30$ ($s = (v^2 - u^2)/(2a) = 300/10$) 12. $t = 5$ ($t = (v - u)/a$) 13. $r = 7$ ($r^2 = A/\pi = 49$) 14. $C = 10$ ($C = 5(F - 32)/9$) 15. $x = 2$ ($x = (y + 1)/(y - 1)$) 16. $m = 5$ ($m = 2E/v^2 = 180/36$)

15 Solving equations

○○○○○ One unknown

1. $x = 5$ 2. $x = 4$ 3. $x = 6$ 4. $x = 14$ 5. 40 ml

○○○○○ Two unknowns

6. $x = 7$, $y = 3$ 7. $x = 2$, $y = 3$ 8. pen £1, pad £2 9. endless (same line) 10. about 25.5 ml and 44.5 ml

○○○○○○ Quadratics (Δ first)

11. ±7 12. Δ = 1: x = 2 or 3 13. Δ = 36: x = 2 or −4 14. x = 0 or 7 15. $(x + 4)^2 - 6$ 16. Δ = 12: −0.27 or −3.73

○○○○○○○ Methods, words and inequalities

17. x = 2, y = 5 18. x = 1, y = 2 19. $x^2 - x - 12 = 0$: x = 4 or −3 20. $3(x - 2)^2 - 7$ 21. x < 2 (open circle) 22. 6 m (reject −8) 23. x = 1 or 2

○○○○○○ Find the unknown

24. k = ±8 ($k^2 - 64 = 0$) 25. k = 1, other root −4 ($9 + 3k - 12 = 0$) 26. k > 9 ($36 - 4k < 0$) 27. k = 2, other root −1/2 ($8 - 10 + k = 0$) 28. y = 1, a = 3 ($3a + 2 = 11$) 29. k = 4 ($16 - 4k = 0$)

16 Matrices

○○○ Reading

1. 1 × 3 2. 4 × 2 3. 7

○○○ Adding and scaling

4. [[1, 7], [2, 6]] 5. [[6, −3], [0, 12]] 6. [[5, 2], [1, 0]]

○○○○ Multiplying

7. [14] 8. [[6], [12]] 9. [[5, 4], [1, 2]] 10. No: inner numbers 2 and 3 don't match

○○○ Inverse and equations

11. 4 − 6 = −2 12. [[3, −1], [−5, 2]] (determinant 1) 13. x = 2, y = 1

○○○○○○ Find the unknown

14. k = 3 ($4k - 12 = 0$) 15. k = ±3 ($k^2 - 9 = 0$) 16. a = 3, b = 4 ($2 + 3a = 11$, $b + 3 = 7$) 17. x = 3, y = −2 (match entries) 18. k = 2 (top-left of the product: $2k - 3 = 1$) 19. yes: the product is [[1, 0], [0, 1]] = I

17 Sequences and series

○○○○○ Arithmetic

1. 64 2. 20 3. 1275 4. 710 5. 3

○○ Sigma

6. 3 + 5 + 7 + 9 + 11 = 35 7. Σ from i = 1 to 10 of 4i (total 220)

○○○○○ Geometric

8. 4374 9. 1275 10. 24 11. 3/4 12. G.P. with r = 1.5; no (r > 1)

○○○○○○ Find the unknown

13. d = 4, a = 5 ($6d = 24$) 14. n = 12 ($3n^2 + 5n - 492 = 0$; reject the negative root) 15. r = 2, a = 3 ($r^3 = 96/12 = 8$) 16. r = 1/3 ($8/(1 - r) = 12$) 17. n = 9 ($S_8 = 3(2^8 - 1) = 765$, $S_9 = 1533$) 18. x = 3 ($(x + 3)^2 = x(x + 9)$, so $3x = 9$; terms 3, 6, 12)

18 Functions

○○○○ Using the notation

1. 5, −1, −13 2. 6, 12 3. 2, 20 4. 3a + 5

○○○ Domain and image

5. x = 4 6. x ≥ 1 7. [5, ∞): outputs of 5 or more

○○○○ Pieces, chains and inverses

8. £5, £10 9. 10, 18 10. $f^{-1}(x) = (x + 2)/5$ 11. $f^{-1}(x) = 4x - 1$

○○○○○○ Find the unknown

12. $x = 5$ ($5x = 25$) 13. $x = 2$ or $x = -6$ ($(x + 6)(x - 2) = 0$) 14. $a = 3$, $b = 1$ ($3a = 9$)
15. $x = 5$ ($fg(x) = 3(x + 2) - 1 = 3x + 5$) 16. $k = 3$ ($8k + 1 = 25$) 17. $a = 5$ ($f^{-1}(a) = 6$ means $a = f(6)$)

19 Graphs of functions

○○○ Points and plotting

1. 2nd (top-left) 2. $y = -5, -3, -1, 1, 3$ (a straight line) 3. y-int 0; $x = 0$ and $x = 4$

○○○○ Even, odd, neither

4. even 5. odd 6. neither (try $x = 1$ and $x = -1$: 2 and 0) 7. odd

○○○○ Recognise and move

8. negative powers (even: always positive) 9. (1, 5) 10. moved left 4 11. x^4

○○○○○○ Find the unknown

12. $h = 3$, $k = -2$ 13. (2, 1): inside +4 moves left 4 14. (1, 1): left 1, down 4 15. (4, 6): $y \times (-2)$ 16. $a = 2$ ($7 = 4a - 1$) 17. $k = -2$ (denominator $-2 - k = 0$)

20 The straight line

○○○○ Reading y = mx + c

1. $m = 5$, $c = -2$ 2. $m = -3$, $c = 7$ 3. $y = -2x + 5$: $m = -2$, $c = 5$ 4. $y = 30x + 40$

○○○○ Two points

5. 3 6. $y = 3x - 1$ 7. $y = -2x + 3$ 8. $y = 3x + 5$

○○○ Sketch and perpendicular

9. (0, 3) and (5, 0) 10. $-1/4$ 11. $3/2$

○○○○ Parallel, perpendicular or neither?

12. parallel 13. perpendicular 14. neither 15. perpendicular

○○○ Lines through a point

16. $y = 2x - 5$ 17. $y = -x/2 + 2$ 18. $y = -x/2 + 5$

○○○○ Midpoint, distance, intersection

19. (2, 6) 20. 5 21. (5, 7) 22. (3, 7)

○○○○○○ Find the unknown

23. $k = 5$ 24. $k = 4$ 25. $k = -1/4$ 26. $a = 2$ 27. yes 28. $y = -2x + 7$; (3.5, 0)

21 The exponential function

○○○○ Power or exponential?

1. power 2. exponential growth 3. exponential decay 4. power

○○○ Using $y = A \times b^x$

5. $200 \times 2^6 = 12{,}800$ 6. $15{,}000 \times 0.8^3 = £7680$ 7. $500 \times 1.03^4 \approx £562.75$

○○○ e

8. 7.39 and 0.14 9. about 62.5 °C 10. 4 A and $4e^{-1} \approx 1.47$ A

○○○○○○ Find the unknown

11. A = 5, b = 3 ($b^2 = 45/5 = 9$) 12. b = 2, A = 5 ($b^2 = 40/10 = 4$) 13. 12.5 mg (4 half-lives: $200 \times (1/2)^4$) 14. 80 minutes ($4800/300 = 16 = 2^4$) 15. 6% ($m^2 = 1.1236$, $m = 1.06$) 16. a 15% decrease ($1 - 0.85$)

22 Logarithms

○○○○○○ What is it?

1. 2 2. 5 3. −3 4. 0 5. $\log_5 125 = 3$ 6. $2^6 = 64$

○○○ Laws

7. log 20 8. log 2 (since 8/4 = 2) 9. 3 log x + log y − log z

○○○○○ Solve

10. 4 11. 2.29 12. ln 20 ≈ 3.00 13. log 2 / log 1.08 ≈ 9.0 years 14. $t = (1/\lambda) \ln(N_0/N)$

○○○○○○ Find the unknown

15. x = 64 ($x = 2^6$) 16. $x = e^3 \approx 20.09$ 17. x = 5 ($x^2 - 3x - 10 = 0$; reject x = −2) 18. x = 0.557 (2x + 1 = log 30 / log 5 = 2.113) 19. x = 1.756 (x = log 5 / (log 5 − log 2)) 20. k = 0.458 (e^(−2k) = 0.4, k = −ln 0.4 / 2)

23 Tables and charts

○○○○ Which chart?

1. pie 2. histogram 3. scatter plot 4. line graph

○○ Build it

5. 1 | 2 5; 2 | 1 2 2 7; 3 | 0 4; 4 | 1; key 2 | 1 means 21 6. 25/90 × 360 = 100°

○○○ Detective

7. correlation ≠ causation (keen students choose the front) 8. 99 looks 4× bigger than 96; start at 0 9. only if it's an error; check it, and say so if you remove it

○○○○○○ Find the unknown

10. 72/360 × 250 = 50 11. 45° is 1/8 of 360°, so 30 × 8 = 240 12. 3.5 × 10 = 35 (frequency = density × width) 13. 24 ÷ 3 = 8 (density = frequency ÷ width) 14. 2 × 10 + 6 × 5 + 1 × 15 = 20 + 30 + 15 = 65 15. x = 360 − 150 − 120 = 90°; 90/360 × 120 = 30

24 Statistics

○○○○ Averages

1. 62 ÷ 7 ≈ 8.86 2. 8 3. 7 4. the mean (it becomes 20.25; median 9)

○○○○ Spread

5. 12 6. 6th; (6 + 1)/2 = 3.5th from each end (halfway between the 3rd and 4th) 7. 8/3 ≈ 2.67, σ ≈ 1.63 8. √(8/2) = 2

○○ Think

9. a few mansions skew the mean upward 10. the spread (standard deviation / IQR)

○○○○○○ Find the unknown

11. 8 (total 5 × 8 = 40; 40 − 32 = 8) 12. x = 6 (6x = 3 × 12 = 36) 13. 17 (60 + 25 = 85; 85 ÷ 5) 14. 8 (60 − 20 = 40; 40 ÷ 5) 15. (a) mean 12, σ = 2 (b) mean 21, σ = 6 16. k = 4 (Σfx = 10 + 2k = 1.8 × (6 + k))

25 Probability and Bayes

○○○○○ Two dice

1. 6/36 = 1/6 2. 4/36 = 1/9 3. 5/6 4. (18 + 6 − 3)/36 = 21/36 = 7/12 5. 1/6 (only blue 5)

○○○ Cards and tables

6. 4/52 × 3/51 = 1/221 7. 15/30 = 0.5 8. yes; 1/2 × 1/6 = 1/12

○ Bayes

9. 20 ill, 18 positive; 980 well, 49 positive; 18/67 ≈ 0.27

○○○○○○ Find the unknown

10. 0.6 + 0.5 − 0.8 = 0.3; 0.6 × 0.5 = 0.3, so independent 11. 0.4 × 0.25 = 0.1 12. 0.12 ÷ 0.3 = 0.4 13. 0.8 − 0.35 = 0.45 (no overlap) 14. 0.7 = 0.4 + p − 0.4p, so p = 0.5 15. bus 120, late 30; others 180, late 18; 48 late; 30/48 = 0.625

26 The Normal distribution

○○○○ z-scores

1. 1 2. −2 3. −1 4. 100 + 1.645 × 15 ≈ 124.7

○○○○ Areas

5. Φ(2) = 0.977 6. 1 − Φ(1) = 0.159 7. 0.683 (the 68% rule) 8. 0.95 − 0.5 = 0.45

○○ Backwards

9. 60 + 1.28 × 10 ≈ 72.8 10. the average of 20: sums of many independent bits become bell-shaped

○○○○○○ Find the unknown

11. Φ(z) = 0.10, z = −1.2816; 100 − 1.2816 × 15 ≈ 80.8 12. z = ±1.96; 100 ± 29.4: 70.6 to 129.4 13. z = −1.645; μ = 250 + 1.645 × 4 ≈ 256.6 g 14. z = 1.2816; σ = (75 − 55)/1.2816 ≈ 15.6 15. 70 = μ + 2σ and 40 = μ − σ, so σ = 10, μ = 50 16. Φ(z) = 0.9332, z = 1.5; σ = 15/1.5 = 10

27 Angles

○○○○ Degrees

1. 120° 2. 105° (hour hand at 285°, minute at 180°) 3. obtuse 4. a quarter turn clockwise

○○○○ Converting

5. 2π/3 6. 30° 7. 143.2° 8. 5π/4

○○○ Arcs

9. 12 cm 10. 4π ≈ 12.6 m² 11. 6 ÷ 12 = 0.5 rad

○○○○○ Find the unknown

12. r = 6 cm (½r² × 1.5 = 27, so r² = 36) 13. θ = 15 ÷ 6 = 2.5 rad ≈ 143.2° 14. r = 27 cm ((40/360) × 2πr = 6π, so 2πr/9 = 6π) 15. 120° ((θ/360) × 81π = 27π, so θ/360 = 1/3) 16. 750° − 720° = 30°; −5π/3 + 2π = π/3

28 Intro to trigonometry

○○○ Pythagoras

1. 13 2. 8 3. √(4 + 9 + 36) = 7

○○○ Sides

4. 8 sin 40° ≈ 5.14 5. 6 tan 25° ≈ 2.80 6. 4 ÷ sin 55° ≈ 4.88

○○○ Angles

7. $\sin^{-1}(5/13) \approx 22.6°$ 8. $\cos^{-1}(0.7) \approx 45.6°$ 9. $\tan^{-1}(1/12) \approx 4.8°$

○○○○○ Find the unknown

10. sin θ = 5/6, θ ≈ 56.4° 11. 30 tan 28° ≈ 15.95 m 12. 45 ÷ tan 18° ≈ 138.50 m 13. sin 5° = 0.8/H, H = 0.8 ÷ sin 5° ≈ 9.18 m 14. O = 0.6 × 15 = 9; H = √(81 + 225) ≈ 17.49 15. 40 tan 31° − 40 tan 25° ≈ 24.03 − 18.65 = 5.38 m

29 Trig graphs

○○○ Circle

1. (0, 1) 2. negative (left half) 3. −0.5

○○○○ Graphs

4. 2π, 4 5. 2π/3 6. 4π 7. x = π/2 and 3π/2

○○ Wheels

8. h = 12 − 10 cos(2πt), t in minutes 9. 0.5 − 0.125/6 + 0.03125/120 ≈ 0.479427

○○○○ Recipes

10. 1 − 0.25/2 + 0.0625/24 ≈ 0.877604 (true 0.877583) 11. 0.3 + 0.027/3 = 0.309 (true 0.3093) 12. sin 0.866 < arc 1.047 < tan 1.732 13. the upright side, h cos θ: sin grows at cos's rate

○○○○○ Find the unknown

14. cos 225° = −cos 45° = −1/√2 (Q3); tan 300° = −tan 60° = −√3 (Q4) 15. α = $\sin^{-1}$ 0.6 = 36.87°, θ = 180° − 36.87° = 143.13°; cos θ = −0.8, tan θ = −0.75 16. D = (7 + 1)/2 = 4, A = (7 − 1)/2 = 3, B = 360/90 = 4 17. B = 360/72 = 5 18. period 45° = 180°/B, so B = 4

30 Trig identities and equations

○○○ Identities

1. 12/13 2. 3 3. no: sin 180° = 0 but 2 sin 90° = 2

○○○○ Equations

4. 53.13°, 126.87° 5. 60°, 300° 6. 45°, 225° 7. 210°, 330°

○○○ r sin form

8. 5 sin(θ + 53.13°) 9. 5 10. θ + 53.13 = 90, so θ = 36.87°

○○○○○ Find the unknown

11. $\theta_1 = -36.87°$; $180° - (-36.87°) = 216.87°$, $-36.87° + 360° = 323.13°$ 12. $\theta_1 = \cos^{-1}(-0.25) = 1.823$; $2\pi - 1.823 = 4.460$ 13. $\theta_1 = -56.31°$; $+ 180° = 123.69°$, $+ 360° = 303.69°$ 14. $2\theta = 60°, 300°, 420°, 660°$ (range 0°–720°), so $\theta = 30°, 150°, 210°, 330°$ 15. $\sin^2\theta = 1 - 25/169 = 144/169$, $\sin\theta = -12/13$ (Q3); $\tan\theta = 12/5$

31 Solving triangles

○○○ Sine rule

1. $8 \sin 75°/\sin 40° \approx 12.02$ 2. $\sin^{-1}(9 \sin 110°/12) \approx 44.8°$ 3. third angle 80°; $20 \sin 65°/\sin 80° \approx 18.41$

○○○ Cosine rule

4. ≈ 6.13 5. $\cos^{-1}((16 + 49 - 81)/56) \approx 106.6°$ 6. 120°: $\sqrt{(149 + 70)} \approx 14.80$ cm

○○ Choose

7. no, only their ratios (the triangle could be any size) 8. 13.5

○○○○○ Find the unknown

9. $\sin C = 48/80 = 0.6$, $C = 36.87°$ or $143.13°$ 10. $35 = ½ \times 10 \times b \times 0.5$, so $b = 14$ cm 11. $C = 70°$; $a = 15 \sin 40°/\sin 70° \approx 10.26$; area $= ½ \times 10.26 \times 15 \times \sin 70° \approx 72.31$ 12. $c^2 = 49 + 81 - 63 = 67$, $c \approx 8.19$; area $= ½ \times 7 \times 9 \times \sin 60° \approx 27.28$ 13. A is obtuse, so one triangle: $\sin B = 5 \sin 120°/9 \approx 0.481$, $B \approx 28.76°$

32 Vectors

○○○ Trips, adding, lengths

1. $2i + 3j$; $4i - 7j$ 2. $\sqrt{13} \approx 3.61$ 3. $-3i + 15j$; $\sqrt{234} \approx 15.30$

○○○ Dot product

4. $8 - 3 = 5$ 5. $\cos^{-1}(5 \div (\sqrt{5} \times \sqrt{10})) = 45°$ 6. yes: $12 - 12 = 0$

○○○ Cross product

7. $3\times2 - 1\times1 = 5$ 8. $6i - 3j + k$ 9. $6 - 6 + 0 = 0$ and $0 - 3 + 3 = 0$

○ Distance or displacement?

10. 7 km; $\sqrt{(9 + 16)} = 5$ km

○○○○○○ Find the unknown

11. $2k - 4 = 0$, $k = 2$ 12. $t = 2$ (from $3 \to 6$), so $k = 4$ 13. length 13: $(5/13)i - (12/13)j$ 14. $b - a = 6i - 8j$; $\sqrt{(36 + 64)} = 10$; $((1 + 7)/2, (3 - 5)/2) = (4, -1)$ 15. $15 \times (0.6i + 0.8j) = 9i + 12j$ 16. $\tan^{-1}(1) = 45°$; $x < 0$ so $+ 180°$: 225°

33 Polar coordinates

○○ Polar to Cartesian

1. (3, 5.20) 2. (−1.41, 1.41)

○○○○ Cartesian to polar (mind the quadrant!)

3. (5, 53.13°) 4. (5, 126.87°) 5. (2.83, 225°) 6. (7.07, 315°)

○○○○ Curves and equations

7. a circle of radius 3 8. $r = 1$, the point (0, 1) 9. $r \cos\theta = 4$, so $r = 4/\cos\theta$ 10. $r^2 = 25$, so $r = 5$

○ Double angle

11. $2 \times 0.7071 \times 0.7071 = 1$

○○○○○ Find the unknown

12. × r: $r^2 = 6r \sin\theta$, $x^2 + y^2 = 6y$, $x^2 + (y - 3)^2 = 9$: circle, centre (0, 3), radius 3 13. $\tan\theta = \sqrt{3}$, $\theta = \pi/3$ (60°) 14. on the negative y axis: (5, 270°) = (5, 3π/2) 15. $\sin\theta = 4/8 = 0.5$ and $x < 0$, so $\theta = 150°$; $x = 8\cos 150° \approx -6.93$ 16. $y = x\tan 135° = (-3)(-1) = 3$; $r = \sqrt{(9 + 9)} \approx 4.24$

34 Gradients of curves

○ Gradient from scratch

1. $(9.0601 - 9) \div 0.01 = 6.01$ (the exact answer is 6)

○○○○ Power rule

2. $20x^3$ 3. $6x - 7$ 4. ½x^(−½) = 1/(2√x) 5. $-12x^{-4} = -12/x^4$

○○○ Standard derivatives

6. 6e^(2x) 7. 10 cos 5x 8. 1/4 = 0.25

○○ Stationary points

9. $x = 3$, $y = -8$: minimum (second derivative $2 > 0$) 10. (1, −2) minimum ($6 > 0$); (−1, 2) maximum ($-6 < 0$)

○○○ Tangents and normals

11. point (1, 3), $m = 4$: $y = 4x - 1$ 12. $m = -1/4$: $y = -x/4 + 13/4$ 13. $dy/dx = -6/x^2 = -3/2$: $y = -3x/2 + 6$

○○○○○ Find the unknown

14. $a = 2$ ($2a + 3 = 7$) 15. (4, −2) ($2x - 5 = 3$, so $x = 4$) 16. $k = -12$ ($3(2)^2 + k = 0$) 17. (3, −8) and (−3, 10) ($3x^2 - 12 = 15$, so $x^2 = 9$) 18. $b = -3$ ($4 + b = 1$); point (2, −2), tangent $y = x - 4$

35 Chain, product, quotient

○○ Higher derivatives

1. $12x^2 - 6$ 2. $\sin x$

○○○ Chain rule

3. $10(2x + 1)^4$ 4. 6x e^(3x²) 5. $3x^2 \div (2\sqrt{(1 + x^3)})$

○○ Product rule

6. $2x\sin x + x^2\cos x$ 7. $e^x + x\,e^x = (1 + x)e^x$

○○ Quotient rule (or rewrite!)

8. $-2/(x - 1)^2$ 9. $1 - 1/x^2$ (easiest: $x + x^{-1}$ first)

○○ Implicit

10. $-x/y$; $-3/4$ 11. $1/(1 - 3y^2)$

○○○ Exam combinations

12. dy/dx = e^(−2x)(1 − 2x) = 0, $x = ½$; (½, 1/(2e)), maximum 13. $(x^2 + 2x)/(x + 1)^2 = 3/4$ 14. $2x/(x^2 + 1)$; 1

○○○○○ Find the unknown

15. a = 6 (a e^0 = 6) 16. a = 6 (4a(1)³ = 24) 17. k = 3 (2/(1 + k) = ½) 18. c = 4 (c/c² = 1/c = ¼) 19. b = 2 (4 + b³ = 12); dy/dx = −2x/(3y²) = −1/3

36 Integration and areas

○ Trapezium rule

1. 3 and 2.75

○○○○○ Indefinite integrals

2. 2x³ + c 3. x³ − 2x² + 5x + c 4. 2 ln|x| + c 5. ⅓e^(3x) + c 6. 2 sin 2x + c

○ Find the curve

7. y = 3x² + 2x + 5

○○○ Definite integrals

8. 9 9. 14 10. ½ (each side is ¼; the integral itself is 0)

○○○○○ Rewrite first

11. −2/x² + c (4x⁻³ → 4x⁻²/(−2)) 12. 4x^(3/2) + c 13. x³/3 + 2x² + 4x + c 14. x³/3 − 5 ln|x| + c 15. 2 (crosses at x = 1: pieces −⅔ and 4/3)

○○○○○ Find the unknown

16. y = 2x² + 3x − 3 (2 + 3 + c = 2) 17. k = 4 (k² − 1 = 15) 18. a = 2 (9a/2 + 6 = 15) 19. k = 4 (k³ = 64) 20. f(x) = 2x³ + 2/x + 1 (2 + 2 + c = 5)

37 Parts and substitution

○○○○ Integration by parts

1. x sin x + cos x + C 2. ½x e^(2x) − ¼e^(2x) + C 3. e^x(x² − 2x + 2) + C 4. 1

○○○○ Integration by substitution

5. (2x − 5)⁷/14 + C 6. (x² + 4)⁶/2 + C 7. e^(sin x) + C 8. 15/4 (limits become 1 and 2)

○○○ Which technique?

9. substitution: −½cos(x²) + C 10. parts: sin x − x cos x + C 11. table: −⅕cos 5x + C

○○○○ Linear inside shortcut

12. (5x − 2)⁴/20 + C 13. −⅓cos(3x + 1) + C 14. −½e^(−2x) + C 15. ¼ ln|4x + 1| + C

○○○○○ Find the unknown

16. k = 1 ([x e^x − e^x] gives (k − 1)e^k + 1 = 1) 17. a = 2 (1/(5a) = 1/10) 18. k = √3 (((k² + 1)⁴ − 1)/4 = 255/4, so k² + 1 = 4) 19. y = (x − 1)e^x + 4 (−1 + C = 3) 20. k = π/4 (½ sin 2k = ½, so 2k = π/2)

38 Partial differentiation

○○○ Freeze the other letter

1. 3x² + 4y, 4x + 2y 2. 10xy³, 15x²y² 3. ∂z/∂x = −3, ∂z/∂y = 7

○○ With the chain rule

4. 3cos(3x + 2y), 2cos(3x + 2y) 5. y e^(xy), x e^(xy)

○○ The hill h = 12 − x² − 2y²

6. ∂h/∂x = −4, ∂h/∂y = 4 7. equally steep (both size 4); east goes down, north goes up

○ Second partials

8. both are 6xy²

○ Small changes

9. δA ≈ 5(0.1) + 10(0.1) = 1.5 (true: 1.51)

○○○ Product rule and notation

10. (1 + xy)e^(xy), x²e^(xy) 11. 2x sin(xy) + x²y cos(xy) 12. 3x²y², 2x³y, 6x²y

○○○○○ Find the unknown

13. a = 3 (2axy = 4a = 12) 14. b = −1 (3 + 3b = 0) 15. k = 2 (ky e⁰ = 3k = 6) 16. (3, −1) (2x − 6 = 0, 2y + 2 = 0) 17. h = 1 (2πrh = πr², so 4πh = 4π)

39 Exam pocket cards

Worked answers, line by line, are on the last pages of this chapter itself.

Slide Errata

Small slips found on the lecture slides while writing this book. Each one is explained where it comes up; here they are all together so you can correct your own copies. Grey with a red wavy line: what the slide says. Green: what it should say.

L01 **Combining fractions: subtraction** see Chapter 2

slide: $\frac{a}{b} - \frac{c}{d} = \frac{ad + bc}{bd}$ ✗

fix: $\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$

L01 **Significant figures table, 4 s.f.** see Chapter 3

slide: $0.123456789 \to 0.1234$ ✗

fix: 0.1235 (rounded; 0.1234 is truncated)

L02 **Big example: (x+1)^3** see Chapter 11

slide: $(x+1)^3 = x^3 + 4x^2 + 3x + 1$ ✗

fix: $(x+1)^3 = x^3 + 3x^2 + 3x + 1$

L02 **Difference of two squares example** see Chapter 12

slide: Expand: $x^2 - 4y^2$ ✗

fix: Factorise: $x^2 - 4y^2$

L02 **MASSIVE example** see Chapter 11

slide: $\cdots + 21x^2 + \cdots = x^4 + 10x^3 + 44x^2 + 50x + 24$ ✗

fix: $\cdots + 12x^2 + \cdots = x^4 + 10x^3 + 35x^2 + 50x + 24$

L02 **Tricky factorising example** see Chapter 12

slide: $x^2 + 2y^2 + 2x^3 + 4xy = (x^2 + 2y)(x^2 + 2y + 2x)$ ✗

fix: $x^2 + 2y + 2x^3 + 4xy = (x^2 + 2y)(1 + 2x)$

L03 **Compounds 2, step (4) solved** see Chapter 15

slide: $y = \dfrac{20}{0.55} = 364$ ✗

fix: $y = \dfrac{20}{0.055} \approx 364$

L06 **Example 3, Pythagoras step** see Chapter 28

slide: $5^2 - 2.5^2 = a = 4.33$ ✗

fix: $a = \sqrt{5^2 - 2.5^2} = \sqrt{18.75} \approx 4.33$

L07 **Even powers list** see Chapter 9

slide: $x \times \cdots \times x \text{ (8 times)} = x^6$ ✗

fix: $= x^8$

L07 **Laws of logs 2, example for log a^n** see Chapter 22

slide: $\log a^2 + \log b^3 = \log a^2 b^3$ ✗

fix: e.g. $\log a^3 = 3 \log a$

L07 **Standard form definition** see Chapter 9

slide: $a \times 10^n,\ 1 \leq a \leq 10$ ✗

fix: $a \times 10^n,\ 1 \leq a < 10$

L10 **Determinant answer** see Chapter 32

slide: $\mathbf{a} \times \mathbf{b} = -55\mathbf{i} + 22\mathbf{j} - 11\mathbf{k}$ ✗

fix: $\mathbf{a} \times \mathbf{b} = -55\mathbf{i} + 22\mathbf{j} + 11\mathbf{k}$

L10 **Dot product 3: angle form example** see Chapter 32

slide: $|A| = 9.22,\ |B| = 16.12,\ \theta = 60° \Rightarrow C = 50$ ✗

fix: $9.22 \times 16.12 \cos 60° = 74.3;\ \text{for } C = 50,\ \theta \approx 70.3°$

L10 **Finding theta (cross product magnitude)** see Chapter 32

slide: $14.87 \sin 19.7° = 11.18$ ✗

fix: $14.87 \sin 19.7° \approx 5.0$

L11 **Polar angles diagram** see Chapter 27

slide: $225° = \frac{4\pi}{3}$ ✗

fix: $225° = \frac{5\pi}{4}$

L11 **Polar to Cartesian** see Chapter 33

slide: $y = x \sin\theta$ ✗

fix: $y = r \sin\theta$

L12 **Multiplying two matrices, bottom-left entry** see Chapter 16

slide: $3 \times 5 + 4 \times 8 = 47$ ✗

fix: $3 \times 5 + 4 \times 7 = 43$

L12 **Reversing the process** see Chapter 16

slide: Now let's calculate CD ✗

fix: Now let's calculate DC

L13 **GP sum formula** see Chapter 17

slide: $\sum_{i=1}^{n} a\frac{1-r^n}{1-r}$ ✗

fix: $S_n = a\frac{1-r^n}{1-r}$ (no $\sum$ needed)

L13 **GP sum to infinity** see Chapter 17

slide: $S_\infty = \frac{a}{r-1}$ ✗

fix: $S_\infty = \frac{a}{1-r}$

L13 **Geometric progression example (footnote)** see Chapter 17

slide: $3072 = 3 \times 2^{11}$ ✗

fix: $3072 = 3 \times 2^{10}$

L13 **Mapping: the squaring function** see Chapter 6

slide: $\mathbb{R} \to \mathbb{Q}^+$ ✗

fix: $\mathbb{R} \to [0, \infty)$ (non-negative reals)

L13 **Sets: {1,2,3,4} in set-builder form** see Chapter 6

slide: $\{x \mid 1 \leq x \leq 4\}$ ✗

fix: $\{x \mid 1 \leq x \leq 4,\ x \in \mathbb{Z}\}$

L14 **Finding stationary points** see Chapter 34

slide: differentiating finds the zeros ✗

fix: it finds where the gradient is zero (stationary points)

L15 **Combined example, multiplied out** see Chapter 35

slide: $340x^4 + 1728x^3 + \ldots$ ✗

fix: $320x^4 + 1728x^3 + \ldots$

L15 **Double differentiation example** see Chapter 35

slide: $\dfrac{d^2y}{dx^2} = 24x - 4 + \dfrac{1}{x^3}$ ✗

fix: $\dfrac{d^2y}{dx^2} = 24x - 4 + \dfrac{2}{x^3}$

L16 **Integral of 1/x** see Chapter 36

slide: $\displaystyle\int \frac{1}{x}\,dx = \ln x + C$ ✗

fix: $\displaystyle\int \frac{1}{x}\,dx = \ln|x| + C$

L16 **Limits: definite integral of ax^n** see Chapter 36

slide: $\dfrac{a}{n+1}\left(b^{n+1} - c^{n+1}\right)$ ✗

fix: $\dfrac{a}{n+1}\left(c^{n+1} - b^{n+1}\right)$ (top limit minus bottom)

L16 **Trapezium rule** see Chapter 36

slide: $\frac{h}{2}(y_0 + y_1 + \cdots + y_n)$ ✗

fix: $\frac{h}{2}\big(y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n\big)$

L17 **Definite integral example (parts)** see Chapter 37

slide: $\big[-xe^{x}\big]_0^2, \quad (-2e^2 - 0) - (e^2 - (-1))$ ✗

fix: $\big[-xe^{-x}\big]_0^2, \quad (-2e^{-2} - 0) - (e^{-2} - 1) = 1 - 3e^{-2}$

L17 **Substitution, second example** see Chapter 37

slide: $\int \frac{\sin y}{4}\,dy = \frac{\cos y}{4} + c$ ✗

fix: $\int \frac{\sin y}{4}\,dy = -\frac{\cos y}{4} + c = -\frac{\cos 2x^2}{4} + c$